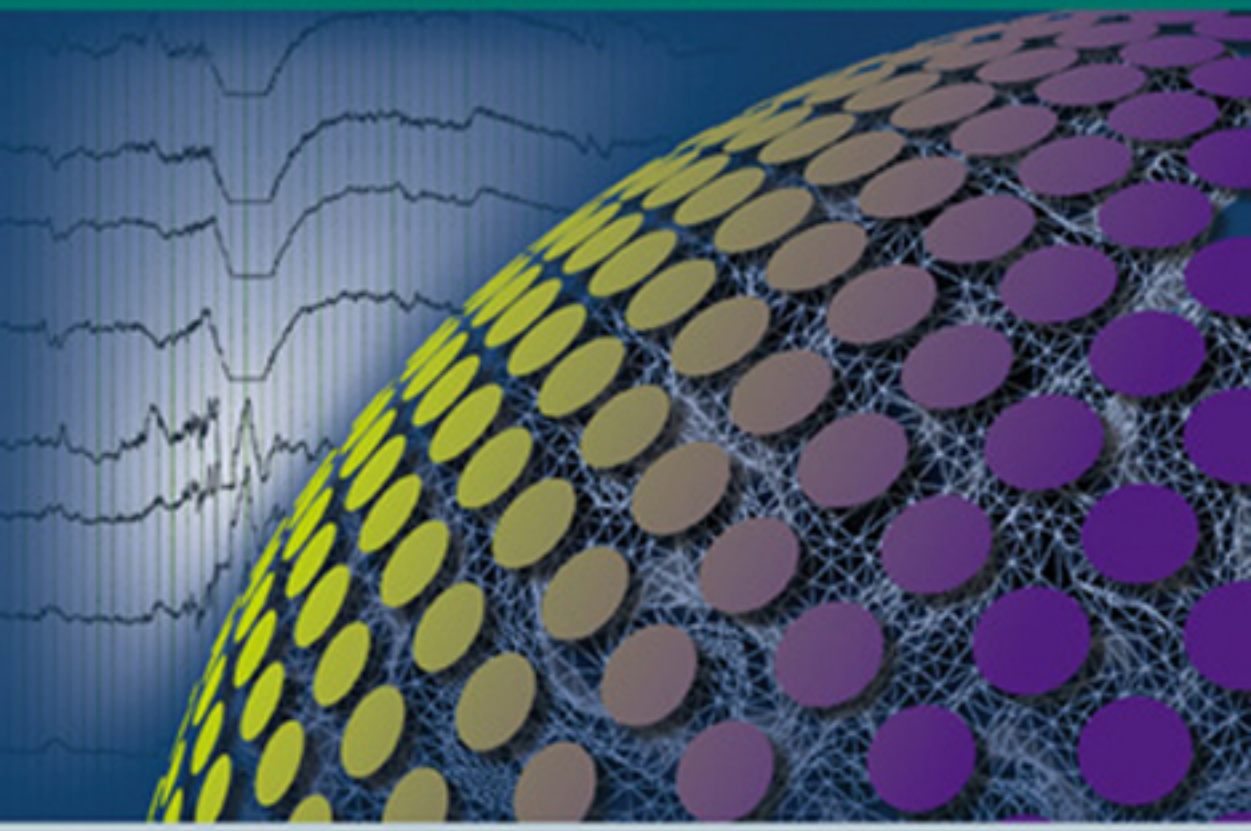
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# Quantitative EEG Analysis Methods and Clinical Applications

Shanbao Tong
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*Editors*



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**Library of Congress Cataloging-in-Publication Data**
A catalog record for this book is available from the U.S. Library of Congress.

**British Library Cataloguing in Publication Data**
A catalogue record for this book is available from the British Library.

ISBN-13: 978-1-59693-204-3

**Cover design by Yekaterina Ratner**

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**685 Canton Street**
**Norwood, MA 02062**

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# Foreword

It has now been 80 years since Hans Berg made the first recordings of the human brain activity using the electroencephalogram (EEG). Although the recording device has been refined, the EEG remains one of the principal methods for extracting information from the human brain for research and clinical purposes. In recent years, there has been significant growth in the types of studies that use EEG and the methods for quantitative EEG analysis. The growth in methodology development has had to keep pace with the growth in the wide range of EEG applications. This timely monograph edited by Shanbao Tong and Nitish V. Thakor provides a much-needed, up-to-date survey of current methods for analysis of EEG recordings and their applications in several key areas of brain research. This monograph covers the topics from the background biophysics and neuroanatomy, EEG signal processing methods, and clinical and research applications to new recording methodologies.

This book begins with a review of essential background information describing the biophysics and neuroanatomy of the EEG along with techniques for recording and preprocessing EEG. The recently developed independent component analysis techniques have made this preprocessing step both more feasible and more accurate. The next chapters of the monograph focus on univariate and bivariate methods for EEG analysis, both in the time and frequency domains. The book nicely assembles in Chapter 3 linear, nonlinear, and information theoretic-based methods for univariate EEG analysis. Chapter 4 presents bivariate extensions to the mutual information analyses and discusses methods for tracking phase synchronization. Chapter 5 concludes with a review of the current state of the art for solving the EEG inverse problem. The topics here include the biophysics of the EEG and single and multiple dipole fitting procedures in addition to the wide range of discrete three-dimensional distributed tomographic techniques.

The applications section starting in Chapter 6 of the monograph explores a broad range of cutting-edge brain research questions to which quantitative EEG analyses are being applied. These include epilepsy detection and monitoring, monitoring brain injury, controlling brain-computer interfaces, monitoring depth of general anesthesia, tracking sleep stages in normal and pathological conditions, and analyzing EEG signatures of depression. In addition to these engaging applications, these application chapters also introduce some additional methodologies including wavelet analyses, the Lyapunov exponents, and bispectral analysis. The final three chapters of the monograph explore three new interesting areas: combined EEG and magnetic resonance imaging studies, functional cortical mapping with high resolution EEG, and cortical mapping with intracranial EEG.

Berg would be quite happy to know that his idea of measuring the electrical field potentials of the human brain has become such a broadly applied tool. Not only has

the EEG technology become more ubiquitous, but its experimental and clinical use has also broadened. This monograph now makes the quantitative methods needed to analyze EEG readily accessible to anyone doing neuroscience, bioengineering, or signal processing. Coverage of quantitative EEG methods applied to clinical problems and needs should also make this book a valuable reference source for clinical neuroscientists as well as experimental neuroscientists. Indeed, this comprehensive book is a welcome reference that has been long overdue.

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# Preface

Since Hans Berger recorded the first electroencephalogram (EEG) from the human scalp and discovered rhythmic alpha brain waves in 1929, EEG has been useful tool in understanding and diagnosing neurophysiological and psychological disorders. For decades, well before the invention of computerized EEG, clinicians and scientists investigated EEG patterns by visual inspection or by limited quantitative analysis of rhythms in the waveforms that were printed on EEG chart papers. Even now, rhythmic or bursting patterns in EEG are classified into $\delta$, $\theta$, $\alpha$, and $\beta$ (and, in some instances, $\gamma$) bands and burst suppression or seizure patterns. Advances in EEG acquisition technology have led to chronic recording from multiple channels and resulted in an incentive to use computer technology, automate detection and analysis, and use more objective quantitative approaches. This has provided the impetus to the field of quantitative EEG (qEEG) analysis.

Digital EEG recording and leaps in computational power have indeed spawned a revolution in qEEG analysis. The use of computers in EEG enables real-time denoising, automatic rhythmic analysis, and more complicated quantifications. Current qEEG analysis methods have gone far beyond the quantification of amplitudes and rhythms. With advances in neural signal processing methods, a wide range of linear and nonlinear techniques have been implemented to analyze more complex nonstationary and nonrhythmic activity. For example, researchers have found more complex phenomena in EEG with the help of nonlinear dynamics and higher-order statistical analysis. In addition, interactions between different regions in the brain, along with techniques for describing correlations, coherences, and causal interactions among different brain regions, have interested neuroscientists as they offer new insights into functional neural networks and disease processes in the brain.

This book provides an introduction to basic and advanced techniques used in qEEG analysis, and presents some of the most successful qEEG applications. The target audience for the book comprises biomedical scientists who are working on neural signal processing and interpretation, as well as biomedical engineers, especially neural engineers, who are working on qEEG analysis methods and developing novel clinical instrumentation. The scope of this book covers both methodologies (Chapters 1–5) as well as applications (Chapters 6–14).

Before we present the qEEG methods and applications, in Chapter 1 we introduce the physiological foundations of the generation of EEG signals. This chapter first explains the fundamentals of brain potential sources and then explains the relation between signal sources at the synaptic level and the scalp EEG. This introduction should also be helpful to readers who are interested in the foundations of source localization techniques.

The first step in any qEEG analysis is to denoise and preprocess the signals recorded on the scalp. Chapter 2 explains how to effectively record the microvolt level EEG signals and remove any artifacts. In particular, different electrode types such as passive and active electrodes, as well as different electrode cap systems and layouts suitable for high-density EEG recordings, are introduced and their potential benefits and pitfalls are described. As one of the most successful techniques for denoising the EEG and decomposing different components, independent component analysis (ICA) is detailed. Thus, Chapter 2 describes the preprocessing of EEG signals as the essential first step before further quantitative interpretation.

Chapter 3 reviews the most commonly used quantitative EEG analysis methods for single-channel EEG signals, including linear methods, nonlinear descriptors, and statistical measures. This chapter covers both conventional spectral analysis methods for stationary processes and time-frequency analysis applied to nonstationary processes. It has been suspected that EEG signals express nonlinear interactions and nonlinear dynamics, especially in signals recorded during pathological disorders. This chapter introduces the methods of higher-order statistical (HOS) analysis and nonlinear dynamics in quantitative EEG (qEEG) analysis. In addition, statistical and information theoretic analyses are also introduced as qEEG approaches.

Even though single-channel qEEG analysis is useful in a large majority of neural signal processing applications, the interactions and correlations between different regions of the brain are also equally interesting topics and of particular usefulness in cognitive neuroscience. Chapter 4 introduces the four most important techniques for analyzing the interdependence between different EEG channels: cross-correlation, coherence, mutual information, and synchronization.

Chapter 5 describes EEG source localization, also called the EEG inverse problem in most literature. A brief historical outline of localization methods, from single and multiple dipoles to distributions, is given. Technical details of the formulation and solution of this type of inverse problem are presented. Readers working on EEG neuroimaging problems will be interested in the technical details of low resolution brain electromagnetic tomography (LORETA) and its variations, sLORETA and eLORETA.

Chapter 6 presents one of the most successful clinical applications of qEEG—the detection and monitoring of epileptic seizures. This chapter describes how wavelets, synchronization, Lyapunov exponents, principal component analysis (PCA), and other techniques could help investigators extract information about impending seizures. This chapter also discusses the possibility of developing a device for detecting and monitoring epileptic seizures.

Global ischemic injury is a common outcome after cardiac arrest and affects a large population. Chapter 7 describes how EEG signals change following hypoxic-ischemic brain injury. This chapter presents the authors' success in using an entropy measure of the EEG signals as a marker of brain injury. The chapter reviews the theory based on various entropy measures and derives novel measures called information quantity and subband information quantity. A suitable animal model and results from carefully conducted experiments are presented and discussed. Experimental results of hypothermia treatment for neuroprotection are evaluated using these qEEG methods to quantitatively evaluate the response to temperature changes.

Brain computer interface (BCI) may emerge as a novel method to control neural prosthetics and human augmentation. Chapter 8 interprets how qEEG techniques could be used as a direct nonmuscular communication channel between the brain and the external world. The approaches in Chapter 8 are based on two types of oscillatory EEG: the steady-state visual evoked potentials from the visual cortex and the sensorimotor rhythms from the sensorimotor cortex. Details of their physiological basis, principles of operation, and implementation approaches are also provided.

Reducing the incidence of unintentional recall of intra-operative events is an important goal of modern patient safety–oriented anesthesiologists. Chapter 9 provides an overview of the clinical utility of assessing the anesthetic response in individual patients undergoing routine surgery. qEEG can predict whether patients are forming memories or can respond to verbal commands. In Chapter 9, the readers will learn about EEG acquisition in the operating room and how the qEEG can be used to evaluate the depth of anesthesia.

Chapter 10 presents an overview of the application of qEEG in one of the most fundamental aspects of everyone's life: sleep. This chapter introduces how qEEG, electromyogram (EMG), electro-oculogram (EOG), and respiratory signals can be used to detect sleep stages and provides clinical examples of how qEEG changes under sleep-related disorders.

Chapter 11 reviews the history of qEEG analysis in psychiatry and presents the application of qEEG as a biomarker for psychiatric disorders. A number of qEEG approaches, including cordance and the antidepressant treatment response (ATR) index, are nearing clinical readiness for treatment management of psychiatric conditions such as major depression. Cautionary concerns about assessing the readiness of new technologies for clinical use are also raised, and criteria that may be used to aid in that assessment are suggested.

EEG has been known to have a high temporal resolution but a low spatial resolution. Combining EEG with functional magnetic resonance imaging (fMRI) techniques may provide high spatiotemporal functional mapping of brain activity. Chapter 12 introduces technologies registering the fMRI and EEG source images based on the volume conduction model. The chapter addresses theoretical and practical considerations for recording and analyzing simultaneous EEG-fMRI and describes some of the current and emerging applications.

Chapter 13 presents a methodology to assess cortical activity by estimating statistically significant sources using noninvasive high-resolution electroencephalography (HREEG). The aim is to assess significant differences between the cortical activities related to different experimental tasks, which is not readily appreciated using conventional time domain mapping procedures.

Chapter 14 reviews how the advantages of intracranial EEG (iEEG) have been exploited in recent years to map human cortical function for both clinical and research purposes. The strengths and limitations of iEEG and its recording techniques are introduced. Assaying cortical function localization and the cortical connectivity based on the quantitative iEEG are described.

This book should primarily be used as a reference handbook by biomedical scientists, clinicians, and engineers in R&D departments of biomedical companies. Engineers will learn about a number of clinical applications and uses, while clini-

cians will become acquainted with the technical issues and theoretical approaches that they may find useful and consider adopting. In view of the strong theoretical framework, along with several scientific and clinical applications presented in many chapters, we also suggest this book as a reference book for graduate students in neural engineering.

As the editors of this book, we invited many leading scientists to write chapters in each qEEG area mentioned and, together, we worked out an outline of these state-of-the-art collections of qEEG methods and applications. We express our sincere appreciation to all the authors for their cooperation in developing this subject, their unique contributions, and the timely manner in which they prepared the contents of their book chapters.

The editors thank the research sponsoring agencies and their institutions for their support during the period when this book was conceived and prepared. Shanbao Tong has been supported by the National Natural Science Foundation of China, and the Science and Technology Commission and Education Commission of Shanghai Municipality; Nitish V. Thakor acknowledges the support of the U.S. National Institutes of Health and the National Science Foundation. The editors thank Dr. Emery Brown for writing the foreword to this book. Dr. Brown is a leading expert in the field of neural signal processing and has uniquely suited expertise in both engineering and medicine to write this foreword. We are also indebted to Miss Qi Yang, who offered tremendous help in preparing and proofreading the manuscript and with the correspondence, communications, and maintaining the digital content of these chapters. We thank the publication staff at Artech House, especially Wayne Yuhasz, Barbara Lovenvirth, and Rebecca Allendorf, for their consideration of this book, and their patience and highly professional support of the entire editorial and publication process. We are eager to maintain an open line of communication with this book's readers. A special e-mail account, qeegeditor@gmail.com, has been set up to serve as a future communication channel between the editors and the readers.

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*March 2009*

CHAPTER 1

# Physiological Foundations of Quantitative EEG Analysis

Paul L. Nunez

Electroencephalography (EEG) involves recording, analysis, and physiological interpretation of voltages on the human scalp. Electrode voltages at scalp locations ($\mathbf{r}_i$, $\mathbf{r}_j$) are typically transformed to new variables according to $V(\mathbf{r}_i, \mathbf{r}_j, t) \rightarrow X(\xi_1, \xi_2, \xi_3, \ldots)$ in order to interpret raw data in terms of brain current sources. These include several reference and bipolar montages involving simple linear combinations of voltages; Fourier-based methods such as power, phase, and coherence estimates; high spatial resolution estimates such as dura imaging and spline-Laplacian algorithms; and so forth. To distinguish transforms that provide useful information about brain sources from methods that only demonstrate fancy mathematics, detailed consideration of electroencephalogram (EEG) physics and physiology is required. To more easily relate brain microsources $s(\mathbf{r}, t)$ at the synaptic level to scalp potentials, we define intermediate scale (mesoscopic) sources $\mathbf{P}(\mathbf{r}', t)$ in cortical columns, making use of known cortical physiology and anatomy. Surface potentials $\Phi(\mathbf{r}, t)$ can then be expressed as

$$\Phi(\mathbf{r},t) = \iint_S \mathbf{G}_S(\mathbf{r},\mathbf{r}') \cdot \mathbf{P}(\mathbf{r}',t)\, dS(\mathbf{r}')$$

Here the Green's function $\mathbf{G}_S(\mathbf{r}, \mathbf{r}')$ accounts for all geometric and conductive properties of the head volume conductor and the integral is over the cortical surface. EEG science divides naturally into generally nonlinear, dynamic issues concerning the origins of the sources $\mathbf{P}(\mathbf{r}', t)$ and linear issues concerning the relationship of these sources to recorded potentials.

## 1.1 Introduction

The electroencephalogram is a record of the oscillations of electric potential generated by brain sources and recorded from electrodes on the human scalp, as illustrated in Figure 1.1. The first EEG recordings from the human scalp were obtained in the early 1920s by the German psychiatrist Hans Berger [1]. Berger's data, recorded mostly from his children, revealed that human brains typically produce

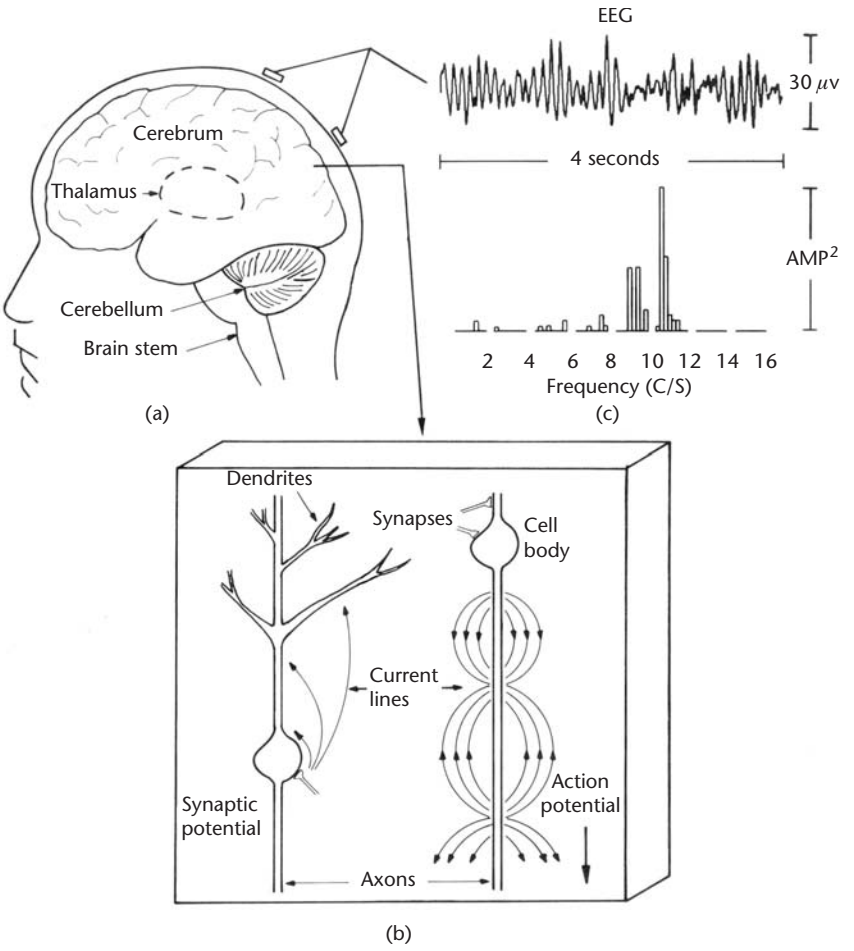



**Figure 1.1** (a) The human brain. (b) Section of cerebral cortex showing microcurrent sources due to synaptic and action potentials. Neurons are actually much more closely packed than shown, about $10^5$ neurons per square millimeter of surface. (c) Each scalp EEG electrode records space averages over many square centimeters of cortical sources. A 4-second epoch of alpha rhythm and its corresponding power spectrum are shown. (*From:* [2]. © 2006 Oxford University Press. Reprinted with permission.)

near-sinusoidal voltage oscillations (alpha rhythms) in awake, relaxed subjects with eyes closed. Early finding that opening the eyes or performing mental calculations often caused substantial reductions in alpha amplitude have been verified by modern studies. Unfortunately, it took more than 10 years for the scientific community to accept these scalp potentials as genuine brain signals. By the 1950s, EEG technology was viewed as a genuine window on the mind, with important applications in neurosurgery, neurology, and cognitive science.

This chapter focuses on the fundamental relationship between scalp recorded potential $V(\mathbf{r}_i, \mathbf{r}_j, t)$, which depends on time $t$ and the electrode pair locations $(\mathbf{r}_i, \mathbf{r}_j)$, and the underlying brain sources. In the context of EEG, brain sources are most conveniently expressed at the millimeter (mesoscopic) tissue scale as current dipole moment per unit volume $\mathbf{P}(\mathbf{r}, t)$.

The relationship between observed potentials $V(\mathbf{r}_i, \mathbf{r}_j, t)$ and brain sources $\mathbf{P}(\mathbf{r}, t)$ depends on the anatomy and physiology of brain tissue (especially the cerebral cortex and its white matter connections) and the physics of volume conduction through the human head. This book is concerned with *quantitative electroencephalography*, consisting of mathematical transformations of recorded potential to new dependent variables $X$ and independent variables $\xi_1, \xi_2, \xi_3, \ldots$; that is,

$$V(\mathbf{r}_i, \mathbf{r}_j, t) \rightarrow X(\xi_1, \xi_2, \xi_3, \ldots) \tag{1.1}$$

The transformations of (1.1) provide important estimates of source dynamics $\mathbf{P}(\mathbf{r}, t)$ that supplement the unprocessed data $V(\mathbf{r}_i, \mathbf{r}_j, t)$. In the case of transformed electrode references, the new dependent variable $X$ retains its identity as an electric potential. With surface Laplacian and dura imaging transformations (high-resolution EEGs), $X$ is proportional to estimated brain surface potential. Other transformations include Fourier transforms, principal/independent components analysis, constrained inverse solutions (source localization), correlation dimension/Lyapunov exponents, and measures of phase locking, including coherence and Granger causality.

Some EEG transformations have clear physical and physiological motivations; others are more purely mathematical. Fourier transforms, for example, are clearly useful across many applications because specific EEG frequency bands are associated with specific brain states. Other transformations have more limited appeal, in some cases appearing to be no more than mathematics in search of application. How does one distinguish mathematical methods that truly benefit EEG from methods that merely demonstrate fancy mathematics? Our evaluation of the accuracy and efficacy of quantitative EEG cannot be limited to mathematical issues; close consideration of EEG physics and physiology is also required. One obvious approach, which unfortunately is substantially underemployed in EEG, is to adopt physiologically based dynamic and volume conduction models to evaluate the proposed transforms $X(\xi_1, \xi_2, \xi_3, \ldots)$. If transformed variables reveal important dynamic properties of the known sources modeled in such simulations, they may be useful with genuine EEG data; if not, there is no apparent justification for the transform. Several examples of appropriate and inappropriate transforms are discussed in [2].

## 1.2 A Window on the Mind

Since the first human recordings in the early 1920s and their widespread acceptance 10 years later, it has been known that the amplitude and frequency content of EEGs reveals substantial information about brain state. For example, the voltage record during deep sleep has dominant frequencies near 1 Hz, whereas the eyes-closed waking alpha state is associated with near-sinusoidal oscillations near 10 Hz. More quantitative analyses allow for identification of distinct sleep stages, depth of anesthesia, seizures, and other neurological disorders. Such methods may also reveal robust EEG correlations with cognitive processes: mental calculations, working memory, and selective attention. Modern methods of EEG are concerned with both temporal and spatial properties given by the experimental scalp potential function

$$V(\mathbf{r}_i, \mathbf{r}_j, t) = \Phi(\mathbf{r}_i, t) - \Phi(\mathbf{r}_j, t) \tag{1.2}$$

Note the distinction between the (generally unknown) potential with respect to infinity $\Phi$ due only to brain sources and the actual recorded potential $V$, which always depends on a pair of scalp electrode locations ($\mathbf{r}_i$, $\mathbf{r}_j$). The distinction between abstract and recorded potentials and the associated reference electrode issue, which often confounds EEG practitioners, is considered in more detail later in this chapter and in Chapter 2.

Electroencephalography provides very large-scale, robust measures of neocortical dynamic function. A single electrode provides estimates of synaptic sources averaged over tissue masses containing between roughly 100 million and 1 billion neurons. The space averaging of brain potentials resulting from extracranial recording is a fortuitous data reduction process forced by current spreading in the head volume conductor. By contrast, intracranial electrodes implanted in living brains provide much more local detail but very sparse spatial coverage, thereby failing to record the "big picture" of brain function. The dynamic behavior of intracranial recordings depends fundamentally on measurement scale, determined mostly by electrode size. Different electrode sizes and locations can result in substantial differences in intracranial dynamic behavior, including frequency content and phase locking. The technical and ethical limitations of human intracranial recording force us to emphasize scalp recordings, which provide synaptic action estimates of sources $\mathbf{P}(\mathbf{r}, t)$ at large scales closely related to cognition and behavior. In practice, *intracranial data provide different information, not more information, than is obtained from the scalp* [2].

## 1.3 Cortical Anatomy and Physiology Overview

The three primary divisions of the human brain are the brainstem, cerebellum, and cerebrum, as shown earlier in Figure 1.1. The *brainstem* is the structure through which nerve fibers relay sensory and motor signals (action potentials) in both directions between the spinal cord and higher brain centers. The *thalamus* is a relay station and important integrating center for all sensory input to the cortex except smell. The *cerebellum*, which sits on top and to the back of the brainstem, is associated with the fine control of muscle movements and certain aspects of cognition.

The large part of the brain that remains when the brainstem and cerebellum are excluded consists of the two halves of the *cerebrum*. The outer portion of the cerebrum, the *cerebral cortex*, is a folded structure varying in thickness from about 2 to 5 mm, with a total surface area of roughly 2,000 $cm^2$ and containing about $10^{10}$ neurons. The cortical folds (fissures and sulci) account for about two-thirds of its surface, but the unfolded gyri provide more favorable geometry for the production of large scalp potentials [2].

Cortical neurons are strongly interconnected. The surface of a large cortical neuron may be densely covered with $10^4$ to $10^5$ synapses that transmit inputs from other neurons. The synaptic inputs to a neuron are of two types: those which produce *excitatory postsynaptic potentials* (EPSPs) across the membrane of the target neuron, thereby making it easier for the target neuron to fire an action potential, and the *inhibitory postsynaptic potentials* (IPSPs), which act in the opposite manner on the output neuron. EPSPs produce local membrane current sinks with correspond-

ing distributed passive sources to preserve current conservation. IPSPs produce local membrane current sources with more distant distributed passive sinks. Much of our conscious experience must involve, in some largely unknown manner, the interaction of cortical neurons. The cortex is also believed to be the structure that generates most of the electric potentials measured on the scalp.

The cortex (or neocortex in mammals) is composed of *gray matter*, so called because it contains a predominance of cell bodies that turn gray when stained, but living cortical tissue is actually pink. Just below the gray matter cortex is a second major region, the so-called *white matter*, composed of myelinated nerve fibers (*axons*). White matter interconnections between cortical regions (*association fibers* or *corticocortical fibers*) are quite dense. Each square centimeter of human neocortex may contain $10^7$ input and output fibers, mostly corticocortical axons interconnecting cortical regions separated by 1 to about 15 cm, as shown in Figure 1.2. A much smaller fraction of axons that enter or leave the underside of human cortical surface radiates from (and to) the thalamus (*thalamocortical fibers*). This fraction is only a few percent in humans, but substantially larger in lower mammals [3, 4].

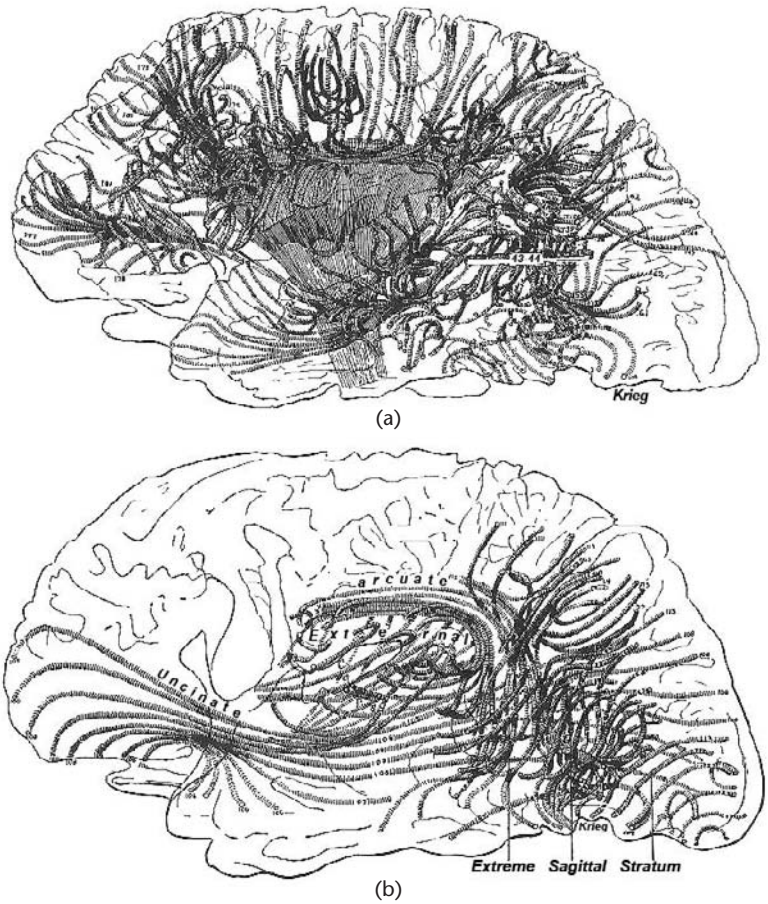



**Figure 1.2** (a) Some of the superficial corticocortical fibers of the lateral aspect of the cerebrum obtained by dissection of a fresh human brain. (b) A few of the deeper corticocortical fibers of the lateral aspect of the cerebrum. The total number of corticocortical fibers is roughly $10^{10}$; for every fiber shown here about 100 million are not shown. (*After:* [5, 6].)

This difference partly accounts for the strong emphasis on thalamocortical interactions (versus corticocortical interactions), especially in physiology literature emphasizing animal experiments. Neocortical neurons within each cerebral hemisphere are connected by short intracortical fibers with axon lengths mostly less than 1 mm, in addition to $10^{10}$ corticocortical fibers. Cross hemisphere interactions occur by means of about $10^8$ *callosal axons* through the corpus callosum and several smaller structures connecting the two brain halves.

Action potentials evoked by external stimuli reach the cerebral cortex in less than 20 ms, and monosynaptic transmission times across the entire cortex are about 30 ms. By contrast, consciousness of external events may take 300 to 500 ms to develop [7]. This finding suggests that consciousness of external events requires multiple feedback signals between remote cortical and subcortical regions. It also implies that substantial functional integration and, by implication, EEG phase locking may be an important metric of cognition [8].

## 1.4 Brain Sources

The relationship between scalp potential and brain sources in an isotropic (but generally inhomogeneous) volume conductor may be expressed concisely by the following form of Poisson's equation:

$$\nabla \cdot \left[\sigma(\mathbf{r})\nabla\Phi(\mathbf{r},t)\right] = -s(\mathbf{r},t) \tag{1.3}$$

Here $\nabla$ is the usual vector operator indicating three spatial derivatives, $\sigma(\mathbf{r})$ is the electrical conductivity of tissue (brain, skull, scalp, and so forth), and $s(\mathbf{r}, t)$ ($\mu$A/mm$^3$) is the neural tissue current source function. A similar equation governs anisotropic tissue; however, the paucity of data on tensor conductivity limits its application to electroencephalography. Figure 1.3 represents a general volume conductor; source current $s(\mathbf{r}, t)$ is generated within the inner circles. In the brain, $s(\mathbf{r}, t)$ dynamic behavior is determined by poorly understood and generally nonlinear

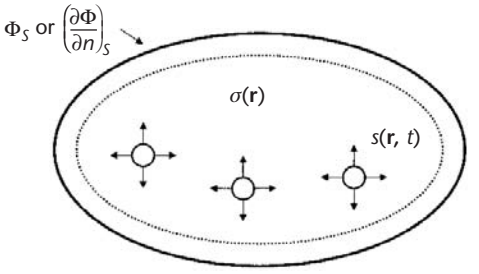



**Figure 1.3** The outer ellipse represents the surface of a general volume conductor; the circles indicate regions where current sources $s(\mathbf{r}, t)$ are generated. The forward problem is well posed if all sources are known, and if either potential $\Phi_s$ or its normal derivative $\left(\frac{\partial\Phi}{\partial n}\right)_S$ is known over the entire surface. In EEG applications, current flow into the surrounding air space and into the neck region is assumed to be zero, that is, the boundary condition $\left(\frac{\partial\Phi}{\partial n}\right)_s \approx 0$ is adopted. In high-resolution EEGs, the potential on some inner surface (dashed line indicating dura or cortex) is estimated from the measured outer surface potential $\Phi_s$.

interactions between cells and cell groups at multiple spatial scales. Poisson's equation (1.3) tells us that scalp dynamics $\Phi(\mathbf{r}, t)$ is produced as a *linear superposition* of source dynamics $s(\mathbf{r}, t)$ with complicated weighting of sources determined by the conductively inhomogeneous head.

In EEG applications, current flow into the surrounding air space and the neck region is assumed to be zero, that is, the boundary condition $\left(\frac{\partial \Phi}{\partial n}\right)_S \approx 0$ is adopted. The forward problem is then well posed, and the potential within the volume conductor (but external to the source regions) may be calculated from Poisson's equation (1.3) if the sources are known. The inverse problem involves estimation of sources $s(\mathbf{r}, t)$ using the recorded surface potential plus additional constraints (typically assumptions) about the sources. The severe limitations on inverse solutions in EEG are discussed in [2]. In high-resolution EEG, no attempt is made to locate sources. Rather, the potential on some inner surface (dashed line indicating dura or cortical surface in Figure 1.3) is estimated from measured outer surface potential $\Phi_S$. In other words, the usual boundary conditions on the outer surface are overspecified by the recorded EEG, and the measured outer potential is projected to an inner surface that is assumed to be external to all brain sources.

Figure 1.4 shows a cortical macrocolumn 3 mm in diameter that contains perhaps $10^6$ neurons and $10^{10}$ synapses. Each synapse generally produces a local membrane source (or sink) balanced by distributed membrane sources required for current conservation; action potentials also contribute to $s(\mathbf{r}, t)$. Brain sources may be characterized at several spatial scales. Intracranial recordings provide distinct measures of neocortical dynamics, with scale dependent on electrode size, which may vary over 4 orders of magnitude in various practices of electrophysiology. By contrast, scalp potentials are largely independent of electrode size after severe space averaging by volume conduction between brain and scalp. Scalp potentials are due mostly to sources coherent at the scale of at least several centimeters with special geometries that encourage the superposition of potentials generated by many local sources.

Due to the complexity of tissue microsources $s(\mathbf{r}, t)$, EEG is more conveniently related to the mesosource function of each tissue mass $W$ by the volume integral

$$\mathbf{P}(\mathbf{r},t) = \frac{1}{W}\iiint_W \mathbf{w}s(\mathbf{r},\mathbf{w},t)dW(\mathbf{w}) \tag{1.4}$$

where $s(\mathbf{r}, t) \rightarrow s(\mathbf{r}, \mathbf{w}, t)$ indicates that the microsources are integrated over the mesoscopic tissue volume $W$ with center located at $\mathbf{r}$, and $\mathbf{P}(\mathbf{r}, t)$ is the current dipole moment per unit tissue volume (or "mesosource" for short) and has units of current density ($\mu$A/mm$^2$). If $W$ is a cortical column and the microsources and microsinks are idealized in depth, $\mathbf{P}(\mathbf{r}, t)$ is the diffuse current density across the column (as suggested in Figure 1.4). More generally, (1.4) provides a useful source definition for millimeter-scale tissue volumes [2].

Equation (1.4) tells us the following:

1. Every brain tissue mass (voxel) containing neurons can generally be expected to produce a nonzero mesosource $\mathbf{P}(\mathbf{r}, t)$.

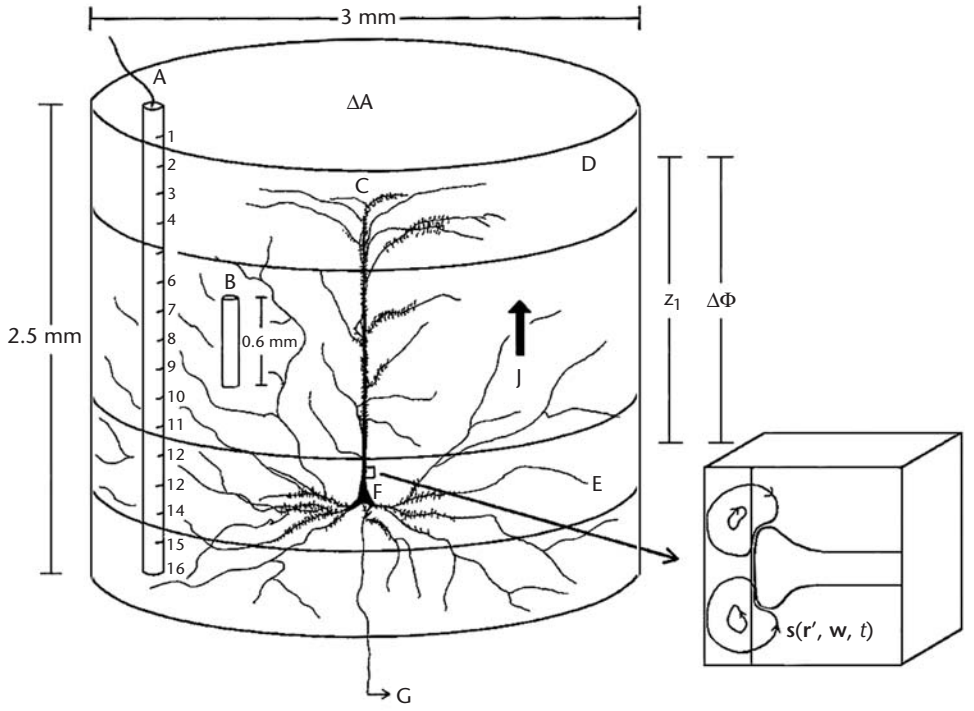



**Figure 1.4** The macrocolumn is defined by the spatial extent of axon branches E that remain within the cortex (recurrent collaterals). The large pyramidal cell C is one of $10^5$ to $10^6$ neurons in the macrocolumn. Nearly all pyramidal cells send an axon G into the white matter; most reenter the cortex at some distant location (corticocortical fibers). Each large pyramidal cell has $10^4$ to $10^5$ synaptic inputs F causing microcurrent sources and sinks $s(\mathbf{r}, \mathbf{w}, t)$. Field measurements can be expected to fluctuate greatly when small electrode contacts A are moved over distances of the order of cell body diameters. Small-scale recordings measure space-averaged potential over some volume B depending on the size of the electrode contact and can be expected to reveal scale-dependent dynamics, including dominant frequency bands. An instantaneous imbalance in sources or sinks in regions D and E will produce a "mesosource," that is, a dipole moment per unit volume $\mathbf{P}(\mathbf{r}, t)$ in the macrocolumn. (*From:* [4]. © 1995 Oxford University Press. Reprinted with permission.)

2. The magnitude of the mesosource depends on the magnitudes of the microsource function $s(\mathbf{r}, \mathbf{w}, t)$ and source separations $\mathbf{w}$ within the mass $W$. Thus, cortical columns with large source-sink separations (perhaps produced by excitatory and inhibitory synapses) may be expected to generate relatively large mesosources. By contrast, random mixtures of sources and sinks within $W$ produce small mesosources, the so-called *closed fields* of electrophysiology.
3. Mesosource magnitude also depends on microsource phase synchronization; large mesosources occur when multiple synapses tend to activate at the same time.

In standard EEG terminology, *synchrony* is a qualitative term normally indicating sources that are approximately phase locked with small or zero phase offsets; sources then tend to add by linear superposition to produce large scalp potentials. In fact, the term *desynchronization* is often used to indicate EEG amplitude reduction, for example, in the case of alpha amplitude reduction during cognitive tasks. The term *coherent* refers to the standard mathematical definition of coherence, equal to the normalized cross spectral density function and a measure of phase locking. With these definitions, all synchronous sources (small phase lags) are expected to produce large coherence estimates, but coherent sources may or may not be synchronous depending on their phase offsets.

## 1.5 Scalp Potentials Generated by the Mesosources

Nearly all EEGs are believed to be generated by cortical sources [2]. Supporting reasons include: (1) cortical proximity to scalp, (2) the large source-sink separations allowed by cortical pyramidal cells (see Figure 1.4), (3) the ability of cortex to produce large dipole layers, and (4) various experimental studies of cortical and scalp recordings in humans and other mammals. Exceptions include the brainstem evoked potential $\langle V(\mathbf{r}_i, \mathbf{r}_j, t)\rangle$, where the angle brackets indicate a time average, in this case over several thousand trials needed to extract brainstem signals from signals due to cortical sources and artifact.

We here view the mesosource function or dipole moment per unit volume $\mathbf{P}(\mathbf{r}, t)$ as a continuous function of cortical location $\mathbf{r}$, in and out of cortical folds. The function $\mathbf{P}(\mathbf{r}, t)$ forms a *dipole layer* (or dipole sheet) covering the entire folded neocortical surface. Localized mesosource activity is then just a special case of this general picture, occurring when only a few cortical regions produce large dipole moments, perhaps because the microsources $s(\mathbf{r}, t)$ are asynchronous or more randomly distributed within most columns. Or more likely, contiguous mesosource regions $\mathbf{P}(\mathbf{r}, t)$ are themselves too asynchronous to generate recordable scalp potentials. Again, the qualitative EEG term *synchronous* indicates approximate phase locking with near zero phase lag; *source desynchronization* then suggests reductions of scalp potential amplitude. In the case of the so-called *focal sources* occurring in some epilepsies, the corresponding $\mathbf{P}(\mathbf{r}, t)$ appears to be relatively large only in selective (centimeter-scale) cortical regions.

Potentials $\Phi(\mathbf{r}, t)$ at scalp locations $\mathbf{r}$ due only to cortical sources can be expressed as the following integral over the cortical surface:

$$\Phi(\mathbf{r}, t) = \iint_S \mathbf{G}_S(\mathbf{r}, \mathbf{r}') \cdot \mathbf{P}(\mathbf{r}', t) dS(\mathbf{r}') \tag{1.5}$$

If subcortical sources contribute, (1.5) may be replaced by a volume integral over the entire brain. All geometric and conductive properties of the volume conductor are accounted for by the *Green's function* $\mathbf{G}_S(\mathbf{r}, \mathbf{r}')$, which weighs the contribution of the mesosource field $\mathbf{P}(\mathbf{r}', t)$ according to source location $\mathbf{r}'$ and the location of the recording point $\mathbf{r}$ on the scalp. Contributions from different cortical regions may or may not be negligible in different brain states. For example, source activity in the

central parts of mesial (underside) cortex and the longitudinal fissure (separating the brain hemispheres) may make negligible contributions to scalp potential in many brain states. Exceptions to this picture may occur in the case of mesial sources contributing to potentials at an ear or mastoid reference, an influence that has sometimes confounded clinical interpretations of EEGs [9, 10].

Green's function $\mathbf{G}_S$ ($\mathbf{r}$, $\mathbf{r}'$) will be small when the *electrical distance* between scalp location $\mathbf{r}$ and mesosource location $\mathbf{r}'$ is large. In an infinite, homogeneous medium electrical distance equals physical distance, but in the head volume conductor, the two measures can differ substantially because of current paths distorted by variable tissue conductivities.

## 1.6 The Average Reference

To facilitate our discussion of relations between brain sources and scalp potentials, two useful transformations of raw scalp potential $V(\mathbf{r}_i, \mathbf{r}_j, t)$ are introduced; the first is the *average reference potential* (or *common average reference*). Scalp potentials are recorded with respect to some reference location $\mathbf{r}_R$ on the head or neck; (1.2) then yields the reference potential

$$V(\mathbf{r}_i, \mathbf{r}_R, t) = \Phi(\mathbf{r}_i, t) - \Phi(\mathbf{r}_R, t) \tag{1.6}$$

Summing recorded potentials over all $N$ (nonreference) electrodes and rearranging terms in (1.6) yield the following expression for the *nominal reference potential with respect to infinity:*

$$\Phi(\mathbf{r}_R, t) = \frac{1}{N}\sum_{i=1}^{N}\Phi(\mathbf{r}_i, t) - \frac{1}{N}\sum_{i=1}^{N}V(\mathbf{r}_i, \mathbf{r}_R, t) \tag{1.7}$$

The term *nominal reference potential* refers to the unknown head potential at $\mathbf{r}_R$ due only to sources located inside the head; that is, we exclude external noise sources that result from, for example, capacitive coupling with power line fields (see Chapter 2). Such external noise should be removed with proper recording methods. The first term on the right side of (1.7) is the nominal average of scalp surface potentials (with respect to infinity) over all recording sites $\mathbf{r}_i$. This term should be small if electrodes are located such that the average approximates a closed head surface integral containing all current within the volume. Apparently only minimal current flows from the head through the neck [2], so to a plausible approximation the head may be considered to confine all current from internal sources. The surface integral of the potential over a volume conductor containing dipole sources must then be zero as a consequence of current conservation [11]. With this approximation, substitution of (1.7) into (1.6) yields an approximation for the nominal potential at each scalp location $\mathbf{r}_i$ with respect to infinity (average reference potential):

$$\Phi(\mathbf{r}_i, t) \approx V(\mathbf{r}_i, \mathbf{r}_R, t) - \frac{1}{N}\sum_{i=1}^{N}V(\mathbf{r}_i, \mathbf{r}_R, t) \tag{1.8}$$

Relation (1.8) provides an estimate of reference-free potential in terms of recorded potentials. Because we cannot measure the potentials over an entire closed surface of an attached head, the first term on the right side of (1.7) will not generally vanish. Due to sparse spatial sampling, *the average reference is expected to provide a very poor approximation if applied with the standard 10–20 electrode system.* As the number of electrodes increases, the error in approximation (1.8) is expected to decrease. Like any other reference, the average reference provides biased estimates of reference-independent potentials. Nevertheless, when used in studies with large numbers of electrodes (say, 100 or more), it often provides a plausible estimate of reference-independent potentials [12]. Because the reference issue is critical to EEG interpretation, *transformation to the average reference is often appropriate before application of other transformations,* as discussed in later chapters.

## 1.7 The Surface Laplacian

The process of relating recorded scalp potentials $V(\mathbf{r}_i, \mathbf{r}_R, t)$ to the underlying brain mesosource function $\mathbf{P}(\mathbf{r}, t)$ has long been hampered by: (1) reference electrode distortions and (2) inhomogeneous current spreading by the head volume conductor. The average reference method discussed in Section 1.6 provides only a limited solution to problem 1 and fails to address problem 2 altogether. By contrast, the *surface Laplacian* completely eliminates problem 1 and provides a limited solution to problem 2. The surface Laplacian is defined in terms of two surface tangential coordinates, for example, spherical coordinates $(\theta, \phi)$ or local Cartesian coordinates $(x, y)$. From (1.6), with the understanding that the reference potential is spatially constant, we obtain the surface Laplacian in terms of (any) reference potential:

$$L_{Si} \equiv \nabla_S^2 \Phi(\mathbf{r}_i, t) = \nabla_S^2 V(\mathbf{r}_i, \mathbf{r}_R, t) = \frac{\partial^2 V(x_i, y_i, \mathbf{r}_R, t)}{\partial x_i^2} + \frac{\partial^2 V(x_i, y_i, \mathbf{r}_R, t)}{\partial y_i^2} \tag{1.9}$$

The physical basis for relating the scalp surface Laplacian to the dura (or inner skull) surface potential is based on Ohm's law and the assumption that skull conductivity is much lower than that of contiguous tissue (by at least a factor of 5 or so). In this case most of the source current that reaches the scalp flows normal to the skull. With this approximation, the following approximate expression for the surface Laplacian is obtained in terms of the local outer skull potential $\Phi_{Ki}$ and inner skull (outer CSF) potential $\Phi_{Ci}$ [2]:

$$L_{Si} \approx A_i(\Phi_{Ki} - \Phi_{Ci}) \tag{1.10}$$

The parameter $A_i$ depends on several tissues thicknesses and conductivities, which are assumed constant over the surface to first approximation. Simulations indicate minimal falloff of potential through the scalp so that $\Phi_K$ reasonably approximates scalp surface potential.

Interpretation of $L_S$ depends critically on the nature of the sources. When cortical sources consist of large dipole layers, the potential falloff through the skull is minimal so $\Phi_K \sim \Phi_C$ and the surface Laplacian is very small. By contrast, when corti-

cal sources consist of single dipoles or small dipole layers, the potential falloff through the skull is substantial such that $\Phi_K << \Phi_C$. Thus for relatively small dipole layers (i.e., diameters of less than a few centimeters), *the negative Laplacian is approximately proportional to cortical (or dura) surface potential.*

## 1.8 Dipole Layers: The Most Important Sources of EEGs

No single macrocolumn (containing about $10^6$ neurons) is expected to generate a dipole moment $\mathbf{P}(\mathbf{r}, t)$ of sufficient strength to produce scalp potentials in the recordable range of EEGs (a few microvolts). As a general "rule of head," about 6 $cm^2$ of cortical gyri tissue (containing about $10^3$ macrocolumns forming a dipole layer) must be "synchronously active" to produce recordable scalp potentials without averaging [9, 13]. In this context, the tissue label *synchronously active* is based on cortical recordings with macroscopic electrodes and is viewed mainly as a qualitative description. In the case of dipole layers in fissures and sulci, tissue areas larger than 6 $cm^2$ are apparently required to produce measurable scalp potentials as a result of partial canceling of opposing dipole vectors and increased distance from scalp [2].

To minimize cumbersome language, the term *dipole layer* is used to indicate cortical regions where the mesosource function $\mathbf{P}(\mathbf{r}, t)$ exhibits relatively high phase synchronization (small phase lags) over its surface, especially in the crowns of contiguous cortical gyri providing large scalp potentials due to superposition of nearly parallel (noncanceling) source vectors. Genuine tissue is not expected to behave in this ideal manner; however, more complex sources can be modeled as multiple overlapping dipole layers.

In Figure 1.5 two cortical dipole layers are defined as follows: $\mathbf{P}_1(\mathbf{r}, t)$ and $\mathbf{P}_2(\mathbf{r}, t)$ are "synchronous" (approximately phase locked with small phase lag) over their respective regions and asynchronous with other cortical tissue. For example, the small region (dashed cylinder) might be internally synchronous in one frequency

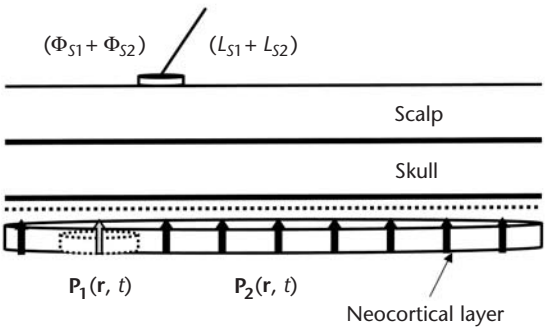



**Figure 1.5** The overlapping dipole layer source regions $\mathbf{P}_1(\mathbf{r}, t)$ and $\mathbf{P}_2(\mathbf{r}, t)$ represent sources in (perhaps multiple) contiguous cortical gyri. The dashed horizontal line indicates a thin CSF layer. Local scalp potential ($\Phi_{S1} + \Phi_{S2}$) and Laplacian ($L_{S1} + L_{S2}$) measures depend on the sum of contributions from each of the two source regions. Smaller cortical regions (small dashed cylinder) tend to make larger relative contributions to the Laplacian, whereas larger regions (large cylinder) contribute more to potential as shown in Figure 1.6. If the two cortical source regions generate dynamics with different dominant frequencies, scalp potential spectra will differ from scalp Laplacian spectra. (*From:* [2]. © 2006 Oxford University Press. Reprinted with permission.)

band $\Delta f_{11}$, while simultaneously producing dynamics in bands $\Delta f_{11}$ and $\Delta f_{12}$, which are asynchronous and synchronous, respectively, with the larger region. In this case the small region may be considered part of the large region for dynamics in band $\Delta f_{12}$, but separate for dynamics in $\Delta f_{11}$.

The scalp potential and Laplacian are generated by the mesosource functions $\mathbf{P}_1(\mathbf{r}, t)$ and $\mathbf{P}_2(\mathbf{r}, t)$ integrated over the surfaces of their respective regions as given by (1.5). Local scalp potential and Laplacian measures depend on the sum of contributions from each of the two mesosource regions. However, the relative contributions of individual regions (dipole layers) can differ substantially. For example, if the small and large regions have diameters in the 2- and 10-cm ranges, respectively, we expect the following relation between surface potentials $\Phi_S$ and Laplacians $L_S$:

$$\left|\frac{\Phi_{S1}}{\Phi_{S2}}\right| << \left|\frac{L_{S1}}{L_{S2}}\right| \tag{1.11}$$

Relation (1.11) indicates that smaller cortical layers tend to make larger relative contributions to the Laplacian, whereas larger regions contribute more to potential. If the two regions generate dynamics with different dominant frequencies, scalp potential spectra will differ from scalp Laplacian spectra, a prediction consistent with experimental observations of spontaneous EEGs [14]. These data indicate that large and small dipole layers can contribute to different frequencies within the alpha band, and may or may not have overlapping frequencies.

My outline of the surface Laplacian in this chapter has been mostly qualitative, but a number of quantitative studies generally support these ideas [2]. For example a *four-sphere head model* (consisting of an inner brain sphere surrounded by three spherical shells: CSF, skull, and scalp) may be used with Poisson's equation (1.3) to estimate the relative sensitivity of the potential and surface Laplacian measures to dipole layer source regions of different sizes. Figure 1.6 shows scalp potential directly above the centers of dipole layers of varying angular extent, forming superficial spherical caps in the four-sphere head model. The four curves shown in each figure correspond to four different ratios of brain-to-skull conductivity. Each curve in the upper part of Figure 1.6 shows scalp potential as a percentage of transcortical potential $V_C$, which is roughly related to the local normal component of cortical mesosource function $P$ through Ohm's law; that is,

$$P \sim \frac{\sigma_C V_C}{d_C} \tag{1.12}$$

**Here** $\sigma_C$ and $d_C$ are the local conductivity and thickness of cortex, respectively. Transcortical potential has been estimated in experiments with mammals, typically $V_C \sim 100 - 300\,\mu$V for spontaneous EEGs [4, 15]. Given this intracranial data, Figure 1.6 suggests maximum scalp potentials of roughly 30 to 150 $\mu$V for dipole layers with spherical cap radii of about 8 cm or gyral surface areas of several hundred square centimeters.

In the lower part of Figure 1.6, the relative scalp Laplacian is plotted due to the same dipole layers. While potentials are shown to be primarily sensitive to broad dipole layers, Laplacians are sensitive to smaller layers, as implied by Figure 1.5. In

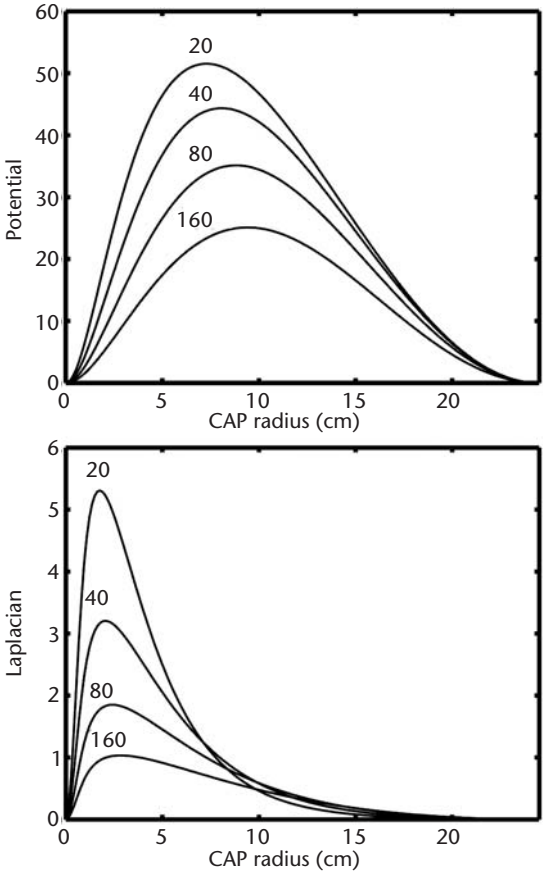



**Figure 1.6** (Top) Maximum scalp potential expressed as a percentage of transcortical potential $V_c$ directly above cortical dipole layers of varying angular extent, forming superficial spherical caps on the inner sphere (cortex) of a four-sphere head model are shown. Each curve is based on the labeled model brain-to-skull conductivity ratio. (Bottom) The relative scalp Laplacian in the same head model due to the same dipole layers is shown. Potentials are primarily sensitive to large dipole layers, whereas Laplacians are sensitive to smaller dipole layers. (*From:* [2]. © 2006 Oxford University Press. Reprinted with permission.)

summary, the surface Laplacian acts as a spatial filter that emphasizes smaller source regions than those likely to make the dominant contributions to scalp potential. Thus, the Laplacian measure supplements but cannot replace the potential measure. The Laplacian is a spatial filter that removes low spatial frequency scalp signals due to both volume conduction and genuine source dynamics, but it cannot distinguish between the two. Further discussion of the strengths and limitations of the surface Laplacian and other high-resolution methods may be found in Chapter 13 and [2].

## 1.9 Alpha Rhythm Sources

Several source properties of spontaneous EEGs are suggested by the following study using the New Orleans spline Laplacian algorithm [2]. Figure 1.7 shows 9 of 111

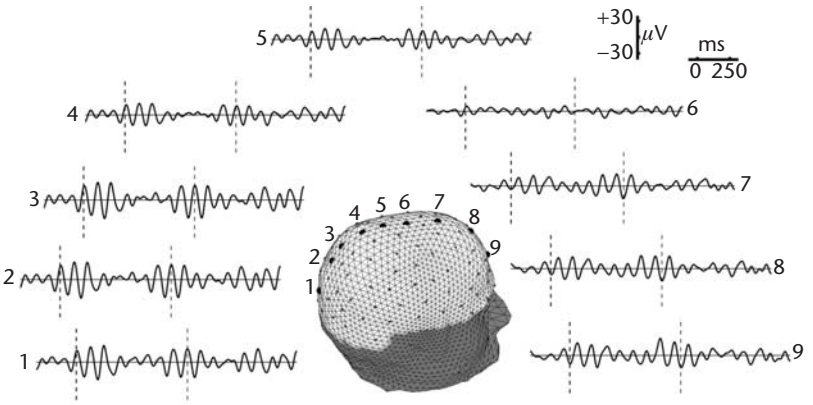



**Figure 1.7** Alpha rhythm (potential) waveforms recorded along the midline with data transformed to the average reference using (1.7). Nine out of a total of 111 channels (electrodes) are shown. The two dashed lines on each waveform indicate fixed time slices and show phase differences recorded from different locations. The sources appear to originate from very large dipole layers, perhaps an anterior-posterior standing cortical wave. (*From:* [2]. © 2006 Oxford University Press. Reprinted with permission.)

channels of averaged reference potentials recorded in the eyes-closed resting state. Two dashed vertical lines are drawn on each waveform to indicate common time slices. At these instances electrode sites over occipital cortex (electrodes 1, 2, and 3) yield a peak positive potential of the alpha rhythm, while electrode sites near the frontal pole (electrodes 8 and 9) show a peak negative potential at the same time slice. Such 180° phase shift between anterior and posterior regions is often observed in alpha rhythm dynamics. The back-to-front spatial distribution of the alpha rhythm suggests very low spatial frequency source activity (associated with large dipole layers), with minimum potential magnitude at electrodes near the vertex. A number of other theoretical and experimental studies suggest that these layers consist of standing waves of cortical source activity along the anterior-posterior direction, perhaps corresponding to a global cortical\white matter fundamental mode of oscillation [2, 4, 16–18].

Figure 1.8 shows the spline Laplacian for identical data at these same 9 midline electrodes (estimated using all 111 electrode sites). The Laplacian oscillations are largest at electrode sites 5 and 6 near the vertex. The waveforms differ at these electrodes, with a peak occurring at the first time slice in electrode 6 with no corresponding peak at electrode 5. The electrode sites over occipital and frontal areas show the strongest alpha signal in the potential plots, but much smaller amplitude Laplacian signals.

Similar differences between potential and Laplacian waveforms were observed in other scalp locations (not shown), again indicating the simultaneous presence of both small and large dipole layers. Occipital cortex, for example, showed several regions (off midline) with large Laplacians and other regions with small Laplacians. Other studies of alpha spectra and coherence measures indicate that the low spatial frequency (large dipole layer) alpha activity typically tends to oscillate near the low end of the alpha band (near 8 Hz). By contrast, the more local alpha activity often

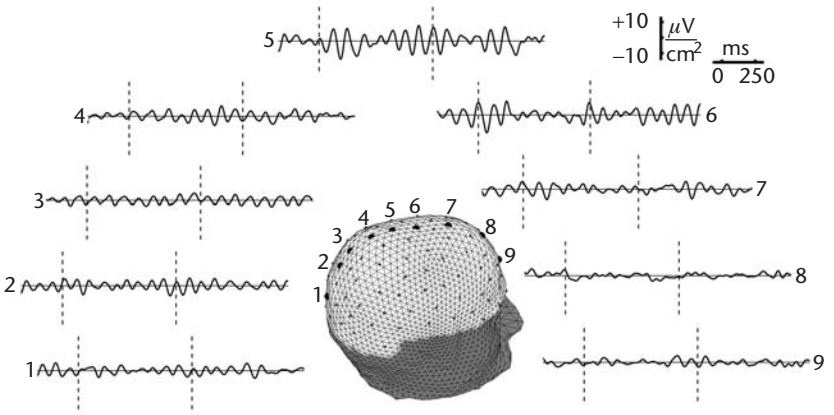



**Figure 1.8** The same data shown in Figure 1.7 have been transformed using the New Orleans spline Laplacian algorithm. The combined information from Figure 1.7 and this figure suggests both global and local (near electrodes 5 and 6) source contributions to alpha rhythm. Other scalp locations (not shown) also indicate regions with both large and smaller dipole layers that produce oscillations with alpha-band frequencies that may or may not be equal or match the dominant oscillation frequency observed with potentials. (*From:* [2]. © 2006 Oxford University Press. Reprinted with permission.)

oscillates near the upper end of the alpha band (near 11 Hz). Furthermore, these distinct source regions react selectively to motor or cognitive experiments involving mental tasks, attention, and so forth [2, 14, 19–21].

The striking difference between potential and Laplacian dynamics (spectra, coherence, and so forth) reflects the principle that these measures are sensitive to different spatial bandwidths of source activity as discussed in Section 1.8. Potentials are dominated by long wavelength source activity that extends from frontal to occipital regions. By contrast, Laplacians are insensitive to this low spatial frequency source activity, but instead reveal multiple small dipole layers, including several sites close to the vertex and occipital cortex. These small source regions, which appear to be *embedded* in the larger scale source regions, are not evident in potential plots, apparently because of much stronger contributions from the large source regions.

These data again emphasize that source activity identified by the surface Laplacian does not generally represent all of the sources that generate scalp potentials. Rather, the Laplacian identifies a subset of sources that are more local, generally within about 2 cm of each electrode. By contrast, the potential map is produced mainly by broadly distributed sources. Contrasting potential and Laplacian signals can clarify the nature of sources because neither measure is as informative when used in isolation. Surface Laplacian or dura image estimates should be used to complement, but not replace, raw scalp potentials.

## 1.10 Neural Networks, Cell Assemblies, and Field Theoretic Descriptions

The physiological origins of the source dynamics $\mathbf{P}(\mathbf{r}, t)$ are poorly understood. Neural network models are able to produce oscillatory behavior that may appear superficially similar to an EEG; however, network models typically depend on many parameters that lack a physiological basis. Furthermore, the multiple mechanisms by which neurons interact may not fit naturally into network models. Any network description must be scale dependent so, for example, macroscopic network elements (centimeter scale) are themselves complex systems containing smaller (*mesoscopic*, millimeter-scale) network elements. The mesoscopic elements are, in turn, composed of still smaller scale elements [22]. Partly for these reasons, neuroscientists often prefer the term *cell assemblies*, originating with the pioneering work of Donald Hebb [23] in 1949. This label denotes a diffuse cell group capable of acting briefly as a single structure, for example, one or more cortical dipole layers that may be functionally connected. We may reasonably postulate cooperative activity within cell assemblies without explicitly specifying interaction mechanisms.

Brain processes may involve the formation of cell assemblies at multiple spatial scales [24–26]. Such neuron groups may produce a wide range of local delays and associated characteristic (or resonant) frequencies [4]. Network models can incorporate some physiologically realistic features; however, field descriptions of brain dynamics may be required to fill the dual role of modeling dynamic behavior and making contact with macroscopic EEG data. In this context, the word *field* refers to mathematical functions expressing, for example, the mesoscopic source function $\mathbf{P}(\mathbf{r}, t)$ or perhaps the numbers of active synapses in each mesoscopic tissue volume. In the view adopted here, cell assemblies are pictured as embedded within synaptic and action potential fields [2, 4, 18–20, 25, 27, 28].

We tentatively view the small alpha dipole layers implied by Figure 1.8 as embedded in the standing wave field implied by Figure 1.7. Electric and magnetic fields (EEGs and MEGs) provide large-scale, short-time measures of the modulations of synaptic and action potential fields around their background levels. These synaptic fields are analogous to common physical fields, for example, sound waves, which are short-time modulations of pressure about background levels. These short-time modulations of synaptic activity are distinguished from long-timescale (seconds to minutes) modulations of brain chemistry controlled by neuromodulators.

## 1.11 Phase Locking

Cell assemblies that form and dissolve on roughly 100-ms timescales in the brain are believed to underlie cognitive processing. They may develop simultaneously at multiple spatial scales, but only vary large scales are observable with scalp recordings. Laplacian measures apply to somewhat smaller spatial structures than potentials, but they are still very large scale compared to intracranial recordings. If cell assemblies are indeed responsible for cognition, one may reasonably expect to observe correlations between EEG phase locking and mental or behavioral activity at some

scales. Here the term *phase locking* is used to indicate "synchronization" with arbitrary phase lag. Several approaches to estimate phase locking are discussed in Chapter 4.

One measure of phase locking between any pair of signals is *coherence*, a correlation coefficient (squared) expressed as a function of frequency band. For example, while performing mental calculations subjects often exhibit increased EEG coherence in the theta (near 5 Hz) and upper alpha (near 10 Hz) bands, whereas the same data may show decreased coherence in the lower alpha band (near 8 Hz) in most electrode pairs [2, 14]. Coherence of *steady-state visually evoked potentials* indicates that mental tasks consistently involve increased 13-Hz coherence in select electrode pairs but decreased coherence in other pairs [29, 30]. Binocular rivalry experiments using steady-state magnetic field recordings show that conscious perception of a stimulus flicker is reliably associated with increased cross hemispheric coherence at 7 Hz [31]. These data are consistent with the formation of large-scale cell assemblies (e.g., cortical dipole layers) at select frequencies with center-to-center cortical separations of roughly 5 to 20 cm.

These cognitive studies emphasize relatively low frequencies (<15 Hz) because of the very low signal-to-noise ratio (SNR) associated with higher frequencies in scalp recordings. By contrast, intracranial studies in lower mammals often emphasize gamma-band phase locking (~ 40 Hz). It is, however, difficult (if not impossible) to record gamma-band spontaneous EEG data from the scalp that are not substantially contaminated by muscle and other artifact. Brain signals above about 15 to 20 Hz have very low scalp amplitudes, often lower than muscle activity in the same general frequency range. By contrast, much higher frequency potentials may be recorded with intracranial electrodes. For example, human studies using subdural electrodes have found increased electrocorticogram (ECoG) power in the 80- to 150-Hz range over auditory and prefrontal cortex when epilepsy patients attended to external stimuli [32].

Past emphasis on the 40-Hz gamma band may have been based partly on two unproven assumptions: (1) Frequency bands in which robust correlations between cognition/behavior and phase locking occur are similar in humans and lower mammals. (2) Physiologically interesting phase locking is mainly confined to frequencies near 40 Hz. Nunez and Srinivasan [2] have challenged these assumptions, suggesting that cognition may easily produce concurrent signatures in multiple frequency bands that may differ in humans and lower mammals. For example, observations of concurrent theta and alpha coherence effects (in three distinct bands) do not preclude additional concurrent effects in multiple bands well above 15 Hz occurring at spatial scales too small to be recorded from the scalp. The most robust effects are likely to be observed in frequency bands that most closely "match" the spatial scale of the recording, as determined by electrode size and distance from sources in intracranial recordings (refer to Figure 1.4).

## 1.12 "Simple" Theories of Cortical Dynamics

Human EEG recordings have been carried out for the past 80 years, but the physiological bases for EEG and $\mathbf{P}(\mathbf{r}, t)$ dynamic behavior remain mostly obscure. We

would like to understand better the complex nonlinear dynamics–associated cognition and behavior, but even the origin of the 100-ms timescale associated with the 10-Hz alpha rhythm remains controversial. Many view the human brain as the preeminent complex system, although one may argue that the human social system consisting of $6 \times 10^9$ interacting brains is far more complex [8]. In any case, the physiological origins of cortical dynamics can be expected to challenge many future generations of scientists.

With this perspective in mind, I tend to favor more emphasis on relatively simple questions about brain dynamics based on the general rule that a person must evidently learn to crawl before he can walk. Thus, I conclude this chapter with a short summary of several "simple" theoretical issues associated with EEG dynamics. Three basic questions can be expressed as follows:

1. Which spatial scale(s) of tissue determine the dominant timescales observed in EEGs? For example, do alpha or gamma oscillations originate from single neurons, from millimeter-scale networks, from the entire cortex, or can they be generated simultaneously at multiple scales?
2. How important are interactions across spatial scales?
3. How important are cortical global boundary conditions? And, do global boundary conditions provide an important top-down influence on small-scale dynamics as typically observed in physical systems including chaotic systems [4]?

To distinguish the various theories of large scale cortical dynamics, I have suggested the label *local theory* to indicate mathematical models of cortical or thalamocortical interactions (feedback loops) for which corticocortical propagation delays are assumed to be zero. The underlying timescales in these theories are typically postsynaptic potentials (PSP) rise and decay times. Thalamocortical networks are also "local" from the viewpoint of scalp electrodes, which cannot distinguish purely cortical from thalamocortical networks. Finally, these theories are "local" in the sense of being independent of global boundary conditions dictated by the size and shape of the cortical-white matter system.

By contrast, I use the label *global theory* to indicate mathematical models in which delays in corticocortical fibers forming most of the white matter in humans provide the important underlying timescales for the large spatial scale EEG dynamics recorded by scalp electrodes. Periodic boundary conditions are generally essential to global theories because the cortical-white matter system of each hemisphere is topologically very close to a spherical shell. One global theory [2, 4, 18, 33] that follows the mesoscopic excitatory synaptic action field $\Psi_e(\mathbf{r}, t)$ has achieved some predictive value in electroencephalography despite its neglect of most network effects. This "toy brain" is presented first as a plausible entry point to more realistic theory in which cell assemblies play a central role in cognition and behavior. Secondly, I conjecture that global synaptic action fields may act (top-down) on local networks in a manner analogous to human cultural influences on social networks, thereby providing a possible solution to the so-called *binding problem* of brain science [2, 8, 18]. Several recent theories of neocortical dynamics include selected

aspects of both local and global theories, but typically with more emphasis on one or the other, as outlined by Nunez and Srinivasan [2, 18].

## 1.13 Summary: Brain Volume Conduction Versus Brain Dynamics

The physical and physiological aspects of electroencephalography are naturally separated into two disparate areas, *volume conduction* and *brain dynamics* (or *neocortical dynamics*). The first area is concerned with the relationships between current sources $\mathbf{P}(\mathbf{r}, t)$, the so-called "EEG generators," and their corresponding scalp potentials. The fundamental laws governing volume conduction, charge conservation, and Ohm's law leading to Poisson's equation (1.3) are well known, although their application to EEG is nontrivial. The time variable in Poisson's equation acts as a parameter such that the time dependence of an EEG at any location is just the weighted space average of the time dependencies of contributing brain sources. The fact that EEG waveforms can look quite different at different scalp locations and be quite different when recorded inside the cranium is due only to the different weights given to each source region in the linear sum of contributions. The resulting simplification of both theory and practice in EEG is substantial.

The issue of brain dynamics, that is, the origins of time-dependent behavior of brain current sources producing EEGs, presents quite a different story. Although a number of plausible, physiologically based mathematical theories have been proposed, we may be far from a proven theory. Nevertheless, even very approximate, speculative, or incomplete dynamic theories can have substantial value in the formation of conceptual frameworks supporting brain function. Such frameworks should provide a rich intellectual environment for designing new experiments and for evaluating quantitative EEG methods. In particular, several dynamic models, with emphasis ranging from more local to more global dynamics, can be combined with volume conduction models as a means of testing the quantitative EEG methods proposed in this book.

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CHAPTER 2

# Techniques of EEG Recording and Preprocessing

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Tzyy-Ping Jung and Scott Makeig[2]

This chapter summarizes the key features of EEG signals, event-related potentials, and event-related oscillations, and then more recent EEG hardware developments are discussed. In particular, different electrode types such as passive and active electrodes, as well as different electrode cap systems and layouts suitable for high-density EEG recordings, are introduced and their potential benefits and pitfalls mentioned. The third part of this chapter focuses on prominent exogenous and endogenous EEG artifacts and on different procedures and techniques of EEG artifact rejection and removal. Specifically, in the final part of this chapter, independent component analysis (ICA) is introduced. ICA can be used for EEG artifact correction and for the spatiotemporal linear decomposition of otherwise mixed neural signatures. In combination with state-of-the-art recording hardware, the advanced analysis of high-density EEG recordings provides access to the neural signatures underlying human cognitive processing.

## 2.1 Properties of the EEG

Recently the dominant role of EEG and MEG in understanding the human brain–behavior relationship has been recognized again. In contrast to functional magnetic resonance imaging (fMRI), EEG and MEG techniques monitor large-scale human brain activity patterns noninvasively and with millisecond precision, which is crucial for understanding the neural foundations of cognitive functions. The following section summarizes some of the assumptions and properties of the EEG signal in event-related brain research before the hardware necessary for EEG recording and preprocessing steps is discussed.

### 2.1.1 Event-Related Potentials

The common way of analyzing event-related EEG signals is the calculation of event-related potentials (ERPs). This is done by repeatedly presenting an event of interest, such as a visual stimulus on a computer screen, and analyzing the small

1. These authors contributed to Sections 2.1, 2.2, and 2.3.
2. These authors contributed to Section 2.4.

fraction of EEG activity that is evoked by this event. Computationally, the ERP is revealed by extracting EEG epochs time-locked to the stimulus presentation and calculating the average over the EEG epochs. The assumptions behind this approach are illustrated in Figure 2.1. A key assumption is that the measured signal consists of the sum of ongoing brain activity and a stimulus-related response that is independent from the ongoing activity. Also, the response is considered invariant over repeated stimulation. Because it is much smaller in amplitude, averaging is necessary, which reveals the phase-locked, evoked portion of brain activity and removes activity that is not time-locked to the event.

ERPs in response to sensory (or cognitive) events usually consist of a number of peaks and deflections, which, if they can be characterized by latency, morphology, topography, and experimental manipulation [1], are called ERP components. Early components typically reflect sensory processing and can be associated with the respective sensory cortical areas, whereas later ERP components can inform about cognitive aspects of brain function. ERP components are usually small in amplitude (1 to 20 $\mu$V), show substantial interindividual variation, and are susceptible to various artifacts. It is therefore necessary to carefully evaluate ERP properties before conclusions can be drawn. One important aspect of evaluating ERPs concerns the SNR, because the average yields a valid estimation of the ERP only to the extent that noise is removed. The ERP SNR improves as a function of the square root of the number of epochs (1/sqrt($N$)). A convenient way of measuring the SNR for an ERP component of interest is to measure the amplitude of this component and divide it by the standard deviation of the prestimulus interval. The prestimulus interval provides a reference period for the estimation of zero potential and, therefore, this SNR definition follows the rationale of the ERP data model. However, because the assumptions of the ERP model may be unrealistic, the estimation of signal and noise based on different intervals could be misleading. An alternative is to compute the difference of ERPs based on odd- and even-numbered epochs. Dividing the difference waveform by 2 reveals what has been known as the plus-or-minus (±) reference [2],

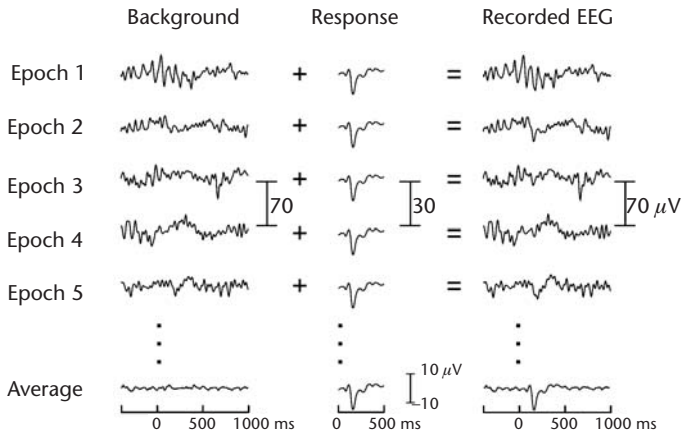



**Figure 2.1** Additive ERP model. The model assumes that ongoing EEG activity sums to zero across repeated events, whereas the brain response is invariant across repeated events and is therefore preserved in the average. Accordingly, the measured signal consists of the sum of ongoing and evoked activity, which are thought to be independent.

which is a noise estimate that is less dependent on the validity of the additive ERP model assumptions.

### 2.1.2 Event-Related Oscillations

Continuous EEG recordings consist largely of oscillations at different frequencies that fluctuate over time and provide valuable information about a subject's brain state. Brain oscillations such as EEG alpha activity (8 to 13 Hz) clearly respond to sensory stimulation (e.g., alpha suppression). To what extent these oscillations contribute to event-related EEG signals such as ERPs is a matter of ongoing research [3, 4]. Notwithstanding this discussion, it has become evident that the ERP does not necessarily capture all event-related information present in the EEG. For instance, oscillations induced by, but not perfectly phase-locked to, an event of interest zero out in the process of ERP calculation. It is therefore helpful to distinguish between evoked, phase-locked oscillations and induced, nonphase-locked signals. In this context, the term *total power* refers to the sum of evoked and induced oscillations. Total power is calculated by summing the values of the frequency transform of the single trials, whereas evoked power is obtained by a frequency transform of the time-domain average, namely, the ERP.

A change in oscillatory power can be due to a change in the size of the neuronal population generating the oscillation, or it can reflect a change in the degree of synchronization of a given neuronal population. With the latter mechanism in mind, Pfurtscheller coined the expression *event-related synchronization* (ERS) for relative power increases and event-related desynchronization (ERD) for relative power decreases [5]. ERS/ERD calculations can be displayed over time (e.g., by using a short-time fast Fourier transform or wavelet decomposition) and expressed as a percentage signal change relative to the pre-event reference period. For instance, the amount of EEG alpha activity prior to the presentation of a visual target predicts, to a substantial extent, whether the target will be consciously perceived [6, 7]. More generally, the ERS/ERD type of analysis and its extension to the broad frequency range [8] reveals important information about brain function without assuming independence between ongoing activity and brain-electrical responses.

### 2.1.3 Event-Related Brain Dynamics

The distinction between evoked (ERP) and induced (event-related oscillations) brain activity suggests that, beyond the consideration of power changes as in the original ERS/ERD analysis, the consistency of phase across epochs provides relevant information. Indeed, ERPs could also be the result of changes in the phase consistency of ongoing oscillations in the absence of a power increase [3], a phenomenon that has been named partial phase resetting (PPR). To investigate these mechanisms, it is necessary to move from the analysis of averaged brain response to the analysis of single epochs or trials. That is, the basis for the consideration of phase concentration is the frequency, or time-frequency, analysis of every single recorded trial. This is expressed in the event-related brain dynamics model [4], which represents a three-dimensional signal space with the axes power change, frequency, and phase consistency (Figure 2.2).

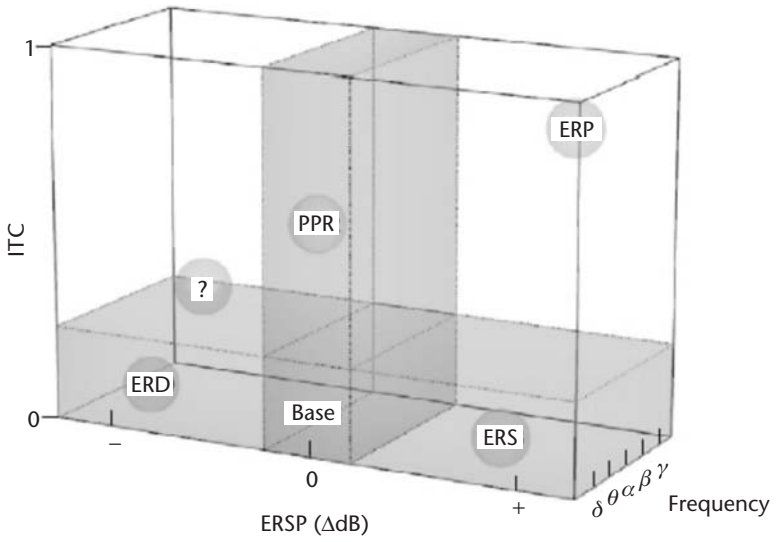



**Figure 2.2** Event-related brain dynamics model. Event-related EEG signals can be described according to amplitude, frequency, and the degree of phase-locking across trials. Relative to a reference period, amplitudes can be decreased or increased (ERS/ERD) in a certain frequency range. The intertrial coherence (ITC) reflects a measure of the phase consistency across trials, with 0 indicating uniform random distribution, and 1 indicating identical phase across trials, specifically for each time and frequency of interest. The evoked, additive response, following the ERP model (Figure 2.1), is located in the upper right-hand corner, whereas ERPs generated by partial phase resetting (PPR) are characterized by phase concentration in the absence of amplitude/power changes. (*From:* [4]. © 2004 Elsevier Ltd. Reprinted with permission.)

In this signal space, event-related EEG responses corresponding to the additive ERP model can be localized as well as ERS/ERD and PPR. The key feature that results from this view is that, more likely than not, further, not-yet-explored spots of response patterns exist. The trend toward EEG/MEG single-trial analysis is very promising for addressing important questions in the field of cognitive neuroscience [9]. We, therefore, consider it crucial to optimize other aspects that determine EEG quality, as discussed in the following section.

## 2.2 EEG Electrodes, Caps, and Amplifiers

### 2.2.1 EEG Electrode Types

Electrodes suitable for EEG recordings can be made from a variety of materials such as tin, stainless steel, gold-plated silver, pure silver, pure gold, and Ag/AgCl. All of these electrode types are actively being used for a variety of clinical and research EEG recording purposes. Because any two or more metals immersed into an electrolyte will result in a dc offset potential being generated and because these dc offsets depend largely on the electrochemical properties of the metals used, one important rule is to never mix recording electrodes made from different materials.

The most commonly used electrode type consists of bonded (sintered) Ag/AgCl, which quickly establishes and then maintains consistent and stable electrochemical potentials against biological tissues, together with low dc offset variability. Moreover, Ag/AgCl electrodes are free from potential allergenic compounds and have excellent long-term electrical stability.

Figure 2.3 shows some common examples of passive EEG electrodes. The term *passive* EEG electrode implies that the electrode itself is not electrically active but instead functions as a passive metal sensor that establishes electrochemical contact to the scalp via the electrolyte used. This sensor then converts the changes in charged ion concentrations on the scalp into an electrical current that is transmitted along the electrode cable and is then measured in the biopotential instrumentation amplifier.

Traditionally, closed (hat-shaped) electrode forms were used in EEG recordings where a low number of individual electrode positions were used and the scalp was abraded before electrode application. Typically, an electrolyte together with collodion has been used to glue the electrode firmly to the scalp, for example, for long-term recordings. However, with the high density and vastly multichannel recordings employed today, placing individual electrodes is neither feasible, nor would exact placement of large quantities of electrodes be practically possible. Therefore, the use of electrode caps has become standard. These caps have electrode holders fixed to the textile fabric and thus establish the approximate electrode positions without need for single electrode position measurements. Scalp preparation is

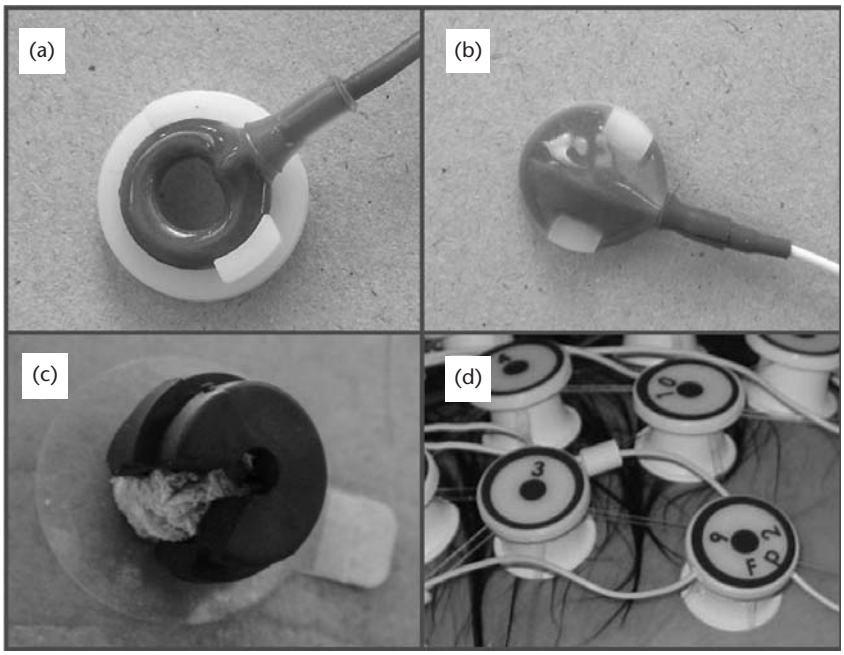



**Figure 2.3** Examples of commonly used passive Ag/AgCl electrodes. (a) Ring-shaped Ag/AgCl electrode in electrode holder, granting easy access to the scalp for skin preparation (EasyCap, Herrsching, Germany). (b) Classical "hat-shaped" electrode typically used for individually placed EEG electrode derivations. (c) QuikCell electrode, cut open to reveal the cellulose sponge element used with an electrolytic solution instead of a gel-based electrolyte (Compumedics, El Paso, Texas). (d) HydroCel electrodes in a Geodesic Sensor Net (EGI, Eugene, Oregon).

performed through openings in the electrode holder. When closed electrodes are used, a process of electrode detachment, scalp preparation, and electrode reattachment is necessary and this is too time-consuming and error-prone to allow for efficient, high-density EEG recordings. Accordingly, preparation time for high electrode counts can be efficiently reduced by using open, ring-shaped electrodes [e.g., as shown in Figure 2.3(a)] that do not require detachment for scalp preparation.

The use of electrode caps and Ag/AgCl ring electrodes does not abolish the need to prepare the scalp by slight mechanical abrasion to establish the impedance between 5 and 10 kΩ that is required by traditional EEG amplifiers for operational stability and noise immunity. With modern EEG amplifiers, homogeneity of electrode impedances is more important for good common mode rejection than achieving low electrode impedances per se. Skin preparation, taken together with gel application and impedance checking, still takes a substantial amount of time, but because this is a critical step in achieving good EEG measurements, one should definitely properly and carefully prepare the scalp, because time saved here invariably means more time can be devoted to data processing efforts. Figure 2.3 shows examples of passive electrodes commonly used today.

Several EEG equipment manufacturers have tried to overcome the limitations imposed by the time-consuming scalp preparation. The most notable example of this is the HydroCel Geodesic Sensor Net (GSN) by Electrical Geodesics Incorporated (EGI, Eugene, Oregon), shown in Figure 2.3(d). This electrode net system does not consist of a traditional textile fabric cap, but instead of a geodesic (i.e., shortest distance between two points on the surface of a sphere) arrangement of flexible rubber-band-like fibers interconnecting the individual electrode holders. The electrode holders themselves contain the electrode pellets. The HydroCel net can be used with sponge inserts soaked in potassium chloride saline for recordings of up to 2 hours or without sponge inserts but with electrolyte applied directly into the electrode wells for recordings of longer than 2 hours. The GSN is then applied to the head much like a wig and the individual electrode holders are straightened out to enable good placement and electrode contact. The entire procedure can be performed in about 10 minutes for 128-channel nets, which makes this an attractive design for fast EEG acquisition preparation. The sponge element holding the saline or electrolyte solution keeps the conductive electrode pellet in contact with the scalp and no further impedance reduction is needed. The resulting scalp impedance typically is on the order of 30 to 70 kΩ, which necessitates the use of a special amplifier with an equally increased input impedance (of 200 kΩ or more). Also, as compelling as the obvious time advantage of using this system is, it must be noted that this electrode system and EEG amplifier may not always be perfectly suited for the optimization of the EEG SNR due to the high impedance measurements employed.

Another variant of this scheme is the Quick Cell system made by Compumedics-Neuroscan (El Paso, Texas), shown in Figure 2.3(c), which consists of classical Ag/AgCl electrodes that can be fitted with cellulose sponges. After the cap is placed on the head, a small amount of a special electrolyte solution is injected into each electrode, thus wetting the sponge, which expands in response to the fluid contact for recordings of up to 3 hours. If an impedance level of 30 to 50 kOhms is deemed sufficient and a high input impedance EEG amplifier is used, then no further

scalp preparation is required to achieve EEG recordings of adequate quality. However, if low impedances are required in order to optimize the data SNR, then further scalp preparation with a blunted needle is required, thus largely abolishing the time advantage obtained when using this electrode type in high impedance mode. However, although this effectively limits the benefit of using this methodology to eliminate the need to apply gel to the subject's hair, this advantage should not be underestimated at least as a motivational factor in today's multichannel recordings.

Figures 2.3(a, b) show typical ring-shaped (EasyCap GmbH, Herrsching, Germany) and closed (hat-shaped) Ag/AgCl electrodes that are commonly used today.

Another way of dealing with the impedance and noise problems inherent in classical passive electrode schemes is to use active electrodes (Figure 2.4). *Active* in this context means that the electrodes themselves contain active electronics circuitry that serve the purpose of treating the incoming high-impedance scalp electrical signal in such a way that three main goals are achieved: (1) stable operation for a much broader range of scalp impedances, (2) tolerance against a wider range of impedance differences across the scalp, and (3) enhanced noise immunity along the cable path toward the amplifier. The electronics principle by which these three goals are achieved is known as impedance conversion. In its simplest form, impedance conversion refers to a circuit consisting of an operational amplifier that is effectively set for

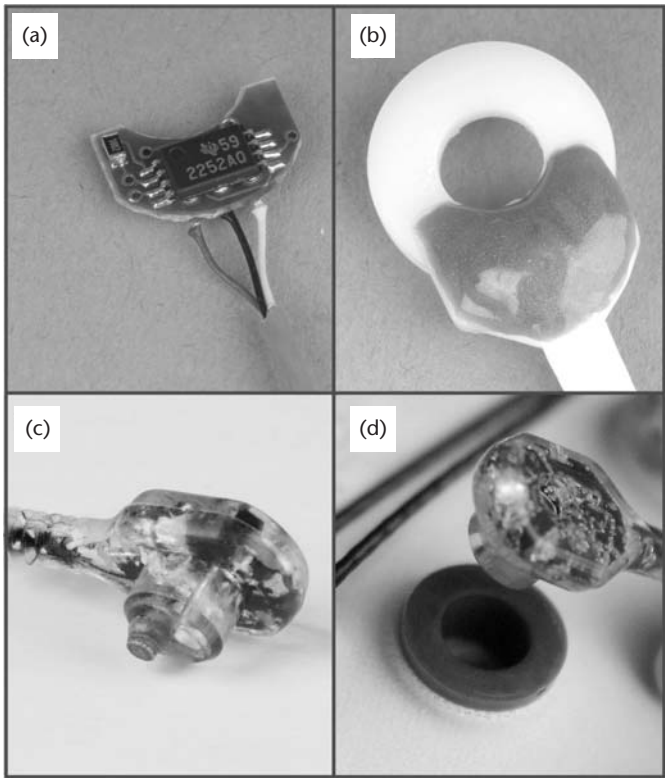



**Figure 2.4** Examples of two commonly used active Ag/AgCl electrodes. (a) Upper side of an EasyCap Active electronics circuit board with operational amplifier. (b) EasyCap Active after completion with Ag/AgCl Pellet and integrated electrode holder (EasyCap, Herrsching, Germany). (c) Upper and (d) lower side views of the actiCap electrode with the LED-based onboard impedance check activated (Brain Products, Gilching, Germany).

no amplification. This makes this circuit a buffer circuit, and the idea is that the operational amplifier has a high input impedance matching that of the scalp signal but also gives low output impedance for the transmission of the signal from the electrode plate to the amplifier input stage. This low impedance immunizes the electrode lead against capacitively coupled ambient noise and thus helps to achieve a good SNR.

Using active electronics on an electrode requires extra leads for the supply of power to the electronics, which potentially increases the bulk of electrode cabling. This can be alleviated by the use of microcabling and is easily made up for by the possibilities the availability of electrical power on the electrode affords. The actiCap system made by Brain Products GmbH (Gilching, Germany) is shown in Figure 2.4(c, d). This system uses an additional data line in order to implement an optical impedance check directly on the electrode itself. During impedance mode a three-color LED (red-yellow-green) shows the impedance state of each electrode directly on the head of the subject, thereby eliminating the need to check the computer display for suboptimal impedances. This new feature allows for extremely efficient and fast electrode preparation.

Another feature, which the actiCap shares with the EasyCap Active by EasyCap GmbH (Herrsching, Germany) shown in Figure 2.4(b), is the ability to connect to virtually any existing EEG amplifier system. Both active electrode systems give all advantages of active electrodes at the sensor level, but also convert the signals so that they can be measured by any connected standard amplifier. This allows for the added advantage of using active electrodes without having to invest in a complete "active" EEG amplifier system such as the ActiveTwo system manufactured by Biosemi (Amsterdam, Netherlands), which was among the first active electrode systems for EEG recordings on the market and is also widely used. A 256-channel Biosemi active electrodes system is shown in Figure 2.5(c).

### 2.2.2 Electrode Caps and Montages

Traditionally, the International 10-20 system defined by [10] has been used to describe the locations of EEG scalp electrodes relative to anatomic landmarks on the

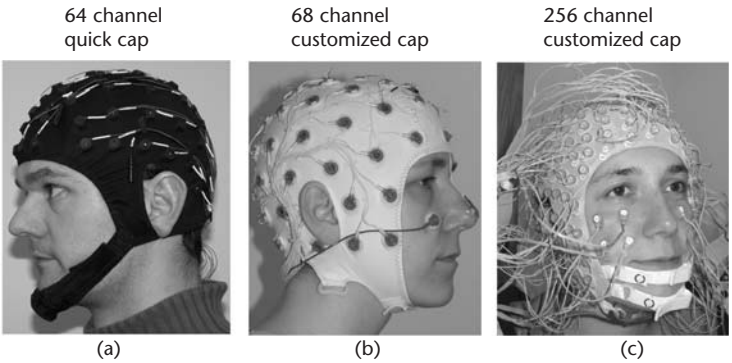



**Figure 2.5** Electrode caps and montages. (a) Commercially available 64-channel electrode cap based on the 10-10 layout (Compumedics, El Paso, Texas). (b) Customized infracerebral 68-channel electrode cap (EasyCap, Herrsching, Germany). (c) 256-channel infracerebral electrode cap developed at the Swartz Center for Computational Neuroscience, San Diego, California, by A. Vankov and S. Makeig, in collaboration with L. Smith (Cortech Solution, Wilmington, North Carolina).

human head. However, because it is limited to only 21 scalp locations, alternatives that providing for a larger number of channels have been proposed. In 1985, the 10-10 system for the placement of up to 74 electrodes was proposed [11]. Oostenveld and Praamstra [12] defined the 10-5 system to further promote the standardization of electrodes in high-resolution EEG studies. In the 10-5 system, a nomenclature and coordinates for up to 345 locations are defined. The system provides great flexibility, because it allows the selection of a subset of homogeneously distributed positions. The interelectrode distance (on a standard head with 58-cm circumference) is typically between 53 and 74 mm for the 10-20 system, and between 28 and 38 mm for a 61-channel montage following the 10-10 system. For a homogenous 128-channel layout based on the 10-5 system, the interelectrode distance would further decrease to approximately 22 to 31 mm. Unfortunately, because both the 10-10 and the 10-5 system are based on the original 10-20 system, none of these systems features equal distances between electrodes. In addition to the matter of interelectrode distance, an important issue is the distribution of spatial sampling. If one simplifies the head as a sphere, the original 10-20 system spatially covers only little more than half of the sphere. In contrast, both the 10-10 and 10-5 system extend the spatial coverage to approximately 64%. Both of these issues, a sufficient electrode density and a maximum coverage of the head sphere, can be considered beneficial for extracting spatial information from an EEG [13]; therefore, many EEG laboratories and some manufacturers have developed equidistant and spatially extended channel montages.

Figure 2.5 shows three different electrode cap systems. Note the different interelectrode distance between the caps as well as the different spatial sampling. The equidistant 68-channel customized cap system features a relatively large interelectrode distance of approximately 38 mm, but covers about 75% of the head sphere and therefore provides a good basis for accurate source localizations [14]. Figure 2.5(c) shows a customized 256-channel cap developed at the Swartz Center for Computational Neuroscience (San Diego, California). This cap provides both a very dense array with 25 mm of distance between electrodes and a significantly extended spatial sampling. In contrast to usual recording traditions, EEG signals are recorded from the face as well, which can provide important additional information [15].

To conclude, with the advent of multichannel EEG recordings, the choice and design of the electrode cap used is a matter of great importance. Caps that extend beyond the traditional 10-20 range can provide significant benefits, among them a better and more comfortable fit, a more evenly distributed weight of electrode cables, and, most importantly, more accurate spatial sampling of the scalp recorded EEG. These are only some of the benefits of modern electrode caps, and we expect further improvements to become commercially available over the next few years.

### 2.2.3 EEG Signal and Amplifier Characteristics

Probably the most important component in optimizing the SNR of EEG signals is the amplification circuitry itself. It is here that noise present in the data can be reduced or eliminated, and much of the reliability and validity of day-to-day EEG research depends on the quality of the amplifier design used.

The most desired property of any EEG amplifier is that it amplifies the EEG signal and disregards or attenuates any undesired signal influences. Three different signal components have to be dealt with by an EEG amplifier: biological signals, electrode offset signals, and mains noise signals. The *EEG biosignals* are the summed potentials measured at a scalp electrode and consist of the cortical and, to a small extent, subcortical activity. However, this signal is invariably compromised by endogenous and exogenous artifacts, such as scalp and sweat potentials, eye movement, and other EMG artifacts as well as artifacts related to the stimulus presentation, for example, electrical pain stimulation spikes. Thus, while the electrocortical signal itself typically only has an amplitude range of approximately ±150 $\mu$V, the total signal referred to here can have an amplitude range of ±2 mV.

The *dc offset* component is inherent to the signal measurement with metallic electrodes of any type. It fluctuates over time and, depending on the type of electrode used, can reach large values of several hundred millivolts. High temporal dc offset stability with low offset values of typically less than 100 mV is a common feature of high-quality Ag/AgCl electrodes, which makes these the favored electrodes in EEG research.

The *mains noise* is another, more obvious noise component consisting of sinusoidal artifacts at the mains frequency (50 or 60 Hz). The prevalence of mains noise in the recording environment depends largely on the presence of mains powered electrical devices in the vicinity of the recording equipment. Also, the type of equipment present has a major influence on the magnitude of mains noise, with devices containing electrical motors such as pumps, razors, and hair driers being particularly "good" emitters of such noise. Mains noise is capacitively coupled into the cables of the EEG electrodes and as such has a more profound influence on the signal measured with high electrode impedances. Also, for the same reason applied to capacitive coupling, keeping all electrodes together in a ribbon or bundle of cables will largely reduce mains noise. Mains noise can be further reduced by operating the EEG amplifier with (rechargeable) batteries.

Another source of this noise lies within the amplifier system itself. For patient safety reasons, the subject has to be kept "floating" with regard to the mains and the earth ground. However, because this would make the difference between the body potential and amplifier potential arbitrary up to the level of the full mains voltage, the body of the subject is typically connected to the patient ground. In this way, the potential difference between the body and the amplifier inputs with reference to mains is kept in a range of typically less than 100 mV. The mains noise signal is present at all inputs (channels) of the amplifier and is therefore oftentimes called *common mode* noise. The common mode signal can also effectively be reduced using a concept called *active shielding*, as discussed later in this chapter.

If we look at the summed potentials resulting from worst case *Biosignal+Offset+Mains* voltages (2 mV + 100 mV + 100 mV = 202 mV), it becomes clear that such levels would be beyond the digitization levels of most modern amplifiers and such a signal could only be amplified by a factor of 25 before an EEG system built from operational amplifiers powered with ±5V would saturate. However, modern EEG amplifiers are built with multichannel instrumentation amplifiers, which are designed to amplify only the *biosignal* portion at the gain set, while passing the *offset* voltages through unamplified and at the same time cancelling the *mains* noise.

This latter characteristic, the amplifier's ability to suppress signal components that are commonly present at all input terminals, is called the common mode rejection ratio (CMRR) and is given in decibels (at a given frequency), with higher decibel values representing better noise suppression.

Another method by which modern amplifiers help to achieve optimal SNR is called *active shielding*, which is as simple as effective. The active shielding circuit consists of a special electrode cable with a shield mesh wrapped around the inner core (the EEG lead). The mains noise (and other ambient noises) is capacitively coupled onto the electrode lead due to the lead's relatively high impedance. However, the output of the amplifier carries the same signal in an amplified and low-impedance variant. This output can then simply be fed onto the shield mesh and effectively insulates the lead core with the EEG signal from the ambient noise. If this method is extended to feeding back the average of all amplifier outputs, then the shield is being driven actively with the common mode signal, which is an efficient mechanism for achieving better SNR from the signals measured.

## 2.3 EEG Recording and Artifact Removal Techniques

### 2.3.1 EEG Recording Techniques

The amplifier parameters chosen to record the EEG have a large impact on the quality of the data derived. Central acquisition parameters are the sample rate, the gain (vertical resolution), the highpass and lowpass filter characteristics, and the notch filter that can be used to eliminate residual mains noise. All of these parameters have to be set with respect to the signals to be derived from the recordings and with respect to the demands of the experimental paradigm at hand.

*Sample Rate*

According to the sampling theorem, the sample rate should be at least twice as high as the highest frequency of interest contained in the signal. However, the question is how this rule relates to the event-related EEG signal of interest. A good rule of thumb for ERPs is to consider the temporal extent of the shortest ERP component of interest and to adjust the sample rate (SR) so that this component is acquired with a minimum of 20 points. For example, if the N1 component of the ERP is the shortest target component with an extent of around 100 ms base to base, then the calculation $SR = (1{,}000/100) \times 20$ would result in a minimally required sample rate of approximately 200 Hz. The sample rate choice should also consider any prior knowledge regarding the temporal extent of typical statistical effects for the components under investigation. However, higher sample rates only make sense within the spectral bounds of the neuronal circuitry under investigation and a trade-off should be sought between information gain and file size.

*Gain*

The gain or vertical resolution of the signal should be chosen with two aspects in mind. First, the gain is directly coupled to the maximum positive and negative voltage the amplifier can resolve without saturation. This is particularly important with dc recordings, where even profound drift of the signal is tolerated. Second, the gain

should be suited to the paradigm used. If the difference between the experimental conditions is very small, such as in EEG gamma-band studies with effects at or below 1 $\mu$V, then the resolution should be high enough to resolve this difference with at least 10 steps to ensure proper quantification of peak values. Also, and equally important, low amplifier gains may not raise the signal level sufficiently above the noise level present at the input stage of the amplifier, resulting in lower SNR. To summarize, the gain should be set as high as possible without risking saturation of the amplifier.

*Highpass Filter*

The choice of the highpass edge frequency and steepness (order) depends on the measures to be acquired. Measurements of slow potentials such as contingent negative variations (CNV) or of lateralized readiness potentials (LRPs) require, or at least benefit from, the use of dc coupled recordings. For all other recording purposes the time constant of this filter should be set long enough to allow passage of the slowest components expected without significant alteration due to the filter.

*Lowpass Filter*

The usable spectrum acquired is generally bounded by the effective bandwidth of the amplifier, which is typically enforced by analog hardware filters in the amplifier input stage. Further, digital lowpass filters can be used to limit the frequency band acquired to the spectral content of interest.

*Mains Notch Filter*

The mains notch filter is a very steep filter designed to specifically filter out a very narrow band of frequency content around the mains frequency. One would generally use a notch filter in EEG recordings, unless its use interferes with the target spectrum. This would be the case, for example, for recordings of gamma-band activity. It would be difficult to give examples of recording parameters that are representative beyond the scope of a specific modality or paradigm since the exact recording parameters have to be honed for the recording task at hand. However, if the parameters are chosen according to the preceding general rules of thumb and with the task and EEG components of interest in mind, it should be easy to achieve the temporal, amplitude, and spectral resolution and accuracy required for the research at hand.

### 2.3.2 EEG Artifacts

As outlined earlier, it is important to create a recording environment that minimizes the potential for ambient artifacts. Typical steps taken to ensure optimal recording environments include the use of an acoustically and electrically shielded EEG cabin and the installation of a separate earth ground band for the laboratory. Further steps should include ensuring that all interconnected devices use the same mains phase and ground and that either centrally or locally installed mains noise filters are used, for example, through a high-quality uninterruptible power supply. However, even though steps taken to reduce artifact sources in the recording environment are usually quite effective, they can only help to reduce, but cannot totally avoid, external

artifact influences. Every EEG recording will therefore typically have a small number of artifacts that need to be dealt with. Also, situations such as bedside or field study recordings do not allow such artifact prevention measures to be taken, so the resulting level of artifacts will clearly be higher.

The exogenous artifacts most commonly seen in EEG recordings are mains noise, spurious electrical noise from other sources such as elevators or engines, and artifacts that result from body or electrode lead movement or brief electrode detachment caused, for example, by contact with the chair or bed against which the head rests. Figure 2.6(d) shows an example of a single-channel artifact. As can be seen, the resulting artifact is brief itself and isolated to one single channel, which makes simple removal (rejection) the method of choice. Most of the exogenous artifacts typically seen are spurious and do not require any action beyond exclusion of the respective data stretches from further analysis. However, some exogenous artifacts have signal properties that allow for correction by time-domain or spatial filtering. The use of a notch filter for mains noise is a good example of the former. A typical example for the correction of an exogenous artifact with a spatial filter would be the use of independent component analysis (ICA) for the decomposition of an EEG signal recorded in the MR and containing helium pump artifacts. ICA typically captures such artifacts in one or two components that can then be selectively removed. For most exogenous artifacts, however, neither of the above is an option and simple removal of the artifact data is required.

Endogenous artifacts are those that have their origin within the subject's body. The most common endogenous artifacts are eye movement–related potential changes and neuromuscular discharges due to movement or muscle tension especially from frontalis and temporalis muscles. Other endogenous artifacts such as EKG intrusions are much less visible, but generally also present in EEG recordings.

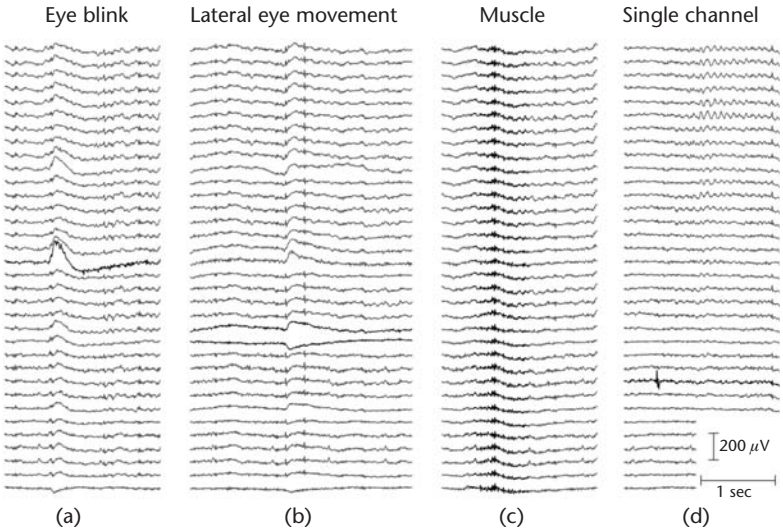



**Figure 2.6** Typical EEG artifacts as represented in multichannel EEG recordings. (a) Eye-blink artifacts are transient signals of characteristic frontopolar topography. (b) Lateral eye movements typically show opposite polarities at lateral frontal electrode sites. (c) Muscle or EMG artifacts contribute power over a broad frequency range to the EEG. (d) Transient single-channel artifact, probably related to electrode movement or sudden changes in the electrical properties of the electrode. Note the absence of a similar artifact at all other channels.

Their magnitude and visibility in the EEG depends on various factors including the EEG reference and montage used [16]. Normally, removal of the EKG influence is not required because it is rarely time-locked to the stimulation and thus will average out during EEG processing. However, if a profound affliction makes EKG artifact removal necessary, this can be done either with a template subtraction based approach or with ICA. A different artifact related to this is the pulse artifact caused by the placement on an electrode directly above a blood vessel, resulting in pulsatile artifacts at the heartbeat frequency, a problem that is particularly present in simultaneous EEG-fMRI recordings [17]. Another source of endogenous EEG artifacts is the respiratory system, which can cause slow variations in scalp impedance resulting in equally slow shifts. Finally, sweating of the scalp can have a profound effect on the EEG since the sodium chloride and other sweat components such as lactic acid react with the electrode metals to produce battery potentials that present in the EEG as slow oscillations (0.1 to 0.5 Hz) of fairly large magnitude.

Figure 2.6 shows examples of three endogenous and one exogenous artifact. Figure 2.6(a) shows a vertical eye blink and how the channels are affected by this to differing degrees. Blink artifacts are quite large in amplitude with typical blink peak values being on the order of several hundred microvolts. Figure 2.6(b) shows a typical horizontal eye movement, with the lateralized activity on the electrodes at the outer canthi of the eyes being clearly visible in the lower middle part of the panel. Both the vertical as well as the horizontal eye movements can easily be detected based on their unique topographies and can subsequently be removed, for example, with a regression-based or, better, an ICA-based method (see Section 2.4). Muscle activity related artifacts, as shown in Figure 2.6(c), typically contaminate the EEG at higher frequencies. EMG artifact reduction is often based on the application of a lowpass filter. Yet to some extent the EEG frequency range of interest may overlap with the broadband contamination that muscle activity contributes, making this artifact notoriously difficult to remove. In addition to the endogenous artifact examples shown in Figure 2.6(a–c), a range of exogenous artifacts can occur as well. The spike-like example shown in Figure 2.6(d) can be classified as an exogenous artifact event, because it is spatially restricted to a single channel, making a brain source highly unlikely due to the lack of volume conductance–related spatial smearing. Indeed, the better spatial sampling of state-of-the-art EEG recordings improves the identification and characterization of exogenous as well as endogenous artifacts.

### 2.3.3 Artifact Removal Techniques

Artifact rejection is required when the artifacts present in the EEG data cannot be removed algorithmically. For this, a great range of different measures and approaches exists, including these:

- *Simple amplitude threshold:* This value defines positive and negative amplitude levels above/below which data is automatically recognized as artifacts.
- *Min-max thresholds:* This measure sets a maximally allowed amplitude difference within a specific length of time. This achieves something very similar to

the amplitude criterion, but is suitable for dc coupled recordings due to its independence from absolute value thresholds.

- *Gradient criterion:* This criterion defines an artifact threshold on the basis of voltage changes from data point to data point relative to intersample time ($\mu$V/ms) and is useful for finding, for example, episodes of supraphysiological rates of change in the data.
- *Low activity:* This criterion defines thresholds of minimally allowed differences between the highest and lowest values in a given length of time and allows detection of, for example, channel saturation or hardware channel failures.
- *Spectral distribution:* These criteria define artifact time stretches based on their spectral composition. One example would be to define episodes as artifacts in which the mains frequency power exceeds a certain threshold.
- *Standard deviation:* With the dynamics of spontaneous EEGs being quite well characterized, one can define artifacts by their dynamics over time as expressed by a moving or segment-based standard deviation index.
- *Joint probability:* This relatively new index determines the probability of the occurrence of a given time point value in a specific channel and segment relative to the global probability of the occurrence of such a value and thus can be used to find improbable data stretches [18].

If artifact rejection is performed on the basis of segmented (epoched) data, the preceding criteria would remove the entire segment afflicted with an artifact. However, if the artifact rejection is done on continuous data, a generous amount of time should be marked as artifact around the actual threshold data because some artifacts may show some time of subthreshold but nonetheless affected values leading up to their full artifact manifestation. Also, the artifact criteria used should be chosen to quite rigidly clean the data and should be applied without individual variance to all EEG datasets in the set of data to be analyzed.

Another option is to restrict the artifact rejection to only those channels that actually carry the artifact (individual channel mode). This is sometimes useful when working with populations that produce a large number of artifacts. If rigid artifact rejection were used, none or too few trials would remain for averaging, so the individual channel mode would therefore be a must in these cases. However, note that this option will result in different trial counts per channel and thus also in a different SNR per channel, which is clearly suboptimal. The use of this option should therefore be reserved for those cases where it is absolutely necessary in order to retain sufficient numbers of trials for averaging.

Some artifacts can also be corrected by application of statistical methods. Regression-based correction is common and can, in principle, be performed for any artifact that: (1) can be recorded concurrently with the EEG data, (2) has a linear relationship with the corresponding artifacts in the EEG data, and (3) shows no temporal lag between the events in the artifact channel and in the EEG channels, thus implying volume conductance as its means of propagation. These prerequisites are largely met by eye blinks, saccades, and other vertical and horizontal eye movements. These electrooculogram (EOG) artifacts are caused by a number of concur-

rently active processes such as the retraction and rotation of the eye bulb [19] as well as the closure and reopening of the eyelids [20].

Although the utility and accuracy of regression-based ocular correction is a matter of ongoing discussion [21], it is widely used, and a number of implementations and variations exist. The most commonly used are those by Gratton et al. [22] and Semlitsch [23]. Both algorithms share the general mechanism of first finding blink-afflicted data stretches in the EOG channel(s) with (different) thresholding techniques. Based on the blink stretches found, both algorithms then calculate the regression of the eye channel(s) with each individual EEG data channel, and correct the EEG data with $EEG' = EEG - \beta \times EOG$, where $\beta$ is the regression weight for a given channel.

The Gratton et al. algorithm [22] has two further characteristics that are worth noting: First, the raw averages for each condition are subtracted from each data segment prior to regression calculation and are added back in before the correction is performed. Second, the algorithm calculates and applies separate regression coefficients for data time ranges inside of blink stretches and for those outside of blink stretches, which can easily lead to the creation of step discontinuities during regression, since time ranges corrected with (slightly) different regression coefficients border directly onto one another. The raw average subtraction is done under the assumption that the measured data can be expressed as EEG + ERP + EOG + NOISE and that all four components are uncorrelated. Under the assumption that EEG tends towards 0 $\mu$V with averaging and after subtraction of the raw average ERP, the regression would be calculated on the EOG + NOISE components alone, which is, of course, desirable. However, this also implicitly assumes that the EOG component is temporally stochastic (not stimulus contingent), which clearly is not the case for many paradigms used today (e.g., visual search, emotion induction paradigms). Thus, a varying amount of stimulus-evoked EOG activity is subtracted along with the raw ERP average, and the regression is then based on the residual, non-stimulus-contingent portion of the EOG activity alone.

A more general problem of regression-based artifact correction is that this approach assumes stability of the artifact over time, which is not always given, especially with experimental paradigms that are monotonous and fatigue inducing. Another problem results from the fact that the regression-based correction only works properly if the eye electrodes are placed completely perpendicular to one another. If this is not the case, then the eye artifact data is not completely linearly represented in the artifact time stretches of the EEG channels, which can result in over- or undercorrection. However, the most grave problem inherent to this method is that it assumes a directional relationship between EOG and EEG where it is assumed that the EOG activity alone is causing the commonality found in the "artifacts" in the EEG; in reality, however, the EEG activity present is as likely to influence the EOG channel recordings. Correcting the EEG readings based on the regression weights may therefore remove substantial amounts of desired EEG (effect) activity along with the true eye movement-borne artifacts. For these reasons and from our own experience, independent component analysis is clearly favored over regression-based approaches for the correction of eye blinks and other artifacts. This approach is discussed in detail in the following section.

## 2.4 Independent Components of Electroencephalographic Data

Because of volume conduction through cerebrospinal (CSF) fluid, skull, and scalp, EEG signals collected from the scalp are supervisions of neural and artifactual activities from multiple brain or extra-brain processes occurring within a large volume. Because these processes have overlapping scalp projections, time courses, and spectra, their distinctive features cannot be separated by simple averaging or spectral filtering. Recently, independent component analysis was proposed to reverse the superposition by separating the EEG into statistically independent components, and ICA has provided evident promise and new insights into macroscopic brain dynamics [24]. The following section discusses the applications of ICA to EEG artifact removal and decomposition of event-related brain dynamics in the EEG recordings.

### 2.4.1 Independent Component Analysis

Independent component analysis refers to a family of related algorithms [25–34] that exploit independence to perform blind source separation. Blind source separation is a signal processing approach to separating statistically independent components that underlie sets of measurements or signals, where neither the source statistics nor the mixing process are known. ICA recovers **N** source signals, $\mathbf{s} = \{s_1(t), \ldots, s_N(t)\}$ (e.g., different voice, music, or noise sources), after they are linearly mixed by multiplying by **A**, an unknown matrix, $\mathbf{x} = \{x_1(t), \ldots, x_N(t)\} = \mathbf{As}$, while assuming as little as possible about the natures of **A** or the source signals. Specifically, one tries to recover a version, $\mathbf{u} = \mathbf{Wx}$, of the original sources **s** that is identical except for scaling and permutation by finding a square matrix **W** that specifies spatial filters that linearly invert the mixing process.

Mathematically, ICA, like principal component analysis (PCA), is a method that undoes linear mixing of sources contributing to the recorded data channels by multiplying the data by a matrix as follows:

$$\mathbf{u} = \mathbf{Wx} \tag{2.1}$$

Here, we imagine the data are zero-mean. While PCA only uses second-order statistics (the data covariance matrix) to decorrelate the outputs (using an orthogonal matrix **W**), ICA uses statistics of all orders, thereby pursuing a more ambitious objective. ICA attempts to make the outputs statistically independent, while placing no constraints on the matrix **W** giving the contributions of the component sources to the data. The key assumption used in ICA is that the time courses of activation of the sources are as statistically independent as possible. Statistical independence means the joint probability density function (pdf) of the output factorizes:

$$p(\mathbf{u}) = \prod_{i=1}^{N} p_i(u_i) \tag{2.2}$$

whereas decorrelation means only that $<\mathbf{uu}^T>$, the covariance matrix of **u**, is diagonal (here < > refer to the average). Another way to think of the transform in $<\mathbf{u}>$ is as follows:

$$\mathbf{x} = \mathbf{W}^{-1}\mathbf{u} \tag{2.3}$$

Here, $\mathbf{x}$ is considered the linear superposition or mixture of basis functions (i.e., columns of $\mathbf{W}^{-1}$), each of which is activated by an independent component, $\mathbf{u}_i$. We call the rows of $\mathbf{W}$ *filters* because they extract the independent components. In orthogonal transforms such as PCA, the Fourier transform, and many wavelet transforms, the basis functions and filters are the same (because $\mathbf{W}^T = \mathbf{W}^{-1}$), but in ICA they are different.

The algorithm for learning $\mathbf{W}$ is commonly accomplished by formulating a cost function and running an optimization process. There are many possible cost functions and many more optimization processes. Thus, there are many somewhat different algorithmic approaches to solving the blind source separation problem. Information maximization [31, 35], maximum likelihood [36, 37], FastICA [38], and Joint Approximate Decomposition of Eigen matrices (JADE) [39] are just some of the widely used algorithms whose cost functions and optimization processes are recommended for further reading.

### 2.4.2 Applying ICA to EEG/ERP Signals

More than a decade ago, the authors first explored and reported the application of ICA to multiple-channel EEG and averaged ERP data recorded from the scalp for separating joint problems of source identification and source localization [24]. Figure 2.7(a) presents a schematic illustration of the ICA decomposition. For EEG or ERP data, the rows of the input matrix $\mathbf{x}$ in (2.1) and (2.3) are EEG/ERP signals recorded at different electrodes and the columns are measurements recorded at different time points [Figure 2.7(a), left]. ICA finds an "unmixing" matrix $\mathbf{W}$ that decomposes or linearly unmixes the multichannel scalp data into a sum of temporally independent and spatially fixed components, $\mathbf{u} = \mathbf{W}\mathbf{x}$ [Figure 2.7(a) right]. The rows of this output data matrix, $\mathbf{u}$, called the component activations, are the time courses of relative strengths or levels of activity of the respective independent components through the input data. The columns of the inverse of the unmixing matrix, $\mathbf{W}^{-1}$, give the relative projection strengths of the respective components onto each of the scalp sensors. These may be interpolated to show the scalp map [Figure 2.7(a), far right] associated with each component. These scalp maps provide very strong evidence as to the components' physiological origins (for example, vertical eye movement artifacts project principally to bilateral frontal sites), and may be separately input into any inverse source localization algorithm to estimate the actual cortical distributions of the cortical area or areas generating each source.

Note that each independent component of the recorded data is specified by *both* component activation and a component map—neither alone is sufficient. Note also that ICA does not solve the inverse (source localization) problem. Instead, ICA, when applied to EEG data, reveals *what* distinct, for example, temporally independent activities compose the observed scalp recordings, separating this question from the question of *where* exactly in the brain (or elsewhere) these activities arise. However, ICA facilitates answers to this second question by determining the fixed scalp projection of each component alone.

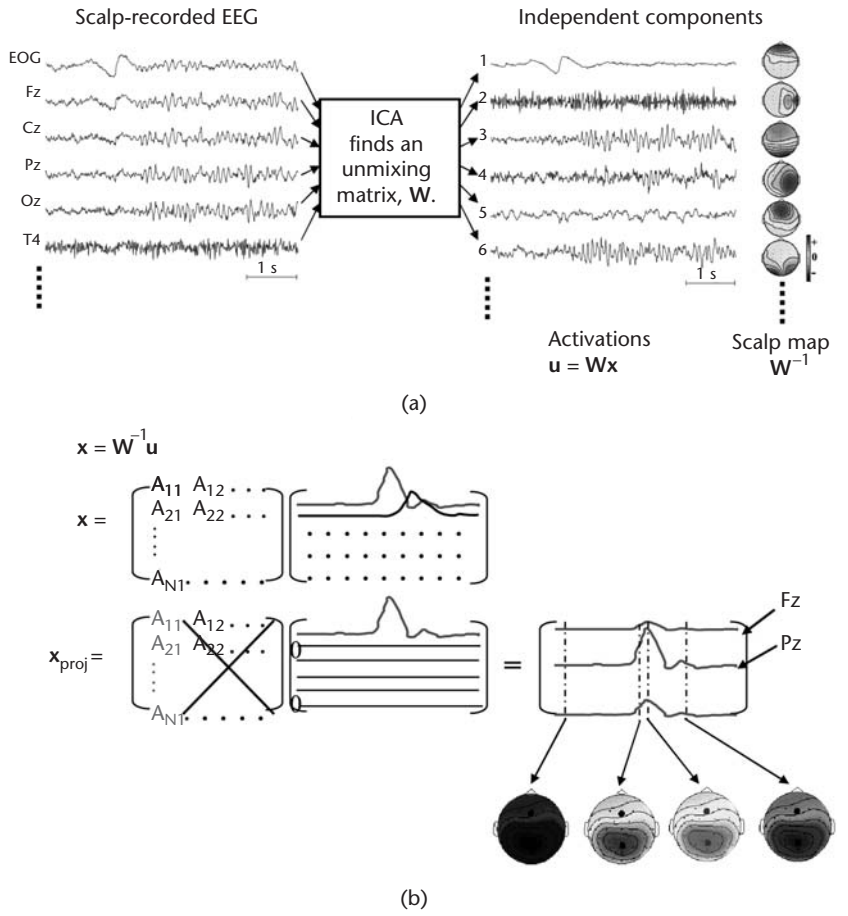



**Figure 2.7** Schematic overview of ICA applied to EEG data. (a) A matrix of EEG data, **x**, recorded at multiple scalp sites (only six are shown), is used to train an ICA algorithm, which finds an "unmixing" weight matrix **W** that minimizes the statistical dependence of the equal number of outputs, **u** = **Wx** (six are shown here). After training, ICA components consist of time series (the rows of **u**) giving the time courses of activation of each component, plus fixed scalp topographies (the columns of $\mathbf{W}^{-1}$) giving the projections of each component onto the scalp sensors. (b) The schematic illustration of the back-projection of a selected component onto the scalp channels.

### 2.4.2.1 Assumptions of ICA Applied to EEG

Standard, so-called complete and instantaneous ICA algorithms are effective for performing source separation in domains where: (1) the summation of different source signals at the sensors is linear, (2) the propagation delays in the mixing medium are negligible, (3) the sources are statistically independent, and (4) the number of independent signal sources is the same as the number of sensors, meaning that if we employ $N$ sensors, the ICA algorithm we can separate $N$ sources [24].

The first two assumptions above, that the underlying sources are mixed linearly in the electrode recordings without appreciable delays, are assured by the biophysics of volume conduction at EEG frequencies [40]. This is the basis for any type of linear decomposition methods including those based on PCA. That is, the EEG mixing process is fortunately linear, although the processes generating it may be highly

nonlinear. In current applications, ICA attempts only to "undo" the linear mixing produced by volume conduction and linear summation of fields at the electrodes.

Assumption (3), of independence or near independence of the underlying source signals, is compatible with physiological models that emphasize the role of anatomically dominant local, short-range intracortical and radial thalamocortical coupling in the generation of local electrical synchronies in the EEG [41]. These facts suggest that synchronous field fluctuations should arise within compact cortical source domains, although they do not in themselves determine the spatial extent of these domains. If we assume, therefore, that the complexity of EEG dynamics can be modeled, in substantial part at least, as summing activities of a number of very weakly linked and, therefore, nearly statistically independent brain processes, EEG data should satisfy assumption (3). However, in practice, it is important to consider which EEG processes may express their independence in the EEG or ERP training data because the assumption of temporal independence used by ICA cannot be satisfied when the training dataset is too small. The number of time points required for training is proportional to the number of variables in the unmixing matrix (the square of the number of channels). Decomposing a single 1-second ERP average (32 channels, 512 time points) from one task condition, for example, is unlikely to obtain comprehensible results. In this case, temporal independence might be achieved or approximated by sufficiently and systematically varying the experimental stimulus and task conditions, creating an ERP average for each stimulus/task condition, and then decomposing the concatenated collection of resulting ERP averages. However, simply varying stimuli and tasks does not always guarantee that all of the spatiotemporally overlapping EEG processes contributing to the averaged responses will be independently activated in the ensemble of input data. These issues imply that results of ICA decomposition of averaged ERPs must be interpreted with caution. A better solution is likely to be obtained by decomposing the concatenated data trials as a single dataset. Because the definition of independence used by many ICA algorithms is based on instantaneous relationships, discontinuities in the data are not an obstacle. Whatever the data ICA decomposition is applied to, converging behavioral or other evidence must be obtained before concluding that spatiotemporally overlapping ICA components measure neurophysiologically or functionally distinct activities.

Assumption (4), that $N$-channel EEG data mixes the activities of $N$ or fewer sources, is certainly questionable, since we do not know in advance the effective number of statistically independent brain signals contributing to the EEG recorded from the scalp. As demonstrated by simulations [42], when training data consist of fewer large source components than channels, plus many more small source components, as might be expected in actual EEG data, large source components are accurately separated into separate output components, with the remaining output components consisting of mixtures of smaller source components. In this sense, performance of the ICA degrades gracefully as the number of smaller sources or the amount of noise in the data increases.

#### 2.4.2.2 Component Projections and Artifact Removal

Brain activities of interest accounted for by single or by multiple components can be obtained by projecting selected ICA component(s) $k$ back onto the scalp,

$$\mathbf{x}_k = \mathbf{W}_k^{-1}\mathbf{u}_k \tag{2.4}$$

where $\mathbf{u}_k$ is the set of the activation matrix rows for components in set $k$ and $\mathbf{W}^{-1}{}_k$ is the scalp map matrix columns $k$. This process is called the back-projection of the component to the data. Back-projected activity is at the original channel locations and in the original recording units (e.g., $\mu$V). Figure 2.7(b) schematically depicts the projection of the first component onto the scalp channels. This is also easily computed by setting the artifactual and/or irrelevant component activations to zero, as interpolated and plotted in Figure 2.7(b) (lower scalp maps). In this case, columns of the inverse unmixing matrix $\mathbf{W}^{-1}$ associated with these components become nonfactors in the back-projection, whereas the column of the inverse unmixing matrix associated with the first component determines the amplitude distribution of the component across scalp channels. For each component, the distribution of current across the scalp electrodes is fixed over time, but the actual potential values (including their polarities) are modulated by the corresponding time course of component activation, the relevant row of the output data matrix, in this case, $u_1(t)$, depicted in the lower panels as the intensity fluctuations of the scalp maps over time.

### 2.4.3 Artifact Removal Based on ICA

As mentioned earlier, one of the most pervasive problems in EEG analysis and interpretation is the interference in the data produced by often large and distracting artifacts arising from eye movements, eye blinks, muscle noise, heart signals, and line noise. Figure 2.8(a) shows a sample 5-second portion of continuous EEG time series data collected from 20 scalp electrodes placed according to the International 10-20 system and from two EOG electrode placements, all referred to the left mastoid. The sampling rate was 256 Hz. In this example, ICA was trained with 10 seconds of spontaneous EEG data. Figure 2.8(b) shows component activations and scalp

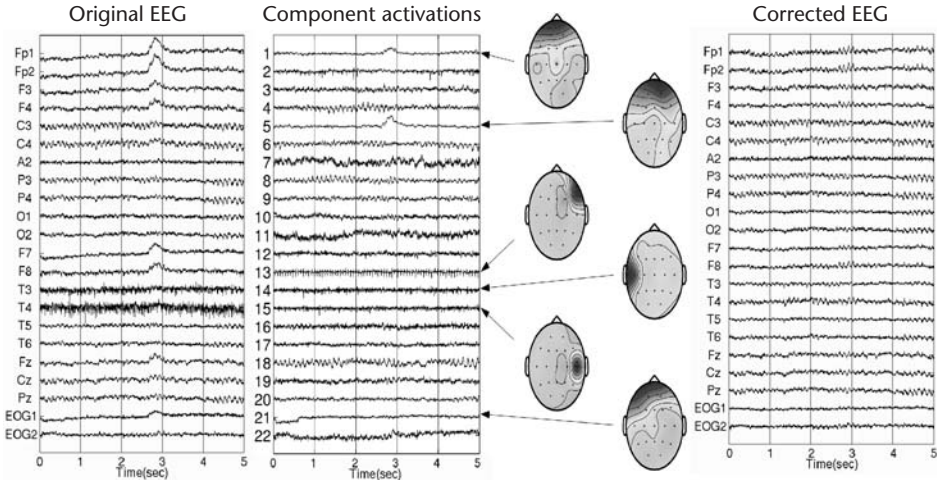



**Figure 2.8** Demonstration of EEG artifact removal by ICA. (a) A 5-second portion of an EEG time series containing a prominent eye movement. (b) Corresponding ICA component activations and scalp maps of six components accounting for horizontal and vertical eye movements and temporal muscle activity. (c) EEG signals corrected for artifacts by removing the six selected ICA components in (b).

topographies of the 22 independent components. The eye movement artifact (between seconds 2 and 3) was isolated by ICA to components IC1, IC5, and IC21. The scalp maps indicate that these components account for the spread of EOG activity to frontal sites. Components IC13, IC14, and IC15 evidently represent muscle noise from temporal muscles. After eliminating these artifactual components, by zeroing out the corresponding rows of the activation matrix **u** and projecting the remaining components to the scalp electrodes, the "corrected" EEG data [Figure 2.8(c)] are free of both EOG and muscle artifacts. Removing EOG activity from frontal channels reveals alpha activity near 8 Hz that occurred during the eye movement but was obscured by the eye movement artifact in the original EEG traces. Close inspection of the EEG records [Figure 2.8(a)] confirms its presence in the raw data. The artifact-corrected data also reveal underlying EEG activity at temporal sites T3 and T4 [Figure 2.8(c)] that was well masked by muscle activity in the raw data [refer to Figure 2.8(a)].

The second example (Figure 2.9) demonstrates that ICA can also be used to remove stimulus-induced eye artifacts from unaveraged event-related EEG data through analysis of a sample data set collected during a selective attention task. The

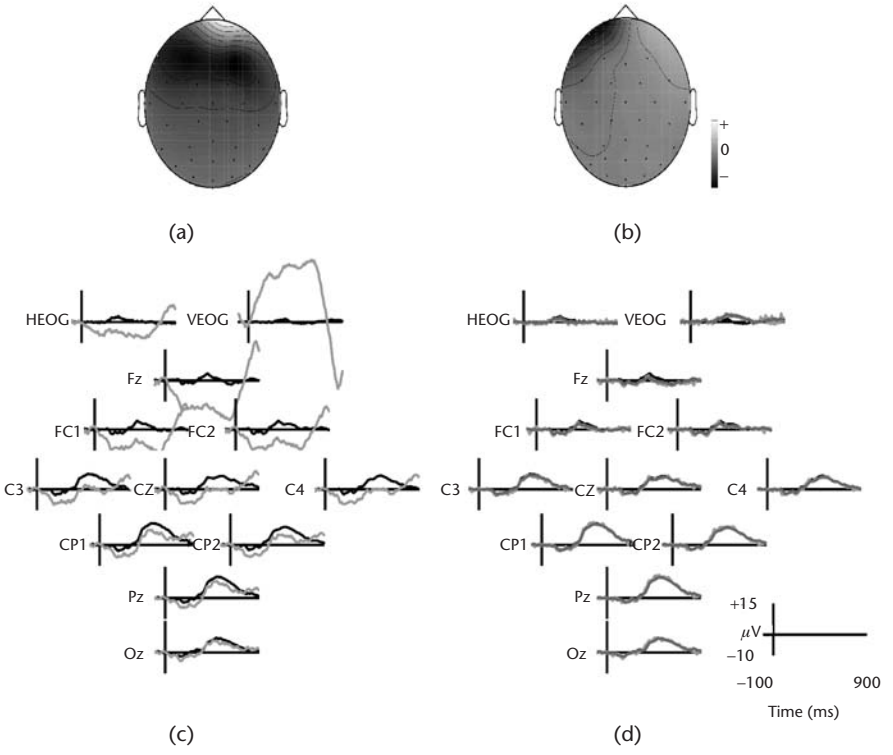



**Figure 2.9** Elimination of eye movement artifacts from ERP data. (a) The scalp topography of an ICA component accounting for blink artifact. This component was separated by ICA from 555 target response trials recorded from a normal subject in a visual selective attention experiment. (*Note:* Because this scalp map interpolation was based on very few frontal electrodes, it is not a representative depiction of an eye blink component map.) (b) The scalp map of a second component accounting for lateral eye movements. (c) Averages of ($N$ = 477) relatively uncontaminated and ($N$ = 78) contaminated single-trial target response epochs from a normal control subject. (d) Averages of ICA-corrected ERPs for the same two trial subgroups overplotted on the average of uncorrected uncontaminated trials.

subject performed a visual-spatial selective attention task during which he covertly attended one of five squares continuously displayed on a black background 0.8 cm above a centrally located fixation point. Four squares were outlined in blue, and one, marking the attended location, was outlined in green. The location of this green square was counterbalanced across 72-second trial blocks. The subject was asked to press a right-hand–held thumb button as soon as possible following target stimulus presentations in the attended location (the green square), and to ignore the similar (nontarget) stimuli presented in the other four boxes. Stimuli were white disks, presented in one of the five boxes at random. EEG data was collected from 29 scalp electrodes mounted in a standard electrode cap (Electrocap, Inc.) at locations based on a modified International 10-20 system, and from two periocular electrodes placed below the right eye and at the left outer canthus. Data was sampled at 512 Hz (downsampled to 256 Hz) with an analog pass band of 0.01 to 50 Hz. Although the subject was instructed to fixate the central cross during each block, he tended to blink or move his eyes slightly toward target stimuli presented at peripheral locations.

After ICA training on 555 concatenated 1-second data trial epochs, independent components that accounted for blinks and eye movements were identified by the procedures detailed in [43] based on the characteristics of time course of component activations, the component scalp topographies, and the locations and orientations of equivalent dipoles obtained using functions available in the freely available EEGLAB environment [44]. Here, ICA successfully isolated blink artifacts to a single independent component [Figure 2.9(a)] whose contributions were removed from the EEG record by subtracting its component projection from the data.

Though the subject was instructed to fixate the central cross during each block, the technician watching the video monitor noticed that the subject's eyes also tended to move slightly toward target stimuli presented at peripheral locations. A second independent component accounted for EEG artifacts produced by these small horizontal eye movements [Figure 2.9(b)]. Its scalp pattern is consistent with that expected for lateral eye movements. Note the overlap in scalp topography between the two independent components accounting for blinks [Figure 2.9(a)] and for lateral eye movements [Figure 2.9(b)]. Again, unlike PCA component maps, ICA component maps need not be orthogonal and may even be nearly spatially coincident.

A standard approach to ERP artifact rejection is to discard eye-contaminated trials containing maximal potentials exceeding some selected value (e.g., $=60\,\mu$V) at periocular sites. For this dataset, this procedure rejected 78 of 555 trials, or 14% of the subject's data. Figure 2.9(c) shows ERP averages of relatively uncontaminated target trials (solid traces) and of the contaminated target trials (faint traces) that would have been rejected by this method. These averages differ most at frontal electrodes. Figure 2.9(d) shows averages of the same uncontaminated (solid traces) and contaminated (solid traces) trials after the independent components accounting for the artifacts were identified and removed, and the summed activities of the remaining components projected back to the scalp electrodes. The two ICA-corrected averages were almost completely coincident, showing that ICA-based artifact removal did not change the neural signals that were not contaminated. Note that the ICA-corrected averages of these two trial groups are remarkably similar to the

average of the uncontaminated trials before artifact removal [Figure 2.9(d)]. This implies that the corrected recordings contained only event-related neural activity and were free of artifacts arising from blinks or eye movements.

#### 2.4.3.1 Cautions Concerning ICA-Based Artifact Removal

ICA-based artifact removal also has some shortcomings. First, it is important to distinguish among artifacts produced by processes associated with stereotyped scalp maps, for example, eye movements, single muscle activity, and single-channel noise. These may be well accounted for by a single independent component if sufficient data is used in the decomposition. At the other end of the scale, nonstereotyped artifacts that produce a long series of noise with varying spatial distributions into the data—for example, artifacts produced by the subject vigorously scratching her scalp—defy the standard ICA model. Here, at each time point, artifacts may be associated with a unique, novel scalp map, posing a severe problem for ICA decomposition. It is by far preferable to eliminate episodes containing nonstereotyped artifacts from the data before decomposition because such artifacts can negatively affect the ICA decomposition even at small amplitudes.

In addition, caution needs to be taken that ICA cannot keep track of sources when processing several time windows of the EEG because the order of resultant independent components is, in general, arbitrary. Therefore, artifact removal requires visual inspection of the components and determination of which components to remove. However, the distributions of spectral power and/or scalp topographies of artifactual components are quite distinct, which suggests that it is feasible to automate procedures for removing these artifacts from contaminated EEG recordings.

### 2.4.4 Decomposition of Event-Related EEG Dynamics Based on ICA

It is noteworthy that ICA is not only effective for removing artifacts from EEG data, but also for direct analysis of distinct EEG components, which arguably represent, in many cases, functionally independent cortical source activities [45]. During the last decade, our laboratory and many others have applied ICA to decompose sets of averaged ERPs, continuous EEG records, and/or sets of event-related EEG data trials and have demonstrated that much valuable information about human brain dynamics contained in event-related EEG data may be revealed using this method. In our experience, ICA decomposition is most usefully applied to a large set of concatenated single-trial data epochs. Simultaneous analysis of a set of hundreds of single-trial EEG epochs gives the concurrently active EEG source processes that contribute to the response and/or the response baseline a far better chance of expressing their temporal independence and thus being separately identified by ICA. ICA algorithms thus can separate the most salient concurrent EEG processes active within the trial time windows. Many studies (including but not limited to [3, 43, 45–47]) have shown that relatively small numbers of independent components exhibited robust event-related activities near stimulus presentation and/or the subject behavioral response. These components tend to have near-dipolar scalp maps, compatible with a compact cortical source area and suggesting that the brain areas

exhibiting responses to these experimental events are indeed spatially stable across epochs and latencies, as is assumed for ICA. However, it cannot be guaranteed that source locations and scalp maps do not change over time (for example, if the subject falls deeply asleep or experiences a seizure). The nature of stability or instability of the spatial EEG source distribution is an open question and the subject of ongoing research. However, new insights about brain function are beginning to emerge from this research that would have been difficult or impossible to obtain without first separating and identifying distinct brain processes combined in noninvasively recorded EEG data. In this sense, ICA has proven to be an effective preprocessing method for EEG analysis and interpretation. For more details about applying ICA to ERP and EEG data, see [43–47].

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CHAPTER 3

# Single-Channel EEG Analysis

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In this chapter, we review the most commonly used quantitative EEG analysis methods for single-channel EEG signals, including linear methods, nonlinear descriptors, and statistics measures: (1) In the linear methods section, we cover conventional spectral analysis methods for stationary signals and the time-frequency distribution property when the EEG is regarded as a nonstationary process; (2) because EEGs have been regarded as nonlinear signals in past years, we also introduce the methods of higher-order statistic (HOS) analysis and nonlinear dynamics in quantitative EEG (qEEG) analysis; and, finally, (3) information theory is introduced to qEEG measurements from the aspect of the randomness in EEG signals.

## 3.1 Linear Analysis of EEGs

An electroencephalograph is a record of the electrical activity generated by a large number of neurons in the brain. It is recorded using surface electrodes attached to the scalp or subdurally or in the cerebral cortex. The amplitude of a human surface EEG signal is in the range of 10 to 100 $\mu$V. The frequency range of the EEG has a fuzzy lower and upper limit, but the most important frequencies from the physiological viewpoint lie in the range of 0.1 to 30 Hz. The standard EEG clinical bands are the delta (0.1 to 3.5 Hz), theta (4 to 7.5 Hz), alpha (8 to 13 Hz), and beta (14 to 30 Hz) bands [1, 2]. EEG signals with frequencies greater than 30 Hz are called gamma waves and have been found in the cerebellar structures of animals [3, 4]. An EEG signal may be considered a random signal generated by a stochastic process and can be represented after digitization as a sequence of time samples [5–9].

EEG signal analysis is helpful in various clinical applications including predicting epileptic seizures, classifying sleep stages, measuring depth of anesthesia, detection and monitoring of brain injury, and detecting abnormal brain states [10–23]. The alpha wave, for example, is observed to be reduced in children and in the elderly, and in patients with dementia, schizophrenia, stroke, and epilepsy [24–26].

1. This author contributed to Section 3.1.
2. These authors contributed to Section 3.2. The work presented in Section 3.2 was supported in part by the American Epilepsy Research Foundation and the Ali Paris Fund for LKS Research and Education, and NSF Grant ECS-0601740.
3. This author contributed to Section 3.3. The work presented in Section 3.3 was supported in part by Shuguang Program of the Education Commission of Shanghai Municipality.

Visual analysis of EEG signals in the time domain is an empirical science and requires a considerable amount of clinical and neurological knowledge. Many brain abnormalities are diagnosed by a doctor or an electroencephalographer after visual inspection of brain rhythms in the EEG signals. However, long-term monitoring and visual interpretation is very subjective and does not lend itself to statistical analysis [27, 28]. Therefore, alternative methods have been used to *quantify* information carried by an EEG signal. Among these are the Fourier transform, the wavelet transform, chaos, entropy, and subband wavelet entropy methods [29–37].

The main goal of this chapter is to provide the reader with a broad perspective of classical and modern spectral estimation techniques and their implementations. The reader is assumed to have some fundamental knowledge of signals and systems that covers continuous and discrete linear systems and transform theory. Practicing engineers and neuroscientists working in neurological engineering will also find this chapter useful in their research work in EEG signal processing. Because all of the EEG spectral analysis techniques are performed using computers, the focus is directed more toward discrete time EEG signals. Furthermore, because most practicing scientists and researchers working with EEG signals use MATLAB, various relevant MATLAB functions are also included.

The remainder of this section is organized into three major sections. In Section 3.1.1, classical spectral analysis is covered, including Fourier analysis, windowing, correlation and estimation of the power spectrum, the periodogram, and Welch's method. This is followed by an illustrative application. In Section 3.1.2, modern spectral techniques using parametric modeling of EEG signals are covered. These models include autoregressive moving average and autoregressive spectrum estimation. In Section 3.1.3, time-frequency analysis techniques are detailed for analyzing nonstationary EEG signals. The techniques included in this section are the short-time Fourier transform and the wavelet transform.

### 3.1.1 Classical Spectral Analysis of EEGs

#### 3.1.1.1 Fourier Analysis

The EEG signal can be represented in several ways including the time and frequency domains. Fourier analysis is the process of decomposing a signal into its frequency components. Fourier analysis is a very powerful method that can be used to reveal information that cannot be easily seen in the time domain. The Fourier transform uses sinusoidal functions or complex exponential signals as basis functions. The Fourier transform of a continuous real-time aperiodic signal $x(t)$ is defined as follows [38–40]:

$$F\{x(t)\} = X(\omega) = \int_{-\infty}^{\infty} x(t)\exp(-j\omega t)dt \tag{3.1}$$

where $\omega = 2\pi f$ is the angular frequency in radians/s, and $F\{\circ\}$ is the Fourier operator. The Fourier transform is complex for real signals.

The inverse Fourier transform is the operator that transforms a signal from the frequency domain into the time domain. It represents the synthesis of signal $x(t)$ as a weighted combination of the complex exponential basis functions. It is defined as

$$F^{-1}\{X(\omega)\} = x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\exp(j\omega t)d\omega \tag{3.2}$$

Because most EEG signal processing is carried out using computers, the signal $x(t)$ is sampled with a sampling frequency of $f_S = \frac{\omega_S}{2\pi} = \frac{1}{T_S}$, where $T_s$ is the sampling time interval. The sampling process generates the sequence $x(n)$ where $n$ denotes the discrete sample time. The discrete time Fourier transform (DTFT) of a discrete-time signal $x(n)$ is defined as

$$DTFT\{x(n)\} = X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)\exp(-j\omega n) \tag{3.3}$$

where $DTFT\{x(n)\}$ is a continuous and periodic function of $\omega$ with period $2\pi$.

If $\omega$ is sampled on the unit circle, then we have the discrete Fourier transform of an $N$-length sequence $x(n)$:

$$DFT\{x(n)\} = X(k) = \sum_{n=0}^{N-1} x(n)\exp\left(-j\frac{2\pi}{N}kn\right) \tag{3.4}$$

where $\omega_k = \frac{2\pi}{N}k, n = 0, 1, ..., N-1$, and $k = 0, 1, ..., N-1$. $N$ is the number of spectral samples in one period of the spectrum $X(e^{j\omega})$. Increasing the sequence length $N$ will improve the frequency resolution of the spectrum by decreasing the discrete frequency spacing of the spectrum. The inverse DFT, which transforms a signal from the discrete frequency domain into the discrete time domain, is

$$\text{IDFT}\{X(k)\} = x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\exp\left(j\frac{2\pi}{N}kn\right) \tag{3.5}$$

where $n = 0, 1, \ldots, N-1$ and $k = 0, 1, \ldots, N-1$.

The fast Fourier transform (FFT) algorithm is used to compute the discrete Fourier transform [38, 40]. The FFT algorithm utilizes some properties of the discrete Fourier transform to perform fast calculations of the transform. The FFT reduces the number of computations from $N^2$ to $N\log(N)$. MATLAB provides several FFT functions for computing spectra.

```
Y = fft(x)
```

for example, returns the complex discrete Fourier transform **Y** of a discrete time vector **x**, computed with the FFT algorithm [41]. The magnitude and phase of the spectrum are computed using

```
MY = abs(Y)
```

and

```
PY= angle(Y)
```

#### 3.1.1.2 Windowing

EEG signals are often divided into finite time segments. Segmentation or truncation in the time domain is equivalent to multiplication of the complete EEG signal with a finite time rectangular window. Because multiplication in time is equivalent to convolution in frequency, the Fourier transform of the signal after windowing is more complex and will leak or extend over a wider frequency range than the original signal [41]. The abrupt transition of the signal values in the case of a rectangular window results in the appearance of ripples in the discrete Fourier transform. These ripples can be reduced using alternative window functions. Many window functions are available in the literature [38–41]. The following examples represent four of the most popular windowing functions:

1. Rectangular:

$$W_R[n] = \begin{cases} 1, & |n| < N \\ 0, & \text{otherwise} \end{cases} \tag{3.6}$$

2. Bartlett:

$$W_B[n] = \begin{cases} \dfrac{N-|n|}{N}, & |n| < N \\ 0, & \text{otherwise} \end{cases} \tag{3.7}$$

3. Hamming:

$$W_H[n] = \begin{cases} 0.54 - 0.46\cos\left(\dfrac{2\pi|n|}{N-1}\right), & |n| < N \\ 0, & \text{otherwise} \end{cases} \tag{3.8}$$

4. Hanning:

$$W_H[n] = \begin{cases} \dfrac{1}{2}\left(1 - \cos\left(\dfrac{2\pi|n|}{N-1}\right)\right), & |n| < N \\ 0, & \text{otherwise} \end{cases} \tag{3.9}$$

The selection of the most appropriate window is not a straightforward matter and depends on the application at hand and may require some trial and error. The window functions while reducing the ripples and tends to reduce sharp variations or resolution of the discrete Fourier transform. For example, if we are interested in resolving two narrowband, closely spaced spectral components, then a rectangular window is appropriate because it has the narrowest mainlobe in the frequency domain. If, on the other hand, we have two signals that are not closely spaced in the frequency domain, then a window with rapidly decaying sidelobes is preferred.

MATLAB provides several windowing functions including these:

```
w = rectwin(L)
```

returns a rectangular window of length **L** in the column vector **w**, $L \in Z^{+}$.

```
w = bartlett(L)
```

returns an **L**-point Bartlett window in the column vector **w**.

```
w = hamming(L)
```

returns an **L**-point symmetric Hamming window in the column vector **w**.

Figures 3.1, 3.2, and 3.3 show the time function of a 65-point window and its Fourier transform magnitude in decibels for rectangular, Bartlett, and Hamming windows, respectively.

### 3.1.1.3 The Autocorrelation Function and Estimation of the Power Spectrum

The spectral characteristics of a deterministic signal can easily be determined using (3.1) to (3.5). However, the EEG is a highly complex signal and can therefore be

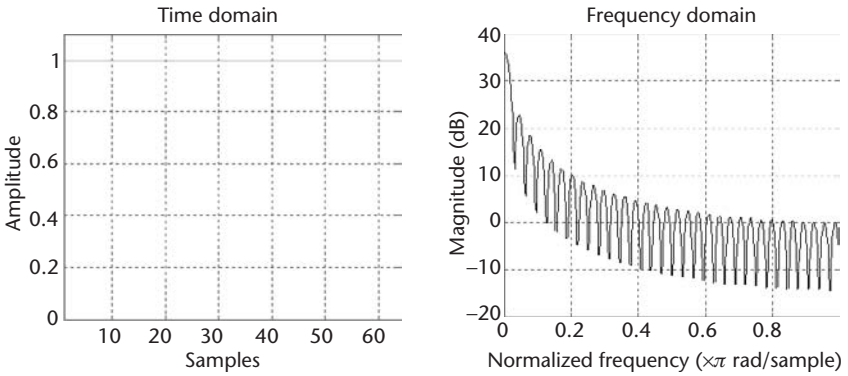



**Figure 3.1** Rectangular window of **L** = 65 and its Fourier transform magnitude in decibels.

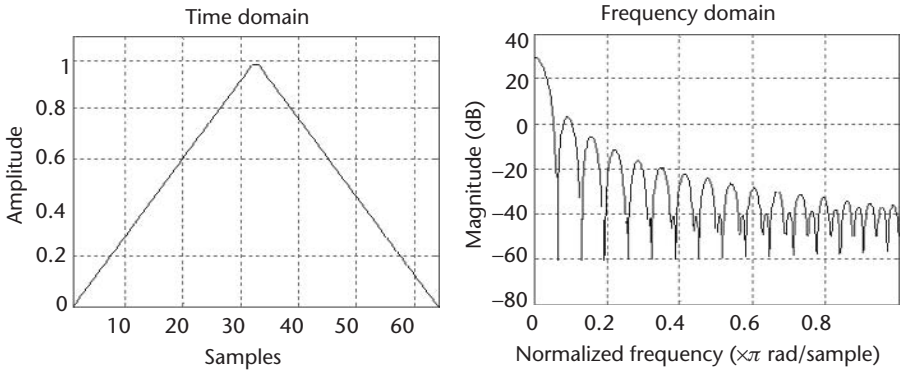



**Figure 3.2** Bartlett window of **L** = 65 and its Fourier transform magnitude in decibels.

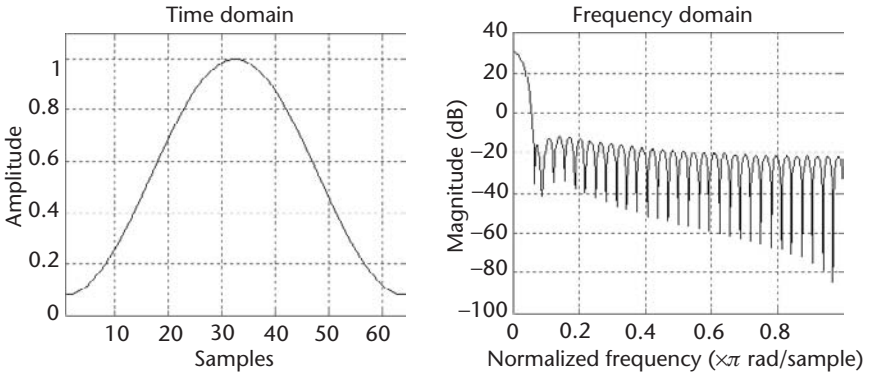



**Figure 3.3** Hamming window of **L** = 65 and its Fourier transform magnitude in decibels.

assumed to be a random signal generated by a stochastic process [5–9]. The direct application of the Fourier transform is not attractive for random processes such as the EEG because the transform may not even exist. If we use power instead of voltage as a function of frequency, then such a spectral function will exist. The power spectrum or the power spectral density (PSD) of a random signal $x(n)$ is defined as the Fourier transform of the autocorrelation function $r_{xx}(m)$. It is defined as follows:

$$\mathrm{PSD}\{x(t)\} = S(\omega) = \sum_{m=-(N-1)}^{N-1} r_{xx}(m)\exp(-j\omega m) \tag{3.10}$$

where

$$r_{xx}(m) = \frac{1}{N}\sum_{n=0}^{N-|m|-1} x(n)x(n+m) \tag{3.11}$$

However, it can be shown that the PSD obtained in (3.10) is equivalent to that obtained using the DFT in (3.3) [39–41]:

$$\mathrm{PSD}\{x(t)\} = S(\omega) = \frac{1}{N}\left|X\left(e^{j\omega}\right)\right|^2 \tag{3.12}$$

The PSD estimation using the DFT is known as the *periodogram*, which can easily be calculated using the FFT method. If we increase $N$, the mean value of the periodogram will converge to the true PSD, but unfortunately, the variance does not decrease to zero. Therefore, the periodogram is a biased estimator. To reduce the variance of the periodogram, ensemble averaging is used. The resultant power spectrum is called the average periodogram. One of the most popular methods for computing the average periodogram is the Welch method, in which windowed overlapping segments are used [39, 40]. The procedure for computing the PSD of a given sequence of $N$ data points is as follows:

1. Divide the data sequence into $K$ segments of $M$ samples each.
2. Compute the periodogram of each windowed segment using the FFT algorithm:

$$S_i(\omega) = \frac{1}{ME}\left|\sum_{n=0}^{M-1} x_i(n)w(n)\exp(-j\omega n)\right|^2 \quad 1 \le i \le K \tag{3.13}$$

where $E = \frac{1}{M}\sum_{n=0}^{M-1} w^2(n)$ is the average power of the window used. Some window functions are listed in the previous section.

3. The average periodogram is then estimated from the ensemble average of $K$ periodograms:

$$S(\omega) = \frac{1}{K}\sum_{i=1}^{K} S_i(\omega) \tag{3.14}$$

MATLAB provides several PSD functions including the Welch technique.

```
[Pxx,w] = pwelch(x)
```

estimates the power spectral density

```
Pxx
```

of the input signal vector **x** using Welch's averaged modified periodogram method of spectral estimation. The vector **x** is segmented into eight sections of equal length, each with 50% overlap. Each segment is windowed with a Hamming window that is the same length as the segment. The PSD is calculated in units of power per radians per sample.

#### 3.1.1.4 Application: Spinal Cord Injury Detection Using Spectral Coherence

Frequency analyses and power spectral estimations of EEG signals have been successfully used in predicting epileptic seizures, classifying sleep stages, detection and monitoring of brain injury, determining the depth of anesthesia, and detecting abnormal brain states [10–23]. In this section, we illustrate an additional application in which power spectral estimation is successfully used in quantitative EEG analysis for the assessment of spinal cord injury.

Millions of patients worldwide are living with the devastating effects of spinal cord injury (SCI). What is required is an objective quantitative assessment method that enables researchers in the area of SCI recovery and rehabilitation to accurately and objectively evaluate possible therapeutic mechanisms to reverse and prevent the devastating effects of SCI. The Basso-Beathe-Bresnahan (BBB) method is a conventional method used to assess spinal cord injuries in the animal model. A 4-minute observation of a rat in an open field is conducted by neurologists and translated into a number on a scale from 0 to 21. The BBB is well accepted and easy to execute. Nevertheless, it is subjective, assesses only motor function, and does not account for the nonwillingness of the rodent to move. One powerful technique used in SCI studies is the evoked potential (EP), which reflects the electrophysiological response of the neural system to an external stimulus. Somatosensory evoked potentials (SEPs) are obtained by electrical stimulation of the median nerve at the wrist or the posterior tibial nerve at the ankle [44]. This technique is used by researchers to evaluate

the ongoing neurophysiological changes throughout the recovery period after SCI. Previous studies using SEP data for SCI detection have used changes in latency and peak amplitude of SEP signals. The inherent disadvantage of time analysis is that spectral changes cannot be detected. Moreover, some SEP signals, as we will demonstrate later in this chapter, do not have a detectable latency or peak amplitude following severe SCI.

A spectral coherence measure method based on SEP signals was successfully used to provide a quantitative measure of SCI [42, 43]. Spectral coherence is the normalized cross-power spectrum computed between two signals. The coherence function gives a measure of similarity between signals and is related to the cross-correlation function. The magnitude-squared spectral coherence $\gamma_{xy}^2(\omega)$ function of two signals $x$ and $y$ is a normalized version of the cross PSD between $x$ and $y$ and is defined as [39, 40]:

$$\gamma_{xy}^2(\omega) = \frac{\left|S_{xy}(\omega)\right|^2}{S_{xx}(\omega)S_{yy}(\omega)} \tag{3.15}$$

where $S_{xy}(\omega)$ is the cross-power spectrum between the $x$ and $y$ signals, $S_{xx}(\omega)$ is the power spectrum of the $x$ signal, and $S_{yy}(\omega)$ is the power spectrum of the $y$ signal.

Spectral coherence was used to study the SEP signals from 15 female adult Fischer rodents before and after SCI [44]. Injury was induced by dropping a 10.0-g rod with a flat circular impact surface onto the exposed spine from heights of 6.25, 12.5, 25, or 50 mm for mild, moderate, severe, and very severe injury. To generate stimulation for SEP, subcutaneous needle electrodes were used for left and right median and tibial nerves (1-Hz frequency, 3.5-mA amplitude, 200-$\mu$s duration, and 50% duty cycle) without direct contact with the nerve bundle. Contralateral SEP recordings were used for the left and right forelimbs, as well as the hind limbs. The recorded SEP signal was then sampled at 5 kHz.

As expected, high coherence was observed to occur at low frequencies. Closer observation of the average of the spectral coherence for the right hind limb baseline with the right forelimb baseline for all rats (Figure 3.4) helped us choose a band of 125 to 175 Hz.

The spectral coherence variations over time before and after injury helped identify the effects of injury on limbs. Because the injury affected primarily the hind limbs, the coherence associated with the forelimbs was relatively high (>0.7) throughout the period of observation. In practice, SEP information is not available before injury. Hence, forelimb signals were used as control signals.

Figure 3.5 shows the spectral coherence (SC) for rats that were subjected to a medium SCI level. It is interesting to note that Rat 10's right hind limb seems to have recovered with an SC reading of 0.4 on day 4 after injury to an SC of 0.82 on day 82. Rats 11 and 13 do not show good right hind limb recovery.

Spectral coherence reveals information specific to each rat that is missing in conventional methods of assessing spinal cord injury. The results for improvement in global spectral coherence over the recovery period may differ among the rats from the same injury group. This could be due to several reasons such as the differences in every individual's recovery or the exact location of injury.

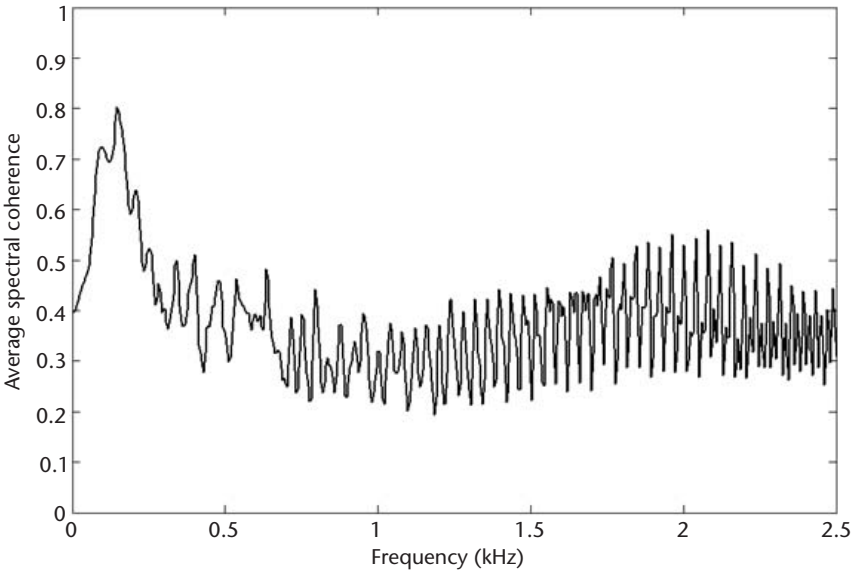



**Figure 3.4** The average of the spectral coherence for the right hind limb baseline with the right forelimb baseline for all rats.

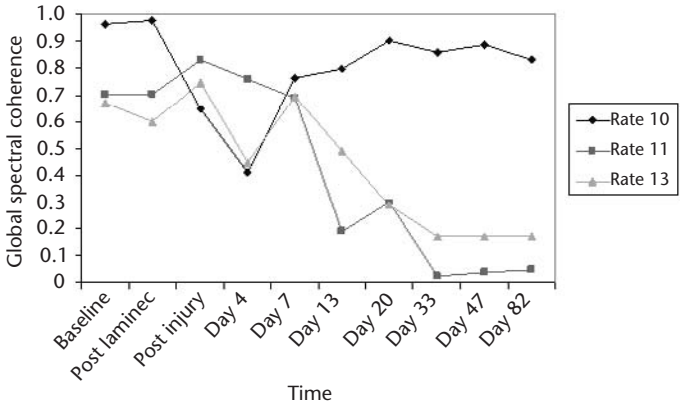



**Figure 3.5** Global coherence of the right hind limb of rats from the injury level of 12.5 mm plotted versus time (control signal baseline right forelimb).

Spectral coherence gave normalized quantifiable results that did not need the baseline and did not require a trained eye. SC also gave information about the existence of an injury in rats that were injured and, therefore, detected no injury for the control group.

### 3.1.2 Parametric Model of the EEG Time Series

Spectral estimation techniques described in the previous section that use the Fourier spectrum are called "classical" spectral estimation methods. The attractive feature of classical methods is that they require very little or no information about the nature of the signal under consideration. In this section, we describe what is known

as the "parametric" spectral estimation techniques. Although these methods are based on time-domain analyses, they are used to characterize and estimate the spectrum of the signal. These methods are very useful when dealing with short segment data sequences [39, 41]. The most popular of the parametric methods is the autoregressive linear model. The input to the model is white noise, which contains all frequencies, whereas the output is compared with the signal being modeled. The model parameters are then adjusted to match the model output to the signal being modeled. The resultant model parameters are then used to estimate the spectrum of the signal under consideration. These models include [39] the autoregressive (AR), moving average (MA), and autoregressive moving average (ARMA) models. The AR method is usually used when the signal being modeled has spectral peaks, whereas the MA method is useful for modeling signals with spectral valleys and without spectral peaks.

The AR model of a single-channel EEG signal is defined as follows [41]:

$$y(n) = -\sum_{k=1}^{p} a_k y(n-k) + x(n) \tag{3.16}$$

where $a_k$, $k = 1, 2, \ldots, p$, are the linear model parameters, $p$ is the model order, $n$ denotes the discrete sample time, and $x(n)$ is white noise input with zero mean and unity variance. The output current value depends on the input signal and previous output samples.

In the ARMA model, the signal is defined as

$$y(n) = -\sum_{k=1}^{p} a_k y(n-k) + \sum_{k=0}^{q} b_k x(n-k) \tag{3.17}$$

where $b_k$, $k = 1, 2, \ldots, q$, are the additional linear model parameters. The parameters $p$ and $q$ are the model orders. Although $p$ and $q$ are usually determined through trial and error, Akaike criterion (AIC) can be used [45, 46] to determine the optimum model order. For an $N$-length data sequence, the optimum AR model order $p$ is obtained by minimizing the AIC criterion:

$$\mathrm{AIC}(p) = N \ln(\rho_p) + 2p \tag{3.18}$$

where $\rho$ is the model error variance defined as

$$\rho_p = \sum_{n=0}^{N-1} |e_p(n)|^2 \tag{3.19}$$

which can be determined using

$$e_p(n) = \sum_{k=0}^{p} a_p(k) x(n-k) \tag{3.20}$$

In any case, the model order should be such that the model estimated spectrum fits the signal spectrum.

The ARMA linear system model is depicted in Figure 3.6 in its discrete form. The frequency-domain transfer function $H$ can be obtained using the $Z$-transform as follows:

$$Y(z) = -\sum_{k=1}^{p} a_k z^{-k} Y(z) + \sum_{k=0}^{q} b_k z^{-k} X(z) \tag{3.21}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{q} b_k z^{-k}}{1 + \sum_{k=1}^{p} a_k z^{-k}} \tag{3.22}$$

The absolute squared value of $H(z)$ evaluated at $z = e^{j\omega}$ is:

$$\left|H\left(e^{j\omega}\right)\right|^2 = \frac{\left|\sum_{k=0}^{q} b_k z^{-k}\right|^2}{\left|1 + \sum_{k=1}^{p} a_k z^{-k}\right|^2} \tag{3.23}$$

Several algorithms are used to estimate the model's coefficients. The most popular are the Yule-Walker, the Burg, and the covariance and modified covariance methods. All of these methods are available in MATLAB.

The following MATLAB functions are used to estimate the AR model parameters (**a**) using the above methods where **x** is the sequence that contains the time-series data and **p** is the order of the AR model [41]:

```
a = arburg(x,p)- Burg's method
a = aryule(x,p)- Yule-Walker
a = arcov(x,p)- covariance
a = armcov(x,p)- modified covariance
```

The following MATLAB functions are used to estimate the PSD of the modeled signal using the above methods where **x** is the sequence that contains the time-series data and **p** is the order of the AR model [41]:

```
Pxx = pburg(x,p) – Burg's method
Pxx = pyulear(x,p) - Yule-Walker
Pxx = pcov(x,p) - covariance
Pxx = pmcov(x,p) - modified covariance
```

The following example illustrates the difference between the classical versus modern spectral estimation techniques. Figure 3.7(a) shows a 2,048-point EEG

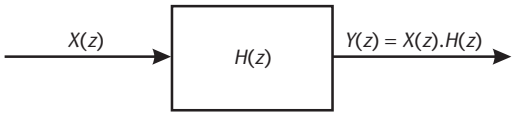



**Figure 3.6** The ARMA linear system model in the discrete form.

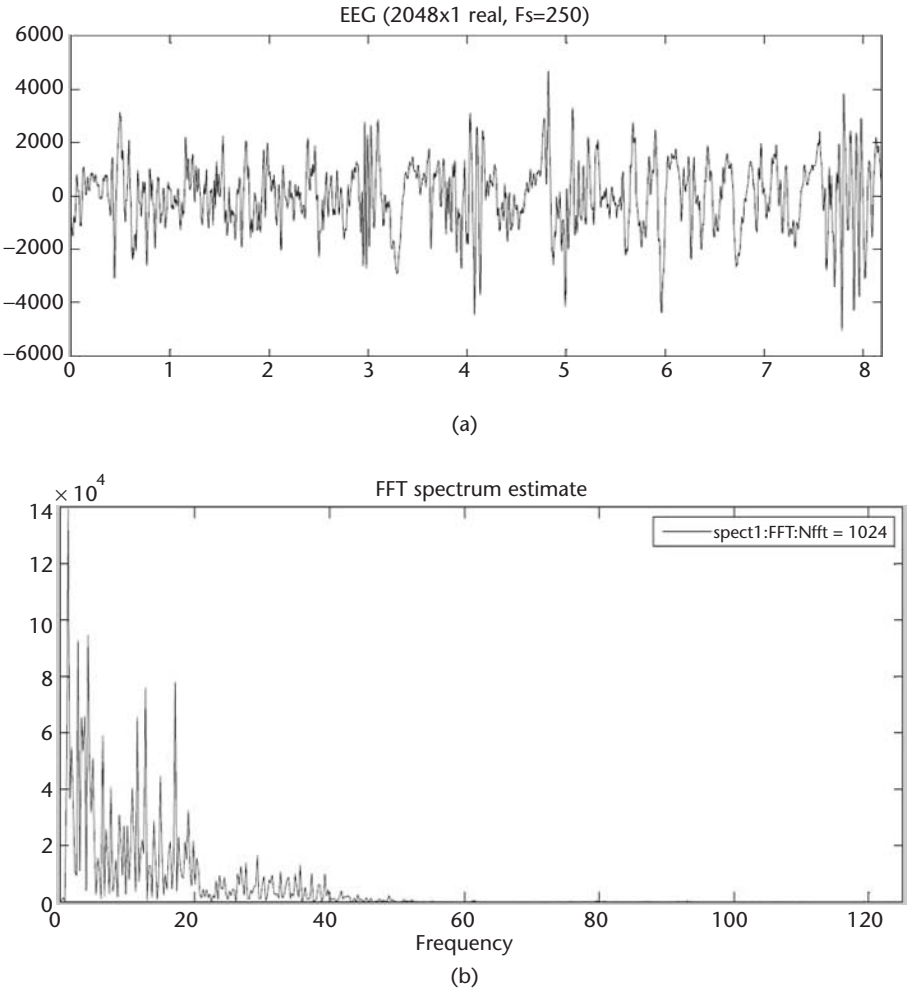



**Figure 3.7** (a) EEG sequence recorded from the left frontoparietal regions of a rat's brain sampled at 250 samples per second. (b) The FFT of the EEG signal.

sequence recorded from the left frontoparietal regions of a rat's brain sampled at 250 samples per second. Figure 3.7(b) shows the FFT of the EEG signal, which, as expected, reflects the statistical variations of the signal.

Figure 3.8 shows the estimated average PSD or periodogram of the same EEG signal using the Welch technique with a Hamming window and a 1,024-point FFT; 64 data points overlap and 125 windows. The estimated PSDs using an AR model of order 10 and 20 of the same EEG signal are shown in Figure 3.9(a, b), respectively.

The preceding example emphasizes the importance of model order selection. An AR model of a relatively low order of 10 produced a "smoothed" spectrum and was not adequate to show the details of the EEG spectrum estimate. We needed to raise the order to at least 20 before seeing any resemblance between classical and modern spectral estimates. If we keep raising the order, less accurate estimates of the signal spectrum with spurious peaks will result. Akaike criterion can be used to determine an optimum order model [45].

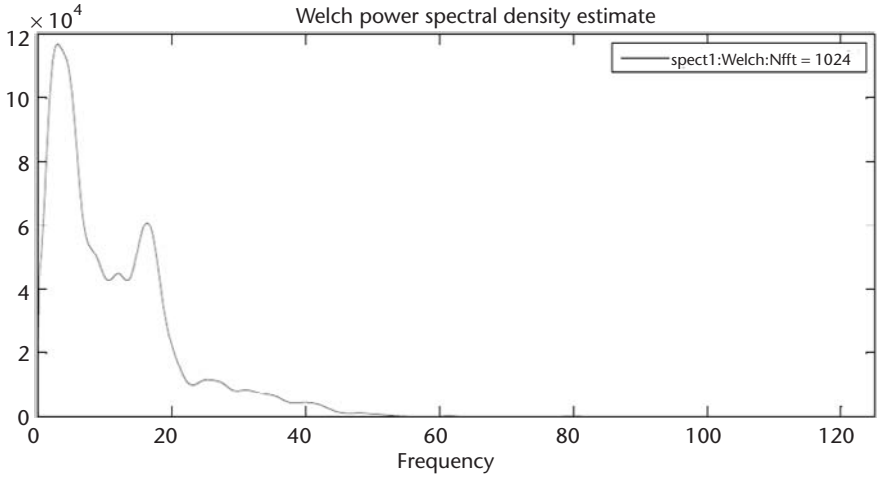


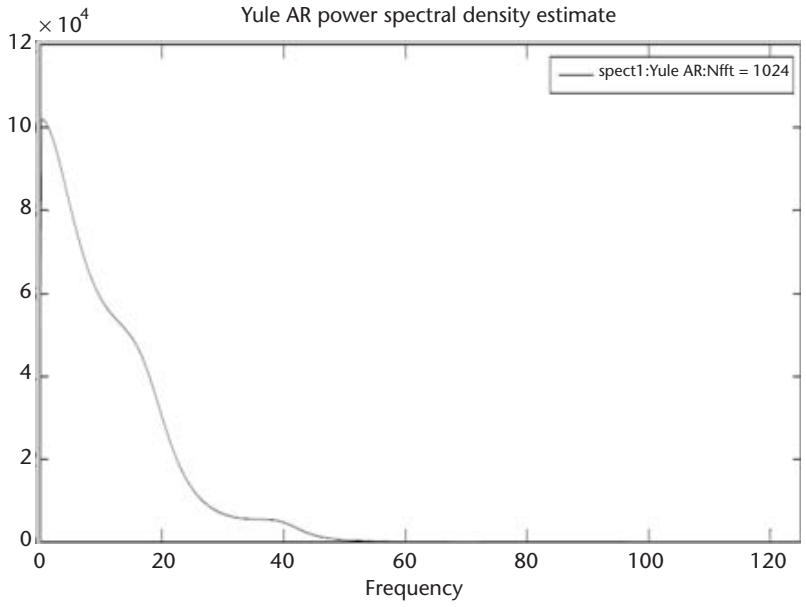

**Figure 3.8** The estimated average PSD or periodogram of the EEG signal from Figure 3.7 using a Hamming window and a 1,024-point FFT; 64 data points overlap and 125 windows.

Classical spectral estimation using a Fourier transform and FFT algorithm is a robust and computationally efficient technique. The main disadvantages of this technique are that it is unsuitable for short data segments and sidelobe leakage results due to windowing of short or finite datasets. This is in addition to the need for averaging to improve the statistical stability of the estimate. The AR model improves the spectral resolution in short data signals, but the order of the model has to be carefully selected.

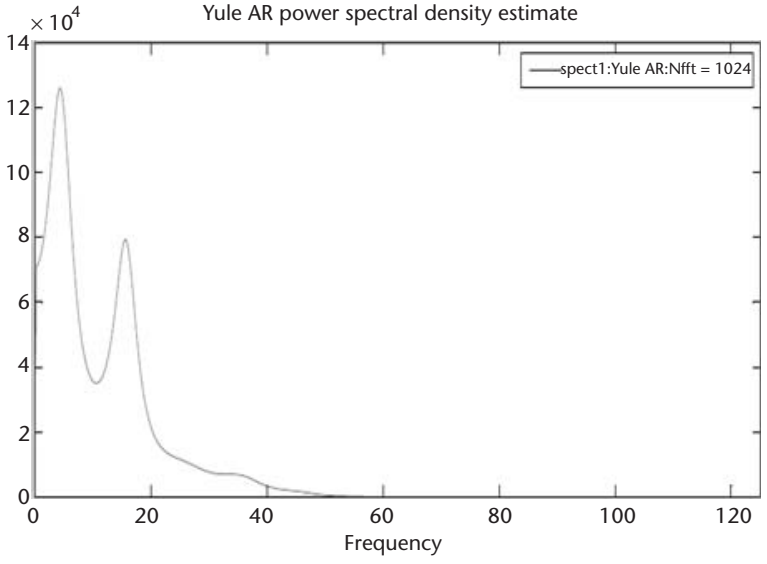
### 3.1.3 Nonstationarity in EEG and Time-Frequency Analysis

The classical spectral analysis techniques described earlier are very useful when dealing with statistically stationary signals. Because most biomedical signals including EEGs are nonstationary, the Fourier transform has the serious disadvantage of being unable to provide information about the time evolution of the signal frequencies. Remember that the statistical and spectral variations of the EEG signals are due to the dynamic mental state of the subject during sleep, intense mental activity, alertness, stress, eyes open or closed, and sensory stimulus. Figure 3.7(a) and Figure 3.10 show two EEG sequences recorded from the left frontoparietal regions of a rat's brain before and after brain injury, respectively.

Figure 3.11 depicts a comparison between the PSD of an EEG signal obtained from a rat model before (spect1) and after brain injury (spect2). The power spectra are distinctly different. If the PSD of the whole signal (before and after injury) is computed, the resultant PSD will not be able to reveal the temporal variations of the EEG spectra. Changes in frequency contents of the EEG signal will result in global change in the time domain. Consequently, any localized change in the time-domain signal will cause changes to all Fourier coefficients. Therefore, the Fourier transform reveals what frequencies exist in a time signal but fails to localize the times at which these frequencies occur. In quantitative EEG analysis, such PSD variations as a function of time are extremely important for the detection and monitoring of brain injury following, for example, a cardiac arrest.




(a)




(b)

**Figure 3.9** The estimated PSDs using an AR model of order (a) 10 and (b) 20 for the same EEG signal used in Figure 3.7.

One solution to this problem is to divide the long-term signal into blocks or windows of short time duration. The Fourier transform is then computed for each of these "short" signal blocks. One problem that may arise is that a short window will lead to a poor spectral resolution. If the window width is increased, the frequency resolution will improve while the time resolution will deteriorate. The time-frequency trade-off is associated with the Heisenberg uncertainty principle, which

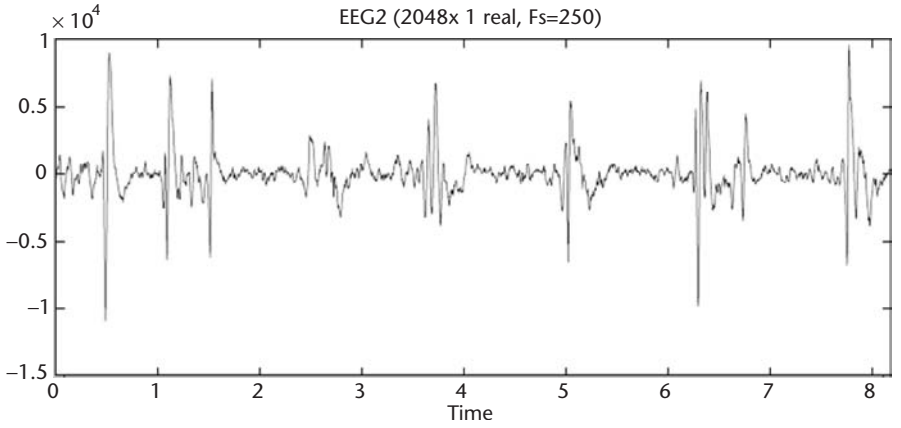



**Figure 3.10** EEG signal obtained from a rat model after brain injury.

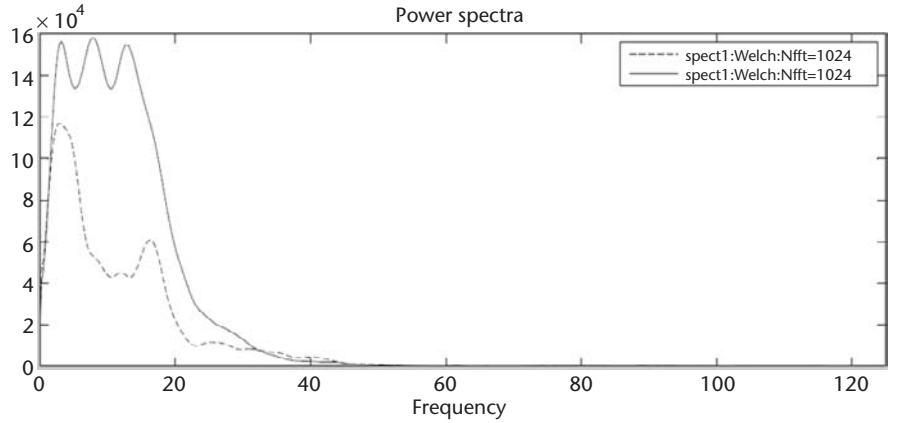



**Figure 3.11** Comparison of the PSDs of EEG signals obtained from a rat model before (spect1) and after brain injury (spect2).

states that arbitrary good time and frequency resolutions at the same location cannot be achieved [47]:

$$\Delta f \Delta t \geq \frac{1}{4\pi} \tag{3.24}$$

where $\Delta f$ and $\Delta t$ are the frequency and time resolutions, respectively. Nevertheless, the time-frequency representation of the EEG signal will provide a better alternative than having EEG information in either the time or frequency domain. Several techniques have been proposed to solve this problem. We will describe the short-time Fourier transform (STFT) and the wavelet transform (WT).

#### 3.1.3.1 Short-Time Fourier Transform

The starting point with the STFT is to slice the EEG signal into short "stationary" segments. This is usually performed by multiplying the EEG signal with a slid-

ing window. The STFT is defined as the DFT applied to the "windowed" segments. The discrete STFT of a discrete-time signal $x(n)$ at time instant $n$ is defined as [38, 41, 45, 46]:

$$\begin{aligned} &\mathrm{STFT}\{x(n)\} = \\ &X(n,k) = \sum_{m=0}^{N-1} x(n+m)W(m)\exp\left(-j\frac{2\pi}{N}km\right) \quad k = 0, 1, 2, \ldots, N-1 \end{aligned} \tag{3.25}$$

where $n$ and $k$ are the discrete time and frequency variables, respectively. The preceding equation is interpreted as the Fourier transform of $x(n+m)$ as viewed through a window $w(m)$ that has a stationary origin and $n$ changes. The signal is shifted past the window so that at each $n$ a different portion of the signal is viewed [38]. The time variable $n$ can be incremented in steps of $\Delta$ with $1 \leq \Delta \leq N$.

The following MATLAB function is used to estimate the STFT of signal $\boldsymbol{x}(n)$:

```
S = spectrogram(x,window,noverlap,nfft,fs)
```

where

**window** is a Hamming window of length nfft.

**noverlap** is the number of overlapping segments that produces 50% overlap between segments.

**nfft** is the FFT length and is a maximum of 256 or the next power of 2 greater than the length of each segment of $x$. (Instead of nfft , you can specify a vector of frequencies, **F**.)

**fs** is the sampling frequency, which defaults to normalized frequency.

Figure 3.12 shows the concatenation of EEG signals described earlier before and after injury and their STFTs.

### 3.1.3.2 Wavelet Transform

Fourier analysis uses sines and cosines as the orthogonal basis functions. These basis functions are localized in frequency but not in time. A small change in frequency will result in a global change in the time domain. Furthermore, any localized change in the time-domain signal will cause changes to all Fourier coefficients. Therefore, the Fourier transform reveals what frequencies exist in a time signal, but fails to localize the times at which these frequencies occur. This problem was resolved by using the STFT as explained earlier. Recall, however, that the time and frequency resolutions of the STFT are determined by the width of the analysis window. The time length of the analysis window is usually selected at the beginning of the analysis, which yields constant time and frequency resolutions. This is depicted by squares in the time–frequency analysis shown in Figure 3.13. STFTs with short time windows will lead to improved time resolution, but poor spectral resolution. If the time window is increased, the frequency resolution will improve, but the time resolution will deteriorate. This conflict between time and frequency resolution is resolved by the wavelet

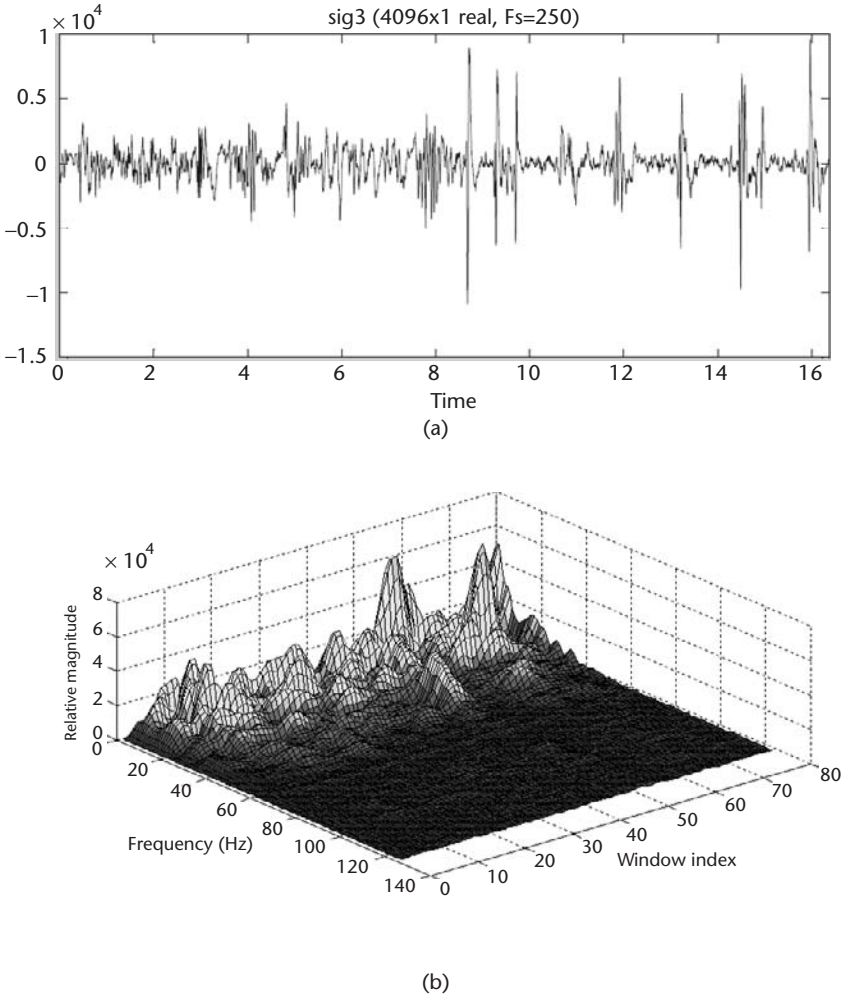



**Figure 3.12** (a) EEG signals before and after brain injury and (b) their STFTs.

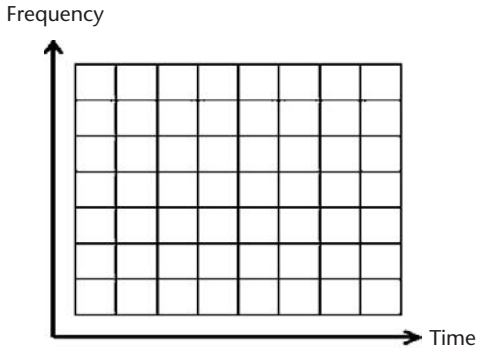



**Figure 3.13** STFT time-frequency resolution.

transform. Grossmann and Morlet introduced the wavelet transform in order to overcome the problem of time-frequency localization of time signals [48]. The

wavelet transform uses wide and narrow windows for slow and fast frequencies, respectively [49–51], thus leading to an optimal time-frequency resolution in all frequency ranges as depicted in Figure 3.14.

Notice that the area of all boxes in the time-frequency plane is constant. Therefore, the area of all boxes in both the STFT and wavelet transform must satisfy the Heisenberg inequality principle.

In wavelet transform analysis, a variety of probing functions is used, but these functions must originate from a basic and unique function known as the "mother wavelet" $\psi(t)$. The term "wavelet" is used because all probing functions have an oscillatory form. Figure 3.15 depicts examples of two popular wavelets: the Morlet and the Mexican hat, which are defined, respectively, as follows:

$$\psi(t) = \exp\left(-t^2/2\right)\cos(5t) \tag{3.26}$$

$$\psi(t) = \left(\frac{2}{\sqrt{3}}\pi^{-1/4}\right)\left(1-t^2\right)\exp\left(-t^2/2\right) \tag{3.27}$$

The basis functions of the wavelet transform should be of finite energy, able to represent signal features locally, and able to adapt to slow and fast variations of the signal. The mother wavelet must at least satisfy the following two conditions [2]:

$$\int_{-\infty}^{\infty}\psi(t)dt = 0 \tag{3.28}$$

$$\int_{-\infty}^{\infty}\left|\psi(t)\right|^2 dt < \infty \tag{3.29}$$

Once this mother wavelet is selected, all wavelets are just *dilations* and *translations* of this mother wavelet.

If $\Psi(t) \in L^2(\Re)$ (square integrable functions) is a basic mother wavelet function, then the continuous wavelet transform (CWT) of a finite energy signal or function $x(t)$ is defined as the convolution between that function and the wavelet functions $\psi_{a,b}$ [47, 52]:

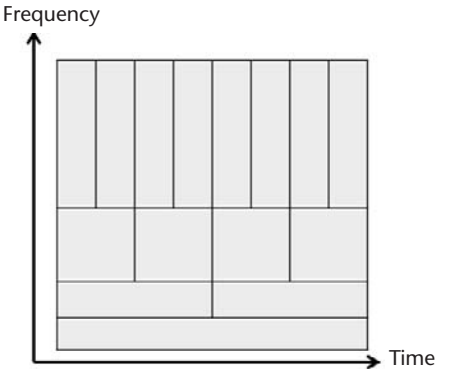



**Figure 3.14** Wavelet time-frequency resolution.

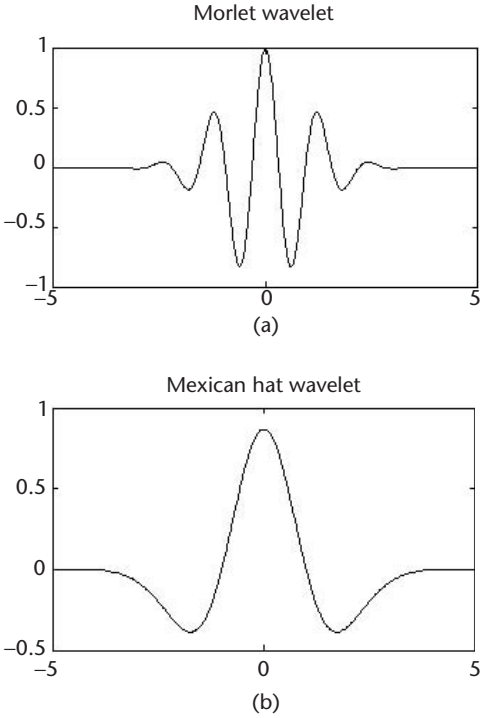



**Figure 3.15** Two popular wavelets: (a) the Morlet and (b) the Mexican hat.

$$\mathrm{WT}\left\{x(t);a,b\right\} = \frac{1}{\sqrt{|a|}}\int_{-\infty}^{\infty} x(t)\psi^{*}_{a,b}\left(\frac{t-b}{a}\right)dt \tag{3.30}$$

where $a, b \in \Re$, $a \neq 0$ represent the scale and translation parameters, respectively; $t$ is the time; and the asterisk stands for complex conjugation. If $a > 1$, then $\psi$ is stretched along the time axis and if $0 < a < 1$, then $\psi$ is contracted. If $b = 0$ and $a = 1$, then the wavelet is termed the mother wavelet. The wavelet coefficients describe the correlation or similarity between the wavelet at different dilations and translations and the signal $x$. As an example of a CWT, Figure 3.16 shows the continuous wavelet transform using the Morlet wavelet of the EEG signal depicted earlier in Figure 3.12(a).

### 3.1.3.3 Discrete Wavelet Transform

If we are dealing with digitized signals, then to reduce the number of redundant wavelet coefficients, $a$ and $b$ must be discretized. The discrete wavelet transform (DWT) attains this by sampling $a$ and $b$ along the dyadic sequence: $a = 2j$ and $b = k^2j$, where $j, k \in Z$ and represent the discrete dilation and translation numbers, respectively. The discrete wavelet family becomes

$$\left\{\psi_{j,k}(t) = 2^{-j/2}\,\psi\left(2^{-1}t - k\right),\ j,k \in Z\right\} \tag{3.31}$$

The scale $2^{-j/2}$ normalizes $\psi_{j,k}$ so that $\|\psi_{j,k}\| = \|\psi\|$.

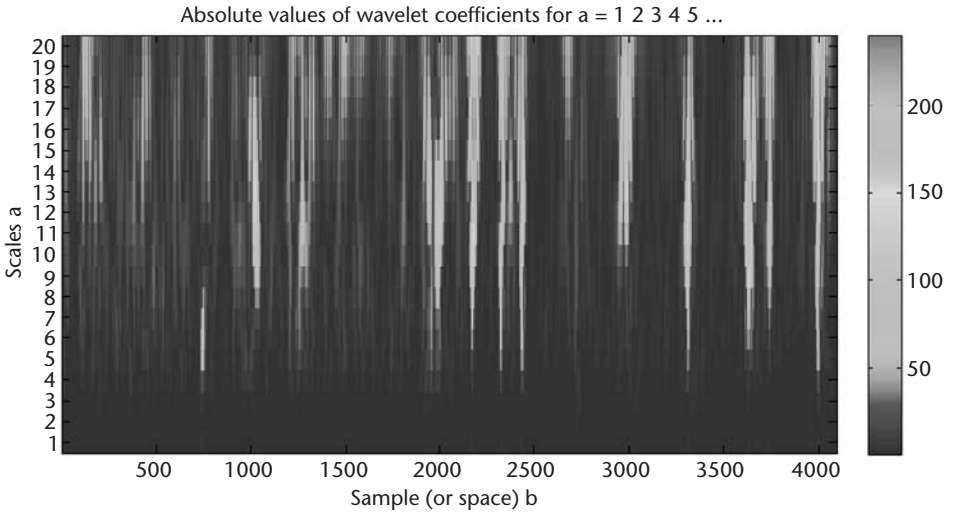



**Figure 3.16** The continuous wavelet transform of the EEG signal depicted earlier in Figure 3.12(a).

The DWT is the defined as

$$\mathrm{DWT}\{x(t);a,b\} = d_{j,k} \cong \int x(t)\cdot\psi^{*}_{j,k}(t)dt \tag{3.32}$$

The original signal can be recovered using the inverse DWT:

$$x(t) = \sum_{j\in Z}\sum_{k\in Z} d_{j,k}\,2^{-j/2}\,\psi\left(2^{-j}t-k\right) \tag{3.33}$$

where the $d_{j,k}$ are the WT coefficients sampled at discrete points $j$ and $k$. Note that the time variable is not yet discretized.

### 3.1.3.4 Multiresolution Wavelet Analysis

Multiresolution wavelet analysis (MRWA) decomposes a signal into scales with different time and frequency resolutions. Consider a finite energy time signal $x(t) \in L^2(R)$. The MRWA of $L^2(R)$ is defined as a sequence of nested subspaces $\{V_j \in L^2(R), j \in Z\}$, which satisfy the following properties [48]:

- Every function falls in some $V_j$ and no function belongs to all $V_j$ except the null function.
- $V_j \subset V_{j-1}$
- Under time shift, if $v(t-k) \in V_0$, then $v(2^{-j}t-k) \in V_j$.

The *scaling function*, sometimes called the *father wavelet*, is $\phi(t) \in V_0$ such that the integer translates set$\{\phi(t)=\phi(t-k): k \in Z\}$ forms a basis of $V_0$. If the dyadic scaling function $\phi_{j,k}(t) = 2^{-j/2}\phi(2^{-j}t-k): j, k \in Z$ is the basis function of $V_j$, then all elements of $V_j$ can be defined as a linear combination of $\phi_{j,k}(t)$.

Now, let us define $W_j$ as the orthogonal compliment of $V_j$ in $V_{j-1}$ such that

$$V_{j-1} = V_j \oplus W_j \quad j \in Z \tag{3.34}$$

where $\oplus$ refers to concatenation. Thus, we have

$$\begin{aligned} V_0 &= W_1 \oplus V_1 \\ V_0 &= W_1 \oplus W_2 \oplus V_2 \\ V_0 &= W_1 \oplus W_2 \oplus W_3 \oplus V_3 \\ &\vdots \end{aligned} \tag{3.35}$$

Thus, the closed subspaces $V_j$ at level $j$ are the sum of the whole function space $L^2(R)$:

$$V_j = W_{j+1} \oplus W_{j+2} \oplus W_{j+3} \oplus \ldots \oplus \quad j \in Z \tag{3.36}$$

Figure 3.17 depicts the MRWA described by (3.32).

Consequently, $\phi(t) \in V_1 \subset V_0$ and $\psi(t/2) \in V_1 \subset V_0$ can be expressed as linear combinations of the basis function of $V_0$, $\{\phi(t) = \phi(t-k): k \in Z\}$, that is:

$$\phi(t) = \sqrt{2}\sum_k h(k)\phi(2t-k) \tag{3.37}$$

and

$$\psi(t) = \sqrt{2}\sum_k g(k)\phi(2t-k) \tag{3.38}$$

where the coefficients $h(k)$ and $g(k)$ are defined as the inner products $\langle \phi(t), \sqrt{2}\phi(2t-k) \rangle$ and $\langle \psi(t), \sqrt{2}\phi(2t-k) \rangle$, respectively. The sequences $\{h(k), k \in Z\}$ and $\{g(k), k \in Z\}$ are coefficients of a lowpass filter $H(\omega)$ and a highpass filter $G(\omega)$, respectively. They form a pair of quadrature mirror filters that is used in the MRWA [52]. There are many scaling functions in the literature including the Haar, Daubechies, biorthogonal, Coiflets, Symlets, Morlet, Mexican hat and Meyer func-

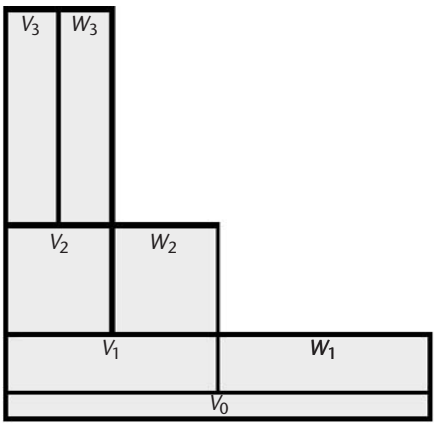



**Figure 3.17** Multiresolution wavelet analysis.

tions. Figure 3.18 depicts Daubechies 4 scaling and wavelet functions. The choice of the wavelet depends on the application at hand.

The process of wavelet decomposition is shown in Figure 3.19. It is the process of successive highpass and lowpass filtering of the function $x(t)$ or EEG signal.

1. The signal is sampled with sampling frequency $f_s$ forming a sequence $x(n)$ of length $N$.
2. The signal is then highpass filtered with filter $G(e^{j\omega})$ and downsampled by 2. The resultant sequence is the "details" wavelet coefficients $D_1$ of length $N/2$. The bandwidth of $d_1$ sequence is $(f_s/4, f_s/2)$.

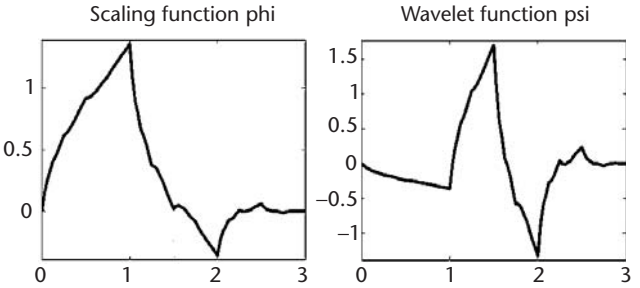



**Figure 3.18** Daubechies 4 scaling and wavelet functions.

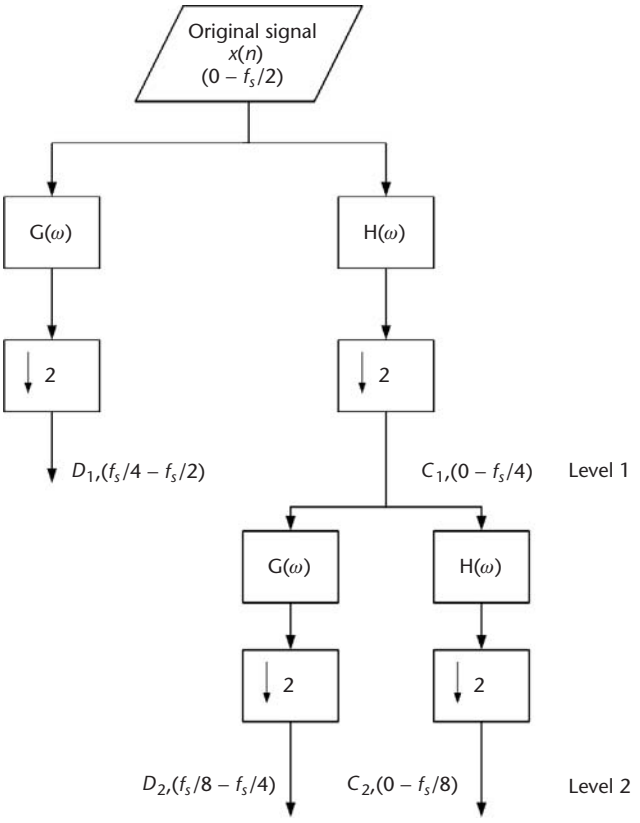



**Figure 3.19** The process of successive highpass and lowpass filtering.

3. The signal is also lowpass filtered with filter $H(e^{j\omega})$ and downsampled by 2. The resultant sequence is the "smoothed" coefficients $C_1$ of length $N/2$. The bandwidth of $C_1$ sequence is $(0, f_s/4)$.
4. The smoothed sequence $C_1$ is further highpass filtered with filter $G(e^{j\omega})$ and downsampled by 2, and lowpass filtered with filter $H(e^{j\omega})$ and downsampled by 2, to generate $D_2$ and $C_2$ of length $N/4$. The bandwidth of $C_2$ sequence is $(0, f_s/8)$ and of $D_1$ sequence is $(f_s/8, f_s/4)$.
5. The process of lowpass filtering, highpass filtering, and downsampling is repeated until the required resolution $j$ is reached.

The signal $x(n)$ can be reconstructed again with the preceding coefficients using the following formula:

$$x(n) = \sum_j C_{j,k} \cdot \phi_{j,k} + \sum_j \sum_k D_{j,k} \cdot \psi_{j,k}(t) \tag{3.39}$$

MATLAB provides several MRWA functions: **[C,L] = wavedec(x,N,'wname')** returns the wavelet decomposition of the signal **x** at level **N**, using **'wname'**. Note that **N** must be a strictly positive integer. Several wavelets are available in MATLAB including Haar, Daubechies, biorthogonal, Coiflets, Symlets, Morlet, Mexican hat, and Meyer. The function **x waverec(C,L,'wname')** reconstructs the signal **x** based on the multilevel wavelet decomposition structure **[C,L]** and wavelet **'wname'**.

For an EEG sampled at 250 Hz, a five-level decomposition results in a good match to the standard clinical bands of interest [20]. The basis functions of the wavelet transform should be able to represent signal features locally and adapt to slow and fast variations of the signal. Another requirement is that the wavelet functions should satisfy the finite support constraint and differentiability to reconstruct smooth changes in the signal symmetry to avoid phase distortions [20, 27, 28]. Figure 3.20 shows the MRWA of the 4,096-point EEG data segment described earlier and shown in Figure 3.12(a). The signal is decomposed into five levels using the Daubechies 4 wavelet.

## 3.2 Nonlinear Description of EEGs

Nonlinear methods of dynamics provide a useful set of tools for the analysis of EEG signals, which by their very nature are nonlinear. Even though these methods are less well understood than their linear counterparts, they have proven to generate new information that linear methods cannot reveal, for example, about nonlinear interactions and the complexity and stability of underlying brain sites [38]. We support this assertion by applying some of the well-known methods to EEGs and epilepsy in this chapter. For a reader to further understand and develop an intuition for these approaches, it is advisable to apply them to simulations with known, well-defined coupled nonlinear systems. Such systems exist, for example, the logistic and Henon maps (discrete-time nonlinear), and the Lorenz, Rossler, and Mackey-Glass systems (continuous-time nonlinear).

The dynamics of highly complex, nonlinear systems in nature [53], medicine [54, 55], and economics [56] has been of much scientific interest recently. A strong

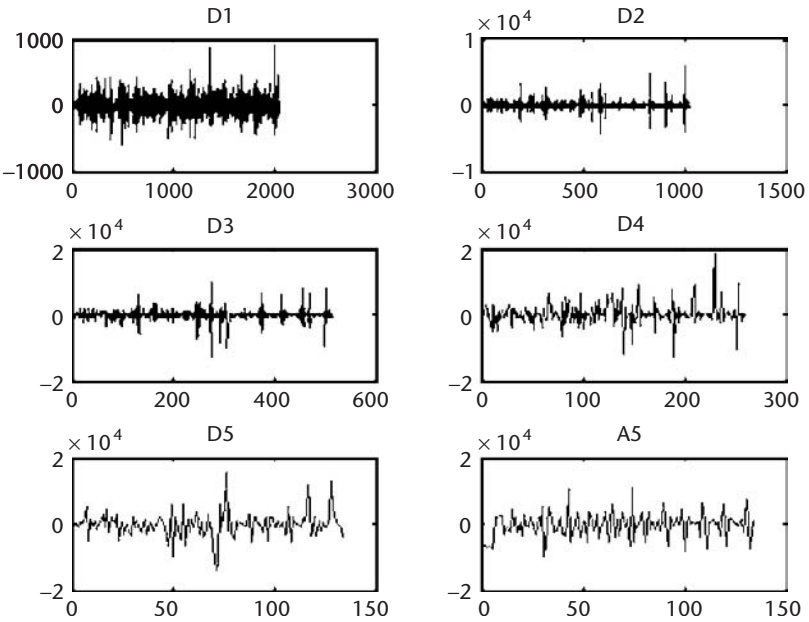



**Figure 3.20** A five-level MRWA for a 4,096-point EEG data segment using the Daubechies 4 wavelet.

motivation is that a successful study of such complex systems may have a significant impact on our ability to forecast their future behavior and intervene in time to control catastrophic crises.

In principle, the dynamics of complex nonlinear systems can be studied both by analytical and numerical techniques. In the majority of these systems, analytical solutions cannot be found following mathematical modeling, because either exact nonlinear equations are difficult to derive from the data or to subsequently solve in closed form. Given our inadequate knowledge of their initial conditions, individual components, and intercomponent connections, mathematical modeling seems to be a formidable task. Therefore, it appears that time-series analysis of such systems is a viable alternative.

Although traditional linear time-series techniques appeared to enjoy initial success in the study of several problems [57], it has progressively become clear that additional information provided by employment of techniques from nonlinear dynamics may be crucial to satisfactorily address these problems. Theoretically, even simple nonlinear systems can exhibit extremely rich (complicated) behavior (e.g., chaotic dynamics). Furthermore, standard linear methods, such as power spectrum analyses, Fourier transforms, and parametric linear modeling, may fail to capture and, in fact, may lead to erroneous conclusions about those systems' behavior [58]. Thus, employing existing and developing new methods within the framework of nonlinear dynamics and higher order statistics for the study of complex nonlinear systems is of practical significance, and could also be of theoretical significance for the fields of signal processing and time-series analysis.

Nonlinear dynamics has opened a new window for understanding the behavior of the brain. Nonlinear dynamic measures of complexity (e.g., the correlation dimension) and stability (e.g., the Lyapunov exponent and Kolmogorov entropy)

quantify critical aspects of the dynamics of the brain as it evolves over time in its state space. Higher order statistics, such as cumulants and bispectrum (straightforward extensions of the traditional linear signal processing concepts of second-order statistics and power spectrum), measure nonlinear interactions between the components of a signal or between signals. In the following, we apply these concepts to the analysis of EEGs. EEG data recorded using depth and subdural electrodes from one patient with temporal lobe epilepsy will be utilized for this purpose.

A brief introduction to higher order statistics is given in Section 3.2.1. We describe the estimation of higher order statistics in the time and frequency domains. In particular, we estimate the cumulants and the bispectrum of EEG data segments before, during, and after an epileptic seizure. Section 3.2.2 introduces the correlation dimension and Lyapunov exponents as nonlinear descriptors of the dynamics of EEG. We utilize the correlation dimension to characterize the complexity of EEGs during an epileptic seizure, and the maximum Lyapunov exponent and its temporal evolution at electrode sites to characterize the stability before, during, and after a seizure.

### 3.2.1 Higher-Order Statistical Analysis of EEGs

The information contained in the power spectrum of a stochastic signal is the result of second-order statistics (e.g., the Fourier transform of the autocorrelation of the signal in the time domain). The power spectrum, in the case of linear Gaussian processes and when phase is not of interest, is a useful and sufficient representation. This is not the case with a nonlinear process, for example, when the process is the output of a nonlinear system excited by white noise. When we deal with nonlinear systems and their affiliated signals, analyses must be performed beyond second-order statistics of the involved signals in order, for example, to accurately detect phase differences (locking) and nonlinear relations or to test for deviation from Gaussianity.

#### 3.2.1.1 Time-Domain Higher-Order Statistics: Moments and Cumulants

Higher order statistics in the time domain are defined in terms of moments and cumulants [59]. Moments and cumulants of a random process can be obtained from the moment and cumulant generating functions, respectively.

Consider a random (stochastic) scalar process $s = \{s_1, s_2, ..., s_n\}$, where $s_i = \{s(t_i): i = 1, ..., n\}$ are different realizations of $s$. The moment generating function (also called the characteristic function) $M$ of $s$ is then defined as

$$M(\lambda_1, \lambda_2, \ldots, \lambda_n) = E\{\exp j(\lambda_1 s_1 + \lambda_2 s_2 + \ldots + \lambda_n s_n)\} \tag{3.40}$$

where $E\{.\}$ denotes the expectation operator on the values of random variable $s_n$.

The $r$ moments of $s$ ($r \geq 1$) can be generated by differentiating the moment generating function $M(\lambda_1, \lambda_2, ..., \lambda_n)$ with respect to $\lambda s$, and estimating it at $\lambda_1 = \lambda_2 = \ldots \lambda_n = 0$, provided these derivatives exist. For example, the $r$th (joint) moment of $s$, where $\gamma = k_1 + k_2 \ldots + k_n$, is given by

$$m_{r=k_1+k_2+\ldots+k_n} = (-j)^r \left.\frac{\partial^r M(\lambda_1, \lambda_2, \ldots, \lambda_n)}{\partial \lambda_1^{k_1} \ldots \partial \lambda_n^{k_n}}\right|_{\lambda_1=\lambda_2=\ldots=\lambda_n=0} = E\left\{ s_1^{k_1} s_2^{k_2} \ldots s_n^{k_n} \right\} \tag{3.41}$$

If we assume that $s$ is a stationary and ergodic process, for the first-order ($k_1 = 1$) moment $m_1(t_1) \overset{(2)}{=} E\{s(t_1)\} = E\{s(t)\} = \text{constant}$, and for the second-order ($k_1 = 1$, $k_2 = 1$) moment $E\{s(t_1)s(t_2)\} = m_2(t_1, t_2) = E\{s(t)s(t+\tau)\} = m_2(\tau)\ \forall \tau \in R$. Of note here is that $m_1 = E\{s(t)\}$ is the mean of $s$, and $m_2(\tau) = E\{s(t)s(t+\tau)\}$ is the autocorrelation function of $s$. Then, the $r$th-order joint moment (i.e., $k_1 = 1, k_2 = 1, \ldots, k_r = 1$; the rest of the $k$s are zeros) can be written as

$$E\left\{ s(t_1)s(t_2)\ldots s(t_n) \right\} \overset{(2)}{=} m_{r=k_1+k_2+\ldots k_n}$$
$$= E\left\{ s(t)^{k_1}\, s(t+\tau_1)^{k_2} \ldots s(t+\tau_{r-1})^{k_r} \cong m_r(\tau_1, \tau_2, \ldots, \tau_{r-1}) \right\}$$

The third-order moment is then $m_3(\tau_1, \tau_2) = E\{s(t)s(t+\tau_1)s(t+\tau_2)\}$.

The cumulant generating function $C$ is defined by taking the natural logarithm of the moment generating function $M$. Then we have

$$C(\lambda_1, \lambda_2, \ldots, \lambda_{r-1}) = \ln\left( E\left\{ \exp\left[ j(\lambda_1 s_1 + \lambda_2 s_2 + \ldots + \lambda_{r-1} s_{r-1}) \right] \right\} \right) \tag{3.42}$$

Along similar lines as above, if we take the $r$th derivative of the cumulant generating function about the origin, we obtain the $r$th-order (joint) cumulant of $s$ (which also is the coefficient in the Taylor expansion of $C$ around 0):

$$c_r(k_1, k_2, \ldots, k_n) \cong c_r(\tau_1, \tau_2, \ldots, \tau_{r-1}) = (-j)^r \left.\frac{\partial^r C(\lambda_1, \lambda_2, \ldots, \lambda_n)}{\partial \lambda_1^{k_1} \ldots \partial \lambda_n^{k_n}}\right|_{\lambda_1=\lambda_2=\ldots\lambda_n=0} \tag{3.43}$$

The first-order cumulant $c_1$ of $s$ is equal to the mean value of $s$, and hence it is equal to the first-order moment $m_1$. The second-order cumulant is equal to the autocovariance function of $s$, that is, $c_2(\tau) - m_2(\tau) - (m_1)^2$. The third-order cumulant is $c_3(\tau_1, \tau_2) = m_3(\tau_1, \tau_2) - m_1\,[m_2(\tau_1) + m_2(\tau_2) + m_2(\tau_1 - \tau_2)] + 2(m_1)^3$. So, $c_2(\tau) = m_2(\tau)$ and $c_3(\tau_1, \tau_2) = m_3(\tau_1, \tau_2)$. when $m_1 = 0$, that is, if the signal is of zero mean. The cumulants are preferred over the moments of the signal to work with for several reasons, one of which is that the cumulant of the sum of independent random processes equals the sum of the cumulants of the individual processes, a property that is not valid for the moments of corresponding order.

If $s$ is a Gaussian process, the third- and higher order cumulants are zero. Also, the third-order cumulant would be zero if the random process is of high order, but its probability distribution is symmetrical around $s = 0$. In this case, we have to estimate higher than an order of three cumulants to better characterize it. For a detailed description of the properties of cumulants, the reader is referred to [59].

#### 3.2.1.2 Estimation of Cumulants from EEGs

To estimate the third-order cumulant from an EEG data segment $s$ of length $N$, sampled with sampling period $Dt$, the following steps are performed:

1. The data segment $s$ is first divided into $R$ smaller segments $s^i$, with $i = 1, \ldots, R$, each of length $M$ such that $R \cdot M = N$.
2. Subtract the mean from each data segment $s^i$.
3. If the data in each segment $i$ is $s^i(n)$ for $n = 0, 1, \ldots, M - 1$, and with a sampling period $Dt$ such that $t_n = n \cdot Dt$, an estimate of the third-order cumulant per segment $s^i$ is given by

$$c_3^i(l_1, l_2) = \frac{1}{M} \sum_{n=u}^{v} s^i(n) s^i(n + l_1) s^i(n + l_2) \tag{3.44}$$

   where $u = \max(0, -l_1, -l_2)$; $v = \min(M - 1, M - 1 - l_1, M - 1 - l_2)$; $l_1 \cdot Dt = \tau_1$, and $l_2 \cdot Dt = \tau_2$. Higher-order cumulants can be estimated likewise [7].
4. Average the computed cumulants across the $R$ segments:

$$c_3(l_1, l_2) = \frac{1}{R} \sum_{i=1}^{R} c_3^i(l_1, l_2) \tag{3.45}$$

   Thus, $c_3(l_1, l_2)$ is the average of the estimated third-order cumulants per short EEG segment $s^i$.

The preceding steps can be performed per EEG segment $i$ over the available time of recording to obtain the cumulants over time.

#### 3.2.1.3 Frequency-Domain Higher-Order Statistics: Bispectrum and Bicoherence

Higher-order spectra (polyspectra) are defined by taking the multidimensional Fourier transform of the higher order cumulants. Thus, the $r$th-order polyspectra are defined as follows:

$$S_r(\omega_1, \omega_2, \ldots, \omega_{r-1}) = \sum_{l_1=-\infty}^{\infty} \cdots \sum_{l_{r-1}=-\infty}^{\infty} c_r(l_1, l_2, \ldots, l_{r-1}) \exp\left[-j \sum_{i=1}^{r-1} \omega_i l_i\right] \tag{3.46}$$

Therefore, the $r$th-order cumulant must be absolutely summable for the $r$th-order spectra to exist.

Substituting $r = 2$ in (3.46), we get

$$S_2(\omega_1) = \sum_{l_1=-\infty}^{\infty} c_2(l_1) \exp(-j\omega_1 l_1) \quad \text{(power spectrum)} \tag{3.47}$$

Substituting $r = 3$ in (3.46), we instead get

$$S_3(\omega_1, \omega_2) = \sum_{l_1=-\infty}^{\infty} \sum_{l_2=-\infty}^{\infty} c_3(l_1, l_2) \exp(-j\omega_1 l_1 - j\omega_2 l_2) \quad \text{(bispectrum)} \tag{3.48}$$

For a real signal $s(t)$, the power spectrum is real and nonnegative, whereas bispectra and the higher order spectra are, in general, complex. For a real, discrete, zero-mean, stationary process $s(t)$, we can determine the third-order cumulant as we did for (3.44). Subsequently, the bispectrum in (3.48) becomes:

$$S_3(\omega_1,\omega_2) = \sum_{l_1=-\infty}^{\infty}\sum_{l_2=-\infty}^{\infty} E\{s(n)s(n+l_1)s(n+l_2)\}\exp(-j\omega_1 l_1 - j\omega_2 l_2) \tag{3.49}$$

Equation (3.49) shows that the bispectrum is a function of $\omega_1$ and $\omega_2$, and that it does not depend on a linear time shift of $s$. In addition, the bispectrum quantifies the presence of quadratic phase coupling between any two frequency components in the signal. Two frequency components are said to be quadratically phase coupled (QPC) when a third component, whose frequency and phase are the sum of the frequencies and phases of the first two components, shows up in the signal's bispectrum [see (3.50)]. Whereas the power spectrum gives the product of two identical Fourier components (one of them taken with complex conjugation) at one frequency, the bispectrum represents the product of a tuple of three Fourier components, in which one frequency equals the sum of the other two [60]. Hence, a peak in the bispectrum indicates the presence of QPC. If there are no phase-coupled harmonics in the data, the bispectrum (and, hence, the second-order cumulant) is essentially zero. Interesting properties of the bispectrum, besides its ability to detect phase couplings, are that the bispectrum is zero for Gaussian signals and that it is constant for linearly related signals. These properties have been used as test statistics to rule out the hypothesis that a signal is Gaussian or linear [59]. Under conditions of symmetry, only a small part of the bispectral space would have to be further analyzed. Examples of such symmetries are [60]:

$$S_3(\omega_1,\omega_2) = S_3(\omega_2,\omega_1) = S_3^*(-\omega_2,-\omega_1) = S_3(-\omega_2-\omega_1,\omega_1) = S_3(\omega_2,-\omega_2-\omega_1)$$

For a detailed discussion on the properties of the bispectrum, we refer the reader to [59].

#### 3.2.1.4 Estimation of Bispectrum

The bispectrum can be estimated using either parametric or nonparametric estimators. The nonparametric bispectrum estimation can be further divided into the indirect method and the direct method. The direct and indirect methods discussed herein have been shown to be more reliable than the parametric estimators for EEG signal analysis. The bias and consistency of the different estimators for bispectrum are addressed in [60].

*Indirect Method*

We first estimate the cumulants as described in Section 3.2.1.2. The first three steps therein are followed to obtain the cumulant $c_3^i(k,l)$ per segment $s^i$. Then, the two-dimensional Fourier transform $S_3^i(\omega_1, \omega_2)$ of the cumulant is obtained. The average of $S_3^i(\omega_1, \omega_2)$ over all segments $i = 1, \ldots, R$, gives the bispectrum estimate $S_3(\omega_1, \omega_2)$.

*Direct Method*

The direct method estimates the bispectrum directly from the frequency domain. It involves the following steps:

1. Divide the EEG data of length $N$ into $R$ segments, each of length $M$, such that $R \cdot M = N$. Let each segment be denoted by $s^i$.
2. In each segment $s^i$ subtract the mean.
3. Compute the one-dimensional FFT for each of these segments to obtain $Y^i(\omega)$.
4. The bispectrum estimate for the segment $s^i$ is obtained by

$$S_3^i(\omega_1, \omega_2) = Y^i(\omega_1) Y^i(\omega_2) Y^{i*}(\omega_1 + \omega_2) \tag{3.50}$$

   for all combinations of $\omega_1$ and $\omega_2$, with the asterisk denoting complex conjugation.
5. As in a periodogram, the bispectrum estimate of the entire data is obtained by averaging the bispectrum estimate of individual segments:

$$S_3(\omega_1, \omega_2) = \frac{1}{R} \sum_{i=1}^{R} S_3^i(\omega_1, \omega_2) \tag{3.51}$$

It is clear from (3.50) that the bispectrum can be used to study the interaction between the frequency components $\omega_1$, $\omega_2$, and $\omega_1 + \omega_2$. A drawback in the use of polyspectra is that they need long datasets to reduce the variance associated with estimation of higher order statistics.

The bispectrum is also influenced by the power of the signal at its components; therefore, it is not only a measure of quadratic phase coupling. The bispectrum could be normalized in order to make it sensitive only to changes in phase coupling (as we do for spectrum in order to generate coherence). This normalized bispectrum is then known as bicoherence [60]. To compute the bicoherence (BIC) of a signal, we define the real triple product RTP($\omega_1$, $\omega_2$) of the signal as follows:

$$\text{RTP}(\omega_1, \omega_2) = P(\omega_1) P(\omega_2) P(\omega_1 + \omega_2) \tag{3.52}$$

where $P(\omega)$ is the power spectrum of the signal at angular frequency $\omega$. The bicoherence is then defined as the ratio of the bispectrum of the signal to the square root of its RTP:

$$\text{BIC}(\omega_1, \omega_2) = \frac{S_3(\omega_1, \omega_2)}{\sqrt{\text{RTP}(\omega_1, \omega_2)}} \tag{3.53}$$

If all frequencies are completely phase coupled to each other (identical phases), $S_3(\omega_1, \omega_2) = \sqrt{\text{RTP}(\omega_1, \omega_2)}$, and $\text{BIC}(\omega_1, \omega_2) = 1$. If there is no QPC coupling at all, the bispectrum will be zero in the $(\omega_1, \omega_2)$ domain. If $|\text{BIC}(\omega_1, \omega_2)| \neq 1$ for some $(\omega_1, \omega_2)$, the signal is a nonlinear process. The variance of the bicoherence estimate is directly proportional to the amount of statistical averaging performed during the

computation of the bispectrum and RTP. Therefore, the choice of segment size and amount of overlap are important to obtaining good estimates.

#### 3.2.1.5 Application: Estimation of Bispectra from Epileptic EEGs

An example of the application of bispectrum to EEG recording follows. Intracranial EEG recordings were obtained from implanted electrodes in the hippocampus (depth EEG) and over the inferior temporal and orbitofrontal cortex (subdural EEG). Figure 3.21 shows the 28-electrode montage used for these recordings. Continuous EEG signals were sampled with a sampling frequency of 256 Hz and lowpass filtered at 70 Hz. Figure 3.22 depicts a typical ictal EEG recording, centered about the time of the onset of an epileptic seizure.

Figure 3.23 shows the cumulant structure of the EEG recorded from one electrode placed on the epileptogenic focus (RTD2) before, during, and after the seizure depicted in Figure 3.22. From Figure 3.23(a), it is clear that there are strong correlations at short timescales/shifts $\tau$ (about $\pm 0.5$ second) in the preictal period (before a seizure), which spread to longer timescales $\tau$ in the ictal period, and switch back to short timescales $\tau$ in the postictal period. Figure 3.24 depicts the bispectrum derived from Figure 3.23. It shows that the main bispectral peaks in the bifrequency domain $(f_1, f_2)$ are interacting in the alpha frequency range in the ictal period, versus in the low-frequency range in the preictal and postictal periods. Because this bispectrum is neither zero nor constant, it implies the presence of nonlinearities and higher than second-order interactions. This information cannot be extracted from traditional linear (or second-order statistics) signal processing techniques and shows the potential to assist in addressing open questions in epilepsy, such as epileptogenic focus localization and seizure prediction.

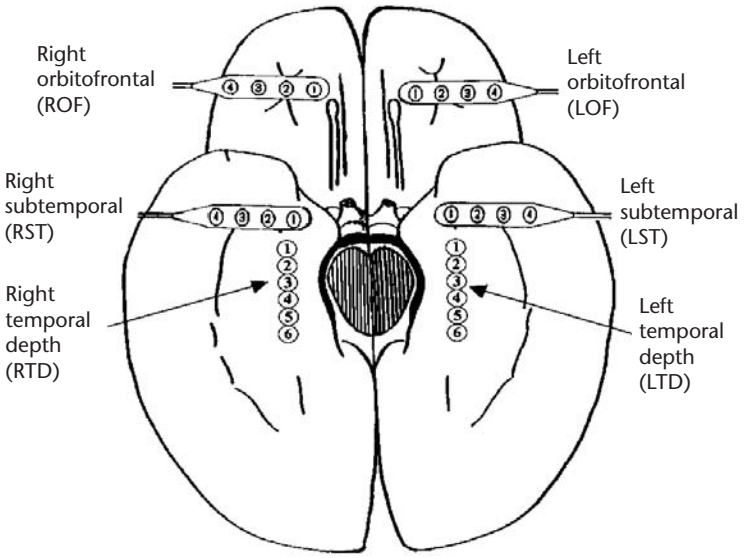



**Figure 3.21** Schematic diagram of the depth and subdural electrode placement. This view from the inferior aspect of the brain shows the approximate location of depth electrodes, oriented along the anterior-posterior plane in the hippocampi (RTD, right temporal depth; LTD, left temporal depth), and subdural electrodes located over the orbitofrontal and subtemporal cortical surfaces (ROF, right orbitofrontal; LOF, left orbitofrontal; RST, right subtemporal; LST, left subtemporal).

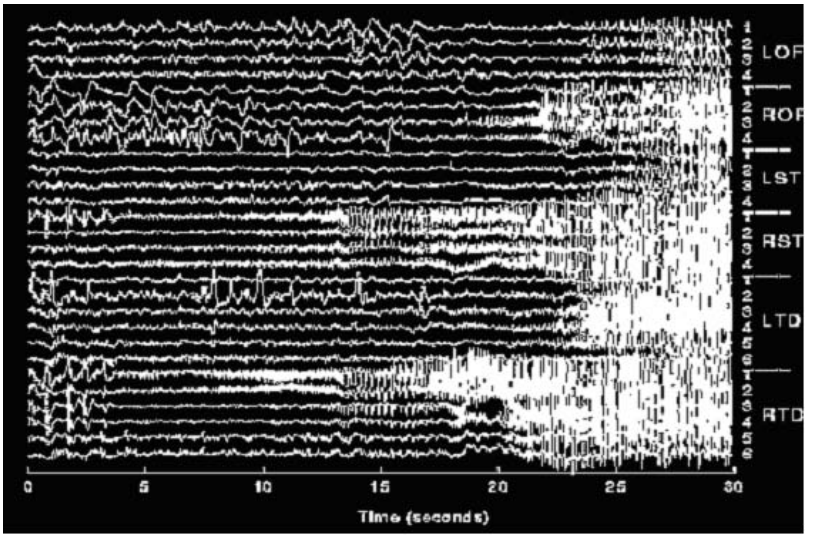



**Figure 3.22** A 30-second EEG segment at the onset of a right temporal lobe seizure, recorded from 12 bilaterally placed depth (hippocampal) electrodes, 8 subdural temporal electrodes, and 8 subdural orbitofrontal electrodes (according to nomenclature in Figure 3.21). The ictal discharge begins as a series of low-amplitude sharp and slow wave complexes in the right depth electrodes (RTD 1–3, more prominently RTD2) approximately 5 seconds into the record. Within seconds, it spreads to RST1, the rest of the right hippocampus, and the temporal and frontal lobes. The seizure lasted for 80 seconds (the full duration of this seizure is not shown in this figure).

### 3.2.2 Nonlinear Dynamic Measures of EEGs

From the dynamic systems theory perspective, a nonlinear system may be characterized by steady states that are chaotic attractors in the state space [55, 61, 62]. A state space is created by treating each time-dependent variable of a system as one of the components of a time-dependent state vector. For most dynamic systems, the state vectors are confined to a subspace of the state space and create an object commonly referred to as an attractor. The geometric properties of these attractors provide information about the dynamics of a system. Among the well-known methods used to study systems in the state space [63–65], the Lyapunov exponents and correlation dimension are discussed further below and applied to EEG.

#### 3.2.2.1 Reconstruction of the State Space: Embedding

A well-known technique for visualizing the dynamics of a multidimensional system is to generate the state space portrait of the system. A state space portrait [66] is created by treating each time-dependent variable of a system as a component of a vector in a vector space. Each vector represents an instantaneous state of the system. These time-dependent vectors are plotted sequentially in the state space to represent the evolution of the state of the system with time. One of the problems in analyzing multidimensional systems in nature is the lack of knowledge of which observable (variables of the system that can be measured) should be analyzed, as well as the limited number of observables available due to experimental constraints. It turns out that when the behavior over time of the variables of the system is related, which

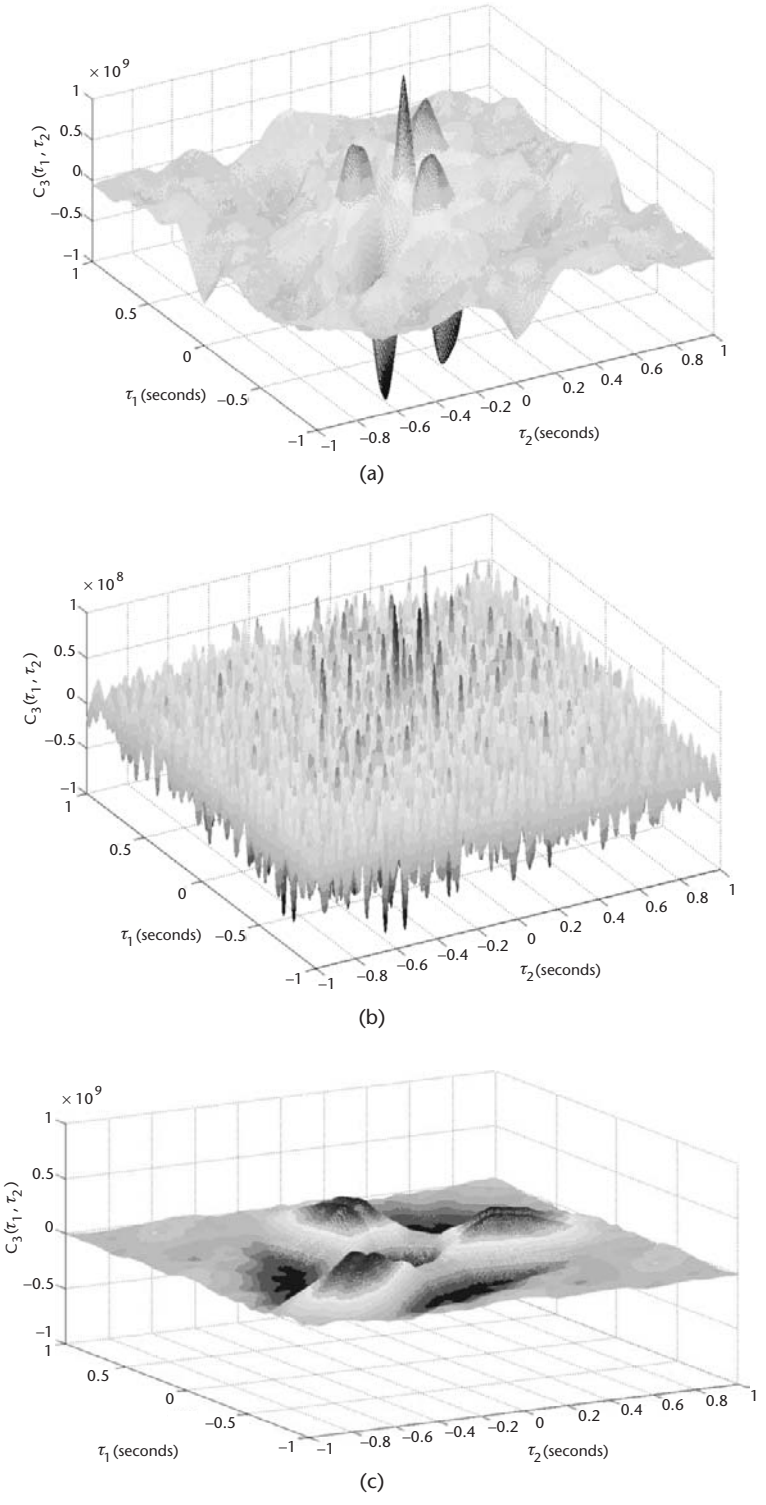



**Figure 3.23** Cumulant $C_3(\tau_1, \tau_2)$ estimated from 10-second EEG segments located at one focal electrode (a) 10 seconds prior to, (b) 20 seconds after, and (c) 10 seconds after the end of an epileptic seizure of temporal lobe origin. The positive peaks observed in the ictal cumulant are less localized in the $(\tau_1, \tau_2)$ space than the ones observed during the preictal and postictal periods.

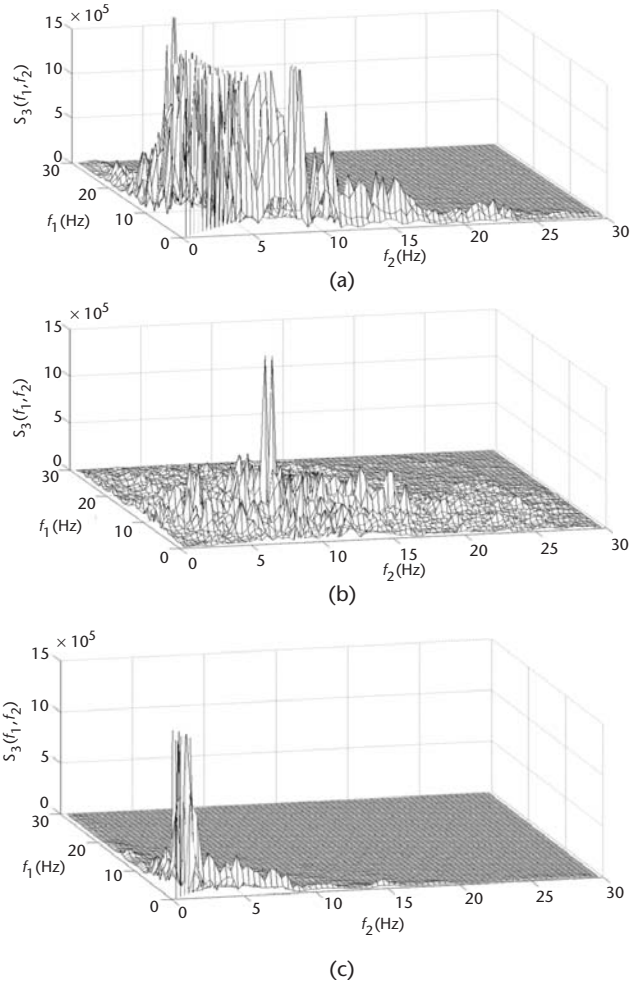



**Figure 3.24** Magnitude of bispectra $S_3(f_1, f_2)$ of the EEG segments with cumulants $C_3(\tau_1, \tau_2)$ depicted in Figure 3.23 for (a) preictal, (b) ictal, and (c) postictal periods of a seizure. The two-dimensional frequency $(f_1, f_2)$ domain of the bispectra has units in hertz. Bispectral peaks occur in the neighborhood of 10 Hz in the ictal period, and at lower frequencies in the preictal and postictal periods.

is typically the case for a system to exist, the analysis of a single observable can provide information about all related variables to it. The technique of obtaining a state space representation of a system from a single time series is called state space reconstruction, or embedding of the time series, and it is the first step for a nonlinear dynamic analysis of the system under consideration.

A time series is obtained by sampling a single observable of a system usually with a fixed sampling period $Dt$:

$$s_n = s\big(x(n \cdot Dt)\big) + \psi_n \tag{3.54}$$

where $t = n \cdot Dt$ and the signal $x(t)$ is measured through some measurement function $s$ and under the influence of some random fluctuation $\psi_n$ (measurement noise). An

$m$-dimensional state space reconstruction with the method of delays is then performed by

$$\mathbf{s}_n = \left(s_n, s_{n-l}, \ldots, s_{n-(m-2)l}, s_{n-(m-1)l}\right) \tag{3.55}$$

The time difference $\tau = l \cdot Dt$ between the successive components of the state vector $\mathbf{s}_n$ is referred to as the lag or delay time, and $m$ is the embedding dimension [67, 68]. The sequence of points (vectors) in the state space given by (3.55) forms a trajectory in the state space as $n$ increases.

The value $m$ of the state space [68] is chosen so that the dynamic invariants of the system in the state space are preserved. According to Taken's theorem [66] and Packard et al. [68], if the underlying state space of a system has $d$ dimensions, the invariants of a system are preserved by reconstructing the time series with an embedding dimension $m = 2d + 1$. The delay time $\tau$ should be as small as possible to capture the shortest change (e.g., high-frequency component) present in the data. Also $\tau$ should be large enough to generate the maximum possible independence between the components of the vectors in the state space. In practice, these two conditions are usually addressed by selecting $\tau$ as the first minimum of the mutual information between the components of the vectors in the state space or as the first zero of the time-domain autocorrelation of the data [69]. Theoretically, the time span $(m-1)\cdot\tau$ should be almost equal to the period of the maximum power (or dominant) frequency component in the data. For example, a sine wave (or a limit cycle) has $d = 1$, then an $m = 2 \cdot 1 + 1 = 3$ is needed for the embedding and $(m-1)\cdot\tau = 2 \cdot \tau$ should be equal to the period of the sine wave. Such a value for $\tau$ would then correspond to the Nyquist sampling period of the sine wave in the time domain.

The state space analysis [55, 70, 71] of the EEG reveals the presence of ever-changing attractors with nonlinear characteristics. To visualize this point, an epileptic EEG signal $s(t)$ recorded preictally (10 seconds before to 20 seconds into a seizure) from a focal electrode [see Figure 3.25(a)] is embedded in a three-dimensional space. The vectors $\mathbf{s}(t) = (s(t), s(t-\tau), s(t-2\tau)$ are constructed with $\tau - l \cdot Dt = 4 \cdot 5$ ms $= 20$ ms and are illustrated in Figure 3.25(b). The state space portraits of the preictal and the ictal EEG segments are strikingly different. The geometric properties and dynamics of such state space portraits can be quantified using invariants of dynamics, such as the correlation dimension and the Lyapunov exponents, to study their complexity and stability, respectively.

#### 3.2.2.2 Measures of Self-Similarity/Complexity: Correlation Integrals and Dimension

Estimating the dimension $d$ of an attractor from a corresponding time series has attracted considerable attention in the past. It is noteworthy that "strange" attractors have a fractal dimension, which is a measure of their complexity. An estimate of $d$ of an attractor is the correlation dimension $\nu$. The correlation dimension quantifies the self-similarity (complexity) of a geometric object in the state space. Thus, given a scalar time series $s(t)$, state space is reconstructed using the embedding procedure described in Section 3.2.1. Once the data vectors have been constructed, the estimation of the correlation dimension is performed in two steps. First, one has to deter-

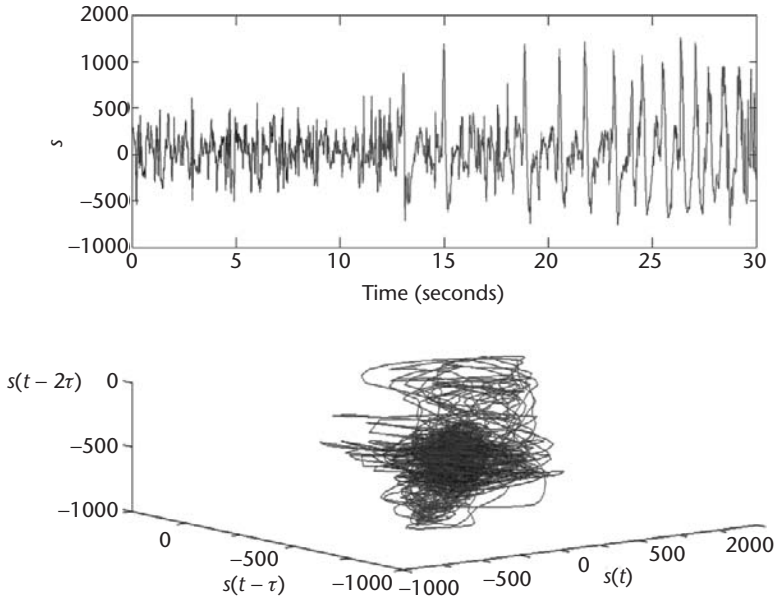



**Figure 3.25** An EEG segment from a focal right temporal lobe cortical electrode, before and after the onset of an epileptic seizure in the time domain and in the state space. (a) A 30-second epoch $s(t)$ of EEG (voltage in microvolts) of which 10 seconds are from prior to the onset of a seizure and 20 seconds from during the seizure. (b) The three-dimensional state space representation of $s(t)$ ($m = 3$, $\tau =$ 20 ms).

mine the correlation integral (sum) $C(m, \varepsilon)$ for a range of $\varepsilon$ (radius in the state space that corresponds to a multidimensional bin size) and for consecutive embedding dimensions $m$. Another way to interpret $C(m, \varepsilon)$ in the state space is in terms of an underlying multidimensional probability distribution. It is the self-similarity of this distribution that $\nu$ and $d$ quantify.

We define the correlation sum for a collection of points $\mathbf{s}_i = \mathbf{s}(i \cdot Dt)$ in the vector space to be the fraction of all possible pairs of points closer than a given distance $\varepsilon$, using a particular norm $\|\cdot\|$ (e.g., the Euclidean or max) to measure this distance. Thus, the basic formula for $C(m, \varepsilon)$ is [64]

$$C(m,\varepsilon) = \frac{2}{N(N-1)} \sum_{i=1}^{N} \sum_{j=i+1}^{N} \Theta\left(\varepsilon - \left\|\mathbf{s}_i - \mathbf{s}_j\right\|\right) \tag{3.56}$$

where $\Theta$ is the Heaviside step function, $\Theta(s) = 0$ if $s = 0$ and $\Theta(s) = 1$ for $s > 0$. The summation counts the pairs of points $(\mathbf{s}_i, \mathbf{s}_j)$ whose distance is smaller than $\varepsilon$. In the limit of an infinite amount of data ($N \rightarrow 8$) and for small $\varepsilon$, we theoretically expect $C$ to scale with $\varepsilon$ exponentially, that is, $C(\varepsilon) \approx \varepsilon^D$ and we can then define $D$ and $\nu$ by

$$D(m) = \lim_{\varepsilon \to 0} \lim_{N \to \infty} \frac{\partial \ln C(m,\varepsilon)}{\partial \ln \varepsilon} \text{ and then } \nu = \lim_{m \to \infty} D(m) \tag{3.57}$$

It is obvious that the limits of (3.57) cannot be satisfied in real data and approximations have to be made. In finite data, $N$ is limited by the size and stationarity of

the data, whereas $\varepsilon$ is limited from below by the finite accuracy of the data, noise, and the inevitable lack of near neighbors in the state space at small length scales. In addition, for large $m$ and for a finite $D$ to exist, we theoretically expect that $D$ would converge to $\nu$ for large values of $m$ (e.g., for $m > 2d + 1$). Also, the previous estimator of correlation dimension is biased toward small values when the pairs included in the correlation sum are statistically dependent simply because of oversampling of the continuous signal in the time domain and/or inclusion of common components in successive state vectors [e.g., $s(t - \tau)$ is a common component in the vectors $\mathbf{s}(t)$ and $\mathbf{s}(t - \tau)$]. Then, it is highly probable that the embedded vectors $\mathbf{s}(t)$ at successive times $t$ are nearby in the state space. In the process of estimation of the correlation dimension, the presence of such temporal correlations may lead to serious underestimation of $\nu$. A solution to this problem is to exclude such pairs of points in (3.56). Thus, the lower limit in the second sum in (3.56) is changed, taking in consideration a correlation time $t_{\min} = n_{\min} \cdot Dt$ (Theiler's correction) [72] as follows:

$$C(m, \varepsilon) = \frac{2}{(N - n_{\min})(N - n_{\min} - 1)} \sum_{i=1}^{N} \sum_{j=i+n_{\min}}^{N} \Theta\left(\varepsilon - \left\|\mathbf{s}_i - \mathbf{s}_j\right\|\right) \tag{3.58}$$

Note that $t_{\min}$ is not necessarily equal to the average correlation time [i.e., the time lag at which the autocorrelation function of $s(t)$ has decayed to $1/e$ of its value at lag zero]. It has rather to do with the time spanned by a state vector's components, that is, with $(m - 1)\tau$.

*Application: Estimation of Correlation Integrals and Dimensions from EEGs*

A reliable estimation of the correlation dimension $\nu$ requires a large number of data points [73, 74]. However, due to the nonstationarity of EEGs, a maximum length $T$ for the EEG segment under analysis (typically on the order of 10 seconds), which also depends on the patient's state and could be derived by measure(s) of nonstationarity, has to be considered in the estimation of $\nu$ [74]. A scaling region of $\ln C(m, \varepsilon)$ versus $\ln\varepsilon$ for the estimation of $D(m)$ is considered true if it occurs for $\varepsilon << \sigma$, where $\sigma$ is the standard deviation either of the one-dimensional data, or the size of the attractor in the $m$-dimensional state space. If the thus estimated $D(m)$ versus $m$ reaches a plateau with increasing $m$, the value of the plateau is a rough estimate of $\nu$.

We show the application of the correlation dimension for the estimation of the complexity of the EEG attractor during an epileptic seizure. The procedure for estimating the correlation dimension of an EEG segment described earlier is applied to a 10-second EEG segment recorded from a focal electrode within a seizure. The TISEAN software package [73] was used to produce the results shown in Figure 3.26.

Figure 3.26(a) shows the $\ln C(m, \varepsilon)$ versus $\ln\varepsilon$ for $m = 2$ up to $m = 20$. The raw EEG data were normalized to ±1 before the estimation of $C$, and therefore $0 < \varepsilon < 1$. Figure 3.26(b) shows the local slopes $D(m, \varepsilon)$ versus $\ln\varepsilon$, estimated in local regions of $\varepsilon$, for $m = 2$ up to $m = 20$ in step 2. It is relatively clear that, as $\varepsilon$ increases from zero, the first plateau of $D(m, \varepsilon)$ with $m$ is observed in the $\varepsilon$ range of $-3.0 = \ln\varepsilon < -2.0$ (i.e., $0.05 = \varepsilon < 0.14$) for $m$ larger than 10. The fact that the plateau is not readily discernible reflects the influence of the limited number of data points and of possible superimposed noise to the data. Nevertheless, the value $\nu$ of the formed plateau is in the

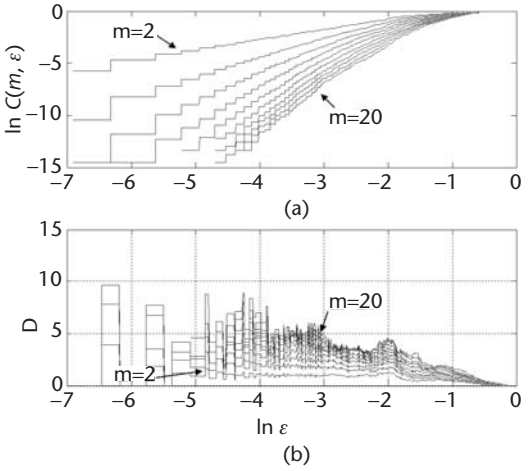



**Figure 3.26** Estimation of the correlation dimension $\nu$ from an ictal EEG segment. (a) ln-ln plots of the correlation integrals $C(\varepsilon)$ versus the hypersphere radius $\varepsilon$ for embedding dimensions $m = 2, 4, 6, \ldots, 20$ obtained from a 20-second (4,096-point) EEG segment inside a seizure of temporal lobe origin and recorded by a focal hippocampal electrode RTD (see Figure 3.1 for the recording montage). (b) Local slope $D$ versus $\ln\varepsilon$ for embedding dimensions $m = 2, 4, 6, \ldots, 20$. A plateau of these curves is observed for $-3.0 = \ln\varepsilon < -2.0$ for large $m$, with a value of $D \approx 3.5$.

neighborhood of 3.5; that is, the dimension of the attractor within the seizure (and therefore its complexity) and the required embedding dimension $m$ ($m > 2 \cdot 3.5 + 1 = 8$) are relatively small. The value of $\nu = 3.5$ for an ictal EEG segment is in good agreement with those reported elsewhere [75–77], and it is much smaller than the one (when it exists) in the nonseizure (interictal) periods (not shown here), thus implying that more complex interictal "attractors" evolve to less complex ones ictally.

#### 3.2.2.3 Measures of Stability: Lyapunov Exponents

In a chaotic attractor, on average, trajectories originating from similar initial conditions (nearby points in the state space) diverge exponentially fast (expansion process), that is, they stay close together only for a short time. If these trajectories belong to an attractor of a finite size, they will have to fold back into it as time evolves (folding process). The result of this expansion and folding process is the attractor's layered structure, which is a characteristic of a strange attractor (a chaotic attractor is always strange, but a strange attractor is not necessarily chaotic). The measures that quantify the chaoticity [61] of an attractor are the Lyapunov exponents. For an attractor to be chaotic, at the very least the maximum Lyapunov exponent $L_{\max}$ should be positive.

The Lyapunov exponents measure the average rate of expansion and folding that occurs along the local eigen-directions within an attractor in state space [70]. A positive $L_{\max}$ means that the rate of expansion is greater than the rate of folding and, therefore, essentially a production rather than destruction of information. If the state space is of $m$ dimensions, we can theoretically measure up to $m$ Lyapunov

exponents. The estimation of the largest Lyapunov exponent $L_{max}$ in a chaotic system has been shown to be the most reliable and reproducible measure [78]. Our algorithm to estimate $L_{max}$ from nonstationary data is described in [79–81]. We have called such an estimate $STL_{max}$ (short-term maximum Lyapunov exponent). For completion purposes, the general guidelines for estimation of $L_{max}$ from stationary data are given next (see also [63]).

First, construction of the state space from a data segment $s(t)$ of duration $T = N \cdot Dt$ is made with the method of delays, that is, the vector $\mathbf{s}(t)$ in an $m$-dimensional state space is constructed as

$$\mathbf{s}(t) = \left[ s(t), s(t-\tau), \ldots, s\left(t-(m-1)\tau\right) \right] \tag{3.59}$$

In the case of the EEG, this method can be used to reconstruct a multidimensional state space of the brain's electrical activity from a single EEG channel at the corresponding brain site. The largest Lyapunov exponent $L_{max}$ is then given by

$$L_{\max} = \frac{1}{N_a \Delta t} \sum_{i=1}^{N_a} \log_2 \frac{\left|\delta \mathbf{s}_{i,j}(\Delta t)\right|}{\left|\delta \mathbf{s}_{i,j}(0)\right|} \tag{3.60}$$

with $\delta \mathbf{s}_{i,j}(0) = \mathbf{s}(t_i)\mathbf{s}(t_j)$ and $\delta \mathbf{s}_{i,j}(\Delta t) = \mathbf{s}(t_i + \Delta t) - \mathbf{s}(t_j + \Delta t)$; $\mathbf{s}(t_i)$ is a point on the fiducial trajectory $\varphi(s(t_0))$; and $t_0$ is the initial time in the fiducial trajectory, that is, usually the time point of the first data in the data segment $s(t)$ of analysis. The vector $s(t_j)$ is properly chosen to be adjacent to $\mathbf{s}(t_i)$ in the state space; $\delta \mathbf{s}_{i,j}(0)$ is the displacement vector at $t_i$, that is a perturbation of the fiducial orbit at $t_i$, and $\delta \mathbf{s}_{i,j}(\Delta t)$ is the evolution of this perturbation after time $\Delta t$; $t_i = t_0 + (i-1) \cdot \Delta t$ and $t_j = t_0 + (j-1) \cdot \Delta t$, where $i \in [1, N_a]$ and $j \in [1, N]$ with $j \neq i$. If the evolution time $\Delta t$ is given in seconds, then $L_{max}$ is given in bits per second. For a better estimation of $L_{max}$, a complete scan of the attractor can be performed by allowing $t_0$ to vary within $[0, \Delta t]$. The term $N_a$ represents the number of local $L_{max}$'s estimated every $\Delta t$, within a duration $T$ data segment. Therefore, if $Dt$ is the sampling period for the time domain data, then $T = (N-1)Dt = N_a \Delta t + (m-1)\tau$.

*Application: Estimation of the Maximum Lyapunov Exponent from EEGs*

The short-term largest Lyapunov exponent $STL_{max}$ is computed by a modified version of the $L_{max}$ procedure above. It is called short-term to differentiate it from the global Lyapunov exponent $L_{max}$ in stationary dynamic systems/signals. For short data segments with transients, as in EEGs from epileptic patients where transients such as epileptic spikes may be present, $STL_{max}$ measures a quantity similar to $L_{max}$, that is, stability and information rate in bits per second, without assuming data stationarity. This is achieved by appropriately modifying the searching procedure for a replacement vector at each point of a fiducial trajectory. For further details about this algorithm, we refer the reader to [79–81].

The brain, being nonstationary, is never in a steady state in the strictly dynamic sense at any location. Arguably, activity at brain sites is constantly moving through steady states, which are functions of the brain's parameter values at a given time.

According to bifurcation theory, when these parameters change slowly over time, or the system is close to a bifurcation, dynamics slow down and conditions of stationarity are better satisfied.

Theoretically, if the state space is of $d$ dimensions, we can estimate up to $d$ Lyapunov exponents. However, as expected, only $d + 1$ of these will be real. The rest are spurious [61]. The estimation of the largest Lyapunov exponent ($L_{max}$) in a chaotic system has been shown to be more reliable and reproducible than the estimation of the remaining exponents [78], especially when $d$ is unknown and changes over time, as in the case of high-dimensional and nonstationary data such as EEGs.

Before we apply the $STL_{max}$ to the epileptic EEG data, we need to determine the dimension of the embedding of an EEG segment in the state space. In the ictal state, temporally ordered and spatially synchronized oscillations in the EEG usually persist for a relatively long period of time (in the range of minutes for seizures of focal origin). Dividing the ictal EEG into short segments ranging from 10.24 to 50 seconds in duration, estimation of $\nu$ from ictal EEGs has given values between 2 and 3. These values stayed relatively constant (invariant) with the shortest duration EEG segments of 10.24 seconds [79, 80]. This implies the existence of a low-dimensional manifold in the ictal state, which we have called an epileptic attractor. Therefore, an embedding dimension $d$ of at least 7 has been used to properly reconstruct this epileptic attractor. Although $d$ for interictal (between seizures) EEGs is expected to be higher than that for ictal states, we have used a constant embedding dimension $d = 7$ to reconstruct all relevant state spaces over the ictal and interictal periods at different brain locations. The strengths in this approach are that: (1) the existence of irrelevant information in dimensions higher than 7 might not have much influence on the estimated dynamic measures, and (2) reconstruction of the state space with a low $d$ suffers less from the short length of moving windows used to handle nonstationary data. A possible drawback is that related information to the transition to seizures in higher dimensions will not be accurately captured.

The $STL_{max}$ algorithm is applied to sequential EEG epochs of 10.24 seconds in duration recorded from electrodes in multiple brain sites. A set of $STL_{max}$ profiles over time (one $STL_{max}$ profile per recording site) is thus created that characterizes the spatiotemporal chaotic signature of the epileptic brain. A typical $STL_{max}$ profile, obtained by analysis of continuous EEGs at a focal site, is shown in Figure 3.27(a). This figure shows the evolution of $STL_{max}$ as the brain progresses from preictal (before a seizure) to ictal (seizure) to postictal (after seizure) states. There is a gradual drop in $STL_{max}$ values over tens of minutes preceding the seizure at this focal site. The seizure is characterized by a sudden drop in $STL_{max}$ values with a subsequent steep rise in $STL_{max}$ that starts soon after the seizure onset, continues to the end of the seizure, and remains high thereafter until the preictal period of the next seizure.

This dynamic behavior of $STL_{max}$ indicates a gradual preictal reduction in chaoticity at the focal site, reaching a minimum within the seizure state, and a postictal rise in chaoticity that corresponds to the brain's recovery toward normal, higher rates of information exchange. What is more consistent across seizures and patients is an observed synchronization of $STL_{max}$ values between electrode sites prior to a seizure. This is shown in Figure 3.27(b). We have called this phenomenon preictal dynamic entrainment (dynamic synchronization), and it has constituted the basis for the development of the first prospective epileptic seizure prediction algo-

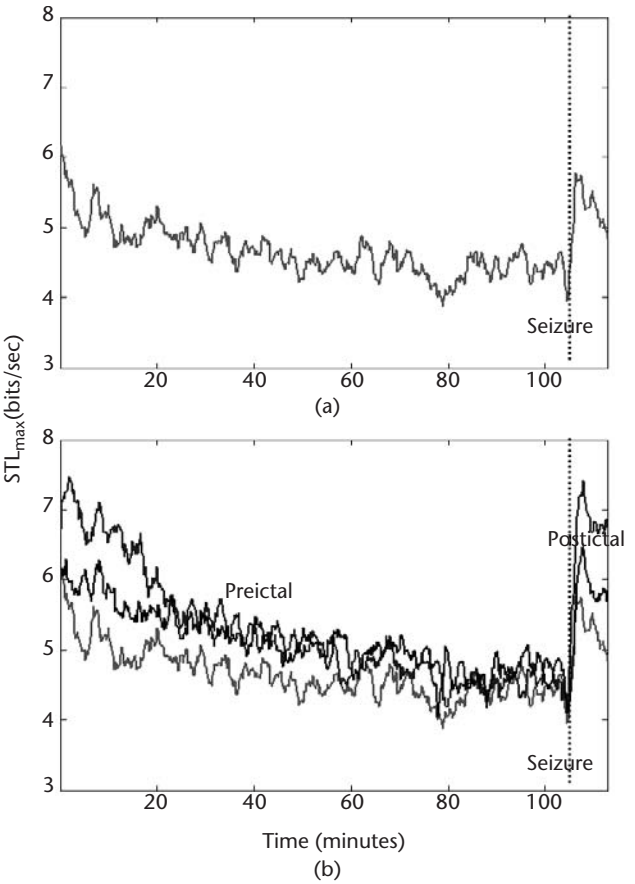



**Figure 3.27** Unsmoothed $STL_{max}$ (bps) over time, estimated per 10.24-second sequential EEG segments before, during, and after an epileptic seizure (a) at one focal site and (b) at critical focal and nonfocal sites. The lowest $STL_{max}$ values occur at seizure's onset. The seizure starts at the vertical black dotted line and lasts for only 2.5 minutes. The trend toward low $STL_{max}$ values is observed long (tens of minutes) before the seizure. Spatial convergence or dynamic entrainment of the $STL_{max}$ profiles starts to appear about 80 minutes before seizure onset. A plateau of low $STL_{max}$ values and entrainment of a critical mass of electrodes start to appear about 20 minutes before seizure onset. Postictal $STL_{max}$ values are higher than the preictal ones, are dynamically disentrained, and they move fast toward their respective interictal values. (Embedding in a state space of $m = 7$, $\tau = 20$ ms.)

rithms [82–85]. This phenomenon has also been observed in simulation models with coupled nonlinear systems, as well as biologically plausible thalamocortical models, where the interpopulation coupling is the parameter that controls the route toward "seizures," and the changes in coupling are effectively captured by the entrainment of the systems' $STL_{max}$ profiles [86–90].

## 3.3 Information Theory-Based Quantitative EEG Analysis

### 3.3.1 Information Theory in Neural Signal Processing

Information theory in communication systems, founded in 1948 by Claude E. Shannon [91], was initially used to quantify the information, that is, the uncertainty, in a

system by the minimal number of bits required to transfer the data. Mathematically, the information quantity of a random event $A$ is the logarithm of its occurrence probability ($P_A$), that is, $\log_2 P_A$. Therefore, the number of bits needed for transferring $N$-symbol data ($A_i$) with probability distribution $\{P_i, i = 1, ..., N\}$ is the averaged information of each symbol:

$$SE = -P_i \log_2 P_i \tag{3.61}$$

A straight conclusion from (3.61) is that $SE$ reaches its global maximum under uniform distribution, that is, $SE_{\max} = \log_2(N)$ when $P_1 = P_2 = ... = P_N$. Therefore, $SE$ measures the extent to which the probability distribution of a random variable diverges from a uniform one, and can be implemented to analyze the variation distribution of physiological signals, such as EEG and electromyogram (EMG).

#### 3.3.1.1 Formality of Entropy Implementation in EEG Signal Processing

Entropy has been used in EEG signal analysis in different formalities, including: (1) approximate entropy ($ApEn$), a descriptor of the changing complexity in embedding space [92, 93]; (2) Kolmogorov entropy ($K_2$), another nonlinear measure capturing the dynamic properties of the system orbiting within the EEG attractor [94]; (3) spectral entropy, evaluating the energy distribution in wavelet subspace [95] or uniformity of spectral components [96]; and (4) amplitude entropy, a direct uncertainty measure of the EEG signals in the time domain [97–99]. In applications, entropy has also been used to analyze spontaneous regular EEG [95, 96], epileptic seizures [100], and EEG from people with Alzheimer's disease [101] and Parkinson's disease [102]. Compared with other nonlinear methods, such as fractal dimension and Lyapunov components, entropy does not require a huge dataset and, more importantly, it can be used to investigate the interdependence across the cerebral cortex [103, 104].

#### 3.3.1.2 Beyond the Formalism of Shannon Entropy

The classic formalism in (3.61) has been shown to be restricted to the domain of validity of Boltzmann-Gibbs statistics (BGS), which describes a system in which the effective microscopic interactions and the microscopic memory are of short range. Such a BGS-based entropy is generally applicable to extensive or additive systems. For two independent subsystems $A$ and $B$, their joint probability distribution is equal to the product of their individual probability, that is,

$$P_{i,j}(A \cup B) = P_i(A)P_j(B) \tag{3.62}$$

where $P_{i,j}(A \cup B)$ is the probability of the combined system $A \cup B$, and $P_i(A)$ and $P_j(B)$ are the probability distribution of systems $A$ and $B$, respectively. Combining (3.62)and (3.61), we can easily conclude additivity in such a combined system:

$$SE(A \cup B) = SE(A) + SE(B) \tag{3.63}$$

In practice, however, the neuronal system consists of multiple subsystems called lobes in neurophysiologic terminology, and works in a way of interaction and memory. For such a neuronal system with long-range correlation, memory, and interactions, a more generalized entropy formalism was proposed by Tsallis [105]:

$$TE = \frac{1 - \sum_{i=1}^{i=N} P_i^q}{q - 1} \tag{3.64}$$

Tsallis entropy ($TE$) degrades to conventional Shannon entropy ($SE$) when the entropic index $q$ converges to 1. Under the nonextensive entropy framework, for two interactive systems $A$ and $B$, the nonextensive entropy of the combined system $A \cup B$ will follow the quasi-additivity:

$$\frac{TE(A \cup B)}{k} = \frac{TE(A)}{k} + \frac{TE(B)}{k} + (1-q)\frac{TE(A)}{k}\frac{TE(B)}{k} \tag{3.65}$$

where $k$ is the Boltzmann constant. When $q \to 1$, (3.65) becomes (3.63). For $q < 1$, $q = 1$, and $q > 1$, we can induce $TE(A \cup B) > TE(A) + TE(B)$, $TE(A \cup B) = TE(A) + TE(B)$, and $TE(A \cup B) < TE(A) + TE(B)$ from (3.65) corresponding to superextensive, extensive, and subextensive systems, respectively.

Although Tsallis entropy has been frequently recommended as the generalized statistical measure in past years [105–108], it is not unique. As the literature shows, we can use other generalized forms of entropy [109]. One of them is the well-known Renyi entropy [110], which is defined as follows:

$$RE = \frac{1}{1-q}\log\left(\sum_{i-1}^{M} P_i^q\right) \tag{3.66}$$

when $q \to 1$, it also recovers to the usual Shannon entropy. This expression of entropy adopts power law–like distribution $x^{-\beta}$. The exponent $\beta$ is expressed as a function $\beta(q)$ of the Renyi parameter $q$ [111]. Renyi entropy of scalp EEG signals has been proven to be sensitive to the rate of recovery of neurological injury following global ischemia [98].

In the remaining part of this section, we introduce the methods of using time-dependent entropy to describe the different rhythmic activities in EEG, and how to use entropy to quantify the nonstationary level in neurological signals.

### 3.3.2 Estimating the Entropy of EEG Signals

EEG signals have been conventionally considered to be random processes, or stochastic signals obeying an autoregressive (and moving averaging) model, also known as AR and ARMA models. Although the parametric methods, such as the AR model, have obtained some success in describing EEG signals, the model selection has always been a critical and time-intensive procedure in these conventional analyses. On the other hand, the amplitude or frequency distribution of EEG signals is strongly physiologically state dependent, for example, in epilepsy seizure and bursting activities following hypoxic-ischemic brain injury. Figure 3.28 shows some typi-

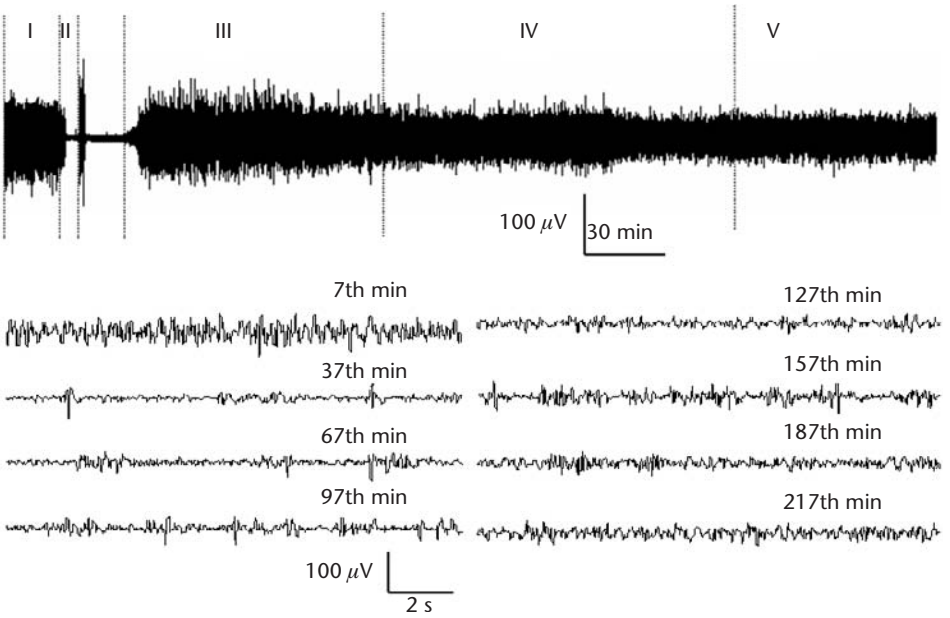



**Figure 3.28** A 4-hour EEG recording in a rat brain injury experiment. Five regions (I–V) correspond to different phases of the experiment. I: baseline (20 minutes); II: asphyxia (5 minutes); III: silent phase after asphyxia (15 minutes); IV: early recovery (90 minutes); and V: late recovery (110 minutes). The high-amplitude signal preceding period III is an artifact due to cardiopulmonary resuscitation manipulations. The lower panel details waveforms at the indicated time, 10 seconds each, from the EEG recording above. (*From:* [97]. © 2003 Biomedical Engineering Society. Reprinted with permission.)

cal EEG waveforms following a hypoxic-ischemic brain injury. Taking the amplitudes in the time domain, we demonstrate how to estimate the entropy from an raw EEG data $s(n)$, where $n = 1, ..., N$, which could be easily extended to frequency- and time-frequency domains. The probability distribution, $\{P_i\}$ in (3.61), (3.64), and (3.66), can be estimated simply by a normalized histogram or more accurate kernel functions.

#### 3.3.2.1 Histogram-Based Probability Distribution

A histogram is the simplest way to obtain the approximate probability distribution. The range of EEG signals is usually equally divided into $M$ interconnected and nonoverlapping intervals, and the probability $\{P_i\}$ of the $i$th bin $(I_i)$ is simply defined as the ratio of the number of samples falling into $I_i$ to the length of the signal $N$:

$$P_i = \frac{N(I_i)}{N}, \text{ for } i = 1, \ldots, M \tag{3.67}$$

This histogram-based method is simple and easy for computer processing. The distribution $\{P_i\}$ is strongly dependent on the number of bins and the partitioning approaches.

#### 3.3.2.2 Kernel Function–Based Probability Density Function

For a short dataset, we recommend the parametric and kernel methods instead of a histogram, which provides an unreliable probability. Because the parametric method is too complicated, we usually use kernel function evolution for accurate probability density function (PDF) estimation. For EEG segment $\{s(k),\ k = 1, ..., N\}$, the PDF is the combination of the kernel function $K(u)$:

$$\hat{p}(x) = \frac{1}{nh}\sum_{i=1}^{N} K\left(\frac{x_i - x}{h}\right) \tag{3.68}$$

where $h$ is the scaling factor of the kernel function. Commonly used kernel function shapes can be rectangular, triangular, Gaussian, or sinusoidal. The difference between histogram and kernel methods is that a histogram provides a probability distribution ($\{P_i\}$) for a discrete variable ($I_i$), whereas the kernel function method approximates PDF ($p(x)$) for a continuous random variable. The entropy calculated from a PDF is usually called differential entropy. The differential Shannon entropy can be written as

$$se = -\int_R p(x)\log_2\left(p(x)\right)dx \tag{3.69}$$

Accordingly, the nonextensive formalism is

$$te = -\int_r p(x)\frac{\left[p(x)\right]^{q=1}}{q-1}dx \tag{3.70}$$

The difference between $SE$ and $se$ is a constant [22].

### 3.3.3 Time-Dependent Entropy Analysis of EEG Signals

Entropy itself represents the average uncertainty in signals, which is not sensitive to transient irregular changes, like the bursting or spiky activities in EEG signals. To describe such localized activities, we introduce time-dependent entropy, in which the time-varying signal is analyzed with a short sliding time window to capture the transient events. For an $N$-sample EEG signal $\{s(k),\ k = 1, ..., N\}$, the $w$-sample sliding window $W(m, w, \Delta)$ is defined as follows:

$$W(m, w, \Delta) = \left\{s(k), k = 1 + m\Delta, \ldots, w + m\Delta\right\} \tag{3.71}$$

where $\Delta$ is the sliding lag, usually satisfying $\Delta \le \omega$ for not missing a sample. The total number of the sliding windows is approximately $[(N - w)/\Delta]$ where $[x]$ denotes the integer part of the variable $x$.

Within each sliding window, the amplitude probability distributions are approximated with the normalized $M$-bin histogram. The amplitude range $D$ within the sliding window $W(m, w, \Delta)$ is equally partitioned into $M$ bins $\{I_i, i = 1, ..., M\}$:

$$\cup I_i = D \text{ and } \cap I_i = \phi \tag{3.72}$$

The amplitude probability distributions $\{P^m(I_i)\}$ within $W(m, w, \Delta)$ then are the ratios of the number of samples falling into bins $\{I_i\}$ to the window size ($w$). Accordingly, the Shannon entropy ($SE(m)$) corresponding to the window $W(m, w, \Delta)$ will be

$$SE(m) = -\sum_{i=1}^{M} P^m(I_i)\log_2\left(P^m(I_i)\right) \tag{3.73}$$

By sliding the window $w$, we eventually obtain the time-dependent entropy (TDE) of the whole signal. Figure 3.29 demonstrates the general procedure for calculating time-dependent entropy. One of the advantages of TDE is that it can detect the transient changes in the signals, particularly the spiky components, such as the seizures in epilepsy or the bursting activities in the EEG signals during the early recovery stage following hypoxic-ischemic brain injury. When such a seizure-like signal enters the sliding window, the probability distribution of the signal amplitudes within that window will change and become sharper, resulting in more diversion from the uniform distribution. Therefore, a short transient activity causes a lower value for the TDE. We demonstrate such a spike-sensitive property of TDE with the synthesized signal shown in Figure 3.30. Figure 3.30(a) is the simulated signal consisting of a real EEG signal recorded from a normal anesthetized rat and three spiky components. The amplitudes of the spikes have been deliberately rescaled such that one of them was even unnoticeable in the compressed waveforms. By using a 128-sample sliding window ($w = 128$, $\Delta = 1$), Figures 3.30(b, c) show that TDE successfully detected the three transient events.

The choices of the parameters, such as windows size ($w$), window lag ($\Delta$), partitioning of the probability ($I_i$ and $P_i$), and entropic index $q$, directly influence the performance of TDE. Nevertheless, parameter selection should always consider the rhythmic properties in the signals.

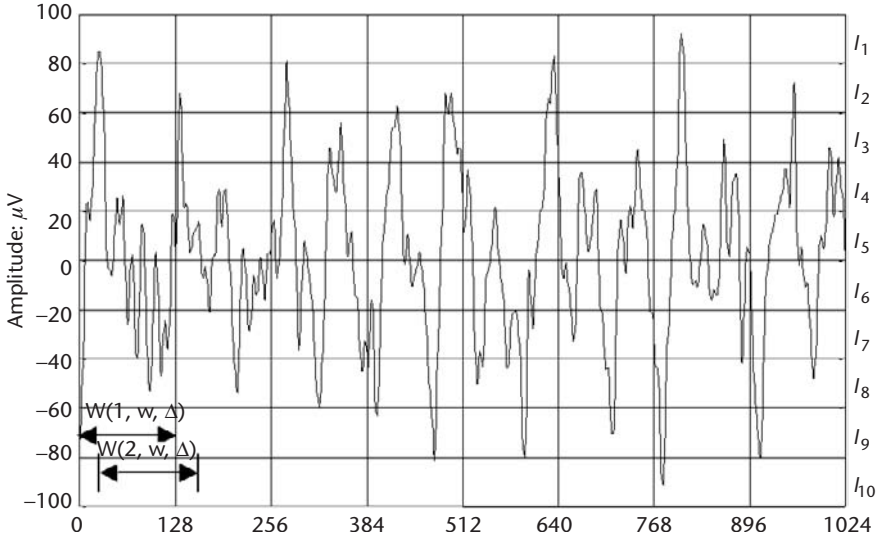



**Figure 3.29** Time-dependent entropy estimation paradigm. The 1,024-point signal is partitioned into 10 disjoint amplitude intervals. The window size is $w = 128$ and it slides every $\Delta = 32$ points. (*From:* [97]. © 2003 Biomedical Engineering Society. Reprinted with permission.)

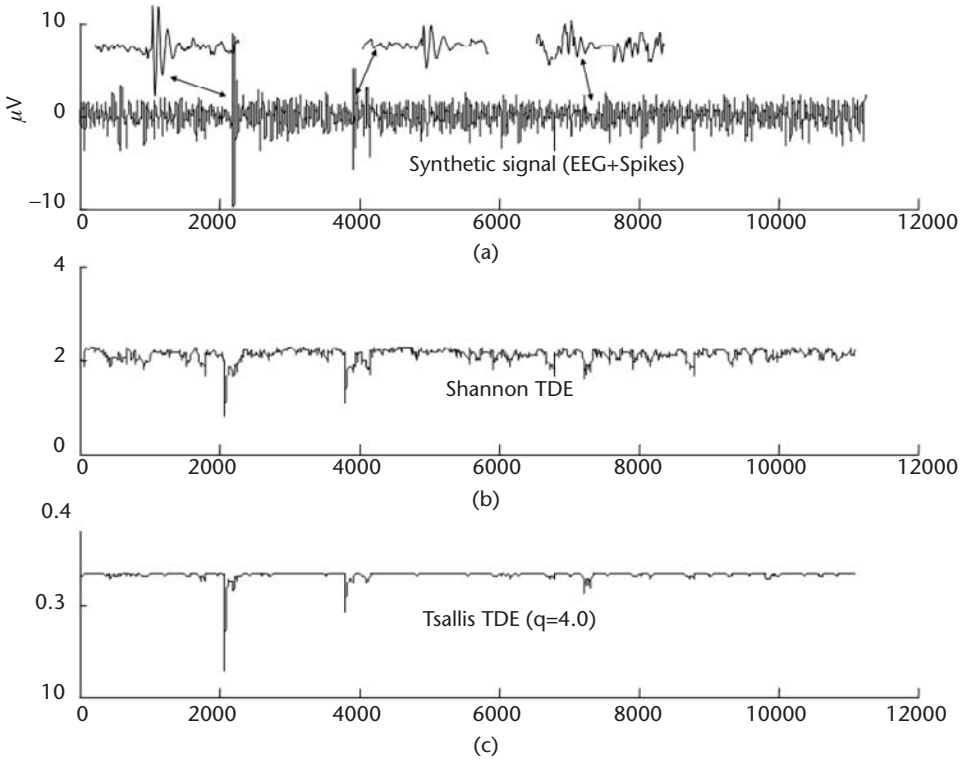



**Figure 3.30** Sensitivity of time-dependent entropy in describing the transient burst activity: (a) synthetic signal (baseline EEG mixed with three bursts of different amplitudes), (b) time-dependent Shannon entropy, and (c) time-dependent Tsallis entropy ($q$ = 4.0). The parameters of the sliding window are $w$ = 128 samples and $\Delta$ = 1 sample.

#### 3.3.3.1 Window Size ($w$)

When studying the short spiky component, for a fixed window lag ($\Delta$), the larger the window size ($w$), the more windows that will include the spike; that is, window size ($w$) determines the temporal resolution of TDE. A smaller window size results in a better temporal localization of the spiky signals. Figure 3.31 illustrates the TDE analysis with different window sizes ($w$ = 64, 128, and 256) for a typical EEG segment following hypoxic-ischemic brain injury, punctuated with three spikes. The TDE results demonstrate the detection of the spikes, but the smaller window size yields better temporal resolution.

Even though a smaller window size provides better temporal localization for spiky signals as shown in Figure 3.31, short data will result in an unreliable PDF, which leads to a bias of entropy estimation and unavoidable errors. By far, however, there is no theoretical conclusion about the selection of window size. In EEG studies, we empirically used a 0.5-second window. Figure 3.32 illustrates the Shannon TDE analysis of typical spontaneous EEG segments ($N$ = 1,024 samples) for window sizes from 64 to 1,024 samples. The figure clearly shows that when the window size is more than 128 samples, the TDE value reaches a stable value.

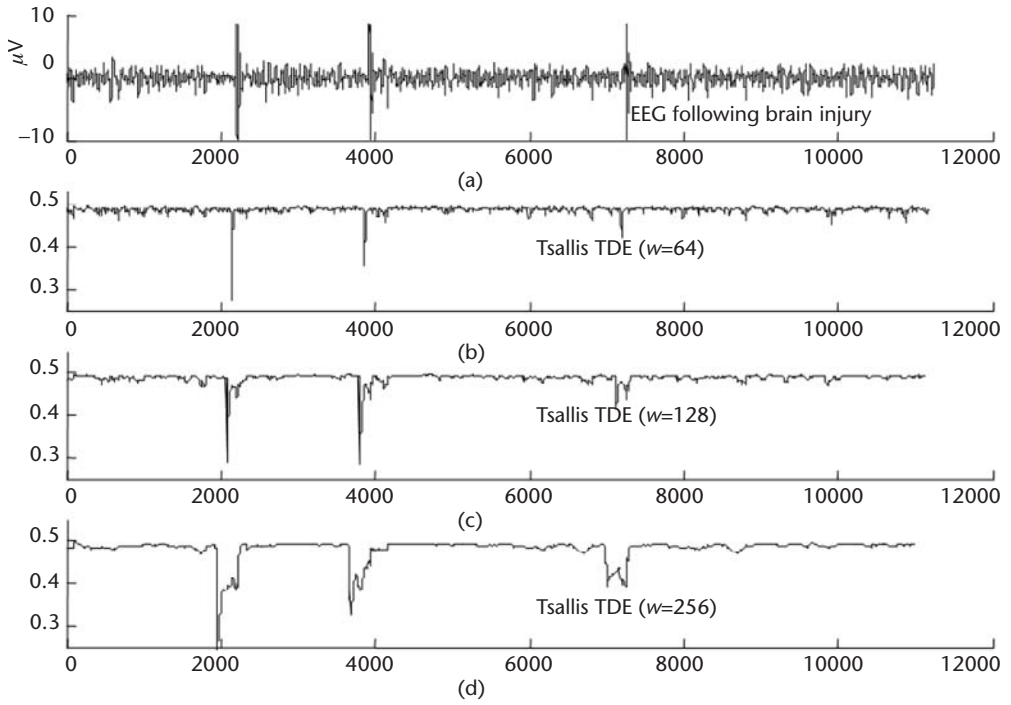



**Figure 3.31** The role of window size in TDE: (a) 40-second EEG segment selected from the recovery of brain asphyxia, which includes three typical spikes; and (b–d) TDE plots for different window size ($w$ = 64, 128, and 256 samples). The sliding step is set to one sample ($\Delta = 1$). The nonextensive parameter $q = 3.0$. Partition number $M = 10$.

#### 3.3.3.2 Window Lag

Because TDE is usually implemented with overlapping sliding windows, the window lag $\Delta$ defines the minimal time interval between two TDE values. Therefore, $\Delta$ is actually the downsampling factor in TDE, where usually $\Delta = 1$ by default. Figure 3.33 illustrates the influence of window lag on the TDE for the same EEG shown in Figure 3.32. Comparing Figure 3.33(b–d), we see that Figure 3.33(c, d) actually selected the TDE values in Figure 3.33(b) every other 64 or 128 samples, respectively.

#### 3.3.3.3 Partitioning

One of the most important steps in TDE analysis is partitioning the signals to get the probability distribution $\{P_i\}$, particularly in histogram-based PDF estimation. The three issues discussed next should be considered in partitioning.

*Range of the Partitioning*

To obtain the probability distribution $\{P_i\}$, the EEG amplitudes should be partitioned into a number ($M$) of bins. By default, some toolboxes, such as MATLAB, create the histogram binning according to the range of the EEG, that is, the maximum and minimum, of the signal. Obviously, such a partitioning is easily affected

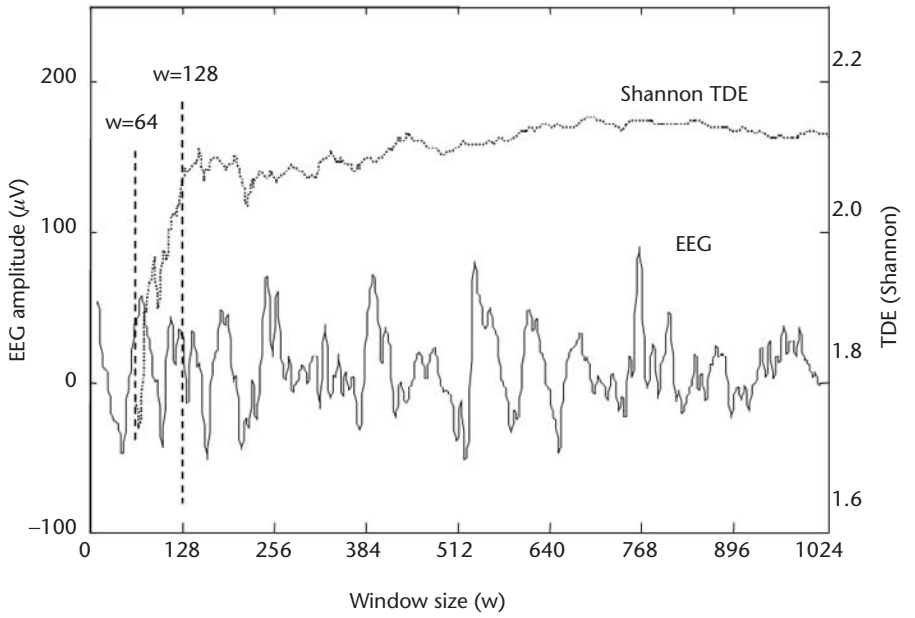



**Figure 3.32** Effect of window size on time-dependent entropy. Four seconds of a typical baseline EEG were chosen to calculate the time-dependent entropies with windows of different sizes. The plot illustrates that the entropy at $w = 128$ is very close to the stable value. (*From:* [97]. © 2003 Biomedical Engineering Society. Reprinted with permission.)

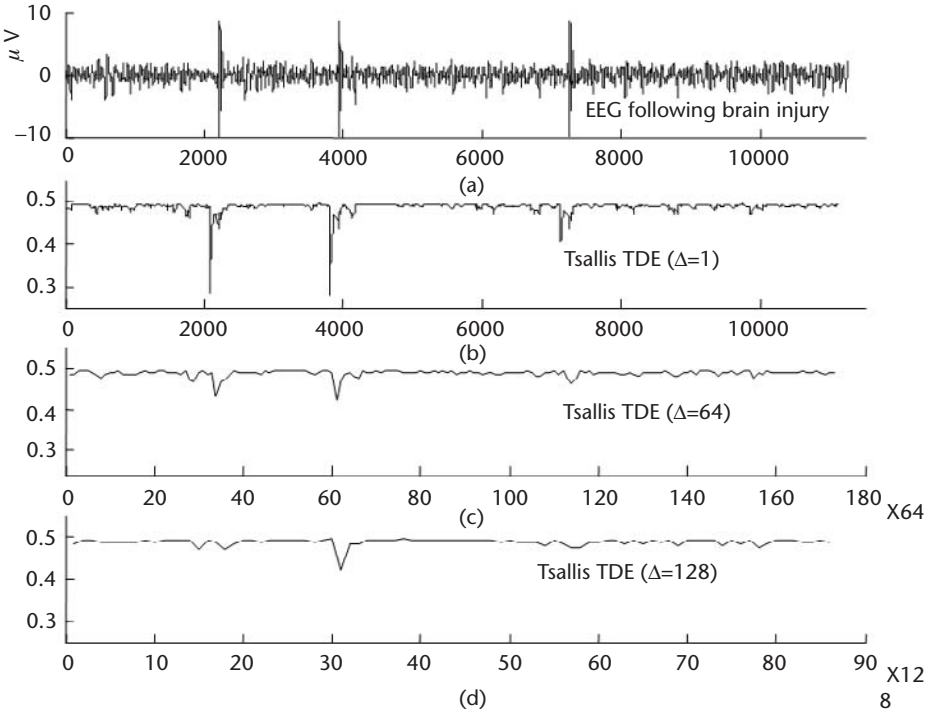



**Figure 3.33** The role of sliding step Δ in TDE: (a) 40-second EEG segment selected from the recovery of brain asphyxia, which includes three typical spikes in the recovery phase; and (b–d) TDE plots for different sliding steps (Δ = 1, 64, and 128). The size of sliding window is fixed at $w = 128$. The nonextensive parameter $q = 3.0$. Partition number $M = 10$.

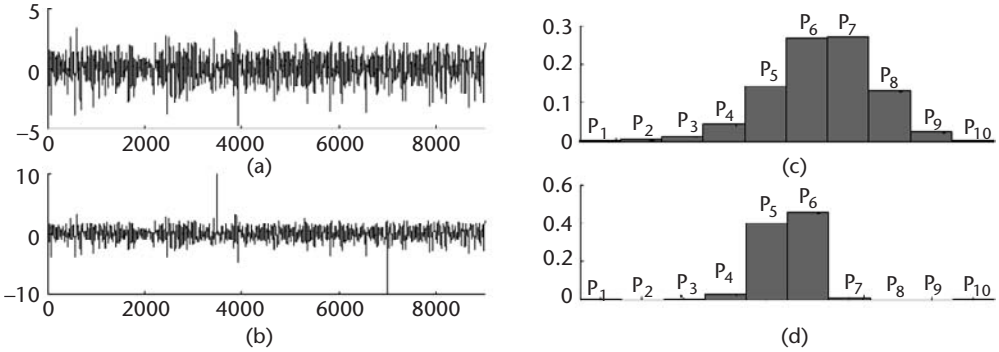



**Figure 3.34** Influences of artifacts in histogram-based probability distribution estimation. (a) Normalized baseline EEG signal (30 seconds); (b) two high-amplitude artifacts mixed in (a); and (c, d) Corresponding histogram-based probability distributions of (a, b) by the MATLAB toolbox. The Shannon entropy values estimated from the probability distributions in (c, d) are 1.6451 and 0.8788, respectively.

by high amplitude and transient noise. Figure 3.34(c, d) are the normalized histogram (i.e., $P_i$, $M = 10$) for the EEG signals in Figure 3.34(a, b), respectively. The signal in Figure 3.34(b) is created from Figure 3.34(a) by introducing two noise artifacts around the 3,500th and 7,000th samples. However, the MATLAB histogram function, *hist*(*x*), generates totally different histograms as shown in Figure 3.34(c, d) that have distinctly different entropy values.

To avoid the spurious range of the EEG signal due to the noise, we recommended a more reliable partitioning range by the standard deviation (std) and mean value ($m$) of the signal so that the histogram or the probability of the signal will be limited to the range of [$m$ – 3 std, $m$ + std] instead of its extremities.

*Partitioning Method*

After the partitioning range is determined, two partitioning methods can be used: (1) fixed partitioning and (2) adaptive partitioning. Fixed partitioning will apply the same partitioning range, usually of the baseline EEG, to all sliding windows of the EEG signals, regardless of the possible changes of the std and $m$ between the sliding windows, whereas the std and $m$ for the adaptive partitioning will be recalculated from the EEG data within each sliding window. Figure 3.35 shows the two partitioning methods for data with 1,000 samples. For the same data, fixed partitioning and adaptive partitioning resulted in different TDEs, as shown in Figure 3.35(c).

Comparing the TDE results in Figure 3.35(c), we can argue that fixed partitioning is useful in detecting changes in long-term trends, whereas adaptive partitioning will focus on the transient changes in amplitude. Both partitioning methods are useful in EEG analysis. For example, EEG signals following hypoxic-ischemic brain injury present evident rhythmics, that is, spontaneous slow waves and spiky bursting EEG in the early recovery phase, both of which are related to the outcome of the neurological injury. Therefore, fixed partitioning and adaptive partitioning can be used to describe changes of different rhythmic activities [113].

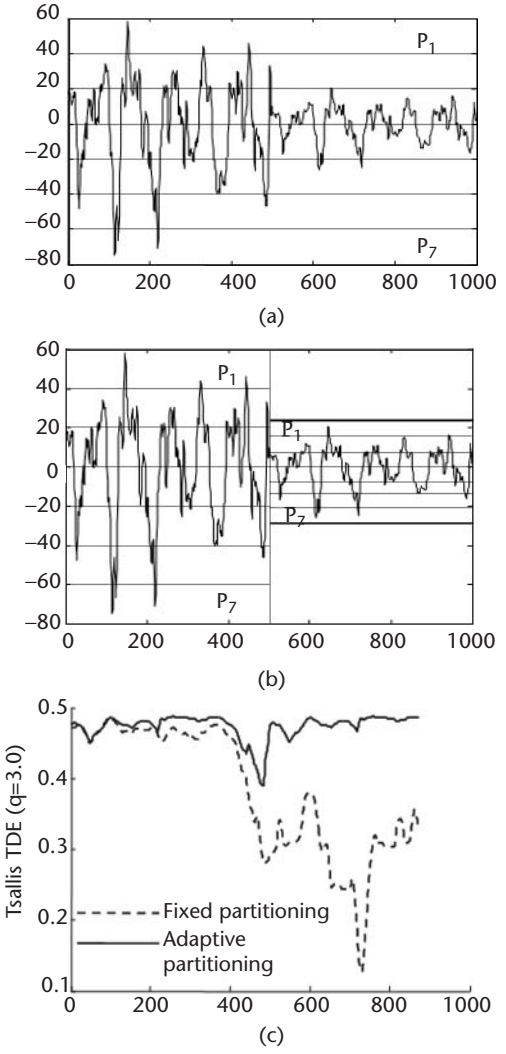



**Figure 3.35** Two approaches to partitioning: A 4-second baseline EEG was scaled to 0.35 original amplitudes in its second half so that an evident amplitude change is clearly shown. Two approaches to partitioning are applied: (a) fixed partitioning for all sliding windows ($M = 7$ in this case), and (b) adaptive partitioning ($M = 7$) dependent on the amplitude distribution within each sliding window.

*Number of Partitions*

The partitions, or bins, correspond to the microstates in (3.61), (3.64), and (3.66). To obtain a reliable probability distribution $\{P_i\}$ for smaller windows (e.g., $w = 128$), we recommend a partitioning number of less than 10. When analyzing long-term activity with large sliding windows (e.g., $w > 2{,}048$), partitions could be up to $M = 30$.

### 3.3.3.4 Entropic Index

Before implementing the nonextensive entropy of (3.64), the entropic index $q$ has to be determined. The variable $q$ represents the degree of nonextensivity of the system,

which is determined by the statistical properties. Discussions in the literature cover the estimation of $q$ [114]; however, it is still not clear how to extract the value of $q$ from recorded raw data such as that found in an EEG. Capurro and colleagues [115] found that $q$ was able to enhance the spiky components in the EEG; that is, a larger $q$ will result in a better signal (spike) to noise (background slow waves) ratio. For the same EEG signal in Figure 3.31(a), Figures 3.36(b–d) show TDE changes under different entropic indexes ($q = 1.5$, 3.0, and 5.0). Regardless of the scale of the TDE, we can still find the change of comparison between the "spike" and background "slow waves." Therefore, by tuning the value of $q$, we are able to make the TDE focus on "slow waves" (smaller $q$) or "spiky components" (larger $q$). Empirically, we recommend a medium value of $q = 3.0$ in the study of EEG signals following hypoxic-ischemic brain injury when both slow wave and spiky activities are present; whereas for the spontaneous EEG signal, smaller entropic index (e.g., $q = 1.5$) or Shannon entropy is suggested.

#### 3.3.3.5 Quantitative Analysis of the Spike Detection Performance of Tsallis Entropy

To quantify the performance of Tsallis entropy in "spike detection," we introduce a measure called spike gain improvement (SGI):

$$\mathrm{SGI} = \frac{M_{\mathrm{sig}} - P_v}{S_{\mathrm{sig}}} \tag{3.74}$$

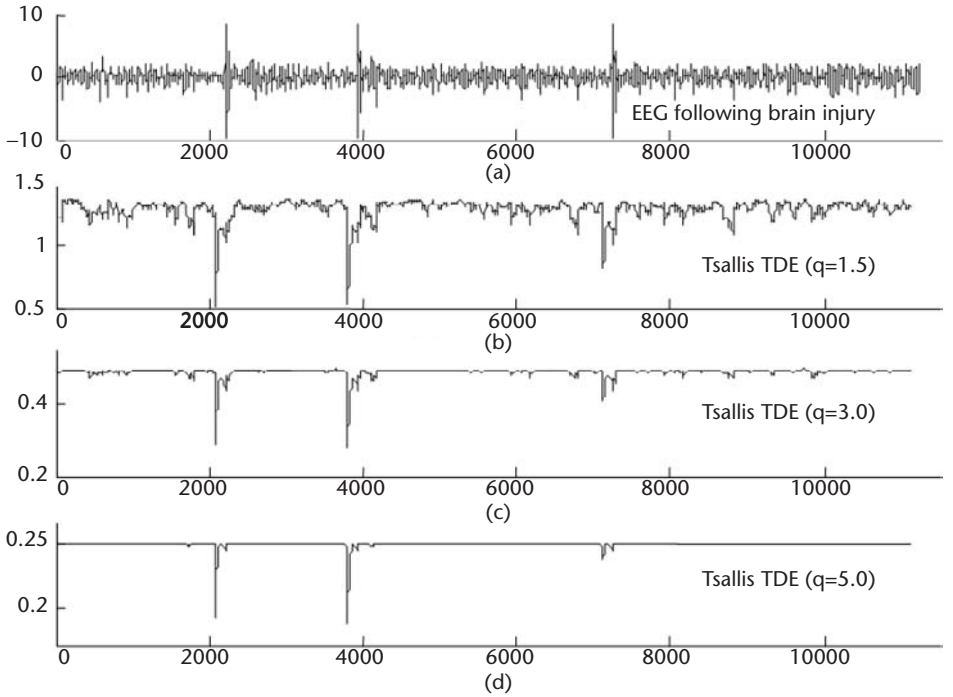



**Figure 3.36** The role of nonextensive parameter $q$ in TDE: (a) 40-second EEG segment selected from the recovery of brain asphyxia, which includes three typical bursts in recovery phase; and (b–d) TDE plots for different nonextensive parameter ($q = 1.5$, 3.0, and 5.0). The size of the sliding window is fixed at $w = 128$. The sliding step is one sample ($\Delta = 1$). Partition number $M = 10$.

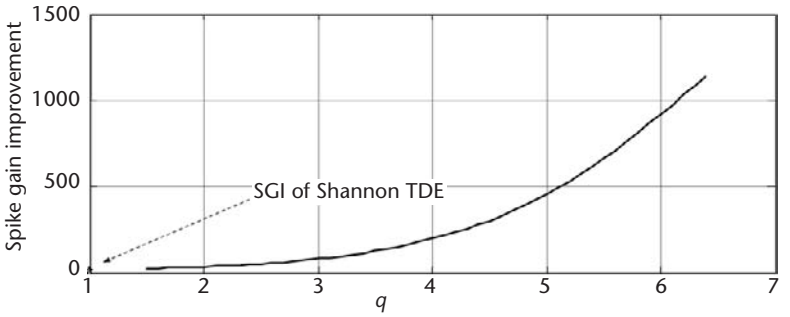



**Figure 3.37** Spike gain improvement by Shannon entropy and Tsallis entropy with different $q$.

where $M_{sig}$ and $S_{sig}$ are corresponding to the mean and standard deviation of the background signal (sig), respectively; and $P_v$ represents the amplitude of the transient spiky components. SGI indicates the significance level of the "spike" component over the background "slow waves." By applying the SGI to both raw EEGs and TDEs in Figure 3.36 under different entropic indexes $q$, we are able to obtain the influence of $q$ on the SGI. Figure 3.37 clearly shows the monotonic change of SGI with the increase of entropic index $q$.

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CHAPTER 4

# Bivariable Analysis of EEG Signals

Rodrigo Quian Quiroga

The chapters thus far have described quantitative tools that can be used to extract information from single EEG channels. In this chapter we describe measures of synchronization between different EEG recordings sites. The concept of synchronization goes back to the observation of the interaction between two pendulum clocks by the Dutch physicist Christiaan Huygens in the seventeenth century. Since the times of Huygens, the phenomenon of synchronization has been largely studied, especially for the case of oscillatory systems [1].

Before getting into technical details of how to measure synchronization, we first consider why it is important to measure synchronization between EEG channels. There are several reasons. First, synchronization measures can let us assess the level of functional connectivity between two areas. It should be stressed that functional connectivity is not necessarily the same as anatomical connectivity, since anatomical connections between two areas may be active only in some particular situations—and the general interest in neuroscience is to find out which situations lead to these connectivity patterns. Second, synchronization may have clinical relevance for the identification of different brain states or pathological activities. In particular, it is well established that epilepsy involves an abnormal synchronization of brain areas [2]. Third, related to the issue of functional connectivity, synchronization measures may show communication between different brain areas. This may be important to establish how information is transmitted across the brain or to find out how neurons in different areas interact to give rise to full percepts and behavior. In particular, it has been argued that perception involves massive parallel processing of distant brain areas, and the binding of different features into a single percept is achieved through the interaction of these areas [3, 4].

Even if outside the scope of this book, it is worth mentioning that perhaps the most interesting use of synchronization measures in neuroscience is to study how neurons encode information. There are basically two views. On the one hand, neurons may transmit information through precise synchronous firing; on the other hand, the only relevant information of the neuronal firing may be the average firing rate. Note that rather than having two extreme opposite views, one can also consider coding schemes in between these two, because the firing rate coding is more similar to a temporal coding when small time windows are used [5].

As beautifully described by the late Francisco Varela [6], synchronization in the brain can occur at different scales. For example, the coordinated firing of a large population of neurons can elicit spike discharges like the ones seen in Figure 4.1(b, c). The sole presence of spikes in each of these signals—or oscillatory activity as in the case of the signal shown in Figure 4.1(a)—is evidence for correlated activity at a smaller scale: the synchronous firing of single neurons.

The recordings in Figure 4.1 are from two intracranial electrodes in the right and left frontal lobes of male adult WAG/Rij rats, a genetic model for human absence epilepsy [7]. Signals were referenced to an electrode placed at the cerebellum, they were then bandpass filtered between 1 and 100 Hz and digitized at 200 Hz. The length of each dataset is 5 seconds long, which corresponds to 1,000 data points. This was the largest length in which the signals containing spikes could be visually judged as stationary.

As we mentioned, spikes are a landmark of correlated activity and the question arises of whether these spikes are also correlated across both hemispheres. The first guess is to assume that bilateral spikes may be a sign of generalized synchronization. It was actually this observation done by a colleague that triggered a series of papers by the author of this chapter showing how misleading it could be to establish synchronization patterns without proper quantitative measures [8]. For example, if we are asked to rank the synchronization level of the three signals of Figure 4.1, it seems

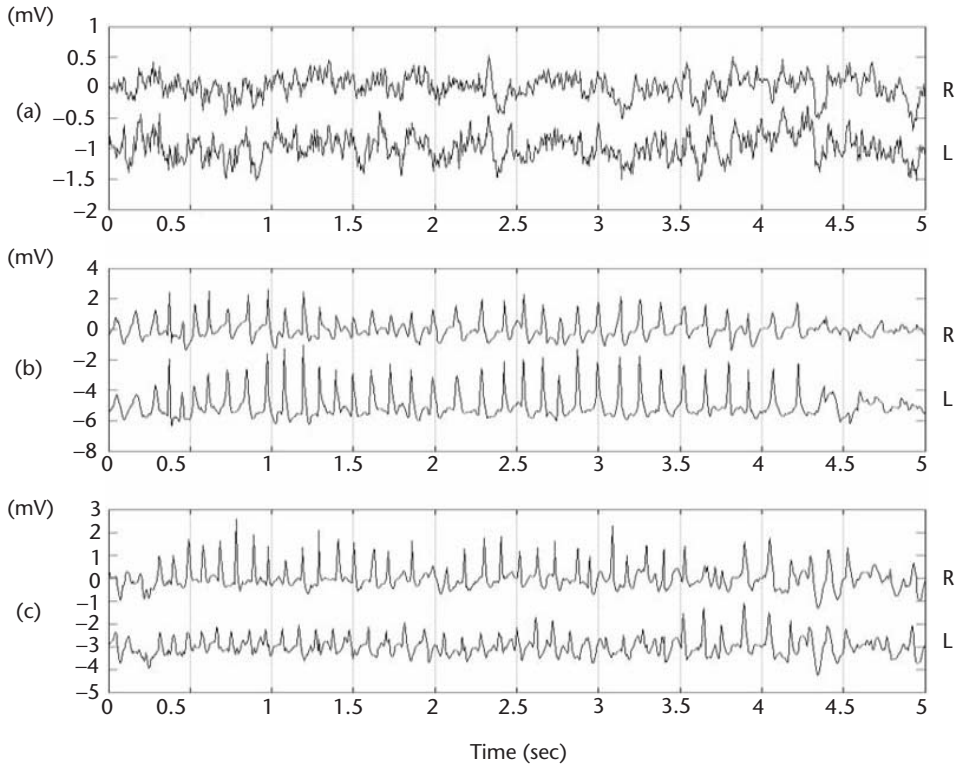



**Figure 4.1** Three exemplary datasets of left and right cortical intracranial recordings in rats. (a) Normal looking EEG activity and (b, c) signals with bilateral spikes, a landmark of epileptic activity. Can you tell by visual inspection which of the examples has the largest and which one has the lowest synchronization across the left and right channels?

that the examples in Figure 4.1(b, c) should have the highest values, followed by the example of Figure 4.1(a). Wrong!

A closer look at Figure 4.1(c) shows that the spikes in both channels have a variable time lag. Just picking up the times of the maximum of the spikes in the left and right channels and calculating the lag between them, we determined that for Figure 4.1(b) the lag was very small and stable, between –5 and +5 ms—of the order of the sampling rate of these signals—and the standard deviation was of 4.7 ms [8]. In contrast, for Figure 4.1(c) the lag was much more variable and covered a range between –20 and 50 ms, with a standard deviation of 14.9. This clearly shows that in example B the simultaneous appearance of spikes is due to a generalized synchronization across hemispheres, whereas in Figure 4.1(c) the bilateral spikes are not synchronized and they reflect local independent generators for each hemisphere.

Interestingly, the signal of Figure 4.1(a) looks very noisy, but a closer look at both channels shows a strong covariance of these seemingly random fluctuations. Indeed, in a comprehensive study using several linear and nonlinear measures of synchronization, it was shown that the synchronization values ranked as follows: $\text{Sync}_B > \text{Sync}_A > \text{Sync}_C$. This stresses the need for optimal measures to establish correlation patterns.

Throughout this chapter, we will use these three examples to illustrate the use of some of the correlation measures to be described. These examples can be downloaded from http://www.le.ac.uk/neuroengineering.

## 4.1 Cross-Correlation Function

The cross-correlation function is perhaps the most used measure of interdependence between signals in neuroscience. It has been, and continues to be, particularly popular for the analysis of similarities between spike trains of different neurons.

Let us suppose we have two simultaneously measured discrete time series $x_n$ and $y_n$, $n = 1, \ldots, N$. The cross-correlation function is defined as

$$c_{xy}(\tau) = \frac{1}{N-\tau} \sum_{i=1}^{N-\tau} \left( \frac{x_i - \bar{x}}{\sigma_x} \right) \left( \frac{y_{i+\tau} - \bar{y}}{\sigma_y} \right) \tag{4.1}$$

where $\bar{x}$ and $\sigma_x$ denote mean and variance and $\tau$ is the time lag. The cross-correlation function is basically the inner product between two normalized signals (that is, for each signal we subtract the mean and divide by the standard deviation) and it gives a measure of the linear synchronization between them as a function of the time lag $\tau$. Its value ranges from –1, in the case of complete inverse correlation (that is, one of the signals is an exact copy of the other with opposite sign), to +1 for complete direct correlation. If the signals are not correlated, then the cross-correlation values will be around zero. Note, however, that noncorrelated signals will not give a value strictly equal to zero and the significance of nonzero cross-correlation values should be statistically validated, for example, using surrogate tests [9]. This basically implies generating signals with the same autocorrelation of the original ones but independent from each other. A relatively simple way of doing this is to shift one

of the signals with respect to the other and assume that they will not be correlated for large enough shifts [8].

Note that formally only the zero lag cross correlation can be considered to be a symmetric descriptor. Indeed, the time delay in the definition of (4.1) introduces an asymmetry that could, in principle, establish whether one of the signals leads or lags the other in time. It should be mentioned, however, that a time delay between two signals does not necessarily prove a certain driver-response causal relationship between them. In fact, time delays could be caused by a third signal driving both with a different delay or by internal delay loops of one of the signals [10].

Figure 4.2 shows the cross-correlation values for the three examples of Figure 4.1 as a function of the time delay $\tau$. To visualize cross-correlation values with large time delays, we used here a slight variant of (4.1) by introducing periodic boundary conditions. The zero lag cross-correlation values are shown in Table 4.1. Here we see that the tendency is in agreement with what we expect from the arguments of the previous section; that is, $\text{Sync}_B > \text{Sync}_A > \text{Sync}_C$. However, the difference between examples A and B is relatively small. In principle, one expects that for long enough lags between the two signals the cross-correlation values should be close to zero. However, fluctuations for large delays are still quite large.

Taking these fluctuations as an estimation of the error of the cross-correlation values, one can infer that cross correlation cannot distinguish between the synchronization levels of examples A and B. This is mainly due to the fact that cross correlation is a linear measure and can poorly capture correlations between nonlinear signals, as is the case for examples B and C with the presence of spikes. More advanced nonlinear measures that are based on reconstruction of the signals in a phase space could indeed clearly distinguish between these two cases [8].

## 4.2 Coherence Estimation

The coherence function gives an estimation of the linear correlation between two signals as a function of the frequency. The main advantage over the cross-correlation function described in the previous section is that coherence is sensitive to interdependences that can be present in a limited frequency range. This is particularly interesting in neuroscience to establish how coherence oscillations may interact in different areas.

Let us first define the sample cross spectrum as the Fourier transform of the cross-correlation function, or by using the Fourier convolution theorem, as

$$C_{xy}(\omega) = (Fx)(\omega)\,(Fy)^{*}(\omega) \tag{4.2}$$

where $(Fx)$ is the Fourier transform of $x$, $\omega$ are the discrete frequencies ($-N/2 < \omega < N/2$), and the asterisk indicates complex conjugation. The cross spectrum can be estimated, for example, using the Welch method [11]. For this, the data is divided into $M$ epochs of equal length, and the spectrum of the signal is estimated as the average spectrum of these $M$ segments. The estimated cross spectrum $\langle C_{xy}(\omega) \rangle$ is a complex number, whose normalized amplitude

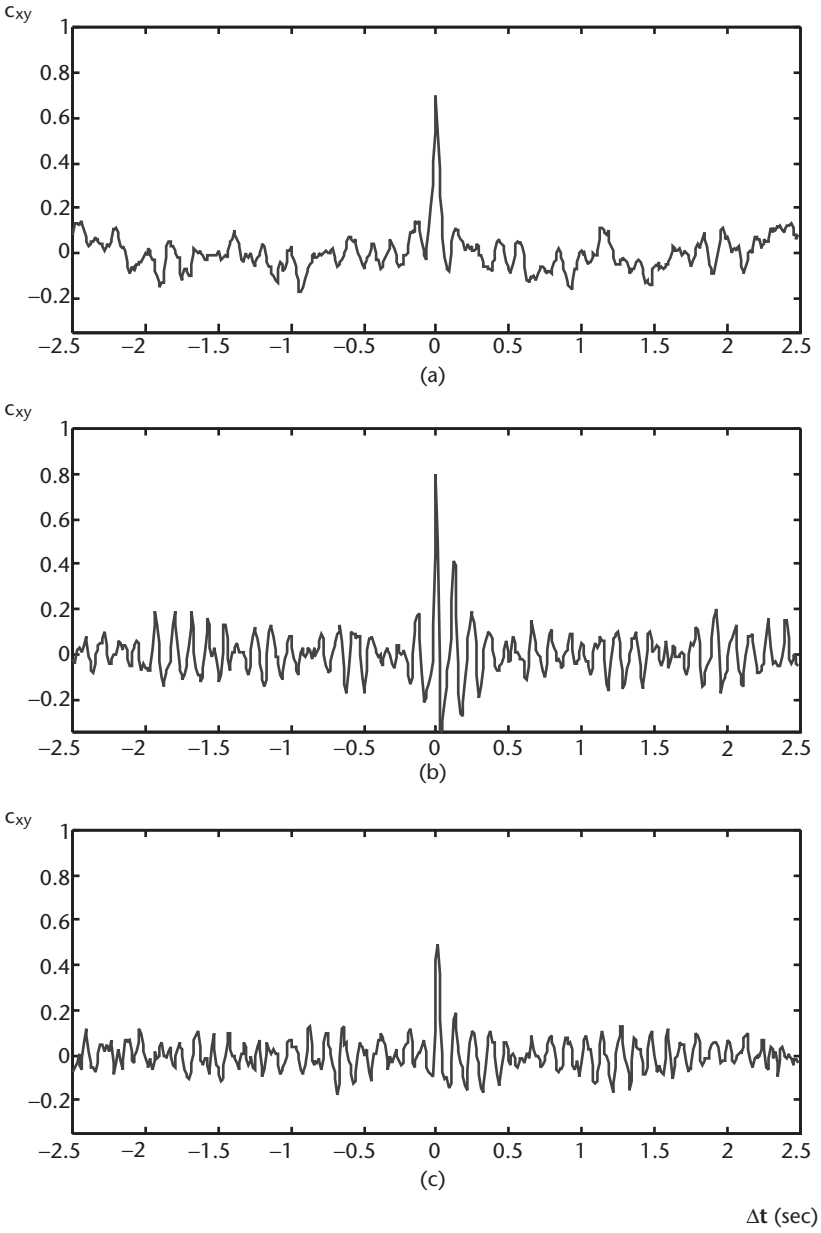



**Figure 4.2** (a–c) Cross-correlation values for the three signals of Figure 4.1 as a function of the time delay $\tau$ between both signals.

$$\Gamma_{xy}(\omega) = \frac{\left|\langle C_{xy}(\omega)\rangle\right|}{\sqrt{\langle C_{xx}(\omega)\rangle}\sqrt{\langle C_{xx}(\omega)\rangle}} \tag{4.3}$$

is named the coherence function. As mentioned earlier, this measure is particularly useful when synchronization is limited to some particular EEG frequency band (for a review, see [12]). Note that without the segmentation of the data introduced to

**Table 4.1** Cross-Correlation, Coherence, and Phase Synchronization Values for the Three Examples of Figure 4.1

| *Example* | $c_{xy}$ | $\Gamma_{xy}$ | $\gamma$ |
|---|---|---|---|
| A | 0.70 | 0.88 | 0.59 |
| B | 0.79 | 0.86 | 0.71 |
| C | 0.42 | 0.40 | 0.48 |

estimate each auto spectrum and cross spectrum, the coherence function of (4.3) gives always a trivial value of 1.

Figure 4.3 shows the power spectra and coherence values for the three examples of Figure 4.1. For the spectral estimates we used half-overlapping segments of 128 data points, tapered with a Hamming window in order to diminish border effects [11]. In the case of example A, the spectrum resembles a power-law distribution with the main activity concentrated between 1 and 10 Hz. This range of frequencies had the largest coherence values. For examples B and C, a more localized spectral distribution is seen, with a peak around 7 to 10 Hz and a harmonic around 15 Hz. These peaks correspond to the frequency of the spikes of Figure 4.1.

It is already clear from the spectral distribution that there is a better matching between the power spectra of the right and left channels of example B than for example C. This is reflected in the larger coherence values of example B, with a significant synchronization for this frequency range. In contrast, coherence values are much lower for example C, seeming significant only for the low frequencies (below 6 Hz). In Table 4.1 the coherence values at a frequency of 9 Hz—the main frequency of the spikes of examples B and C—are reported. As it was the case for the cross correlation, note that the coherence function does not distinguish well between examples A and B. From Figure 4.3, there is mainly a difference for frequencies larger than about 11 Hz, but this just reflects the lack of activity at this frequency range for example A, whereas in example B it reflects the synchronization between the high-frequency harmonics of the spikes. Even then, it is difficult to assess which frequency should be taken to rank the overall synchronization of the three signals (but some defenders of coherence may still argue that an overall synchronization value is meaningless).

## 4.3 Mutual Information Analysis

The cross-correlation and coherence functions evaluate linear relationships between two signals in the time and frequency domains, respectively. These measures are relatively simple to compute and interpret but have the main disadvantage of being linear and, therefore, not sensitive to nonlinear interactions. In this section we describe a measure that is sensitive to nonlinear interactions, but with the caveat that it is usually more difficult to compute, especially for short datasets.

Suppose we have a discrete random variable $X$ with $M$ possible outcomes $X_1, \ldots, X_M$, which can, for example, be obtained by partitioning of the $X$ variables into $M$ bins. Each outcome has a probability $p_i$, $i = 1, \ldots, M$, with $p_i \geq 0$ and $\Sigma p_i = 1$. A first estimate of these probabilities is to consider $p_i = n_i/N$, where $n_i$ is the probability of

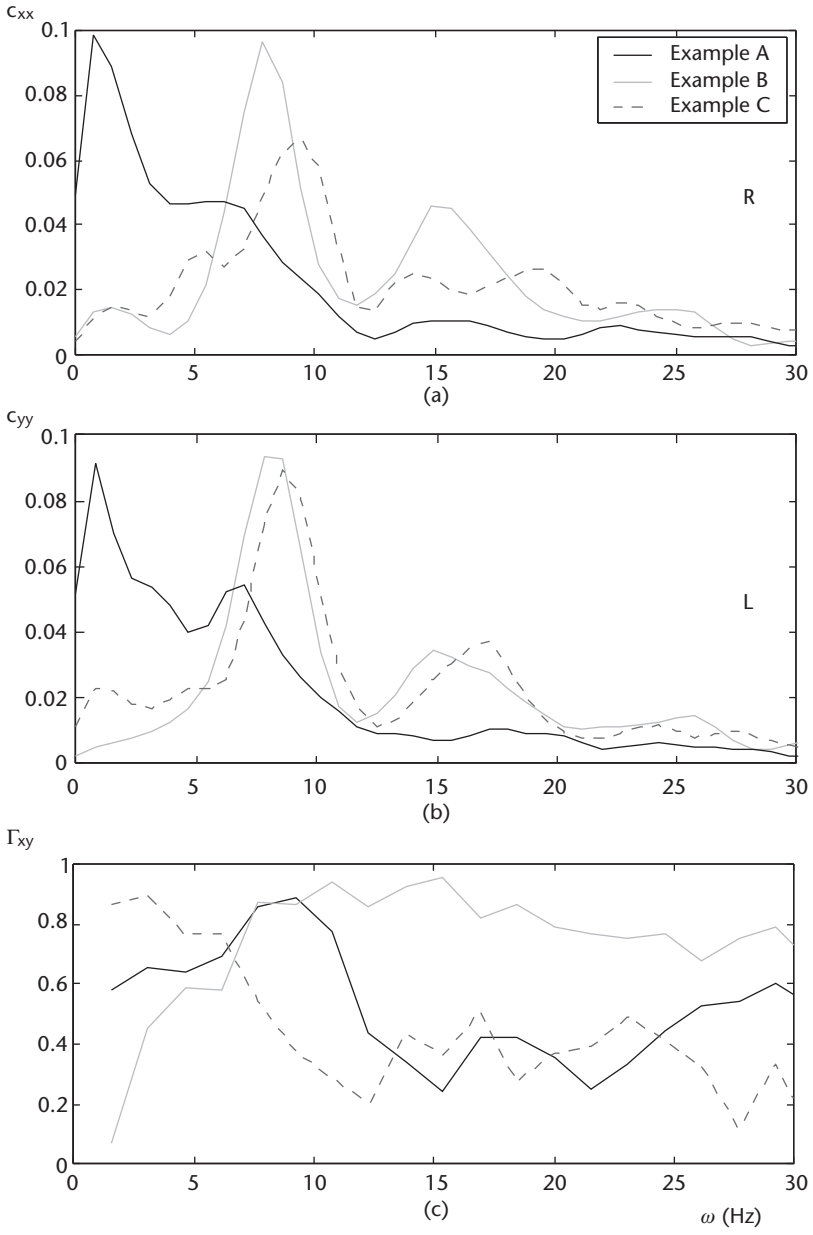



**Figure 4.3** (a–c) Power spectral estimation for the three signals of Figure 4.1 and the corresponding coherence estimation as a function of frequency.

occurrence of $X_i$ after $N$ samples. Note, however, that for a small number of samples this naïve estimate may not be appropriate and it may be necessary to introduce corrections terms [8]. Given this set of probabilities, we can define the Shannon entropy as follows:

$$I(X) = -\sum_{i=1}^{M} p_i \log p_i \tag{4.4}$$

The Shannon entropy is always positive and it measures the information content of $X$, in bits, if the logarithm is taken with base 2.

Next, suppose we have a second discrete random variable $Y$ and that we want to measure its degree of synchronization with $X$. We can define the joint entropy as

$$I(X,Y) = -\sum_{i,j} p_{ij}^{XY} \log p_{ij}^{XY} \tag{4.5}$$

in which $p_{ij}^{XY}$ is the joint probability of obtaining an outcome $X = X_i$ and $Y = Y_i$. For independent systems, one has $p_{ij}^{XY} = p_i^X p_j^Y$ and therefore, $I(X,Y) = I(X) + I(Y)$. Then, the mutual information between $X$ and $Y$ is defined as

$$MI(X,Y) = I(X) + I(Y) - I(X,Y) \tag{4.6}$$

The mutual information gives the amount of information of $X$ one obtains by knowing $Y$ and vice versa. For independent signals, $MI(X,Y) = 0$; otherwise, it takes positive values with a maximum of $MI(X,Y) = I(X) = I(Y)$ for identical signals.

Alternatively, the mutual information can be seen as a Kullback-Leibler entropy, which is an entropy measure of the similarity of two distributions [13, 14]. Indeed, (4.6) can be written in the form

$$MI(X,Y) = \sum p_{ij}^{XY} \log \frac{p_{ij}^{XY}}{p_i^X p_j^Y} \tag{4.7}$$

Then, considering a probability distribution $q_{ij}^{XY} = p_i^X p_j^Y$, (4.7) is a Kullback-Leibler entropy that measures the difference between the probability distributions $p_{ij}^{XY}$ and $q_{ij}^{XY}$. Note that $q_{ij}^{XY}$ is the correct probability distribution if the systems are independent and, consequently, the mutual information measures how different the true probability distribution $p_{ij}^{XY}$ is from another one in which independence between $X$ and $Y$ is assumed.

Note that it is not always straightforward to estimate $MI$ from real recordings, especially since an accurate estimation requires a large number of samples and small partition bins (a large $M$). In particular, for the joint probability densities $p_{ij}^{XY}$ there will usually be a large number of bins that will not be filled by the data, which may produce an underestimation of the value of $MI$. Several different proposals have been made to overcome these estimation biases whose description is outside the scope of this chapter. For a recent review, the reader is referred to [15]. In the particular case of the examples of Figure 4.1, the estimation of mutual information depended largely on the partition of the stimulus space used [8].

## 4.4 Phase Synchronization

All the measures described earlier are sensitive to relationships both in the amplitudes and phases of the signals. However, in some cases the phases of the signals may be related but the amplitudes may not. Phase synchronization measures are particularly suited for these cases because they measure any phase relationship between

signals independent of their amplitudes. The basic idea is to generate an analytic signal from which a phase, and a phase difference between two signals, can be defined.

Suppose we have a continuous signal $x(t)$, from which we can define an analytic signal

$$Z_x(t) = x(t) + j\tilde{x}(t) = A_x(t)e^{j\phi_x(t)} \tag{4.8}$$

where $\tilde{x}(t)$ is the Hilbert transform of $x(t)$:

$$\tilde{x}(t) \equiv (Hx)(t) = \frac{1}{\pi} P.V. \int_{-\infty}^{+\infty} \frac{x(t')}{t - t'} dt' \tag{4.9}$$

where P.V. refers to the Cauchy principal value. Similarly, we can define $A_y$ and $\phi_y$ from a second signal $y(t)$. Then, we define the $(n,m)$ phase difference of the analytic signals as

$$\phi_{xy}(t) \equiv n\phi_x(t) - m\phi_y(t) \tag{4.10}$$

with $n$, $m$ integers. We say that $x$ and $y$ are $m$:$n$ synchronized if the $(n,m)$ phase difference of (4.10) remains bounded for all $t$. In most cases, only the (1:1) phase synchronization is considered. The phase synchronization index is defined as follows [16–18]:

$$\gamma \equiv \left| \left\langle e^{j\phi_{xy}(t)} \right\rangle_t \right| = \sqrt{\left\langle \cos\phi_{xy}(t) \right\rangle_t^2 + \left\langle \sin\phi_{xy}(t) \right\rangle_t^2} \tag{4.11}$$

where the angle brackets denote average over time. The phase synchronization index will be zero if the phases are not synchronized and will be one for a constant phase difference. Note that for perfect phase synchronization the phase difference is not necessarily zero, because one of the signals could be leading or lagging in phase with respect to the other. Alternatively, a phase synchronization measure can be defined from the Shannon entropy of the distribution of phase differences $\phi_{xy}(t)$ or from the conditional probabilities of $\phi_x(t)$ and $\phi_y(t)$ [19].

An interesting feature of phase synchronization is that it is parameter free. However, it relies on an accurate estimation of the phase. In particular, to avoid misleading results, broadband signals (as it is usually the case of EEGs) should be first bandpass filtered in the frequency band of interest before calculating phase synchronization.

It is also possible to define a phase synchronization index from the wavelet transform of the signals [20]. In this case the phases are calculated by convolving each signal with a Morlet wavelet function. The main difference with the estimation using the Hilbert transform is that a central frequency $\omega_0$ and a width of the wavelet function should be chosen and, consequently, this measure is sensitive to phase synchronization in a particular frequency band. It is of particular interest to mention that both approaches—either defining the phases with the Hilbert or with the wavelet transform—are intrinsically related (for details, see [8]).

Figure 4.4 shows the time evolution of the (1:1) phase differences $\phi_{xy}(t)$ estimated using (4.10) for the three examples of Figure 4.1. It is clear that the phase differences of example B are much more stable than the one of the other two examples. The values of phase synchronization for the three examples are shown in Table 4.1 and are in agreement with the general tendency found with the other measures; that is, $\text{Sync}_B > \text{Sync}_A > \text{Sync}_C$. Given that with using the Hilbert transform, we can extract an instantaneous phase for each signal, (the same applies to the wavelet transform) we can see how phase synchronization varies with time, as shown in the

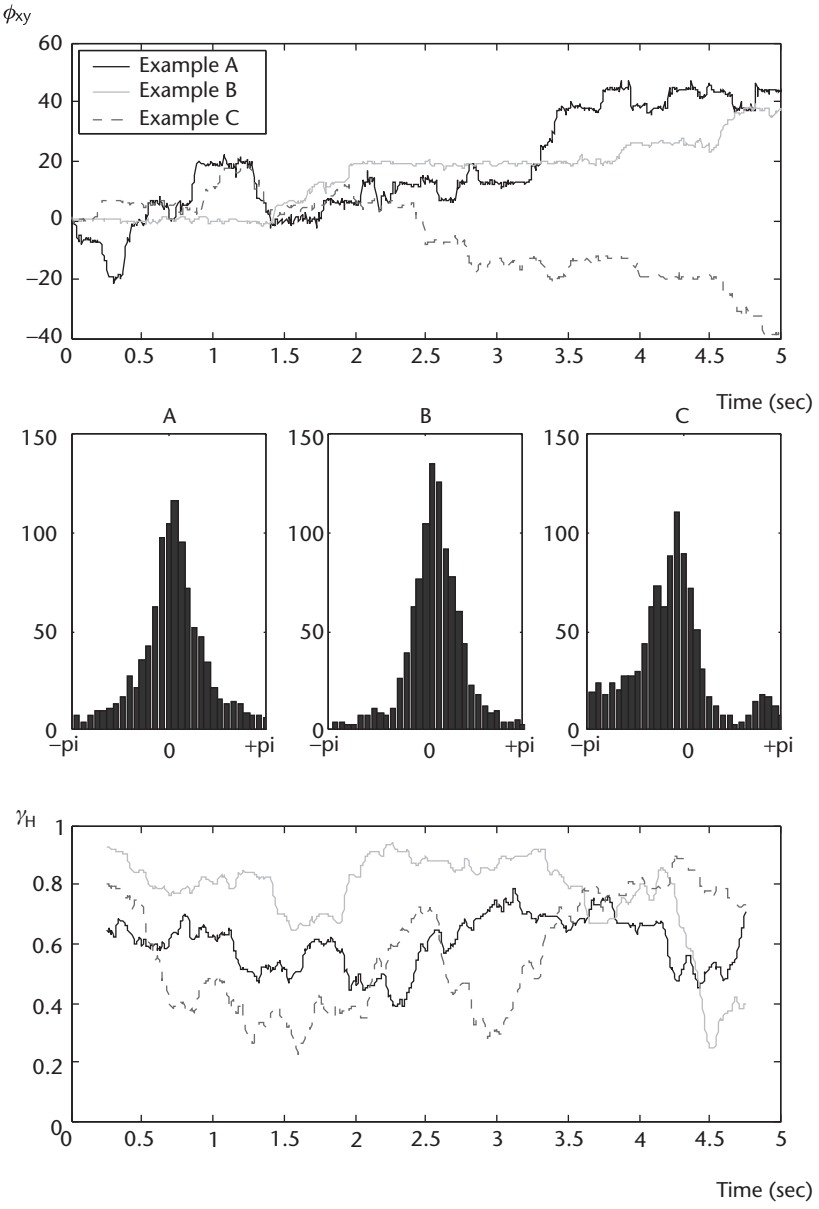



**Figure 4.4** (Top) (1:1) phase difference for the three examples of Figure 4.1. (Middle) Corresponding distribution of the phase differences. (Bottom) Time evolution of the phase synchronization index.

bottom panel of Figure 4.4. Note the variable degree of synchronization, especially for example C, which has a large increase of synchronization after second 3.

## 4.5 Conclusion

In this chapter we applied several linear and nonlinear measures of synchronization to three typical EEG signals. The first measure we described was the cross-correlation function, which is so far the most often used measure of correlation in neuroscience. We then described how to estimate coherence, which gives an estimation of the linear correlation as a function of the frequency. In comparison to cross correlation, the advantage of coherence is that it is sensitive to correlations in a limited frequency range. The main limitation of cross correlation and coherence is that they are linear measures and are therefore not sensitive to nonlinear interactions.

Using the information theory framework, we showed how it is possible to have a nonlinear measure of synchronization by estimating the mutual information between two signals. However, the main disadvantage of mutual information is that it is more difficult to compute, especially with short datasets. Finally, we described phase synchronization measures to quantify the interdependences of the phases between two signals, irrespective of their amplitudes. The phases can be computed using either the Hilbert or the wavelet transform, with similar results.

In spite of the different definitions and sensitivity to different characteristics of the signals of different synchronization methods, we saw that all of these measures gave convergent results and that naïve estimations based on visual inspection can be very misleading. It is not possible in general to assert which is the best synchronization measure. For example, for very short datasets mutual information may be not reliable, but it could be very powerful if long datasets are available. Coherence may be very useful for studying interactions at particular frequency bands, and phase synchronization may be the measure of choice if one wants to focus on phase relationships. In summary, the "best measure" depends on the particular data and questions at hand.

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CHAPTER 5

# Theory of the EEG Inverse Problem

Roberto D. Pascual-Marqui

In this chapter we deal with the EEG neuroimaging problem: Given measurements of scalp electric potential differences, find the three-dimensional distribution of the generating electric neuronal activity. This problem has no unique solution. Particular solutions with optimal localization properties are of primary interest, because neuroimaging is concerned with the correct localization of brain function. A brief historical outline of localization methods is given: from the single dipole, to multiple dipoles, to distributions. Technical details on the formulation and solution of this type of inverse problem are presented. Emphasis is placed on linear, discrete, three-dimensional distributed EEG tomographies having a simple mathematical structure that allows for a complete evaluation of their localization properties. One particular noteworthy member of this family is exact low-resolution brain electromagnetic tomography [1], which is a genuine inverse solution (not merely a linear imaging method, nor a collection of one-at-a-time single best fitting dipoles) with zero localization bias in the presence of measurement and structured biological noise.

## 5.1 Introduction

Hans Berger [2] reported as early as 1929 on the human EEG, which consists of time-varying measurements of scalp electric potential differences. At that time, using only one posterior scalp electrode with an anterior reference, he measured the alpha rhythm, an oscillatory activity in the range of 8 to 12 Hz, that appears when the subject is awake, resting, with eyes closed. He observed that by simply opening the eyes, the alpha rhythm would disorganize and tend to disappear. Such observations led Berger to the belief that the EEG was a window into the brain. Through this "window," one can "see" brain function, for example, what posterior brain regions are doing when changing state from eyes open to eyes closed.

The concept of "a window into the brain" already implies the localization of different brain regions, each one with certain characteristics and functions. From this point of view, Berger was already performing a very naïve type of low spatial resolution, a low spatial sampling form of neuroimaging, by assuming that the electrical activity recorded at a scalp electrode was determined by the activity of the underlying brain structure. To this day, many published research papers still use the

same technique, in which brain localization inference is based on the scalp distribution of electric potentials (commonly known as topographic scalp maps).

We must emphasize from the outset that this topographic-based method is, in general, not correct. In the case of EEG recordings, scalp electric potential differences are determined by electric neuronal activity from the entire cortex and by the geometrical orientation of the cortex. The cortical orientation factor alone has a very dramatic effect: An electrode placed over an active gyrus or sulcus will be influenced in extremely different ways. The consequence is that a scalp electrode does not necessarily reflect activity of the underlying cortex.

The route toward EEG-based neuroimaging must rely on the correct use of the physics laws that connect electric neuronal generators and scalp electric potentials. Formally, the EEG inverse problem can be stated as follows: Given measurements of scalp electric potential differences, find the three-dimensional distribution of the generators, that is, of the electric neuronal activity.

However, it turns out that in its most general form, this type of inverse problem has no unique solution, as was shown by Helmholtz in 1853 [3]. The curse of nonuniqueness [4] informally means that there is insufficient information in the scalp electric potential distribution to determine the actual generator distribution. Equivalently, given scalp potentials, there are infinitely different generator distributions that comply with the scalp measurements. The apparent consequence is that there is no way to determine the actual generators from scalp electric potentials.

This seemingly hopeless situation is not very true. The general statement of Helmholtz applies to arbitrary distributions of generators. However, the electric neuronal generators in the human brain are not arbitrary, and actually have properties that can be combined into the inverse problem statement, narrowing the possible solutions. In addition to endowing the possible inverse solutions with certain neuroanatomical and electrophysiological properties, we are interested only in those solutions that have "good" localization properties, because that is what neuroimaging is all about: the localization of brain function.

Several solutions are reviewed in this chapter, with particular emphasis on the general family of linear imaging methods.

## 5.2 EEG Generation

Details on the electrophysiology and physics of EEG/MEG generation can be found in publications by Mitzdorf [5], Llinas [6], Martin [7], Hämäläinen et al. [8], Haalman and Vaadia [9], Sukov and Barth [10], Dale et al. [11], and Baillet et al. [12]. The basic underlying physics can be studied in [13].

### 5.2.1 The Electrophysiological and Neuroanatomical Basis of the EEG

It is now widely accepted that scalp electric potential differences are generated by cortical pyramidal neurons undergoing postsynaptic potentials (PSPs). These neurons are oriented perpendicular to the cortical surface. The magnitude of experimentally recorded scalp electric potentials, at any given time instant, is due to the spatial summation of the impressed current density induced by highly synchronized

PSPs occurring in large clusters of neurons. A typical cluster size must cover at least 40 to 200 $mm^2$ of cortical surface in order to produce a measurable scalp signal.

Summarizing, there are two essential properties:

1. The EEG sources are confined to the cortical surface, which is populated mainly by pyramidal neurons (constituting approximately 80% of the cortex), oriented perpendicular to the surface.
2. Highly synchronized PSPs occur frequently in spatial clusters of cortical pyramidal neurons.

This information can be used to narrow significantly the nonuniqueness of the inverse solution, as explained later in this chapter.

The reader should keep in mind that there is a very strict limitation on the use of the equivalent terms *EEG generators* and *electric neuronal generators*. This is best illustrated with an example, such as the alpha rhythm. Cortical pyramidal neurons located mainly in occipital cortical areas are partly driven by thalamic neurons that make them beat synchronously at about 11 Hz (a thalamocortical loop). But the EEG does not "see" all parts of this electrophysiological mechanism. The EEG only sees the final electric consequence of this process, namely, that the alpha rhythm is *electrically* generated in occipital cortical areas. This raises the following question: Are scalp electric potentials only due to electrically active cortical pyramidal neurons? The answer is no. All active neurons contribute to the EEG. However, the contribution from the cortex is overwhelmingly large compared to all other structures, due to two factors:

1. The number of cortical neurons is much larger than that of subcortical neurons.
2. The distance from subcortical structures to the scalp electrodes is larger than from cortical structures to the electrodes.

This is why EEG recordings are mainly generated by electrically active cortical pyramidal neurons.

It is possible to manipulate the measurements in order to enhance noncortical generators. This can be achieved by averaging EEG measurements appropriately, as is traditionally done in average ERPs. Such an averaging manipulation usually reduces the amplitude of the background EEG activity, enhancing the brain response that is phase locked to the stimulus. When the number of stimuli is very high, the average scalp potentials might be mostly due to noncortical structures, as in a brain stem auditory evoked potential [14].

### 5.2.2 The Equivalent Current Dipole

From the physics point of view, a cortical pyramidal neuron undergoing a PSP will behave as a current dipole, which consists of a current source and a current sink separated by a distance in the range of 100 to 500 $\mu$m. This means that both poles (the source and the sink) are always paired, and extremely close to each other, as seen from the macroscopic scalp electrodes. For this reason, the sources of the EEG can be modeled as a distribution of dipoles along the cortical surface.

Figure 5.1 illustrates the equivalent current dipole corresponding to a cortical pyramidal neuron undergoing an excitatory postsynaptic potential (EPSP) taking place at a basal dendrite. The cortical pyramidal neuron is outlined in black. Notice the approximate size scale (100-$\mu$m bar in lower right). An incoming axon from a presynaptic neuron terminates at a basal dendrite. The event taking place induces specific channels to open, allowing (typically) an inflow of $Na^+$, which gives rise to a sink of current. Electrical neutrality must be conserved, and a source of current is produced at the apical regions.

This implies that it would be very much against electrophysiology to model the sources as freely distributed, nonpaired monopoles of current. An early attempt in this direction can be found in [15]. Those monopolar inverse solutions were not pursued any further because, as expected, they simply were incapable of correct localization when tested with real human data such as visual, auditory, and somatosensory ERPs, for which the localization of the sensory cortices is well known.

Keep in mind that a single active neuron is not enough to produce measurable scalp electric potential differences. EEG measurements are possible due to the existence of relatively large spatial clusters of cortical pyramidal cells that are geometrically arranged parallel to each other, and that simultaneously undergo the same type

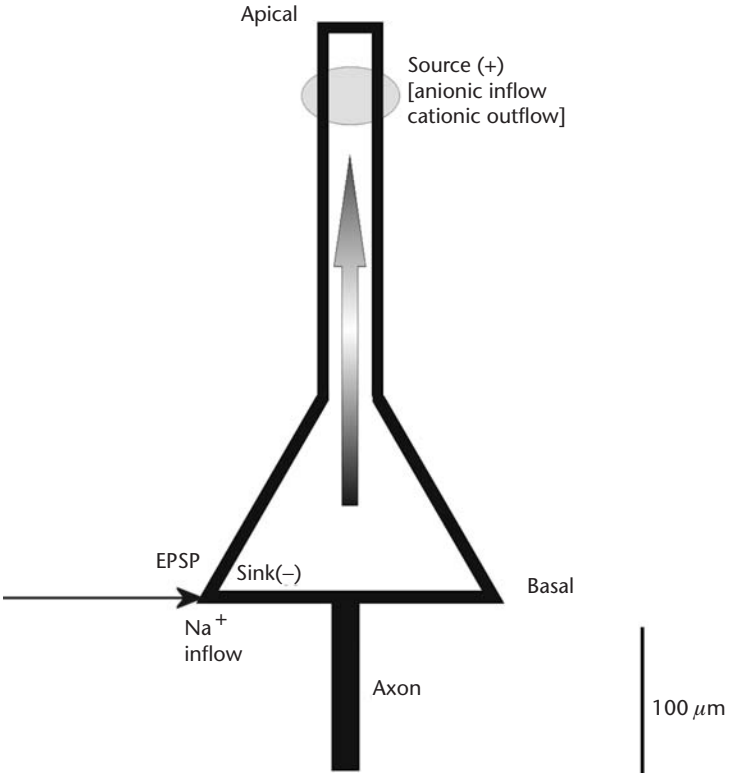



**Figure 5.1** Schematic representation of the generators of the EEG: the equivalent current dipole corresponding to a cortical pyramidal neuron undergoing an EPSP taking place at a basal dendrite. The cortical pyramidal neuron is outlined in black. The incoming axon from a presynaptic neuron terminates at a basal dendrite. Channels open, allowing (typically) an inflow of $Na^+$, which gives rise to a sink of current. Due to the conservation of electrical neutrality, a source of current is produced at the apical regions.

of postsynaptic potential (synchronization). If these conditions are not met, then the total summed activity is too weak to produce nonnegligible extracranial fields.

## 5.3 Localization of the Electrically Active Neurons as a Small Number of "Hot Spots"

An early attempt at the localization of the active brain region responsible for the scalp electric potential distribution was performed in a semiquantitative manner by Brazier in 1949 [16]. It was suggested that electric field theory be used to determine the location and orientation of the current dipole from the scalp potential map. This can be considered to be the starting point for what later developed into "dipole fitting."

Immediately afterward, using a spherical head model, the equations were derived that relate electric potential differences on the surface of a homogeneous conducting sphere due to a current dipole within [17, 18]. About a decade later, an improved, more realistic head model considered the different conductivities of neural tissue, skull, and scalp [19]. Use was made of these early techniques by Lehmann et al. [20] to locate the generator of a visual evoked potential.

Note that in the single-current dipole model, it is assumed that brain activity is due to a single small area of active cortex. In general, this model is very simplistic and nonrealistic, because the whole cortex is never totally "quiet" except for a single small area. Nevertheless, the dipole model does produce reasonable results under some particular conditions. This was shown very convincingly by Henderson et al. [21], both in an experimentally simulated head (a head phantom) and with real human EEG recordings. The conditions under which a dipole model makes sense are limited to cases where electric neuronal activity is dominated by a small brain area. Two examples where the model performs very well are in some epileptic spike events, and in the description of the early components of the average brain stem auditory evoked potential [14]. However, it would seem that the localization of higher cognitive functions could not be reliably modeled by dipole fitting.

### 5.3.1 Single-Dipole Fitting

Single-dipole fitting can be seen as the localization of the electrically active neurons as a single "hot spot." Consider the case of a single current dipole located at position $\mathbf{r}_v \in R^{3\times1}$ with dipole moment $\mathbf{j}_v \in R^{3\times1}$, where

$$\mathbf{r}_v = \begin{pmatrix} x_V & y_V & z_V \end{pmatrix}^T \tag{5.1}$$

denotes the position vector, with the superscript $T$ denoting vector/matrix transposition, and

$$\mathbf{j}_v = \begin{pmatrix} j_x & j_y & j_z \end{pmatrix}^T \tag{5.2}$$

To introduce the basic form of the equations, consider the nonrealistic, simple case of the current dipole in an infinite homogenous medium with conductivity $\sigma$. Then the electric potential at location $\mathbf{r}_e \in R^{3\times 1}$ for $\mathbf{r}_e \neq \mathbf{r}_v$ is

$$\phi(\mathbf{r}_e, \mathbf{r}_v, \mathbf{j}_v) = \mathbf{k}_{e,v}^T \mathbf{j}_v + c \tag{5.3}$$

where

$$\mathbf{k}_{e,v} = \frac{1}{4\pi\sigma} \frac{(\mathbf{r}_e - \mathbf{r}_v)}{\|\mathbf{r}_e - \mathbf{r}_v\|^3} \tag{5.4}$$

denotes what is commonly known as the lead field. In (5.3), $c$ is a scalar accounting for the physics nature of electric potentials, which are determined up to an arbitrary constant.

A slightly more realistic head model corresponds to a spherical homogeneous conductor in air. The lead field in this case is

$$\mathbf{k}_{e,v} = \frac{1}{4\pi\sigma}\left[2\frac{(\mathbf{r}_e - \mathbf{r}_v)}{\|\mathbf{r}_e - \mathbf{r}_v\|^3} + \frac{\mathbf{r}_e\|\mathbf{r}_e - \mathbf{r}_v\| + (\mathbf{r}_e - \mathbf{r}_v)\|\mathbf{r}_e\|}{\|\mathbf{r}_e\|\|\mathbf{r}_e - \mathbf{r}_v\|\left[\|\mathbf{r}_e\|\|\mathbf{r}_e - \mathbf{r}_v\| + \mathbf{r}_e^T(\mathbf{r}_e - \mathbf{r}_v)\right]}\right] \tag{5.5}$$

in which this notation is used:

$$\|\mathbf{X}\|^2 = tr(\mathbf{X}^T\mathbf{X}) = tr(\mathbf{X}\mathbf{X}^T) \tag{5.6}$$

and where *tr* denotes the trace, and $\mathbf{X}$ is any matrix or vector. If $\mathbf{X}$ is a vector, then this is the squared Euclidean $L_2$ norm; if $\mathbf{X}$ is a matrix, then this is the squared Frobenius norm.

The equation for the lead field in a totally realistic head model (taking into account geometry and full conductivity profile) is not available in closed form, such as in (5.4) and (5.5). Numerical methods for computing the lead field can be found in [22].

Nevertheless, in general, the components of the lead field $\mathbf{k}_{e,v} = (k_x \quad k_y \quad k_z)^T$ have a very simple interpretation: $k_x$ corresponds to the electric potential at position $\mathbf{r}_e$, due to unit strength current dipole $j_x = 1$ at position $\mathbf{r}_v$; and similarly for the other two components.

Formally, we are now in a position to state the single-dipole fitting problem. Let $\hat{\phi}_e$ (for $e = 1, \ldots, N_E$) denote the scalp electric potential measurement at electrode $e$, where $N_E$ is the total number of cephalic electrodes. All measurements are made using the same reference. Let $\phi_e(\mathbf{r}_v, \mathbf{j}_v)$ (for $e = 1, \ldots, N_E$) denote the theoretical potential at electrode $e$, due to a current dipole located at $\mathbf{r}_v$ with moment $\mathbf{j}_v$. Then the problem consists of finding the unknown dipole position $\mathbf{r}_v$ and moment $\mathbf{j}_v$ that *best* explain the actual measurements. The simplest way to achieve this is to minimize the *distance* between theoretical and experimental potentials.

Consider the functional:

$$F = \sum_{e=1}^{N_E} \left[ \phi_e(\mathbf{r}_v, \mathbf{j}_v) - \hat{\phi}_e \right]^2 \tag{5.7}$$

This expresses the distance between measurements and model, as a function of the two main dipole parameters: its location and its moment. The aim is to find the values of the parameters that minimize the functional, that is, the *least squares* solution. Many algorithms are available for finding the parameters, as reviewed in [10, 14].

### 5.3.2 Multiple-Dipole Fitting

A straightforward generalization of the previous case consists of attempting to explain the measured EEG as being due to a small number of active brain spots. Based on the principle of superposition, the theoretical potential due to $N_V$ dipoles is simply the sum of potentials due to each individual dipole. Therefore, the functional in (5.7) generalizes to

$$F = \sum_{e=1}^{N_E} \left[ \sum_{v=1}^{N_V} \phi_e(\mathbf{r}_v, \mathbf{j}_v) - \hat{\phi}_e \right]^2 \tag{5.8}$$

and the least squares problem for this multiple-dipole fitting case consists of finding all dipole positions $\mathbf{r}_v$ and moments $\mathbf{j}_v$, for $v = 1 \ldots N_V$ that minimize $F$.

Two major problems arise when using multiple-dipole fitting:

1. The number of dipoles $N_V$ must be known beforehand. The dipole locations vary greatly for different values of $N_V$.
2. For realistic measurements (which includes measurement noise), and for a given fixed value of $N_V > 1$, the functional in (5.8) has many local minima, with several of them very close in value to the absolute minimum, but all of them with very different locations for the dipoles. This makes it very difficult to choose objectively the correct solution.

## 5.4 Discrete, Three-Dimensional Distributed Tomographic Methods

The principles that will be used in this section are common to other tomographies, such as structural X-rays (i.e., CAT scans), structural MRI, and functional tomographies such as fMRI and positron emission tomography (PET).

For the EEG inverse problem, the solution space consists of a distribution of points in three-dimensional space. A classical example is to construct a three-dimensional uniform grid throughout the brain and to retain the points that fall on the cortical surface (mainly populated by pyramidal neurons). At each such point, whose coordinates are known by construction, a current density vector with unknown moment components is placed. The current density vector (i.e., the equivalent current dipole) at a grid point represents the total electric neuronal activity of the volume immediately around the grid point, commonly called a voxel.

The scalp electric potential difference at a given electrode receives contributions, in an additive manner, from all voxels. The equation relating scalp potentials and current density can be conveniently expressed in vector/matrix notation as:

$$\mathbf{\Phi} = \mathbf{KJ} + c\mathbf{1} \tag{5.9}$$

where the vector $\mathbf{\Phi} \in R^{N_E \times 1}$ contains the instantaneous scalp electric potential differences measured at $N_E$ electrodes with respect to a single common reference electrode (e.g., the reference can be linked earlobes, the toe, or one of the electrodes included in $\mathbf{\Phi}$); the matrix $\mathbf{K} \in R^{N_E \times (3N_V)}$ is the lead field matrix corresponding to $N_V$ voxels; $\mathbf{J} \in R^{(3N_V) \times 1}$ is the current density; $c$ is a scalar accounting for the physics nature of electric potentials, which are determined up to an arbitrary constant; and $\mathbf{1}$ denotes a vector of ones, in this case $\mathbf{1} \in R^{N_E \times 1}$. Typically $N_E << N_V$, and $N_E \geq 19$. In (5.9), the structure of the lead field matrix $\mathbf{K}$ is

$$\mathbf{K} = \begin{pmatrix} \mathbf{k}_{11}^T & \mathbf{k}_{12}^T & \dots & \mathbf{k}_{1N_V}^T \\ \mathbf{k}_{21}^T & \mathbf{k}_{22}^T & \dots & \mathbf{k}_{2N_V}^T \\ \dots & & & \\ \mathbf{k}_{N_E 1}^T & \mathbf{k}_{N_E 2}^T & \dots & \mathbf{k}_{N_E N_V}^T \end{pmatrix} \tag{5.10}$$

where $\mathbf{k}_{ev} \in R^{3 \times 1}$ (for $e = 1, \ldots, N_E$ and for $v = 1, \ldots, N_V$) corresponds to the scalp potentials at the $e$th electrode due to three orthogonal unit strength dipoles at voxel $v$, each one oriented along the coordinate axes $x$, $y$, and $z$. Equations (5.4) and (5.5) are examples of the lead field that can be written in closed form, although they correspond to head models that are too unrealistic.

Note that $\mathbf{K}$ can also be conveniently written as

$$\mathbf{K} = \left(\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3, \ldots, \mathbf{K}_{N_V}\right) \tag{5.11}$$

where $\mathbf{K} \in R^{N_E \times 3}$ (for $v = 1, \ldots, N_V$) is defined as follows:

$$\mathbf{K}_v = \begin{pmatrix} \mathbf{k}_{1v}^T \\ \mathbf{k}_{2v}^T \\ \dots \\ \mathbf{k}_{N_E v}^T \end{pmatrix} \tag{5.12}$$

In (5.9), $\mathbf{J}$ is structured as

$$\mathbf{J} = \begin{pmatrix} \mathbf{j}_1 \\ \mathbf{j}_2 \\ \dots \\ \mathbf{j}_{N_V} \end{pmatrix} \tag{5.13}$$

where $\mathbf{j}_v \in R^{3 \times 1}$ denotes the current density at the $v$th voxel, as in (5.2).

At this point, the basic EEG inverse problem for the discrete, three-dimensional distributed case consists of solving (5.9) for the unknown current density **J** and constant $c$, given the lead field **K** and measurements $\Phi$.

### 5.4.1 The Reference Electrode Problem

As a first step, the reference electrode problem is solved by estimating $c$ in (5.9). Given $\Phi$ and **KJ,** the reference electrode problem is

$$\min_{c} \left\| \Phi - \mathbf{KJ} - c\mathbf{1} \right\|^2 \tag{5.14}$$

The solution is

$$c = \frac{1^T}{1^T 1}\left(\Phi - \mathbf{KJ}\right) \tag{5.15}$$

Plugging (5.15) into (5.9) gives

$$\mathbf{H}\Phi = \mathbf{HKJ} \tag{5.16}$$

where

$$\mathbf{H} = \mathbf{I} - \frac{11^T}{1^T 1} \tag{5.17}$$

is the average reference operator, also known as the centering matrix, and $\mathbf{I} \in R^{N_E \times N_E}$ is the identity matrix. This result establishes the fact that any inverse solution will not depend on the reference electrode. This applies to any form of the EEG inverse problem, including the inverse dipole fitting problems in (5.7) and (5.8).

Henceforth, it will be assumed that the EEG measurements and the lead field are average reference transformed, that is,

$$\left\{ \begin{array}{l} \Phi \leftarrow \mathbf{H}\Phi \\ \mathbf{K} \leftarrow \mathbf{HK} \end{array} \right\} \tag{5.18}$$

and (5.9) is then rewritten as follows:

$$\Phi = \mathbf{KJ} \tag{5.19}$$

Note that **H** plays the role of the identity matrix for EEG data. It actually is the identity matrix, except for a null eigenvalue corresponding to an eigenvector of ones, accounting for the reference electrode constant.

### 5.4.2 The Minimum Norm Inverse Solution

In 1984 Hämäläinen and Ilmoniemi [23] published a technical report with a particular solution to the inverse problem corresponding to the forward equation of the

type shown in (5.19). As the name of the method implies, this particular solution is the one that has a minimum norm. The problem in its simplest form is stated as follows:

$$\left\{\begin{array}{l}\min\limits_{\mathbf{J}} \mathbf{J}^T\mathbf{J} \\ \text{such that}: \boldsymbol{\Phi} = \mathbf{K}\mathbf{J}\end{array}\right\} \tag{5.20}$$

The solution is

$$\hat{\mathbf{J}} = \mathbf{T}\boldsymbol{\Phi} \tag{5.21}$$

with

$$\mathbf{T} = \mathbf{K}^T\left(\mathbf{K}\mathbf{K}^T\right)^+ \tag{5.22}$$

The superscript + denotes the Moore-Penrose generalized inverse [24]. The minimum norm inverse solution of (5.21) and (5.22) is a genuine solution to the system of (5.19).

If the measurements are contaminated with noise, it is typically more convenient to change the statement of the inverse problem in such a way as to avoid the current density being influenced by errors. The new inverse problem now is

$$\min_{\mathbf{J}} F \tag{5.23}$$

with

$$F = \left\|\boldsymbol{\Phi} - \mathbf{K}\mathbf{J}\right\|^2 + \alpha\mathbf{J}^T\mathbf{J} \tag{5.24}$$

In (5.24), the parameter $\alpha > 0$ controls the relative importance between the two terms on the right-hand side: a penalty for being unfaithful to the measurements and a penalty for a large current density norm. This parameter is known as the Tikhonov regularization parameter [25]. The solution is

$$\hat{\mathbf{J}} = \mathbf{T}\boldsymbol{\Phi} \tag{5.25}$$

with

$$\mathbf{T} = \mathbf{K}^T\left(\mathbf{K}\mathbf{K}^T + \alpha\mathbf{H}\right)^+ \tag{5.26}$$

The current density estimator in (5.25) and (5.26) does not explain the measurements of (5.19) when $\alpha > 0$. In the limiting case $\alpha \to 0$, the solution is again the (nonregularized) minimum norm solution.

The main property of the original minimum norm method [23] was illustrated by showing correct, blurred localization of test point sources. The simulations corresponded to MEG sensors distributed on a plane, and with the cortex represented as a square grid of points on a plane located below the sensor plane. The test point source (i.e., the equivalent current dipole) was placed at a cortical voxel, and the the-

oretical MEG measurements were computed, which were then used in (5.25) and (5.26) to obtain the estimated minimum norm current density, which showed maximum activity at the correct location, but with some spatial dispersion. These first results were very encouraging. However, there was one essential omission: The method does not localize deep sources. In a three-dimensional cortex, if the actual source is deep, the method misplaces it to the outermost cortex. The reason for this behavior was explained in Pascual-Marqui [26], where it was noted that the EEG/MEG minimum norm solution is a harmonic function [27] that can only attain extreme values (maximum activation) at the boundary of the solution space, that is, at the outermost cortex.

### 5.4.3 Low-Resolution Brain Electromagnetic Tomography

The discrete, three-dimensional distributed, linear inverse solution that achieved low localization errors (in the sense defined earlier by Hämäläinen and Ilmoniemi [23]) even for deep sources was the method known as low-resolution electromagnetic tomography (LORETA) [28].

Informally, the basic property of this particular solution is that the current density at any given point on the cortex be maximally similar to the average current density of its neighbors. This "smoothness" property (see, e.g., [29, 30]) must hold throughout the entire cortex. Note that the smoothness property approximates the electrophysiological constraint under which the EEG is generated: Large spatial clusters of cortical pyramidal cells must undergo simultaneously and synchronously the same type of postsynaptic potentials.

The general inverse problem that includes LORETA as a particular case is stated as

$$\min_{\mathbf{J}} F_W \tag{5.27}$$

with

$$F_W = \left\|\Phi - \mathbf{K}\mathbf{J}\right\|^2 + \alpha \mathbf{J}^T \mathbf{W} \mathbf{J} \tag{5.28}$$

The solution is

$$\hat{\mathbf{J}}_W = \mathbf{T}_W \Phi \tag{5.29}$$

with the pseudoinverse given by

$$\mathbf{T}_W = \mathbf{W}^{-1}\mathbf{K}^T\left(\mathbf{K}\mathbf{W}^{-1}\mathbf{K}^T + \alpha \mathbf{H}\right)^+ \tag{5.30}$$

where the matrix $\mathbf{W} \in R^{(3N_V)\times(3N_V)}$ can be tailored to endow the inverse solution with a particular property.

In the case of LORETA, the matrix $\mathbf{W}$ implements the squared spatial Laplacian operator discretely. In this way, maximally synchronized PSPs at a relatively large macroscopic scale will be enforced. For the sake of simplicity, lead field normalization has not been mentioned in this description, although it is an integral part of the

weight matrix used in LORETA. The technical details of the LORETA method can be found in [26, 28].

When LORETA is tested with point sources, low-resolution images with very low localization errors are obtained. These results were shown in a nonpeer-reviewed publication [31] that included discussions with M. S. Hämäläinen, R. J. Ilmoniemi, and P. L. Nunez. The mean localization error of LORETA with EEG was, on average, only one grid unit, which happened to be three times smaller than that of the minimum norm solution. These results were later reproduced and validated by an independent group [32].

It is important to take great care when implementing the Laplacian operator. For instance, Daunizeau and Friston [33] implemented the Laplacian operator on a cortical surface consisting of 500 vertices, which are very irregularly sampled, as can be unambiguously appreciated from their Figure 2 in [33]. Obviously, the Laplacian operator is numerically worthless, and yet they conclude rather abusively that "the LORETA method gave the worst results." Because their Laplacian is numerically worthless, it is incapable of correctly implementing the smoothness requirement of LORETA. When this is done properly with a regularly sampled solution space, as in [31, 32], LORETA localizes with a very low localization error.

At the time of this writing, LORETA has been extensively validated, such as in studies combining LORETA with fMRI [34, 35], with structural MRI [36], and with PET [37]. Further LORETA validation has been based on accepting as ground truth localization findings obtained from invasive implanted depth electrodes, in which case there are several studies in epilepsy [38–41] and cognitive ERPs [42].

### 5.4.4 Dynamic Statistical Parametric Maps

The inverse solutions previously described correspond to methods that estimate the electric neuronal activity directly as current density. An alternative approach within the family of discrete, three-dimensional distributed, linear imaging methods is to estimate activity as statistically standardized current density.

This approach was introduced by Dale et al. in 2000 [43], and is referred to as the dynamic statistical parametric map (dSPM) approach or the noise-normalized current density approach. The method uses the ordinary minimum norm solution for estimating the current density, as given by (5.25) and (5.26). The standard deviation of the minimum norm current density is computed by assuming that its variability is exclusively due to noise in the measured EEG.

Let $\mathbf{S}_{\Phi}^{\text{Noise}} \in R^{N_E \times N_E}$ denote the EEG noise covariance matrix. Then the corresponding current density covariance is

$$\mathbf{S}_{\hat{\mathbf{J}}}^{\text{Noise}} = \mathbf{T}\mathbf{S}_{\Phi}^{\text{Noise}}\mathbf{T}^{T} \tag{5.31}$$

with $\mathbf{T}$ given by (5.26). This result is based on the quadratic nature of the covariance in (5.31), as derived from the linear transform in (5.19) (see, e.g., Mardia et al. [44]). From (5.31), let $\left[\mathbf{S}_{\hat{\mathbf{J}}}^{\text{Noise}}\right]_v \in R^{3\times 3}$ denote the covariance matrix at voxel $v$. Note that this is the $v$th $3 \times 3$ diagonal block matrix in $\mathbf{S}_{\hat{\mathbf{J}}}^{\text{Noise}}$, and it contains current density

noise covariance information for all three components of the dipole moment. The noise-normalized imaging method of Dale et al. [43] then gives

$$\mathbf{q}_v = \frac{\hat{\mathbf{j}}_v}{\sqrt{tr\left[\mathbf{S}_{\hat{\mathbf{j}}}^{\text{Noise}}\right]_v}} \tag{5.32}$$

where $\hat{\mathbf{j}}_v$ is the minimum norm current density at voxel $v$. The squared norm of $\mathbf{q}_v$

$$q_v^T q_v = \frac{\hat{j}_v^T \hat{j}_v}{tr\left[\mathbf{S}_{\hat{j}}^{\text{Noise}}\right]_v} \tag{5.33}$$

is an *F*-distributed statistic.

Note that the noise-normalized method in (5.32) is a linear imaging method in the case when it uses an estimated EEG noise covariance matrix based on a set of measurements that are thought to contain no signal of interest (only noise) and that are independent from the measurements whose generators are sought.

Pascual-Marqui [45] and Sekihara et al. [46] showed that this method has significant nonzero localization error, even under quasi-ideal conditions of negligible measurement noise.

### 5.4.5 Standardized Low-Resolution Brain Electromagnetic Tomography

Another discrete, three-dimensional distributed, linear statistical imaging method is standardized low-resolution brain electromagnetic tomography (sLORETA) [45]. The basic assumption in this method is that the current density variance receives contributions from possible noise in the EEG measurements, but more importantly, from biological variance, that is, variance in the actual electric neuronal activity. The biological variance is assumed to be due to electric neuronal activity that is independent and identically distributed all over the cortex, although any other a priori hypothesis can be accommodated. This implies that all of the cortex is equally likely to be active. Under this hypothesis, *s*LORETA produces a linear imaging method that has exact, zero-error localization under ideal conditions, as shown empirically in [45] and theoretically in [46] and [47].

In this case, the covariance matrix for EEG measurements is

$$\mathbf{S}_\Phi = \mathbf{K}\mathbf{S}_\mathbf{J}\mathbf{K}^T + \mathbf{S}_\Phi^{\text{Noise}} \tag{5.34}$$

where $\mathbf{S}_\Phi^{\text{Noise}}$ corresponds to noise in the measurements, and $\mathbf{S}_\mathbf{J}$ to the biological source of variability, that is, the covariance for the current density. When $\mathbf{S}_\mathbf{J}$ is set to the identity matrix, it is equivalent to allowing an equal contribution from all cortical neurons to the biological noise. Typically, the covariance of the noise in the measurements $\mathbf{S}_\Phi^{\text{Noise}}$ is taken as being proportional to the identity matrix. Under these conditions, the current density covariance is given by

$$\mathbf{S}_{\hat{\mathbf{J}}} = \mathbf{T}\mathbf{S}_{\Phi}\mathbf{T}^{T} = \mathbf{T}\left(\mathbf{K}\mathbf{S}_{\mathbf{J}}\mathbf{K}^{T} + \mathbf{S}_{\Phi}^{\text{Noise}}\right)\mathbf{T}^{T} = \mathbf{T}\left(\mathbf{K}\mathbf{K}^{T} + \alpha\mathbf{H}\right)\mathbf{T}^{T} = \mathbf{K}^{T}\left(\mathbf{K}\mathbf{K}^{T} + \alpha\mathbf{H}\right)\mathbf{K} \tag{5.35}$$

The sLORETA linear imaging method then is

$$\sigma_{v} = \left[\mathbf{S}_{\hat{\mathbf{J}}}\right]_{v}^{-1/2}\hat{\mathbf{j}}_{v} \tag{5.36}$$

where $\left[\mathbf{S}_{\hat{\mathbf{J}}}\right]_{v} \in R^{3\times 3}$ denotes the $v$th $3 \times 3$ diagonal block matrix in $\mathbf{S}_{\hat{\mathbf{J}}}$ (5.35), and $\left[\mathbf{S}_{\hat{\mathbf{J}}}\right]_{v}^{-1/2}$ is its symmetric square root inverse (as in the Mahalanobis transform; see, for example, Mardia et al. [44]). The squared norm of $\sigma_v$, that is,

$$\sigma_{v}^{T}\sigma_{v} = \hat{j}_{v}^{T}\left[S_{\hat{j}}\right]_{v}^{-1}\hat{j}_{v} \tag{5.37}$$

can be interpreted as a pseudostatistic with the form of an *F*-distribution.

It is worth emphasizing that Sekihara et al. [46] and Greenblatt et al. [47] showed that sLORETA has no localization bias in the absence of measurement noise; but in the presence of measurement noise, sLORETA has a localization bias. They did not consider the more realistic case where the brain in general is always active, as modeled here by the biological noise. A recent result [1] presents proof that sLORETA has no localization bias under these arguably much more realistic conditions.

### 5.4.6 Exact Low-Resolution Brain Electromagnetic Tomography

It is likely that the main reason for the development of EEG functional imaging methods in the form of standardized inverse solutions (e.g., dSPM and sLORETA) was that up to very recently all attempts to obtain an actual solution with no localization error have been fruitless. This has been a long-standing goal, as testified by the many publications that endlessly search for an appropriate weight matrix [refer to (5.27) to (5.30)]. For instance, to correct for the large depth localization error of the minimum norm solution, one school of thought has been to give more importance (more weight) to deeper sources. A recent version of this method can be found in Lin et al. [48]. That study showed that with the best depth weighting, the average depth localization error was reduced from 12 to 7 mm.

The inverse solution denoted as exact low-resolution brain electromagnetic tomography (eLORETA) achieves this goal [1, 49]. Reference [1] shows that eLORETA is a genuine inverse solution, not merely a linear imaging method, and endowed with the property of no localization bias in the presence of measurement and structured biological noise.

The eLORETA solution is of the weighted type, as given by (5.27) to (5.30). The weight matrix $\mathbf{W}$ is block diagonal, with subblocks of dimension $3 \times 3$ for each voxel. The eLORETA weights satisfy the system of equations:

$$\mathbf{W}_v = \left[\mathbf{K}_v^T \left(\mathbf{K}\mathbf{W}^{-1}\mathbf{K}^T + \alpha\mathbf{H}\right)^+ \mathbf{K}_v\right]^{1/2} \tag{5.38}$$

where $\mathbf{W}_v \in R^{3\times 3}$ is the $v$th diagonal subblock of $\mathbf{W}$.

As shown in [1], eLORETA has no localization bias in the presence of measurement noise and biological noise with variance proportional to $\mathbf{W}^{-1}$.

The screenshot in Figure 5.2 shows a practical example for the eLORETA current density inverse solution corresponding to a single-subject visual evoked potential to pictures of flowers. The free academic eLORETA-KEY software and data are publicly available from the appropriate links at the home page of the KEY Institute for Brain-Mind Research, University of Zurich (http://www.keyinst.uzh.ch). Maximum total current density power occurs at about 100 ms after stimulus onset (shown in panel A). Maximum activation is found in Brodmann areas 17 and 18

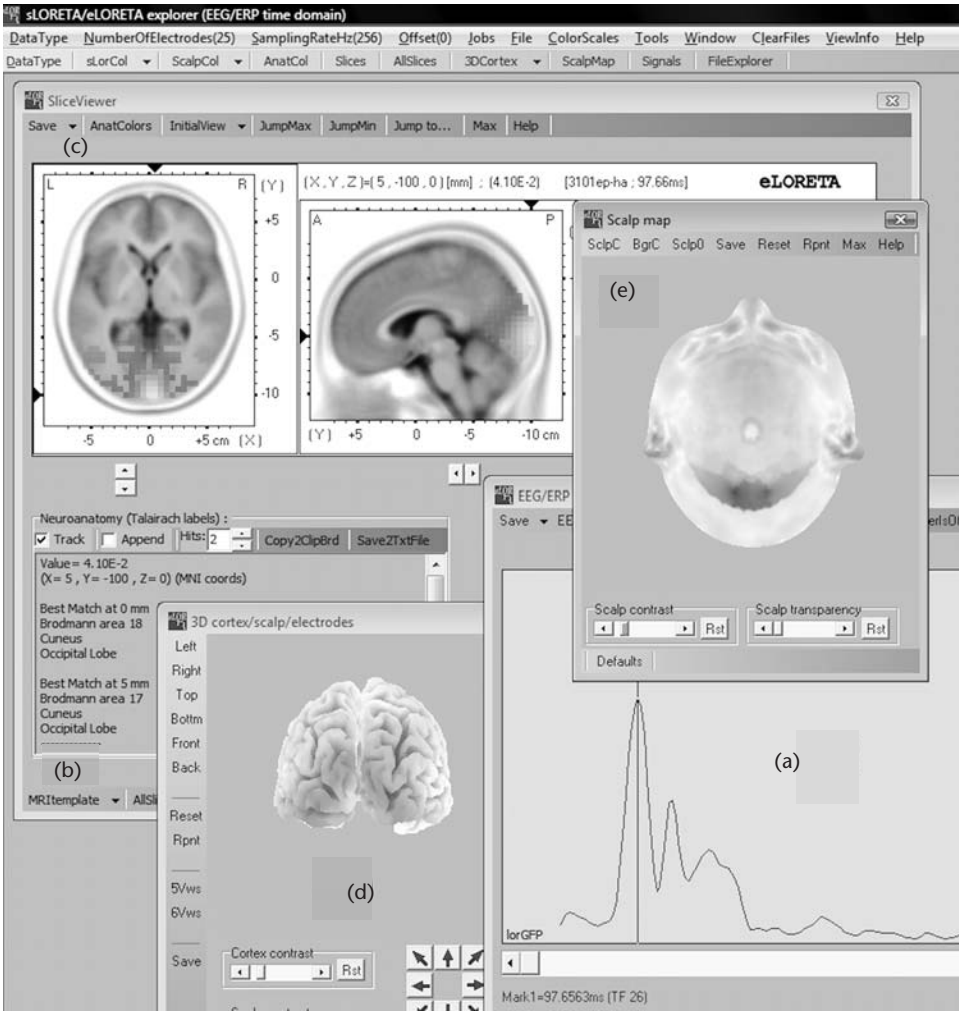



**Figure 5.2** Three-dimensional eLORETA inverse solution displaying estimated current density for a visual evoked potential to pictures of flowers (single-subject data). (a) Maximum current density occurs at about 100 ms after stimulus onset. (b) Maximum activation is found in Brodmann areas 17 and 18. (c) Orthogonal slices through the point of maximum activity. (d) Posterior three-dimensional cortex. (e) Average reference scalp map.

(panel B). Panel C shows orthogonal slices through the point of maximum current density. Panel D shows the posterior three-dimensional cortex. Panel E shows the average reference scalp electric potential map.

### 5.4.7 Other Formulations and Methods

A variety of very fruitful approaches to the inverse EEG problem exist that lie outside the class of discrete, three-dimensional distributed, linear imaging methods. In what follows, some noteworthy exemplary cases are mentioned.

The beamformer methods [46, 47, 50, 51] have mostly been employed in MEG studies, but are readily applicable to EEG measurements. Beamformers can be seen as a spatial filtering approach to source localization. Mathematically, the beamformer estimate of activity is based on a weighted sum of the scalp potentials. This might appear to be a linear method, but the weights require and depend on the time-varying EEG measurements themselves, which implies that the method is not a linear one. The method is particularly well suited to the case in which EEG activity is generated by a small number of dipoles whose time series have low correlation. The method tends to fail in the case of correlated sources. It must also be stressed that this method is an imaging technique that does not estimate the current density, which means that there is no control over how well the image complies with the actual EEG measurements.

The functionals in (5.24) and (5.28) have a dual interpretation. On the one hand, they are conventional forms studied in mathematical functional analyses [25]. On the other hand, they can be derived from a Bayesian formulation of the inverse problem [52]. Recently, the Bayesian approach has been used in setting up very complicated and rich forms of the inverse problem, in which many conditions can be imposed (in a soft or hard fashion) on the properties of the inverse solution at many levels. An interesting example with many layers of conditions on the solution and its properties can be studied in [53]. In general, this technique does not directly estimate the current density, but instead gives some probability measure of the current density. In addition, these methods are nonlinear and are very computer intensive (a problem that is less important with the development of faster CPUs).

Another noteworthy approach to the inverse problem is to consider models that take into account the temporal properties of the current density. If the assumptions on dynamics are correct, the model will very likely perform better than the simple instantaneous models considered in the previous sections. One example of such an approach is [54].

## 5.5 Selecting the Inverse Solution

We are in a situation in which many possible tomographies are available from which to choose. The question of selecting the best solution is now essential. For instance:

1. Is there any way to know which method is correct?
2. If we cannot answer the first question, then at least is there any way to know which method is best?

The first question is the most important one, but it is so ill posed that it does not have an answer: There is no way to be certain of the validity of a given solution, unless it is validated by independent methods. This means that the best we can do is to validate the estimated localizations with some "ground truth," if available.

The second question is also difficult to answer, because there are different criteria for judging the quality of a solution. Pascual-Marqui and others [1, 26, 31, 45] used the following arguments for selecting the "least worst" (as opposed to the possibly nonexistent "best") discrete, three-dimensional distributed, linear tomography:

1. The "least worst" linear tomography is the one with minimum localization error.
2. In a linear tomography, the localization properties can be determined by using test-point sources, based on the principles of linearity and superposition.
3. If a linear tomography is incapable of zero-error localization for test-point sources that are active one at a time, then the tomography will certainly be incapable of zero-error localization to two or more simultaneously active sources.

Based on these criteria, sLORETA and eLORETA are the only linear tomographies that have no localization bias, even under nonideal conditions of measurement and biological noise.

These criteria are difficult to apply to nonlinear methods, for the simple reason that in such a case the principles of linearity and superposition do not hold. Unlike the case of simple linear methods, in the case of nonlinear methods, uncertainty will remain if the method localizes well in general.

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CHAPTER 6

# Epilepsy Detection and Monitoring

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Epilepsy is one of the world's most common neurological diseases, affecting more than 40 million people worldwide. Epilepsy's hallmark symptom, seizures, can have a broad spectrum of debilitating medical and social consequences. Although antiepileptic drugs have helped treat millions of patients, roughly a third of all patients are unresponsive to pharmacological intervention.

As our understanding of this dynamic disease evolves, new possibilities for treatment are emerging. An area of great interest is the development of devices that incorporate algorithms capable of detecting early onset of seizures or even predicting them hours before they occur. This lead time will allow for new types of interventional treatment. In the near future a patient's seizure may be detected and aborted before physical manifestations begin. In this chapter we discuss the algorithms that will make these devices possible and how they have been implemented to date. We investigate how wavelets, synchronization, Lyapunov exponents, principal component analysis, and other techniques can help investigators extract information about impending seizures. We also compare and contrast these measures, and discuss their individual strengths and weaknesses. Finally, we illustrate how these techniques can be brought together in a closed-loop seizure prevention system.

## 6.1 Epilepsy: Seizures, Causes, Classification, and Treatment

Epilepsy is a common chronic neurological disorder characterized by recurrent, unprovoked seizures [1, 2]. Epilepsy is the most common neurological condition in children and the third most common in adults after Alzheimer's and stroke. The World Health Organization estimates that there are 40 to 50 million people with epilepsy worldwide [3]. Seizures are transient epochs due to abnormal, excessive, or synchronous neuronal activity in the brain [2]. *Epilepsy* is a generic term used to define a family of seizure disorders. A person with recurring seizures is said to have epilepsy. Currently there is no cure for epilepsy. Many patients' seizures can be controlled, but not cured, with medication. Those resistant to the medication may become candidates for surgical intervention. Not all epileptic syndromes are lifelong conditions; some forms are confined to particular stages of childhood. Epilepsy

should not be understood as a single disorder, but rather as a group of syndromes with vastly divergent symptoms all involving episodic abnormal electrical activity in the brain.

Roughly 70% of cases present with no known cause. Of the remaining 30%, the following are the most frequent causes: brain tumor and/or stroke; head trauma, especially from automobile accidents, gunshot wounds, sports accidents, and falls and blows; poisoning, such as lead poisoning, and substance abuse; infection, such as meningitis, viral encephalitis, lupus erythematosus and, less frequently, mumps, measles, diphtheria, and others; and maternal injury, infection, or systemic illness that affects the developing brain of the fetus during pregnancy.

All people inherit varying degrees of susceptibility to seizures. The genetic factor is assumed to be greater when no specific cause can be identified. Mutations in several genes have been linked to some types of epilepsy. Several genes that code for protein subunits of voltage-gated and ligand-gated ion channels have been associated with forms of generalized epilepsy and infantile seizure syndromes [4]. One interesting finding in animals is that repeated low-level electrical stimulation (kindling) to some brain sites can lead to permanent increases in seizure susceptibility. Certain chemicals can also induce seizures. One mechanism proposed for this is called excitotoxicity.

Epilepsies are classified in five ways: their etiology; semiology, observable manifestations of the seizures; location in the brain where the seizures originate; identifiable medical syndromes; and the event that triggers the seizures, such as flashing lights. This classification is based on observation (clinical and EEG) rather than underlying pathophysiology or anatomy. In 1989, the International League Against Epilepsy proposed a classification scheme for epilepsies and epileptic syndromes. It is broadly described as a two-axis scheme having the cause on one axis and the extent of localization within the brain on the other.

There are many different epilepsy syndromes, each presenting with its own unique combination of seizure type, typical age of onset, EEG findings, treatment, and prognosis. *Temporal lobe epilepsy* is the most common epilepsy of adults. In most cases, the epileptogenic region is found in the mesial temporal structures (e.g., the hippocampus, amygdala, and parahippocampal gyrus). Seizures begin in late childhood or adolescence. There is an association with febrile seizures in childhood, and some studies have shown herpes simplex virus (HSV) DNA in these regions, suggesting this epilepsy has an infectious etiology. Most of these patients have complex partial seizures sometimes preceded by an aura, and some temporal lobe epilepsy patients also suffer from secondary generalized tonic-clonic seizures. *Absence epilepsy* is the most common childhood epilepsy and affects children between the ages of 4 and 12 years of age. These patients have recurrent absence seizures that can occur hundreds of times a day. On their EEG, one finds the stereotypical generalized 3-Hz spike and wave discharges.

The first line of epilepsy treatment is anticonvulsant medication. In some cases the implantation of a vagus nerve stimulator or a special ketogenic diet can be helpful. Neurosurgical operations for epilepsy can be palliative, reducing the frequency or severity of seizures; however, in some patients, an operation can be curative. Although antiepileptic drug treatment is the standard therapy for epilepsy, one third of all patients remain unresponsive to currently available medication. There is gen-

eral agreement that, despite pharmacological and surgical advances in the treatment of epilepsy, seizures cannot be controlled in many patients and there is a need for new therapeutic approaches [5–7]. Of those unresponsive to anticonvulsant medication, 7% to 8% may profit from epilepsy surgery. However, about 25% of people with epilepsy will continue to experience seizures even with the best available treatment [8]. Unfortunately for those responsive to medication, many antiepileptic medicines have significant side effects that have a negative impact on quality of life. Some side effects can be of particular concern for women, children, and the elderly.

For these reasons, the need for more effective treatments for pharmacoresistant epilepsy was among the driving force behind a White House–initiated Curing Epilepsy: Focus on the Future (Cure) Conference held in March 2000 that emphasized specific research directions and benchmarks for the development of effective and safe treatment for people with epilepsy. There is growing awareness that the development of new therapies has slowed, and to move toward new and more effective therapies, novel approaches to therapy discovery are needed [9]. A growing body of research indicates that controlling seizures may be possible by employing a seizure prediction, closed-loop treatment strategy. If it were possible to predict seizures with high sensitivity and specificity, even seconds before their onset, therapeutic possibilities would change dramatically [10]. One might envision a simple warning system capable of decreasing both the risk of injury and the feeling of helplessness that results from seemingly unpredictable seizures. Most people with epilepsy seize without warning. Their seizures can have dangerous or fatal consequences especially if they come at a bad time and lead to an accident. In the brain, identifiable electrical changes precede the clinical onset of a seizure by tens of seconds, and these changes can be recorded in an EEG.

The early detection of a seizure has many potential benefits. Advanced warning would allow patients to take action to minimize their risk of injury and, possibly in the near future, initiate some form of intervention. An automatic detection system could be made to trigger pharmacological intervention in the form of fast-acting drugs or electrical stimulation. For patients, this would be a significant breakthrough because they would not be dependent on daily anticonvulsant treatment. Seizure prediction techniques could conceivably be coupled with treatment strategies aimed at interrupting the process before a seizure begins. Treatment would then only occur when needed, that is, on demand and in advance of an impending seizure. Side effects from treatment with antiepileptic drugs, such as sedation and clouded thinking, could be reduced by on-demand release of a short-acting drug or electrical stimulation during the preictal state.

Paired with other suitable interventions, such applications could reduce morbidity and mortality as well as greatly improve the quality of life for people with epilepsy. In addition, identifying a preictal state would greatly contribute to our understanding of the pathophysiological mechanisms that generate seizures. We discuss the most available seizure detection and prediction algorithms as well as their potential use and limitations in later sections in this chapter. First, however, we review the dynamic aspects of epilepsy and the most widely used approached to detect and predict epileptic seizures.

## 6.2 Epilepsy as a Dynamic Disease

The EEG is a complex signal. Its statistical properties depend on both time and space [11]. Characteristics of the EEG, such as the existence of limit cycles (alpha activity, ictal activity), instances of bursting behavior (during light sleep), jump phenomena (hysteresis), amplitude-dependent frequency behavior (the smaller the amplitude the higher the EEG frequency), and existence of frequency harmonics (e.g., under photic driving conditions), are among the long catalog of properties typical of nonlinear systems. The presence of nonlinearities in EEGs recorded from an epileptogenic brain further supports the concept that the epileptogenic brain is a nonlinear system. By applying techniques from nonlinear dynamics, several researchers have provided evidence that the EEG of the epileptic brain is a nonlinear signal with deterministic and perhaps chaotic properties [12–14].

The EEG can be conceptualized as a series of numerical values (voltages) over time and space (gathered from multiple electrodes). Such a series is called a *multivariate time series*. The standard methods for time-series analysis (e.g., power analysis, linear orthogonal transforms, and parametric linear modeling) not only fail to detect the critical features of a time series generated by an autonomous (no external input) nonlinear system, but may falsely suggest that most of the series is random noise [15]. In the case of a multidimensional, nonlinear system such as the EEG generators, we do not know, or cannot measure, all of the relevant variables. This problem can be overcome mathematically. For a dynamical system to exist, its variables must be related over time. Thus, by analyzing a single variable (e.g., voltage) over time, we can obtain information about the important dynamic features of the whole system. By analyzing more than one variable over time, we can follow the dynamics of the interactions of different parts of the system under investigation. Neuronal networks can generate a variety of activities, some of which are characterized by rhythmic or quasirhythmic signals. These activities are reflected in the corresponding local EEG field potential. An essential feature of these networks is that variables of the network have both a strong nonlinear range and complex interactions. Therefore, they belong to a general class of nonlinear systems with complex dynamics. Characteristics of the dynamics depend strongly on small changes in the control parameters and/or the initial conditions. Thus, real neuronal networks behave like nonlinear complex systems and can display changes between states such as small-amplitude, quasirandom fluctuations and large-amplitude, rhythmic oscillations. Such dynamic state transitions are observed in the brain during the transition between interictal and epileptic seizure states.

One of the unique properties of the brain as a system is its relatively high degree of plasticity. It can display adaptive responses that are essential to implementing higher functions such as memory and learning. As a consequence, control parameters are essentially plastic, which implies that they can change over time depending on previous conditions. In spite of this plasticity, it is necessary for the system to stay within a stable working range in order for it to maintain a stable operating point. In the case of the patient with epilepsy, the most essential difference between a normal and an epileptic network can be conceptualized as a decrease in the distance between operating and bifurcation points.

In considering epilepsies as dynamic diseases of brain systems, Lopes da Silva and colleagues proposed two scenarios of how a seizure could evolve [11]. The first is that a seizure could be caused by a sudden and abrupt state transition, in which case it would not be preceded by detectable dynamic changes in the EEG. Such a scenario would be conceivable for the initiation of seizures in primary generalized epilepsy. Alternatively, this transition could be a gradual change or a cascade of changes in dynamics, which could in theory be detected and even anticipated.

In the sections that follow, we use these basic concepts of brain dynamics and review the state-of-the-art seizure detection and seizure prediction methodologies and give examples using real data from human and rat epileptic time series.

## 6.3 Seizure Detection and Prediction

The majority of the current state-of-the-art techniques used to detect or predict an epileptic seizure involve linearly or nonlinearly transforming the signal using one of several mathematical black boxes, and subsequently trying to predict or detect the seizure based off the output of the black box. These black boxes include some purely mathematical transformations, such as the Fourier transform, or some class of machine learning techniques, such as artificial neural networks, or some combination of the two. In this section, we review some of the techniques for detection and prediction of seizures that have been reported in the literature.

Many techniques have been used in an attempt to detect epileptic seizures in the EEG. Historically, a visual confirmation was used to detect seizures. The onset and duration of a seizure could be identified on the EEG by a qualified technician. Figure 6.1 is an example of a typical spontaneous seizure in a laboratory animal model. Recently much research has been put into trying to predict or detect a seizure based off the EEG. The majority of these techniques use some kind of time-series analysis method to detect seizures offline. Time-series analysis of an EEG in general falls under one of the following two groups:

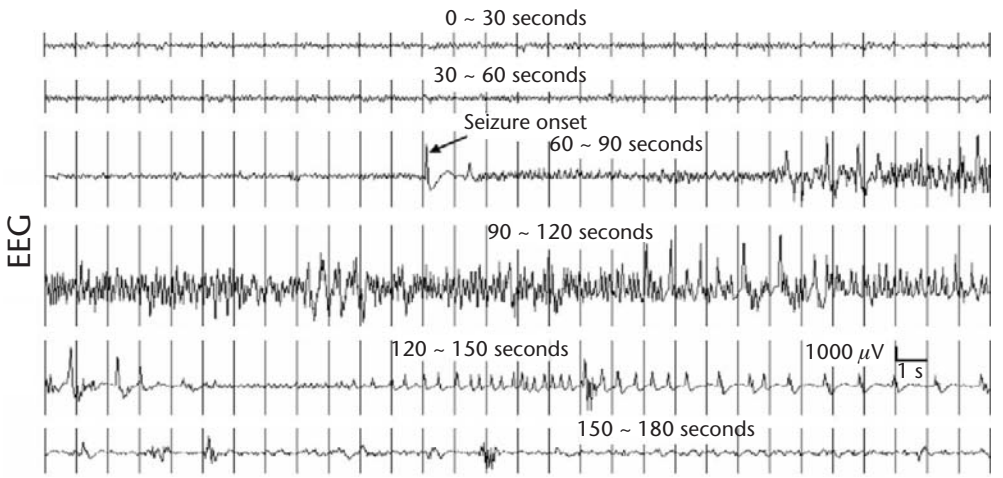



**Figure 6.1** Three minutes of EEG (demonstrated by six sequential 30-second segments) data recorded from the left hippocampus, showing a sample seizure from an epileptic rat.

1. *Univariate time-series analyses* are time-series analyses that consist of a single observation recorded sequentially over equal time increments. Some examples of univariate time series are the stock price of Microsoft, daily fluctuations in humidity levels, and single-channel EEG recordings. Time is an implicit variable in the time series. Information on the start time and the sampling rate of the data collection can allow one to visualize the univariate time series graphically as a function of time over the entire duration of data recording. The information contained in the amplitude value of the recorded EEG signal sampled in the form of a discrete time series $x(t) = x(t_i) = x(i\Delta t)$, ($i = 1, 2, ..., N$ and $\Delta t$ is the sampling interval) can also be encoded through the amplitude and the phase of the subset of harmonic oscillations over a range of different frequencies. Time-frequency methods specify the map that translates between these representations.
2. *Multivariate time-series analyses* are time-series analyses that consist of more than one observation recorded sequentially in time. Multivariate time-series analysis is used when one wants to understand the interaction between the different components of the system under consideration. Examples include records of stock prices and dividends, concentration of atmospheric CO and global temperature, and multichannel EEG recordings. Time again is an implicit variable.

In the following sections some of the most commonly used measures for EEG time-series analysis will be discussed. First, a description of the linear and nonlinear univariate measures that operate on single-channel recordings of EEG data is given. Then some of the most commonly utilized multivariate measures that operate on more than a single channel of EEG data are described.

The techniques discussed next were chosen because they are representative of the different approaches used in seizure detection. Time–frequency analysis, nonlinear dynamics, signal correlation (synchronization), and signal energy are very broad domains and could be examined in a number of ways. Here we review a subset of techniques, examine each, and discuss the principles behind them.

## 6.4 Univariate Time-Series Analysis

### 6.4.1 Short-Term Fourier Transform

One of the more widely used techniques for detecting or predicting an epileptic seizure is based on calculating the power spectrum of one or more channels of the EEG. The core hypothesis, stated informally, is that the EEG signal, when partitioned into its component periodic (sine/cosine) waves, has a signature that varies between the ictal and the interictal states. To detect this signature, one takes the Fourier transform of the signal and finds the frequencies that are most prominent (in amplitude) in the signal. It has been shown that there is a relationship between the power spectrum of the EEG signal and ictal activity [16]. Although there appears to be a correlation between the power spectrum and ictal activity, the power spectrum is not used as a stand-alone detector of a seizure. In general, it is coupled with some other time-series prediction technique or machine learning to detect a seizure.

The Fourier transform is a generalization of the Fourier series. It breaks up any time-varying signal into its frequency components of varying magnitude and is defined in (6.1).

$$F(k) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i k x} dx \tag{6.1}$$

Due to Euler's formula, this can also be written as shown in (6.2) for any complex function $f(t)$ where $k$ is the $k$th harmonic frequency:

$$F(k) = \int_{-\infty}^{\infty} f(t) \cos(-2\pi k x) dx + \int_{-\infty}^{\infty} f(t) i \sin(-2\pi k x) dx \tag{6.2}$$

We can represent any time-varying signal as a summation of sine and cosine waves of varying magnitudes and frequencies [17]. The Fourier transform is represented with the power spectrum. The power spectrum has a value for each harmonic frequency, which indicates how strong that frequency is in the given signal. The magnitude of this value is calculated by taking the modulus of the complex number that is calculated from the Fourier transform for a given frequency ($|F(k)|$).

Stationarity is an issue that needs to be considered when using the Fourier transform. A stationary signal is one that is constant in its statistical parameters over time, and is assumed by the Fourier transform to be present. A signal that is made up of different frequencies at different times will yield the same transform as a signal that is made up of those same frequencies for the entire time period considered. As an example, consider two functions $f_1$ and $f_2$ over the domain $0 \le t \le T$, for any two frequencies $\omega_1$ and $\omega_2$ shown in (6.3) and (6.4):

$$f_1(t) = \sin(2\pi\omega_1 t) + \cos(2\pi\omega_2 t) \text{ if } 0 \le t < T \tag{6.3}$$

and

$$f_2(t) = \begin{cases} \sin(2\pi\omega_1 t) \text{ if } 0 \le t < T/2 \\ \cos(2\pi\omega_2 t) \text{ if } T/2 \le t < T \end{cases} \tag{6.4}$$

When using the short-term Fourier transform, the assumption is made that the signal is stationary for some small period of time, $T_s$. The Fourier transform is then calculated for segments of the signal of length $T_s$. The short-term Fourier transform at time $t$ gives the Fourier transform calculated over the segment of the signal lasting from $(t - T_s)$ to $t$. The length of $T_s$ determines the resolution of the analysis. There is a trade-off between time and frequency resolution. A short $T_s$ yields better time resolution, but it limits the frequency resolution. The opposite of this is also true; a long $T_s$ increases frequency resolution while decreasing the time resolution of the output. Wavelet analysis overcomes this limitation, and offers a tool that can maintain both time and frequency resolution. An example of Fourier transform calculated prior to, during, and following an epileptic seizure is given in Figure 6.2.

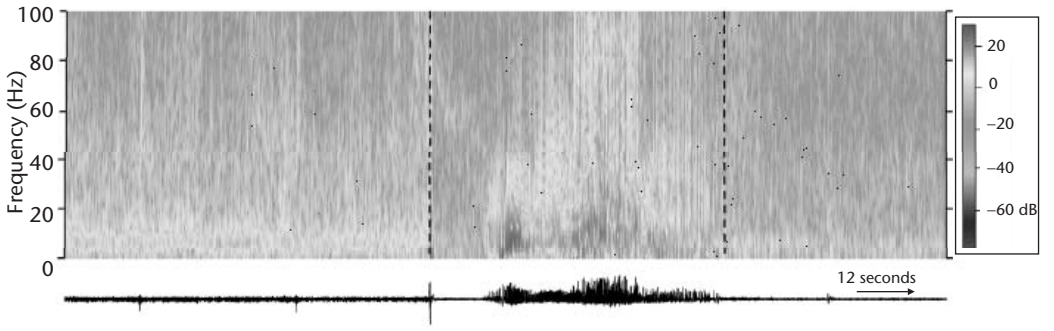



**Figure 6.2** Time-frequency spectrum plot for 180-second epoch of seizure. Black dotted lines mark the onset time and the offset times of the seizure.

### 6.4.2 Discrete Wavelet Transforms

*Wavelets* are another closely related method used to predict epileptic seizures. Wavelet transforms follow the principle of superposition, just like Fourier transforms, and assume EEG signals are composed of various elements from a set of parameterized basis functions. Rather than being limited to sine and cosine wave functions, however, as in a Fourier transform, wavelets have to meet other mathematical criteria, which allow the basis functions to be far more general than those for simple sine/cosine waves. Wavelets make it substantially easier to approximate choppy signals with sharp spikes, as compared to the Fourier transform. The reason for this is that sine (and cosine) waves have infinite support (i.e., stretch out to infinity in time), which makes it difficult to approximate a spike. Wavelets are allowed to have finite support, so a spike in the EEG signal can be easily estimated by changing the magnitude of the component basis functions.

The discrete wavelet transform is similar to the Fourier transform in that it will break up any time-varying signal into smaller uniform functions, known as the basis functions. The basis functions are created by scaling and translating a single function of a certain form. This function is known as the mother wavelet. In the case of the Fourier transform, the basis functions used are sine and cosine waves of varying frequency and magnitude. Note that a cosine wave is just a sine wave translated by $\pi/2$ radians, so the mother wavelet in the case of the Fourier transform could be considered to be the sine wave. However, for a wavelet transform the basis functions are more general. The only requirements for a family of functions to be a basis is that the functions are both complete and orthonormal under the inner product. Consider the family of functions $\Psi = \{\psi_{ij} | -\infty < i,j < \infty\}$ where each $i$ value specifies a different scale and each $j$ value specifies a different translation based off of some mother wavelet function. Note that $\Psi$ is considered to be complete if any continuous function $f$, defined over the real line $x$, can be defined by some combination of the functions in $\Psi$ as shown in (6.5) [17]:

$$f(x) = \sum_{i,j=-\infty}^{\infty} c_{ij}\psi_{ij}(x) \tag{6.5}$$

In order for a family of functions to be orthonormal under the inner product, they must meet two criteria. It must be the case that for any $i$, $j$, $l$, and $m$ where $i \neq l$ and $j \neq m$ that $<\psi_{ij}, \psi_{lm}> \geq 0$ and $<\psi_{ij}, \psi_{ij}> \geq 1$, where $<f, g>$ is the inner product and is defined as in (6.6) and $f(x)^*$ is the complex conjugate of $f(x)$:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)^* g(x) dx \tag{6.6}$$

The wavelet basis is very similar to the Fourier basis, with the exception that the wavelet basis does not have to be infinite. In a wavelet transform the basis functions can be defined over a certain window and then be zero everywhere else. As long as the family of functions defined by scaling and translating the mother wavelet is orthonormally complete, that family of functions can be used as the basis. With the Fourier transform, the basis is made up of sine and cosine waves that are defined over all values of $x$ where $-\infty < x < \infty$.

One of the simplest wavelets is the *Haar wavelet* (Daubechies 2 wavelet). In a manner similar to the Fourier series, any continuous function $f(x)$ defined on [0, 1] can be represented using the expansion shown in (6.7). The $h_{j,k}(x)$ term is known as the Haar wavelet function and is defined as shown in (6.8); $p_{j,k}(x)$ is known as the Haar scaling function and is defined in (6.9) [17]:

$$f(x) = \sum_{j=J}^{\infty} \sum_{k=0}^{2^j - 1} \langle f, h_{j,k} \rangle h_{j,k}(x) + \sum_{k=0}^{2^J - 1} \langle f, p_{J,k} \rangle p_{J,k}(x) \tag{6.7}$$

$$h_{j,k}(x) = \begin{cases} 2^{j/2} & \text{if } 0 \leq 2^j x - k < 1/2 \\ -2^{j/2} & \text{if } 1/2 \leq 2^j x - k < 1 \\ 0 & \text{otherwise} \end{cases} \tag{6.8}$$

$$p_{J,k}(x) = \begin{cases} 2^{J/2} & \text{if } 0 \leq 2^j x - k < 1 \\ 0 & \text{otherwise} \end{cases} \tag{6.9}$$

The combination of the Haar scaling function at the largest scale, along with the Haar wavelet functions, creates a set of functions that provides an orthonormal basis for functions in $\mathbb{R}^2$.

Wavelets and short-term Fourier transforms also serve as the foundation for other measures. Methods such as the spectral entropy method calculate some feature based on the power spectrum. Entropy was first used in physics as a thermodynamic quantity describing the amount of disorder in a system. Shannon extended its application to information theory in the late 1940s to calculate the entropy for a given probability distribution [18]. The entropy measure that Shannon developed can be expressed as follows:

$$H = -\sum p_k \log p_k \tag{6.10}$$

Entropy is a measure of how much information there is to learn from a random event occurring. Events that are unlikely to occur yield more information than events that are very probable. For spectral entropy, the power spectrum is consid-

ered to be a probability distribution. This insinuates that the random events would be that the signal was made up of a sine or cosine wave of a given frequency. The spectral entropy allows us to calculate the amount of information there is to be gained from learning the frequencies that make up the signal. When the Fourier transform is used, nonstationary signals need to be accounted for. To do this, the short-term Fourier transform is used to calculate the power spectrum over small parts of the signal rather than the entire signal itself. The spectral entropy is an indicator of the number of frequencies that make up a signal. A signal made up of many different frequencies (white noise, for example) would have a uniform distribution and therefore yield high spectral entropy, whereas a signal made up of a single frequency would yield low spectral entropy.

In practice, wavelets have been applied to electrocorticogram (ECoG) signals in an effort to try to predict seizures. In one report, the authors first partitioned the ECoG signal into seizure and nonseizure components using a wavelet-based filter. This filter was not specifically predictive of seizures. It flagged any increase in power or shift in frequency, whether this change in the signal was caused by a seizure, an interictal epileptiform discharge, or merely normal activity. After the filter decomposed the signal down into its components, it was passed through a second filter that tried to isolate the seizures from the rest of the events. By decomposing the ECoG signal into components and passing it through the second step of isolating the seizures, the authors were able to detect all seizures with an average of 2.8 false positives per hour [19]. Unfortunately, this technique did not allow them to predict (as opposed to detect) seizures.

### 6.4.3 Statistical Moments

When a cumulative distribution function for a random variable cannot be determined, it is possible to describe an approximation to the distribution of this variable using moments and functions of moments [20]. Statistical moments relate information about the distribution of the amplitude of a given signal. In probability theory, the $k$th moment is defined as in (6.11) where $E[x]$ is the expected value of $x$:

$$\mu_{k'} = E\left[x^k\right] = \int x^k p(x) \tag{6.11}$$

The first statistical moment is the mean of the distribution being considered.

In general, the statistical moments are taken about the mean. This is also known as the $k$th central moment and is defined by (6.12) where $\mu$ is the mean of the dataset considered [20]:

$$\mu_k = E\left[(x-\mu)\right]^k = \int (x-\mu)^k p(x) \tag{6.12}$$

The second moment about the mean would give the variance. The third and fourth moments about the mean would produce the skew and kurtosis, respectively. The skew of a distribution indicates the amount of asymmetry in that distribution, whereas the kurtosis shows the degree of peakedness of that distribution. The absolute value of the skewness $|\mu_3|$ was used for seizure prediction in a review by Mormann et al. [14]. The paper showed that skewness was not able to significantly

predict a seizure by detecting the state change from interictal to preictal. Although unable to predict seizures, statistical moments may prove valuable as seizure detectors in recordings with large amplitude seizures.

### 6.4.4 Recurrence Time Statistics

The recurrence time statistic (RTS), T1, is a characteristic of trajectories in an abstract dynamical system. Stated informally, it is a measure of how often a given trajectory of the dynamical system visits a certain neighborhood in the phase space. T1 has been calculated for ECoG data in an effort to detect seizures, with significant success. With two different patients and a total of 79 hours of data, researchers were able to detect 97% of the seizures with only an average of 0.29 false negatives per hour [21]. They did not, however, indicate any attempts to predict seizures. Results from our preliminary studies on human EEG signals showed that the RTS exhibited significant change during the ictal period that is distinguishable from the background interictal period (Figure 6.3). In addition, through the observations over multichannel RTS features, the spatial pattern from channel to channel can also be traced. Existence of these spatiotemporal patterns of RTS suggests that it is possible to utilize RTS to develop an automated seizure-warning algorithm.

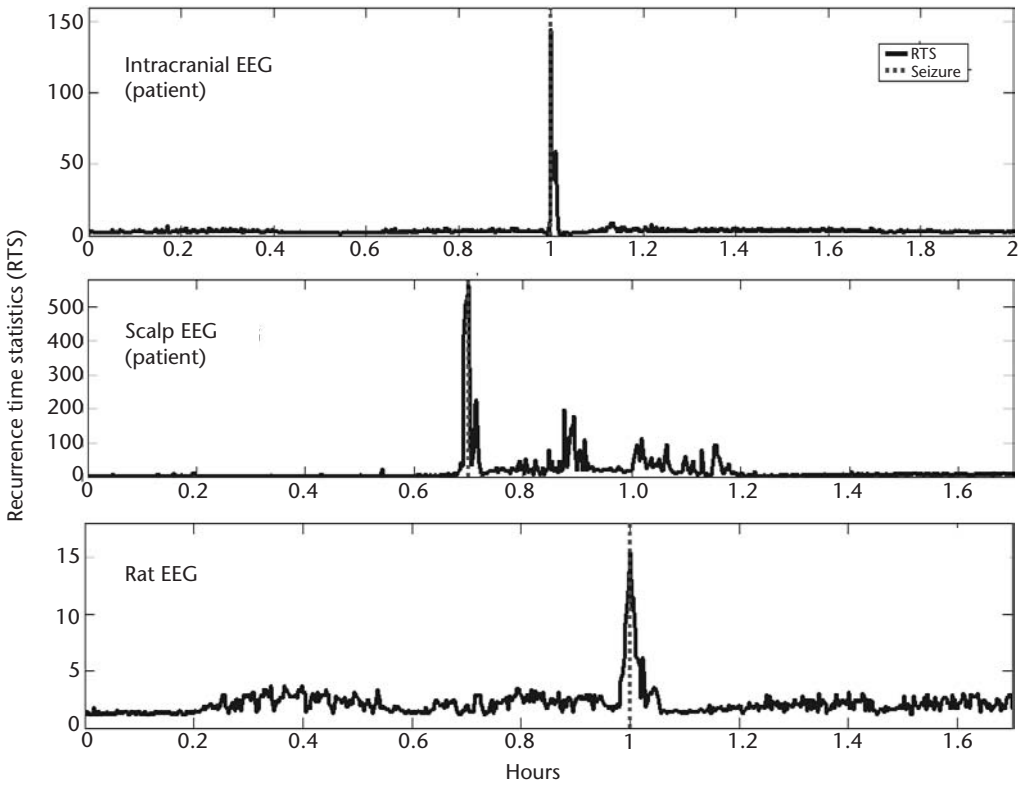



**Figure 6.3** Studies on human EEG signals show that the recurrence time statistics exhibit changes during the ictal period that is distinguishable from the background interictal period.

### 6.4.5 Lyapunov Exponent

During the past decade, several studies have demonstrated experimental evidence that temporal lobe epileptic seizures are preceded by changes in dynamic properties of the EEG signal. A number of nonlinear time-series analysis tools have yielded promising results in terms of their ability to reveal preictal dynamic changes essential for actual seizure anticipation.

It has been shown that patients go through a preictal transition approximately 0.5 to 1 hour before a seizure occurs, and this preictal state can be characterized by the *Lyapunov exponent* [12, 22–29]. Stated informally, the Lyapunov exponent measures how fast nearby trajectories in a dynamical system diverge. The noted approach therefore treats the epileptic brain as a dynamical system [30–32]. It considers a seizure as a transition from a chaotic state (where trajectories are sensitive to initial conditions) to an ordered state (where trajectories are insensitive to initial conditions) in the dynamical system. The Lyapunov exponent is a nonlinear measure of the average rate of divergence/convergence of two neighboring trajectories in a dynamical system dependent on the sensitivity of initial conditions. It has been successfully used to identify preictal changes in EEG data [22–24]. Generally, Lyapunov exponents can be estimated from the equation of motion describing the time evolution of a given dynamical system. However, in the absence of the equation of motion describing the trajectory of the dynamical system, Lyapunov exponents are determined from observed scalar time-series data, $x(t_n) = x(n\,\delta t)$, where $\delta t$ is the sampling rate for the data acquisition. In this situation, the goal is to generate a higher dimensional vector embedding of the scalar data $x(t)$ that defines the state space of the multivariate brain dynamics from which the scalar EEG data is derived. Heuristically, this is done by constructing a higher dimensional vector $\mathbf{x}_i$ from the data segment $x(t)$ of given duration $T$, as shown in (6.13) with $\tau$ defining the embedding delay used to construct a higher dimensional vector $\mathbf{x}$ from $x(t)$ with $d$ as the selected dimension of the embedding space and $t_i$ being the time instance within the period $[T - (d-1)\tau]$:

$$\mathbf{x}_i = \left[x(t_i), x(t_i - \tau), \ldots, x\left(t_i - (d-1)\tau\right)\right] \tag{6.13}$$

The geometrical theorem of [33] tells us that for an appropriate choice of $d > d_{\min}$, $x_i$ provides a faithful representation of the phase space for the dynamical systems from which the scalar time series was derived. A suitable practical choice for $d$, the embedding dimension, can be derived from the "false nearest neighbor" algorithm. In addition, a suitable prescription for selecting the embedding delay, $\tau$, is also given in Abarbanel [34]. From $\mathbf{x}_i$ a most stable short-term estimation of the largest Lyapunov exponent can be performed that is referred to as the *short-term largest Lyapunov exponent* ($\text{STL}_{\max}$) [24]. The estimation $L$ of $\text{STL}_{\max}$ is obtained using (6.14) where $\delta x_{ij}(0) = x(t_i) - x(t_j)$ is the displacement vector, defined at time points $t_i$ and $t_j$ and $\delta x_{ij}(\Delta t) = x(t_i + \Delta t) - x(t_j + \Delta t)$ is the same vector after time $\Delta t$, and where $N$ is the total number of local $\text{STL}_{\max}$ that will be estimated within the time period $T$ of the data segment, where $T = N\Delta t + (d-1)\tau$:

$$L = \frac{1}{N\Delta t}\sum_{i=1}^{N}\log_2\left|\frac{\delta x_{ij}(\Delta t)}{\delta x_{ij}(0)}\right| \tag{6.14}$$

A decrease in the Lyapunov exponent indicates this transition to a more ordered state (Figure 6.4). The assumptions underlying this methodology have been experimentally observed in the $STL_{max}$ time-series data from human patients [18, 26] and rodents [35]. For instance, in an experimental rat model of temporal lobe epilepsy, changes in the phase portrait of $STL_{max}$ can be readily identified for the preictal, ictal, and postictal states, during a spontaneous limbic seizure (Figure 6.5). This characterization by the Lyapunov exponent has, however, been successful only for

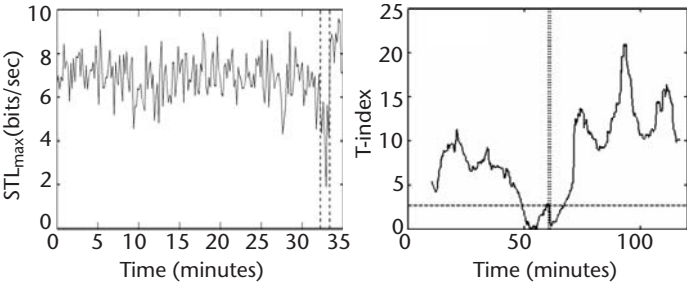



**Figure 6.4** Sample $STL_{max}$ profile for a 35-minute epoch including a grade 5 seizure from an epileptic rat. Seizure onset and offset are indicated by dashed vertical lines. Note the drop in the $STL_{max}$ value during the seizure period. (b) T-index profiles calculated from $STL_{max}$ values of a pair of electrodes from rat A. The electrode pair includes a right hippocampus electrode and a left frontal electrode. Vertical dotted lines represent seizure onset and offset. The horizontal dashed line represents the critical entrainment threshold. Note a decline in the T-index value several minutes before seizure occurrence.

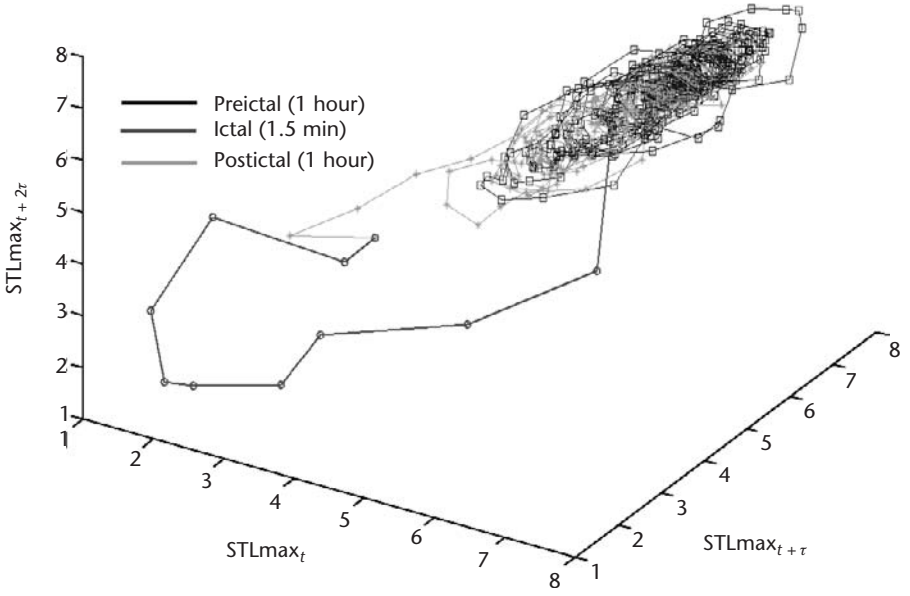



**Figure 6.5** Phase portrait of $STL_{max}$ of a spontaneous rodent epileptic seizure (grade 5).

EEG data recorded from particular areas in the neocortex and hippocampus and has been unsuccessful for other areas. Unfortunately, these areas can vary from seizure to seizure even in the same patient. The method is therefore very sensitive to the electrode sites chosen. However, when the correct sites were chosen, the preictal transition was seen in more than 91% of the seizures. On average, this led to a prediction rate of 80.77% and an average warning time of 63 minutes [28]. Sadly, this method has been plagued by problems related to finding the critical electrode sites because their predictive capacity changes from seizure to seizure.

## 6.5 Multivariate Measures

Multivariate measures take more than one channel of EEG into account simultaneously. This is used to consider the interactions between the channels and how they correlate rather than looking at channels individually. This is useful if there is some interaction (e.g., synchronization) between different regions of the brain leading up to a seizure. Of the techniques discussed in the following sections, the simple synchronization measure and the lag synchronization measure fall under a subset of the multivariate measures, known as bivariate measures. Bivariate measures only consider two channels at a time and define how those two channels correlate. The remaining metrics take every EEG channel into account simultaneously. They do this by using a dimensionality reduction technique called principal component analysis (PCA). PCA takes a dataset in a multidimensional space and linearly transforms the original dataset to a lower dimensional space using the most prominent dimensions from the original dataset. PCA is used as a seizure detection technique itself [36]. It is also used as a tool to extract the most important dimensions from a data matrix containing pairwise correlation information for all EEG channels, as is the case with the correlation structure.

### 6.5.1 Simple Synchronization Measure

Several studies have shown that areas of the brain synchronize with one other during certain events. During seizures abnormally large amounts of highly synchronous activity are seen, and it has been suggested this activity may begin hours before the initiation of a seizure.

One multivariate method that has been used to calculate the synchronization between two EEG channels is a technique suggested by Quiroga et al. [37]. It first defines certain "events" for a pair of signals. Once the events have been defined in the signals, this method then counts the number of times the events in the two signals occur within a specified amount of time ($\tau$) of each other [37]. It then divides this count by a normalizing term equivalent to the maximum number of events that could be synchronized in the signals.

For two discrete EEG channels $x_i$ and $y_i$, $i = 1, \ldots, N$, where $N$ is the number of points making up the EEG signal for the segment considered, event times are defined to be $t_i^x$ and $t_i^y$ ($i = 1, \ldots, m_x$; $j = 1, \ldots, m_y$). An event can be defined to be anything; however, events should be chosen so that the events appear simultaneously across the signals when they are synchronized. Quiroga et al. [37] define an event to be a

local maximum over a range of $K$ values. In other words, the $i$th point in signal $x$ would be an event if $x_i > x_{i \pm k}$, $k = 1, \ldots, K$. The term $\tau$ represents the time within which events from $x$ and $y$ must occur in order to be considered synchronized, and it must be less than half of the minimum interevent distance; otherwise, a single event in one signal could be considered to be synchronized with two different events in the other signal.

Finally, the number of events in $x$ that appear "shortly" (within $\tau$) after an event in $y$ is counted as shown in (6.15) when $J_{ij}\tau$ is defined as in (6.16):

$$c^{\tau}(x|y) = \sum_{i=1}^{m_x} \sum_{j=1}^{m_y} J_{ij}^{\tau} \tag{6.15}$$

$$J_{ij}^{\tau} = \begin{cases} 1 & \text{if } 0 < t_i^x - t_j^y \\ 1/2 & \text{if } t_i^x = t_j^y \\ 0 & \text{else} \end{cases} \tag{6.16}$$

Similarly, the number of events in $y$ that appear shortly after an event in $x$ can also be defined in an analogous way. This would be denoted $c^{\tau}(y|x)$. With these two values, the synchronization measure $Q_{\tau}$ can be calculated. This measure is shown in (6.17):

$$Q_{\tau} = \frac{c^{\tau}(x|y) + c^{\tau}(y|x)}{\sqrt{m_x m_y}} \tag{6.17}$$

The metric is normalized so that $0 \leq Q_{\tau} \leq 1$and $Q_{\tau}$ is 1 if and only if $x$ and $y$ are fully synchronized (i.e., always have corresponding events within $\tau$).

### 6.5.2 Lag Synchronization

When two different systems are identical with the exception of a shift by some time lag $\tau$, they are said to be lag synchronized [38]. This characteristic was tested by Mormann et al. [39] when applied to EEG channels in the interictal and preictal stage. To calculate the similarity of two signals they used a normalized cross-correlation function (6.18) as follows:

$$C(s_a, s_b)(\tau) = \left| \frac{corr(s_a, s_b)(\tau)}{\sqrt{corr(s_a, s_a)(0) \cdot corr(s_b, s_b)(\tau)}} \right| \tag{6.18}$$

where $\text{corr}(s_a, s_b)(\tau)$ represents the linear cross-correlation function between the two time series $s_a(t)$ and $s_b(t)$computed at lag time $\tau$ as defined here:

$$corr(s_a, s_b)(\tau) = \int_{-\infty}^{\infty} s_a(t+\tau) s_b(t) dt \tag{6.19}$$

The normalized cross-correlation function yields a value between 0 and 1, which indicates how similar the two signals ($s_a$ and $s_b$) are. If the normalized

cross-correlation function produces a value close to 1 for a given $\tau$, then the signals are considered to be lag synchronized by a phase of $\tau$. Hence the final feature used to calculate the lag synchronization is the largest normalized cross correlation over all values of $\tau$, as shown in (6.20). A $C_{max}$ value of 1 indicates totally synchronized signals within some time lag $\tau$ and unsynchronized signals produce a value very close to 0.

$$C_{\max} = \max_{\tau}\left\{C(s_a, s_b)(\tau)\right\} \tag{6.20}$$

## 6.6 Principal Component Analysis

Principal component analysis attempts to solve the problem of excessive dimensionality by combining features to reduce the overall dimensionality. By using linear transformations, it projects a high dimensional dataset onto a lower dimensional space so that the information in the original dataset is preserved in an optimal manner when using the least squared distance metric. An outline of the derivation of PCA is given here. The reader should refer to Duda et al. [40] for a more detailed mathematical derivation.

Given a $d$-dimensional dataset of size $n$ ($\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$), we first consider the problem of finding a vector $\mathbf{x}_0$ to represent all of the vectors in the dataset. This comes down to the problem of finding the vector $\mathbf{x}_0$, which is closest to every point in the dataset. We can find this vector by minimizing the sum of the squared distances between $\mathbf{x}_0$ and all of the points in the dataset. In other words, we would like to find the value of $\mathbf{x}_0$ that minimizes the criterion function $J_0$ shown in (6.21):

$$J_0(\mathbf{x}_0) = \sum_{k=1}^{n} \|\mathbf{x}_0 - \mathbf{x}_k\|^2 \tag{6.21}$$

It can be shown that the value of $\mathbf{x}_0$ that minimizes $J_0$ is the sample mean ($1/N$ $\Sigma\mathbf{x}_i$) of the dataset [40]. The sample mean has zero dimensionality and therefore does not give any information about the spread of the data, because it is a single point. To represent this information, the dataset would need to be projected onto a space with some dimensionality. To project the original dataset onto a one-dimensional space, we need to project it onto a line in the original space that runs through the sample mean. The data points in the new space can then be defined by $\mathbf{x} = \mathbf{m} + a\mathbf{e}$. Here, $\mathbf{e}$ is the unit vector in the direction of the line and $a$ is a scalar, which represents the distance from $\mathbf{m}$ to $\mathbf{x}$. A second criterion function $J_1$ can now be defined that calculates the sum of the squared distances between the points in the original dataset and the projected points on the line:

$$J_1(a_1, \ldots, a_n, \mathbf{e}) = \sum_{k=1}^{n} \|(\mathbf{m} + a_k + \mathbf{e}) - \mathbf{x}_k\|^2 \tag{6.22}$$

Taking into consideration that $\|\mathbf{e}\| = 1$, the value of $a_k$ that minimizes $J_1$ is found to be $a_k = \mathbf{e}^t(\mathbf{x}_k - \mathbf{m})$. To find the best direction $\mathbf{e}$ for the line, this value of $a_k$ is substi-

tuted back into (6.22) to get (6.23). Then $J_1$ from (6.23) can be minimized with respect to **e** to find the direction of the line. It turns out that the vector that minimizes $J_1$ is one that satisfies the equation $\mathbf{Se} = \lambda\mathbf{e}$, for some scalar value $\lambda$, where **S** is the scatter matrix of the original dataset as defined in (6.24).

$$J_1(\mathbf{e}) = \sum_{k=1}^{n} a_k^2 - 2\sum_{k=1}^{n} a_k^2 + \sum_{k=1}^{n} \|\mathbf{x}_k - \mathbf{m}\|^2 \tag{6.23}$$

$$\mathbf{S} = \sum_{k=1}^{n} (\mathbf{x}_k - \mathbf{m})(\mathbf{x}_k - \mathbf{m})^t \tag{6.24}$$

Because **e** must satisfy $\mathbf{Se} = \lambda\mathbf{e}$, it is easy to realize that **e** must be an eigenvector of the scatter matrix **S**. In addition to **e** being an eigenvector of **S**, Duda et al. [40] also showed that the eigenvector that yields the best representation of the original dataset is the one that corresponds to the largest eigenvalue. By projecting the data onto the eigenvectors of the scatter matrix that correspond to the *d'* highest eigenvalues, the original dataset can be projected down to a space with dimensionality *d*.

## 6.7 Correlation Structure

One method of seizure analysis is to consider the correlation over all of the recorded EEG channels. To do this, a correlation is defined over the given channels. To define the correlation matrix, a segment of the EEG signal is considered for a given window of a specified time. The EEG signal is then channel-wise normalized within this window. Given *m* channels, the correlation matrix **C** is defined as in (6.25), where $w_l$ specifies the length of the given window ($w$) and $\text{EEG}_i$ is the *i*th channel. The value of $\text{EEG}_i$ has also been normalized to have zero mean and unit variance [6]. The $C_{ij}$ term will yield a value of 0 when $\text{EEG}_i$ and $\text{EEG}_j$ are uncorrelated, a value of 1 when they are perfectly correlated, and a value of –1 when they are anticorrelated. Note also that the correlation matrix is symmetrical since $C_{ij} = C_{ji}$. In addition, $C_{ii} = 1$ for all values of *i* because any signal will be perfectly correlated with itself. It follows that the trace of the matrix ($\Sigma\ C_{ii}$) will always equal the number of channels ($m$).

$$C_{ij} = \frac{1}{w_l} \sum_{t=w} EEG_i(t) \cdot EEG_j(t) \tag{6.25}$$

To simplify the representation of the correlation matrix, the eigenvalues of the matrix are calculated. The eigenvalues reveal which dimensions of the original matrix have the highest correlation. When the eigenvalues ($\lambda_1, \lambda_2, \ldots, \lambda_m$) are sorted so that $\lambda_1 \leq \lambda_2 \leq \ldots \leq \lambda_{\max}$, they can then be used to produce a spectrum of the correlation matrix **C** [41]. This spectrum is sorted by intensity of correlation. The spectrum is then used to track how the dynamics of all *m* EEG channels are affected when a seizure occurs.

## 6.8 Multidimensional Probability Evolution

Another nonlinear technique that has been used for seizure detection is based on a multidimensional probability evolution (MDPE) function. Using the probability density function, changes in the nature of the trajectory of the EEG signal, as it evolves, can be detected. To accomplish the task of detection, the technique tracks how often various parts of the state space are visited when the EEG is in the nonictal state. Using these statistics, anomalies in the dynamics of the system can then be detected, which usually implies the occurrence of a seizure. In one report, when MDPE was applied to test data, it was able to detect all of the seizures that occurred in the data [42]. However, there was no mention of the number of false positives, false negatives, or if the authors had tried to predict seizures at all.

## 6.9 Self-Organizing Map

The techniques just described are all based on particular mathematical transformations of the EEG signal. In contrast, a machine learning–based technique that has been used to detect seizures is the self-organizing map (SOM). The SOM is a particular kind of an artificial neural network that uses unsupervised learning to classify data; that is, it does not require training samples that are labeled with the class information (in the case of seizure detection, this would correspond to labeling the EEG signal as an ictal/interictal event); it is merely provided the data and the network learns on its own. Described informally, the SOM groups inputs that have "similar" attributes by assigning them to close by neurons in the network. This is achieved by incrementally rewarding the activation function of those artificial neurons in the network (and their neighbors) that favor a particular input data point. Competition arises because different input data points have to jockey for position on the network. One reported result transformed the EEG signal using a FFT, and subsequently used the FFT vector as input to a SOM. With the help of some additional stipulations on the amplitudes and frequencies, the SOM was able to detect 90% of the seizures with an average of 0.71 false positives per hour [43]. However, the report did not attempt to apply the technique to predicting seizures, which would most definitely have produced worse results.

## 6.10 Support Vector Machine

A more advanced machine learning technique that has been used for seizure detection is a support vector machine (SVM). As opposed to an SOM, an SVM is a reinforcement learning technique—it requires data that is labeled with the class information. A support vector machine is a classifier that partitions the feature space (or the kernel space in the case of a kernel SVM) into two classes using a hyperplane. Each sample is represented as a point in the feature space (or the kernel space, as the case may be) and is assigned a class depending on which side of the hyperplane it lies. The classifier that is yielded by the SVM learning algorithm is the optimal hyperplane that minimizes the expected risk of misclassifying unseen samples. Ker-

nel SVMs have been applied to EEG data after removing noise and other artifacts from the raw signals in the various channels. In one report, the author was able to detect 97% of the seizures using an online detection method that used a kernel SVM. Of the seizures that were detected, the author reported that he was able to predict 40% of the ictal events by an average of 48 seconds before the onset of the seizure [44].

## 6.11 Phase Correlation

Methods of measuring phase synchrony include methods based on spectral coherence. These methods incorporate both amplitude and phase information, detection of maximal values after filtering. For weakly coupled nonlinear equations, phases are locked, but the amplitudes vary chaotically and are mostly uncorrelated. To characterize the strength of synchronization, Tass [45] proposed two indices, one based on Shannon entropy and one based on conditional probability. This approach aims to quantify the degree of deviation of the relative phase distribution from a uniform phase distribution.

All of the techniques that have been described thus far approach the problem of detecting and predicting seizures from a traditional time-series prediction perspective. In all such cases, the EEG signal is viewed like any other signal that has predictive content embedded in it. The goal, therefore, is to transform the signal using various mathematical techniques so as to draw out this predictive content. The fact that an EEG signal is generated in a particular biological context, and is representative of a particular physical aspect of the system, does not play a significant role in these techniques.

## 6.12 Seizure Detection and Prediction

Seizure anticipation (or warning) can be classified into two broad categories: (1) *early seizure detection* in which the goal is to use EEG data to identify seizure onset, which typically occurs a few seconds in advance of the observed behavioral changes or during the period of early clinical manifestation of focal motor changes or loss of patient awareness, and (2) *seizure prediction* in which the aim is to detect preictal changes in the EEG signal that typically occur minutes to hours in advance of an impending epileptic seizure.

In seizure detection, since the aim of these algorithms is to causally identify an ictal state, the statistical robustness of early seizure detection algorithms is very high [46, 47]. The practical utility of these schemes in the development of an online seizure abatement strategy depends critically on the few seconds of time between the detection of an EEG seizure and its actual manifestation in patients in terms of behavioral changes. Recently Talathi et al. [48] conducted a review of a number of nonparametric early seizure detection algorithms to determine the critical role of the EEG acquisition methodology in improving the overall performance of these algorithms in terms of their ability to detect seizure onset early enough to provide a suitable time to react and intervene to abate seizures.

In seizure prediction, the effectiveness of seizure prediction techniques tends to be lower in terms of statistical robustness. This is because the time horizon of these methods ranges from minutes to hours in advance of an impending seizure and because the preictal state is not a well-defined state across multiple seizures and across different patients. Some studies have shown evidence of a preictal period that could be used to predict the onset of an epileptic seizure with high statistical robustness [13, 49]. However, many of these studies use a posteriori knowledge or do not use out-of-sample training [14]. This leads to a model that is "overfit" for the data being used. When this same model is applied to other data, the accuracy of the technique typically decreases dramatically.

A number of algorithms have been developed solely for seizure detection and not for seizure prediction. The goal in this case is to identify seizures from EEG signals offline. Technicians spend many hours going through days of recorded EEG activity in an effort to identify all seizures that occurred during the recording. A technique that could automate this screening process would save a great amount of time and money. Because the purpose is to identify every seizure, any part of the EEG data may be used. Particularly a causal estimation of algorithmic measures can be used to determine the time of seizure occurrence. Algorithms designed for this purpose typically have better statistical performance and can only be used as an offline tool to assist in the identification of EEG seizures in long records of EEG data.

## 6.13 Performance of Seizure Detection/Prediction Schemes

With so many seizure detection and prediction methods available, there needs to be a way to compare them so that the "best" method can be used. Many statistics that evaluate how well a method does are available. In seizure detection, the technique is supposed to discriminate EEG signals in the ictal (seizure) state from EEG signals in the interictal (nonseizure) state. In seizure prediction, the technique is supposed to discriminate EEG signals in the preictal (before the seizure) state from EEG signals in the interictal (nonseizure) state. The classification an algorithm gives to a particular segment of EEG for either seizure detection or prediction can be placed into one of four categories:

- *True positive* (*TP*): A technique correctly classifies an ictal segment (preictal for prediction) of an EEG as being in the ictal state (preictal for prediction).
- *True negative* (*TN*): A technique correctly classifies an interictal segment of an EEG as being in the interictal state.
- *False positive* (*FP*): A technique incorrectly classifies an interictal segment of an EEG as being in the ictal state (preictal for prediction).
- *False negative* (*FN*): A technique incorrectly classifies an ictal segment (preictal for prediction) of an EEG as being in the interictal state.

Next we discuss how these classifications can be used to create metrics for evaluating how well a seizure prediction/detection technique does. In addition, we also discuss the use of a posteriori information. A posteriori information is used by certain algorithms to improve their accuracy. However, in most cases, this information

is not available when using the technique in an online manner so it cannot be generalized to online use.

### 6.13.1 Optimality Index

From these four totals (TP, TN, FP, FN) we can calculate two statistics that give a large amount of information regarding the success of a given technique. The first statistic is the sensitivity ($S$), which is defined in (6.26). In detection this indicates the probability of detecting an existent seizure and is defined by the ratio of the number of detected seizures to the number of total seizures. In prediction this indicates the probability of predicting an existent seizure and is defined by the ratio of the number of predicted seizures to the number of total seizures.

$$S = \frac{TP}{TP + FN} \tag{6.26}$$

In addition to the sensitivity, the specificity ($K$) is also used and is defined in (6.27). This indicates the probability of not incorrectly detecting/predicting a seizure and is defined by the ratio of the number of interictal segments correctly identified in comparison to the number of interictal segments.

$$K = \frac{TN}{TN + FP} \tag{6.27}$$

A third metric used to measure the quality of a given algorithm is the predictability. This indicates how far in advance of a seizure the seizure can be predicted or how long after the onset of the seizure it can be detected. In other words, the predictability ($\Delta T$) is defined by $\Delta T = T_a - T_e$ where $T_a$ is the time at which the given algorithm detects the seizure and $T_e$ is the time at which the onset of the seizure actually occurs according to the EEG.

Note that either of these metrics alone is not a sufficient measure of quality for a seizure detection/prediction technique. Consider a detection/prediction algorithm that always said the signal was in the ictal or preictal state, respectively. Such a method would produce a sensitivity of 1 and a specificity of 0. On the other hand, an algorithm that always said the signal was in the interictal state would produce a sensitivity of 0 and a specificity of 1. The ideal algorithm would produce a value of 1 for each. To accommodate this, Talathi et al. [48] defined the optimality index ($O$), a single measure of goodness, which takes all three of these metrics into account. It is defined in (6.28), where $D^*$ is the mean seizure duration of the seizures in the dataset:

$$O = \frac{S + K}{2} - \frac{\Delta T}{D^*} \tag{6.28}$$

### 6.13.2 Specificity Rate

The specificity rate is another metric used to assess the performance of a seizure prediction/detection algorithm [50]. It is calculated by taking the number of false pre-

dictions or detections divided by the length of time of the recorded data (FP/*T*). It gives an estimate of the number of times that the algorithm under consideration would produce a false prediction or detection in a unit time (usually an hour).

Morman et al. [50] also point out that the prediction horizon is important when considering the specificity rate of prediction algorithms. The prediction horizon is the amount of time before the seizure for which the given algorithm is trying to predict it. The reason is false positives are more costly as the prediction horizon increases. A false positive for an algorithm with a larger prediction horizon causes the patient to spend more time expecting a seizure that will not occur. This is in opposition to an algorithm with a smaller prediction horizon. Less time is spent expecting a seizure that will not occur when a false positive is given. To correct this, they suggest using a technique that reports the portion of time from the interictal period during which a patient is not in the state of falsely awaiting a seizure [50].

Another issue that should be considered when assessing a particular seizure detection/prediction technique is whether or not a posteriori information is used by the technique in question. A posteriori information is information that can be used to improve an algorithm's accuracy, but is specific to the dataset (EEG signal) at hand. When the algorithm is applied to other datasets where this information is not known, the accuracy of the algorithm can drop dramatically. In-sample optimization is one example of a posteriori information used in some algorithms [14, 50]. With in-sample optimization, the same EEG signal that is used to test the given technique is also used to train the technique. When training a given algorithm, certain parameters are adjusted in order to come up with a general method that can distinguish two classes. When training the technique, the algorithm is optimized to classify the training data. Therefore, when the same data that is used to test a technique is used to train the technique, the technique is optimized ("overfit") for the testing data. Although this produces promising results as far as accuracy, these results are not representative of what would be produced when the algorithm is applied to nontraining, that is, out-of-sample, data.

Another piece of a posteriori information that is used in some algorithms is optimal channel selection. When testing, other algorithms are given the channel of the EEG that produces the best results. It has been shown that out of the available EEG channels, not every channel provides information that can be used to predict or detect a seizure [48, 50]. Other channels provide information that would produce false positives. So when an optimal channel is provided to a given algorithm, the results produced from this technique again will be biased. Therefore, the algorithm does not usually generalize well to the online case when the optimal channel is not known.

## 6.14 Closed-Loop Seizure Prevention Systems

The majority of patients with epilepsy are treated with chronic medication that attempts to balance cortical inhibition and excitation to prevent a seizure from occurring. However, anticonvulsant drugs only control seizures for about two-thirds of patients with epilepsy [51]. Electrical stimulation is an alternative treatment that has been used [52]. In most cases, open-loop simulation is used. This

type of treatment delivers electrical stimulus to the brain without any neurological feedback from the system. The stimulation is delivered on a preset schedule for predetermined lengths of time (Figure 6.6).

Electrically stimulating the brain on a preset schedule raises questions about the long-term effects of such a treatment. Constant stimulation of the neurons could cause long-term damage or totally alter the neuronal architecture. Because of this, recent research has been aimed at closed-loop and semi-closed-loop prevention systems. Both of these systems take neurological feedback into consideration when delivering the electrical stimulation. In semi-closed-loop prevention systems, the stimulus is supplied only when a seizure has been predicted or detected by some algorithm (Figure 6.7). The goal is to reduce the severity of or totally stop the oncoming seizure. In closed loop stimulation the neurological feedback is used to create an optimal stimulation pattern that is used to reduce seizure severity.

In general, an online seizure detection algorithm is used rather than a prediction algorithm. Although a technique that could predict a seizure beforehand would be ideal, in practice, prediction algorithms leave much to be desired as far as statistical accuracy goes when compared to seizure detection algorithms. As the prediction horizon increases, the correlation between channels tends to decrease. Therefore, the chance of accurately predicting a seizure decreases as well. However, the downside of using an online detection algorithm is that it does not always detect the seizure in enough time to give the closed-loop seizure prevention system sufficient warning to prevent the seizure from occurring.

Finally, factors concerning the collection of the EEG data also play a significant role in the success of seizure detection algorithms [48]. Parameters such as the location of EEG electrodes, the type of the electrode, and the sampling rate of the electrodes can play a vital role in the success of a given online detection algorithm. By increasing the sampling rate, the detection technique is supplied with more data points for a given time period. This gives the detector more chances to pick up on any patterns that would be indicative of a seizure

## 6.15 Conclusion

Epilepsy is a dynamic disease, characterized by numerous types of seizures and presentations. This has led to a rich set of electrographical records to analyze. To understand these signals, investigators have started to employ various signal processing techniques. Researchers have a wide assortment of both univariate and

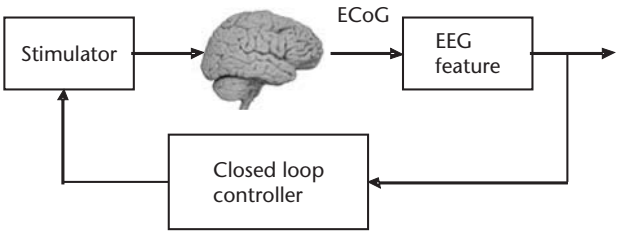



**Figure 6.6** Schematic diagram for seizure control.

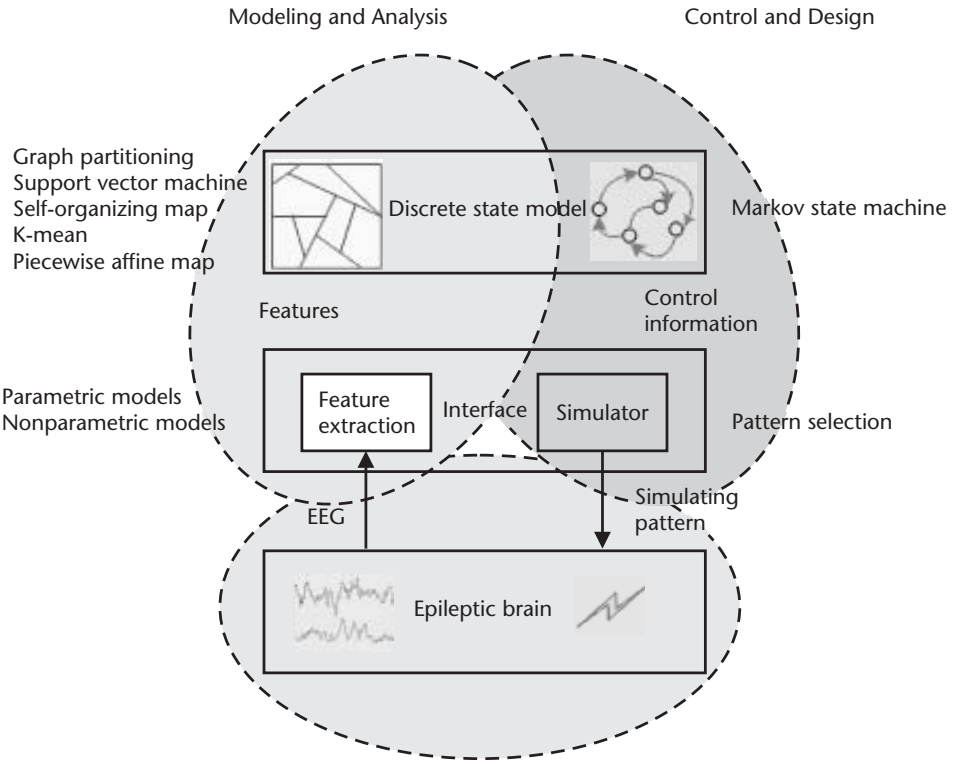



**Figure 6.7** A hybrid system that is composed of four parts of modeling phases: modeling, analysis, control, and design.

multivariate tools at their disposal. Even with these tools, the richness of the datasets has meant that these techniques have been met with limited success in predicting seizures. To date, there has been limited amount of research into comparing techniques on the same datasets. Oftentimes the initial success of a measure has been difficult to repeat because the first set of trials was the victim of overtraining. No measure has been able to reliably and repeatedly predict seizures with a high level of specificity and sensitivity.

While the line between seizure prediction, early detection, and detection can sometimes blur, it is important to note they do comprise three different questions. While unable to predict a seizure, many of these measures can detect a seizure. Seizures often present themselves as electrical storms in the brain, which are easily detectable, by eye, on an EEG trace. Seizure prediction seeks to tease out minute changes in the EEG signal. Thus far the tools that are able to detect one of these minor fluctuations often fall short when trying to replicate their success in slightly altered conditions. Coupled with the proper type of intervention (e.g., chemical stimulation or directed pharmacological delivery) early detection algorithms could usher in a new era of epilepsy treatment. The techniques presented in this chapter need to be continually studied and refined. They should be tested on standard datasets in order for their results to be accurately compared. Additionally, they need to be tested on out-of-sample datasets to determine their effectiveness in a clinical setting.

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CHAPTER 7

# Monitoring Neurological Injury by qEEG

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The EEG provides a measure of the continuous neurological activity on multiple spatial scales. It should, therefore, be useful in monitoring the brain's response to a global injury. The most prevalent situation of this nature arises when the brain becomes globally ischemic after cardiac arrest (CA). Fortunately, timely intervention with resuscitation and therapeutic hypothermia may provide neuroprotection. Currently, no clinically acceptable means of monitoring the brain's response after CA and resuscitation is available because monitoring is impeded by the ability to interpret the complex EEG signals.

Novel methodologies that can evaluate the complexity of the transient and time-varying responses in EEG, such as quantitative EEG (qEEG), are required. qEEG methods that employ entropy and information measures to determine the degree of brain injury and the effects of hypothermia treatment are well suited to evaluate changes in EEG. Two such measures—the information quantity and the subband information quantity—are presented here that can quantitatively evaluate the response to a graded ischemic injury and response to temperature changes. A suitable animal model and results from carefully conducted experiments are presented and the results discussed. Experimental results of hypothermia treatment are evaluated using these qEEG methods.

## 7.1 Introduction: Global Ischemic Brain Injury After Cardiac Arrest

Cardiac arrest affects between 250,000 and 400,000 people annually and remains the major cause of death in the United States [1]. Only a small fraction (17%) of patients resuscitated from CA survive to hospital discharge [2]. Of the initial 5% to 8% of out-of-hospital CA survivors, approximately 40,000 patients reach an intensive care unit for treatment [3]. As many as 80% of these remain comatose in the immediate postresuscitative period [2]. Very few patients survive the hospitalization, and even among the survivors significant neurological deficits prevail [3]. Among survivors, neurological complications remain as the leading cause of disability [4, 5]. CA leads to a drastic reduction in systemic blood circulation that causes a catastrophic diminution of cerebral blood flow (CBF), resulting in oxygen deprivation and subsequent changes in the bioelectrical activity of the brain [6]. The neurological impairment stemming from oxygen deprivation adversely affects syn-

aptic transmission, axonal conduction, and cellular action potential firing of the neurons in the brain [7].

Controlled animal studies can be helpful in elucidating the mechanisms and developing the methods to monitor brain injury. This chapter reviews the studies done in an animal model of global ischemic brain injury, monitoring brain response using EEG and analyzing the response using qEEG methods. These studies show that the rate of return of EEG activity after CA is highly correlated with behavioral outcome [8–10]. The proposed EEG monitoring technique is based on the hypothesis that brain injury reduces the entropy of the EEG, also measured by its information content (defined classically as bits per second of information rate [11]) in the signal. As brain function is impaired, its ability to generate complex electrophysiologic activity is diminished, leading to a reduction in the entropy of EEG signals. Given this observation, recent studies support the hypothesis that neurological recovery can be predicted by monitoring the recovery of entropy, or equivalently, a derived measure called information quantity (IQ) [12, 13] of the EEG signals. Information can be quantified by calculating EEG entropy [11, 14]. This information theory–based qEEG analysis method has produced promising results in predicting outcomes from CA [15–18].

### 7.1.1 Hypothermia Therapy and the Effects on Outcome After Cardiac Arrest

The neurological consequences of CA in survivors are devastating. In spite of numerous clinical trials, neuroprotective agents have failed to improve outcome statistics after CA [19, 20]. Recent clinical trials using therapeutic hypothermia after CA showed a substantial improvement in survival and functional outcomes compared to normothermic controls [19, 21, 22]. As a result, the International Liaison Committee on Resuscitation and the American Heart Association recommended cooling down the body temperature to 32°C to 34°C for 12 to 24 hours in out-of-hospital patients with an initial rhythm of ventricular fibrillation who remain unconscious even after resuscitation [23].

Ischemic brain injury affects neurons at many levels: synaptic transmission, axonal conduction, and cellular action potential firing. Together these cellular changes contribute to altered characteristics of EEGs [24]. Cellular mechanisms of neuroprotective hypothermia are complex and may include retarding the initial rate of ATP depletion [25, 26], reduction of excitotoxic neurotransmitter release [27], alteration of intracellular messengers [28], reduction of inflammatory responses [29], and alteration of gene expression and protein synthesis [30, 31]. Hypothermia reduces the excitatory postsynaptic potential (EPSP) slope in a temperature-dependent manner [32]. A recent study done on parietal cortex slice preparation subjected to different temperatures showed greater spontaneous spike amplitude and frequency in the range of mild hypothermia (32°C to 34°C) [32]. However, more detailed cellular information about neural activity in different brain regions is not available and the neural basis to the effects of hypothermia therapy remains poorly understood.

The ischemic brain is sensitive to temperature and even small differences can critically influence neuropathological outcomes [33]. Hyperthermia, for example, has been demonstrated to worsen the ischemic outcome and is associated with

increased brain injury in animal models [33, 34] and clinical studies [35–37]. On the other hand, induced hypothermia to 32°C to 34°C is shown to be beneficial and hence recommended for comatose survivors of CA [23, 38].Therapeutic hypothermia was recently shown to significantly mitigate brain injury in animal models [39–41] and clinical trials [21, 42–44].

The effects of changes in brain temperature on EEG have been described as far back as the 1930s. Among the reported studies, Hoagland found that hyperthermic patients showed faster alpha rhythms (9 to 10 Hz) [45–47], whereas Deboer demonstrated that temperature changes in animals and humans had an influence on EEG frequencies and that the changes were similar in magnitude in the different species [48, 49]. More recently, hypothermia has been shown to improve EEG activity with reperfusion and reoxygenation [50–52]. Most of these results have been based on clinical observations and neurologists' interpretations of EEG signals—both of which can be quite subjective.

## 7.2 Brain Injury Monitoring Using EEG

Classically, EEG signals have been analyzed using time, frequency, and joint time-frequency domains. Time-domain analysis is useful in interpreting the features in EEG rhythms such as spikes and waves indicative of nervous system disorders such as epilepsy. Frequency-domain analysis is useful for interpreting systematic changes in the underlying rhythms in EEG. This is most evident when spectral analysis reveals changes in the constituent dominant frequencies of EEG during different sleep stages or after inhalation or administration of anesthetics. Brain injury, however, causes markedly different changes in the EEG signal. First of all, there is a significant reduction in signal power, with the EEG reducing to isoelectric soon after cardiac arrest (Figure 7.1). Second, the response tends to be nonstationary during the recovery period. Third, a noteworthy feature of the experimental EEG recordings during the recovery phase after brain injury is that the signals contain both predictable or stationary and unpredictable or nonstationary patterns. The stationary component of the EEG rhythm is the gradual recovery of the underlying baseline rhythm, generally modeled by parametric models [16]. The nonstationary part of the EEG activity includes seizure activity, burst-suppression patterns, nonreactive $\alpha$ or $\theta$ patterns, and generalized suppression. Quite possibly, the nonstationary part of the EEG activity may hold information in the form of unfavorable EEG patterns after CA.

Time-frequency, or wavelet, analysis provides a mathematically rigorous way of looking at the nonstationary components of the EEG. However, in conditions resulting from brain injury, neither time-domain nor frequency-domain approaches are as effective due to nonstationary and unpredictable or transient signal patterns. Injury causes unpredictable changes in the underlying statistical distribution of EEG signal samples. Thus, EEG signal changes resulting from injury may be best evaluated by using statistical measures that quantify EEGs as a random process.

Measures designed to assess the randomness of the signals should provide more objective analysis of such complex signals. Signal randomness can be quantitatively assessed with the entropy analysis. The periodic and predictable signals should

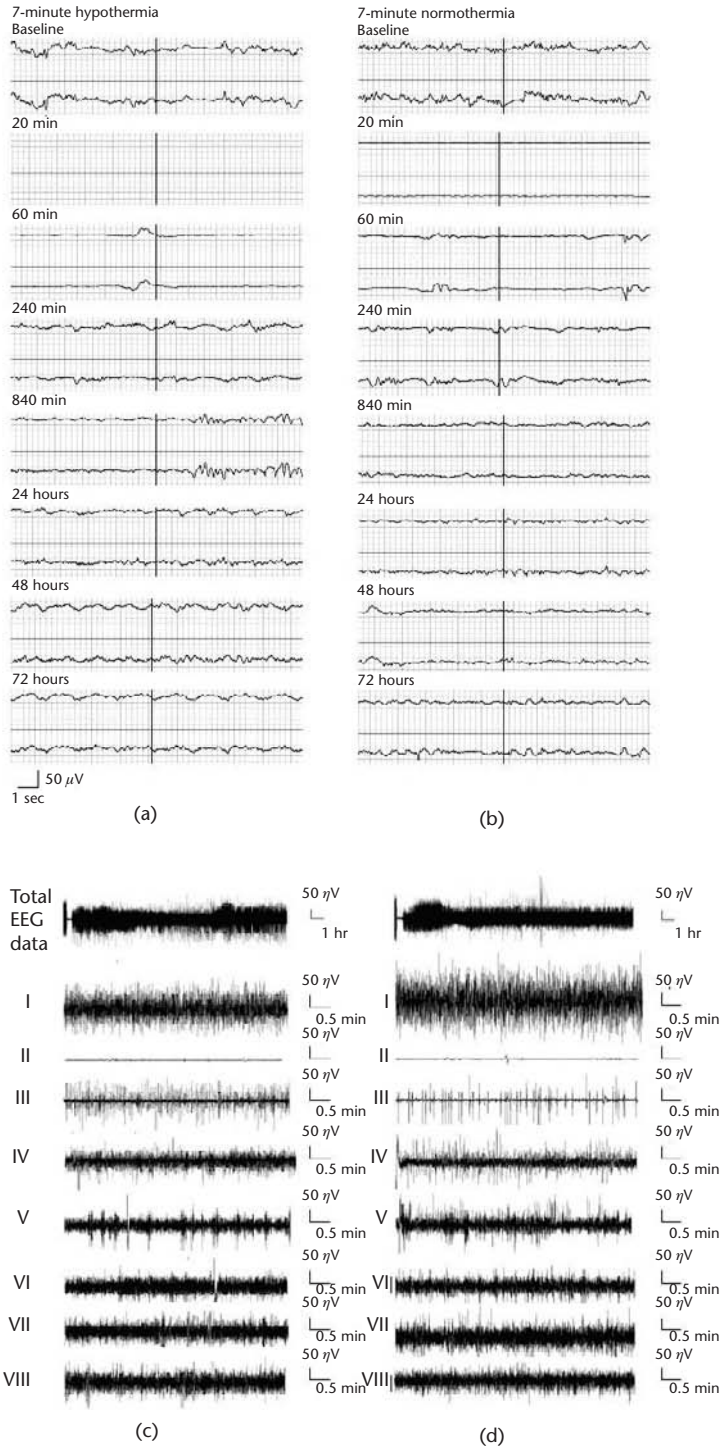



**Figure 7.1** Raw EEG data for representative animals at various time points with 7-minute asphyxial CA: (a) real-time raw EEG under hypothermia; (b) real-time raw EEG under normothermia; (c) raw compressed EEG under hypothermia, (I) baseline prior to CA, 0 minute, (II) early stage after CA, 20 minutes, (III) initiation of hypothermia, 60 minutes, (IV) hypothermia maintenance period, 4 hours, (V) initiation of rewarming, 12 hours, (VI) late recovery, 24 hours, (VII) late recovery, 48 hours, (VIII) late recovery, 72 hours; and (d) raw compressed EEG under normothermia. (*From:* [64]. © 2006 Elsevier. Reprinted with permission.)

score low on entropy measures. Reactive EEG patterns occurring during the recovery periods render the entropy measures more sensitive to detecting improvements in EEG patterns after CA. Therefore, we expect entropy to reduce soon after injury and at least during the early recovery periods. Entropy should increase with recovery following resuscitation, reaching close to baseline levels of high entropy upon full recovery.

## 7.3 Entropy and Information Measures of EEG

The classical entropy measure is the Shannon entropy (SE), which results in useful criteria for analyzing and comparing probability distribution and provides a good measure of the information. Calculating the distribution of the amplitudes of the EEG segment begins with the sampled signal. One approach to create the time series for entropy analysis is to partition the sampled waveform amplitudes into $M$ segments. Let us define the raw sampled signal as $\{x(k), \text{for } k = 1, ..., N\}$. The amplitude range $A$ is therefore divided into $M$ disjointed intervals $\{I_i, \text{for } i = 1, ..., M\}$. The probability distribution of the sampled data can be obtained from the ratio of the frequency of the samples $N_i$ falling into each bin $I_i$ and the total sample number $N$:

$$p_i = N_i / N \tag{7.1}$$

The distribution $\{p_i\}$ of the sampled signal amplitude is then used to calculate one of the many entropy measures developed [16]. Entropy can then be defined as

$$SE = -\sum_{i=1}^{M} p_i (\log p_i) \tag{7.2}$$

This is the definition of the traditional Shannon entropy [11]. Another form of entropy, postulated in 1988 in a nonlogarithm format by Tsallis, which is also called Tsallis entropy (TE) [17, 18], is

$$TE = -(q-1)^{-1} \sum_{i=1}^{M} \left( p_i^q - 1 \right) \tag{7.3}$$

where $q$ is the entropic index defied by Tsallis, which empirically allows us to scale the signal by varying the $q$ parameter. This method can be quite useful in calculating entropy in the presence of transients or long-range interactions as shown in [18]. Shannon and Tsallis entropy calculations of different synthetic and real signals are shown in Figure 7.2. It is clear that the entropy analysis is helpful in discriminating the different noise signals and EEG brain injury states.

To analyze nonstationary signals such as EEGs after brain injury, the temporal evolution of SE must be determined from digitized signals (Figure 7.3). So, an alternative time-dependent SE measure based on a sliding temporal window technique is applied [15, 18]. Let $\{s(i), \text{for } i = 1, ..., N\}$ denote the raw sampled signal and set the sliding temporal window as

$$W(n, w, \Delta) = \{ s(i), i = 1 + n\Delta, \ldots, w + n\Delta \} \text{ of length } w \le N \tag{7.4}$$

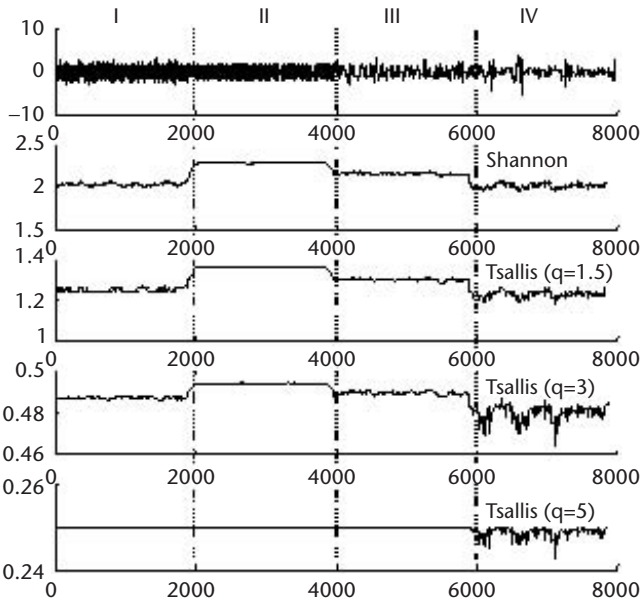



**Figure 7.2** An EEG trace composed by sequencing (I…IV), going from (I) Gaussian, (II) uniform distributed noises, and segments of EEG from experiments: baseline (III) and early recovery EEG after brain injury (IV and V). The Shannon and Tsallis entropy for different $q$ index values ($q = 1.5, 3, 5$) are shown. Tsallis entropy is sensitive enough to distinguish between signals with different probability distribution functions and to differentiate the baseline EEG from early recovery EEG. (*From:* [15]. © 2003. Biomedical Engineering Society. Reprinted with permission.)

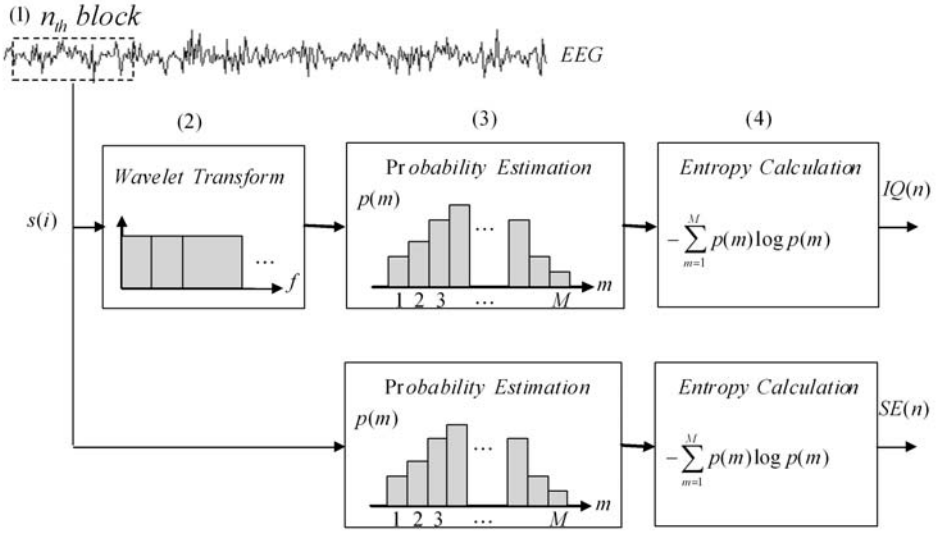



**Figure 7.3** Block diagram of EEG signal processing using the conventional Shannon entropy (SE) measure and the proposed information quantity (IQ) measure.

Here $\Delta \leq w$ is the sliding step and $n = 0, 1, ..., [n/\Delta] - w + 1$, where $[x]$ denotes the integer part of $x$. To calculate the probability $p_n(m)$ within each window $W(n; w; \Delta)$, we introduce intervals $W(n; w; \Delta) = \bigcup_{m=1}^{M} I_m$. The probability $p_n(m)$ of the sampled signal belonging to the interval $I_m$ is the ratio of the number of the signals found within interval $I_m$ and the total number of signals in $W(n; w; \Delta)$. The value of $SE(n)$ is defined using $p_n(m)$ as

$$SE(n) = -\sum_{m=1}^{M} p_n(m) \log_2 p_n(m) \tag{7.5}$$

### 7.3.1 Information Quantity

Although it is common to use the distribution of signal amplitudes to calculate the entropy, there is no reason why other signal measures could not be employed. For example, Fourier coefficients reflect the signal power distribution, whereas the wavelet coefficients reflect the different signal scales, roughly corresponding to coarse and fine time scales or correspondingly low- and high-frequency bands. Instead of calculating entropy of the amplitude of the sampled signals, entropy of the wavelet coefficients of the signal may be calculated to get an estimate of the entropy in different wavelet subbands. Wavelet analysis decomposes the signal into its different scales, from coarse to fine. Wavelet analysis of the signal is carried out to decompose the EEG signals into wavelet subbands, which can be interpreted as frequency subbands.

We calculate the IQ information theoretic analysis on the wavelet subbands. First the discrete wavelet transform (DWT) coefficients within each window are obtained as $WC(r; n; w; \Delta) = \text{DWT}[W(n; w; \Delta)]$. The wavelet coefficients are obtained from the DWT, and the IQ is obtained from the probability distribution of the wavelet coefficients as follows:

$$IQ(n) = -\sum_{m=1}^{M} p_n^{wc}(m) \log_2 p_n^{wc}(m) \tag{7.6}$$

where $p_n(m)$ is an estimated probability that the wavelet-transformed signal belongs to the $m$th bin where $M$ is the number of bin.

We calculate IQ from a temporal sliding window block of the EEG signal as explained earlier. Figure 7.4 shows the IQ trend plots for two experimental subjects. IQ trends accurately indicate the progression of recovery after CA injury. The time trends indicate the changing values of IQ during the various phases of the experiments following injury and during recovery. The value of these trends lies in comparing the differences in the response to hypothermia and normothermia. There are evident differences in the IQ trends for hypothermia versus normothermia. Hypothermia improves the IQ levels showing quicker recovery under hypothermia and over the 72-hour duration. The final IQ level is closer to the baseline (hatched line) under hypothermia. These results support the idea of using IQ trends to monitor brain electrical activity following injury by CA.

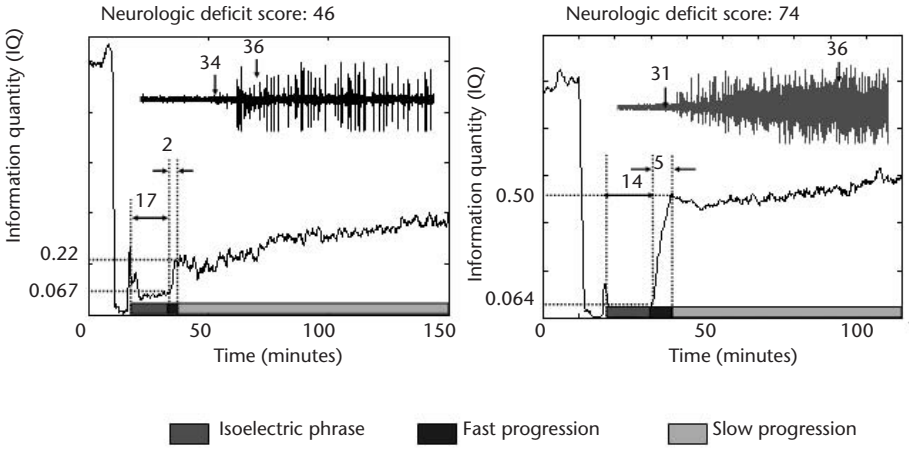



**Figure 7.4** IQ characteristic comparison of poor and good outcomes after 7-minute injury. The small figure inside each figure is a compressed EEG. We quantify IQ evolution from various perspectives mainly in three different phases: isoelectric phase just after CA injury, fast increase, and slow increase phases.

Indeed, there is a very good relationship between the IQ levels obtained and the eventual outcome of the animal as assessed by the neurological deficit scoring (NDS) evaluation [39, 43, 53, 54]. A low NDS value reflects a poor outcome and a high NDS a better outcome. As seen in Figure 7.4, the IQ level recovery takes place faster and equilibrates to a higher level for the animal with the greater NDS. What we discovered is that the recovery patterns are quite distinctive, with periods of isoelectricity, fast progression, and slow progression. In addition, in the poor outcome case, there is a period of spiking and bursting, while in the good outcome case there is a rapid progression to a fused, more continuous EEG.

### 7.3.2 Subband Information Quantity

Although IQ is a good measure of EEG signals, it has the limitation that EEG recovery in each clinical band ($\delta, \theta, \alpha, \beta, \gamma$) is not characterized [55]. Therefore, we extend the IQ analysis method and propose another measure that separately calculates IQ in different subbands ($\delta, \theta, \alpha, \beta, \gamma$)? This subband method, SIQ, is similar to IQ but separately estimates the probability in each subband. The probability that $p_n^k(m)$ in the $k$th subband for that the sampled EEG belongs to the interval $I_m$ is the ratio between the number of the samples found within interval $I_m$ and the total number of samples in the $k$th subband. Using $p_n^k(m)$, $SIQ^k(n)$ in $k$th subband is defined as

$$SIQ^k(n) = -\sum_{m=1}^{M} p_n^k(m) \log_2 p_n^k(m) \tag{7.7}$$

Thus, we can now evaluate the evolution of SIQ for the whole EEG data $\{s(i)$, for $i = 1, \ldots, N\}$. Figure 7.5 clearly indicates that recovery differs among subbands. The subband analysis of signal trends might lead to better stratification of injury and recovery and identification of unique features within each subband. This wavelet

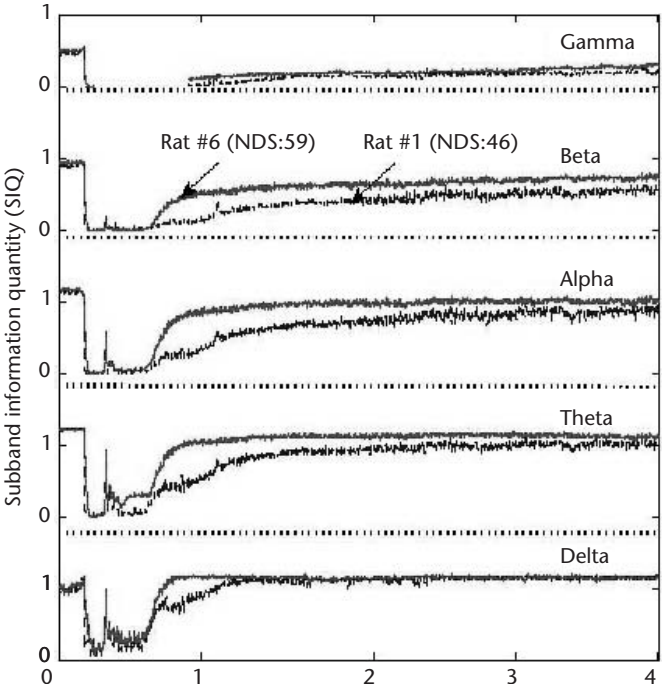



**Figure 7.5** Subband information quantity (SIQ) of rat #1 (NDS: 46) and rat #6 (NDS: 59) for five clinical bands: gamma (>30 Hz), beta (16–30 Hz), alpha (8–15 Hz), theta (4–8 Hz), delta (<4 Hz). (*From:* [71]. © 2008 IEEE. Reprinted with permission.)

entropy subband analysis is analogous to how EEG is analyzed and interpreted by looking for power in different spectral subbands ($\delta$, $\theta$, $\alpha$, $\beta$).

## 7.4 Experimental Methods

The experiments reported in this chapter were carried out on rodents. Global ischemic brain injury and hypothermic neuroprotection response were evaluated. EEG recording was done using a conventional monitoring technique involving scalp electrodes, signal amplification and acquisition, and eventual qEEG evaluation. In a typical experiment, EEG epidural screw electrodes (Plastics One, Roanoke, Virginia) were implanted 1 week before the experiment. Two channels of EEGs were recorded in the right and left parietal areas throughout the experiment using the DI700 Windaq system. Using stereotactic guidance, electrodes were placed 2 mm lateral to and 2 mm anterior or posterior to the bregma. A ground electrode was placed 2 mm posterior to the lambda in the midline. Recording was continued throughout the hypothermia treatment and during the rewarming periods and the recovery hours. Serial 30-minute EEG recordings were also done in free-roaming, unanesthetized rats at 24, 48, and 72 hours after return of spontaneous circulation (ROSC). Two-channel EEG signals were recorded and analyzed.

After CA in humans and various animal models, several common EEG patterns are seen that may be predictive of poor neurological recovery including generalized EEG suppression, persistent burst suppression, generalized unreactive $\alpha$ or $\theta$ activ-

ity, and epileptiform discharges [56–61]. Figure 7.6 shows example traces of EEG signals during various phases of the experiment. Figure 7.6 (top) shows an EEG during hypothermic recovery, whereas Figure 7.6 (bottom) shows an EEG during normothermic recovery. First of all, a very distinct evolution of the EEG waveforms after CA is evident, beginning with an isoelectric period, followed by the period with spikes and burst suppression, and continuous activity in the subsequent phases. The difference between hypothermic and normothermic EEG is not so obvious by visual examination alone. This is why a more objective quantitative, or qEEG, approach is needed to provide a measurable, serial analysis that follows the trends in the signal evolution.

### 7.4.1 Experimental Model of CA, Resuscitation, and Neurological Evaluation

We have developed a CA animal model using the rat. This model produces graduated levels of brain injury by controlling the duration of asphyxial CA. In this model, different physiological parameters, short-term and long-term neurobehavioral outcomes, EEG recovery, and postmortem histological results are measured [9, 53, 62–65]. Briefly, rats are endotracheally intubated and mechanically ventilated at 50 breaths per minute (Harvard Apparatus model 683, South Natick, Massachusetts) with 1.0% Halothane in $N_2/O_2$ (50%/50%). Ventilation is adjusted to maintain physiological pH, $pO_2$, and $pCO_2$. A body temperature of 37.0±0.5°C is maintained throughout the experiment. Venous and arterial catheters are inserted into the femoral vessels to continuously monitor mean arterial pressure (MAP), intermittently sample arterial blood gas (ABG), and administer fluid and drugs. After 5 minutes of baseline recording, vecuronium 2 mg/kg is infused and the inhaled anesthetic is discontinued for 5 minutes to reduce residual effects of the anesthetics on the EEG signals [8]. No sedative or anesthetic agents are subsequently administered throughout the remainder of the experiment to avoid confounding effects on EEGs [66]. CA is initiated via asphyxia by cessation of mechanical ventilation after neuromuscular blockade for a period of 7 minutes. CA is defined by pulse pressure <10 mm Hg and asystole. Cardiopulmonary resuscitation (CPR) is performed with resumption of ventilation and oxygenation (100% $FIO_2$), infusion of epinephrine (0.005 mg/kg), $NaHCO_3$ (1 mmol/kg), and sternal chest compressions (200/min) until ROSC with

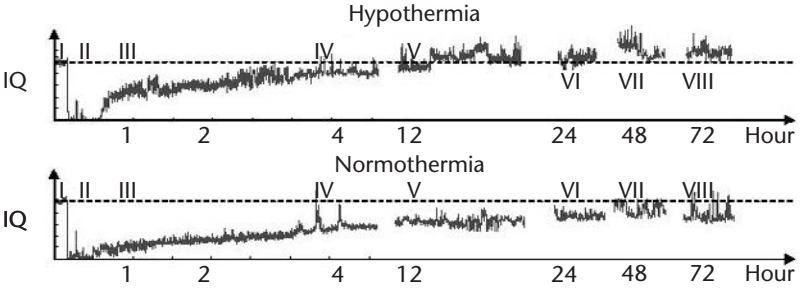



**Figure 7.6** IQ for hypothermia and normothermia over 72-hour period. The various phases of the experiment are as follows: I: beginning of the experiment, II: cardiac arrest and resuscitation, III: early recovery phase, IV: late recovery phase, V–VIII: intermittent recordings during the 72-hour recovery period. (*From:* [64]. © 2006 Elsevier B. V. Reprinted with permission.)

MAP > 60 mm Hg and the presence of the pulse waveform. Ventilator adjustments are made to normalize ABG findings. The animals are allowed to recover after resuscitation and are subsequently extubated along with removal of all invasive catheters.

Our group has developed a standardized NDS that takes cues from the procedures used in human and animal behavioral studies [39, 43, 53, 54]. This score has been validated in a rat model for global ischemic brain injury following CA and published in various works [8, 9, 62–64, 67]. The NDS is determined after any temperature manipulation and during the recovery period on the first day, after the first 8 hours post-ROSC, and subsequently repeated at 24, 48, and 72 hours after ROSC. The NDS measures level of arousal, cranial nerve reflexes, motor function, and simple behavioral responses and has a range of 0 to 80 (where the best outcome equals 80 and the worst outcome equals 0) [9, 63, 64]. The standardized NDS examination is performed by a trained examiner blinded to temperature group assignment, and the primary outcome measure of this experiment, indicative of neurological recovery, is the 72-hour NDS score.

### 7.4.2 Therapeutic Hypothermia

The rats in the experimental protocol are randomly selected into groups of animals subjected to 7- and 9-minute asphyxia times, thus stratifying the degree of injury. Similarly, one-half of each group is randomly subjected to hypothermia ($T = 33°C$ for 12 hours) and the other half to normothermia ($T = 37°C$). Continuous physiological monitoring of blood pressure and EEG, core body temperature monitoring, and intermittent ABG analysis are undertaken. Neurological recovery after resuscitation is monitored using serial NDS calculation and qEEG analysis.

A temperature sensor (G2 E-mitter 870-0010-01, Mini Mitter, Oregon) implanted in the peritoneum is used to monitor the core body temperature while a rectal temperature sensor is used as a reference. To allow for animal recovery, the sensor is implanted into the peritoneal cavity about 1 week before the experiment. Hypothermia is induced 45 minutes after ROSC by external cooling with a cold water and alcohol mist, aided by an electric fan, to achieve the target temperature of 33°C within 15 minutes [62, 64]. An automatic warming lamp (Thermalet TH-5, model 6333, Phyritemp, New Jersey), on the other hand, is used to prevent precipitous temperature decline. Abrupt or extreme reduction in temperature has been associated with complications such as bleeding and arrhythmias. In this experiment, the core temperature is maintained by manual control between 32°C and 34°C for 12 hours. Rewarming is initiated 13 hours after ROSC. Rats are gradually rewarmed from 33.0°C to 37.0°C over 2 hours using a heating pad and heating lamp. For control animals, temperature after CPR is maintained at 36.5°C to 37.5°C throughout the sham-hypothermia period. To ensure that no temperature fluctuation occurred after the resuscitation, such as the spontaneous hypothermia previously reported [34], all animals are then kept inside a neonatal incubator (Isolette infant incubator model C-86, Air-Shields Inc., Pennsylvania) for the first 24-hour post-ROSC. This procedure has been adequate to maintain the animals within the desired target range of 36.5°C to 37.5°C.

## 7.5 Experimental Results

Figure 7.7 presents the experimental data for eight segments of EEG signals recorded following hypothermic treatment, and eight IQ measurements derived after qEEG analysis: baseline (0–5 minutes, IQ1), CA period (10–40 minutes, IQ2), hypothermia starting period (60–90 minutes, IQ3), hypothermia maintenance period (3–5 hours, IQ4), hypothermia end period (rewarming period) (13–15 hours, IQ5), 24 hours after CPR (30 minutes, IQ6), 48 hours after CPR (30 minutes, IQ7), and 72 hours after CPR (30 minutes, IQ8).

It is evident that IQ levels follow the expected course during each phase of injury and recovery, beginning with a normalized level of 1 during the baseline period. Phase 2, immediately after CA and resuscitation, as expected, shows a significant drop, and the subsequent phases show a gradual course of recovery of the IQ levels.

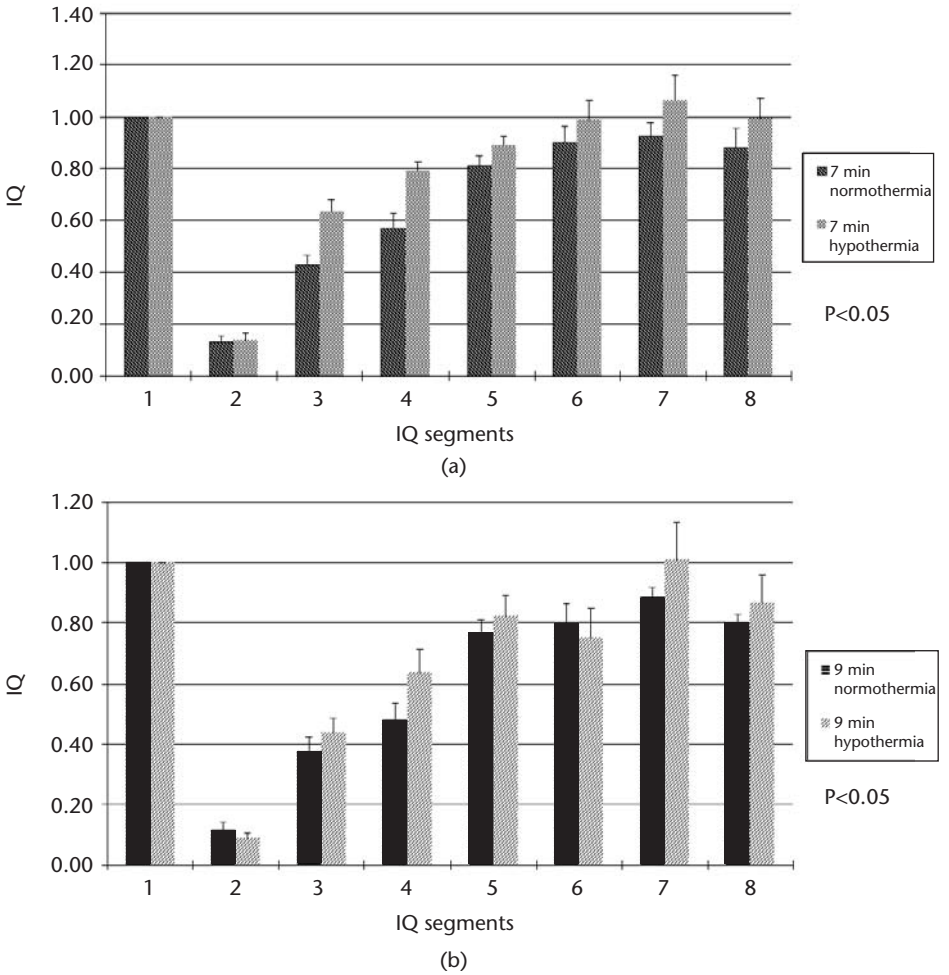



**Figure 7.7** Comparison of IQ between hypothermic and normothermic rats subjected to (a) 7-minute and (b) 9-minute ischemia at different time periods: IQ segment 1, baseline; 2, CA period; 3, hypothermia starting period; 4, hypothermia maintenance period; 5, rewarming period; 6–24 hours after ROSC; 7–48 hours after ROSC; 8–72 hours after ROSC. (*From:* [64]. © 2006 Elsevier. Reproduced with permission.)

It also compares the recovery for relatively mild (7-minute) versus severe (9-minute) asphyxial arrest durations. The experiment demonstrates greater recovery of IQ in rats treated with hypothermia compared to normothermic controls for both the injury groups ($p < 0.05$) (Figure 7.7). Not surprisingly, the 9-minute CA results in a slower progression of recovery.

Further, IQ levels in the normothermia cohort are compared with the hypothermia cohort, and in each case the recovery of the hypothermia cohort is greater; that is, the IQ levels restore toward the normalized baseline. Baseline characteristics of animals in the hypothermia and normothermia groups were similar, including weight on the day of CA, duration of asphyxia prior to CA, duration of CPR prior to ROSC, and baseline ABG data. The baseline control measures include arterial pH, $HCO_3^-$, $PCO_2$, $PO_2$, and $O_2$ saturation.

Another way to assess the outcome is through NDS. In our studies, the 72-hour NDS scores, an estimate of the long-term outcome of the animals, of the control and hypothermia groups were compared. NDS significantly improved under hypothermia compared to the normothermia group ($p < 0.05$). There was a trend toward improved survival rates and mean duration of survival hours in animals treated with hypothermia in both the 7-minute and 9-minute groups. Another interesting question worth asking is whether IQ values measured early (at 4 hours) correlate with the primary neurological outcome measure later on (72 hours). The animal studies presented here demonstrate the potential utility of qEEG-IQ to track the response to neuroprotective hypothermia during the early phase of recovery from CA.

### 7.5.1 qEEG-IQ Analysis of Brain Recovery After Temperature Manipulation

Previous studies showed that temperature maintenance of the brain has profound effects on the neurological recovery and survival of animals. It is evident that hypothermia has a neuroprotective effect and, conversely, hyperthermia should have harmful effects. Using the asphyxial CA rodent model, we tracked qEEG of 6-hour immediate postresuscitation hypothermia ($T = 33°C$), normothermia ($T = 37°C$), or hyperthermia ($T = 39°C$) ($N = 8$ per group). While hypothermia was implemented as before, hyperthermia was achieved using a warming blanket and an automatic warming lamp (Thermalet TH-5, model 6333, Physitemp, New Jersey) to achieve a target temperature of 39°C within 15 minutes and the temperature was maintained at 38.5°C to 39.5°C for 6 hours. NDS cutoff for good outcome was NDS = 60 (characterized as independently functioning animals) and poor outcome was NDS < 60 (characterized as sluggish to unresponsive animals) [9, 10, 67].

To study the temperature effects on neurological outcomes, three groups of animals were evaluated: (1) cohort 1: 6 hours of hypothermia ($T = 33°C$); (2) cohort 2: normothermia ($T = 37°C$); and (3) cohort 3: hyperthermia ($T = 39°C$) immediately postresuscitation from 7-minute CA. Temperature was maintained as before using surface cooling. Neurological recovery was defined by a 72-hour NDS assessment. The key observations were that burst frequency was higher during the first 90 minutes in rats treated with hypothermia (25.6 ± 12.2/min) and hyperthermia (22.6 ± 8.3/min) compared to normothermia (16.9 ± 8.5/min) ($p < 0.001$). The burst frequency correlated strongly with 72-hour NDS in normothermic rats ($p < 0.05$), but

not in hypothermic or hyperthermic rats. The 72-hour NDS of the hypothermia group was significantly higher than that of the normothermia and hyperthermia groups ($p < 0.001$) [67].

No significant difference was observed in IQ values during the periods of hypothermia in sham animals (no CA injury). Unnormalized IQ values also did not show significant differences between baseline halothane anesthesia and the washout periods. Hence, the eventual differences in IQ were attributed to the injury and temperature in the three experimental groups. Hypothermia produced greater recovery of IQ ($0.74 \pm 0.03$) compared to normothermia ($0.60 \pm 0.03$) ($p < 0.001$) or hyperthermia ($0.56 \pm 0.03$) ($p = 0.016$). This study demonstrates beneficial effects of hypothermia and harmful effects of hyperthermia post-CA resuscitation.

EEG monitoring may be used early, preferably immediately after resuscitation. The basis for that suggestion is that early monitoring may provide a prognostic indication of eventual outcome and serve as a guide for therapeutic interventions. More specifically, the question is whether recovery assessed using qEEG can be correlated with the NDS. Figure 7.8(a) shows that there is a significant difference ($p < 0.01$) in the IQ values of the animals in each of the three groups within the first 2 hours after ROSC. In fact, these differences are seen consistently at 30 minutes, 1 hour, 2 hours, 24 hours, and 72 hours. Importantly, the IQ values obtained as early as 30 minutes after ROSC correlate well with the NDS evaluation done at the end of the study duration (72 hours) [Figure 7.8(b)].

### 7.5.2 qEEG-IQ Analysis of Brain Recovery After Immediate Versus Conventional Hypothermia

The therapeutic benefits of hypothermia are becoming increasingly evident. However, out of practical consideration or historic reasons, hypothermia is conventionally applied quite some time (up to several hours) after CA and resuscitation. One impediment is access to the subject during out-of-hospital resuscitation, delaying the application of hypothermia. However, in certain situations, such as when the patient is in the intensive care unit, it may be possible to deliver hypothermia in a timely manner. Most clinical studies delay the initiation of hypothermia by 2 or more hours after resuscitation [21, 42, 43, 68]. It may also be possible to monitor the efficacy of hypothermia. Previous studies have shown that cooling may be successful even if it is delayed by 4 to 6 hours [23]. Our studies show that mild to moderate hypothermia (33°C to 34°C) mitigates brain injury when induced before [39], during [39, 40], or after ROSC [40, 41, 69].

Additional results demonstrating the effect of immediate (upon restoration of spontaneous circulation) therapeutic hypothermia on brain recovery after CA are further examined. Immediate initiation of 6-hour hypothermia (IH) upon successful resuscitation was compared to conventional hypothermia (CH) initiated at 1-hour postresuscitation. The conventional hypothermia group was designed to mimic the delay of intervention, as noted in most clinical cases in actual practice. The EEG signals are analyzed with the help of real-time qEEG tracking, and the study is terminated following functional outcome assessment and histological assessment. Two groups of animals, receiving 7- and 9-minute CA were studied with animals divided into two groups of CH (1 hour after ROSC) and IH. Hypothermia is induced by

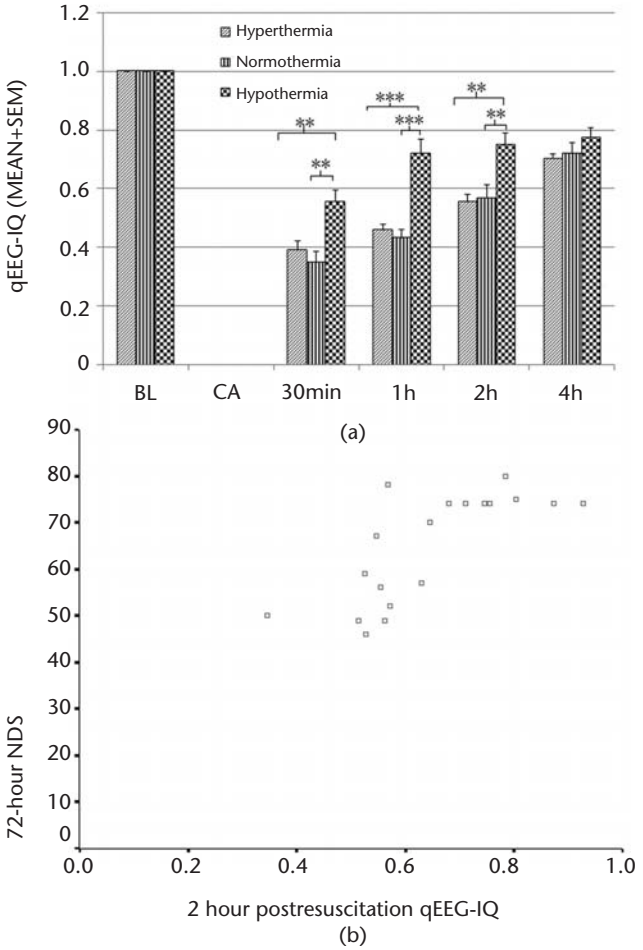



**Figure 7.8** Comparison of (a) qEEG-IQ at different intervals by temperature groups and (b) correlation between 2-hour qEEG-IQ and 72-hour NDS. Aggregate comparisons show that IQ was significantly different among hypothermia (0.74 ± 0.03), normothermia (0.60 ± 0.03), and hyperthermia (0.56 ± 0.03) groups ($p < 0.001$). IQ values correlated well with 72-hour NDS 2 hours after cardiac arrest (*$p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). (*From:* [62]. © 2008 Elsevier. Reprinted with permission.)

cooling with cold mist to achieve the target core temperature of 33°C. Further details are given in [64]. Immediate hypothermia is initiated within15 minutes of ROSC and again maintained at 32°C to 34°C for 6 hours. The experimental subjects are gradually rewarmed from 33.0°C to 37.0°C over 2 hours. Animals are maintained in an incubator to prevent post-ROSC hypothermia [70].

Figure 7.9 presents the results of qEEG analysis using the subband analysis algorithm, SIQ, presented previously. The motivation of using SIQ also is to separately characterize recovery trends in different EEG bands. Figure 7.9 and the results in [71] confirm that EEG recovery is better for immediate hypothermia over the conventional hypothermia. Analogously, NDS after 72 hours is also significantly improved in the IH group compared to the CH group ($p < 0.001$) (Figure 7.10). These studies also compared the IQ score measured at 24 hours with the NDS

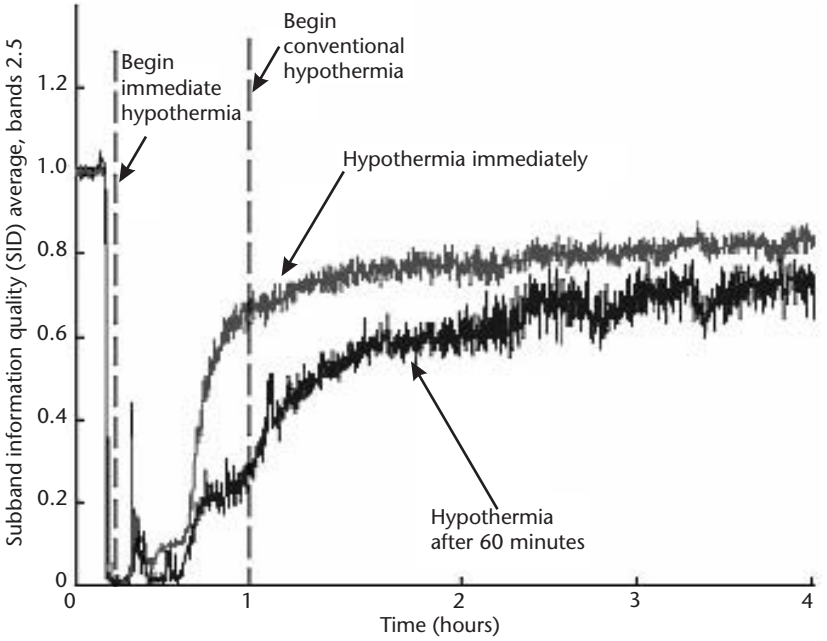



**Figure 7.9** Subband IQ of two representative animals receiving 9-minute asphyxial CA, followed by conventional hypothermia after 60 minutes and immediate hypothermia, and subsequently monitored for the first 4 hours. Immediate hypothermia results in rapid and early recovery of IQ levels.

values at 72 hours (Pearson correlation, 0.408; two-tailed significance, 0.025). In conclusion, early initiation of hypothermia improves not only the electrical response but also functional brain recovery in rats after CA.

### 7.5.3 qEEG Markers Predict Survival and Functional Outcome

The final question considered is whether hypothermia improves eventual survival. The actual question posed is this: Are the IQ values contrasted for the survivors and the animals that died prematurely? IQ values for animals that died were $0.48 \pm 0.04$, whereas values for the survivor group were $0.66 \pm 0.02$). The differences are statistically significant ($p < 0.001$) [Figure 7.11(a)]. IQ values are determined every 30 minutes starting from 30 minutes post-ROSC up until 4 hours. The animals that died prematurely showed significantly lower IQ values during each 30-minute interval studied compared with an average of the first 4 hours for the survivor group ($p < 0.05$) [Figure 7.11(b)].

Finally, IQ levels of rats with good and bad functional outcomes are compared. Rats with good outcomes, defined as NDS = 60, had a higher IQ level ($0.56 \pm 0.02$) than those with poor outcomes, defined as NDS < 60 ($0.56 \pm 0.02$). This difference is statistically significant ($p < 0.001$). These differences are consistent throughout the recovery periods from 30 minutes to 48 hours [Figure 7.11(c)]. To evaluate the overall performance, receiver operating characteristics (ROC) were calculated to determine IQ cutoff points. For an IQ value >0.523, the sensitivity was 81.8% sensitivity and specificity was 100% for predicting good outcomes, giving an area under the ROC curve of 0.864 [Figure 7.11(d)].

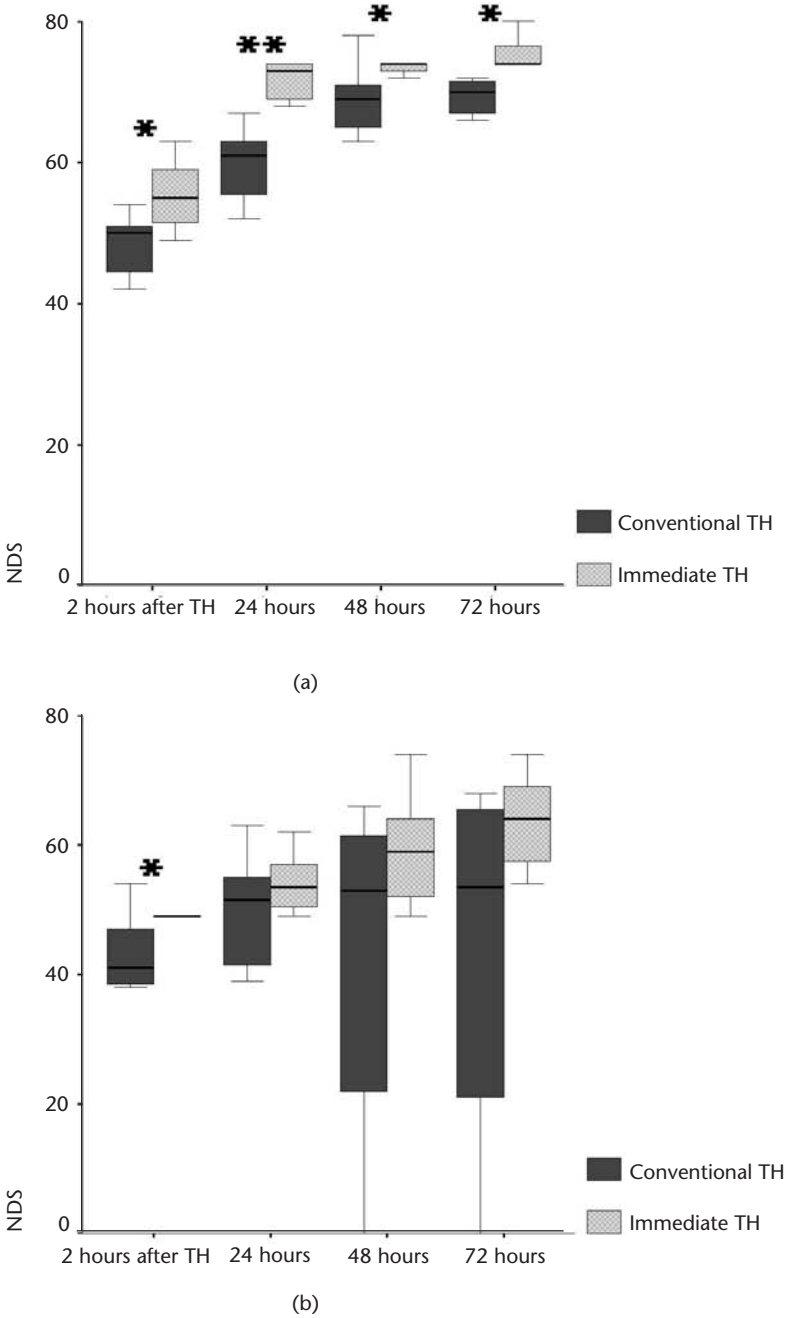



**Figure 7.10** NDS score by hypothermia and injury groups, Median (25th to 75th percentile): best outcome = 80; worst outcome = 0. TH: therapeutic hypothermia. A significant difference was noted over the 72-hour experiment in (a) 7-minute immediate hypothermia (7IH) versus conventional hypothermia (7CH) ($p = 0.001$) and (b) 9-minute IH (9IH) versus CH (9CH) ($p = 0.022$) asphyxial CA. Significant differences existed in all periods between the 7-minute groups and at 2 hour posthypothermia between the 9-minute groups (*$p < 0.05$, **$p < 0.01$). Note that qEEG was able to detect the significant difference as early as 30 minutes between the 9-minute groups, and qEEG values correlated well with 72-hour NDS values as early as 1 hour after CA. (*From:* [62]. © 2008 Elsevier. Reprinted with permission.)

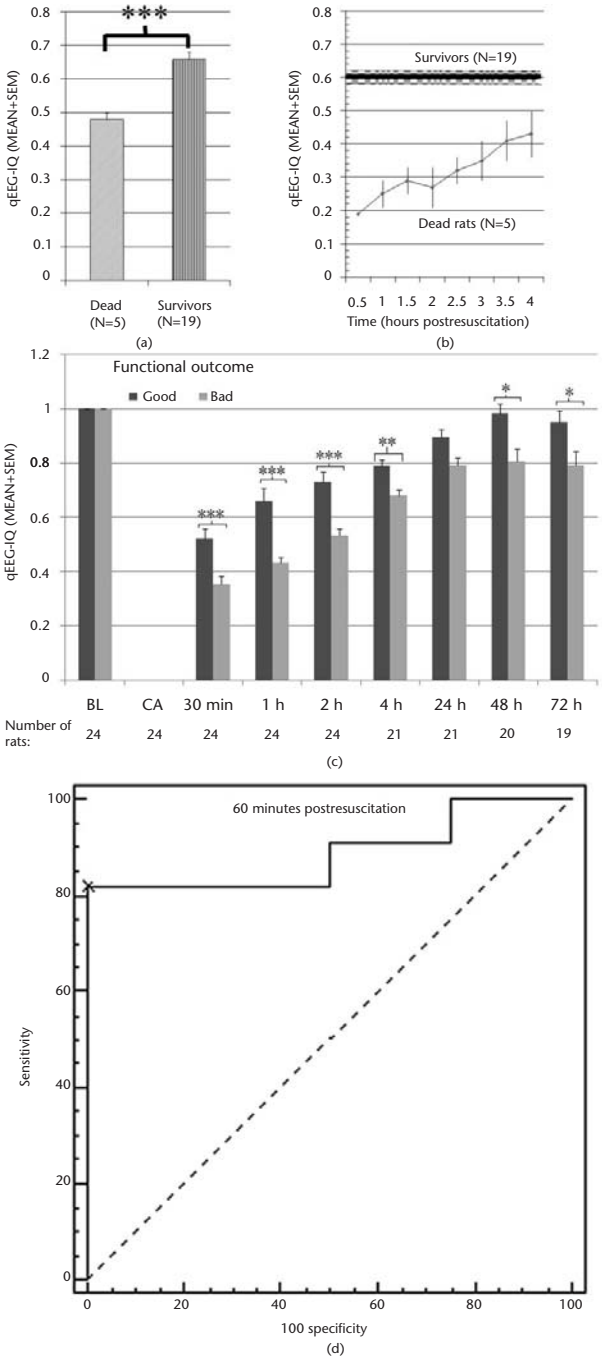



**Figure 7.11** Comparison of qEEG- IQ for (a) nonsurviving and surviving animals; rats that died within 72-hour postresuscitation had lower IQ values over the 72-hour experiment than survivors. (b) Rats that died prematurely showed significantly lower IQ during each 30-minute interval compared with an average of the first 4 hours (0.60 ± 0.02). The black line is the MEAN and the shadow is the SEM for survivors ($p < 0.05$). (c) Rats with a bad functional outcome (NDS < 60) had significantly lower qEEG-IQ values over 72 hours than those with a good functional outcome (NDS = 60). (d) This 1-hour postresuscitation ROC curve demonstrates the IQ value ($x$) with optimal sensitivity and specificity for good neurological outcomes. A cutpoint of >0.523 yielded 81.8% sensitivity and 100% specificity for good outcomes, with an area under the ROC curve of 0.886. (*From:* [63]. © 2008 Lippincott Williams & Wilkins. Reprinted with permission.)

## 7.6 Discussion of the Results

Cardiac arrest results in global ischemic brain injury leading to low survival and very poor neurological outcome in the majority of patients. Comatose CA survivors are typically cared for by nurses and physicians in general or cardiac intensive care units who may have little specialized training in neurological examination. Currently, the ability to monitor and detect even major or outcome-modifying changes in brain function in comatose patients is limited. Recent clinical studies have shown that hypothermia treatment, involving cooling the brain and the whole body, is able to ameliorate the neurological injury of global ischemia, leading to better survival and neurological function in CA survivors. At the present time, therapeutic hypothermia is provided within an empirically determined range of temperatures. However, a real-time method of tracking the effects of temperature on neurological recovery after CA has not been developed. Neurological monitoring may provide the means to evaluate the brain's response to global ischemic injury and to titrate the effects of hypothermic therapy.

The experimental studies of global ischemic brain injury following CA and the utility of the qEEG analysis based on the entropy measure IQ have been presented. The results show that the entropy measure IQ is an early marker of injury and neurological recovery after asphyxial CA. IQ accurately predicted the impact of temperature on recovery of cortical electrical activity, functional outcomes, and mortality soon after resuscitation. With the use of sham animals, we also demonstrated that it is not the temperature itself that alters the EEG, but the response of the injured brain to hypothermia or hyperthermia as manifested in the qEEG results. This review further validates the value of the IQ measure for predicting 72-hour NDS. Essentially, as early as 30-minute post-ROSC, a sufficient indication of the long-term outcome—as measured by NDS—is evident in the EEG signals. The most significant observations occurred within the first 2 hours while rats remain unresponsive and when clinical evaluation would be least reliable.

Based on ROC analysis, the optimal IQ cutoff point may be used as a threshold to predict the eventual good neurological outcomes. These studies demonstrate that IQ thresholds can be determined as early as 60 minutes after ROSC with the goal of reliably predicting the downstream neurological outcomes. From a neurological monitoring perspective, this review highlights the importance of the immediate postresuscitation period when brain injury may be most amenable to therapeutic interventions [72]. Recording the brain's electrical response and its rapid analysis by qEEG during the first 2 hours postresuscitation may thus prove to be clinically useful. Such monitoring during the early hours, coupled with hypothermia therapy, may help with efforts to protect the brain.

Hypothermia treatment in the studies reported led to better functional outcomes and EEG recovery quantified by IQ. IQ levels are significantly greater in rats treated with hypothermia compared to normothermic controls. The separation of IQ values between the treatment and control groups is noticeable within 1 hour of ROSC and persists throughout the 72-hour experiment. Better IQ values are associated with significant improvements in neurological function as measured by NDS throughout the experiment. IQ measure is able to detect the acceleration of neuro-

logical recovery as measured by NDS in animals treated with hypothermia. IQ also predicts the 72-hour functional outcomes as early as 4 hours after CA.

The qEEG methodology presented here has the potential for early prognostication after resuscitation. Early correlation of NDS and IQ at 4 hours may provide an opportune time for intervention or injury stratification. The strong correlation at 4 hours likely reflects the great variability between rats destined for good or poor outcomes during this period [9]. These early hours after ROSC are characterized by increasing frequency and complexity of bursts with a concomitant decrease in the duration of EEG suppression. Animals that proceed more quickly to a continuous EEG pattern have higher IQ values and better functional outcomes. The strong correlation between the 72-hour IQ and final NDS likely reflects the reemergence of EEG reactivity during this period in the group with good outcomes, whereas those with poor outcomes tend to have nonreactive $\alpha$ or $\theta$ patterns and lower IQ values.

Our research also shows that the earlier administration of therapeutic hypothermia after CA not only leads to better functional outcome compared to conventional hypothermia administration, but allows for reduction of treatment duration by half (6 hours versus 12 hours). We also showed that the effect of therapeutic hypothermia on brain recovery was detected by the qEEG measure. Our experiments lend further support to the theory that cooling should begin as soon as possible after ROSC. Hypothermia may have a greater impact during the early period of recovery, and injured neurons that are immediately treated may have a better chance of recovering.

One of the major goals of our group is to develop experimental approaches that can easily be translated clinically. The development of a noninvasive strategy to track the course of recovery early after resuscitation from CA has a number of readily translatable functions in humans. EEG technology is readily available in most hospitals and is familiar to staff, rather than being restricted to tertiary academic centers. Entropy analysis as exemplified by IQ simplifies interpretation of EEGs by translating complicated and subjective waveform analysis into an objective measure that can be displayed in real time, allowing physicians to monitor the response to potential neuroprotective strategies.

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CHAPTER 8

# Quantitative EEG-Based Brain-Computer Interface

Bo Hong, Yijun Wang, Xiaorong Gao, and Shangkai Gao

The brain-computer interface (BCI) is a direct (nonmuscular) communication channel between the brain and the external world that makes possible the use of neural prostheses and human augmentation. BCI interprets brain signals, such as neural spikes and cortical and scalp EEGs in an online fashion. In this chapter, BCIs based on two types of oscillatory EEG, the steady-state visual evoked potential from the visual cortex and the sensorimotor rhythm from the sensorimotor cortex, are introduced. Details of their physiological bases, principles of operation, and implementation approaches are provided as well.

For both of the BCI systems, the BCI code is embedded in an oscillatory signal, either as its amplitude or its frequency. With the merits of robust signal transmission and easy signal processing, the oscillatory EEG-based BCI shows a promising perspective for real applications as can be seen in the example systems described in this chapter. Some challenging issues in real BCI application, such as subject variability in EEG signals, coadaptation in BCI operation, system calibration, effective coding and decoding schemes, robust signal processing, and feature extraction, are also discussed.

## 8.1 Introduction to the qEEG-Based Brain-Computer Interface

### 8.1.1 Quantitative EEG as a Noninvasive Link Between Brain and Computer

In the past 15 years, many research groups have explored the possibility of establishing a direct (nonmuscular) communication channel between the brain and the external world, by interpreting brain signals, such as neural spikes and cortical and scalp EEGs, in an online fashion [1–3]. This communication channel is now widely known as the brain–computer interface. BCI research originally was aimed at being the next generation of neural prostheses, to help people with disabilities, especially locked-in patients, interact with their environment. Besides potential application in clinics, BCI has been adopted as a new way of human–computer interaction as well, which can provide healthy people with an augmentative means of operating a computer when it is inconvenient for some reason to use the hands, or for computer gaming.

Basically, our brain functions—from sensation to motor control to memory and decision making—originate from microvolt-level electrical pulses, the firing (action potential) of hundreds of billions of neurons. If all or part of the neuron firings could be captured, theoretically we would be able to interpret ongoing brain activity. With the help of microelectrode arrays and computational power advancements, this kind of system has been implemented. With tens of years of exploration of motor cortex function on primates, several neurophysiology groups have been able to teach a monkey to control a computer cursor and robotic arms by using its neuron activities [4–7]. More recently, human patients have been coupled with this kind of BCI and have been able to use direct brain control to guide external devices [8]. At this level, the BCI system is dealing with the neural activity at the resolution of a single neuron, that is, at the micrometer scale. This high resolution gives neuron-based BCI a remarkable information transfer rate, which ensures real-time control of the motion trajectory of a computer cursor or a robotic arm.

Because of the invasiveness and the technical difficulty of maintaining a long-term stable recording of neuron activity, the intracranial BCI has a long way to go before it is widely accepted by paralyzed patients. This obstacle holds true for the cortical EEG-based BCI [9], which places grid and/or strip electrodes under the dura, recording local field potentials from a large population of neurons.

The electrical activity from populations of neurons not only spreads inside the dura and skull, but also propagates to the surface of the scalp, which makes it possible to conduct noninvasive recording and interpreting of neural electrical signals and, hence, possibly a noninvasive BCI [2]. However, because of volume conduction, the EEG signal captured on the scalp is a blurred version of local field potentials inside the dura. In addition, the muscle activity, eye movement, and other recording artifacts contaminate the signal more, which make it impossible to conduct a direct interpretation of such signals. As discussed in other chapters of this book, numerous efforts have been made to improve the SNR of qEEG signals. Here, in the context of BCI, the challenge of interpreting noisy qEEG signals is even harder, because a BCI system requires real-time online processing [10].

### 8.1.2 Components of a qEEG-Based BCI System

As shown in Figure 8.1, a qEEG-based BCI system usually consists of three essential components: (1) intent "encoding" by the human brain, (2) control command "decoding" by a computer algorithm, and (3) real-time feedback of control results. The decoding component is the kernel part of a BCI system, linking the brain and external devices. It usually consists of three steps in the process: EEG acquisition, EEG signal processing, and pattern classification.

#### 8.1.2.1 BCI Input: Intent "Encoding" by Human Brain

In the neuron-based BCI system, the expression of subject's voluntary intent is straightforward. If the subject wants the computer cursor to move following a desired trajectory, he or she just needs to think about it as controlling his or her own hand [8]. In an EEG-based BCI system, however, there is not enough information contained in noisy EEGs for such explicit decoding and control. Typically, the con-

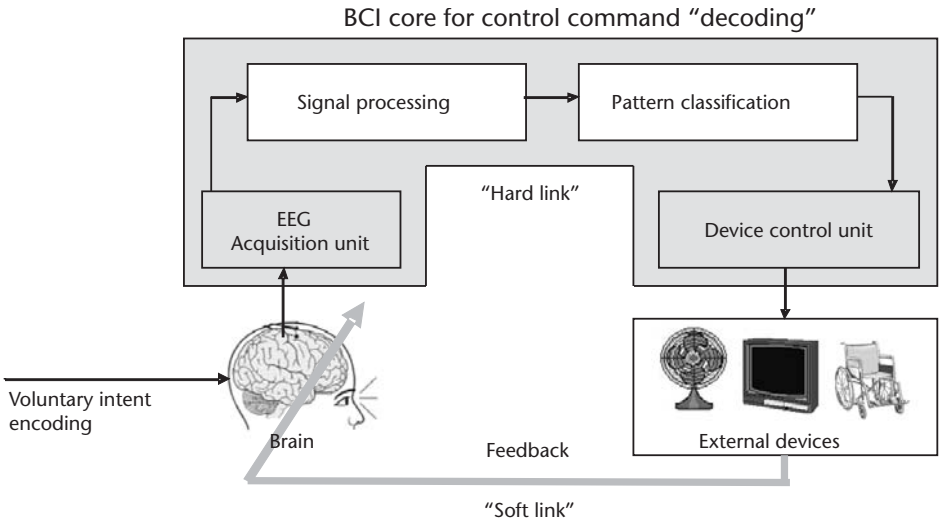



**Figure 8.1** Components of BCI system.

trol command, such as moving a cursor up or down, is assigned a specific mental state beforehand. The subject needs to perform the corresponding mental task to "encode" the desired control command, either through attention shift or by voluntary regulation of his EEG [2]. Currently, several types of EEG signals exist—such as sensorimotor rhythm (SMR; also known as $\mu/\beta$ rhythm) [11–13], steady-state visual evoked potential (SSVEP) [14, 15], slow cortical potential (SCP) [16, 17], and P300 [18, 19]—that can be used as neural media in the qEEG-based BCI system. Among these EEG signals, SMR and SCP can be modulated by the user's voluntary intent after training, whereas the SSVEP and P300 can be modulated by the user's attention shift. In fact, the design of the EEG-based BCI paradigm is largely about how to train or instruct the BCI user to express ("encode") his or her voluntary intent efficiently [20]. The more efficient the user's brain encodes voluntary intent, the stronger the target EEG signal we may have for further decoding.

#### 8.1.2.2 BCI Core: Control Command "Decoding" with a BCI Algorithm

Feeding the BCI system with a clear input is the function of a biological intelligent system—the brain, whereas translating input EEG signals into output control commands is the purpose of an artificial intelligent system—the BCI algorithm. Besides a high-quality EEG recording, appropriate signal processing (SP) and robust pattern classification are two major parts of a successful BCI system. Because scalp EEGs are weak and noisy, and the target EEG components are even weaker in a BCI context, various SP methods have been employed to improve the SNR and to extract meaningful features for classification in BCI [10].

Basically, these methods can be categorized into three domains: time, frequency, and space. In the time domain, for example, ensemble averaging is a widely used temporal processing technique to enhance the SNR of target EEG components, as in P300-based BCI. In the frequency domain, Fourier transform and wavelet analyses are very effective to find target frequency components, as in SMR and SSVEP-based BCI. In the space domain, spatial filter techniques such as common

spatial pattern (CSP) [21] and independent component analysis (ICA) methods [22] have been proved to be very successful in forming a more informative virtual EEG channel by combining multiple real EEG channels, as has been done for SMR-based BCI.

For most of the cases, the output of the signal processing is a set of features that can be used for further pattern classification. The task of pattern classification of a BCI system is to find a suitable classifier and to optimize it for classifying the EEG data into predefined brain states, that is, a logical value of class label. The process usually consists of two phases: offline training phase and online operating phase. The parameters of the classifier are trained offline with given training samples with class labels and then tested in the online BCI operating session. Various classifiers have been exploited in BCI research [23], among which the Fisher discriminant analysis and SVM classifiers bear the merit of robustness and better generalization ability. When considering pattern classification methods, keep in mind that the brain is an adaptive and dynamic system during interaction with computer programs. Basically, a linear classifier with low complexity is more likely to have good generalization ability and be more stable than nonlinear ones, such as a multilayer neural network.

#### 8.1.2.3 BCI Output: Real-Time Feedback of Control Results

As shown in Figure 8.1, two links are used to interface the brain and external devices. The BCI core as described earlier comprised of a set of amplifier and computer equipment with the proper program installed can be considered as a "hard link." Meanwhile, the feedback of control results is perceived by one of the BCI user's sensory pathway, such as the visual, auditory, or tactile pathway, which serves as a "soft link" to help the user adjust the brain activity for facilitating the BCI operation.

As discussed before, the BCI user needs to produce specific brain activity to drive the BCI system. The feedback tells the user how to modify their brain's encoding in order to improve the output, as happens during a natural movement control through the normal muscular pathway. It is the feedback that closes the loop of the BCI, resulting in a stable control system. Many experimental data have shown that, without feedback, BCI performance and robustness are much lower than in the feedback case [12, 24]. From this perspective, the performance of a BCI system is not only determined by the quality of the BCI translation algorithm, but also greatly affected by the BCI user's skill of modulating his or her brain activity. Thus, a proper design for the presentation of feedback could be a crucial point that can make a difference in terms of BCI performance.

### 8.1.3 Oscillatory EEG as a Robust BCI Signal

Evoked potentials, early visual/auditory evoked potentials like P100 or late potentials like P300, are low-frequency components, typically in the range of tens of microvolts in amplitude. As a transient brain response, an evoked potential is usually phase locked to the onset of an external stimulus or event [25], although oscillatory EEG, such as SSVEP or SMR, has a relatively higher frequency and larger

amplitude of several hundreds of microvolts. As a steady-state response, oscillatory EEG is usually time locked to the onset of an external stimulus or internal event, without strict phase locking [25].

Some transient evoked potential-based BCIs, such as the P300 speller [18, 19], show promising performance for real application with locked-in patients [26]. However, from the perspective of signal acquisition and processing, the oscillatory EEG-based BCI has several advantages over the ERP-based BCI: (1) The oscillatory EEG has a larger amplitude and needs no dc amplification, which greatly reduce the requirement of the EEG amplifier; (2) the oscillatory EEG is much less sensitive to low-frequency noise caused by eye movement and electrode impedance change, comparing with ERP; (3) the oscillatory EEG is a sustained response and requires merely coarse timing, which allows for the flexibility of asynchronous control, whereas for ERP-based BCI, stimulus synchrony is crucial for EEG recording and analysis; and (4) with amplitude and phase information easily obtained by robust signal processing methods, such as the FFT and Hilbert transform, there are more flexible ways of analyzing oscillatory EEGs than ERPs in a single trial fashion.

For these reasons, the oscillatory EEG-based BCI will be the focus of the following two sections of this chapter. Two major oscillatory EEG-based BCIs, SSVEP and SMR-based BCI, are introduced, along with details of their physiological mechanism, system configuration, alternative approaches, and related issues.

## 8.2 SSVEP-Based BCI

### 8.2.1 Physiological Background and BCI Paradigm

Visual evoked potentials (VEPs) reflect the visual information processing along the visual pathway and primary visual cortex. VEPs corresponding to low stimulus rates or rapidly repetitive stimulations are categorized as transient VEPs (TVEPs) and steady-state VEPs (SSVEPs), respectively [27]. Ideally, a TVEP is a true transient response to a visual stimulus that does not depend on any previous trial. If the visual stimulation is repeated with intervals shorter than the duration of a TVEP, the response evoked by each stimulus will overlap each other, and thus an SSVEP is generated. The SSVEP is a response to a visual stimulus modulated at a frequency higher than 6 Hz [25]. SSVEPs can be recorded from the scalp over the visual cortex, with maximum amplitude at the occipital region (around EEG electrode Oz).

Among brain signals recorded from the scalp, VEPs may be the first kind used as a BCI control. After Vidal's pilot VEP-based BCI system in the 1970s [28] and Sutter's VEP-based word processing program with a speed of 10 to 12 words/minute in 1992 [29], Middendorf et al. [15] and Gao et al. [30] independently reported the method for using SSVEPs to determine gaze direction.

Two physiological mechanisms underlie SSVEP-based BCI. The first one is the photic driving response [25], which is characterized by an increase in amplitude at the stimulus frequency, resulting in significant fundamental and second harmonics. Therefore, it is possible to detect the stimulus frequency based on measurement of SSVEPw. The second one is the central magnification effect [25]. Large areas of the visual cortex are allocated to processing the center of our field of vision, and thus the amplitude of the SSVEP increases enormously as the stimulus is moved closer to

the central visual field. For these two reasons, different SSVEP patterns can be produced by gazing at one of a number of frequency-coded stimuli. This is the basic principle of an SSVEP-based BCI.

As shown in Figure 8.2, in a typical SSVEP-based BCI setup, 12 virtual keyboard buttons appear on a screen and flash at different frequency, while the user gazes at a button labeled with the desired number/letter. The system determines the frequency of the SSVEP over visual cortex by means of spectral analysis and looks up the predefined table to decide which number/letter the user wants to select. In the example paradigm shown in Figure 8.2, when the BCI user directs his attention or gaze at the digit button "1" flashing at 13 Hz, a 13-Hz rhythmic component will appear in the EEG signal recorded over the occipital area of scalp, and can be detected by proper spectral analysis. Thus, the predefined command "1" will be executed. Although other flashing buttons may cause interference, because of the central magnification effect, 13-Hz components are very likely to dominate the power spectrum, compared with the flashing frequencies of other buttons. In this paradigm, the rhythmic SSVEP is modulated by the BCI user's gaze direction (attention) and the conveyed information is encoded in the frequency contents of occipital EEG.

With careful optimization of the system, an average information transfer rate (ITR) of more than 40 bits per second can be achieved [30, 31], which is relatively higher than most other BCI paradigms [2]. Besides a high information transfer rate, the recognized advantages of SSVEP-based BCI include easy system configuration, little user training, and robustness of system performance. This is the reason why it has received remarkably increased attention in BCI research [14, 15, 28–35].

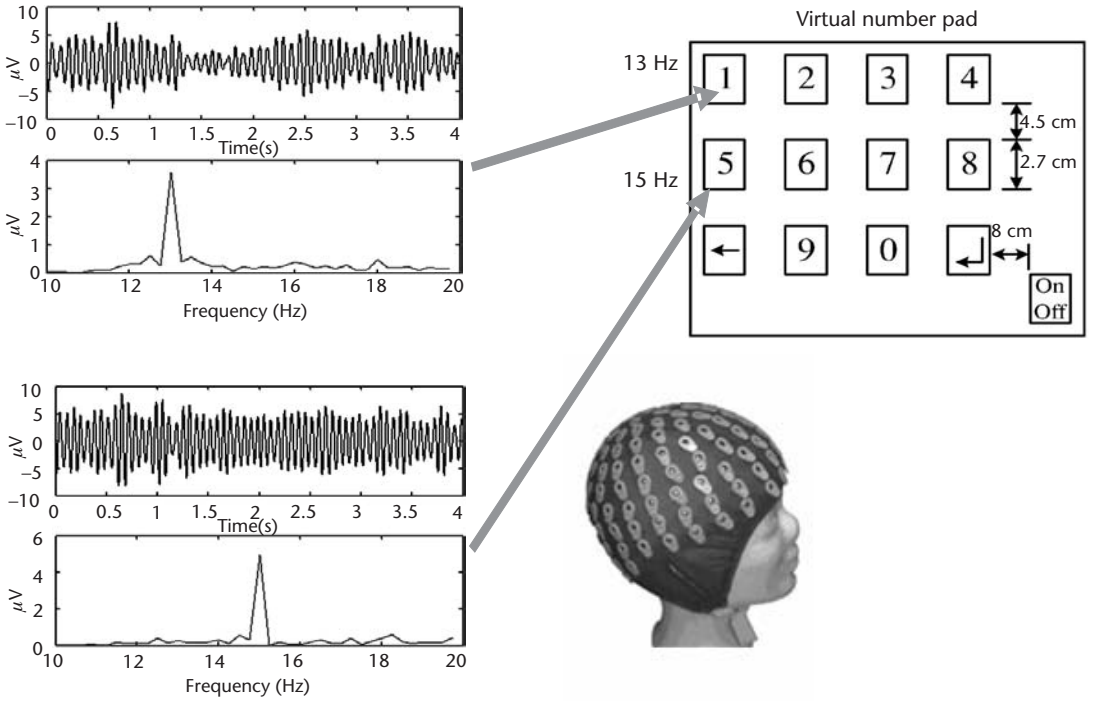



**Figure 8.2** Principle of SSVEP-based BCI.

Although various studies have been done to implement and evaluate SSVEP-based BCI demonstration systems in laboratories, the challenge facing the development of a practical BCI system for real-life application is still worth emphasizing. In the following section, a practical SSVEP-based BCI system implemented in our BCI group is introduced.

### 8.2.2 A Practical BCI System Based on SSVEP

#### 8.2.2.1 System Configuration

Our BCI system is composed of a visual stimulation and feedback unit (VSFU), an EEG data acquisition unit (EDAU), and a personal computer (Figure 8.3). In the VSFU, compact LED modules flickering at predefined frequency bands were employed as visual stimulators. For a typical setting, 12 LEDs in a 4-by-3 array formed an external number pad with numbers 0 through 9 and Backspace and Enter keys [Figure 8.3(b)]. When the user focused his/her visual attention on the flickering LED labeled with the number that he/she wanted to input, the EDAU and the software running on a PC identified the number by analyzing the EEG signal recorded from the user's head surface. By this means, the computer user was able to input numbers (0 through 9) and other characters with proper design of the input method. In the mode of mouse cursor control, four of the keys were assigned the UP, DOWN, LEFT, and RIGHT movements of the cursor. Real-time feedback of input characters was provided by means of a visual display and voice prompts.

Aiming at a PC peripheral device with standard interface, the hardware of a BCI system was designed and implemented as a compact box containing both an EEG data acquisition unit and a visual stimulation and feedback unit. Two USB ports are used for real-time data streaming from the EDAU and online control of the VSFU, respectively. In the EDAU, a pair of bipolar Ag/AgCl electrodes was placed over the

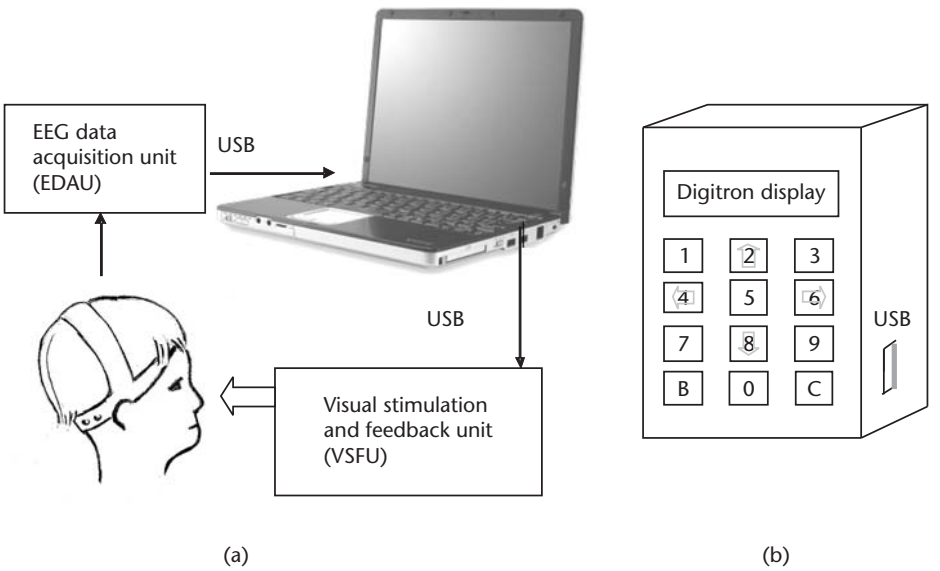



**Figure 8.3** System configuration of a practical BCI using SSVEP. (a) System configuration and main components; and (b) external number pad for visual stimulation and feedback.

user's occipital region on the scalp, typically on two sites around Oz in the 10-20 EEG electrode system. A tennis headband was modified to harness the electrodes on the head surface.

The EEG signal was amplified by a customized amplifier and digitized at a sampling rate of 256 Hz. After a 50-Hz notch filtering to remove the power line interference, the digital EEG data were streamed to PC memory buffer through a USB port. For the precision of frequency control, the periodical flickering of each LED was controlled by a separate lighting module, which downloads the frequency setting from the PC through the USB port. In one of the demonstrations, our BCI system was used for dialing a phone number. In that case, a local telephone line was connected to the RJ11 port of an internal modem on the PC.

#### 8.2.2.2 BCI Software and Algorithm

The main software running on the PC consists of key parts of the EEG translation algorithm, including signal enhancing, feature extraction, and pattern classification. The following algorithms were implemented in C/C++ and compiled into a stand-alone program. The real-time EEG data streaming was achieved by using a customized dynamic link library.

In the paradigm of SSVEP, the target LED evokes a peak in the amplitude spectrum at its flickering frequency. After a band filtering of 4 to 35 Hz, the FFT was applied on the ongoing EEG data segments to obtain the running power spectrum. If a peak value was detected over the frequency band of 4 to 35 Hz, the frequency corresponding to the peak was selected as the candidate of target frequency. To avoid a high false-positive rate, a crucial step was taken to ensure that the amplitude of a given candidate's frequency was higher than the mean power of the whole band. Herein, the ratio between the peak power and the mean power was defined as

$$Q = P_{peak} / P_{mean} \tag{8.1}$$

Basically, if the power ratio $Q$ was higher than the predefined threshold $T$, then the peak power was considered to be significant. For each individual, the threshold $T$ was estimated beforehand in the parameter customization phase. The optimal selection of the threshold balanced the speed and accuracy of the BCI system. Detailed explanation of this power spectrum threshold method can be found in previous studies [30, 31].

Due to the nonlinearity that occurs during information transfer in the visual system, strong harmonics are often found in the SSVEPs. Muller-Putz et al. investigated the impact of using SSVEP harmonics on the classification result of a four-class SSVEP-based BCI [32]. In their study, the accuracy obtained with combined harmonics (up to the third harmonic) was significantly higher than that obtained with only the first harmonic. In our experience, for some subjects, the intensity of the second harmonic may sometimes be even stronger than that of the fundamental component. Thus, analysis of the frequency band should cover at least the second harmonic, and the frequency feature has to be taken as the weighted sum of their powers, namely,

$$P(i) = \alpha P_{f1}(i) + (1-\alpha) P_{f2}(i) \quad i = 1, \ldots, N \tag{8.2}$$

where $N$ is the number of targets and, $P_{f1}(i)$ and $P_{f2}(i)$ are, respectively, the spectrum peak values of fundamental and second harmonics of $i$th frequency (i.e., $i$th target) and $\alpha$ is the optimized weighting factor that varies between subjects. Its empirical value may be taken as

$$\alpha = \frac{1}{N}\sum_{i=1}^{N} P_{f1}(i) \Big/ \left(P_{f1}(i) + P_{f2}(i)\right) \tag{8.3}$$

#### 8.2.2.3 Parameter Customization

To address the issue of individual diversity and to improve the subject applicability, a procedure of parameter customization was conducted before BCI operation. Our previous study suggests that the crucial system parameters include EEG electrode location, the visual stimulus frequency band, and the threshold ($T$) for target frequency determination [31]. To maintain the simplicity of operation and efficiency of parameter selection, a standard procedure was designed to help the system customization. It consists of the steps discussed next.

*Step 1: Frequency Scan*
Twenty-seven frequencies in the range of 6 to 19 Hz (0.5-Hz spacing) were randomly divided into three groups and the 9 frequencies in each group were randomly assigned to numbers 1 through 9 on the above-mentioned LED number pad. Then the frequency scan was conducted by presenting the numbers 1 through 9 on the digitron display one by one and each for 7 seconds. During this time period, the user was asked to gaze at the LED number pad corresponding to the presented number. This kind of scan was repeated for three sessions containing all 27 frequencies. There was a 2-second resting period between each number and a 1-minute resting period between groups. It took about 8 minutes for a complete frequency scan. The 7-second SSVEP response during each frequency stimulus was saved for the following offline analysis. In the procedure of frequency scanning, the bipolar EEG electrodes were placed at Oz (center of the occipital region) and one of its surrounding sites (3 cm apart on the left or right side). According to our previous study [31, 36], this electrode configuration was the typical one for most users.

*Step 2: Simulation of Online Operation*
The saved EEG segments were analyzed using the FFT to find the optimal frequency band with relatively high $Q$ values. The suitable value of the threshold $T$ and the weight coefficients $\alpha$ were estimated in a simulation of online BCI operation, in which the saved EEG data were fed into the algorithm in a stream.

*Step 3: Electrode Placement Optimization*
Only one bipolar lead was chosen as an input in our system. For some of the subjects, when the first two steps did not provide reasonable performance, an advanced electrode placement optimization method was employed to find the optimal bipolar electrodes. The best electrode pair for bipolar recording with the highest SNR was selected by mapping the EEG signal and noise amplitude over all possible elec-

trodes. Generally, the electrode giving the strongest SSVEP, which is generally located in the occipital region, is selected as the signal channel.

The location of the reference channel is searched under the following considerations: The amplitude of this channel's SSVEP should be lower and its position should lie in the vicinity of the signal channel such that the noise component is similar to that in the signal channel. A high SNR can then be gained when the potentials of the two electrodes are subtracted. Figure 8.4 shows an example of a significant enhancement of the SSVEP SNR derived from the lead selection method.

Most of the spontaneous background activities are eliminated after the subtraction; the SSVEP component, however, is retained. Details of this method can be found in previous studies [31, 36]. According to our observations, although the selection varies across subjects, it is relatively stable for each subject over time. This finding makes the electrode selection method feasible for practical BCI application. For a new subject, the multichannel mapping only needs to be done once to optimize the lead position.

In tests of the system based on frequency features (dialing a telephone number), with optimized system parameters for five participants, an average ITR of 46.68 bits/min was achieved.

### 8.2.3 Alternative Approaches and Related Issues

#### 8.2.3.1 SSVEP Feature: Amplitude Versus Phase

In the SSVEP BCI system based on frequency coding, the flickering frequencies of the targets are not the same. To ensure sufficiently high classification accuracy, a sufficient interval should be kept between two different frequencies such that the number of targets is restricted. If phase information embedded in SSVEPs is added, the number of flickering targets may be increased and a higher ITR should be expected. An SSVEP BCI based on phase coherent detection was proposed [37], in which two stimuli with the same frequency but different phases were discriminated

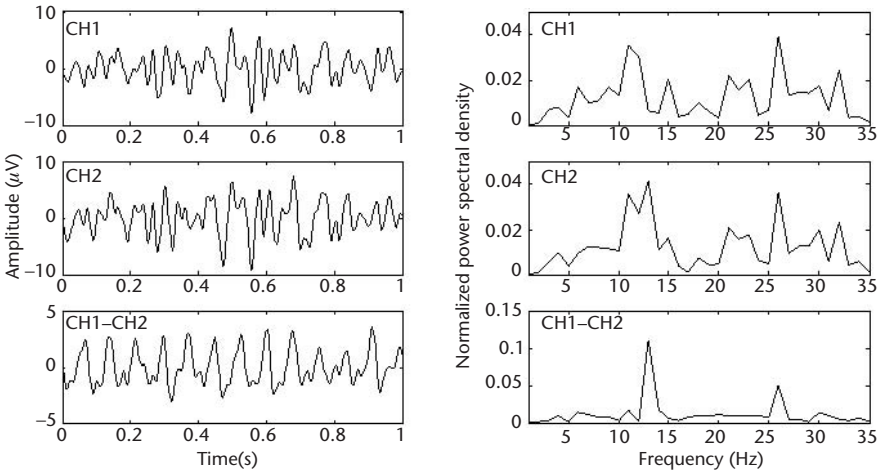



**Figure 8.4** Monopolar and optimal bipolar EEGs and their normalized power spectral density of one subject. The frequency of target stimulation is 13 Hz, shown as a clear peak in the bipolar electrode derivation, along with its harmonics at 26 Hz.

successfully in their demonstration. Inspired by this work, we tried to further the work by designing a BCI system with stimulating signals of six different phases under the same frequency. Initial testing indicates the feasibility of this method.

For a phase-encoded SSVEP BCI, flickering targets on a computer screen at the same frequency with strictly constant phase difference are required. We use the relatively stable computer screen refreshing signal (60 Hz) as a basic clock, and six stable 10-Hz signals are obtained by frequency division as shown in Figure 8.5. They are used for the stimulating signal of the flickering spots on the screen to control the flashing moment of the spots. The flashing moments [shadow areas along the time axis in Figure 8.5(a)] of the spots are interlaced by one refreshing period of the screen (1/60 second). In other words, because the process repeats itself every six times, the phase difference of the flashing is strictly kept at 60 degrees (taking the flashing cycle of all the targets as 360 degrees). Six targets flickering at the same frequency with different phases are thus obtained.

During the experiment, the subject was asked to gaze at the six targets respectively. The spectrum value at the characteristic frequency ($f_0$=10 Hz) was calculated simply by the following formula:

$$y(f_0) = \frac{1}{N}\sum_{n=1}^{N} x(n)\exp\left[-j2\pi(f_0/f_s)n\right] \tag{8.4}$$

where $f_s$ is the sampling frequency (1,000 Hz) and data length $N$ is determined by the length of the time window. The complex spectrum value at 10 Hz can be displayed on a plane of complex value as shown in Figure 8.5(b). With a data length of 1 second, six phase clusters are clearly shown. The SSVEP and visual stimulus signal are stably phase locked, sharing the same phase difference of 60 degrees between targets. This makes it possible to set up several visual targets flickering under the same frequency but with different phases so as to increase the number of targets for choice. As an example, we used the system described to implement an

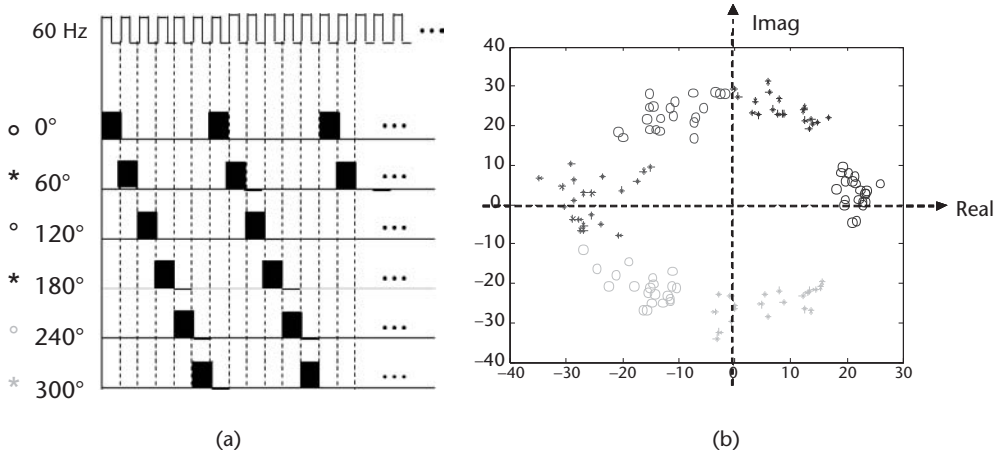



**Figure 8.5** An SSVEP phase interlacing design for discrimination of multiple screen targets. (a) Timing scheme for phase interlacing of six screen targets, with shadow areas indicating the ON time of each screen target, with a reference to the cycles of CRT screen refreshing; and (b) phase clustering pattern on the complex plane indicates a discriminability among the six targets.

EEG-operated TV controller, demonstrating the practicability of phase coding in SSVEP- based BCI systems.

#### 8.2.3.2 Coding Approach: Frequency Domain Versus Temporal Domain

According to the VEP signals used for information coding, VEP-based BCIs fall into two categories: transient VEPs and SSVEPs. The first category uses TVEPs to detect gaze direction. Spatial distributions of TVEPs elicited by a stimulus located in different visual fields were used by Vidal in the 1970s to identify visual fixation [28]. According to the approach for information coding, the SSVEP-based BCIs can be further divided into time-coded and frequency-coded subgroups. Hereafter, we refer to them as *t*SSVEP and *f*SSVEP, respectively. The BCI system described in Section 8.2.2 employs the *f*SSVEP approach. Instead of using a periodic flashing with fixed time interval between flashes, in Sutter's VEP-based BCI system, the occurrence time of visual flashes was not periodic (although it has a short interval as required by SSVEPs). The varying temporal patterns of these flashing sequences make it possible to discriminate among targets, thereby falling into the category of *t*SSVEP.

So far, both frequency decoding and temporal decoding strategies have been employed in VEP-based BCI research. Feature extraction of the TVEP is based on waveform detection in the temporal domain [38, 39]. Similarly, a template matching approach by cross-correlation analysis was used to detect the *t*SSVEP in the BRI system [29]. For a frequency-coded design, the amplitude of the *f*SSVEP from multiple flashing targets is modulated by gaze or spatial attention, and detected by using power spectral density estimation. Note that analysis of the TVEP and *t*SSVEP methods needs accurate time triggers from the stimulator, which can be omitted in frequency amplitude-based detection of the SSVEP.

#### 8.2.3.3 Muscular Dependence: Dependent Versus Independent BCI

According to the necessity of employing the brain's normal output pathways to generate brain activity, BCIs are divided into two classes: dependent and independent [2, 40]. The VEP system based on gaze detection falls into the dependent class. The generation of the desired VEP depends on gaze direction controlled by the motor activity of extraocular muscles. Therefore, this BCI is inapplicable for people with severe neuromuscular disabilities who may lack reliable extraocular muscle control.

Totally different from amplitude modulation by gaze control, recent studies on visual attention also reveal that the VEP is modulated by spatial attention and feature-based attention independent of neuromuscular function [41, 42]. These findings make it possible to implement an independent BCI based on attentional modulation of VEP amplitude. Only a few independent SSVEP-based BCIs have been reported, in which the amplitude of SSVEPs elicited by two flashing stimuli were covertly modulated by the subject's visual attention, without shifting gaze [34, 40, 43]. Compared with the dependent type, this attention-based BCI needs more subject training, attention, and concentration. The amplitude of SSVEP elicited by attention shifting is much lower than that elicited by gaze shifting, which poses a challenge when pursuing a high information transfer rate [34, 40].

#### 8.2.3.4 Stimulator: CRT Versus LED

In an SSVEP-based BCI, the visual stimulator serves as a visual response modulator and a virtual control panel, thus it is a crucial aspect of system design. The visual stimulator commonly consists of flickering targets in the form of color alternating or checkerboard reversing. Usually, the CRT/LCD monitor or flashtube/LED is used for stimulus display. A computer monitor is convenient for target alignment and feedback presentation by programming. But for a frequency-coded system, the number of targets is limited due to the refresh rate of the monitor and poor timing accuracy of the computer operating system. Therefore, an LED stimulator is preferable for a multiple-target system. The flickering frequency of each LED can be controlled independently by a programmable logic device. Using such a stimulator, a 48-target BCI was reported in [30].

The number of stimulation targets can be up to 64, leading to various system performances. Generally, the system with more targets can achieve a higher information transfer rate. For example, in tests of a 13-target system, the subjects had an average information transfer rate of 43 bits/min [31]. However, due to the fact that a stimulator with more targets is also more exhausting for users, the number of targets should be considered by evaluating the trade-off between system performance and user comfort.

#### 8.2.3.5 Optimization of Electrode Layout: Bipolar Versus Multielectrode

As we know, using a small number of electrodes can reduce the cost of hardware while improving the convenience of system operation. The Oz, O1, and O2 electrode positions of the international 10-20 system are widely used in SSVEP-based BCI. As shown in Section 8.2.2.3, in our system, we use a subject-specific electrode placement method to achieve a high SNR for the SSVEPs, especially for the subjects with strong background brain activities over the area of the visual cortex [31, 36].

In the near future, more convenient electrode designs, for example, the dry electrode [44], will be highly desirable as replacements for the currently used wet electrode. Under this circumstance, it is acceptable to use more electrodes to acquire more sufficient data to fulfill detection of SSVEP signals with multichannel data analysis approaches, for example, spatial filtering techniques described in [45] and the canonical correlation analysis method presented in [46]. An additional advantage of multiple-channel recording is that no calibration for electrode selection is needed.

## 8.3 Sensorimotor Rhythm-Based BCI

### 8.3.1 Physiological Background and BCI Paradigm

In scalp EEGs, the occipital alpha rhythm (8 to 13 Hz) is a prominent feature especially when the subject is in the resting wakeful state. This kind of spontaneous alpha rhythm is usually called “idling” activity. Besides visual alpha rhythm, a distinct alpha-band rhythm, in some circumstance with a beta-band accompaniment (around 20 Hz), can be measured over the sensorimotor cortex, which is called sensorimotor rhythm (SMR) [47, 48]. The mu and beta rhythms are commonly con-

sidered as EEG indicators of motor cortex and adjacent somatosensory cortex functions [49]. When the subject is performing a limb movement, thinking about a limb movement, or receiving a tactile/electrical stimulation on a limb, a prominent attenuation of ongoing mu rhythm can be observed over the rolandic area on the contralateral hemisphere [47, 48].

Following Pfurtscheller's classical work in the 1970s [50], this SMR attenuation is usually termed event-related desynchronization (ERD), whereas the increases in SMR amplitude are termed event-related synchronization (ERS). Moreover, the spatial distribution of ERD/ERS is closely related to the body map on the sensorimotor cortex. For example, the left hand and right hand produce the most prominent ERD/ERS pattern in the corresponding hand area in the contralateral sensorimotor cortex (Figure 8.6).

Thinking about, or imagining, a limb movement generates SMR patterns that are similar to those generated during real movement. These real/imagined movement patterns make up the physiological basis for SMR-based BCI (in some of the literature, this is also termed motor imagery-based BCI, or mu rhythm-based BCI) [13, 47, 48, 51].

In recent years, BCI systems based on classifying single-trial EEGs during motor imagery have developed rapidly. Most of the current SMR-based BCIs are based on characteristic ERD/ERS spatial distributions corresponding to different motor imagery states, such as left-hand, right-hand, or foot movement imagination. The first motor imagery-based BCI was developed by Pfurtscheller et al. and was based on the detection of EEG power changes caused by ERD/ERS of mu and beta rhythms during imagination of left- and right-hand movements [47]. As shown in Figure 8.6, for example, imagination of left-hand movement causes a localized decrease of

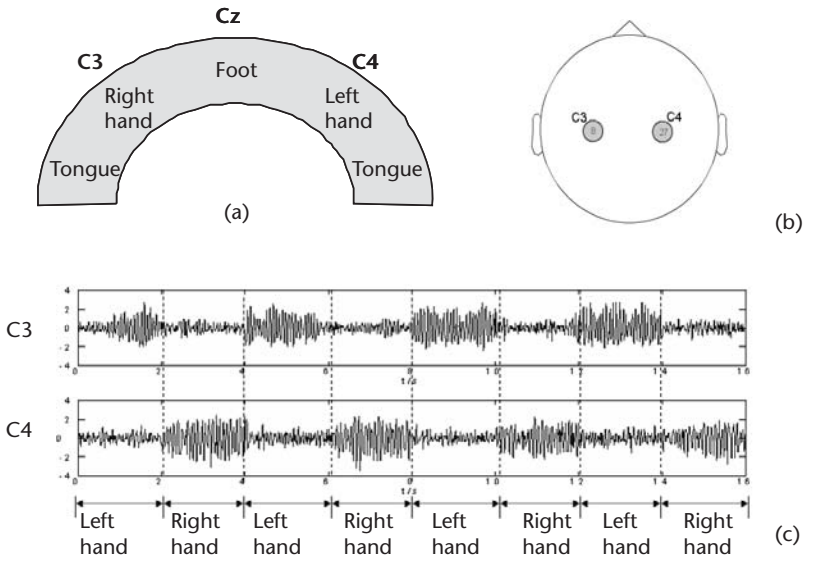



**Figure 8.6** Basic principle of SMR-based BCI. (a) Approximate representation areas of body parts shown in the coronal section of the sensorimotor cortex; (b) position of C3/C4 electrode on the scalp; and (c) typical EEG (bandpass filtered at 4 to 30 Hz, covering the mu and beta bands) during imagination of left- or right-hand movement, which shows a distinct temporal pattern on the C3/C4 electrode.

mu-band power around electrode C4 (over the corresponding cortex area of the left hand). Accordingly, right-hand imagery causes a similar mu-band power decrease on electrode C3. This makes it possible for a classifier to discriminate the states of left- or right-hand motor imagery just by using spatial distribution of mu-band power.

Another SMR-based BCI approach proposed by Wolpaw et al. was to train the users to regulate the amplitude of mu and/or beta rhythms to realize two-dimensional control of cursor movement [12]. Two linear equations were used to transform the sum and the difference of EEG power over left and right motor areas into vertical and horizontal movement of screen cursors.

### 8.3.2 Spatial Filter for SMR Feature Enhancing

In SMR-based BCI, localized spatial distribution of SMR is a crucial feature other than its temporal power change. Because EEG has very poor spatial resolution due to volume conduction, constructing virtual EEG channels using a weighted combination of original EEG recordings is a commonly used technique to get a clear local EEG activity, or "source activity" [21, 52]. The general idea of spatial filtering can be denoted by the following equation:

$$\mathbf{Y} = \mathbf{F} \cdot \mathbf{X} \tag{8.5}$$

where **X** is the original EEG data matrix, containing recordings from each electrode in its rows; and **F** is a square transformation matrix to project the original recordings to virtual channels in the new data matrix **Y**. Each row in **Y**, as a virtual channel, is a weighted combination of all (or part of) the original recordings. The filtered data matrix **Y** is supposed to be better than **X**, for extraction of task-related features.

So far, for SMR signal enhancement, two categories of spatial filters have been explored. One category is based on EEG electrode placement, such as common average reference (CAR) and Laplacian methods [53]. CAR virtual channels are obtained by subtracting the average signal across all EEG electrodes from each original channel, as shown in the following formula of weighted combination:

$$V_i^{CAR} = V_i^{ER} - \frac{1}{n}\sum_{j=1}^{n} V_j^{\mathrm{ER}} \quad i = 1, \ldots, n \tag{8.6}$$

where $n$ is the number of electrodes and $V_i^{\mathrm{ER}}$ is the original EEG recording. Similarly, the Laplacian channels are constructed by removing contributions of neighboring electrodes from the central electrode as follows:

$$\begin{aligned} V_i^{\mathrm{LAP}} &= V_i^{\mathrm{ER}} - \sum_{j \in S_i} g_{ij} V_j^{\mathrm{ER}} \\ g_{ij} &= \left(1/d_{ij}\right) \Big/ \sum_{k \in S_i} \left(1/d_{ik}\right) \end{aligned} \tag{8.7}$$

where $S_i$ is a subset of neighboring electrodes of the $i$th electrode and $d_{ij}$ denotes the geometric distance between electrode $i$ and electrode $j$. If $S_i$ consists of the near-

est-neighbor electrodes, the method is called small Laplacian. If the elements in $S_i$ are the next-nearest-neighbor electrodes, which have a larger distance to the central electrode, it is called large Laplacian. Both CAR and Laplacian methods serve as a spatial highpass filter, which enhances the local activity beneath the current electrodes. A comparison between these spatial filter shows that both the CAR and large Laplacian methods provide a better extraction of mu rhythm in SMR-based BCI [2, 53] than the small Laplacian method. This implies that although the SMR activity is a local one, it has a fairly broad spread.

The other category is the data-driven subject-specific spatial filter, which includes PCA, ICA, and common spatial pattern (CSP). Among these three filters, the PCA and ICA spatial filters are obtained through unsupervised learning, under certain statistic assumptions. Although they have been employed in some EEG-based BCI studies [22], manual intervention of the component selection is always a problem. Up to now, the CSP method is considered to be the most effective spatial filtering technique for enhancing SMR activity, and it has been successfully applied in many BCI studies [21, 52, 54].

Similar to the spatial filtering function described in (8.5), the main idea of CSP is to use a linear transform to project the multichannel EEG data into low-dimensional spatial subspace with a projection matrix, each row of which consists of the weights corresponding to each channel. This transformation can maximize the variance of two-class signal matrices. The EEG signals under two tasks A and B can be modeled as the combination of task-related components specific to each task and nontask components common to both tasks. In the case of discrimination of left- and right-hand imagery through EEGs, the aim of the CSP method is to design two spatial filters ($\mathbf{F}_L$ and $\mathbf{F}_R$), which led to the estimations of task-related source activities ($\mathbf{Y}_L$ and $\mathbf{Y}_R$) corresponding to left hand and right hand, respectively. Then, spatial filtering is performed to eliminate the common components and extract the task-related components. The $\mathbf{Y}_L$ and $\mathbf{Y}_R$ terms are estimated by $\mathbf{Y}_L = \mathbf{F}_L \cdot \mathbf{X}$ and $\mathbf{Y}_R = \mathbf{F}_R \cdot \mathbf{X}$, where $\mathbf{X}$ is the data matrix of preprocessed multichannel EEGs.

The calculation of the spatial filter matrix $\mathbf{F}_L$ and $\mathbf{F}_R$ is based on the simultaneous diagonalization of the covariance matrices of both classes. The EEG data of each trial is first bandpass filtered in the desired mu or beta band and then used to form matrix $\mathbf{X}_L$ and $\mathbf{X}_R$ of size $N * M$, where $N$ is the number of EEG channels and $M$ is the data samples for each channel. The normalized spatial covariance can be calculated as

$$\mathbf{R}_L = \frac{\mathbf{X}_L \mathbf{X}_L^T}{\text{trace}\left(\mathbf{X}_L \mathbf{X}_L^T\right)} \quad \mathbf{R}_R = \frac{\mathbf{X}_R \mathbf{X}_R^T}{\text{trace}\left(\mathbf{X}_R \mathbf{X}_R^T\right)} \tag{8.8}$$

Then $\mathbf{R}_L$ and $\mathbf{R}_R$ are averaged across all trials, respectively, for left and right imagery cases, to get more robust estimates of the spatial covariance $\overline{\mathbf{R}_L}$ and $\overline{\mathbf{R}_R}$. The composite spatial covariance $\mathbf{R}$, as the sum of $\overline{\mathbf{R}_L}$ and $\overline{\mathbf{R}_R}$, can be diagonalized by singular value decomposition (SVD):

$$\mathbf{R} = \overline{\mathbf{R}_L} + \overline{\mathbf{R}_R} = \mathbf{U}_0 \Sigma \mathbf{U}_0^T \tag{8.9}$$

where $\mathbf{U}_0$ is the eigenvector matrix, and $\Sigma$ is a diagonal matrix with corresponding eigenvalues as its diagonal elements. The variance in the space spanned by $\mathbf{U}_0$ components can be equalized by the following whitening matrix **P**:

$$\mathbf{P} = \Sigma^{-1/2}\mathbf{U}_0^T \tag{8.10}$$

It can be shown that, if $\overline{\mathbf{R}_L}$ and $\overline{\mathbf{R}_R}$ are transformed into $\mathbf{S}_L$ and $\mathbf{S}_R$ by whitening matrix **P**:

$$\mathbf{S}_L = \mathbf{P}\overline{\mathbf{R}_L}\mathbf{P}^T \quad \mathbf{S}_R = \mathbf{P}\overline{\mathbf{R}_R}\mathbf{P}^T \tag{8.11}$$

then $\mathbf{S}_L$ and $\mathbf{S}_R$ will share common eigenvalues. This means, given the SVD of $\mathbf{S}_L$ and $\mathbf{S}_R$,

$$\mathbf{S}_L = \mathbf{U}_L\Sigma_L\mathbf{U}_L^T \quad \mathbf{S}_R = \mathbf{U}_R\Sigma_R\mathbf{U}_R^T \tag{8.12}$$

the following equation holds true:

$$\mathbf{U}_L = \mathbf{U}_R = \mathbf{U} \quad \Sigma_L + \Sigma_R = \mathbf{I} \tag{8.13}$$

Thus, $\Sigma_L$ and $\Sigma_R$ may look like the following diagonal matrix:

$$\begin{aligned}\Sigma_L &= \mathrm{diag}\left[\overbrace{1\cdots1}^{m_L}\ \overbrace{\sigma_1\cdots\sigma_{m_C}}^{m_C}\ \overbrace{0\cdots0}^{m_R}\right]\\ \Sigma_R &= \mathrm{diag}\left[\overbrace{0\cdots0}^{m_L}\ \overbrace{\delta_1\cdots\delta_{m_C}}^{m_C}\ \overbrace{1\cdots1}^{m_R}\right]\end{aligned} \tag{8.14}$$

Because the sum of corresponding eigenvalues in $\Sigma_L$ and $\Sigma_R$ is always 1, the biggest eigenvalue of $\mathbf{S}_L$ corresponds to the smallest eigenvalue of $\mathbf{S}_R$. The eigenvectors in $\Sigma_L$ corresponding to the first **m** eigenvalues in $\Sigma_L$ are used to form a new transform matrix $\mathbf{U}_l$, which makes up the spatial filter with whitening matrix **P**, for extracting the so-called *source activity* of left-hand imagery. The spatial filters for the left and the right cases are constructed as follows:

$$\mathbf{F}_L = \mathbf{U}_l^T\mathbf{P} \quad \mathbf{F}_R = \mathbf{U}_r^T\mathbf{P} \tag{8.15}$$

Then the source activities $\mathbf{Y}_L$ and $\mathbf{Y}_R$ are derived by applying the preceding spatial filter on bandpass-filtered EEG data matrix **X**, that is,

$$\mathbf{Y}_L = \mathbf{F}_L \cdot \mathbf{X} \quad \mathbf{Y}_R = \mathbf{F}_R \cdot \mathbf{X} \tag{8.16}$$

Because of the way in which the spatial filter is derived, the filtered source activities $\mathbf{Y}_L$ and $\mathbf{Y}_R$ are expected to be better features for discriminating these two imagery tasks, compared with the original EEGs. Usually, the following inequation holds

true, which means the variance of the spatial filtered signal can be a good feature for classification purposes:

$$\mathrm{var}(\mathbf{F}_L \cdot \mathbf{X}_L) > \mathrm{var}(\mathbf{F}_R \cdot \mathbf{X}_L) \quad \mathrm{var}(\mathbf{F}_R \cdot \mathbf{X}_R) > \mathrm{var}(\mathbf{F}_L \cdot \mathbf{X}_R) \tag{8.17}$$

Alternatively, the band powers of $\mathbf{Y}_L$ and $\mathbf{Y}_R$ are more straightforward features. As shown in Figure 8.7, a more prominent peak difference can be seen on the power spectrum of the CSP-filtered signal than on the original power spectrum.

For the purpose of visualization, the columns of the inverse matrix of $\mathbf{F}_L$ and $\mathbf{F}_R$ can be mapped onto each EEG electrode to get a spatial pattern of CSP source distribution. As shown in the right-hand panel of Figure 8.7, the spatial distribution of $\mathbf{Y}_L$ and $\mathbf{Y}_R$ resembles the ERD topomap, which shows a clear focus in the left- and right-hand area over the sensorimotor cortex.

### 8.3.3 Online Three-Class SMR-Based BCI

#### 8.3.3.1 BCI System Configuration

In this study, three states of motor imagery were employed to implement a multiclass BCI. Considering the reliable spatial distributions of ERD/ERS in sensorimotor cortex areas, imagination of body part movements including those of the left hand, right hand, and foot were considered as mental tasks for generating detectable brain patterns. We designed a straightforward online feedback paradigm, where real-time visual feedback was provided to indicate the control result of three

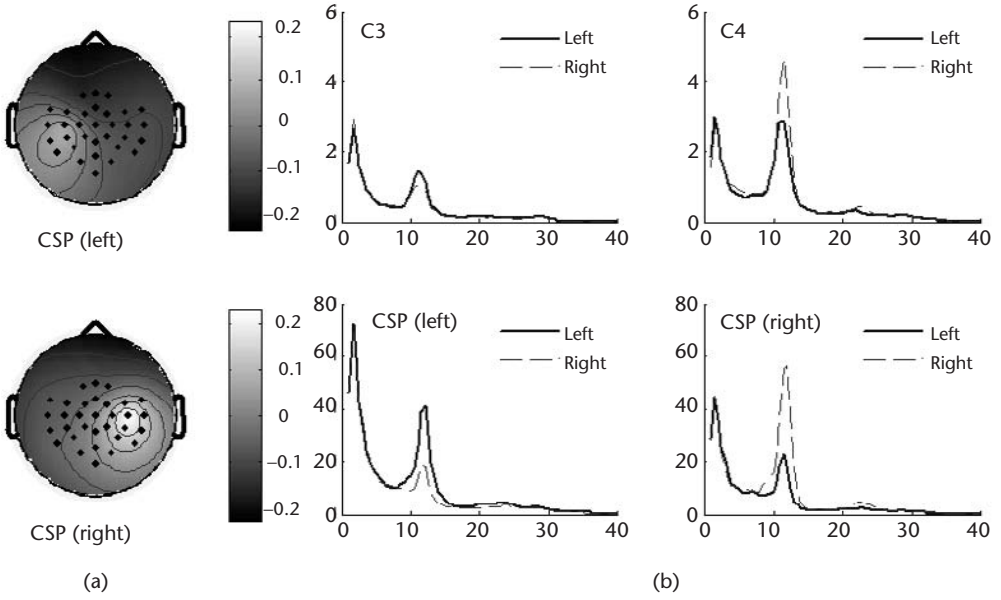



**Figure 8.7** CSP spatial filtering enhances the SMR power difference between left- and right-hand motor imagery. (a) CSP spatial pattern of left- and right-hand imagery; and (b) the PSD of the temporal signal, with a solid line for left imagery and a dashed line for right imagery. Upper row: PSD of raw EEG from electrodes C3 and C4; lower row: PSD of derived CSP temporal signal.

directional movements, that is, left-hand, right-hand, and foot imagery for moving left, right, and forward, respectively.

Five right-handed volunteers (three males and two females, 22 to 27 years old) participated in the study. They were chosen from the subjects who could successfully perform two-class online BCI control in our previous study [55]. The recording was made using a BioSemi ActiveTwo EEG system. Thirty-two EEG channels were measured at positions involving the primary motor area (M1) and the supplementary motor area (SMA) (see Figure 8.8). Signals were sampled at 256 Hz and preprocessed by a 50-Hz notch filter to remove the power line interference, and a 4- to 35-Hz bandpass filter to retain the EEG activity in the mu and beta bands.

Here we propose a three-phase approach to allow for better adaptation between the brain and the computer algorithm. The detailed procedure is shown in Figure 8.9. For phase 1, a simple feature extraction and classification method was used for

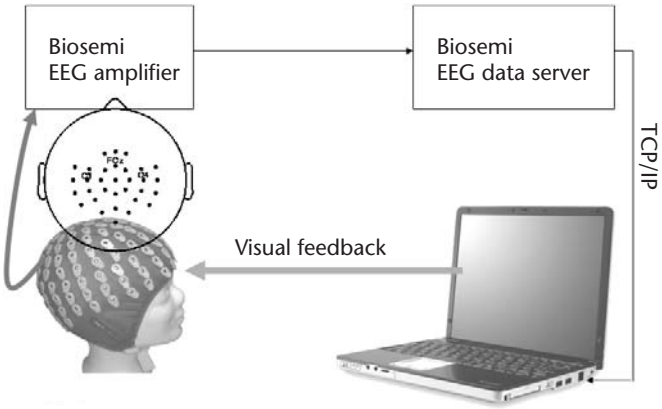



**Figure 8.8** System configurations for an online BCI using the motor imagery paradigm. EEG signals were recorded with electrodes over sensorimotor and surrounding areas. The amplified and digitized EEGs were transmitted to a laptop computer, where the online BCI program translated it into screen cursor movements for providing visual feedback for the subject.

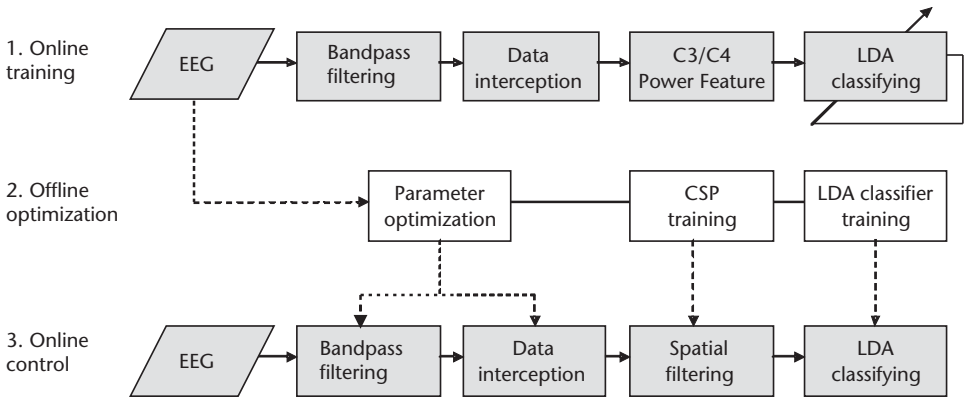



**Figure 8.9** Flowchart of three-phase brain computer adaptation. The brain and BCI algorithm were first coadapted in an initial training phase, then the BCI algorithm was optimized in the following phase for better online control in the last phase.

online feedback training, allowing for the initial adaptation of both the human brain and the BCI algorithm. For phase 2, the recorded data from phase 1 were employed to optimize the feature extraction and to refine the classifier parameters for each individual, aiming at a better BCI algorithm through refined machine learning. For the real testing phase, phase 3, three-class online control was achieved by coupling the trained brain and optimized BCI algorithm.

#### 8.3.3.2 Phase 1: Simple Classifier for Brain and Computer Online Adaptation

Figure 8.10 shows the paradigm of online BCI training with visual feedback. The "left hand," "right hand," and "foot" movement imaginings were designated to control three directional movements: left, right, and upward, respectively. The subject sat comfortably in an armchair, opposite a computer screen that displayed the visual feedback. The duration of each trial was 8 seconds. During the first 2 seconds, while the screen was blank, the subject was in relaxing state. At second 2, a visual cue (arrow) was presented on the screen, indicating the imagery task to be performed.

The arrow pointing left, right, and upward indicated the task of imagination of left-hand, right-hand, and foot movement, respectively. At second 3, three progress bars with different colors started to increase simultaneously from three different directions. The value of each bar was determined by the accumulated classification results from a linear discriminant analysis (LDA), and it was updated every 125 ms. For example, if the current classification result is "foot," then the "up" bar will increase one step and the values of the other two bars will be retained. At second 8, a true or false mark appeared to indicate the final result of the trial through calculating the maximum value of the three progress bars, and the subject was asked to relax and wait for the next task. The experiment consisted of two or four sessions and each session consisted of 90 trials (30 trials per class). The dataset comprising 360 or 180 trials (120 or 60 trials per class) was used for further offline analysis.

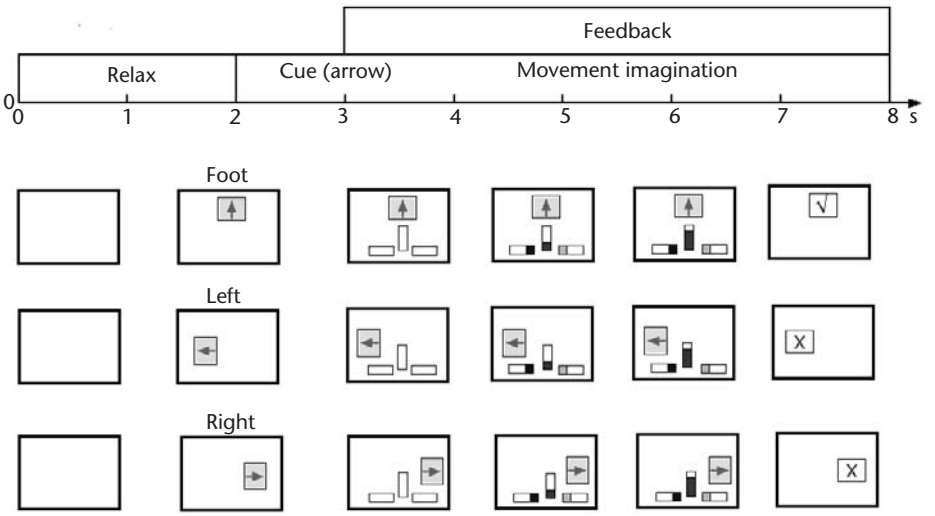



**Figure 8.10** Paradigm of three-class online BCI training with visual feedback.

The features extracted for classification were bandpass power of mu rhythms on left and right primary motor areas (C3 and C4 electrodes). LDA was used to classify the bandpass power features on C3/C4 electrodes referenced to FCz [9]. A linear classifier was defined by a normal vector **w** and an offset **b** as

$$y = \text{sign}\left(\mathbf{w}^T \mathbf{x} + b\right) \tag{8.18}$$

where **x** was the feature vector. The values of **w** and **b** were determined by Fisher discriminant analysis (FDA). The three-class classification was solved by combining three binary LDA discriminant functions:

$$\begin{aligned} \mathbf{x}(t) &= \left[P_{C3}(t) P_{C4}(t)\right]^T \\ y_i(t) &= \text{sgn}\left(\mathbf{w}_i^T \mathbf{x}(t) + b_i\right), i = 1-3 \end{aligned} \tag{8.19}$$

where $P_{C3}(t)$ and $P_{C4}(t)$ are values of the average power in the nearest 1-second time window on C3 and C4, respectively. Each LDA was trained to discriminate two different motor imagery states. The decision rules are listed in Table 8.1, in which six combinations were designated to the three motor imagery states, respectively, with two combinations not classified.

An adaptive approach was used to update the LDA classifiers trial by trial. The initial normal vectors $\mathbf{w}_i^T$ of the classifiers were selected as [+1 –1], [0 –1], and [–1 0] (corresponding to the three LDA classifiers in Table 8.1) based on the ERD distributions. They were expected to recognize the imagery states through extracting the power changes of mu rhythms caused by contralateral distribution of ERD during left- and right-hand imagery, but bilateral power equilibrium during foot imagery over M1 areas [47, 48]. The initial **b** was set to zero.

When the number of samples reached five trials per class, the adaptive training began. Three LDA classifiers were updated trial by trial, gradually improving the generalization ability of the classifiers along with the increase of the training samples. This kind of gradual updating of classifiers provided a chance for initial user brain training and system calibration in an online BCI.

Figure 8.11 shows the probability that three progress bars won during an online feedback session. In each motor imagery task, the progress bar that has the maxi-

**Table 8.1** Decision Rules for Classifying the Three Motor Imagery States Through Combining the Three LDA Classifiers

| *Left Versus Right* | *Left Versus Foot* | *Right Versus Foot* | *Decision* |
|---|---|---|---|
| +1 | +1 | –1 | Left |
| +1 | +1 | +1 | Left |
| –1 | +1 | +1 | Right |
| –1 | –1 | +1 | Right |
| +1 | –1 | –1 | Foot |
| –1 | –1 | –1 | Foot |
| +1 | –1 | +1 | None |

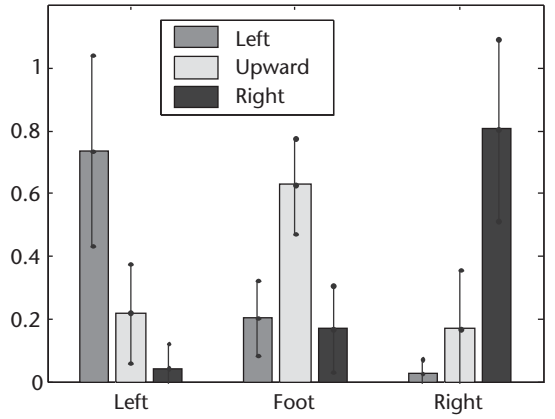



**Figure 8.11** Winning probability of three progress bars in three-class motor imagery (one subject, 120 trials per class).

mum value correctly indicates the true label of the corresponding class. For example, during foot imagination, the "up" bar had a much higher value than the "left" and "right" bars; therefore, for most foot imagery tasks, the final decision was correct although some errors may occur.

#### 8.3.3.3 Phase 2: Offline Optimization for Better Classifier

To improve the classification accuracy, we used the common spatial patterns method, as described earlier, to improve the SNR of the mu rhythm through extracting the task-related EEG components.

The CSP multiclass extensions have been considered in [56]. Three different CSP algorithms were presented based on one-versus-one, one-versus-rest, and approximate simultaneous diagonalization methods. Similar to the design of binary classifiers, the one-versus-one method was employed in our system to estimate the task-related source activities as the input of the binary LDA classifiers. It can be easily understood and with fewer unclassified samples compared to the one-versus-rest method. The design of spatial filters through approximate simultaneous diagonalization requires a large amount of calculation and the selection of the CSP patterns is more difficult than the two-class version.

As illustrated earlier in Figure 8.9, before online BCI control, the CSP-based training procedure was performed to determine the parameters for data preprocessing, the CSP spatial filters, and the LDA classifiers. A sliding window method was integrated to optimize the frequency band and the time window for data preprocessing in the procedure of joint feature extraction and classification. The accuracy was estimated by a $10 \times 10$-fold cross-validation. The optimized parameters, CSP filters, and LDA classifiers were used to implement the online BCI control and ensured a more robust performance compared with the online training procedure.

Table 8.2 lists the parameters for data preprocessing and the classification results for all subjects. The passband and the time window are subject-specific parameters that can significantly improve the classification performance. Average accuracy derived from online and offline analysis was 79.48% and 85.00%, respec-

**Table 8.2** Classification Accuracies of Three Phases

| *Subjects* | *Passband (Hz)* | *Time Window (seconds)* | *Phase 1 Accuracy (%)* | *Phase 2 Accuracy (%)* | *Phase 3 Accuracy (%)* |
|---|---|---|---|---|---|
| S1 | 10–35 | 2.5–8 | 94.00 | 98.11 | 97.03 |
| S2 | 13–15 | 2.5–7.5 | 94.67 | 97.56 | 95.74 |
| S3 | 9–15 | 2.5–7 | 74.71 | 80.13 | 81.32 |
| S4 | 10–28 | 2.5–6 | 68.00 | 77.00 | 68.40 |
| S5 | 10–15 | 2.5–7.5 | 66.00 | 72.22 | 71.50 |
| Mean | — | — | 79.48 | 85.00 | 82.80 |

tively. For subjects S1 and S2, no significant difference existed between the classification results of the three binary classifiers, and a high accuracy was obtained for three-class classification. For the other three subjects, the foot task was difficult to recognize, and the three-class accuracy was much lower than the accuracy of classifying left- and right-hand movements. This result may be caused by less training of the foot imagination, because all of the subjects did more training sessions of hand movement in previous studies of two-class motor imagery classification [55]. The average offline accuracy was about 5% higher than the online training phase due to the employment of parameter optimization and the CSP algorithm applied to multichannel EEG data.

#### 8.3.3.4 Phase 3: Online Control of Three-Direction Movement

In phase 3, a similar online control paradigm as in phase 1 was first employed to test the effect of parameter optimization, and a 3% increase in online accuracy was observed. Then, three of the subjects participated in online control of three-direction movement of robot dogs (SONY, Aibo) for mimicking a brain signal controlled robo-cup game, in which one subject controlled the *goalkeeper* and the other controlled the *shooter*. This paradigm and approach could be used for applications such as wheelchair control [57] and virtual reality gaming [58, 59].

### 8.3.4 Alternative Approaches and Related Issues

#### 8.3.4.1 Coadaptation in SMR-Based BCI

As discussed in Section 8.1.2, the BCI is not just a feedforward translation of brain signals into control commands; rather, it is about the bidirectional adaptation between the human brain and a computer algorithm [2, 6, 60], in which real-time feedback plays a crucial role during coadaptation.

For an SSVEP-based BCI system, the amplitude modulation of target EEG signals is automatically achieved by voluntary direction of the gaze direction and only the primary visual area is involved in the process. In contrast, for an SMR-based BCI system, the amplitude of the mu and/or beta rhythm is modulated by the subject's voluntary manipulation of his or her brain activity over the sensorimotor area, in which secondary, even high-level, brain areas are possibly involved. Thus, the

BCI paradigm with proper consideration of coadaptation feasibility is highly preferred for successful online BCI operation.

As summarized by McFarland et al. [61], there are at least three different paradigms for training (coadaptation) in an SMR-based BCI: (1) the "let the machines learn" approach, best demonstrated by the Berlin BCI group on naive subjects [51]; (2) the "let the brain learn" or "operant-conditioning," best demonstrated by the Tübingen BCI group on well-trained subjects [62]; or (3) the "let the brain and computer learn and coadapt simultaneously," best demonstrated by the Albany BCI group on well-trained subjects [12, 61]. Basically, the third approach fits the condition of online BCI control best, but poses the challenge of online algorithm updating, especially when a more complicated spatial filter is considered.

Alternatively, we have proposed a three-step BCI training paradigm for coadaptation. The brain was first trained for a major adaptation, then the BCI algorithm was trained offline, and finally the trained brain and fine-tuned BCI algorithm were coupled to provide better online operation. This can be best expressed by the statement "let the brain learn first, then the machines learn," which results in a compromise between maintaining an online condition and the more simple task of online algorithm updating.

#### 8.3.4.2 Optimization of Electrode Placement

Different spatial distribution of SMR over sensorimotor areas is the key to discriminating among different imagery brain states. Although the topographic organization of the body map is genetic and conservative, each individual displays considerable variability because of the handiness, sports experience, and other factors that may cause a plastic change in the sensorimotor cortex. To deal with this spatial variability, a subject-specific spatial filter has proven to be very effective in the case of multiple-electrode EEG recordings. For a practical or portable BCI system, placing fewer EEG electrodes is preferred. Thus, it is crucial to determine the optimal electrode placement for capturing SMR activity effectively.

In a typical SMR-based BCI setting [48], six EEG electrodes were placed over the cortical hand areas: C3 for the right hand, C4 for the left hand, and two supplementary electrodes at positions anterior and posterior to C3/C4. Different bipolar settings, such as anterior-central (a-c), central-posterior (c-p), and anterior-posterior (a-p), were statistically compared and a-c bipolar placement was verified as the optimal one for capturing mu-rhythm features for 19 out of 34 subjects.

Instead of this typical setting, for considering the physiological role of the supplementary motor area (SMA), we proposed a novel electrode placement with only two bipolar electrode pairs: C3-FCz and C4-FCz. Functional neuroimaging studies indicated that motor imagery also activates the SMA [63] (roughly under electrode FCz). We investigated the phase synchronization of mu rhythms between the SMA and the hand area in M1 (roughly under electrode C3/C4) and observed a contralaterally increased synchronization similar to the ERD distribution [55]. This phenomenon makes it possible to utilize the signal over the SMA to enhance the significance of the power difference between M1 areas, by considering SMA (FCz) as the reference. It was demonstrated to be optimal for recognizing motor imagery states, which can satisfy the necessity of a practical BCI [64]. This simple and effec-

tive electrode placement can be a default setting for most subjects. For a more subject-specific optimization, ICA can be employed to find the "best" bipolar electrode pairs to retain the mu rhythm relevant signal components and to avoid other noisy components, which is similar with that described in the Section 8.2.2.3.

#### 8.3.4.3 Visual Versus Kinesthetic Motor Imagery

As discussed in Section 8.1.2, an EEG-based BCI system requires the BCI user to generate specific EEG activity associated with the intent he or she wants to convey. The effectiveness of producing the specific EEG pattern by the BCI user largely determines the performance of the BCI system. In SMR-based BCI, for voluntary modulation of the $\mu$ or $\beta$ rhythm, the BCI user needs to do movement imagination of body parts. Two types of mental practice of motor imagery are used: *visual* motor imagery, in which the subject produces a visual image (mental video) of body movements in the mind, and *kinesthetic* imagery, in which the subject rehearses his or her own action performed with imagined kinesthetic feelings.

In a careful comparison of these two categories of motor imagery, the kinesthetic method produced more significant SMR features than the visual one [65]. In our experience with SMR-based BCI, those subjects who get used to kinesthetic motor imagery perform better than those who do not. And usually, given same experiment instructions, most of the naïve subjects tend to choose visual motor imagery, whereas well-trained subjects prefer kinesthetic imagery. As shown in Neuper et al.'s study [65], the spatial distribution of SMR activity on the scalp varies between these two types of motor imagery, which implies the necessity for careful design of the spatial filter or electrode placement to deal with this spatial variability.

#### 8.3.4.4 Phase Synchrony as BCI Features

Most BCI algorithms for classifying EEGs during motor imagery are based on the feature derived from power analysis of SMR. Phase synchrony as a bivariate EEG measurement could be a supplementary, even an independent, feature for novel BCI algorithms. Because phase synchrony is a bivariate measurement, it is subject to the proper selection of electrode pairs for the calculation. Basically, two different approaches are used. One is a random search among all possible electrode pairs with a criteria function related to the classification accuracy [66, 67]; the other is a semi-optimal approach that employs physiological prior knowledge to select the appropriate electrode pairs. Note that the latter approach has the advantage of lower computation costs, robustness, and better generalization ability, which has been shown in our study [55].

We noticed that phase coherence/coupling has been widely used in the physiology community and motor areas beyond primary sensorimotor cortex have been explored to find the neural coupling between these areas. Gerloff et al. demonstrated that, for both externally and internally paced finger extensions, functional coupling occurred between the primary sensorimotor cortex (SM1) of both hemispheres and between SM1 and the mesial premotor (PM) areas, probably including the SMA [68]. The study of event-related coherence showed that synchronization

between mu rhythms occurred in the precentral area and SM1 [69]. Spiegler et al. investigated phase coupling between different motor areas during tongue-movement imagery and found that phase-coupled 10-Hz oscillations were induced in SM1 and SMA [70]. All of this evidence points to the possible neural synchrony between SMA (and/or PM) and SM1 during the motor planning, as well as the motor imagery. Thus, we chose electrode pairs over SM1 and SMA as the candidate for phase synchrony measurement.

In one of our studies [55], a phase-locking value was employed to quantify the level of phase coupling during imagination of left- or right-hand movements, between SM1 and SMA electrodes. To the best of our knowledge, for the first time, use of a phase-locking value between the SM1 and SMA in the band of the mu rhythm was justified as additional features for the classification of left- or right-hand motor imagery, which contributed almost as much of the information as the power of the mu rhythm in the SM1 area. A similar result was also obtained by using a nonlinear regressive coefficient [71].

## 8.4 Concluding Remarks

### 8.4.1 BCI as a Modulation and Demodulation System

In this chapter, brain computer interfaces based on two types of oscillatory EEGs—the SSVEP from the visual cortex and the SMR from the motor cortex—were introduced and details of their physiological bases, example systems, and implementation approaches were given. Both of these BCI systems use oscillatory signals as the information carrier and, thus, can be thought of as modulation and demodulation systems, in which the human brain acts as a modulator to embed the BCI user's voluntary intent in the oscillatory EEG. The BCI algorithm then demodulates the embedded information into predefined codes for devices control.

In SSVEP-based BCI, the user modulates the photonic-driven response of the visual cortex by directing his or her gaze direction (or visual attention) to the target with different flashing frequencies. With an enhanced target frequency component, the BCI algorithm is able to use frequency detection to extract the predefined code, which largely resembles the process of frequency demodulation. Note that the carried information is a set of discrete BCI codes, instead of a continuous value, and the carrier signal here is much more complicated than a typical pure oscillation, covering a broad band of peri-alpha rhythms, along with other spontaneous EEG components. The SMR-based BCI system, however, resembles an amplitude modulation and demodulation system in which the BCI user modulates the amplitude of the mu rhythm over the sensorimotor cortex by doing specific motor imagery, and the demodulation is done by extracting the amplitude change of the mu-band EEG. The difference from typical amplitude modulation and demodulation systems is that two or more modulated EEG signals from specific locations are combined to derive a final code, for example, left, right, or forward.

For both of the BCI systems, the BCI code is embedded in an oscillatory signal, either as its amplitude or its frequency. As stated at the beginning of this chapter, this type of BCI bears the merit of robust signal transmission and easy signal processing. All examples demonstrated and reviewed in previous sections have indi-

cated a promising perspective for real applications. However, it cannot escape from the challenge of nonlinear and dynamic characteristics of brain systems as well, especially in terms of information modulation. The way in which the brain encodes/modulates the BCI code into the EEG activity varies across subjects and changes with time. These factors pose the challenge of coadaptation as discussed in the previous section. This suggests again that BCI system design is not just about the algorithm and that human factors should be considered very seriously.

### 8.4.2 System Design for Practical Applications

For the BCI systems discussed here, many studies have been done to implement and evaluate demonstration systems in the laboratory; however, the challenge facing the development of practical BCI systems for real-life application is still worth emphasizing. According to a survey done by Mason et al. [72], the existing BCI systems could be divided into three classes: transducers, demo systems, and assistive devices. Among the 79 BCI groups investigated, 10 have realized assistive devices (13%), 26 have designed demonstration systems (33%), and the remaining 43 are only in the stage of offline data analysis (54%). In other words, there is still a long way to go before BCI systems can be put into practical use. However, as an emerging engineering research field, if it can only stay in the laboratory for scientific exploration, its influence on human society will certainly be limited. Thus, the feasibility of creating practical applications is a serious challenge for BCI researchers. A practical BCI system must fully consider the user's human nature, which includes the following two key aspects:

1. *A better electrode system is needed that allows for convenient and comfortable use.* Current EEG systems use standard wet electrodes, in which electrolytic gel is required to reduce electrode-skin interface impedance. Using electrolytic gel is uncomfortable and inconvenient, especially if a large number of electrodes are adopted. First of all, preparations for EEG recording before BCI operation are time consuming. Second, problems caused by electrode damage or bad electrode contact can occur. Third, an electrode cap with large numbers of electrodes is uncomfortable for users to wear and then not suitable for long-term recording. Moreover, an EEG recording system with a high number of channels is usually quite expensive and not portable. For all of these reasons, reducing the number of electrodes in a BCI system is a critical issue and, currently, it has become the bottleneck in developing an applicable BCI system. In our system, we use a subject-specific electrode placement optimization method to achieve a high SNR for SSVEP and SMR. Although we demonstrated the applicability of the subject-specific positions in many online experiments, much work is still needed to explore the stationarity of the optimized electrode positions. Alternatively, more convenient electrode designs, for example, one that uses dry electrodes [44, 73], are highly preferable to replace the currently used wet electrode system.
2. *Better signal recording and processing is needed to allow for stable and reliable system performance.* Compared with the environment in an EEG

laboratory, electromagnetic interference and other artifacts (e.g., EMGs and EOGs) are much stronger in daily home life. Suitable measures then need to be applied to ensure the quality of the EEG recordings. Therefore, for data recording in an unshielded environment, the use of active electrodes may be better than the use of passive electrodes. Such usage can ensure that the recorded signal is less sensitive to interference. To remove the artifacts in EEG signals, additional recordings of EMGs and EOGs may be necessary and advanced techniques for online artifact canceling should be applied. Moreover, to reduce the dependence on technical assistance during system operation, ad hoc functions should be provided in the system to adapt to the individual diversity of the user and nonstationarity of the signal caused by changes of electrode impedance or brain state. These functions must be convenient for users to employ. For example, software should be able to detect bad electrode contacts in real time and adjust the algorithms to fit the remaining good channels automatically.

## Acknowledgments

This work was partly supported by the National Natural Science Foundation of China (30630022, S. Gao, 60675029, B. Hong) and the Tsinghua-Yu-Yuan Medical Sciences Fund (B. Hong).

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CHAPTER 9

# EEG Signal Analysis in Anesthesia

Ira J. Rampil

After nearly 80 years of development, EEG monitoring has finally assumed the status of a routine aid to patient care in the operating room. Although the EEG has been used in its raw form for decades in surgery that risks the blood supply of the brain in particular, it is only recently that processed EEG has developed to the point where it can reliably assess the anesthetic response in individual patients undergoing routine surgery and can predict whether they are forming memories or can respond to verbal commands. Reducing the incidence of unintentional recall of intraoperative events is an important goal of modern patient safety–oriented anesthesiologists. This chapter provides an overview of the long gestation of EEG and the algorithms that provide clinical utility.

## 9.1 Rationale for Monitoring EEG in the Operating Room

Generically, patient monitoring is performed to assess a patient's condition and, in particular, to detect physiological changes. The working hypothesis is that early detection allows for timely therapeutic intervention after changes and preservation of good health and outcome. It is, of course, difficult to demonstrate this effect in practice due to many confounding factors. In fact, data demonstrating a positive effect on actual patient outcomes does not exist for electrocardiography, blood pressure monitoring, or even for pulse oximetry. Despite this lack of convincing evidence, monitoring physiological variables is the international standard of care during general anesthesia. Among the available variables, the EEG has been used to target three specific and undesirable physiological states: hypoxia/ischemia, localization of seizure foci, and inadequate anesthetic effect. Many forms of EEG analysis have been proposed for use during anesthesia and surgery over the years, the vast majority as engineering exercises without meaningful clinical trials. Because clinical medicine has become ever more results oriented, the chapter points out, where data are available, which techniques have been tested in a clinical population and with what results.

Consciousness and the spontaneous electric activity of a human brain will begin to change within 10 seconds after the onset of ischemia (no blood flow) or hypoxia (inadequate oxygen transport despite blood flow) [1]. The changes in EEGs are usually described as slowing, but in more detail include frequency-dependent suppression of background activity. Beta (13- to 30-Hz) and alpha (7- to 13-Hz) range activity are depressed promptly, with the transition in activity complete within 30 seconds of a step change in oxygen delivery.

If the deprivation of oxygen is prolonged for more than several minutes, or involves a large volume of brain, theta (3- to 7-Hz) and delta (0.5- to 3-Hz) range activity will also be diminished. Until it is suppressed, the delta range activity will actually increase in amplitude and the raw tracing may appear as a nearly monochromatic low-frequency wave. If left without oxygen, neurons will begin to die or enter apoptotic pathways (delayed death) after about 5 minutes. General anesthesia and concomitant hypothermia in the operating room may extend the window of potential recovery to 10 minutes or more [2]. On the other hand, general anesthesia renders the functioning of the brain rather difficult to assess by conventional means (physical exam). During surgical procedures that risk brain ischemia, the EEG can thus provide a relatively inexpensive, real-time monitor of brain function.

In the context of carotid endarterectomy, EEG has been shown to be sensitive but only moderately specific to ischemic changes that presage new neurological deficits [3]. Somatosensory-evoked responses due to stimulation of the posterior tibial nerve are perhaps more specific to ischemic changes occurring in the parietal watershed area, but are not sensitive to ischemic activity occurring in other locations due to emboli. The practice of EEG monitoring for ischemia is not very common at this time, but persists in cases that risk the cerebrovascular circulation because it is technically simple to perform and retains moderate accuracy in most cases. An EEG technician or, rarely, a neurologist will be present in the operating room (OR) to perform the monitoring. It has been hypothesized that EEG may be useful to guide clinical decision making when ischemia is detected, particularly in the use of intraluminal shunts or modulation of the systemic blood pressure; however, adequately powered, randomized clinical trials are not available to prove utility.

Surgery to remove fixed lesions that generate seizures is an increasingly popular treatment for epilepsy [4, 5]. Although the specific location of the pathological seizure focus is usually well defined preoperatively, its location is confirmed intraoperatively in most centers using electrocorticography and a variety of depth electrodes or electrode array grids.

Drugs that induce general anesthesia, along with many sedatives and anxiolytics, create a state of unconsciousness and amnesia. In fact, from a patient's point of view, amnesia is the primary goal of general anesthesia. During the past two decades, most of the effort in developing EEG monitoring technology has concentrated on assessment of anesthetic drug effect on the brain [6]. In particular, development has focused on the detection of excessive and inadequate anesthetic states. Unintentional postoperative recall of intraoperative events has been established as an uncommon, but potentially debilitating phenomenon [7–9], especially when accompanied by specific recall of paralysis and or severe pain. Several risk factors have consistently appeared in surveys of patients who suffer from postoperative recall. These include several types of high-risk surgery, use of total intravenous (IV)

anesthesia, nondepolarizing muscle relaxants, and female gender. These risk factors, however, account for only about half of the cases of recall. Many other cases occur in the setting of inadvertent failure of intended anesthetic agent delivery, extreme pharmacological tolerance, and even, occasionally, simple errors in anesthetic management.

In unmedicated subjects pain or fear can elicit a substantial increase in blood pressure and heart rate due to activation of the sympathetic nervous system. Several lines of evidence suggest that routine monitoring of vital signs (e.g., blood pressure and heart rate) is insensitive to the patient's level of consciousness in current anesthetic practice [10–12]. In the face of widespread use of opiates, beta-blockers, and central alpha agonists, and the general anesthetic agents themselves, the likelihood of a detectable sympathetic response to painful stimulus or even consciousness is diminished. Domino et al.'s [13] review of the ASA Closed Claims database failed to find a correlation between recorded vital signs and documented recall events. Other attempts to score vital sign plus diaphoresis and tearing have also failed to establish a link between routine vital signs and recall [10–12].

Because existing hemodynamic monitors have definitively failed to detect ongoing recall in the current environment of mixed pharmacology (if they ever did), a new, sensitive monitor could be useful, especially for episodes of recall not predicted by preexisting risk factors. Real-time detection of inadequate anesthetic effect and a prompt therapeutic response with additional anesthetics appear likely to reduce the incidence of overt recall.

With the goal of monitoring anesthetic effect justified, it is now appropriate to review the effect of anesthetic drugs on the human EEG. It is important to first note that anesthesiologists in the OR and intensivists in the critical care unit use a wide range of drugs, some of which alter mentation, but only some of which are true general anesthetics. Invoking the spirit of William Thompson, Lord Kelvin, who said one could not understand a phenomenon unless one could quantify it, the state of general anesthesia is poorly understood, in part because there have been no quantitative measures of its important effects until very recently.

This author has defined general anesthesia as a therapeutic state induced to allow safe, meticulous surgery to be tolerated by patients [14]. The "safe and meticulous" part of the definition refers to the lack of responsiveness to noxious stimulation defined most commonly as surgical somatic immobility, a fancy way of saying that the patient does not move perceptibly or purposefully in response to incision. In practice, this unresponsiveness also mandates stability of the autonomic nervous system and the hormonal stress response. These are the features central to surgeons' and anesthesiologists' view of a quality anesthetic. Patients, on the other hand, request and generally require amnesia for intraoperative events including disrobing, marking, positioning, skin prep, and, of course, the pain and trauma of the surgery itself. Also best to avoid is recall of potentially disturbing intraoperative conversation.

General anesthetics are those agents which by themselves can provide both unresponsiveness and amnesia. Anesthetic drugs include inhaled agents such as diethyl ether, halothane, isoflurane, sevoflurane, and desflurane. Barbiturates such as thiopental and pentobarbital as well as certain $GABA_A$ agonists, including propofol and etomidate, are also general anesthetics. Other $GABA_A$ agonists such as

benzodiazepines are good amnestic agents but impotent in blocking movement or sympathetic responses. Opioids and other analgesics, on the other hand, can at high doses diminish responsiveness, but do not necessarily create amnesia or even sedation. These observations are important in the understanding of EEG during anesthesia since all of these aforementioned drugs have an impact on the EEG and their signatures partly overlap those of true anesthetics.

The effects of sedatives and anesthetics on the EEG were described as early as 1937 by Gibbs et al. [15], less than 10 years after the initial description of human scalp EEGs by Berger [16]. By 1960 certain patterns were described by Faulconer and Bickford that remain the basis of our understanding of anesthetic effect [17]. One such pattern is illustrated in Figure 9.1. The EEG of awake subjects usually contains mixed alpha and beta range activity but is quite variable by most quantitative measures. With the slow administration of an anesthetic, there is an increase in high-frequency activity, which corresponds clinically to the "excitement" phase. The population variance decreases as anesthetic administration continues and the higher frequencies (15 to 30 Hz) diminish, then the mid and lower frequencies (3 to 15 Hz) in a dose-dependent fashion. With sufficient agent, the remaining EEG activity will become intermittent and finally isoelectric.

The pattern associated with opioids differs in the absence of an excitement phase and the presence of a terminal plateau of slow activity and no burst suppression or isoelectricity. There is no current electrophysiological theory to adequately relate what little is known about the molecular actions of anesthetics with what is seen in the scalp waveforms. Therefore, all intraoperative EEG analysis is empirically based. Empirically then, after the excitement phase, the multitude of generators of EEGs appear to synchronize in phase and frequency and the dominant frequencies slow. We will see later that the major systems quantifying anesthetic drug effect all target these phenomena of synchronization and slowing.

In the next section, technical issues involved in EEG monitoring and recent updates in commercially available monitors are discussed, followed by a brief review of interesting recent clinical literature.

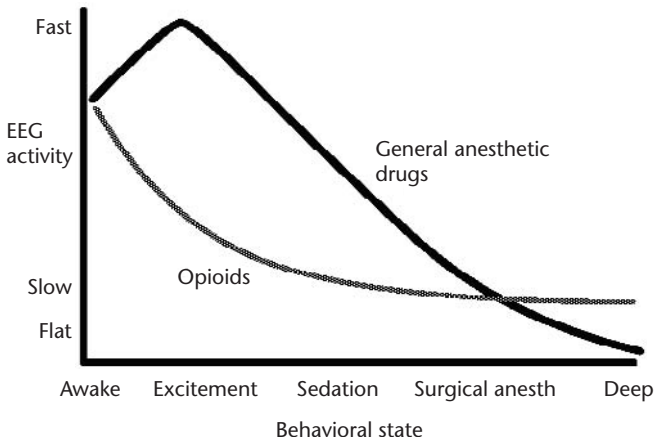



**Figure 9.1** EEG for anesthetic dose response. The large variance across a population in awake EEG activity tends to diminish with increasing anesthetic effect.

## 9.2 Nature of the OR Environment

The term "electrically hostile" scarcely does just justice to the operating room environment. The signal of interest, the EEG, is one or two orders of magnitude lower amplitude than the electrocardiogram (ECG) and shares its frequency spectral range with another biological signal cooriginating under the scalp, the electromyogram (EMG). While the patient is awake or only lightly sedated, facial grimacing is associated with an EMG signal amplitude many times the EEG signal. In fact, the desynchronization of EEGs seen in anxious patients leads to an EEG with a broader frequency spectrum and diminished amplitude, even worsening the EEG SNR. Fortunately, EMG activation usually mirrors activation in the EEG and accentuates the performance of EEG-derived variables predicting inadequate anesthesia. Other cranial sources of biological artifact include the electro-oculogram (from movement of the retina-corneal dipole) and swallowing and the ECG as its projected vector sweeps across the scalp.

The electrochemistry of silver/silver chloride electrodes creates a relatively stable electrode potential of several hundred millivolts (depending on temperature and molar concentration of electrolyte) at each skin contact [18, 19]. Any change in contact pressure or movement of the skin/electrode interface will provoke changes in the electrode potential that will be orders of magnitude larger than the EEG signal because it is unlikely that a change at one site will be exactly balanced and thus canceled out by the electrode potential at the other end of an electrode pair. Silver/silver chloride electrode potentials are also sensitive to ambient light.

The next source of artifact to contend with in the OR is pickup of the existing electromagnetic field in the environment. The two dominant sources are the power-line frequency field that permeates all buildings that are wired for electricity and the transmitted output of electrosurgical generators. Power-line frequencies are fairly easily dealt with using effective common-mode rejection in the input stage amplifiers and narrow bandpass filtering. Electrosurgical generators (ESUs) are a far greater problem for EEG recordings. ESU devices generate large spark discharges with which the surgeon cuts and cauterizes tissue. Several different types of ESUs are in use and the output characteristics vary, but in general, the output voltage will be in the range of hundreds of volts, the frequency spectrum very broad and centered on about 0.5 MHz, with additional amplitude modulation in the subaudio range.

Some ESU devices feature a single probe whose current flow proceeds through volume conduction of the body to a remote "ground" pad electrode. This "unipolar" ESU is associated with the worst artifact at the scalp that will exceed the linear dynamic range of the input stages producing rail-to-rail swings on the waveform display. The EEG data is usually lost during the surgeon's activation of a unipolar ESU. Other ESUs are bipolar in that the surgeon uses a tweezer-like pair of electrodes that still radiates enormous interference, but most of the current is contained in the tissue between the tweezer's tips. Some monitoring companies have gone to great lengths to reduce the time during which ESU artifacts render their product offline.

## 9.3 Data Acquisition and Preprocessing for the OR

Voltage signals, like the EEG, are always measured as a difference in potential between two points, thus a bioelectric amplifier has two signal inputs, a plus and a minus. Bioelectric amplifiers also have a third input for a reference electrode, which is discussed later. Because the electrical activity of the cortex is topographically heterogeneous, it is generally advantageous to measure this activity at several locations on the scalp. In diagnostic neurology, several systems of nomenclature for electrode placement have evolved. The most commonly used at present is the International 10-20 system [20].

Practicality in the operating room environment requires an absolute minimum of time be spent securing scalp electrodes. Production pressure in the form of social and economic incentives to minimize time between surgeries will react negatively to time-consuming electrode montages and doom products that require them to failure unless absolutely required for clinical care. Nonresearch, intraoperative EEG monitoring for drug effect is now performed exclusively with preformed strips of electrodes containing one or two channels. Many of these strips are designed to self-prep the skin, eliminating the time-consuming separate step of local skin cleaning and abrasion.

Note that this environment is quite distinct from that of the neurology clinic where diagnostic EEGs are seldom recorded with fewer than 16 channels (plus-minus pairs of electrodes) in order to localize abnormal activity. Monitoring 8 or 16 channels intraoperatively during carotid surgery is often recommended, although there is a paucity of data demonstrating increased sensitivity for the detection of cerebral ischemia when compared with the 2- or 4-channel computerized systems more commonly available to anesthesia personnel. Although regional changes in EEG occur during anesthesia [21, 22], there is little evidence that these topographic features are useful markers of clinically important changes in anesthetic or sedation levels [23], and most monitors of anesthetic drug effect use only a single frontal channel.

### 9.3.1 Amplifiers

As noted previously, the EEG signal is but one of several voltage waveforms present on the scalp. Although all of these signals may contain interesting information in their own right, if present, they distort the EEG signal. An understanding of the essential characteristics of specific artifacts can be used to mitigate them [24]. A well-designed bioelectric amplifier can remove or attenuate some of these signals as the first step in signal processing. For example, consider power-line radiation. This artifact possesses two characteristics useful in its mitigation: At its very low frequency, it is in the same amplitude phase over the entire body surface and it is a single characteristic frequency (50 or 60 Hz). Because EEG voltage is measured as the potential difference between two electrodes placed on the scalp, both electrodes will have the same power-line artifact (i.e., it is a common-mode signal). Common-mode signals can be nearly eliminated in the electronics stage of an EEG machine by using a differential amplifier that has connections for three electrodes: plus (+), minus (–),

and a reference. This type of amplifier detects two signals—the voltage between plus and the reference, and between minus and the reference—and then subtracts the second signal from the first.

The contribution of the reference electrode is common to both signals and thus cancels. Attenuation of common-mode artifact signals will be complete only if each of the plus and minus electrodes is attached to the skin with identical contact impedance. If the electrodes do not have equal contact impedances, the amplitude of the common-mode signal will differ between the plus and minus electrodes, and they will not cancel exactly. Most commonly, the EEG is measured (indirectly) between two points on the scalp with a reference electrode on the ear or forehead. If the reference electrode is applied far from the scalp (i.e., on the thorax or leg), there is always a chance that large common-mode signals such as the ECG will not be ideally canceled, leaving some degree of a contaminating artifact.

Some artifacts, like the EMG, characteristically have most of their energy in a frequency range different from that of the EEG. Hence, the amplifier can bandpass filter the signal, passing the EEG and attenuating the nonoverlapping EMG. However, it is not possible to completely eliminate EMG contamination when it is active. At present, most commercial EEG monitors quantify and report EMG activity on the screen.

### 9.3.2 Signal Processing

Signal processing of an EEG is done to enhance and aid the recognition of information in the EEG that correlates with the physiology and pharmacology of interest. Metaphorically, the goal is to separate this "needle" from an electrical haystack. The problem in EEG-based assessment of the anesthetic state is that the characteristics of this needle are unknown, and since our fundamental knowledge of the central nervous system (CNS) remains relatively limited, our models of these "needles" will, for the foreseeable future, be based on empirical observation.

Assuming a useful target is identified in the raw EEG waveform, it must be measured and reduced to a qEEG parameter. The motivation for quantitation is threefold: to reduce the clinician's workload in analyzing intraoperative EEGs, to reduce the level of specialized training required to take advantage of EEG, and finally to develop a parameter that might, in the future, be used in an automated closed-loop titration of anesthetic or sedative drugs. The following section introduces some of the mechanics and mathematics behind signal processing.

Although it is possible to perform various types of signal enhancement on analog signals, the speed, flexibility, and economy of digital circuits has produced revolutionary changes in the field of signal processing. To use digital circuits, it is, however, necessary to translate an analog signal into its digital counterpart.

Analog signals are continuous and smooth. They can be measured or displayed with any degree of precision at any moment in time. The EEG is an analog signal: The scalp voltage varies smoothly over time. Digital signals are fundamentally different in that they represent discrete points in time and their values are quantified to a fixed resolution rather than continuous. The binary world of computers and digi-

tal signal processors operates on binary numbers that are sets of bits. A bit is quantal; it contains the smallest possible chunk of information: a single ON or OFF signal. More useful binary numbers are created by aggregating between 8 and 80 bits. The accuracy or resolution ($q$) of binary numbers is determined by the number of bits they contain: An 8-bit binary number can represent $2^8$ or 1 of 256 possible states at any given time; a 16-bit number, $2^{16}$ or 65,536 possible states. If one were using an 8-bit number to represent an analog signal, the binary number would have, at best, a resolution of approximately 0.4% (1/256) over its range of measurement. Assuming, for example, that a converter was designed to measure voltages ranging from –1.0 to +1.0V, the step size of an 8-bit converter would be about 7.8 mV and a 16-bit converter about 30 $\mu$V. Commercial EEG monitoring systems use 12 to 16 bits of resolution. More bits also create a wider dynamic range with the possibility of recovery from more artifact; however, more bits increase the expense dramatically.

Digital signals are also quantized in time, unlike analog signals, which vary smoothly from moment to moment. When translation from analog to digital occurs, it occurs at specific points in time, whereas the value of the resultant digital signal at all other instants in time is indeterminate. Translation from the analog to digital world is known as sampling, or digitizing, and in most applications is set to occur at regular intervals. The reciprocal of the sampling interval is known as the sampling rate ($fs$) and is expressed in hertz (Hz or samples per second). A signal that has been digitized is commonly written as a function of a sample number, $i$, instead of analog time, $t$. An analog voltage signal written as $v(t)$, would be referred to, after conversion, as $v(i)$. Taken together, the set of sequential digitized samples representing a finite block of time is referred to as an epoch.

When sampling is performed too infrequently, the fastest sine waves in the epoch will not be identified correctly. When this situation occurs, aliasing distorts the resulting digital data. Aliasing results from failing to meet the requirement of having a minimum of two points within a single sinusoid. If sampling is not fast enough to place at least two sample points within a single cycle, the sampled wave will appear to be slower (longer cycle time) than the original.

Aliasing is familiar to observers of the visual sampled-data system known as cinema. In a movie, where frames of a scene are captured at rate of approximate 24 Hz, rapidly moving objects such as wagon wheel spokes often appear to rotate slowly or even backwards. Therefore, it is essential to always sample at a rate more than twice the highest expected frequency in the incoming signal (Shannon's sampling theorem [25]). Conservative design actually calls for sampling at a rate 4 to 10 times higher than the highest expected signal, and to also use a lowpass filter prior to sampling to eliminate signals that have frequency components that are higher than expected. Lowpass filtering reduces high-frequency content in a signal, just like turning down the treble control on a stereo system. In monitoring work, EEG signals have long been considered to have a maximal frequency of 30 or 40 Hz, although 70 Hz is a more realistic limit. In addition, other signals present on the scalp include power-line interference at 60 Hz and the EMG, which, if present, may extend above 100 Hz. To prevent aliasing distortion in the EEG from these other signals, many digital EEG systems sample at a rate above 250 Hz (i.e., a digital sample every 4 ms).

## 9.4 Time-Domain EEG Algorithms

Analysis of the EEG can be accomplished by examining how its voltage changes over time. This approach, known as time-domain analysis, can use either a strict statistical calculation (i.e., the mean and variance of the sampled waveform, or the median power frequency) or some ad hoc measurement based on the morphology of the waveform. Most of the commonly used time-domain methods are grounded in probabilistic analysis of "random" signals and, therefore, some background on statistical approaches to signals is useful.

Of necessity, the definitions of probability functions, expected values, and correlation are given mathematically as well as descriptively. However, the reader need not feel compelled to attain a deep understanding of the equations presented here to continue on. A more detailed review of the statistical approach to signal processing can be obtained from Chapter 3 or one of the standard texts [26–28]. At present the only two time-domain statistical qEEGs in clinical use in anesthesia are entropy and the burst suppression ratio. The family of entropy qEEG parameters derived from communications theory is used to estimate the degree of chaos, or lack of predictability, in a signal. Entropy is discussed further later in this chapter.

A few definitions related to the statistical approach to time-related data are called for. The EEG is not a deterministic signal, which means that it is not possible to exactly predict future values of the EEG. Although the exact future values of a signal cannot be predicted, some statistical characteristics of certain types of signals are predictable in a general sense. These roughly predictable signals are termed *stochastic*. The EEG is such a nondeterministic, stochastic signal because its future values can only be predicted in terms of a probability distribution of amplitudes already observed in the signal. This probability distribution, $p(x)$, can be determined experimentally for a particular signal, $x(t)$, by simply forming a histogram of all observed values over a period of time. A signal such as that obtained by rolling dice has a probability distribution that is rectangular or uniform [i.e., the likelihood of all face values of a throw are equal and in the case of a single die, $p(x) = 1/6$ for each possible value]; a signal with a bell-shaped, or normal probability distribution is termed Gaussian. As illustrated in Figure 9.2, EEG amplitude histograms may have a nearly Gaussian distribution. The concept of using statistics, such as the mean, standard deviation, skewness, and so forth, to describe a probability distribution will be familiar to many readers.

If the probability function $p(x)$ of a stochastic signal $x(i)$ does not change over time, that process is deemed stationary. The EEG is not strictly stationary because its statistical parameters may change significantly within seconds, or it may be stable for tens of minutes (quasistationary) [29, 30]. If the EEG is at least quasistationary, then it may be reasonable to check it for the presence of rhythmicity, where rhythmicity is defined as repetition of patterns in the signal. Recurring patterns can be identified mathematically using the concept of correlation. Correlation between two signals measures the likelihood of change in one signal leading to a consistent change in the other. In assessing the presence of rhythms, autocorrelation is used, testing the match of the original signal against different starting time points of the same signal. If rhythm is present, then at a particular offset time (equal to the interval of the rhythm), the correlation statistic increases, sug-

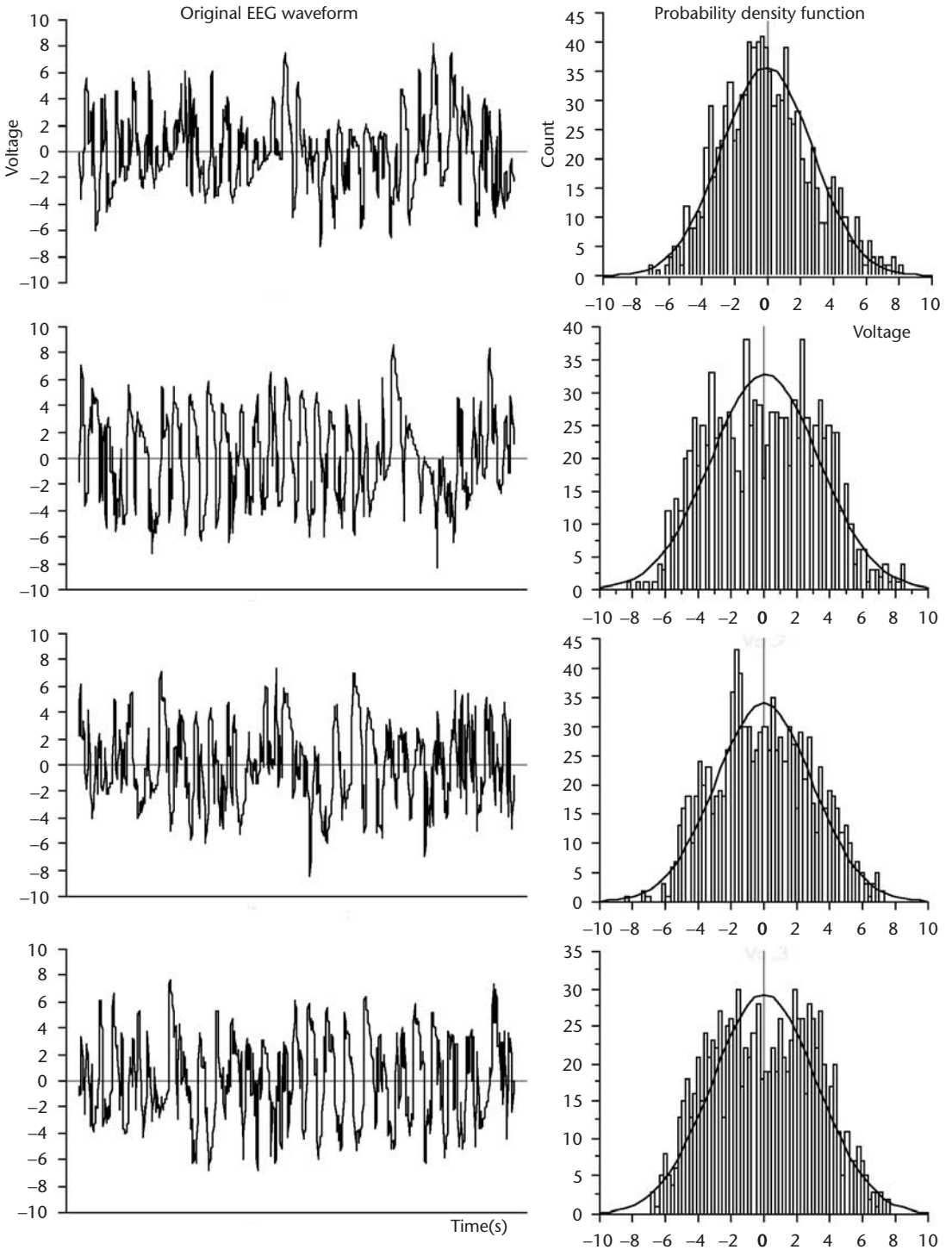



**Figure 9.2** EEG amplitude values sampled over time exhibit a normal distribution. Data recorded from anesthetized rats by the author at 256 Hz with a gain of 500 and analyzed as sequential 4-second-long epochs.

gesting a repetition of the original signal voltage. The autocorrelation of signal $x$ (i.e., correlation of $x$ versus $x$) is denoted as $\gamma_{XX}(\tau)$ where $\tau$ is the offset time interval or lag.

Empirically, it is known that the EEG has a mean voltage of zero, over time: It is positive as often as it is negative. However, the EEG and its derived statistical measurements seldom have a true Gaussian probability distribution. This observation complicates the task of a researcher or of some future automated EEG alarm system that seeks to identify changes in the EEG over time. Strictly speaking, non-Gaussian signals should not be compared using the common statistical tests, such as *t*-tests or analysis of variance that are appropriate for normally distributed data. Instead, there are three options: nonparametric statistical tests, a transform to convert non-Gaussian EEG data to a normal (Gaussian) distribution, or higher order spectral statistics (see later discussion).

Transforming non-Gaussian data by taking its logarithm is frequently all that is required to allow analysis of the EEG as a normal distribution [31]. For example, a brain ischemia detection system may try to identify when slow wave activity has significantly increased. A variable such as "delta" power (described later), which measures slow wave activity, has a highly non-Gaussian distribution; thus, directly comparing this activity at different times requires the nonparametric Kruskal-Wallis or Friedman's test. However, a logarithmic transform of delta power may produce a nearly normal $p(x)$ curve; therefore, the more powerful parametric analysis of variance with repeated measures could be used appropriately to detect changes in log(delta power) over time. Log transformation is not a panacea, however, and whenever statistical comparisons of qEEG are to be made, the data should be examined to verify the assumption of normal distribution.

### 9.4.1 Clinical Applications of Time-Domain Methods

Historically (predigital computer), intraoperative EEG analysis used analog, time domain–based methods. In 1950 Faulconer and Bickford noted that the electrical power in the EEG (power = voltage × current = voltage$^2$/resistance) was associated with changes in the rate of thiopental or diethyl ether administration. Using analog technology, they computed a power parameter as (essentially) a moving average of the square of EEG voltage and used it to control the flow of diethyl ether to a vaporizer. This system was reported to successfully control depth of anesthesia in 50 patients undergoing laparotomy [17]. Digital total power (TP = sum of the squared values of all the EEG samples in an epoch) was later used by several investigators, but it is known have several problems, including its sensitivity to electrode location and its insensitivity to important changes in frequency distribution as well as a highly non-Gaussian distribution. Arom et al. reported that a decrease in TP may predict neurological injury following cardiac surgery [32], but this finding has not been replicated.

More comprehensive time domain–based approaches to analysis of the EEG were reported by Burch [33], and by Klein [34] who estimated an "average" frequency by detecting the number of times the EEG voltage crosses the zero voltage level per second. Investigators have not reported strong clinical correlations with zero crossing frequency (ZXF). The ZXF does not correlated with depth of anesthe-

sia in volunteers [35]. While simple to calculate in the era before inexpensive computer chips, the ZXF parameter is not simply related to frequency-domain estimates of frequency content as demonstrated in Figure 9.3, because not all frequency component waves in the signal will cross the zero point.

Demetrescu refined the zero crossing concept to produce what he termed *aperiodic analysis* [36]. This method simply splits the EEG into two frequency bands (0.5 to 7.9 and 8 to 29.9 Hz) and the filtered waveforms from the high-and low-frequency bands are each separately sent to a relative minima detector. Here, a wavelet is defined as a voltage fluctuation between adjacent minima, and its frequency is defined as the reciprocal of the time between the waves. Wavelet amplitude is defined as the difference between the intervening maxima and the average of the two minima voltages. Aperiodic analysis produces a spectrum-like display which plots a sampling of detected wavelets as an array of "telephone poles" whose height represents measured wave amplitude, distance from the left edge frequency (in a logarithmic scale), and distance from the lower edge time since occurrence. The Lifescan Monitor (Diatek, San Diego, California) was an implementation of aperiodic analysis; it is not commercially available at present but the algorithms are described in detail in paper by Gregory and Pettus [37]. Reports in the literature have used this

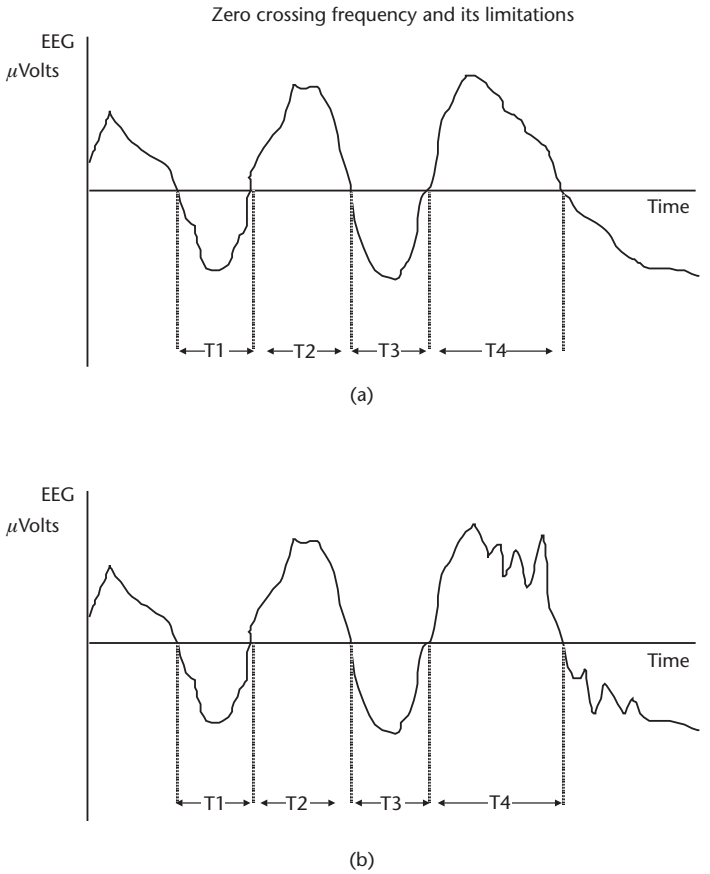



**Figure 9.3** (a, b) Failure of zero crossing algorithm to be sensitive to all components of EEG waveform. In interval T4 and beyond, the high-frequency, low-amplitude activity in waveform B is ignored.

technology as a marker of pharmacological effects in studies of certain drugs, but there were no reports of the Lifescan having an impact on patient outcome.

### 9.4.2 Entropy

Most commonly, entropy is considered in the context of physics and thermodynamics where it connotes the energy lost in a system due to disordering. In 1948 Claude Shannon of Bell Labs developed a theory of information concerned with the efficiency of information transfer [38]. He coined a term known as *information entropy*, now known as *Shannon entropy*, which in our limited context can simply be considered the amount of information (i.e., bits) per transmitted symbol.

Many different specific algorithms have been applied to calculate various permutations of the entropy concept in biological data (Table 9.1). Recently, a commercial EEG monitor based on the concept of entropy has become available. The specific entropy algorithm used in the GE Healthcare EEG monitoring system is described as "time-frequency balanced spectral entropy," which is nicely described in an article by Viertiö-Oja et al. [39]. This particular entropy uses both time- and frequency-domain components. In brief, this algorithm starts EEG data sampling at 400 Hz followed by FFT-derived power spectra derived from several different length sampling epochs ranging from about 2 to 60 seconds. The spectral entropy, $S$, for any desired frequency band ($f_1 - f_2$) is the sum:

$$S[f_1, f_2] = \sum_{f_i = f_1}^{f_2} P_n(f_i) \log\left(1 / P_n(f_i)\right)$$

where $P_n(f_i)$ is a normalized power value at frequency $f_i$. The spectral entropy of the band is then itself normalized, $S_N$, to a range of 0 to 1.0 via:

$$S_N[f_1, f_2] = \frac{S[f_1, f_2]}{\log\left(n[f_1, f_2]\right)}$$

where $n[f_1, f_2]$ is the number of spectral data points in the range $f_1 - f_2$. This system actually calculates two separate, but related entropy values, the state entropy (SE) and the response entropy (RE). The SE value is derived from the 0.8- to 34-Hz frequency range and uses epoch lengths from 15 to 60 seconds to attempt to emphasize the relatively stationary cortical EEG components of the scalp signal. The RE, on the other hand, attempts to emphasize shorter term, higher frequency components of the scalp signal, generally the EMG and faster cortical components, which rise and fall faster than the more stationary cortical signals. To accomplish this, the RE

**Table 9.1** Entropy Algorithms Applied to EEG Data

| | |
|---|---|
| Approximate entropy [76–78] | Kolmogorov entropy [79] |
| Spectral entropy [80, 81] | Lempel-Ziv entropy [80, 82, 83] |
| Shannon entropy [82, 84] | Maximum entropy [85] |
| Tsallis entropy [86] | Sample entropy [87] |
| Wavelet entropy [88, 89] | Time-frequency balanced spectral entropy [39, 40] |

algorithm uses the frequency range of 0.8 to 47 Hz and epoch lengths from 2 to 15 seconds. The RE was designed to detect those changes in the scalp signal that might reflect transient responses to noxious stimulation, whereas the SE reflects the more steady-state degree of anesthetic-induced depression of cortical activity. To simplify the human interface, there is additional scaling in the algorithm to ensure that the RE value is nearly identical to the SE, except when there are rapid transients or EMG activity, in which case the RE values will be higher.

By 2007, more than 30 peer-reviewed papers had been published on the GE M-Entropy monitor. Of these, 16 were clinical trials, mostly comparing it against the Aspect Medical Systems BIS monitor (the present gold standard).Several studies appear to confirm the relative sensitivity of RE to nociception. Vakkuri et al. compared the accuracy of M-Entropy and BIS in predicting whether patients were conscious during the use of three different anesthetic agents(sevoflurane, propofol, or thiopental) [40]. They found that the entropy variables were approximately equal in predictive performance to BIS, and that both monitors performed slightly better during sevoflurane and propofol usage than during thiopental usage. The area under the curve of the receiver operating characteristics curve exceeded 0.99 in all cases. In another study of 368 patients, use of SE to titrate propofol administration allowed for fast patient recovery and the use of less drug compared to a control (no EEG) group [41].

Although epileptiform spikes, seizures, and certain artifacts are detected by waveform pattern matching, very little anesthetic-related EEG activity can be assessed by detection of specific patterns in the voltage waveforms. In fact, only one class of ad hoc pattern matching time-domain method, burst suppression quantitation, is in current use in perioperative monitoring systems.

As noted earlier, during deep anesthesia the EEG may develop a peculiar pattern of activity that is evident in the time-domain signal. This pattern, known as burst suppression, is characterized by alternating periods of normal to high voltage activity changing to low voltage or even isoelectricity rendering the EEG "flat-line" in appearance. Of course, the actual measured voltage is never actually zero for any length of time due to the presence of various other signals on the scalp as noted earlier. Following head trauma or brain ischemia, this pattern carries a grave prognosis; however, it may also be induced by large doses of general anesthetics, in which case burst suppression has been associated with reduced cerebral metabolic demand and possible brain "protection" from ischemia. Titration to a specific degree of burst suppression has been recommended as a surrogate end point against which to titrate barbiturate coma therapy.

The burst suppression ratio (BSR) is a time-domain EEG parameter developed to quantify this phenomena [42, 43]. To calculate this parameter, suppression is recognized as those periods longer than 0.50 second during which the EEG voltage does not exceed approximately $\pm 5.0\,\mu$V. The total time in a suppressed state is measured, and the BSR is calculated as the fraction of the epoch length where the EEG meets the suppression criteria (Figure 9.4). The random character of the EEG dictates that the qEEG parameters extracted will exhibit a moment-to-moment variation without discernible change in the patient's state. Thus, output parameters are often smoothed by a moving average prior to display. Due to the particularly vari-

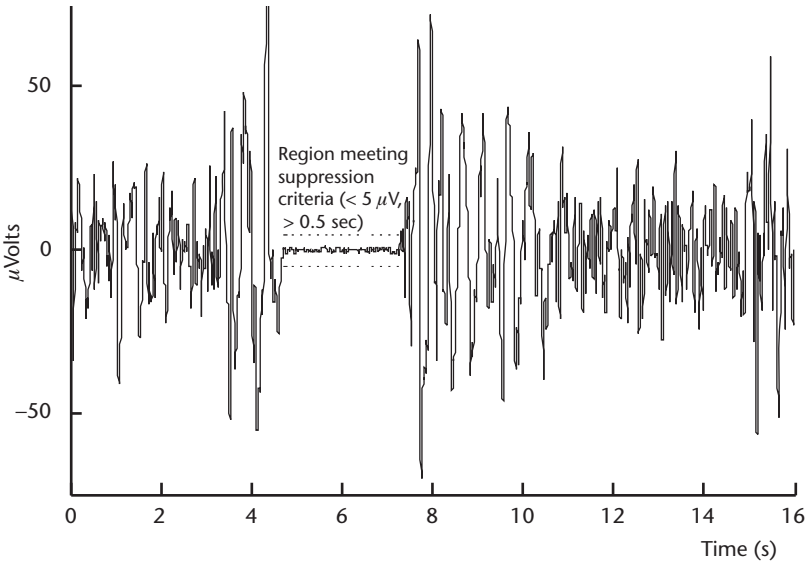



**Figure 9.4** The BSR algorithm [42, 43].

able (nonstationary) nature of burst suppression, the BSR should be averaged over at least 60 seconds.

At present there are about 40 publications referring to the use of burst suppression in EEG monitoring during anesthesia or critical care.

## 9.5 Frequency-Domain EEG Algorithms

Like all complex time-varying voltage waveforms, EEGs can be viewed as many simple, harmonically related sine waves superimposed on each other. An important alternative approach to time-domain analysis examines signal activity as a function of frequency. So-called frequency-domain analysis has evolved from the study of simple sine and cosine waves by Jean Baptiste Joseph Fourier in 1822. Fourier analysis is covered in detail in Chapter 3. Here we concentrate on its applications.

### 9.5.1 Fast Fourier Transform

The original integral-based approach to computing a Fourier transform is computationally laborious, even for a computer. In 1965, Cooley and Tukey published an algorithm for efficient computation of Fourier series from digitized data [44]. This algorithm is known as the fast Fourier transform. More information about the implementation of FFT algorithms can be found in the text by Brigham [45] or in any current text on digital signal processing. Whereas the original calculation of the discrete Fourier transform of a sequence of $N$ data points requires $N^2$ complex multiplications (a relatively time-consuming operation for a microprocessor), the FFT requires only $N(\log 2N)/2$ complex multiplications. When the number of points is large, the difference in computation time is significant, for example, if $N = 1{,}024$, the FFT is faster by a factor of about 200.

In clinical monitoring applications, the results of a EEG Fourier transform are graphically displayed as a power versus frequency histogram and the phase spectrum has been traditionally discarded as uninteresting. Whereas the frequency spectrum is relatively independent of the start point of an epoch (relative to the waveforms contained), the Fourier phase spectrum is highly dependent on the start point of sampling and thus very variable. Spectral array data from sequential epochs are plotted together in a stack (like pancakes) so that changes in frequency distribution over time are readily apparent. Raw EEG waveforms, because they are stochastic, cannot be usefully stacked together since the results would be a random superposition of waves. However, the EEG's quasistationarity in the frequency domain creates spectral data that is relatively consistent from epoch to epoch, allowing enormous visual compression of spectral data by stacking and thus simplified recognition of time-related changes in the EEG. Consider that raw EEG is usually plotted at a rate of 30 mm/s or 300 pages per hour on a traditional strip recorder used by a neurologist, whereas the same hour of EEG plotted as a spectral array could be examined on a single screen for relevant trends by an anesthesiologist occupied by several different streams of physiological data.

Two types of spectral array displays are available in commercial instruments: the compressed spectral array (CSA) and the density spectral array (DSA). The CSA presents the array of power versus frequency versus time data as a pseudo three-dimensional topographic perspective plot (Figure 9.5) [46], and the DSA presents the same data as a grayscale-shaded or colored two-dimensional contour plot [47]. Although both convey the same information, the DSA is more compact, whereas the CSA permits better resolution of the power or amplitude data.

Early in his survey of human EEG, Hans Berger identified several generic EEG patterns that were loosely correlated with psychophysiological state [16]. These types of activity, such as the alpha rhythms seen during awake periods with eyes closed, occurred within a stereotypical range of frequencies that came to be known as the alpha band. Eventually, five such distinct bands came to be familiar and widely accepted: delta, theta, alpha, beta, and gamma.

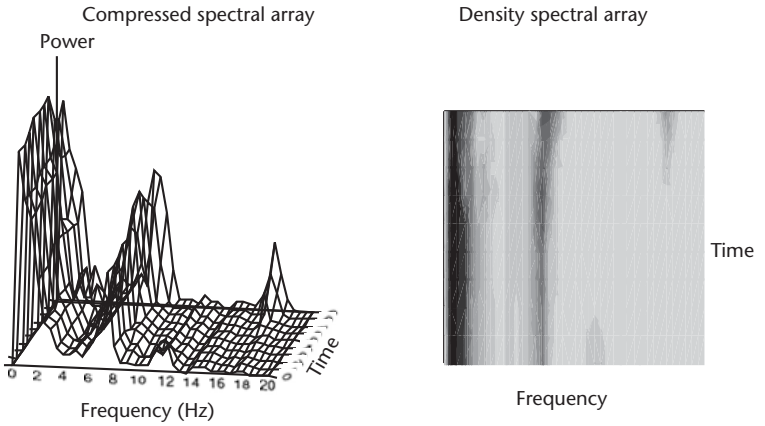



**Figure 9.5** Comparison of EEG spectral display formats. The creation of a spectral array display involves the transformation of time-domain raw EEG signals into the frequency domain via the FFT. The resulting spectral histograms are smoothed and plotted in perspective with hidden line suppression for CSA displays (left) or by converting each histogram value into a gray value for the creation of a DSA display (right).

Using an FFT, it is a simple matter to divide the resulting power spectrum from an epoch of EEG into these band segments, then summate all power values for the individual frequencies within each band to determine the "band power." Relative band power is simply band power divided by power over the entire frequency spectrum in the epoch of interest.

In the realm of anesthesia-related applications, traditional band power analysis is of limited utility, because these bands were defined for the activity of the awake or natural sleep-related EEG without regard for the altered nature of brain activity during anesthesia. Drug-related EEG oscillations can often be observed to alter their central frequency and to pass smoothly through the "classic" band boundaries as the drug dose changes. Familiarity with band analysis is still useful, however, because of the extensive neurological literature utilizing it.

In an effort to improve the stability of plotted band-related changes, Volgyesi introduced the augmented delta quotient (ADQ) [48]. This value is approximately the ratio of power in the 0.5- to 3.0-Hz band to the power in the 0.5- to 30.0-Hz range. This definition is an approximation because the author used analog bandpass filters with unspecified, but gentle roll-off characteristics that allowed them to pass frequencies outside the specified band limits with relatively little attenuation. The ADQ was used in a single case series that was looking for cerebral ischemia in children [49], but was never tested against other EEG parameters or formally validated.

Jonkman et al. [50] applied a normalizing transformation [31] to render the probability distribution of power estimates of the delta frequency range close to a normal distribution in the CIMON EEG analysis system (Cadwell Laboratories, Kennewick, Washington). After recording a baseline "self-norm" period of EEGs, increases in delta-band power that are larger than three standard deviations from the self norm were considered to represent an ischemic EEG change [51]. Other investigators have concluded this indicator may be nonspecific [52] because it yielded many false-positive results in control (nonischemic) patients.

Another approach to simplifying the results of a power spectral analysis is to find a parameter that describes a particular characteristic of the spectrum distribution. The first of these descriptors was the peak power frequency (PPF), which is simply the frequency in a spectrum at which the highest power in that epoch occurs. The PPF has never been the subject of a clinical report. The median power frequency (MPF) is that frequency which bisects the spectrum, with half the power above and the other half below. There are approximately 150 publications regarding the use of MPF in EEG monitoring. Although the MPF has been used as a feedback variable for closed-loop control of anesthesia, there is little evidence that specific levels of MPF correspond to specific behavioral states, that is, recall or the ability to follow commands.

The spectral edge frequency (SEF) [53] is the highest frequency in the EEG, that is, the high-frequency edge of the spectral distribution. The original SEF algorithm utilized a form of pattern detection on the power spectrum in order to emulate mechanically visual recognition of the "edge." Beginning at 32 Hz, the power spectrum is scanned downward to detect the highest frequency where four sequential spectral frequencies are above a predetermined threshold of power. This approach provides more noise immunity than the alternative computation, SEF95 [I. J.

Rampil and F. J. Sasse, unpublished results, 1977]. SEF95 is the frequency below which 95% of the power in the spectrum resides. Clearly, either approach to SEF calculation provides a monitor that is only sensitive to changes in the width of the spectral distribution (there is always some energy in the low-frequency range).

Approximately 260 peer-reviewed papers describe the use of SEF. The field of pharmacodynamics (analysis of the time-varying effects of drugs on physiology) of anesthetics and opioids benefited enormously from access to the relatively sensitive, specific and real-time SEF. Many of the algorithms driving open-loop anesthetic infusion systems use population kinetic data derived using SEF.

Similar to MPF, few of the existing published trials examine the utility of SEF in reducing drug dosing while ensuring clinically adequate anesthesia. In our hands, neither the SEF nor the F95 seems to predict probability of movement response to painful stimulus or verbal command in volunteers [35] at least in part due to the biphasic characteristic of its dose response curve. While the SEF is quite sensitive to anesthetic effect, there is also substantial variation across patients and across drugs. Therefore, a specific numeric value for SEF that indicates adequate anesthetic effect in one patient may be not be adequate in the same patient using a different drug. A rapid decline in SEF (>50% decrease sustained below prior baseline within <30 seconds) in a patient being monitored has, however, been reliably correlated with the onset of cerebral ischemia during carotid artery surgery [3, 54–57].

Many commonly used general anesthetics produce burst suppression EEG patterns without slowing the waves present during the remaining bursts, thus pure SEF of the epoch would not reflect the additional anesthetic-induced depression. Combining the SEF with the BSR parameter to form the burst-compensated SEF (BcSEF) creates a parameter that appears to smoothly track changes in the EEG due to either slowing or suppression from isoflurane or desflurane [43, 58]:

$$BcSEF = SEF\left(1 - \frac{BSR}{100}\right)$$

Spectral qEEG parameters such as MPF or SEF compress into a single variable the 60 or more spectral power estimates that constitute the typical EEG spectrum. As Levy pointed out [59], a single feature may not be sensitive to all possible changes in spectral distribution. Frequency domain–based qEEG parameters, like their time domain–based relatives, are generally averaged over time prior to display. The author uses nonlinear smoothing when computing SEF that strongly filters small variations, but passes large changes with little filtering. This approach, also known as a tracking filter, diminishes noise, but briskly displays major changes, such as those that can occur secondary to ischemia or following a bolus injection of anesthetics.

Figure 9.6(a) illustrates a sample of an anesthetized human EEG as it is transformed by common analytical processes. The original EEG waveform is $x(i)$ (after analog antialias filtering with a bandpass of 0.3 to 30 Hz) digitized at 256 Hz for 16 seconds. The tracing below $x(i)$ demonstrates the effect of windowing on the original signal. Windowing is a technique employed to reduce distortion from epoch end artifacts in subsequent frequency-domain processing. A window consists of a set of digital values with the same number of members as the data epoch. In this case, a Blackman window was employed:

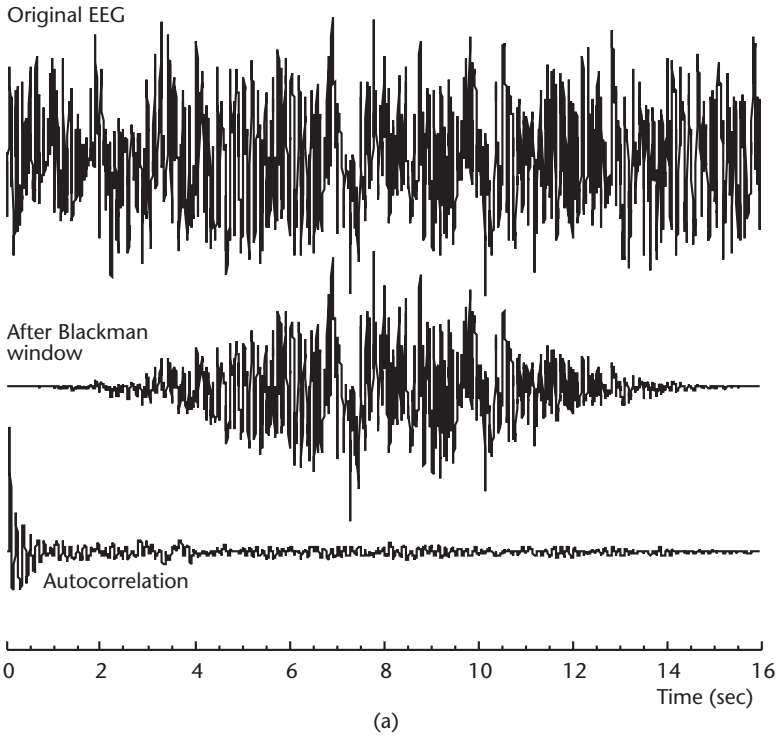



**Figure 9.6** (a) An illustration of time-domain–based EEG processing. The top waveform is the original signal of anesthetized rat scalp recording following analog antialias filtering with a bandpass of 0.3–30 Hz and digitizing at 256 Hz. The middle tracing demonstrates the effect of windowing on the original signal. Windowing is a technique employed to reduce distortion from epoch end artifacts in subsequent frequency domain processing. A window consists of a set of digital values with the same number of members as the data epoch. In this case, a Blackman window was employed. The window operation multiplies each data sample value against its corresponding window value, that is, the resulting waveform $z(i) = x(i) * w(i)$ for each value of $i$ in the epoch. The bottom tracing is the autocorrelation function of this epoch of EEG. The autocorrelation provides much of the same information as a frequency spectrum in that it can identify rhythmicities in the data. In this case, the strongest autocorrelation is at time = 0 as might be anticipated, and there are some weak rhythmicities which taper off as the lag increases above 1 second. (b) Continuing with the same epoch of digitized EEG, the top two tracings are the real and imaginary component spectra respectively resulting from the Fourier transform. The middle trace is the phase spectrum, which is classically has been discarded due to the present lack of known clinically useful correlation. The bottom tracing is the power spectrum. It is calculated as the sum of the squared real and imaginary components at each frequency [i.e., measuring the squared magnitude for each frequency value of the complex spectrum, $X(f)$]. Recall that power equals squared voltage. Note that the power spectrum, by reflecting only spectral magnitude, has explicitly removed whatever phase versus frequency information was present in the original complex spectrum. From the power spectrum, the QEEG and relative band powers are calculated as described in the text.

$$w_{Blackman}(i) = 0.42 - 0.5\cos\left(\frac{2\pi i}{n-1}\right) + 0.8\cos\left(\frac{4\pi i}{n-1}\right)$$

$$x_{windowed}(i) = x(i) * w(i) \text{ for each } i \text{ in the epoch}$$

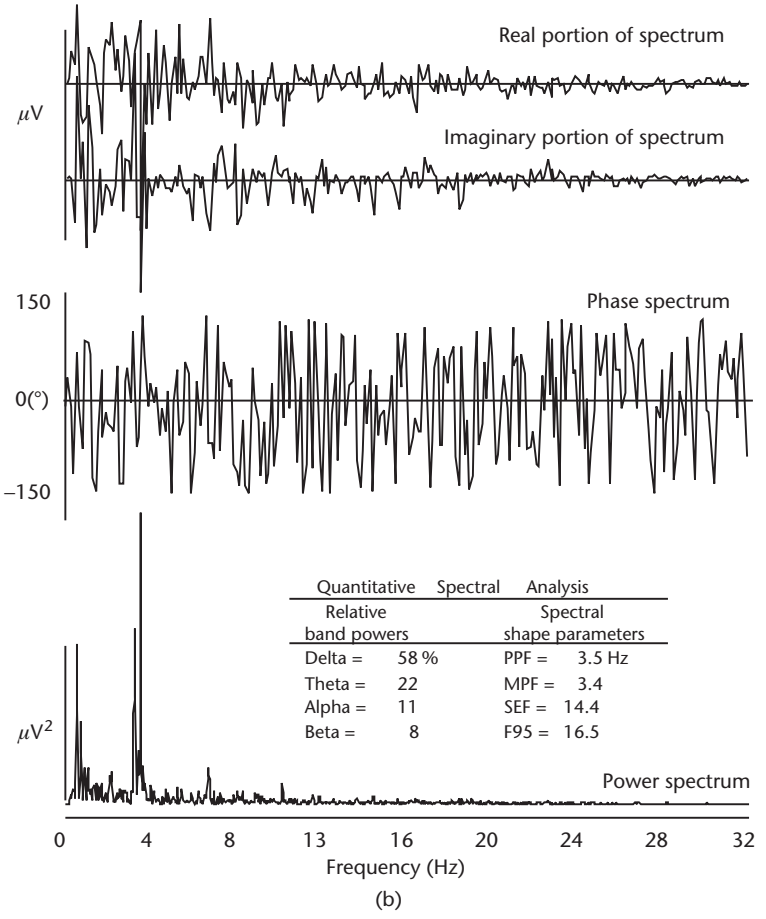



**Figure 9.6** (continued)

Below the windowed EEG is an illustration of autocorrelation. Because the only large peak is at time zero, there are no strongly repetitive patterns. The weak oscillation near the beginning of the correlation curve suggests the presence of harmonics. Figure 9.6(b) is the result of a Fourier transform. The top of Figure 9.6(b) illustrates the raw output of the transform, below that is the resulting phase spectrum, and finally the power spectrum with its associated band powers and other spectral qEEG is shown. Because there was no burst suppression phenomena in the original EEG epoch, the BcSEF would equal the SEF at 14.4 Hz.

The qEEG variables described to this point were all created to measure patterns apparent by visual inspection in the raw waveform or the power spectrum of the EEG. Although many of these qEEG variables detect changes in the EEG due to anesthetic drugs, as noted earlier, they suffer from the lack of calibration to behavioral end points that are the gold standards of anesthetic effect. Their performance as anesthetic monitors also suffers due to their sensitivity to the different EEG patterns induced by different drugs.

### 9.5.2 Mixed Algorithms: Bispectrum

The effort to glean useful information from the EEG has led from first-order (mean and variance of the amplitude of the signal waveform) to second-order (power spectrum, or its time-domain analog, autocorrelation) statistics, and now to higher order statistics (HOS). HOS include the bispectrum and trispectrum (third- and fourth-order statistics, respectively). Little work has been published to date on trispectral applications in biology, but there are currently well over 1,000 peer-reviewed papers to date related to bispectral analysis of the EEG. Whereas the phase spectrum produced by Fourier analysis measures the phase of component frequencies relative to the start of the epoch, the bispectrum measures the correlation of phase between different frequency components as described later. What exactly these phase relationships mean physiologically is uncertain. One simple teleological model holds that strong phase relationships relate inversely to the number of independent EEG generator elements.

Bispectral analysis has several additional characteristics that may be advantageous for processing EEG signals: Gaussian sources of noise are suppressed, thus enhancing the SNR for the non-Gaussian EEG, and bispectral analysis can identify nonlinearities that may be important in the signal generation process. A complete treatment of higher order spectra may be found in the text by Proakis et al. [60]. As described later in this chapter, the commercial exemplar of bispectral EEG processing, the Aspect BIS monitor, actually mixes parameters from both the time, frequency, and HOS domains to produce its output.

As noted earlier, the bispectrum quantifies the relationship among the underlying sinusoidal components of the EEG. Specifically, bispectral analysis examines the relationship between the sinusoids at two primary frequencies, $f_1$ and $f_2$, and a modulation component at the frequency $f_1 + f_2$. This set of three frequency components is known as a triplet ($f_1$, $f_2$, and $f_1 + f_2$). For each triplet, the bispectrum, $B(f_1, f_2)$, a quantity incorporating both phase and power information, can be calculated as described next. The bispectrum can be decomposed to separate out the phase information as the bicoherence, $BIC(f_1, f_2)$, and the joint magnitude of the members of the triplet, as the real triple product, $RTP(f_1, f_2)$. The defining equations for bispectral analysis are described in detail next.

A high bicoherence value at ($f_1$, $f_2$) indicates that there is a phase coupling within the triplet of frequencies $f_1$, $f_2$ and $f_1 + f_2$. Strong phase coupling implies that the sinusoidal components at $f_1$ and $f_2$ may have a common generator, or that the neural circuitry they drive may, through some nonlinear interaction, synthesize a new, dependent component at the modulation frequency $f_1 + f_2$. An example of such phase relationships and the bispectrum is illustrated in Figure 9.7.

Calculation of the bispectrum, $B(f_1, f_2)$, of a digitized epoch, $x(i)$, begins with an FFT to generate complex spectral values, $X(f)$. For each possible triplet, the complex conjugate of the spectral value at the modulation frequency $X^*(f_1 + f_2)$ is multiplied against the spectral values of the primary frequencies of the triplet:

$$B(f_1, f_2) = \left| X(f_1) \cdot X(f_2) \cdot X^*(f_1 + f_2) \right|$$

This multiplication is the heart of the bispectral determination: If, at each frequency in the triplet, there is a large spectral amplitude (i.e., a sinusoid exists for

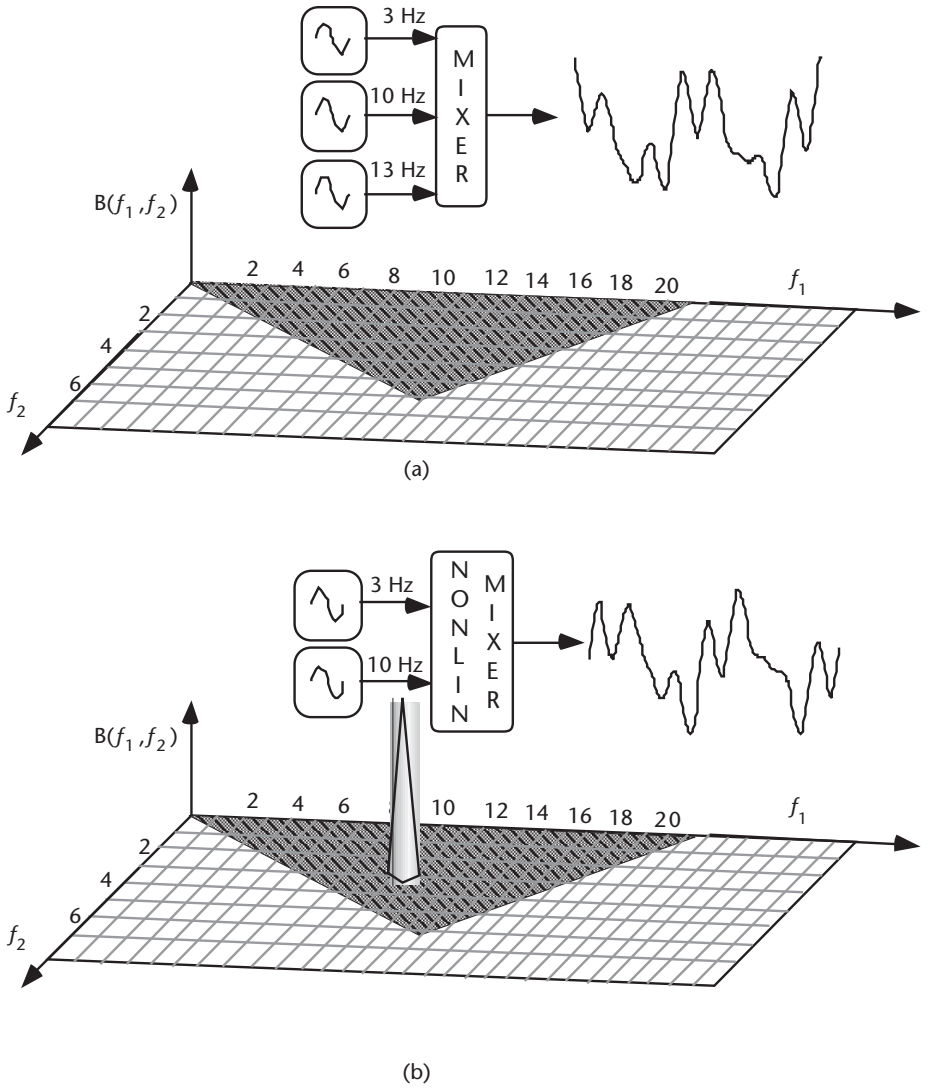



**Figure 9.7** The bispectrum is calculated in a two-dimensional space of frequency$_1$ versus frequency$_2$ as represented by the coarsely crosshatched area. Due to the symmetric redundancy noted in the text and the limit imposed by the sampling rate, the bispectrum need only be calculated for the limited subset of frequency combinations illustrated by the darkly shaded triangular wedge. A strong phase relationship between $f_1$, $f_2$ and $f_1 + f_2$ creates a large bispectral value $B(f_1, f_2)$ represented as a vertical spike rising out of the frequency versus frequency plane. (a) Three waves having no phase relationship are mixed together to produce the waveform shown at upper right. The bispectrum of this signal is equal to zero everywhere. (b) Two independent waves at 3 and 10 Hz are combined in a nonlinear fashion, creating a new waveform that contains the sum of the originals plus a wave at 13 Hz, which is phase locked to the 3- and 10-Hz components. In this case, computation of the bispectrum reveals a point of high bispectral energy at $f_1$ = 3 Hz and $f_2$ = 10 Hz.

that frequency), and the phase angles for each are aligned, then the resulting product will be large; if one of the component sinusoids is small or absent, or the phase

angles are not aligned, the product will be small. Finally, the complex bispectrum is converted to a real number by computing the magnitude of the complex product.

If one starts by sampling EEGs at 128 Hz into 4-second epochs, then the resulting Fourier spectrum will extend from 0 to 64 Hz at 0.25-Hz resolution, or a total of 256 spectral points. If all triplets were to be calculated, there would be 65,536 ($256 \times 256$) points. Fortunately, it is unnecessary to calculate the bispectrum for all possible frequency combinations. The minimal set of frequency combinations to calculate a bispectrum can be visualized as a wedge of frequency versus frequency space (Figure 9.7). The combinations outside this wedge need not be calculated because of symmetry [i.e., $B(f_1, f_2) = B(f_2, f_1)$] and because the range of allowable modulation frequencies, $f_1 + f_2$, is limited to frequencies less than or equal to half of the sampling rate. Still, because this calculation must be performed, using complex number arithmetic, for at least several thousand triplets, it is easy to see that it is a major computational burden.

As noted earlier, computation of the bispectrum itself is only the beginning for complete higher order spectral analysis. If one is interested in isolating and examining solely the phase relationships, as noted earlier, the bispectrum must have the existing variations in signal amplitude normalized. Recall that the amplitude of a particular Fourier spectral element $X(f)$ is determined by the magnitude or the length of its complex number vector. The RTP is formed from the multiplied product of the squared magnitudes of the three spectral values in the triplet:

$$RTP(f_1, f_2) = |X(f_1)|^2 * |X(f_2)|^2 * |X(f_1 + f_2)|^2$$

The square root of the RTP yields the joint amplitude of the triplet, the factor that is used to normalize the bispectrum into the bicoherence. The bicoherence, $BIC(f_1, f_2)$ is a number that varies from 0 to 1 in proportion to the degree of phase coupling in the frequency triplet:

$$BIC(f_1, f_2) = \frac{BIC(f_1, f_2)}{\sqrt{RTP(f_1, f_2)}}$$

Figure 9.8 illustrates some representative data during bispectral analysis. Computing the bispectrum of a stochastic biological signal such as the EEG generally requires that the signal be divided into relatively short epochs for calculation of the bispectrum and bicoherence, which are then averaged over a number of epochs to provide a relatively stable estimate of the true bispectral values. Figure 9.8(a) is a two-dimensional plot of bispectrum $B(f_1, f_2)$; Figure 9.8(b) is a three-dimensional perspective illustration of the same data. Figure 9.8(c) is a three-dimensional illustration of the bicoherence $BIC(f_1, f_2)$.

### 9.5.3 Bispectral Index: Implementation

The BIS is a complex qEEG parameter, composed of a combination of time-domain, frequency-domain, and higher order spectral subparameters. It was the first among the qEEG parameters reviewed here to integrate several disparate descriptors of the EEG into a single variable based on the post hoc analysis of a large volume of clini-

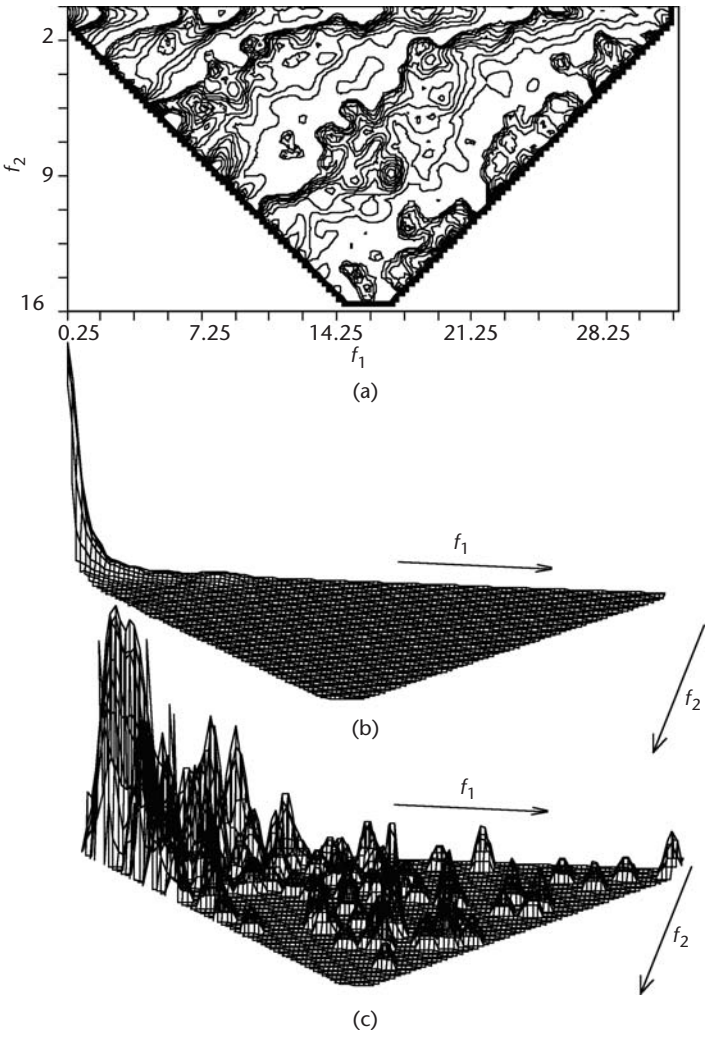



**Figure 9.8** (a) Two-dimensional plot of bispectrum $B(f_1, f_2)$; (b) three-dimensional perspective illustration of bispectrum $B(f_1, f_2)$; and (c) three-dimensional illustration of the bicoherence $BIC(f_1, f_2)$.

cal data to synthesize a combination that, by design, correlates behavioral assessments of sedation and hypnosis yet is insensitive to the specific anesthetic or sedative agents chosen. Further, devices that implement BIS are the only ones currently approved by the Food and Drug Administration for marketing claims to reduce the incidence of unintended postoperative recall.

The particular (proprietary) mixture of subparameters in BIS version 3 was derived empirically [6, 61] from a prospectively collected database of EEG and behavioral scales representing approximately 1,500 anesthetic administrations and 5,000 hours of recordings that employed a variety of anesthetic protocols. BIS was then tested prospectively in other populations and the process iterated. At present, the commercial device incorporates the fourth major revision of the index.

The calculation of the BIS (Figure 9.9) begins with a sampled EEG that is filtered to exclude both high- and low-frequency artifacts and divided into epochs of 2-sec-

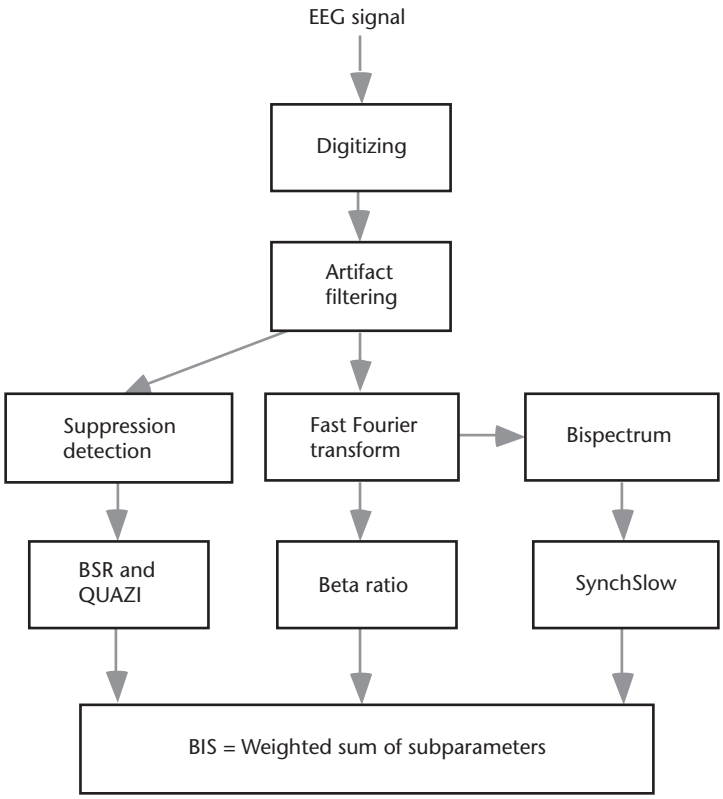



**Figure 9.9** Overview of BIS algorithm.

ond duration. A series of algorithms then detects and attempts to remove or ignore artifacts.

The first phase of artifact handling uses a cross-correlation of the EEG epoch with a template pattern of an ECG waveform. If ECG or pacer spikes are detected, they are removed from the epoch and the missing data estimated by interpolation. Epochs repaired in this phase are still considered viable for further processing.

Next eye-blink events are detected, again relying on their stereotypical shape to match a template with cross-correlation. Epochs with blink artifacts are considered to have unrepairable noise and are not processed further. Surviving epochs are checked for a wandering baseline (low-frequency electrode noise) and if this state is detected, additional filtering to reject very low frequencies is applied. In addition, the variance (i.e., the second central moment) of the EEG waveform for each epoch is calculated. If the variance of an epoch of raw EEG changes markedly from an average of recent prior epochs, the new epoch is marked as "noisy" and not processed further; however, the new variance is incorporated into an updated average. If the variance of new incoming epochs continues to be different from the previous baseline, the system will slowly adapt as the prior average changes to the new variance.

Presuming the incoming EEG epoch is artifact free, or is deemed repaired, the time-domain version of the epoch is used to calculate the degree of burst suppression with two separate algorithms: BSR and QUAZI [6]. The BSR algorithm used by the BIS calculation is quite similar to that described in the preceding section. The

QUAZI suppression index was designed to detect burst suppression in the presence of a wandering baseline voltage. QUAZI incorporates slow wave (<1.0-Hz) information derived from the frequency domain to detect burst activity superimposed on these slow waves that would "fool" the original BSR algorithm by exceeding the voltage criteria for electrical "silence."

The waveform data in the current epoch is prepared for conversion to the frequency domain by a Blackman window function, as illustrated in Figure 9.6(a). Then the FFT and the bispectrum of the current EEG epoch are calculated. The resulting spectrum and bispectrum are smoothed using a running average against those calculated in the prior minute, then the frequency domain–based subparameters "SynchFastSlow" and "BetaRatio" are computed. The BetaRatio subparameter is the log ratio of power in two empirically derived frequency bands: log[(P30, 47 Hz)/(P11, 20 Hz)]. The SynchFastSlow subparameter is the contribution from high-order (bispectral) analysis. SynchFastSlow is defined as another log ratio. Here the log of the ratio of the sum of all bispectra peaks in the area from 0.5 to 47 Hz over the sum of the bispectrum in the area from 40 to 47 Hz. The resulting BIS is defined as a proprietary combination of these qEEG subparameters.

Each of the component subparameters was chosen to have a specific range of anesthetic effect where they perform best; that is, the SynchFastSlow (HOS) parameter is well correlated with behavioral responses during moderate sedation or light anesthesia. The combination algorithm that determines BIS therefore weights the Beta Ratio (FFT) most heavily when the EEG has the characteristics of light sedation. The SynchFastSlow (bispectral component) predominates during the phenomena of EEG activation (excitement phase) and during surgical levels of hypnosis, and the BSR and QUAZI detect very deep anesthesia.

The subparameters are combined using a nonlinear function whose coefficients were determined by the iterative data collection and tuning process. Two key features of the Aspect BIS multivariate model are, first, that it accounts for the nonlinear stages of EEG activity by allowing different subparameters to dominate the resulting BIS as the EEG changes its character with increasing anesthesia. Second, the model framework is extensible, so new subparameters can be added to improve performance, if needed, in the presence of new anesthetic regimes. The combination of the four subparameters produces a single number, BIS, which decreases monotonically with decreasing level of consciousness (hypnosis). As described earlier, computation of a bispectral parameter (SynchFastSlow) requires averaging several epochs; therefore, the BIS value reported on the front panel of the monitor represents an average value derived from the previous 60 seconds of usable data.

### 9.5.4 Bispectral Index: Clinical Results

BIS has been empirically demonstrated to correlate with behavioral measures of sedation and light anesthesia [62–67] due to a variety of anesthetics including isoflurane [67, 68] sevoflurane [69, 70] and desflurane [71] vapors, and propofol [67] and midazolam [67], which are parenteral anesthetics of different classes. The BIS parameter is not sensitive to the effects of ketamine because that agent's dominant effect on the EEG is in the theta-band range. Although it is difficult to test memory formation in pediatric patients, it appears, based on dose-response, that the BIS

appears to work in all but infants (=12 months of age). This is due to the development changes in the EEG of children. It has been demonstrated that close titration of anesthetic effect using BIS improves some measures of patient outcome or operating suite efficiency [72, 73]. Finally, and most significant to patients is that anesthetic titration using BIS has been demonstrated in two large clinical trials to reduce the incidence of unintended recall of intraoperative events [74, 75].

## 9.6 Conclusions

Divining the anesthetic effect message within the EEG has long been sought. General anesthetic vapors, propofol, barbiturates, and benzodiazepines induce synchronization and eventual slowing of cortical electrical activity. EEG itself is not the end point clinicians desire, but it may serve as a surrogate for the behaviors which are important. The recent availability of devices including Aspect's BIS and GE Medical's Entropy monitors, which claim to link the behaviors of awareness and recall to the EEG, is a significant step forward in this endeavor. Although there is currently no theoretical or mechanistic link proposed between neural network pharmacology in the cerebral cortex and the intrafrequency coupling notion of the BIS, or the channel complexity implicit in entropy analysis, the empirical correlations have been confirmed. The exact role and limitations of this new technology will be determined through additional clinical experience. With the attainment of this present benchmark level of clinical correlation, further refinements in signal processing can now be reasonably expected to create increasingly useful tools for a wide range of clinical settings.

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CHAPTER 10

# Quantitative Sleep Monitoring

Paul R. Carney, Nicholas K. Fisher, William Ditto, and James D. Geyer

Sleep is a major part of everyone's life and has been studied thoroughly. Sleep is made up of nonrapid eye movement (NREM) sleep and rapid eye movement (REM) sleep. NREM sleep is further broken down into four stages. In this chapter, we discuss much of the terminology used in polysomnography as well as some of the techniques and issues involved in recording polysomnographic data. The physiological characteristics measured by the polysomnograph as well as how they differ throughout the various sleep stages are explained as well. In addition, some of the quantitative characteristics of the polysomnograph are discussed. Techniques that try to automatically detect the sleep stage based on these quantitative characteristics are also expounded upon. Finally, we delve into examples of sleep-related disorders and their causes.

## 10.1 Overview of Sleep Stages and Cycles

Sleep is not homogeneous and is characterized by sleep stages based on EEG or electrical brain wave activity, EOG or eye movements, and EMG or muscle electrical activity [1–3]. The basic terminology and methods involved with monitoring each of these types of activity are discussed below. Sleep is composed of NREM and REM sleep. NREM sleep is further divided into stages 1, 2, and 3/4. Stages 1 and 2 are called *light sleep*, and stages 3 and 4 are called *deep* or *slow-wave sleep*. During the night there are usually four or five cycles of sleep, each composed of a segment of NREM sleep followed by REM sleep. Periods of wake may also interrupt sleep. As the night progresses, the length of REM sleep in each cycle usually increases. The hypnogram (Figure 10.1) is a convenient method of graphically displaying the organization of sleep during the night. Each stage of sleep is characterized by a level on the vertical axis of the graph, with time of night on the horizontal axis. REM sleep is often highlighted by a dark bar.

Most sleep recording is performed digitally, but the convention of scoring sleep in 30-second epochs or windows is still the standard. If there is a shift in sleep stage

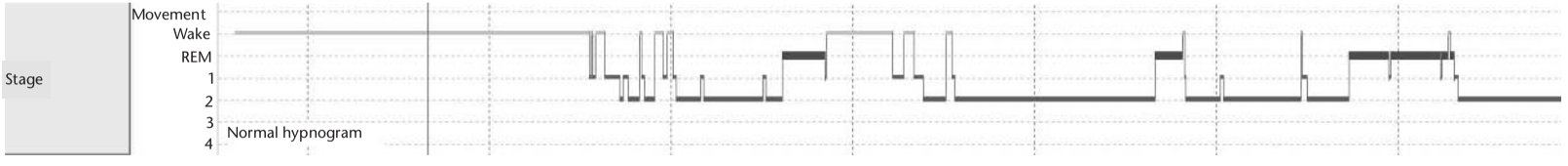



**Figure 10.1** Hypnogram showing the various stages of sleep, represented by levels on the vertical axis; time of night is shown on the horizontal axis. In this patient there were several sleep cycles, each composed of a segment of NREM sleep followed by REM sleep. (Courtesy of James Geyer and Paul Carney.)

during a given epoch, the stage present for the majority of the time names the epoch. When the tracings used to stage sleep are obscured by artifact for more than one-half of an epoch, it is scored as movement time (MT). When an epoch of what would otherwise be considered MT is surrounded by epochs of wake, the epoch is also scored as wake. Some sleep centers consider MT to be wake and do not tabulate it separately.

## 10.2 Sleep Architecture Definitions

The term *sleep architecture* describes the structure of sleep. Common terms used in sleep monitoring are listed in Table 10.1.

The normal range of the percentage of sleep spent in each sleep stage varies with age [2, 3] and is impacted by sleep disorders (Table 10.2). Chronic insomnia (difficulty initiating or maintaining sleep) is characterized by a long sleep latency and increased WASO. The amount of stages 3 and 4 and REM sleep is commonly decreased as well. The REM latency is also affected by sleep disorders and medications.

## 10.3 Differential Amplifiers, Digital Polysomnography, Sensitivity, and Filters

EEG, EOG, and EMG activity is recorded by differential ac amplifiers that amplify the difference in voltage between two inputs. Signals common to both inputs are not amplified (common mode rejection). This permits the recording of very small signals that are superimposed upon larger scalp-voltage changes and 60-cycle interference from nearby ac power lines. Common mode rejection depends on the impedance at input 1 and input 2 being relatively equal [4, 5]. A poorly conducting electrode (high impedance) will result in a large amount of 60-Hz artifact being present. By convention in EEG recording, if input 1 is negative relative to input 2, the deflection is upward (up polarity). In modern digital sleep monitoring, one may record the activity of numerous electrodes against a common electric reference (ref-

**Table 10.1** Sleep Architecture Definitions

| |
|---|
| Lights out—start of sleep recording |
| Lights on—end of sleep recording |
| TBT (total bedtime time)—time from lights out to lights on |
| TST (total sleep time)—minutes of stages 1, 2, 3, 4, and REM |
| WASO (wake after sleep onset)—minutes of wake after first sleep but before the final awakening |
| SPT (sleep period time)—TST + WASO |
| Sleep latency—time from lights out until the first epoch of sleep |
| REM latency—time from first epoch of sleep to the first epoch of REM sleep |
| Sleep efficiency—(TST × 100)/TBT |
| Stages 1, 2, 3, 4, and REM as % TST—percentage of TST occupied by each sleep stage |
| Stages 1, 2, 3, 4, and REM, WASO as % SPT—percentage of SPT occupied by sleep stages and WASO |

**Table 10.2** Representative Changes in Sleep Architecture

| | *20-Year-Old* | *60-Year-Old* | *Severe Sleep Apnea* |
|---|---|---|---|
| WASO% SPT | 5 | 15 | 20 |
| 1% SPT | 5 | 5 | 10 |
| 2% SPT | 50 | 55 | 60 |
| 3 and 4% SPT | 20 | 5 | 0 |
| REM% SPT | 25 | 20 | 10 |

**Table 10.3** Montages for Sleep Monitoring

| *Bipolar* | | | *Referential* |
|---|---|---|---|
| *Minimal* | *Typical* | *Recording (Each Against Reference Electrode)* | *Displays*[a] |
| C4-A1 (C3-A2) | C4-A1 | C4 | C4-A1 |
| ROC-A1[b] | C3-A2 | C3 | C3-A2 |
| LOC-A2[b] | O2-A1 | O2 | O1-A2 |
| Chin EMG1-EMG2 | O1-A2 | O1 | O2-A1 |
| | ROC-A1 | ROC | ROC-A1 |
| | LOC-A2 | LOC | LOC-A2 |
| | Chin EMG1-EMG2 | A1 | Chin EMG1-EMG2 |
| | | A2 | |
| | | EMG1 | |
| | | EMG2 | |
| | | EMG3 | |

[a]Any combination of referentially recorded electrodes can be displayed.
[b]ROC = right outer canthus; LOC = left outer canthus.

erential recording). Any combination of various tracings of interest can be obtained by digital subtraction (electrode A-reference) – (electrode B-reference) = electrode A – electrode B either during recording or during review (see Table 10.3) [5, 6]. Digital recording also allows for the mixture of referential (EEG, eye electrodes, EMG electrodes), true bipolar (chest, abdominal movement, airflow), and dc (oxygen saturation) recording.

The sampling rate must be more than twice the frequencies being recorded to avoid signal distortion (aliasing). In addition, signals with a frequency higher than one-half the sampling rate must be filtered out, because they can cause aliasing distortion [5].

Time windows of 60 to 240 seconds may be used to view and score respiratory events. Alternatively, viewing data in 10-second windows (equivalent to 30 mm/s) is the usual practice for viewing clinical EEG and displaying interictal or epileptic activity. It also can be useful for measuring the frequency of a complex of oscillations or viewing the EKG.

## 10.4 Introduction to EEG Terminology and Monitoring

EEG activity is characterized by the frequency in cycles per second or hertz (Hz), amplitude (voltage), and the direction of major deflection (polarity). The classically described frequency ranges are delta (<4 Hz), theta (4 to 7 Hz), alpha (8 to 13 Hz), and beta (>13 Hz). Alpha waves (8 to 13 Hz) are commonly noted when the patient is in an awake but relaxed state with the eyes closed (Figure 10.2). They are best recorded over the occiput and are attenuated when the eyes are open. Bursts of alpha waves also are seen during brief awakenings from sleep—called *arousals*. Alpha activity can also be seen during REM sleep. Alpha activity is prominent during drowsy eyes-closed wakefulness. This activity decreases with the onset of stage 1 sleep. Near the transition from stage 1 to stage 2 sleep, vertex *sharp waves*—high-amplitude negative waves (upward deflection on EEG tracings) with a short duration—occur. They are more prominent in central than in occipital EEG tracings. A sharp wave is defined as deflection of 70 to 200 ms in duration.

Sleep spindles are oscillations of 12 to 14 Hz with a duration of 0.5 to 1.5 seconds. They are characteristic of stage 2 sleep. They may persist into stages 3 and 4 but usually do not occur in stage REM. The K complex is a high-amplitude, biphasic wave of at least 0.5-second duration. As classically defined, a K complex consists of an initial sharp, negative voltage (by convention, an upward deflection) followed by a positive-deflection (down) slow wave. Spindles frequently are superimposed on K complexes. Sharp waves differ from K complexes in that they are narrower, not biphasic, and usually of lower amplitude.

As sleep deepens, slow (delta) waves appear. These are high-amplitude, broad waves. Whereas delta EEG activity is usually defined as <4 Hz, in human sleep scoring the slow-wave activity used for staging is defined as EEG activity slower than 2 Hz (longer than 0.5-second duration) with a peak-to-peak amplitude of >75 $\mu$V. The amount of slow-wave activity *as measured in the central EEG tracings* is used to determine if stage 3/4 is present [1]. Because a K complex resembles slow-wave activity, differentiating the two is sometimes difficult. However, by definition, a K complex should stand out (be distinct) from the lower-amplitude, background EEG activity. Therefore, a continuous series of high-voltage slow waves would not be considered to be a series of K complexes.

Sawtooth waves (Figure 10.2) are notched-jagged waves of frequency in the theta range (3 to 7 Hz) that may be present during REM sleep. Although they are not part of the criteria for scoring REM sleep, their presence is a clue that REM sleep is present.

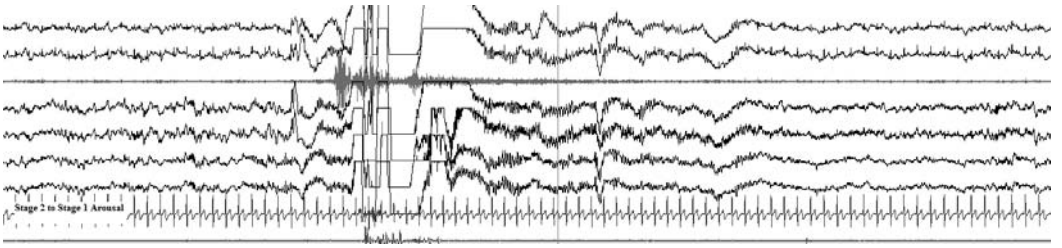



**Figure 10.2** Stage 2 sleep is shown. The EEG shows a K complex. (Courtesy of James Geyer and Paul Carney.)

## 10.5 EEG Monitoring Techniques

The traditional Rechtschaffen and Kales (R&K) guidelines for human sleep staging were based on central EEG monitoring [1]. However, most sleep recording today also includes occipital electrodes. Alpha activity is more prominent in occipital tracings. The terminology for the electrodes adheres to the International 10-20 nomenclature in which the electrodes are placed at 10% or 20% of the distance between structural landmarks on the head. Even subscripts refer to electrodes on the right side of the head and odd to electrodes on the left side. The usual derivations use the central or occipital electrodes referenced to the opposite mastoid electrode (C4-A1, O1-A2). The greater distance between electrodes increases the voltage difference. A minimum of one central EEG derivation must be recorded for sleep staging. In modern digital recording, typically all of the electrodes (C4, C3, O2, O1, A1, A2) are recorded. Of note, additional electrodes may be added if one suspects seizure activity. This is discussed in detail in later chapters.

## 10.6 Eye Movement Recording

The main purpose of recording eye movements is to identify REM sleep. EOG (eye movement) electrodes typically are placed at the outer corners of the eyes—at the right outer canthus (ROC) and the left outer canthus (LOC). In a common approach, two eye channels are recorded and the eye electrodes are referenced to the opposite mastoid (ROC-A1 and LOC-A2). To detect vertical as well as horizontal eye movements, one electrode is placed slightly above and one slightly below the eyes [4–7].

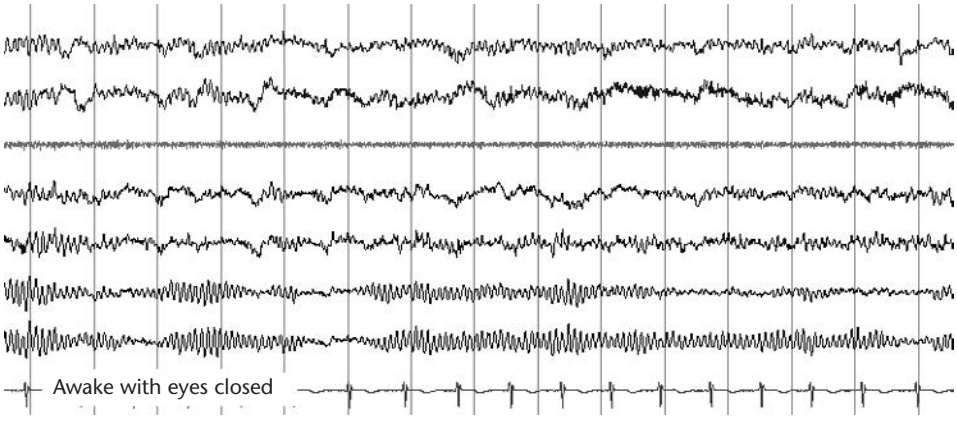
Recording of eye movements is possible because a potential difference exists across the eyeball: front positive (+), back negative (–). Eye movements are detected by EOG recording of voltage changes. When the eyes move toward an electrode, a positive voltage is recorded.

There are two common patterns of eye movements (Figure 10.3). Slow eye movements (SEMs), also called *slow-rolling* eye movements, are pendular oscillating movements that are seen in drowsy (eyes closed) wakefulness and stage 1 sleep. By stage 2 sleep, SEMs usually have disappeared. REMs are sharper (more narrow deflections), which are typical of eyes-open wake and REM sleep.

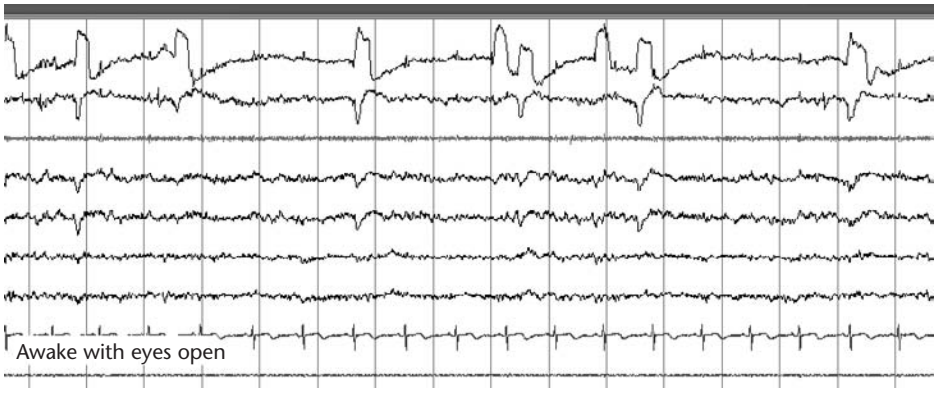
In the two-tracing method of eye movement recording, large-amplitude EEG activity or artifact reflected in the EOG tracings usually causes *in-phase deflections*. In Figure 10.4, a K complex causes an in-phase deflection in the eye tracings, while REM result in *out-of-phase* deflections.

## 10.7 Electromyographic Recording

Usually, three EMG leads are placed in the mental and submental areas. The voltage between two of these three is monitored (for example, EMG1-EMG3). If either of these leads fails, the third lead can be substituted. The gain of the chin EMG is adjusted so that some activity is noted during wakefulness. The chin EMG is an




(a)




(b)

**Figure 10.3** Typical patterns of eye movements. (a) SEMs are pendular and are common in drowsy wake and stage 1 sleep. (b) REMs are sharper (shorter duration) and are seen in eyes-open wake or REM sleep. (Courtesy of James Geyer and Paul Carney.)

essential element only for identifying stage REM sleep. In stage REM, the chin EMG is relatively reduced—the amplitude is equal to or lower than the lowest EMG amplitude in NREM sleep. If the chin EMG gain is adjusted high enough to show some activity in NREM sleep, a drop in activity is often seen on transition to REM sleep. The chin EMG may also reach the REM level long before the onset of REM sleep or an EEG-meeting criteria for stage REM. Depending on the gain, a reduction in the chin EMG amplitude from wakefulness to sleep and often a further reduction on transition from stage 1 to 4 may be seen. However, a reduction in the chin EMG is not required for stages 2 to 4. The reduction in the EMG amplitude during REM sleep is a reflection of the generalized skeletal-muscle hypotonia present in this sleep stage. Phasic brief EMG bursts still may be seen during REM sleep. In Figure 10.4, there is a fall in chin EMG amplitude just before the REMs occur. The combination of REMs, a relatively reduced chin EMG, and a low-voltage mixed-frequency EEG is consistent with stage REM.

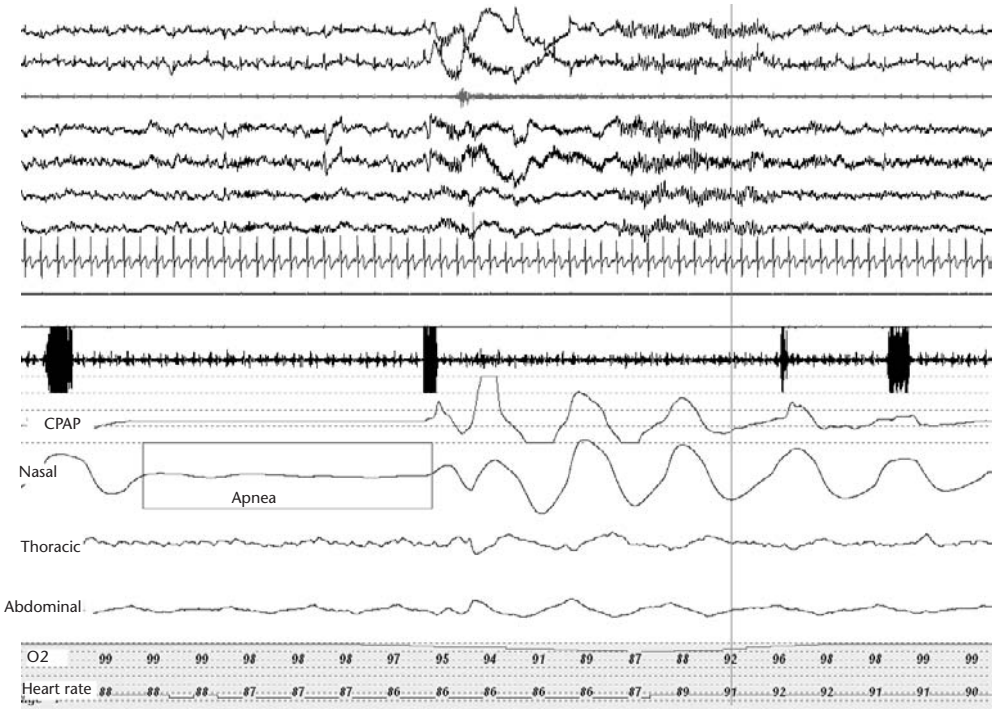



**Figure 10.4** There is an arousal from NREM sleep associated with the apnea. An increase in EMG is noted, but this is not required to score an arousal from NREM sleep. The shift in EEG frequency is best seen in this example in the central electrodes. (See text for arousal definitions.) (Courtesy of James Geyer and Paul Carney.)

## 10.8 Sleep Stage Characteristics

The basic rules for sleep staging are summarized in Table 10.4. Note that some characteristics are required and some are helpful but not required [8, 9]. The typical patterns associated with each sleep stage are outlined.

### 10.8.1 Atypical Sleep Patterns

Four special cases in which sleep staging is made difficult by atypical EEG, EOG, and EMG patterns are briefly mentioned. In alpha sleep, prominent alpha activity persists into NREM sleep [10–13]. The presence of spindles, K complexes, and slow-wave activity allows sleep staging despite prominent alpha activity. Causes of the pattern include pain, psychiatric disorders, chronic pain syndromes, and any cause of nonrestorative sleep [12, 13]. Patients taking benzodiazepines may have very prominent "pseudo-spindle" activity (14 to 16 rather than the usual 12 to 14 Hz) [14]. SEMs are usually absent by the time stable stage 2 sleep is present. However, patients on some serotonin reuptake inhibitors (fluoxetine and others) may have prominent slow and rapid eye movements during NREM sleep [15]. Although a reduction in the chin EMG is required for staging REM sleep, patients with the REM sleep behavior disorder may have high chin activity during what otherwise appears to be REM sleep [16].

**Table 10.4** Summary of Sleep Stage Characteristics

| | *Characteristic*[a, b] | | |
|---|---|---|---|
| *Stage* | *EEG* | *EOG* | *EMG* |
| Wake (eyes open) | Low-voltage, high-frequency, attenuated alpha activity | Eye blinks, REMs | Relatively high |
| Wake (eyes closed) | Low-voltage, high-frequency >50% alpha activity | Slow-rolling eye movements | Relatively high |
| Stage 1 | **Low-amplitude mixed-frequency <50% alpha activity NO spindles, K complexes** Sharp waves near transition to stage 2 | Slow-rolling eye movements | May be lower than wake |
| Stage 2 | **At least one sleep spindle or K complex <20% slow-wave activity**[b] | | May be lower than wake |
| Stage 3 | 20–50% Slow-wave activity | [c] | Usually low |
| Stage 4 | **>50% Slow-wave activity** | [c] | Usually low |
| Stage REM | **Low-voltage mixed-frequency** Sawtooth waves—may be present | **Episodic REMs** | **Relatively reduced** (equal to or lower than the lowest in NREM) |

[a]Boldface type indicates required characteristics.
[b]Slow-wave activity, frequency < 2 Hz; peak-to-peak amplitude > 75 $\mu$V; >50% means slow-wave activity present in more than 50% of the epoch.
[c]Slow waves usually seen in EOG tracings.

### 10.8.2 Sleep Staging in Infants and Children

Newborn term infants do not have the well-developed adult EEG patterns to allow staging according to R&K rules. The following is a brief description of terminology and sleep staging for the newborn infant according to the state determination of Anders, Emde, and Parmelee [17]. Infant sleep is divided into active sleep (corresponding to REM sleep), quiet sleep (corresponding to NREM sleep), and indeterminant sleep, which is often a transitional sleep stage. Behavioral observations are critical. Wakefulness is characterized by crying, quiet eyes open, and feeding. Sleep is often defined as sustained eye closure. Newborn infants typically have periods of sleep lasting 3 to 4 hours interrupted by feeding, and total sleep in 24 hours is usually 16 to 18 hours. They have cycles of sleep with a 45- to 60-minute periodicity with about 50% active sleep. In newborns, the presence of REM (active sleep) at sleep onset is the norm. In contrast, the adult sleep cycle is 90 to 100 minutes, REM occupies about 20% of sleep, and NREM sleep is noted at sleep onset.

The EEG patterns of newborn infants have been characterized as low-voltage irregular, tracé alternant, high-voltage slow, and mixed (Table 10.5). Eye movement monitoring is used as in adults. An epoch is considered to have high or low EMG if more than one-half of the epoch shows the pattern. The characteristics of active sleep, quiet sleep, and indeterminant sleep are listed in Table 10.6. The change from active to quiet sleep is more likely to manifest indeterminant sleep. Nonnutritive sucking commonly continues into sleep.

As children mature, more typically adult EEG patterns begin to appear. Sleep spindles begin to appear at 2 months and are usually seen after 3 to 4 months of age [18]. K complexes usually begin to appear at 6 months of age and are fully developed by 2 years of age [19]. The point at which sleep staging follows adult rules in

**Table 10.5** EEG Patterns Used in Infant Sleep Staging

| *EEG Pattern* | *Characteristics* |
|---|---|
| Low-voltage irregular (LVI) | Low-voltage (14 to 35 $\mu$V), little variation theta (5 to 8 Hz) predominates |
| | Slow activity (1 to 5 Hz) also present |
| Tracé alternant (TA) | Bursts of high-voltage slow waves (0.5 to 3 Hz) with superimposition of rapid low-voltage sharp waves (2 to 4 Hz) |
| | In between the high-voltage bursts (alternating with them) is low-voltage mixed-frequency activity of 4 to 8 seconds in duration |
| High-voltage slow (HVS) | Continuous moderately rhythmic medium- to high-voltage (50 to 150 $\mu$V) slow waves (0.5 to 4 Hz) |
| Mixed (M) | High-voltage slow and low-voltage polyrhythmic activity |
| | Voltage lower than in HVS |

**Table 10.6** Characteristics of Active and Quiet Sleep

| | *Active Sleep* | *Quiet Sleep* | *Indeterminant* |
|---|---|---|---|
| *Behavioral* | Eyes closed | Eyes closed | Not meeting criteria for active or quiet sleep |
| | Facial movements: smiles, grimaces, frowns | No body movements except startles and phasic jerks | |
| | Burst of sucking | Sucking may occur | |
| | Body movements: small digit or limb movements | | |
| *EEG* | LVI, M, HVS (rarely) | HVS, TA, M | |
| *EOG* | REMs | No REMs | |
| | A few SEMs and a few dysconjugate movements may occur | | |
| *EMG* | Low | High | |
| *Respiration* | Irregular | Regular | |
| | | Postsigh pauses may occur | |

not well defined but usually is possible after age 6 months. After about 3 months, the percentage of REM sleep starts to diminish and the intensity of body movements during active (REM) sleep begins to decrease. The pattern of NREM at sleep onset begins to emerge. However, the sleep cycle period does not reach the adult value of 90 to 100 minutes until adolescence.

Note that the sleep of premature infants is somewhat different from that of term infants (36 to 40 weeks' gestation). In premature infants quiet sleep usually shows a pattern of tracé discontinu [20]. This differs from tracé alternant in that there is electrical quiescence (rather than a reduction in amplitude) between bursts of high-voltage activity. In addition, delta brushes (fast waves of 10 to 20 Hz) are superimposed on the delta waves. As the infant matures, delta brushes disappear and tracé alternant pattern replaces tracé discontinu.

## 10.9 Respiratory Monitoring

The three major components of respiratory monitoring during sleep are airflow, respiratory effort, and arterial oxygen saturation [21, 22]. Many sleep centers also find a snore sensor to be useful. For selected cases, exhaled or transcutaneous $PCO_2$ may also be monitored.

Traditionally, airflow at the nose and mouth was monitored by thermistors or thermocouples. These devices actually detect airflow by the change in the device temperature induced by a flow of air over the sensor. It is common to use a sensor in or near the nasal inlet and over the mouth (nasal–oral sensor) to detect both nasal and mouth breathing. Although temperature-sensing devices may accurately detect an absence of airflow (apnea), their signal is not proportional to flow, and they have a slow response time [23]. Therefore, they do not accurately detect decreases in airflow (hypopnea) or flattening of the airflow profile (airflow limitation).

Exact measurement of airflow can be performed by use of a pneumotachograph. This device can be placed in a mask over the nose and mouth. Airflow is determined by measuring the pressure drop across a linear resistance (usually a wire screen). However, pneumotachographs are rarely used in clinical diagnostic studies. Instead, monitoring of nasal pressure via a small cannula in the nose connected to a pressure transducer has gained in popularity for monitoring airflow [23, 24]. The nasal pressure signal is actually proportional to the square of flow across the nasal inlet [25]. Thus, nasal pressure underestimates airflow at low flow rates and overestimates airflow at high flow. In the midrange of typical flow rates during sleep, the nasal pressure signal varies fairly linearly with flow. The nasal pressure versus flow relationship can be completely linearized by taking the square root of the nasal pressure signal [26]. However, in clinical practice, this is rarely performed.

In addition to changes in magnitude, changes in the shape of the nasal pressure signal can provide useful information. A flattened profile usually means that airflow limitation is present (constant or decreasing flow with an increasing driving pressure) [23, 24]. The unfiltered nasal pressure signal also can detect snoring if the frequency range of the amplifier is adequate. The only significant disadvantage of nasal pressure monitoring is that mouth breathing often may not be adequately detected (10%–15% of patients). This can be easily handled by monitoring with both nasal pressure and a nasal–oral thermistor.

An alternative approach to measuring flow is to use respiratory inductance plethysmography. The changes in the sum of the ribcage and abdomen band signals (RIPsum) can be used to estimate changes in tidal volume [27, 28]. During positive-pressure titration, an airflow signal from the flow-generating device is often recorded instead of using thermistors or nasal pressure. This flow signal originates from a pneumotachograph or other flow-measuring device inside the flow generator.

In pediatric polysomnography, exhaled $CO_2$ is often monitored. Apnea usually causes an absence of fluctuations in this signal, although small expiratory puffs rich in $CO_2$ can sometimes be misleading [7, 22]. The end-tidal $PCO_2$ (value at the end of exhalation) is an estimate of arterial $PCO_2$ . During long periods of hypoventilation

that are common in children with sleep apnea, the end-tidal $PCO_2$ will be elevated (>45 mm Hg) [22].

Respiratory effort monitoring is necessary to classify respiratory events. A simple method of detecting respiratory effort is detecting movement of the chest and abdomen. This may be performed with belts attached to piezoelectric transducers, impedance monitoring, respiratory-inductance plethysmography (RIP), or monitoring of esophageal pressure (reflecting changes in pleural pressure). The surface EMG of the intercostal muscles or diaphragm can also be monitored to detect respiratory effort. Probably the most sensitive method for detecting effort is monitoring of changes in esophageal pressure (reflecting changes in pleural pressure) associated with inspiratory effort [24]. This may be performed with esophageal balloons or small fluid-filled catheters. Piezoelectric bands detect movement of the chest and abdomen as the bands are stretched and the pull on the sensors generates a signal. However, the signal does not always accurately reflect the amount of chest/abdomen expansion. In RIP, changes in the inductance of coils in bands around the rib cage (RC) and abdomen (AB) during respiratory movement are translated into voltage signals. The inductance of each coil varies with changes in the area enclosed by the bands. In general, RIP belts are more accurate in estimating the amount of chest/abdominal movement than piezoelectric belts. The sum of the two signals [RIPsum = $(a \times RC) + (b \times AB)$] can be calibrated by choosing appropriate constants $a$ and $b$. Changes in the RIPsum are estimates of changes in tidal volume [29]. During upper-airway narrowing or total occlusion, the chest and abdominal bands may move paradoxically. Of note, a change in body position may alter the ability of either piezoelectric belts or RIP bands to detect chest/abdominal movement. Changes in body position may require adjusting band placement or amplifier sensitivity. In addition, very obese patients may show little chest/abdominal wall movement despite considerable inspiratory effort. Thus, one must be cautious about making the diagnosis of central apnea solely on the basis of surface detection of inspiratory effort.

Arterial oxygen saturation ($SaO_2$) is measured during sleep studies using pulse oximetry (finger or ear probes). This is often denoted as $SpO_2$ to specify the method of $SaO_2$ determination. A desaturation is defined as a decrease in $SaO_2$ of 4% or more from baseline. Note that the nadir in $SaO_2$ commonly follows apnea (hypopnea) termination by approximately 6 to 8 seconds (longer in severe desaturations). This delay is secondary to circulation time and instrumental delay (the oximeter averages over several cycles before producing a reading). Various measures have been applied to assess the severity of desaturation, including computing the number of desaturations, the average minimum $SaO_2$ of desaturations, the time below 80%, 85%, and 90%, as well as the mean $SaO_2$ and the minimum saturation during NREM and REM sleep. Oximeters may vary considerably in the number of desaturations they detect and their ability to discard movement artifact. Using long averaging times may dramatically impair the detection of desaturations.

## 10.10 Adult Respiratory Definitions

In adults, apnea is defined as absence of airflow at the mouth for 10 seconds or longer [21, 22]. If one measures airflow with a very sensitive device, such as a

pneumotachograph, small expiratory puffs can sometimes be detected during an apparent apnea. In this case, there is "inspiratory apnea." Many sleep centers regard a severe decrease in airflow (to <10% of baseline) to be an apnea.

An obstructive apnea is cessation of airflow with persistent inspiratory effort. The cause of apnea is an obstruction in the upper airway. A mixed apnea is defined as an apnea with an initial central portion followed by an obstructive portion. A hypopnea is a reduction in airflow for 10 seconds or longer [21]. The apnea + hypopnea index (AHI) is the total number of apneas and hypopneas per hour of sleep. In adults, an AHI of <5 is considered normal.

Hypopneas can be further classified as obstructive, central, or mixed. If the upper airway narrows significantly, airflow can fall (obstructive hypopnea). Alternatively, airflow can fall from a decrease in respiratory effort (central hypopnea). Finally, a combination is possible (mixed hypopnea), with both a decrease in respiratory effort and an increase in upper airway resistance. However, unless accurate measures of airflow and esophageal or supraglottic pressure are obtained, such differentiation is usually not possible. In clinical practice, one usually identifies an obstructive hypopnea by the presence of airflow vibration (snoring), chest–abdominal paradox (increased load), or evidence of airflow flattening (airflow limitation) in the nasal pressure signal. A central hypopnea is associated with an absence of snoring, a round airflow profile (nasal pressure), and absence of chest–abdominal paradox. However, in the absence of esophageal pressure monitoring, a central hypopnea cannot always be classified with certainty. In addition, obstructive hypopnea may not always be associated with chest–abdominal paradox. Because of the limitations in exactly determining the type of hypopnea, most sleep centers usually report only the total number and frequency of hypopneas.

The exact requirements for an event to be classified as a hypopnea are a source of controversy [30, 31]. A task force of the AASM recommended that if an accurate measure of airflow is used, a 50% reduction in airflow for 10 seconds or longer would qualify as a hypopnea [28]. Alternatively, any reduction in flow associated with an arousal or a 3% or greater drop in the $SaO_2$ (desaturation) would also meet the criteria. In contrast, the Clinical Practice Review Committee of the same organization defined a hypopnea as a 30% reduction in airflow of 10 seconds or longer, associated with a 4% or greater desaturation [32]. The presence or absence of arousal is not a factor in their definition. The rationale for this recommendation is that there is considerable variability in scoring arousals, and studies using an associated 4% drop in the $SaO_2$ to define hypopnea have shown an association between an increased AHI and cardiovascular risk.

The new requirements for an event to be classified as a hypopnea are as follows. A hypopnea should be scored only if all of the following criteria are present:

- The nasal pressure signal excursions (or those of the alternative hypopnea sensor) drop by >30% of baseline.
- The event duration is at least 10 seconds.
- There is a >4% desaturation from preevent baseline.
- At least 90% of the event's duration must meet the amplitude reduction of criteria for hypopnea.

Alternatively, a hypopnea can also be scored if all of these criteria are present:

- The nasal pressure signal excursions (or those of the alternative hypopnea sensor) drop by >50% of baseline.
- The duration of the event is at least 10 seconds.
- There is a >3% desaturation from preevent baseline or the event is associated with arousal.
- At least 90% of the event's duration must meet the amplitude reduction of criteria for hypopnea.

Respiratory events that do not meet criteria for either apnea or hypopnea can induce arousal from sleep. Such events have been called upper-airway resistance events (UARS), after the upper-airway resistance syndrome [11]. An AASM task force recommended that such events be called *respiratory effort-related arousals* (*RERAs*). The recommended criteria for a RERA is a respiratory event of 10 seconds or longer followed by an arousal that does not meet criteria for an apnea or hypopnea but is associated with a crescendo of inspiratory effort (esophageal monitoring) [28]. Typically, following arousal, there is a sudden drop in esophageal pressure deflections. The exact definition of hypopnea that one uses will often determine whether a given event is classified as a hypopnea or a RERA.

One can also detect flow-limitation arousals (FLA) using an accurate measure of airflow, such as nasal pressure. Such events are characterized by flow limitation (flattening) over several breaths followed by an arousal and sudden, but often temporary, restoration of a normal-round airflow profile. One study suggested that the number of FLA per hour corresponded closely to the RERA index identified by esophageal pressure monitoring [33]. Some centers compute a respiratory arousal index (RAI), determined as the arousals per hour associated with apnea, hypopnea, or RERA/FLA events [10]. The AHI and respiratory disturbance index (RDI) are often used as equivalent terms. However, in some sleep centers the RDI = AHI + RERA index, where the RERA index is the number of RERAs per hour of sleep and RERAs are arousals associated with respiratory events not meeting criteria for apnea or hypopnea.

One can use the AHI to grade the severity of sleep apnea. Standard levels include normal (<5), mild (5 to <15), moderate (15 to 30), and severe (>30) per hour. Many sleep centers also give separate AHI values for NREM and REM sleep and various body positions. Some patients have a much higher AHI during REM sleep or in the supine position (REM-related or postural sleep apnea). Because the AHI does not always express the severity of desaturation, one might also grade the severity of desaturation. For example, it is possible for the overall AHI to be mild but for the patient to have quite severe desaturation during REM sleep.

## 10.11 Pediatric Respiratory Definitions

Periodic breathing is defined as three or more respiratory pauses of at least 3 seconds in duration separated by less than 20 seconds of normal respiration. Periodic breath-

ing is seen primarily in premature infants and mainly during active sleep [34]. Although controversial, some feel that the presence of periodic breathing for >5% of TST or during quiet sleep in term infants is abnormal. Central apnea in infants is thought to be abnormal if the event is >20 seconds in duration or associated with arterial oxygen desaturation or significant bradycardia [34–37].

In children, a cessation of airflow of any duration (usually two or more respiratory cycles) is considered an apnea when the event is obstructive [34–37]. Of note, the respiratory rate in children (20 to 30 per minute) is greater than that in adults (12 to 15 per minute). In fact, 10 seconds in an adult is usually the time required for two to three respiratory cycles. Obstructive apnea is very uncommon in normal children. Therefore, an obstructive AHI >1 is considered abnormal. In children with obstructive sleep apnea, the predominant event during NREM sleep is obstructive hypoventilation rather than a discrete apnea or hypopnea. Obstructive hypoventilation is characterized by a long period of upper-airway narrowing with a stable reduction in airflow and an increase in the end-tidal $PCO_2$. There is usually a mild decrease in the arterial oxygen desaturation. The ribcage is not completely calcified in infants and young children. Therefore, some paradoxical breathing is not necessarily abnormal. However, worsening paradox during an event would still suggest a partial airway obstruction. Nasal pressure monitoring is being used more frequently in children and periods of hypoventilation are more easily detected (reduced airflow with a flattened profile). Normative values have been published for the end-tidal $PCO_2$. One paper suggested that a peak end-tidal $PCO_2$ > 53 mm Hg or end-tidal $PCO_2$ > 45 mm Hg for more than 60% of TST should be considered abnormal [35].

Central apnea in infants was discussed above. The significance of central apnea in older children is less certain. Most do not consider central apneas following sighs (big breaths) to be abnormal. Some central apnea is probably normal in children, especially during REM sleep. In one study, up to 30% of normal children had some central apnea. Central apneas, when longer than 20 seconds, or those of any length associated with $SaO_2$ below 90%, are often considered abnormal, although a few such events have been noted in normal children [38]. Therefore, most would recommend observation alone unless the events are frequent.

## 10.12 Leg Movement Monitoring

The EMG of the anterior tibial muscle (anterior lateral aspect of the calf) of both legs is monitored to detect leg movements (LMs) [39]. Two electrodes are placed on the belly of the upper portion of the muscle of each leg about 2 to 4 cm apart. An electrode loop is taped in place to provide strain relief. Usually each leg is displayed on a separate channel. However, if the number of recording channels is limited, one can link an electrode on each leg and display both leg EMGs on a single tracing. Recording from both legs is required to accurately assess the number of movements. During biocalibration, the patient is asked to dorsiflex and plantarflex the great toe of the right and then the left leg to determine the adequacy of the electrodes and amplifier settings. The amplitude should be 1 cm (paper recording) or at least one-half of the channel width on digital recording.

An LM is defined as an increase in the EMG signal of a least one-fourth the amplitude exhibited during biocalibration that is 0.5 to 5 seconds in duration [39]. Periodic LMs (PLMs) should be differentiated from bursts of spike-like phasic activity that occur during REM sleep. To be considered a PLM, the movement must occur in a group of four or more movements, each separated by more than 5 and less than 90 seconds (measured onset to onset). To be scored as a PLM in sleep, an LM must be preceded by at least 10 seconds of sleep. In most sleep centers, LMs associated with termination of respiratory events are not counted as PLMs. Some may score and tabulate this type of LM separately. The PLM index is the number of PLMs divided by the hours of sleep (TST in hours). Rough guidelines for the PLM index are as follows: >5 to <25 per hour, mild; 25 to <50, moderate; and =50, severe [40]. A PLM arousal is an arousal that occurs simultaneously with or following (within 1 to 2 seconds) a PLM. The PLM arousal index is the number of PLM arousals per hour of sleep. A PLM arousal index of >25 per hour is considered severe. LMs that occur during wake or after an arousal are either not counted or tabulated separately. For example, the PLMW (PLMwake) index is the number of PLMs per hour of wake. Of note, frequent LMs during wake, especially at sleep onset, may suggest the presence of the restless legs syndrome. The latter is a clinical diagnosis made on the basis of patient symptoms.

## 10.13 Polysomnography, Biocalibrations, and Technical Issues

A summary of the signals monitored in polysomnography is listed in Table 10.7. In addition, body position (using low-light video monitoring) and treatment level (continuous positive airway pressure, bilevel pressure) are usually added in comments by the technologists. In most centers, a video recording is also made on traditional videotape or digitally as part of the digital recording. It is standard practice to perform amplifier calibrations at the start of recording. In traditional paper recording, a calibration voltage signal (square wave voltage) was applied and the resulting pen deflections, along with the sensitivity, polarity, and filter settings on each channel, were documented on the paper. Similarly, in digital recording, a voltage is applied, although it is often a sine-wave voltage. The impedance of the head electrodes is also

**Table 10.7** Polysomnography—Respiratory Variables

| *Variables* | *Purpose* | *Methods* |
|---|---|---|
| Airflow | Classify apneas and hypopneas | Nasal–oral thermistor<br>Nasal pressure<br>RIPsum (changes approximate tidal volume)<br>Exhaled $CO_2$ |
| Respiratory effort | Classify apneas and hypopneas | Chest and abdominal bands (RIP, piezo bands)<br>Intercostal EMG<br>Esophageal pressure |
| Pulse oximetry | Arterial oxygen saturation | Pulse oximetry |
| End-tidal $PCO_2$ | Estimate of arterial $PCO_2$ (detect hypoventilation) | Capnography—exhaled $CO_2$ |
| Transcutaneous $PCO_2$ | Estimate of arterial $PCO_2$ (detect hypoventilation) | Transcutaneous $PCO_2$ |

checked prior to recording. An ideal impedance is <5,000Ω. Electrodes with higher impedances should be changed.

A biocalibration procedure is performed (Table 10.8) while signals are acquired with the patient connected to the monitoring equipment [4, 5]. This procedure permits checking of amplifier settings and integrity of monitoring leads/transducers. It also provides a record of the patient's EEG and eye movements during wakefulness with eyes closed and open. A summary of typical commands and their utility is listed in Table 10.8.

## 10.14 Quantitative Polysomnography

As mentioned previously, the polysomnograph is traditionally used to conduct sleep studies. The polysomnograph records electrical activity that represents specific physiological characteristics during sleep. It can be made up of bioelectrical potentials, transduced signals, and signals that are derived from ancillary equipment [41].

The EEG, EOG, and EMG are some of the measurements taken in a polysomnograph. These measurements can be used for sleep state detection. Each of these devices monitors different physiological characteristics: The EEG measures brain activity, the EOG measures eye movement, and the EMG measures muscle activity. Each device gives insight into the sleep stage that the patient is currently in.

### 10.14.1 EEG

According to the Rechtschaffen and Kales [1] sleep-staging system, certain bandwidths appear or disappear within the EEG signal depending on the sleep state the patient is in. In addition, certain neurophysiological activity (sleep spindles and K complexes) can be used to distinguish the stage of sleep from the EEG.

The main points of the R&K classification [1] system were discussed previously. To summarize, during wakefulness, alpha activity exists, as well as low-voltage mixed-frequency activity. The alpha waves exist in the wake state and decrease when the patient enters the first stage of NREM sleep. Sleep spindles and K complexes are an indication of stage 2 NREM sleep. Then delta waves appear in stages 3

**Table 10.8** Biocalibration Procedure

| | |
|---|---|
| Eyes closed | EEG: alpha EEG activity<br>EOG: slow eye movements |
| Eyes open | EEG: attenuation of alpha rhythm<br>EOG: REMs, blinks |
| Look right, look left, look up, look down | Integrity of eye leads, polarity, amplitude<br>Eye movements should cause out-of-phase deflections |
| Grit teeth | Chin EMG |
| Breathe in, breathe out | Airflow, chest, abdomen movements adequate gain? Tracings in phase? (Polarity of inspiration is usually upward) |
| Deep breath in, hold breath | Apnea detection |
| Wiggle right toe, left toe | Leg EMG, amplitude reference to evaluate LMs |

and 4 of NREM sleep, and finally theta waves are an indication of REM sleep [42]. Examples of EEG signals in each of the stages have been adapted from Geyer et al. [43] (Figures 10.5 to 10.9). Figure 10.5 shows a 30-second segment of EEG when the patient is awake. Alpha rhythms become more accentuated after the patient closes her eyes, indicated by the dashed line. Figure 10.6 is an example of what an EEG signal may look like when the patient is in stage 1 of NREM sleep. The alpha activity that was present in the wake state eventually turns into theta waves. Stage 2 of NREM sleep is shown in Figure 10.7. Stage 2 is denoted by the K complexes (solid line) and sleep spindles (dotted lines). The final stages of NREM sleep, stages 3 and 4, are depicted in Figure 10.8 and are revealed by the delta activity. Finally, the sawtooth theta waves, shown by the solid line in Figure 10.9, are a clear indication that the patient is in REM sleep.

Although a visual inspection of the EEG signal can be an indicator of the frequencies that make up the signal, a clearer representation of the frequencies can be achieved by transforming the signal into the frequency domain by using the Fourier transform. The Fourier transform is a mathematical technique that can transform any time series into a spectrum of the frequencies that produce it. It is a generalization of the Fourier series that breaks up any time-varying signal into the frequency components of varying magnitude that make it up. The Fourier transform is defined in (10.1), where $f(t)$ is any complex function and $k$ is the $k$th harmonic frequency.

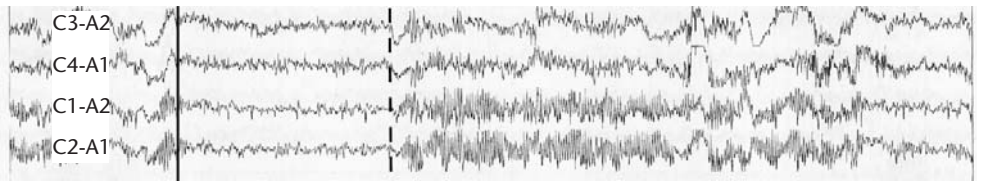



**Figure 10.5** A 30-second segment of EEG when the patient is awake. Alpha rhythms become more accentuated after the patient closes her eyes, indicated by the dashed line. (Courtesy of James Geyer and Paul Carney.)

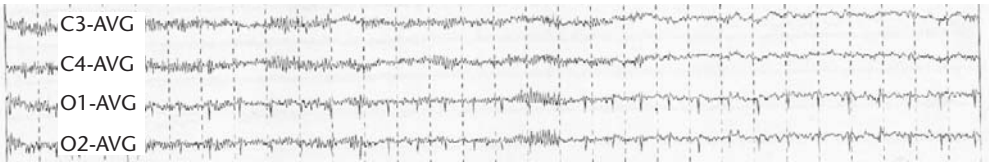



**Figure 10.6** An example of what an EEG signal may look like when the patient is in stage 1 of NREM sleep. The alpha activity that was present in the wake state eventually turns into theta waves. (Courtesy of James Geyer and Paul Carney.)

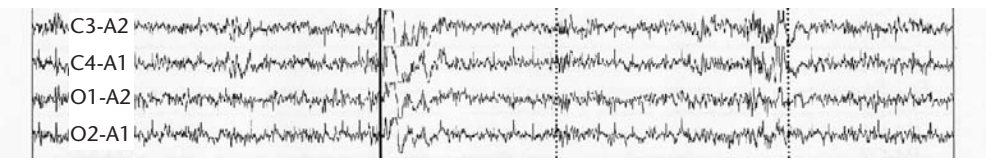



**Figure 10.7** Stage 2 of NREM sleep, denoted by the K complexes (solid line) and sleep spindles (dotted lines). (Courtesy of James Geyer and Paul Carney.)

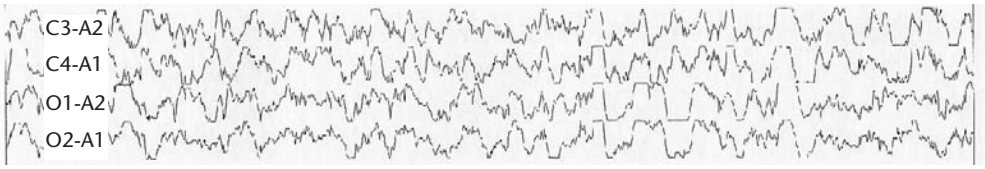



**Figure 10.8** Stages 3 and 4 of NREM sleep are depicted by delta activity. (Courtesy of James Geyer and Paul Carney.)




**Figure 10.9** Sawtooth theta waves, shown by the solid line, are a clear indication that the patient is in REM sleep. (Courtesy of James Geyer and Paul Carney.)

$$F(k) = \int_{-\infty}^{\infty} f(t)\cos(-2\pi kx)dx + \int_{-\infty}^{\infty} f(t)i\sin(-2\pi kx)dx \tag{10.1}$$

Due to Euler's formula, this can also be written as shown in (10.2):

$$F(k) = \int_{-\infty}^{\infty} f(t)e^{-2\pi ikx}dx \tag{10.2}$$

Any time-varying signal can be represented as a summation of sine and cosine waves of varying magnitude and frequencies [44]. The Fourier transform is represented with the power spectrum (or spectral density). The power spectrum has a value for each harmonic frequency that indicates how strong that frequency is in the given signal. The magnitude of this value is calculated by taking the modulus of the complex number that is calculated from the Fourier transform for a given frequency [$|F(k)|$].

After the EEG signal has been transformed to the frequency domain using the Fourier transform, the actual frequencies that create the original signal can be noted. The existence (or lack thereof) of certain bandwidths can then be used to determine the stage of sleep the patient was in at the time the EEG segment was recorded. Alpha waves (8 to 13 Hz) are an indication that the patient is in the wake state or stage 1 of NREM sleep. Delta waves (<4 Hz) are an indication of being in stage 3 or 4 of NREM sleep. Theta waves (4 to 7 Hz) are an indication of REM sleep.

These characteristics are shown in Figures 10.10 through 10.13. Figure 10.10 shows a normalized power spectrum for a segment of an EEG channel monitoring stage 1 of NREM sleep. Sleep spindles cause a relative increase in the 12- to 14-Hz range when the patient is in stage 2 of NREM sleep (Figure 10.11). Then an increase in the delta frequency band, specifically those bands less than 2 Hz, can be seen in Figure 10.12, when the patient is in stages 3 and 4 of NREM sleep. Finally, after the

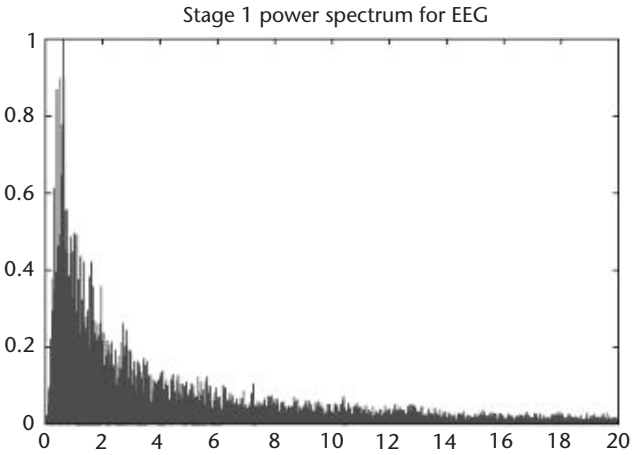



**Figure 10.10** A normalized power spectrum for a segment of an EEG channel monitoring stage 1 of NREM sleep.

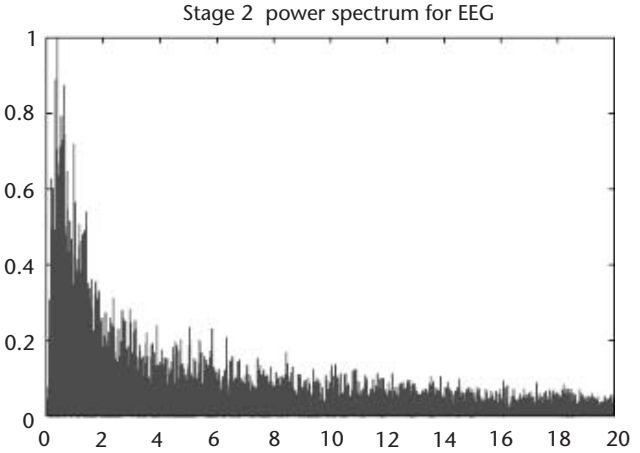



**Figure 10.11** The power spectrum for stage 2 of NREM sleep shows a relative increase in the 12- to 14-Hz range because of sleep spindles.

patient enters REM sleep, the sawtooth waves are denoted by the increase in the theta frequency band (4 to 7 Hz) (Figure 10.13).

### 10.14.2 EOG

The EOG measures the potential difference between the front and back of the ocular globe and is able to detect movements of the eyes [45]. The eye movement is an indication of whether the patient is in the REM sleep stage or NREM sleep. SEMs appear in stage 1 of NREM sleep and are usually gone by stage 2 of NREM sleep. NREM stages 3 and 4 do not contain any eye movement. REMs, which appear as much sharper impulses, show up in wakefulness and REM sleep. Figures 10.14 and 10.15 have been adapted from Geyer et al. [43]. Figure 10.14 shows an example of slow eye movements in stage 1 of NREM sleep, whereas Figure 10.15 shows exam-

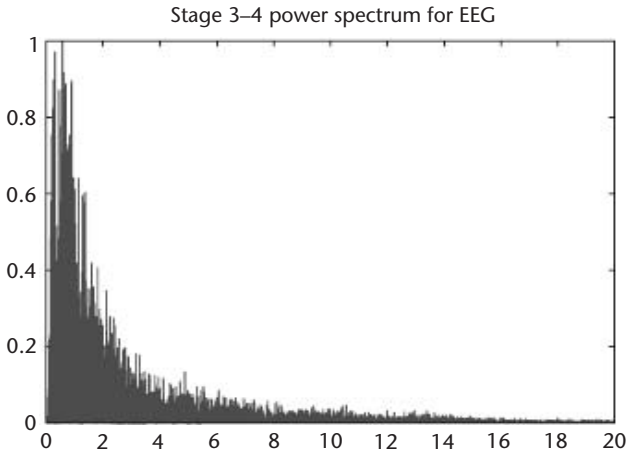



**Figure 10.12** An increase in the delta frequency band, specifically those bands less than 2 Hz, can be seen when the patient is in stages 3 and 4 of NREM sleep.

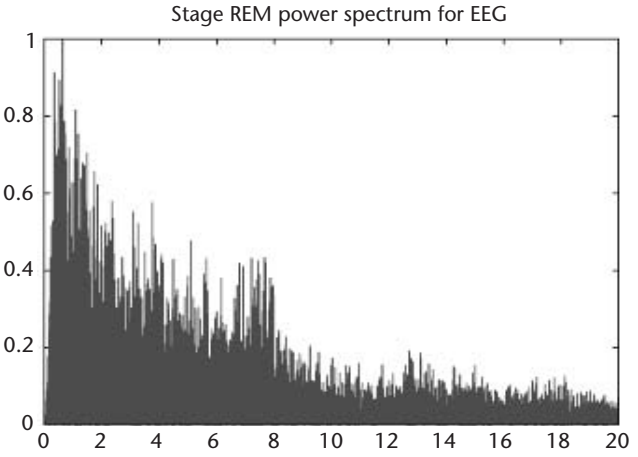



**Figure 10.13** After the patient enters REM sleep, the sawtooth waves are denoted by the increase in the theta frequency band (4 to 7 Hz).

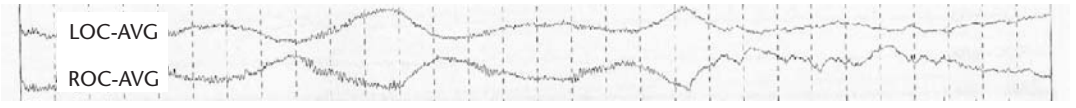



**Figure 10.14** Slow eye movements from the EOG recording stage 1 of NREM sleep. (Courtesy of James Geyer and Paul Carney.)

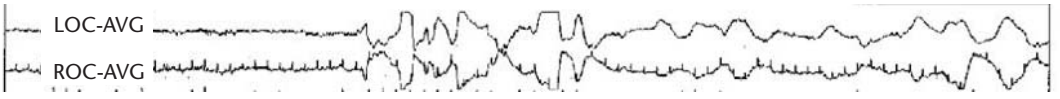



**Figure 10.15** Rapid eye movements from the EOG taken during REM sleep. (Courtesy of James Geyer and Paul Carney.)

ples of rapid movements during REM sleep. The frequency of the eye movements can also be shown by transforming the EOG channel into the frequency domain using the Fourier transform. Figure 10.16 shows the resulting power spectrum from an epoch of EOG recording stage 1 of NREM sleep. The presence of SEMs is indicated by the high spectral values in the lower-frequency ranges. Figure 10.17, which is the resulting power spectrum from an epoch of EOG recording REM sleep, shows a decrease in the amount of SEMs; however, no REM activity can be seen in the resulting spectrum. Epochs of REM sleep exist that do not contain REMs. These are referred to as *tonic REM* sleep.

### 10.14.3 EMG

The EMG measures the potential difference of electrodes placed on the chin, and indicates the chin's muscle tone [45]. There are high levels of activity when the

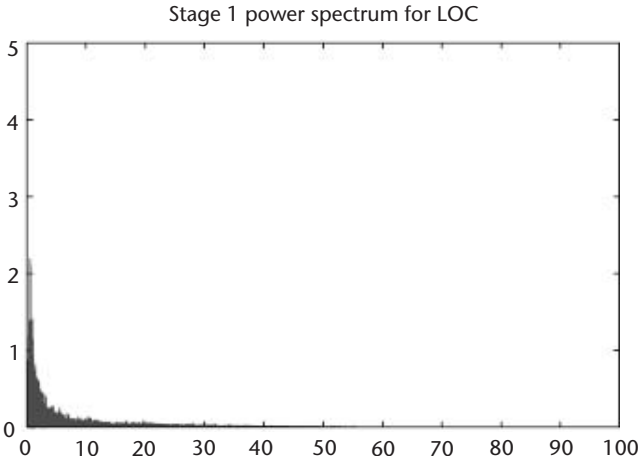



**Figure 10.16** The power spectrum from an epoch of EOG showing stage 1 of NREM sleep. An increase in the 0 to 2-Hz range is an indicator of slow oscillations.

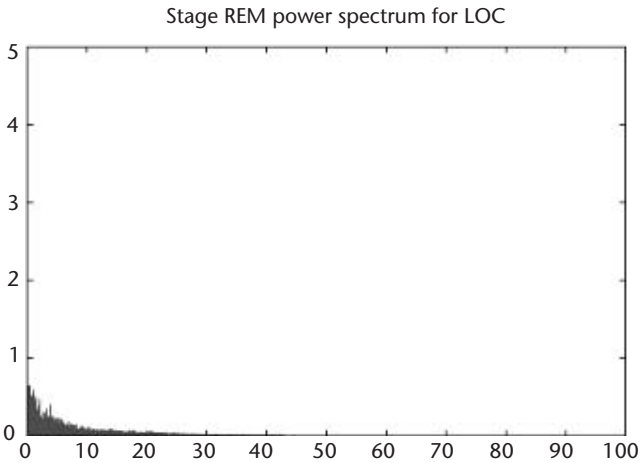



**Figure 10.17** The resulting power spectrum from an epoch of EOG taken during REM sleep shows a decrease in the amount of SEMs (0 to 2-Hz range).

patient is awake. However, this activity decreases when the patient is in NREM sleep, and the activity almost disappears after the patient enters the REM sleep stage. Figures 10.18 and 10.19, adapted from Geyer et al. [43], show 30-second epochs of an EMG channel when the patient is in stages 3/4 of NREM sleep and in REM sleep, respectively. A decrease in the amount of activity in the EMG can be seen as the patient progresses to REM sleep.

A sliding window variance analysis could be used to show the decrease in EMG activity. A sliding window variance analysis over time is shown in Figure 10.20 for stages 3 and 4 of NREM sleep. The variance is much larger in NREM stages relative to a sliding window variance analysis of the EMG channel in REM sleep, as shown in Figure 10.21.

The variance for a given set of points is defined as in (10.3), where $\mu$ is the mean of the sampled points. At a given time $t$, a sliding window variance analysis calculates the variance for all points sampled within the last $T$ seconds, where $T$ is some preset constant of time:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2 \tag{10.3}$$

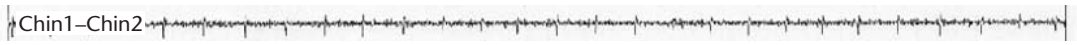



**Figure 10.18** 30-second epoch of an EMG channel when the patient is in stages 3/4 of NREM sleep. (Courtesy of James Geyer and Paul Carney.)




**Figure 10.19** 30-second epoch of an EMG channel when the patient is in REM sleep. (Courtesy of James Geyer and Paul Carney.)

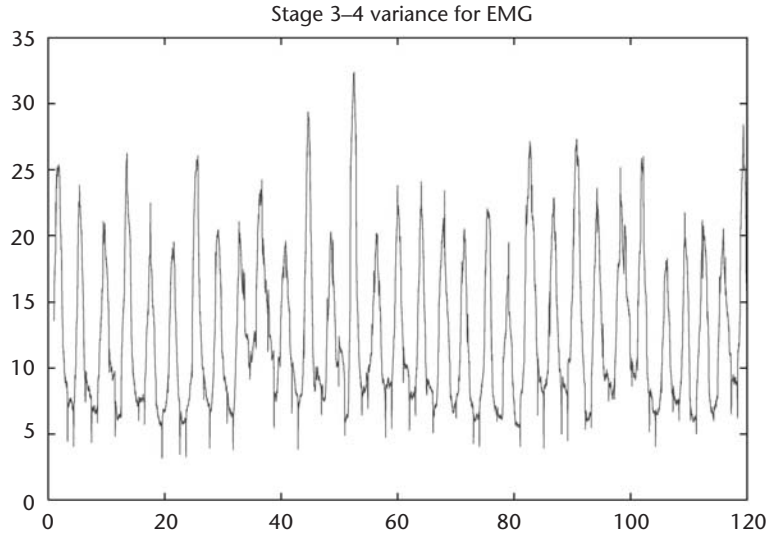



**Figure 10.20** A sliding window variance analysis over time is shown for stages 3 and 4 of NREM sleep.

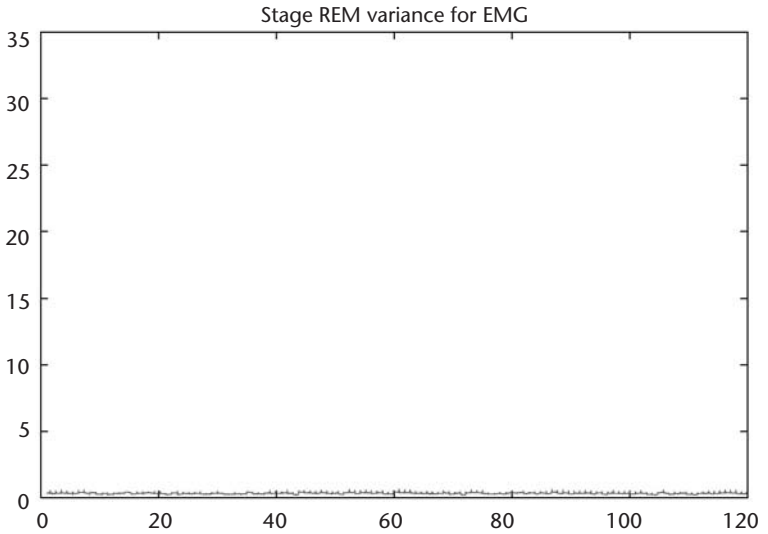



**Figure 10.21** A sliding window variance analysis over time is shown for REM sleep.

The polysomnograph has been used for sleep stage detection since R&K published the standardized manual [1]. It contains different physiological measurements, such as the EEG, the EMG, and the EOG, that can be used to facilitate the classification process. The actual classification is done by identifying certain characteristics of the signal itself, such as frequency bands in the EEG, eye movement frequency in the EOG, and the amount of muscle movement in the EMG.

## 10.15 Advanced EEG Monitoring

The process of manually categorizing a segment of a polysomnograph into its respective sleep state classification is long and tedious. Any technique that could automatically detect sleep states would be very beneficial in clinical practice and research.

Various techniques have been presented to either automatically detect sleep states or at the very least to assist in the classification process. When a technician manually classifies a polysomnograph into its respective sleep stages, the EMG and EOG are mainly used to facilitate the classification of sleep into REM or NREM sleep. However, the EEG depicts distinct characteristics of each stage of sleep, and therefore classification could be done based on the EEG alone. The majority of techniques use only the EEG signal. The stages of the R&K sleep state classification system can usually be distinguished by considering the bandwidths of the waves that form the EEG signal or by looking for specific waveforms within the signal, for example, K complexes and sleep spindles. Because of this, the majority of sleep stage detection methods involve a time-frequency analysis of the EEG signal or, in certain cases, some other signal of the polysomnograph. Nonlinear analysis of the EEG for sleep stage detection has also been used [46, 47]; however, Shen et al. have shown that there are weak nonlinear signatures in the sleep stages of the EEG, which is an argument for using linear methods [48]. Techniques that are used

include spectral analysis [49], wavelet analysis [50], and matching pursuit [51]. The following sections discuss some of the more recent time-frequency analysis techniques used to automatically detect the sleep state from EEG using the aforementioned approaches.

### 10.15.1 Wavelet Analysis

Wavelet analysis is a generalization of the short-term Fourier transform that allows for basis functions that are more general than a sine or cosine wave. Rather than considering certain frequency bands present in a given stage of sleep, these can be used to determine the existence of certain physiological wave forms (K complexes or sleep spindles) that are introduced in certain stages of sleep [50, 52]. Akin and Akgul attempt to detect sleep spindles by using a discrete wavelet transform [52].

The discrete wavelet transform is similar to the Fourier transform in that it will break up any time-varying signal into smaller uniform functions, known as the *basis* functions. The basis functions are created by scaling and translating a single function of a certain form. This function is known as the Mother wavelet. In the case of the Fourier transform, the basis functions used are sine and cosine waves of varying frequency and magnitude. Since a cosine wave is just a sine wave translated by $\pi/2$ radians; the mother wavelet in the case of the Fourier transform could be considered to be the sine wave.

However, for a wavelet transform the basis functions are more general. The only requirements for a family of functions to be a basis are that the functions are both complete and orthonormal under the inner product. Consider the family of functions $\Psi = \{\Psi_{ij} | -\infty < i,j < \infty\}$, where each $i$ value specifies a different scale and each $j$ value specifies a different translation based on some mother wavelet function. $\Psi$ is considered to be complete if any continuous function $f$, defined over the real line $x$, can be defined by some combination of the functions in $\Psi$ as shown in (10.4) [44]:

$$f(x) = \sum_{i,j=-\infty}^{\infty} c_{ij} \Psi_{ij}(x) \tag{10.4}$$

For a family of functions to be orthonormal under the inner product, they must meet two criteria: It must be the case that for any $i$, $j$, $l$, and $m$ where $i \neq l$ and $j \neq m$ that $\langle \psi_{ij}, \psi_{lm} \rangle = 0$ and $\langle \psi_{ij}, \psi_{ij} \rangle = 1$, where $\langle f, g \rangle$ is the inner product, defined as in (10.5), and $f(x)^*$ is the complex conjugate of $f(x)$:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x)^* g(x) dx \tag{10.5}$$

The wavelet basis is very similar to the Fourier basis, with the exception that the wavelet basis does not have to be infinite. In a wavelet transform the basis functions can be defined over a certain window and then be zero everywhere else. As long as the family of functions defined by scaling and translating the mother wavelet is orthonormally complete, that family of functions can be used as the basis. With the Fourier transform, the basis is made up of sine and cosine waves that are defined over all values of $x$ where $-\infty < x < \infty$.

Akin and Akgul attempt to detect sleep spindles by using the Daubechie mother wavelet to create the family of functions for the basis [52]. The discrete wavelet transform would easily detect sleep spindles if the mother wavelet had the same form as a sleep spindle. Hence, the Daubechie wavelet was chosen to best approximate the form of a sleep spindle so that when the mother wavelet is scaled and translated, it is possible to detect sleep spindles of different sizes occurring at different times. This technique, however, works only for sleep spindle detection. Therefore, it can be used to identify only stage 2 of NREM sleep.

### 10.15.2 Matching Pursuit

Matching pursuit provides a solution to the adaptive approximation problem. It was first suggested by Mallat and Zhang [53] as a signal processing tool. It is similar in concept to the Fourier transform or the wavelet transform in that it represents some signal $x$ by using a linear summation of functions from some group of functions, termed a *dictionary*.

The matching pursuit algorithm attempts to find a solution to the linear expansion problem:

$$x = \sum_{n=1}^{N} a_n g_n \tag{10.6}$$

Here, $g_n$ belongs to some family of functions known as a dictionary, $D$. The matching pursuit algorithm attempts to find the $g_n$ that best approximate the original function $x$. When the dictionary $D$ is an orthonormal basis, the matching pursuit algorithm yields the same results as the wavelet transform.

The matching pursuit algorithm has been applied to the problem of sleep stage detection in various ways [51, 54]. In these studies, the Gabor functions were the family of functions used as the dictionary. Gabor functions are a mixture of a sinusoidal and a Gaussian and have the form shown in (10.7). In addition to a traditional time-frequency analysis, matching pursuit is better equipped than wavelets to identify transients (waveforms such as K complexes or sleep spindles) [55]. Because there are fewer restrictions on the dictionary used in matching pursuit than there are on an orthonormal wavelet basis, a Gabor function closely resembling these waveforms can easily be chosen from the dictionary.

$$g_\lambda(t) = K(\lambda) e^{-\pi\left(\frac{t-u}{s}\right)^2} \cos\left(\omega(t-u) + \phi\right) \tag{10.7}$$

## 10.16 Statistics of Sleep State Detection Schemes

With so many sleep state detection methods available, there needs to be a way to compare them so that the "best" method can be used. However, a challenge facing such a metric is that sleep state detection is a multicategory classification problem, as opposed to a binary classification problem. In sleep state detection there are five possible classifications for a feature point that is extracted from a given epoch of the polysomnograph (or EEG). It can be a feature from either the wake state, NREM

stage 1, NREM stage 2, NREM stages 3/4, or REM sleep. Because of this, traditional binary classification metrics, such as the sensitivity and specificity, cannot be applied directly to the multicategory classification. Some statistical models such as analysis of variance and multivariate analysis of variance have been used [47]. Others use clustering as a technique to classify the data points, but no statistical evaluation of how well the clustering algorithm classified the points is given [49].

Although there may be problems relating to scalability, it is possible to break an $M$-category classification problem into $M$ separate binary classification problems. This as well as the use of a contingency table will now be discussed.

### 10.16.1 *M* Binary Classification Problems

Although it is not a binary classification problem, an $M$-category classification problem can be broken up into $M$ different binary classification problems. To do this, for each class of points $X_i$, where $1 \leq i \leq M$, two new classes $P_i$ and $N_i$ are created, such that $P_i$ is made up of all the points in $X_i$ and $N_i$ is made up of all of the points not in $X_i$. With these new classes, the original $M$-category classification problem has been partitioned into $M$ separate binary classification problems, where $P_i$ and $N_i$ for $1 \leq i \leq M$ are the two possible classes for each problem.

With binary classification problems many statistics exist that can be used for performance evaluation. The classification an algorithm gives to a particular segment of EEG or polysomnograph can be placed into one of four categories:

- *True positive* ($TP_i$). A technique correctly classifies a data point from class $P_i$ as being in class $P_i$.
- *True negative* ($TN_i$). A technique correctly classifies a data point from class $N_i$ as being in class $N_i$.
- *False positive* ($FP_i$). A technique incorrectly classifies a data point from class $N_i$ as being in class $P_i$.
- *False negative* ($FN_i$). A technique incorrectly classifies a data point from class $P_i$ as being in class $N_i$.

From these four values ($TP_i$, $TN_i$, $FP_i$, and $FN_i$), two statistics that give a large amount of information on the success of a given technique can be calculated. The first statistic is the sensitivity ($S_i$), which is defined in (10.8). The sensitivity indicates the probability of detecting a point from class $X_i$ and is defined by the ratio of the number of points in $X_i$ that are detected divided by the total number of points in $X_i$:

$$S_i = \frac{TP_i}{TP_i + FN_i} \tag{10.8}$$

In addition to the sensitivity, the specificity ($K_i$) can also be used as defined in (10.9). This indicates the probability of not incorrectly classifying a point that is not in $X_i$ as being in $X_i$. It is defined by the ratio of the number of points classified as not being in $X_i$ divided by the total number of points that are not in $X_i$. Both the specificity and sensitivity are used by Tsuji [56] when working with the binary classification

problem of REM detection. In this case, the classifier needs to decide only whether the segment contains REMs or not:

$$K_i = \frac{TN}{TN_i + FP_i} \tag{10.9}$$

It should be noted that either the specificity or the sensitivity alone is not a sufficient measure of goodness. Consider a classification algorithm that always classified the point as being in $X_i$. Such a method would produce a sensitivity of 1 and a specificity of 0. Similarly, an algorithm that always classified the point as not being in $X_i$ would produce a specificity of 1 and a sensitivity of 0. The ideal algorithm would produce a value of 1 for each, so some combination of the two should be used when creating a performance metric.

When applied to sleep detection there are five different categories. A data point can be from either the wake state, NREM stage 1, NREM stage 2, NREM stages 3/4, or REM sleep. The above method would then yield five different specificity and sensitivity values. A performance metric could be created by using any one of the 10 values (such as the minimum or maximum) or some combination of the 10 values (such as the average).

### 10.16.2 Contingency Table

Another method used to evaluate the performance of a multicategory classification technique is a contingency table. This is used specifically for sleep detection by Estévez [45] and Sinha [57]. A contingency table indicates the relationship between two or more variables. In this case the variables are the classifications given to the data points by the technique and the classifications given to the same data points by an expert (the polysomnographer). A contingency table would be similar to the one shown in Table 10.9. Each cell of the table represents the number of points that the system and the expert classified under the respective row and column. The cell value at the intersection of the NREM-I column and the WAKE row indicates that there were 11 points that the expert classified as being from NREM stage 1 sleep and the system classified as being from the wake state. In a contingency table the cells along the main diagonal indicate correct classifications, and all other cells are incorrect classifications.

In addition to the contingency table, Estévez also extracts further information from the contingency table such as the kappa index. The kappa index compares the

**Table 10.9** Example of What a Contingency Table Might Look Like

| | *Expert's Classifications* | | | | | |
|---|---|---|---|---|---|---|
| *System's Classifications* | *WAKE* | *NREM-I* | *NREM-II* | *NREM-III & IV* | *REM* | *Total* |
| WAKE | 543 | 11 | 18 | 6 | 3 | 581 |
| NREM-I | 23 | 456 | 6 | 9 | 22 | 516 |
| NREM-II | 21 | 21 | 348 | 52 | 21 | 463 |
| NREM-III and IV | 43 | 12 | 2 | 632 | 43 | 732 |
| REM | 12 | 7 | 12 | 23 | 312 | 366 |
| Total | 642 | 507 | 386 | 722 | 401 | 2,658 |

observed agreement among the system and the expert to the chance agreement. The chance agreement measures the amount of agreement to be expected by chance alone [45].

Although evaluation of the performance of a sleep detection algorithm is more complicated than evaluating the performance of a binary classification problem, many metrics do exist that could be used. Unfortunately, no standard evaluation has been used across techniques, so it is difficult to say which technique works the best. It would be very beneficial to the field of automated sleep detection to apply a standard metric, such as a contingency table or the kappa index, to the different techniques for comparison purposes.

## 10.17 Positive Airway Pressure Treatment for Obstructive Sleep Apnea

The primary treatment for obstructive sleep apnea is positive airway pressure (PAP). This treatment is available in a number of forms, including continuous PAP (CPAP), bilevel PAP (BPAP), autotitrating PAP (APAP), and flex settings with patient-selected pressure releases. All of these systems work under the same principle—pneumatic splinting of the retropharyngeal space and upper airway.

Airway resistance (AR) is proportional to the inverse of the radius $r$ of the airway to the fourth power:

$$AR \propto 1/r^4$$

Therefore, even small improvements in the airway radius result in very significant improvements in airway resistance. PAP increases the radius by the above described pneumatic splinting.

Most PAP devices run at a given setting with little or no variation. Autotitrating systems attempt to match the apparent need for pressure to a given setting on the machine. This theory is very appealing, but the machines do not fully achieve this goal.

### 10.17.1 APAP with Forced Oscillations

The use of the forced oscillation technique (FOT) measures airway impedance and was initially applied to the measurement of obstruction in the lower airway [58]. The technique can also be applied to obstruction involving the upper airway, for example, obstructive sleep apnea. The impedance measured by FOT correlates with esophageal pressure recordings.

Forced oscillations of externally applied airflow are used to determine the mechanical response of the respiratory system. The system requires the use of low-amplitude oscillations to maintain linearity. It is possible that this limitation could mask important and as yet incompletely described nonlinear components.

### 10.17.2 Measurements for FOT

The impedance ($Z$) and the spectral relationship between pressure ($P$) and airflow ($V$) comprise the key variables in forced oscillation analysis. Depending on the sites

of the $P$ and $V$ measurements and of the application of the forced oscillations, various subtypes of respiratory system impedance can be defined. This allows the system to identify the type of apnea and, at least in theory, create the appropriate alteration in the pressure delivered to the patient.

In the most common monitoring system, the forced oscillations are applied at the airway opening, and the central airflow ($V_{ao}$) is measured with a pneumotachograph attached to the mask. Pressure is measured at the airway opening ($P_{ao}$) with reference to the local atmospheric pressure ($P_a$). The input respiratory impedance ($Z_{in}$) is the spectral relationship between transrespiratory pressure ($P_{rs} = P_{ao} - P_a$) and $V_{ao}$: $Z_{in}(f) = P_{rs}(f)/V_{ao}(f)$ [59].

Subsequently the impedance $Z$ can be divided into pulmonary ($Z_p$) and chest wall ($Z_w$) impedance based on the intraesophageal pressure ($P_{es}$) The derivation of $Z_p$ and $Z_w$ are as follows: $Z_p = (P_{ao} - P_{es})/V_{ao}$ and $Z_w = (P_{es} - P_a)/V_{ao}$ [59].

With this set of data, the APAP system has the data necessary to identify the need for pressure adjustment, at least from a linear dynamics perspective. Feedback systems involving monitoring technologies and PAP systems provide the best hope for truly effective PAP treatment systems. The present autotitrating systems account for only some linear respiratory variables. Nonlinear respiratory variables have not traditionally been analyzed, and nonrespiratory variables have been excluded from the analysis altogether.

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CHAPTER 11

# EEG Signals in Psychiatry: Biomarkers for Depression Management

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Monitoring brain function with qEEG is the focus of much research on psychiatric disorders. Because the fundamental, underlying neurobiological defects for major depression and other common psychiatric disorders are incompletely understood, most work continues to be descriptive, unlike the case for many neurological disorders where theory and research evidence both support the migration of qEEG methods from research to clinical spheres. Nonetheless, a number of qEEG approaches may be nearing readiness for clinical application in aiding treatment management decisions in psychiatry, for example, in major depression, and this chapter focuses on these methods. We also raise cautionary concerns about assessing the readiness of new technologies for clinical use and suggest criteria that may be used to aid in that assessment.

## 11.1 EEG in Psychiatry

### 11.1.1 Application of EEGs in Psychiatry: From Hans Berger to qEEG

Although brain electrical activity had been observed in animals as early as the 1870s by the British scientist Richard Caton, it was not until the 1920s that the first human EEG was recorded by the German psychiatrist and neuroanatomist Hans Berger, as part of his quest for understanding "mental energy." Electroencephalography was embraced relatively rapidly for the study of neurological illnesses, with compelling utility in several realms: the diagnosis of seizure disorders, the localization of brain tumors before the advent of tomographic neuroimaging, the detection of brain dysfunction in delirium and dementia, and the determination of "brain death," to name a few. Despite the initial discovery by a psychiatrist, several decades elapsed before the application of EEG methods to studying psychiatric illness gained acceptance, and its use still remains controversial in some quarters.

Although computerized analysis of EEG signals was reported as early as the 1960s [1], qEEG via digital computer analysis methods did not become widespread until the 1980s in conjunction with the declining costs of digital microcomputers.

The new possibilities for spectral analysis came within reach of an increasing number of clinicians and investigators, which led to cautions about the need for specific training and expertise prior to adoption of these techniques [2].

It is important to note that the approach to clinical electrophysiology in psychiatric illnesses continues to be largely correlative, primarily because the neurophysiology and pathophysiology of most psychiatric illnesses are not well characterized. In contrast to illnesses such as seizure disorders, in which the pathophysiological findings of a clearly circumscribed excitatory focus, an inhibitory surrounding area, and their respective firing patterns are well described, studies of patients with psychiatric disorders such as major depression have implicated abnormal activity in large regions of the brain—many parts of the limbic system, or dorsolateral prefrontal cortex, for example—and the circuits that link them. Although some features, such as sensory gating, have been studied neurophysiologically and clinically in conditions such as schizophrenia (as recently reviewed by Potter and colleagues [3]), this situation remains the exception rather than the rule for psychiatric disorders overall. If we are to be intellectually honest, we must acknowledge that the underlying pathophysiology of depression is incompletely understood. Because the fundamental defects are not clear, research still remains descriptive and the search for meaningful endophenotypes continues. Nonetheless, qEEG can make important contributions to elucidate and improve the management of depression by informing clinical decision making. This chapter emphasizes this particular application of qEEG measures.

### 11.1.2 Challenges to Acceptance: What Do the Signals Mean?

Although improvements in technology have made it straightforward for scalp EEG signals to be measured digitally and for spectral analysis to be performed, determining the underlying meaning of qEEG measures has been a persistent challenge at both physiological and clinical levels of interpretation. For example, what is the meaning of a finding of diminished alpha power over a specific brain region? Does it depend on clinical context? Brain regions characterized by high levels of alpha power are often described as "deactivated," but what does that mean, neurobiologically or functionally? Does rhythmic activity in the alpha range over the occipital cortex in the resting, eyes-closed state mean the same thing as alpha range activity over a frontal region during a cognitive activation paradigm?

As described in other chapters of this book, rhythmic surface EEG arises from the coordinated activity of large ensembles of neurons, and there are many different permutations of neuronal firing patterns that will give rise to similar surface EEG rhythms. The variation, for example, in alpha range power values explains a statistically significant but small proportion of the variance in regional cerebral metabolism or perfusion [4–12], suggesting that data from EEG signals may be more than serving a reflection of perfusion or metabolism. Furthermore, the correlations between spectral power and regional perfusion or metabolism can be influenced by other phenomena such as hypocapnia [13], structural brain injury from cerebrovascular disease [14], or degenerative disorders [15, 16].

### 11.1.3 Interpretive Frameworks to Relate qEEG to Other Neurobiological Measures

Cerebral glucose uptake and blood flow were hypothesized long ago to be reflections of the brain's energy utilization [17]. In healthy subjects, cerebral glucose uptake and blood flow generally are accepted as tightly coupled measures of cerebral energy utilization [18–20]. Indeed, the mainstays of functional neuroimaging methodologies—positron emission tomography (PET), single photon emission computed tomography (SPECT), and fMRI—have contributed much to our understanding of the physiology of the CNS by providing a window into regional metabolism or blood flow. Given that the brain's electrical activity represents the single greatest demand on cerebral metabolism [21], the measurement of electrical energy also should be coupled to cerebral metabolism and perfusion, an idea that traces back to Berger [22].

To overcome issues about inconsistencies in the methods and results encountered in previous studies, the UCLA Laboratory of Brain, Behavior, and Pharmacology examined the relationship between surface-recorded EEGs in different frequency bands and the perfusion of underlying brain tissue by performing simultaneous qEEG recordings during sessions measuring regional cerebral perfusion with $^{15}$O-PET in healthy adults, at rest and while performing a motor activation task. We established that the relationship between qEEG measures (i.e., absolute and relative power) and regional blood flow was influenced by the recording montage being used [8], with the best correlations being obtained through a "reattributional" montage. With this approach, qEEG power values for each electrode location were computed by taking power values from bipolar pairs of electrodes that share a common electrode and averaging them together to yield the reattributed power (Figure 11.1) [8]. For example, to determine a power value for the brain region underlying the F4 electrode, we first compute power spectra for the neighboring bipolar channels that include the F4 electrode (i.e., F4–F8, F4–AF2, F4–FC2, and F4–FC6) and then average the absolute power values from those channels to obtain the reattributed power for the F4 electrode. This is somewhat similar to the single source method of Hjorth [8, 23, 24], but this approach recombines the power values, whereas Hjorth's method recombines voltage signals by averaging signal amplitudes from pairs of electrodes. The reattributional montage provides a higher association between EEG measures and regional cortical perfusion than does the Hjorth method [8] and so offers an advantage if a researcher's scientific objective focuses on understanding findings within the general functional neuroimaging conceptual framework.

We developed the cordance method to incorporate this re-referencing approach and then to employ normalization and integration of absolute and relative power values from all electrode sites for a given EEG recording in three steps [10]. First, the reattributed absolute power values are calculated at each electrode site, and reattributed relative power is calculated in the conventional manner at each electrode site, as the percentage of reattributed power in each band, relative to the total spectrum considered (in that work, 0.5 to 20 Hz) [25].

Second, these absolute and relative power values for each individual EEG recording are normalized across electrode sites, using a $z$-transformation statistic to assign a value to each electrode site $s$ in each frequency band $f$ [yielding $A_{\text{norm}}(s,f)$

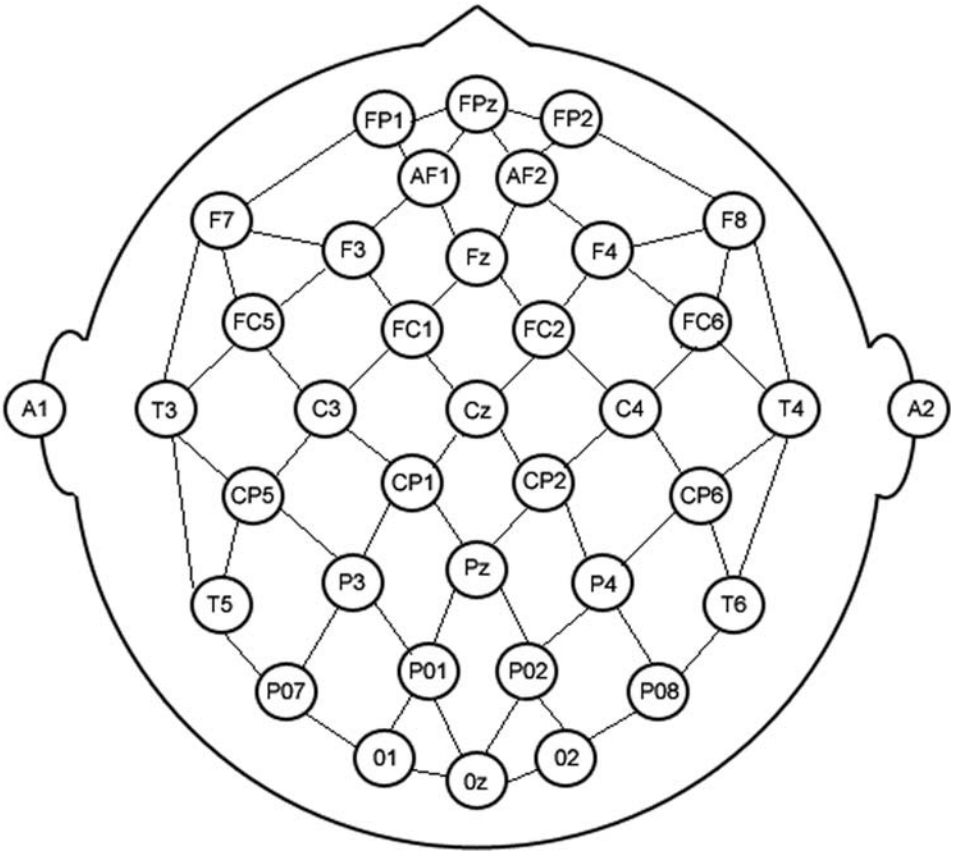



**Figure 11.1** Reattributional electrode montage.

and $R_{\text{norm}}(s,f)$, respectively]. Note that these $z$-scores are based on the average power in each band for all electrodes within a given qEEG recording, and are not $z$-scores referenced to some normative population (e.g., as in the Neurometrics approach [26]). These $z$-score–based values reflect whether a given site is above or below the average power value in each band. The spatial normalization process also places absolute and relative power values into a common unit (standard deviation or $z$-score units), which allows them to be combined.

Third, the cordance values are formed by summing the $z$-scores for normalized absolute and relative power

$$Z(s,f) = A_{\text{norm}}(s,f) + R_{\text{norm}}(s,f)$$

for each electrode site and in each frequency band. Cordance values have been shown to have higher correlations with regional cerebral blood flow than absolute or relative power alone [10]; thus, this combination measure can be placed in context with prior work in depression that employed functional measures of regional brain activity such as PET data.

Other approaches to interpreting qEEG data may offer alternative perspectives. The qEEG coherence measure [27, 28] has been interpreted as reflecting functional connectivity [29–31] in pathways linking parts of neural circuits [32–35], and has

utility in studying circuit function in aging and in dementia; this is discussed at greater length later in this chapter.

Another approach, the quantification of cerebral microstates through adaptive segmentation [36], may help in differentiating healthy brain aging from processes of cognitive decline. Although a reduction in the duration of microstate periods can be interpreted as evidence of a fragmentation of quasistationary EEG periodic activity and, thus, some sort of problem in sustaining coordinated activity among brain regions, it is not clear whether this may arise from disrupted corticocortical connections, from dysfunction of cortical neurons themselves, or from disturbances in neuromodulatory activity. Prichep [37] has reported a novel approach examining both Neurometrics [26] and source localization to differentiate elders with healthy aging from those with cognitive decline; this approach was interpreted in the context of levels of regional brain activity.

Yet other methods can be interpreted within their own particular context. Sleep polysomnography has been used to assess stage of sleep (e.g. the classic manual by Rechtschafen and Kales [38] and Chapter 10 in this book), and abnormalities in sleep architecture have long been reported in many but not all patients with depression [39–42]. Sleep deprivation has been reported to have a transient mood-restoring effect in some individuals with depression [43, 44]. The use of nonlinear methods [45] has also been explored to characterize the sleeping brain in depression, yet physiological monitoring of sleep abnormalities has not become a part of routine clinical care for depression. Additional work may help demonstrate how chronobiological perspectives can contribute to improved patient care for depression [46].

## 11.2 qEEG Measures as Clinical Biomarkers in Psychiatry

### 11.2.1 Biomarkers in Clinical Medicine

The use of *biomarkers* is commonplace in most branches of medicine: Specific biological features of an individual patient provide critical information about that person's diagnosis, prognosis, or predicted response to treatment. Examples include tumor markers in oncology [47–50], alpha-feto-protein in obstetrics [51], troponin and other serum factors in cardiology [52–54], and inflammatory markers and specific serum antibody levels in rheumatology [55]. Additionally, the use of biomarkers may be useful in drug discovery and development, by monitoring response to a test exposure of an experimental medication [56]. Nonetheless, in the field of psychiatry, the biological features of a patient's illness generally continue to be eclipsed by the central role played by clinical signs and symptoms [57].

Although a number of recent research reports suggest that biomarkers may soon be suitable for clinical use in the care of psychiatric patients, the quest for biomarkers to improve the care of mental illnesses is not new in the twenty-first century. For several decades, measurements of specific molecules in cerebrospinal fluid, such as homovanillic acid (HVA) and 5-hydroxy-indoleacetic acid (5-HIAA) [58]; metabolites of neurotransmitters in urine, such as 3-methoxy-4-hydroxyphenylglycol (MHPG) [59]; and serum markers of neuroendocrine dysregulation [60], that is, dexamethasone suppression test (DST) [61]; have been

complemented by studies of sleep architecture [62–64], eye movement abnormalities [65, 66], and electrodermal and other measures of autonomic nervous system activity [67]. Although these approaches have greatly expanded knowledge of the neurobiology of psychiatric disorders by serving as research tools, they have found no significant application to clinical practice or evidence-based practice guidelines because the measures are not sufficiently associated with diagnosis or prognosis to prove useful in clinical decision making [57, 68]. As biological measures ("biomeasures" [68]) and new techniques are reported and considered for use as clinically applicable biomarkers, it is important for researchers and clinicians both to understand how these may or may not be ready for "prime time" and adoption into widespread clinical use.

### 11.2.2 Potential for the Use of Biomarkers in the Clinical Care of Psychiatric Patients

Biomarkers have great potential for improving care for psychiatric patients. Three areas in particular can be identified: enhanced diagnostic accuracy, prognostic information about the natural course of an individual's illness, and prediction of response to treatment.

As described earlier, clinical signs and symptoms are the central basis for establishing psychiatric diagnoses [57]. Yet some symptoms may be present in multiple diagnoses: A reduction in the amount of sleep can be a diagnostic element of a depressive episode, a manic episode, or generalized anxiety disorder. Biomarkers have promise for enhancing diagnostic accuracy in this arena. Consider, for example, a 21-year-old patient with a 3-month bout of depression that has interfered with college classwork and social relationships: Is this depression a component of unipolar major depressive disorder (MDD), or does the person really suffer from bipolar disorder (formerly called manic-depressive illness), but has not yet experienced a clear manic episode, because the patient is early in the course of illness? In an older patient with mild but measurable cognitive impairments, do these problems originate from the neurodegenerative changes of Alzheimer's disease (albeit mild in severity at this point), from ischemic damage to white and gray matter structures as is seen in vascular dementia, or from major depression (previously termed the "pseudodementia" of depression)? In a child, are inattention and disruptive behaviors manifesting the symptoms of attention deficit hyperactivity disorder (ADHD), the early onset of bipolar disorder, or are they simply reflective of coping skills that are overwhelmed by stressful circumstances (e.g., parental divorce)? For most patients, clinical information is sufficient to converge on the salient psychiatric diagnosis rapidly, but for many individuals, diagnostic ambiguity may challenge even expert clinicians. The use of biological markers has potential to assist in this important process, but more work is needed before the field will have useful tools for this application.

Prognostic information is another area where biomarkers could offer valuable insights. In oncology, the elevation of a tumor marker prompts an evaluation for a recurrence of disease and initiation of treatment, even before clinical manifestations would have led to a reevaluation. In contrast, for psychiatric disorders, an impending full relapse of a disorder (e.g., schizophrenia) is heralded principally by the

return of the symptoms themselves (e.g., psychotic symptoms such as hallucinations or delusion). As a patient of one of the authors (IAC) liked to frame the question for her recurrent depressions, "When is a bad day just a 'bad day,' and when is it the start of a new episode?" In the care of older adults with depression, some will likely progress from late-life depression to dementia [68], but identification of this subset of patients remains problematic.

Lastly, many patients with mood disorders experience recurrent thoughts of death and may perceive life as painful and/or meaningless while in the midst of a depressive episode. Although this group of patients has an elevated risk for suicidal behaviors, accurately determining which individuals will go on to harm themselves and which will not cannot be forecast reliably on clinical or historical grounds [69]; some preliminary work suggests measures of brain structure and function [70, 71] or genotyping [72, 73] may be developed to refine this process. Rather than believing that research efforts will eventually identify *the* single, measurable factor that leads to a phenomenon as complex as suicide, it may be more reasonable to anticipate that the greatest clinical utility for this sort of prediction may emerge from a model combining genetic and neurobiological features with current and past clinical features and familial history, though the relative weightings of these factors remains indeterminate at this time.

Prediction of individual treatment response is viewed by many as a critical area for improvement in psychiatry. Whereas treatments are effective for managing psychiatric illnesses in general, no single treatment works for everyone with a given disorder, and selection of the best treatment for each patient remains a challenge. The general standard of care is to embark on a course of treatment that is likely to be effective for that disorder, based on evidence from randomized clinical trials and other data relevant to the individual patient (e.g., clinical experience, past patient response to treatment); one then monitors for a good outcome and allows for treatment adjustment if improvement fails to occur. Both steps fundamentally rely on clinical findings to assess the degree of symptomatic or functional response. Nobel laureate Niels Bohr often is said to have observed that "Prediction is difficult, especially about the future," and this statement rings true in this aspect of psychiatric care. The failure of depressive symptoms to improve early in treatment often heralds poor eventual outcome [74], but what is true on a group level does not necessarily provide useful guidance on a patient-by-patient basis; for instance, some patients simply may take longer than others to respond to treatment that eventually will work well for them [75].

Measurement-based care [76, 77], with its systematic collection of clinical data with rating scales, can improve detection of good or poor response to treatment with greater utility than a clinician's global impression, but fundamentally these are better observations of what is already occurring, rather than predictions of future outcomes. The principle of identifying "the right drug for the right person at the right dose at the right time (phase of illness)" is central to the "personalized medicine" approach [78]. Not inconsequentially, this runs contrary to the "blockbuster medication" school of thought, predicated on the belief that a compound can be developed that will be an effective treatment for nearly all patients with a particular disorder and thus take preeminence in practice and in the marketplace.

#### 11.2.2.1 Cordance and the ATR Index

Several physiologically based biomarker approaches to predicting outcomes have emerged in recent years in the area of depression with peer-reviewed publication and independent replication of findings. These approaches can serve as useful examples for evaluating a candidate biomarker for clinical use. The first measure uses changes in resting-state prefrontal brain activity, assessed with qEEG cordance [10] over the course of a test exposure to an antidepressant medication. This early physiological change has been found to be predictive of later treatment outcome with that agent for the individual patient, in studies using either serotonin reuptake inhibitors (SRIs) or dual-reuptake inhibitor antidepressants [79–83]. Cordance is a measure that combines features of absolute and relative EEG power. Because cordance is better correlated with regional cerebral blood flow than other EEG measures [10], findings with this measure can be interpreted within the same conceptual framework as other functional neuroimaging studies.

A multisite replication and extension project (NCT00375843) has recently closed enrollment and data analysis is now under way. The relationship between early change in cordance and later clinical outcome was independently replicated in an inpatient sample using a variety of medications [84] and in a second inpatient sample using only venlafaxine in Level 1 treatment-resistant depression [85]. Using data from our prior trials [80], a receiver operating characteristic (ROC) curve can be constructed as an example of the use of an early change in prefrontal cordance as a predictor of treatment outcome. Using data from the 2-week assessment qEEG recordings, overall predictive accuracy in differentiating treatment responders from nonresponders was 84%, with sensitivity of 77% and specificity of 92% (Figure 11.2).

These findings sparked additional research in the use of physiological biomarkers to advance the possibilities of personalized medicine. An even larger collaborative, multisite trial, BRITE-MD (Biomarkers for Rapid Identification of Treatment Effectiveness in Major Depression, NCT00289523, $n = 375$), was under-

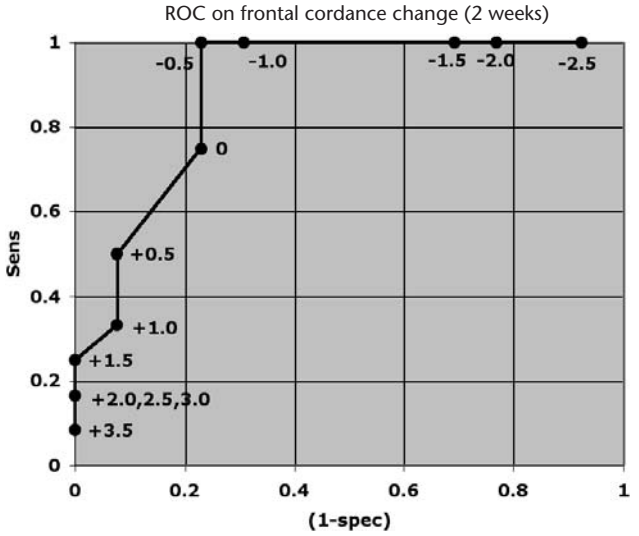



**Figure 11.2** ROC curve using cordance as a treatment outcome predictor.

taken, using a related EEG measure, the antidepressant treatment response (ATR) index [86]. The ATR can be computed using a simplified electrode array with five electrodes placed over prefrontal and frontal brain regions (FPz, AT1, AT2, A1, A2), instead of ~35 to 40 electrodes placed over all scalp locations for measuring cordance ("full-head montage"), making this a technology well suited for use in outpatient physicians' offices and avoiding the need to send patients to a dedicated EEG facility.

The EEG features comprising ATR were derived from the power spectra; ATR (revision 4.1) is a nonlinear combination of three features measured at two time points (in this trial, at baseline and week 1): (1) absolute power in an alpha subband (8.5 to 12 Hz), (2) absolute power in a second alpha subband (9 to 11.5 Hz), and (3) relative power in a combined theta and alpha band (3 to 12 Hz), calculated as the ratio of absolute combined theta and alpha power divided by total power (2 to 20 Hz). ATR is a weighted combination of the relative theta and alpha power at week 1 and the difference of alpha power between baseline (alpha: 8.5 to 12 Hz) and week 1 (alpha: 9 to 11.5 Hz), and is scaled to range from 0 (low probability of response to treatment) to 100 (high probability of response).

In the BRITE-MD study, subjects began with a 1-week test period of escitalopram, and then were randomized to receive either continued escitalopram treatment, a switch to bupropion, or a combination of the two medications. EEG data were recorded before and after the 1-week test period. In outpatients with major depression, individuals who received treatment consistent with their biomarker prediction were significantly more likely to experience response and remission than individuals who were randomized to a treatment not predicted to be useful [86–89]. Further development and replication projects are under way, and must be completed before this paradigm of early physiological change can be considered for clinical application.

#### 11.2.2.2 Loudness-Dependent Auditory Evoked Potential

The second approach utilizes an EEG measure that is proposed to reflect central serotonergic activity, the loudness-dependent auditory evoked potential (LDAEP) [90–92]. In the measurement of LDAEP indices, subjects listen to a set of sine wave tones at a series of loudness levels, while evoked potentials are being recorded [93]. The variation in the ratio of N1 to P2 amplitude values in the primary auditory cortex is measured with dipole source analysis, and the tangential dipole is reported to have a strong signal in the presence of low serotonergic activity; conversely, low-amplitude dipole signals are associated with high central serotonergic activity [94]. This phenomenon has been attributed to the serotonergic innervation of the primary auditory cortex [95]. For use as a biomarker in depression, qEEG data recorded prior to treatment would be interpreted to indicate whether a depressed patient has a low or high level of central serotonergic activity; those with low activity would be predicted to have a favorable response to a serotonergic medication (whereas high activity would be linked to better outcomes with a noradrenergic agent).

This method been examined using treatment with SRIs [96–98] or a noradrenergic agent [99, 100], and the relationship between level of serotonergic

activity and predicted treatment response has been observed in all of these studies. Data presented in these reports, however, generally does not permit evaluation on an individual-case-prediction level that would facilitate evaluation of the LDAEP approach for use in guiding clinical decisions. Furthermore, some other reports have suggested that the interpretation of the LDAEP may be more complex than solely indicating central serotonergic activity levels and that it may not be highly selective for serotonergic versus noradrenergic medications [101, 102]. LDAEP values generally were calculated using dipole source analysis methods and data from full-head EEG electrode arrays.

#### 11.2.2.3 Anterior Cingulate Activity Prior to Treatment

The third approach links resting-state pretreatment measures of activity in the rostral anterior cingulate cortex (rACC) to outcome with a variety of treatments, including sleep deprivation [103, 104] or a number of different medications [105] including the SRI paroxetine [106]. All of these investigations utilized PET methods to study regional brain metabolism, and found that higher rACC activity was significantly associated with good treatment response.

Additionally several studies have used the LORETA EEG method [107] to determine the level of electrical activity (current density) at current sources attributed to rACC [100, 108]. LORETA uses surface EEGs to solve the "inverse problem," and in order to find a unique solution for the three-dimensional distribution among the infinite set of different possible solutions, "this method assumes that neighboring neurons are simultaneously and synchronously activated," [107]; that is, the smoothest distribution of current sources constrained to gray matter is most likely correct.

Using a linear transformation matrix, LORETA yields current vectors at each of 2,394 voxels positioned stereotactically in cortical gray matter according to the Montreal Neurological Institute's model [109]. It has been widely validated with other functional neuroimaging methods, including MRI [110], fMRI [111], PET [12, 112], and subdural electrocorticography [111]. The studies using LORETA in MDD have shown that better response to treatment with nortriptyline [108] and reboxetine or citalopram [100] was associated with higher pretreatment current density in the ACC in the theta band (6.5 to 8 Hz). Furthermore, ACC theta current density correlates positively with glucose metabolism [112].

An inexpensive, noninvasive measure of ACC activity, such as EEG, is an intriguing approach to improving treatment of depression. However, more research is needed with this methodology to evaluate clinical applicability.

#### 11.2.2.4 Pretreatment Hemispheric Asymmetry Measures

The asymmetric nature of cerebral processing is widely recognized, and includes such everyday experiences as preferred handedness for writing. Questions have been raised about lateralization of function and physiological aspects of depression. A large number of studies have given frontal EEG asymmetry a sizable degree of construct validity as a measure of an underlying "approach-withdrawal" related motivational style [113, 114], though, as Allen and Kline [115] observed "the evidence

linking frontal EEG asymmetry to the activity of underlying neural systems involved in the experience, expression, and regulation of emotion is considerably lacking." Although more research in this area is desirable, a number of approaches have already yielded important results.

One measurable manifestation of lateralized processing is the "perceptual preference" for one hemisphere over the other during dichotic listening tasks. In dichotic listening paradigms, different stimuli—a pair of words or of tones—are simultaneously presented to the left and right ears; these stimuli compete with one another for identification, and the advantage for hearing items in the right or left ear is referred to as perceptual asymmetry (PA), an index of which (contralateral) hemisphere is favored for processing this verbal or tonal data. Studies using dichotic listening tests [116] have indicated that pretreatment measures of functional asymmetry of the brain are related to subsequent responsiveness to treatment with the selective SRI (SSRI) fluoxetine [117–119]. Individuals with MDD who responded well to fluoxetine exhibited greater left-hemispheric PA for perceiving dichotic words and less right-hemisphere PA for complex tone stimuli [117].

In a two-sample replication/extension study of PA before and after treatment with fluoxetine, PA did not change with fluoxetine treatment, so this measure may be considered as a stable, enduring, "trait" characteristic [118]. Replication and extension studies found the relationship of PA to treatment outcome to be dependent on gender [118]: Women but not men exhibited a heightened left-hemisphere perceptual advantage for words, while men but not women showed a reduced right-hemisphere advantage for tones among fluoxetine responders.

Perceptual asymmetries may also be related to resting-state asymmetries in the EEG. Bruder and colleagues [120] built on their dichotic listening work and studied patients entering treatment with fluoxetine with EEG as well, and found that responders and nonresponders differed not only in their pretreatment PA measure during dichotic listening, but also in their resting-state EEG alpha asymmetry. Nonresponders showed an alpha asymmetry indicative of overall greater activation of the right hemisphere than the left, whereas responders did not (eyes-open, resting state). This relationship between hemispheric asymmetry and treatment response interacted with gender, being present for female but not male subjects. In a recent project extending that work [121], it was reported that fluoxetine responders were characterized by greater alpha power compared with nonresponders and with healthy control subjects, with the largest differences being detected at occipital sites. They also reported differences in alpha asymmetry between responders and nonresponders at occipital sites, with responders showing greater alpha (less activity) over right than left hemisphere.

As to putative mechanism(s) relating this phenomenon to treatment response, perceptual asymmetry has been found to be significantly associated with plasma cortisol levels in MDD subjects [122], a neuroendocrine abnormality found in many MDD patients. Given that serotonergic activity may be related to arousal [121], it was hypothesized that the increased alpha power found in depressed patients who respond to an SSRI might reflect low arousal associated with low serotonergic activity. Researchers noted that the right temporoparietal and subcortical regions were particularly important in mediating arousal, which might account for their alpha asymmetry observations, and suggest that low serotonergic activity, tied to activity

of mesencephalic raphe nuclei and cortical afferents, could play a role in both the increased alpha and the alpha asymmetry they observed in fluoxetine responders. Additional independent work to expand this line of investigation is desirable.

#### 11.2.2.5 qEEG Measures and Adverse Events

Whereas most research on qEEG predictors in psychiatry has focused on response or remission outcomes, there may well be a place also for predictors of adverse events as well. Two outcomes that have begun to be addressed are antidepressant side effects and treatment-emergent suicidal ideation (TESI).

Medication side effects pose a significant concern in the treatment of depression. Many patients experience adverse effects that may lead to premature discontinuation of antidepressant medication [123], which in turn is associated with a 77% increase in relapse or recurrence of depression [124]. Antidepressant side effects can be highly variable among patients, and the ability to predict vulnerability could be useful in the clinical management of pharmacotherapy for depression. In this regard, an exploratory study found localized changes in the EEG (left lateralized frontal changes in qEEG cordance) related to later side effect burden among subjects randomized to antidepressant medication (fluoxetine 20 mg or venlafaxine 150 mg) [125]. Decreases in left prefrontal theta-band cordance prior to the start of medication were significantly correlated with later side effect burden during antidepressant treatment ($p < 0.0003$). The lateralized EEG pattern was not observed in relation to side effects reported during randomized treatment with placebo, suggesting that the qEEG marker may be specific to medication side effects. Development of qEEG markers along these lines could allow clinicians to prospectively identify patients who are at greatest risk. Those individuals might then benefit from additional support, closer monitoring during the initial weeks of treatment when side effects peak, and slower drug titration.

Although antidepressant medications are overall tremendously valuable in treating symptoms of depression including suicidality [126, 127], it appears that a small subset of persons may be at risk for developing increased suicidal ideation during antidepressant treatment. Findings are of sufficient concern that the U.S. Food and Drug Administration [128] has issued advisories concerning increased suicidal thoughts and behaviors in children, adolescents, and adults during the first few months of antidepressant treatment. Recent large-scale studies have found TESI rates ranging from 6% to 14% among depressed subjects receiving SSRI treatment [72, 73, 129].

Regardless of the cause of TESI, the phenomenon is of keen public and research interest. In a recent qEEG approach to predicting TESI, investigators examined baseline EEGs among 82 depressed patients receiving open-label SSRI treatment [70]. In this sample, 11% of subjects exhibited worsening suicidal ideation on at least one occasion during the first 4 weeks of treatment as assessed using item 3 of the 17-item Hamilton Depression Rating Scale ($HamD_{17}$). Results showed that baseline alpha-theta relative power asymmetry measured from frontal channels was significantly associated with TESI, where greater left-sided dominance was linked to emergent suicidal ideation (ANOVA $F_{1,57} = 8.33$, $p = 0.006$, controlling for gender baseline SI, specific medication and interactions). ROC analyses found the frontal

asymmetry measure to predict any TESI with 65% accuracy ($p = 0.14$). Accuracy, however, was higher (81%, $p = 0.04$) when considering only those subjects who exhibited an increase of 2 or more points in suicidal ideation on item 3 of the $HamD_{17}$. These findings are promising but await replication including replication in a placebo-controlled trial.

#### 11.2.2.6 qEEG and the Placebo Response

Placebo response is of interest in psychiatry from a clinical perspective, as well as from research and drug development perspectives. Placebo response rates are higher for MDD than those for some other medical conditions [130], ranging from 20% to 80% [131]. On one hand, high placebo response rates are evidence of the potential impact of nonpharmacological "placebo-related" factors such as patient expectations, classical conditioning effects, and the patient–physician relationship. On the other hand, high placebo response rates may obscure the efficacy of specific interventions making it difficult to evaluate treatment effects. qEEG imaging may have the potential to address both of these issues.

In a landmark study, serial EEGs were examined over 8 weeks of treatment in 51 MDD subjects assigned randomly to receive antidepressant medication (fluoxetine or venlafaxine) or placebo [132]. Subjects were categorized according to one of four outcome groups: medication responders, medication nonresponders, placebo responders, or placebo nonresponders. Analysis of regional changes in qEEG theta-band cordance revealed significant differences in the prefrontal region in placebo responders as compared to all other groups; placebo responders uniquely showed early increases in prefrontal cordance prior to achieving response. This finding documents neurophysiological changes associated with placebo response in depression, and suggests that drug and placebo response have at least some distinct underlying mechanisms.

Results of this study can be considered alongside a fluorodeoxyglucose (FDG) PET study of depressed males treated with fluoxetine or placebo [133]. PET scans obtained at pretreatment baseline, and again after 1 and 6 weeks of treatment, revealed some regional changes that were common to both medication and placebo responders (i.e., metabolic increases in prefrontal, parietal, and posterior cingulate regions, and decreases in subgenual cingulate), and other changes that were uniquely seen in fluoxetine responders (metabolic changes in subcortical and limbic regions including increases in pons and decreases in striatum, hippocampus, and anterior insula). Considering these data, it is possible that qEEG measures are especially well suited to capturing functional changes that are unique to placebo response.

In another report, pretreatment baseline features of the EEG, as well as other pretreatment neurophysiological and clinical characteristics, were examined for their ability to predict who will respond to placebo [134]. At baseline, those subjects who would later be classified as placebo responders exhibited lower theta-band frontocentral qEEG cordance as compared to all other subjects ($p < 0.006$). An exploratory multiple-variable model including this frontocentral qEEG marker, in addition to measures of cognitive processing time and insomnia, accurately identified 97.6% of eventual placebo responders. The ability to prospectively identify pla-

cebo responders could find utility in the drug development process where it is important to be able to distinguish specific medication effects from nonspecific, that is, "placebo" effects.

In yet another application involving placebo effects, qEEG has been used to examine the interplay between placebo-related neurophysiological changes and subsequent clinical response to antidepressant treatment. A typical design in clinical trials for depression is the use of a 7- to 10-day placebo lead-in phase during which time all subjects receive single-blind treatment with placebo prior to randomized treatment with active medication or placebo for the duration of the trial. Examination of serial EEGs beginning at pretreatment baseline and spanning both the placebo lead-in period and the postrandomization phase may shed light on relationships among brain functional changes, placebo effects, and medication effects in the treatment of depression. To this point, a novel study examined regional changes in qEEG cordance during the placebo lead-in phase in relation to final outcomes for depressed subjects later randomized to antidepressant medication or placebo [135]. Results showed that prefrontal changes during placebo lead-in explained 19% of the variance in final $HamD_{17}$ scores after 8 weeks of antidepressant treatment. This suggests that nonpharmacological treatment factors (i.e., those that are present during placebo lead-in) may act to prime the brain for better antidepressant response. Imaging with qEEG methods may help elucidate the role of placebo mechanisms in determining antidepressant response [83].

### 11.2.3 Pitfalls

Prior biomarker work has encountered numerous pitfalls, and it is vital to learn from past experiences. Perhaps most worrisome is the problem of premature clinical application: Not only is there the risk of doing harm to patients (e.g., being misdirected in treatment decisions), but there is also the risk associated with cynicism about biomarkers in general that this can engender.

The usual vetting of new biomedical innovations—procedures, techniques, medications, and devices—requires peer review of findings and independent replication: What applicability is there to a biomarker if it has only been shown to work in a single laboratory and others researchers are unable to confirm the results? Furthermore, it must be clearly disclosed what patient group was used to develop the biomarker, because this has great relevance to generalizability: In the universe of all patients with any psychiatric disorder, only a minority will have a syndrome that is refractory to multiple treatments, yet this is just the sort of patient who may seek out expert care in desperation and consequently be enrolled in a biomarker discovery research program. The generalizability may be quite limited for a biomarker developed with an idiosyncratic and nonrepresentative sample of patients, and without clear disclosure of these details, it is difficult to evaluate these qualities of a biomarker.

An additional caveat about biomarkers relates to the heterogeneity within a given clinical diagnosis. With our clinically defined diagnostic categories, there is variety both in the patients who seek care and in the individuals enrolled in research projects. A telling example is shown in Table 11.1, in which two individuals who both meet the formal diagnostic criteria for MDD have zero symptoms in common.

**Table 11.1** Heterogeneity Within Diagnoses

| *Patient A* | *Patient B* |
|---|---|
| Depressed mood | Anhedonia |
| Insomnia | Hypersomnia |
| Weight loss | Weight gain |
| Agitation | Psychomotor slowing |
| Reduced concentration | Feelings of worthlessness, guilt |
| Fatigue | Suicidal ideation |

Note: Two patients both meet the criteria for major depression, yet have no symptoms in common.

Thus, development of biomarkers also should disclose the nature of the patient population and consider evaluating whether the accuracy and reliability of the measure are improved or degraded in some subpopulations (e.g., psychotic depression, depression in Bipolar I versus Bipolar II patients).

Although biomarkers should have a high degree of clinical utility in order to be considered for use, there is also a need for them to be interpretable in the context of the rest of neuroscience. What aspect of a patient's pathophysiology is being assessed by a test: the form of a reuptake transporter that is associated with greater or lesser efficiency? the level of activity in a particular brain region? a component of a neuroendocrine feedback loop? Biomarker methods that fail to be comprehensible within or integrated into the extant body of neurobiological knowledge are unlikely to gain clinical acceptance, even if an empiric trial suggests that they might be useful.

Finally, it is worthwhile to note that statistical significance is not the same thing as clinical significance. Studies may report that a result is significant at the $p < 0.05$ level, really meaning that there is less than 1 chance in 20 that their finding arose by chance alone. Given a large-enough sample, even a clinically irrelevant difference (e.g., a very small improvement on a clinical rating scale) might be reported to occur with an impressive $p$-value.

An important measure for evaluating biomarkers includes the "number needed to treat" (NNT) [136], which assesses the number of patients needed to be treated differently (e.g., with biomarker guidance, with a new medication) in order to have one additional patient experience the desired, positive outcome.

Predictive biomarkers are also often characterized by a series of metrics that can help evaluate the usefulness of a potential biomarker: ROC curves and measures such as sensitivity, specificity, and overall predictive accuracy [137–139]. Sensitivity is the ratio of "true positive" tests to the number of individuals with the condition; for an outcome predictor, it would be the number of people in a sample who are predicted to respond to a treatment, divided by the total number of people who actually respond. Specificity is the ratio of "true negative" tests to the number of people who do not have a particular condition; in the outcome predictor context, this would be the number of people predicted not to respond divided by the total number of nonresponders. Overall predictive accuracy is the proportion of predictions that are correct. ROC curves plot the trade-offs between sensitivity and speci-

ficity, as different thresholds (cut points) are used to differentiate between positive and negative tests, for example, between response and nonresponse to a treatment.

### 11.2.4 Pragmatic Evaluation of Candidate Biomarkers

Given the potential for improving care and the pitfalls that may await possible biomarkers, how then can we judge a biomarker for use in psychiatric management? Table 11.2 summarizes some key, desirable characteristics of psychiatric biomarkers. Many of them follow directly from the pitfalls just detailed, but the last three on the list merit special mention.

First, the information provided by the biomarker must be timely, clinically useful, and cost effective. A test that is able to predict 8-week treatment response at week 5 is much less clinically valuable than a test making the prediction at week 1. A biomarker that identifies an individual with a treatment-refractory illness is somewhat less useful than one that points the way to an alterative treatment strategy. It is unlikely that the field would adopt a biomarker that consumes more resources than it saves, either in direct expenses or by wrongly suggesting pursuit of an ineffective treatment.

Second, the technology needed to assess the biomarker must be available and well tolerated by the target patient population. For example, some neuroimaging methods may be very well suited to neuroscience research applications, in which a small number of subjects can be observed with great detail. From a broader perspective, though, if the scanning technology costs too much to be deployed widely in the community, the method may not come to be translated into practice. Similarly, a procedure that is perceived by patients as painful (e.g., lumbar puncture) or challenging (e.g., a prolonged scanning procedure requiring immobility) may have low penetration into the clinical arena for reasons of practicality.

Third, methods that can be seamlessly integrated into existing clinical care practice patterns are more likely to be accepted than those that require major shifts in the delivery of care. For example, sending a patient to a different facility for a biomarker procedure and waiting for test results for a day or two is less desirable than being able to perform a test in one's office or ward.

To summarize, qEEG-based biomarkers have great potential for improving the care of patients with psychiatric disorders, much as other biomarkers have in other medical specialties. Adoption of biomarkers into clinical care, however, requires careful and thorough evaluation, and there is risk to patients if measures are

**Table 11.2** Desirable Characteristics of Biomarkers in Psychiatry

| |
|---|
| Test reliability, accuracy, and limitations are well characterized. |
| Biomarker development process is clearly disclosed. |
| Findings are reproducible with independent replication and peer review. |
| Interpretative framework for the biomarker allows comparison with other neurobiological observations. |
| Information provided by the biomarker is timely, clinically useful, and cost effective. |
| Technology is available and well tolerated by target patient population. |
| Methodology can be integrated into clinical care practice patterns. |

embraced prematurely. The set of criteria we have proposed can be used in the pragmatic evaluation of candidate biomarkers.

## 11.3 Research Applications of EEG to Examine Pathophysiology in Depression

### 11.3.1 Resting State or Task-Related Differences Between Depressed and Healthy Subjects

Coherence is analogous to the squared correlation in the frequency domain between two EEG signals (time series) measured simultaneously at different scalp locations [27]. When interpreting differences in EEG coherence between electrodes with different physical distance within hemispheres (e.g., F3–P3 versus F3–O1) or between them (O1–O2 versus T5–T6), two particularly important factors come into play. Volume conduction may serve to inflate measures of coherence at short (<10-cm) interelectrode distances, while an increasing phase difference may reduce coherence estimates at large distances (>15 cm) [140–142]. Thatcher et al. [28] mapped the physical distinction between "short" and "large" distances between scalp electrodes onto an anatomical model, by employing Braitenberg's [143] two-compartment model of axonal systems in the cerebral cortex. According to Braitenberg, compartment A is composed of the basal dendrites that receive input from the axon collaterals from adjacent pyramidal cells, whereas compartment B consists of the apical dendrites of pyramidal cells that receive input from remotely originating corticocortical projections.

Our laboratory and others have used coherence to study circuit function in patient groups. We examined the use of electrode pairings selected because they could assess connectivity over known neuroanatomic pathways. For example, to study connectivity in the superior longitudinal fasciculus, we used an average of coherence from one anterior pair of channels to three posterior pairings [35]. This approach yielded useful data in differentiating healthy elders from individuals with Alzheimer's disease or vascular dementia [35] and in relating structural damage in white matter tracts to cognitive performance in asymptomatic older adults [32].

We have recently examined a three-dimensional elaboration of the coherence construct to address a fundamental limitation: Coherence is calculated using two EEG signals recorded from separate locations, but conventionally this means two different scalp recording sites. To examine coherence in pathways between cortex near the scalp electrodes and locations deeper within the skull (remote from scalp electrodes), we developed a new method referred to as *current source coherence* (CSC) [144]. Whereas determining electrical sources in three dimensions from surface measurements is inherently ambiguous (i.e., the "inverse problem"), a reasonable estimate can be obtained if some assumptions are made about the distribution of current sources. Our implementation of CSC employs the LORETA algorithm [145] as a solution to the inverse problem, but other approaches could also be used. Instantaneous current vectors can be calculated with LORETA in the 2,394 gray matter voxels of its solution space, using the Montreal Neurological Institute (MNI) standard brain model.

LORETA current waveforms for all voxels and for each 2-second epoch can be generated for use in CSC calculations, first by combining them into a priori defined regions of interest (ROIs) to test hypotheses about circuit function and connectivity. As an example, we examined data recorded from a healthy adult male during the resting state and during a simple motor task of repeatedly squeezing either the dominant or nondominant hand (methods as in [144]). Using ROIs to examine connectivity between the motor strip [Brodmann area (BA) 4] and premotor cortex and supplementary motor cortex (BA 6), we found that the motor task was associated with a change in CSC in the contralateral hemisphere's pathway (BA 6 to BA 4) compared with the resting state (Figure 11.3). We interpret this to indicate that connectivity, as reflected in the shared electrical activity at two anatomically linked brain regions, may vary instantaneously with the demands of task activation. The use of this method to study individuals with MDD is currently under way in our laboratory; preliminarily, there appear to be differences both in the resting state and in task-activated CSC values, and the relationship of these measures to symptomatic and functional treatment response will be examined.

A limitation to the interpretation of coherence is that it cannot easily discriminate between the direct coupling of activity between two regions, and a value arising because both regions are connected to a separate, third area that "drives" the signals in both probed regions. This has been addressed by the construct of partial directed coherence (PDC) [146–150]. This method calculates a decomposition of the ordinary coherence function into two "directed" coherences: one representing the feedforward and the other representing the feedback aspects of the interaction between two structures. Rather than just revealing mutual synchronicity, PDC describes "whether and how two structures are functionally connected" [151]. A refinement of CSC could incorporate such advances in coherence analyses to consider Granger causality [152] and the direction of information flow in these interactions between brain regions.

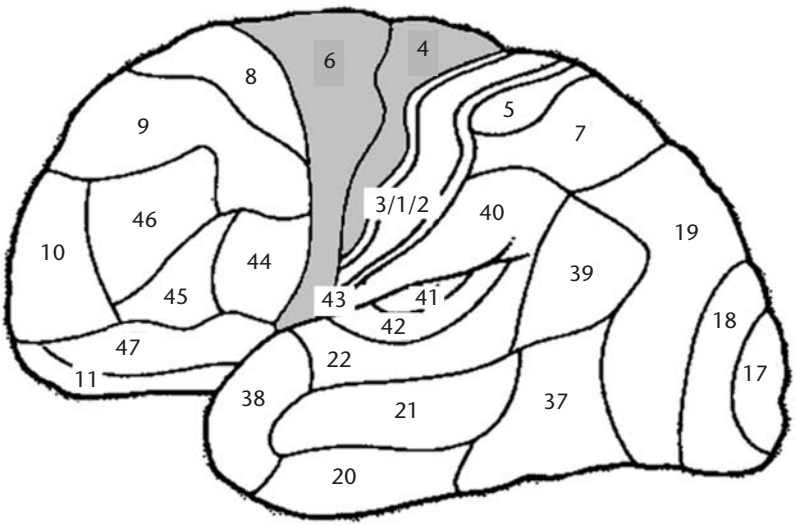



**Figure 11.3** Brain regions of interest.

### 11.3.2 Toward Physiological Endophenotypes

As illustrated by the clinical presentations listed earlier in Table 11.1, there can be considerable heterogeneity in individuals all carrying the same clinical diagnosis. Given the limited historical success in relating symptom-based subtypes to individual response to treatment, it has been suggested that expanding our ability to relate neurobiological discoveries to clinical realities will require a shift in perspective, away from symptom-focused nosologies and toward that of endophenotypes [153], namely, the "measurable components unseen by the unaided eye along the pathway between disease and distal genotype." In the present context, the endophenotype concept can be stated as follows: Within a set of individuals who all meet the diagnostic criteria for unipolar major depressive disorder, there are distinct, discrete subpopulations each of which shares a common set of meaningful physiological characteristics.

The identification of these groups might allow better understanding of many important issues: how genes and the environment may contribute vulnerability to developing depression; why some individuals respond preferentially to one treatment over another; why some patients will have a single, brief bout of depression in their lifetime and others will spend large fractions of the lives disabled by recurrent, chronic depression; why some patients have cognitive disabilities that impair work and social function, while others do not; why some women with past experiences with depression develop significant episodes during pregnancy and in the postpartum period, while others do not; why some patients with depression go on to develop dementia in a few years and others do not, to name a few. Measurements with qEEG techniques may be able to aid in endophenotypic characterization. Additional research along this thematic line will be needed to determine the usefulness of this approach.

## 11.4 Conclusions

Quantitative EEG methods have much to contribute to psychiatry, not only in expanding our understanding of the physiological underpinnings of disorders such as depression, but critically in improving the ability of clinicians to treat patients struggling with psychiatric brain disorders. The use of qEEG-based biomarkers could greatly impact the field, but there are considerable risks to premature adoption of methods and measures that have not met the usual peer-reviewed independent replication standards.

We proposed some useful guidelines in assessing the readiness of any qEEG biomarker for clinical use, particularly in the sphere of major depression. Finally, many new methods are being developed by groups around the world, some of which will come to find clinical application. We described some of the promising approaches being examined in our own laboratory and elsewhere, but note that this rapidly evolving field is best monitored via online databases of peer-reviewed journal publications.

## Acknowledgments

The authors wish to thank the staff of the UCLA Laboratory of Brain, Behavior, and Pharmacology for many years of fruitful and enjoyable collaborative teamwork, and Ms. Jamie Stiner and Ms. Kelly Nielson for technical assistance in the preparation of this manuscript.

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CHAPTER 12

# Combining EEG and MRI Techniques

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Simultaneous use of EEG and MRI offer new methodologies for studying the structure and function of the brain. Section 12.1 outlines the techniques for utilizing anatomical information from the MRI in solving the EEG inverse problem. Section 12.2 addresses the theoretical and practical considerations for recording and analyzing simultaneous EEG-fMRI, as well as some current and emerging applications. We begin by presenting the technical challenges associated with recording EEG within the high-field-strength magnet of the MRI scanner, including artifacts in the EEG unique to the MRI environment as well as distortions in the MRI due to the presence of EEG hardware.

We present some recent approaches for using fMRI techniques to study EEG phenomena such as evoked potentials and rhythms. Concurrently, we review methods that aim to generate more meaningful fMRI images by incorporating information from the EEG into the mathematical models used for generating functional MR images. Some potential clinical applications, such as in studying epilepsy, as well as sleep studies are also presented. The ultimate goal of combining with fMRI is to exploit the complementary information in these two separate datasets to better understand the functional dynamics of the brain.

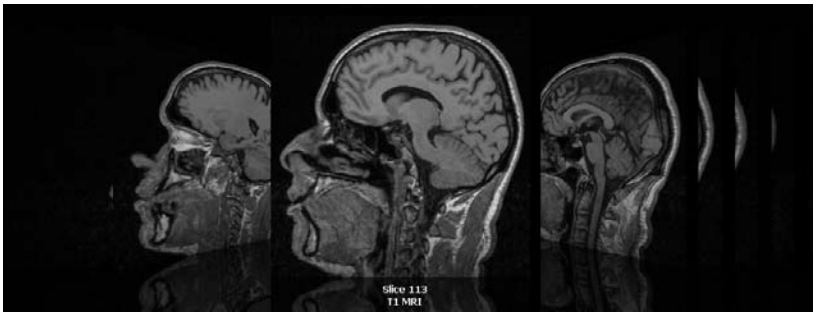
## 12.1 EEG and MRI

Although this chapter appears in the applications part of the book, it is predominantly a methods chapter. The methods proposed here deal with bringing EEG and MRI together, creating realistically shaped volume conductor models (head models) and enhancing EEG source analysis, both by means of information from MRI. After defining the (cortical) source space, cortical current density reconstructions (CDRs) as well as cortical dipole or beamformer scans are possible. The sections on volume conductor models and source analysis techniques build on the information presented in Chapter 5, and the same notation as in Chapter 5 will be used. Finally, sensors, mappings, anatomical structures, and reconstructed sources can be displayed in a common visualization framework.

1. This author contributed to Section 12.1.
2. These authors contributed to Section 12.2.

Three-dimensional structural MRI datasets are stacks of two-dimensional images (slices), each typically $256 \times 256$ pixels in size (rarely, $512 \times 512$). The pixel dimensions of such a $256 \times 256$ image showing a cross section of the head are approximately $1 \times 1$ mm$^2$. The slice distance in the stack of images is somewhere between 1 and 2 mm. The field of view should be adjusted so that the whole head is captured, not only the brain—this allows for the definition of skin landmarks and realistic head models. T1 protocols deliver good gray matter–white matter contrast and short acquisition times. When loaded into the computer, the stack of slices becomes a three-dimensional image (see Figure 12.1) with cuboid (brick-shaped) voxels.

Although some of the benefits of using MRI can also be achieved using a normalized and averaged dataset such as the ICBM152 brain from the Montreal Neurological Institute (MNI) [1], exact head models can be derived only from individual image data, and cortical structures should not be used across subjects at all due to the high intersubject variability of cortical gyri and sulci.




(a)



(b)

**Figure 12.1** (a) Three-dimensional structural MRI datasets are stacks of slices, each typically $256 \times 256$ pixels in size. (b) When loaded into the computer, the stack of two-dimensional slices becomes a three-dimensional image.

### 12.1.1 Coregistration

Coregistration brings EEG electrodes and MRI into the same three-dimensional spatial reference system. Electrode locations can be inferred from their labels or measured using a three-dimensional pointing/tracking device (digitizer), together with anatomical landmarks (fiducials) or the subject's headshape.

#### 12.1.1.1 Label-Based

Because of the advent of electrode caps and the increased number of EEG channels, hardly any lab implements the International 10-20 System [2] or one of the proposed modifications thereof (10% system [3], 5% system [4]) by actually measuring and subdividing distances on the skull, based on the locations of the nasion, the inion, and the preauricular points. The electrode *labeling* scheme brought forth by these systems, however, is in wide use. As a consequence, it can make sense to infer three-dimensional electrode locations based on their labels, especially if a digitization is not available or if data is to be pooled or averaged across subjects on a channel-by-channel basis.

For identifying labeled electrodes in the MRI, one could identify the electrodes actually used in a list of stock locations, in conjunction with matching fiducials and one of the methods described in the next section. Alternatively, one of the measuring schemes defined in the literature could be employed, but performed by a computer algorithm on the segmented skin of the subject's MRI and based on fiducials identified in the MRI beforehand [5]. In either case, only approximate electrode locations can be determined.

#### 12.1.1.2 Landmark-Based

Three or more landmarks are digitized together with the electrodes. Landmarks should be chosen according to two criteria. (1) They should be identifiable on subject's head as well as in MRI. Anatomical landmarks or MRI-visible markers may be used. (2) Robust coregistration is required. The smallest coregistration error occurs for locations close to the landmark's center of gravity. Most EEG labs use three fiducials: the nasion and two points near the ears (ear points). Because the preauricular points used for the 10-20 system are hard to identify in MRI, the tragus, the lower end of the intertragic notch, or the incisura anterior auris are often a better choice. After landmarks have been located in the MRI, digitizer and image coordinates can be matched. For this purpose, least-squares fitting can be used. Because of the one-to-one correspondence between landmarks, it is sufficient to solve the related orthogonal Procrustes problem [6].

However, if nasion and ear points are used, a *landmark-based coordinate system* is preferable. Such a coordinate system can be defined by placing the origin on a line connecting left and right ear points, using an $x$ axis through the right ear point and a $y$ axis through the nasion, with the $z$ axis pointing upward, orthogonal to both $x$ and $y$. The advantage of this approach is threefold. (1) The coordinate system definition is robust against landmark mismatch along the $x$ and $y$ axes, which can occur due to pressure applied during digitization and uncertainties regarding the skin–air boundary in the MRI. (2) Regardless of which information (digitized

electrodes and landmarks or MRI and image data landmarks) is available first, one can start working in a valid coordinate system. (3) No fitting or optimization is involved (see Figure 12.2).

#### 12.1.1.3 Head Shape–Based

In addition to the electrodes and instead of a small number of landmarks, the subject's head shape is digitized, yielding several hundred digitization points [7]. If image data is unavailable, this approach allows rendering of a crude anatomical reference by plotting the head shape together with the electrodes. However, the MRI coregistration problem is more involved here, not only because the skin surface needs to be segmented from MRI first, but because the round shape of the head and the many-to-many correspondence between landmarks and segmented skin points lead to problems with shallow and local minima during the optimization. Fitting just the electrode locations to the skin (without accompanying head shape) is an even more ill-defined problem.

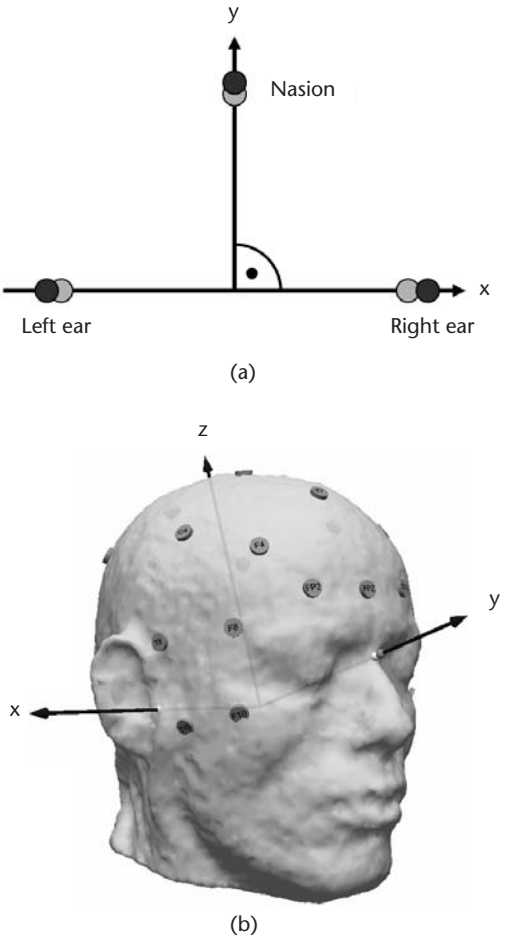



**Figure 12.2** (a) Coordinate system definition based on digitized and image data landmarks. (b) Digitized electrodes and landmarks together with image data landmarks, landmark-based coordinate system axes, and skin surface segmented from MRI.

### 12.1.2 Volume Conductor Models

An exact model of the conducting properties of the human head is a necessary (although not sufficient) condition for accurately localizing brain activity based on the EEG. As outlined in Chapter 5, head modeling is part of the forward problem of source analysis, which implies that head model errors or inaccuracies cannot necessarily be detected by an inspection of the solution of the inverse problem or its associated goodness-of-fit. The typical effect of an inaccurate head model is a mislocalization of brain activity.

#### 12.1.2.1 Spherical Head Models

Spherical head models (see Chapter 5) have been the traditional approach to EEG source localization and have been in use since well before EEG-MRI integration was an option. By using a spherical head model, one makes the assumption that the head has the shape of a sphere, with electrodes located on the surface of that sphere. Electrode locations on the sphere can be obtained by radial projection of the actual electrode locations. Head conductivities are usually modeled as being piecewise isotropic, with concentric subspheres delineating regions of different conductivities. Usually, three or four differently conducting compartments are modeled, mimicking the skull, everything outside the skull (often called the skin), and everything inside the skull (the brain), and, optionally, a cerebrospinal fluid layer surrounding the brain. Thus, the brain is also assumed to be of spherical shape.

When source localization results obtained using spherical head models are overlaid onto MR images and compared with independently obtained information about the true source locations, localization errors in the centimeter range can be observed [8]. Discrepancies are larger in "nonspherical" parts of the head (e.g., the temporal lobes) than they are in "spherical" parts, such as the central sulcus area. Forward calculations using spherical head models can be performed quickly and with high numerical accuracy.

Using MRI, the parameters governing a spherical head model can be fitted to the segmented skin and skull, although in practice they are usually fitted to the electrodes, and predefined percentages of the outer sphere's radius are employed for the outside and the inside of the skull.

Modifications and extensions of the isotropic concentric three-sphere model include the use of eccentric spheres [9], of anisotropic conductivities for the skull (which is known to conduct better tangentially than radially) [10], of ellipsoids instead of spheres [11], and of individually adapted spherical models per sensor [12]. The geometric properties of these extensions can be obtained from an analysis of the MRI.

#### 12.1.2.2 Realistically Shaped Head Models

The sphere and ellipsoid ansatz can be overcome altogether by using tessellated surfaces or volumes: The boundary element method (BEM) can be used for modeling the same compartments of isotropic conductivities (outside the skull, skull, inside the skull) as used by the isotropic sphere models, but with BEM the compartment

boundaries are triangulated surfaces (see Figure 12.3) [13, 14]. The geometric shape of these surfaces can be obtained from MRI.

Several constraints govern the generation of BEM compartment border meshes. (1) Memory requirements of the matrix decomposition algorithms utilized in BEM computations scale with the square of the number of nodes, and their computation times scale with the third power of the number of nodes. For everyday applications with model setup times of a few minutes or less, this limits the total number of nodes per head model to approximately 5,000 and the number of triangles to approximately 10,000. Typical triangle sizes for a 5,000-node model are 6 to 8 mm. (2) Accuracy decreases as source locations or the node locations of another mesh come closer to a BEM mesh than one-half of a triangle side length. Triangle side lengths, therefore, should not be larger than 6 to 9 mm, or, given the triangle side lengths, a minimal distance should be kept between sources and inner skull boundary as well as between boundaries.

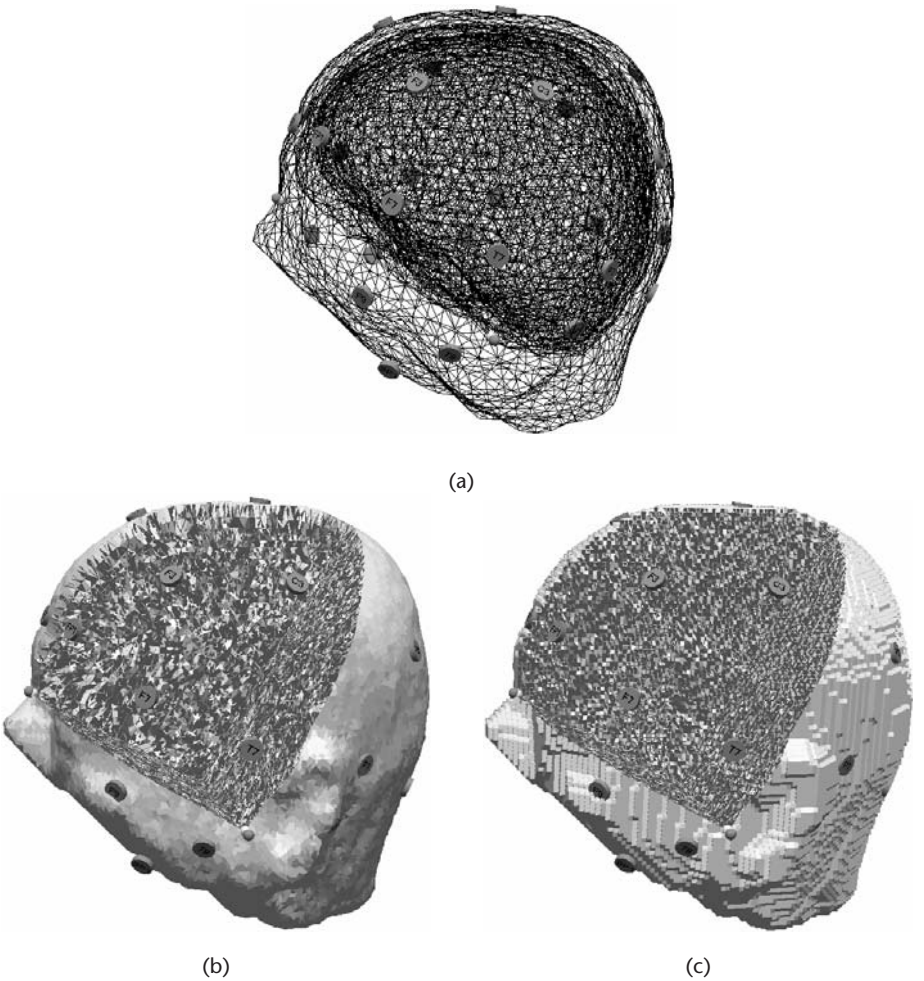



**Figure 12.3** (a) BEM model with a resolution of 6 mm (inner skull), 8 mm (outer skull), and 9 mm (skin) comprising 5,000 nodes and 10,000 triangles. (b) FEM model with a resolution of 2 mm and 1.2 million tetrahedra. (c) FEM model with a resolution of 2 mm and 440,000 cubes.

As one can see, there is not much leeway for designing the model geometry, and triangle sizes will be between 6 and 9 mm [15]. This, in turn, implies that a faithful representation of anatomy is only one requirement of an algorithm that creates BEM triangle meshes from MRI data. Equally important are a minimum distance between compartment borders and a level of smoothness that allows triangles of the required size to adequately represent the boundaries [16].

Finite element models (FEM) [17] use conductivity tensors per tetrahedral or cubic volume element and allow us to overcome the isotropy restrictions imposed by the BEM. However, computation times for the approximately 1 million elements required are only now reaching practical levels [18], the definition of the tensor orientations taken from, for example, diffusion tensor imaging is challenging, and for the absolute conductivities the same literature values are still used as for the BEM and spherical models.

### 12.1.3 Source Space

Large portions of prior knowledge about the sources of surface EEG are closely related to cortical anatomy: locations and orientations of neurons, and spatial connectivity (see Chapter 1). The cortex is a complexly folded two-dimensional structure. To make use of all available information from MRI for EEG source analysis and to allow for advanced visualization, cortical triangle meshes have become a de facto standard: The triangle mesh passes through the middle of the cortical gray matter layers, its nodes represent potential source locations, and its edges encode local neighborhood and allow us to compute surface normals, which represent the orientation of neuronal current flow.

#### 12.1.3.1 Source Locations

To constrain sources to the cortical gray matter, corresponding locations need to be identified in the MRI. In a typical MRI, some 100,000 to 200,000 voxels represent cortical gray matter [see Figure 12.4(a)]. If cortical voxels are taken as potential source locations and source orientations are not taken into account, three unknown source components per potential source location need to be computed, resulting in 300,000 to 600,000 unknowns. Because the spatial resolution of an EEG is more in the order of a few millimeters than of a voxel (approximately 1 mm), some form of data reduction may take place; without such data reduction, computation times would be unjustifiably long.

As a consequence, there is basically the choice between a regular three-dimensional grid with some 5- to 11-mm spacing, and a cortical source space sampled with 2- to 3-mm resolution. When source coupling or extended sources are modeled, the neighborhood relations of the source locations also need to be known. Although this is straightforward for three-dimensional grids, the Euclidean distance is not the correct measure for cortical sources. Functionally quite distinct areas can be as close as two opposing walls of a sulcus while being several centimeters apart on the two-dimensional cortical sheet. For this reason, a triangulation of the cortex is the method of choice for defining the source space, where each vertex of the corti-

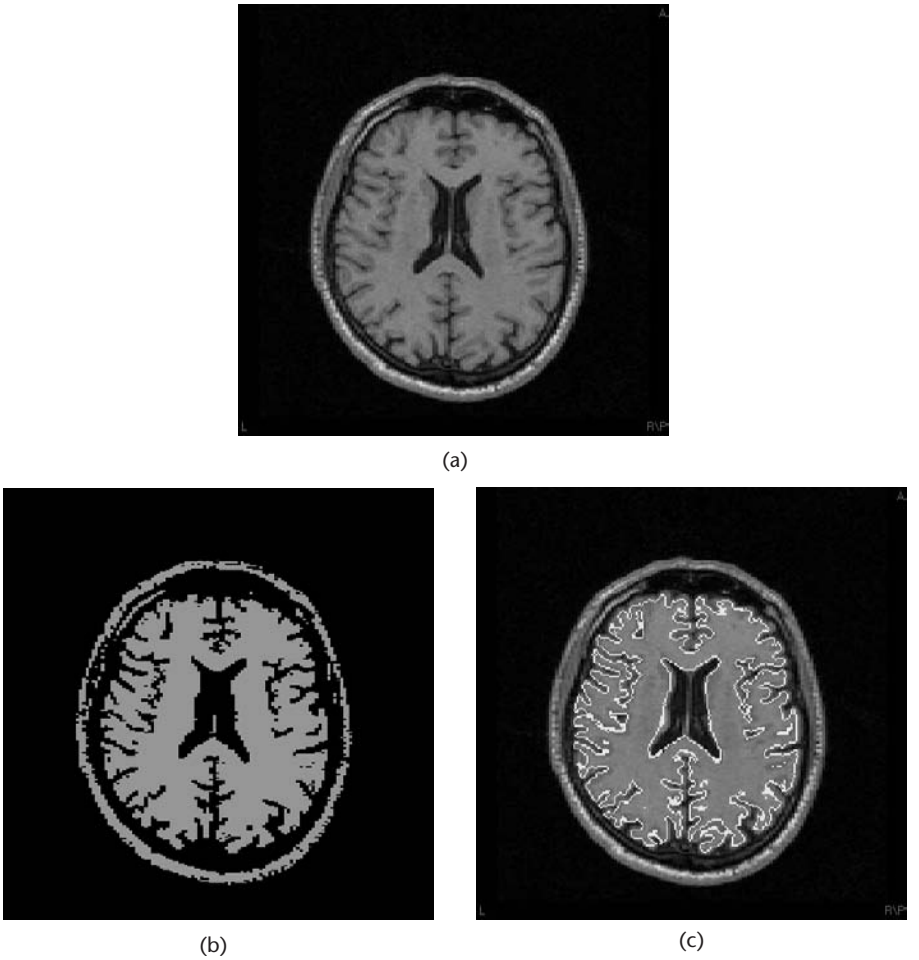


**Figure 12.4** The same T1 MRI slice showing (a) raw image intensities, (b) thresholded image, and (c) segmented cortex.

cal triangle mesh serves as a source location, and source orientations can be fixed to the local surface normal, if desired.

Cortex segmentation and triangulation can be combined as in the marching cubes family of algorithms [19]. Because of their subvoxel accuracy, the resulting source locations and triangles are very densely spaced. To achieve data reduction, mesh reduction techniques have to be employed. A caveat with this approach is that the resulting triangle meshes may contain triangles of largely different sizes and angles. This can make it necessary to explicitly account for the different gray matter volumes that each source location represents in the inverse algorithm, especially in the context of minimum norm least squares (MNLS) and CDR depth-weighting (see Chapter 5).

Another option, taking the two-dimensional topology of the cortex into account, is to perform a three-dimensional region-growing segmentation and identify the cortex as the segmented surface. Several segmentation algorithms designed to deal with the cortical sheet have been proposed, exploiting the connectedness of the white matter or the spherical topology of the cortex [20, 21]. A fully automatic

approach based on thresholds and shape constraints is described in [16]. The result is a set of labeled voxels (see Figure 12.4). A representation of each voxel face by two triangles produces large numbers of triangles and, again, requires a subsequent mesh reduction [see Figure 12.6(b)] [22, 23]. The locally two-dimensional connectivity of the cortical voxels, however, allows us to perform data reduction based on two-dimensional distance, producing a thinned-out set of voxels with a given minimum two-dimensional distance. A subsequent triangulation yields triangles of very similar sizes and angles [see Figures 12.5 and 12.6(a)] [24].

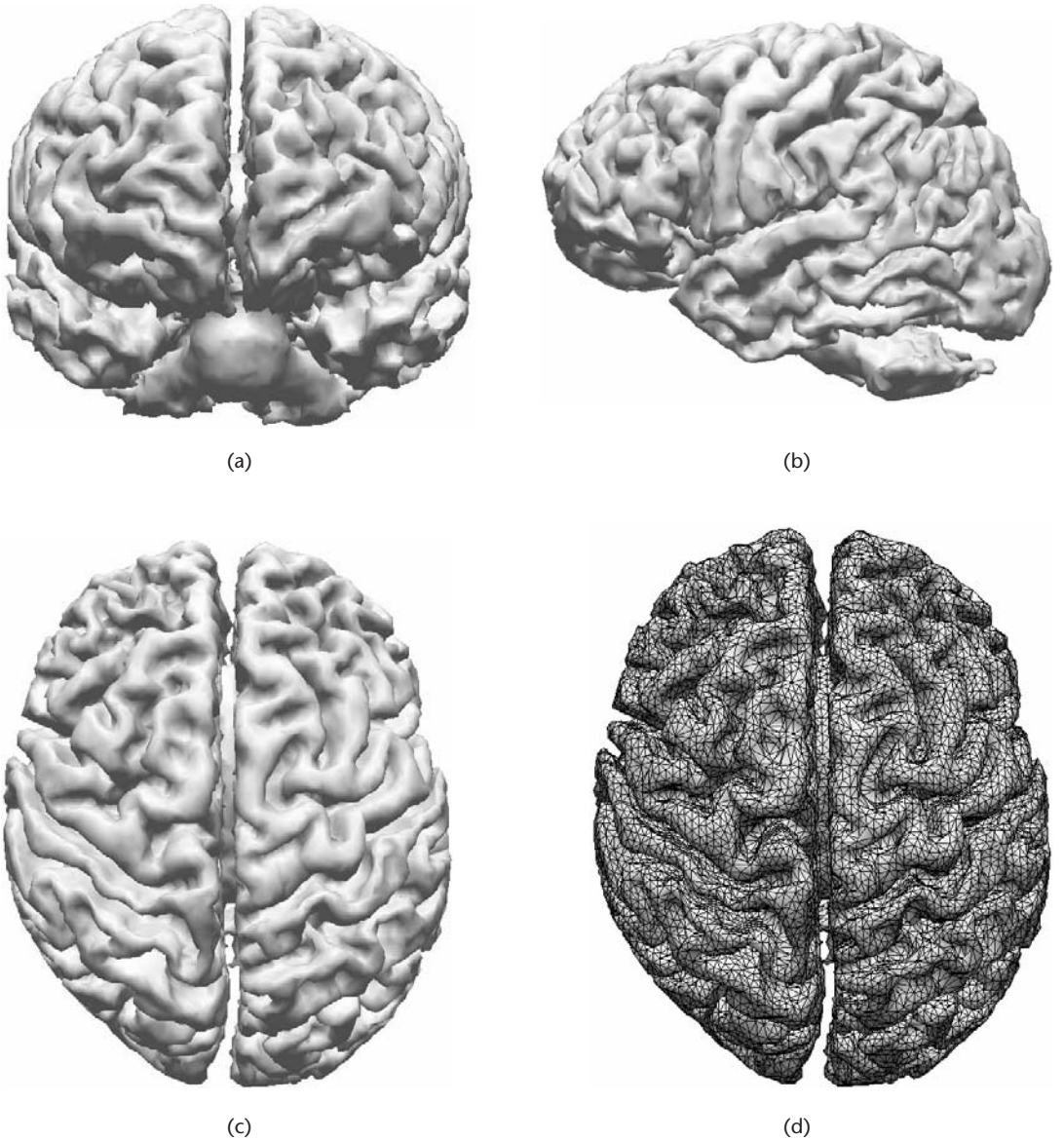
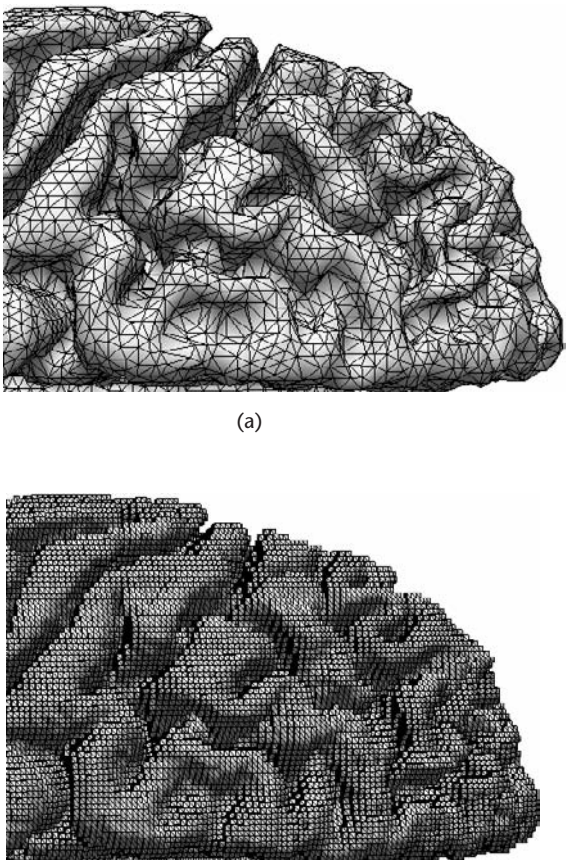



**Figure 12.5** Three-dimensional rendering of triangulated cortex: (a) front view, (b) left view, (c) top view, and (d) cortical triangles.



(a)

(b)

**Figure 12.6** Three-dimensional rendering of triangulated cortex: (a) Delaunay triangulation and (b) voxel face triangulation.

#### 12.1.3.2 Source Orientations

Source orientations may be constrained to be perpendicular to the surface of the cortical sheet, thus modeling the principal orientations of the synapses of the gray matter pyramidal cells that are the sources of the surface EEG (see Chapter 1). On a cortical triangle mesh, source orientations can easily be computed as the vector sum of the normals of all triangles surrounding a given node. If a cortical triangle mesh is not available, the three-dimensional intensity gradient computed from the MRI can be used. With given source orientations, one unknown source component per location needs to be computed.

#### 12.1.3.3 Connectivity

The temporally correlated activity of large populations of nearby and similarly oriented neurons is the basic building block of the EEG. The equivalent current dipole representing activity from several cubic millimeters of cortical tissue is one way to model this basic building block. For a cortical source space, connectivity needs to be exploited. Cortical triangle meshes encode all information necessary for modeling

the spatial correlation of sources. The necessary methods are described in the next section.

### 12.1.4 Source Localization Techniques

The basics of source analysis were introduced in Chapter 5, and most of the techniques described there can easily be adapted to make use of cortical locations and orientations. In this section, some less obvious extensions that become available with the advent of cortical triangle meshes are described, both dealing with the incorporation of local neighborhood information.

#### 12.1.4.1 Spatial Coupling

The LORETA method uses MNLS fitting with spatial coupling between source locations. The spatial Laplacian (second derivative) of the source distribution is used in the model term, rather than the source strengths as in standard MNLS [25]. The effect is that the model term demands minimum curvature of the source distribution rather than minimum norm. Spatial coupling is achieved via a nondiagonal Laplacian weighting matrix **B**, while the diagonal weighting matrix **D** is responsible for removing the depth dependency [24, 26], so that

$$\mathbf{W} = \mathbf{D}^T \mathbf{B}^T \mathbf{B}\mathbf{D} \tag{12.1}$$

In the case of locations on a regular three-dimensional grid, for any location $i$ and its $N_i$ neighbors $j$ (with $N_i \leq 6$) [27],

$$\begin{aligned} B_{i,i} &= -1 \\ B_{i,j} &= \left(6 + N_i\right)/12N_i \end{aligned} \tag{12.2}$$

In the case of cortical sources [28], where $d_{i,j}$ is the distance between locations $i$ and $j$ and the sums loop over all $N_i$ neighbors $j$ of location $i$,

$$\begin{aligned} B_{i,i} &= -\Sigma\left(1/d_i\right)/\Sigma d_i \\ B_{i,j} &= 1/\left(d_{i,j}\Sigma d_i\right) \end{aligned} \tag{12.3}$$

Because **B** is nondiagonal but sparse, it can be favorable to minimize [see (12.1)]

$$\mathbf{J} = \arg\min\left\|\Phi - \mathbf{K}\mathbf{J}\right\| + \lambda \mathbf{J}^T \mathbf{W}\mathbf{J} = \arg\min\left\|\Phi - \mathbf{K}\mathbf{J}\right\| + \lambda \mathbf{J}^T \mathbf{D}^T \mathbf{B}^T \mathbf{B}\mathbf{D}\mathbf{J} \tag{12.4}$$

directly [24], instead of computing J as

$$\mathbf{J} = \mathbf{W}^{-1}\mathbf{K}^T\left(\mathbf{K}\mathbf{W}^{-1}\mathbf{K}^T + \lambda 1\right)^{-1}\Phi \tag{12.5}$$

#### 12.1.4.2 Extended Sources

The methods described until now have used a lead field matrix **K** that encodes the impact of point sources onto the measured data and a model term measuring prop-

erties of the corresponding source vector **J**. However, the smallest unit of brain activity that is visible outside the head is not a cortical point source but an extended cortical patch. The dense 2- to 3-mm discretization of the cortical sheet (corresponding to "point sources" that actually are 2- to 3-mm patches) is necessary to sample the variation of surface orientations with sufficient resolution. The extension of cortical excitation that actually produces an EEG signal above noise is typically larger by a factor of 5 to 10, and so is the spatial resolution of the inverse methods used.

The effects of this discrepancy can best be seen when cortical surface normals are taken into account and cortical source orientations are fixed. In such a case, MNLS will reconstruct activity only in parts of the cortex where locations and surface normals match the measured field distribution. Because the variation of normal orientations within the range of matching locations is rather wide, a fragmented source constellation is reconstructed. Such a constellation might show activity on opposing walls of a gyrus or sulcus but not on the crown, and it is hard to tell whether these distinct activities are distinct sources or an artifact of using point sources and cortical surface normals.

The use of extended sources (cortical patches) instead of point sources in the lead field matrix and the model term promises to remove this ambiguity. The size of the patches should match the resolving power of the inverse method or the size of the actual extension of cortical excitation, whatever is larger (see Figure 12.7).

Patch-based source models have also been proposed as extensions of dipole fit methods, where only one or a few patches are active simultaneously [29, 30], and it is certainly advantageous to perform an exhaustive search optimization for one or a few cortical patches instead of dipoles [31]. Cortical patches can also be used as the basis of CDR:

The relation between $N_J$ point sources **J** and $N_P$ cortical patches **U** can be expressed by the $N_P \times N_J$ weighting matrix **P**, whose columns represent the shape of the patch in terms of the point sources **J** [32]:

$$\mathbf{J} = \mathbf{PU} \tag{12.6}$$

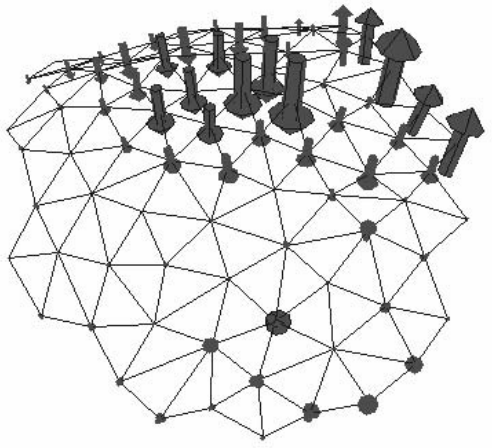
To capture the variability of possible source constellations, it is important that overlapping patches are used. When using cortical patches, further spatial coupling is unnecessary, so that $\mathbf{B} = 1$. The dimension of **W** is $N_P \times N_P$. Favorably, fixed orientations (cortical normals) are used such that each patch is centered around the location of the respective point source, yielding $N_P = N_J = N_V$. Such a cortical patch models the joint activity of sources with a variety of orientations along the folded gray matter sheet.

Exchanging **J** for **U** yields the model term $\mathbf{U}^T\mathbf{WU}$, while the data term is modified according to (12.6)

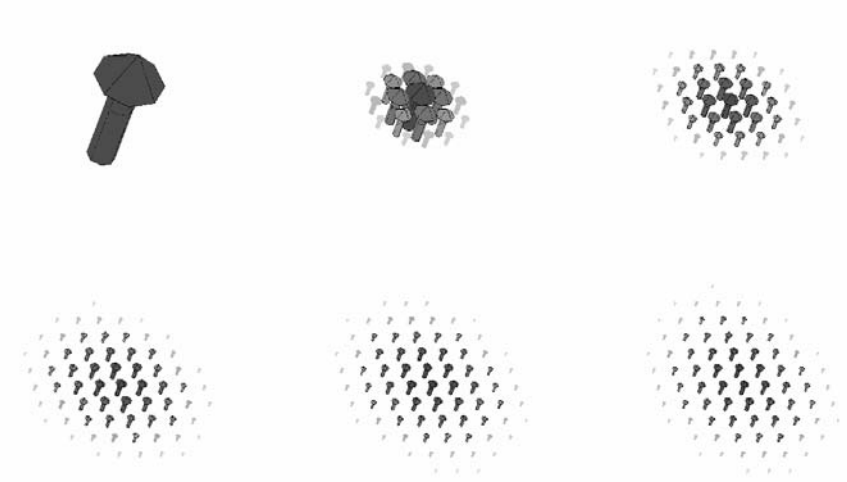
$$\|\Phi - \mathbf{KJ}\| = \|\Phi - \mathbf{KPU}\| \tag{12.7}$$

We obtain a new CDR formulation that can be used with *any* inverse method. In the MNLS case,

$$\mathbf{U} = \arg\min\|\Phi - \mathbf{KJ}\| + \lambda\mathbf{U}^T\mathbf{WU} = \arg\min\|\Phi - \mathbf{KPU}\| + \lambda\mathbf{U}^T\mathbf{WU} \tag{12.8}$$



(a)



(b)

**Figure 12.7** (a) Cortical triangle mesh with connected cortical sources and (b) point source (dipole) and source patches of different sizes with Gaussian strength profiles.

Solving for **U** and inserting (12.6) yields

$$\mathbf{J} = \mathbf{P}\mathbf{W}^{-1}\mathbf{P}^{T}\mathbf{K}^{T}\left(\mathbf{K}\mathbf{P}\mathbf{W}^{-1}\mathbf{P}^{T}\mathbf{K}^{T} + \lambda 1\right)^{-1}\Phi \tag{12.9}$$

### 12.1.5 Communication and Visualization of Results

By visualizing EEG data and the results of EEG source analysis in an anatomical context, the information contained therein can be lifted onto a new level: Mappings (and derived values; see Chapter 4 and [33]) can be compared with the underlying brain anatomy, dipole results can be visualized at their anatomical locations, and CDR and scan maps can be plotted onto the cortical sheet (see Figure 12.8).

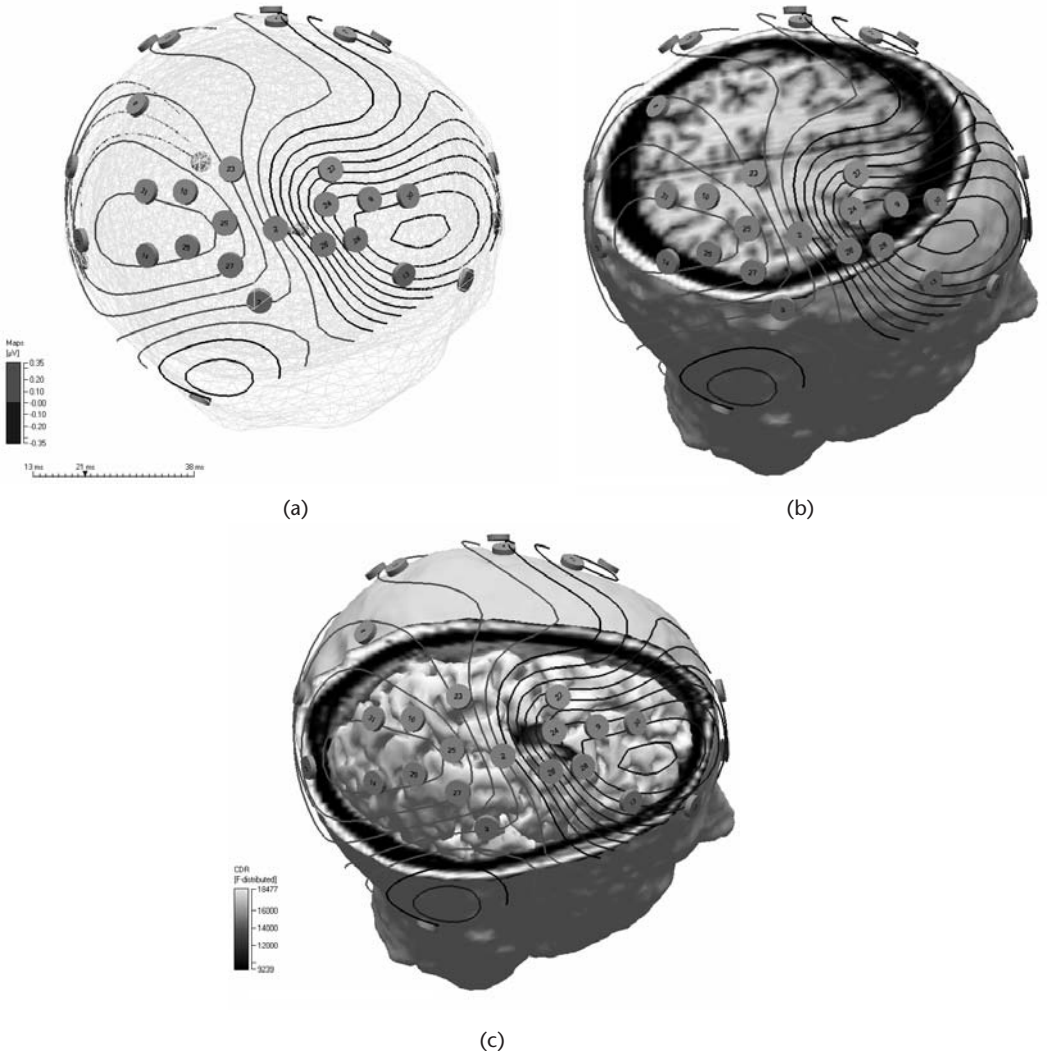
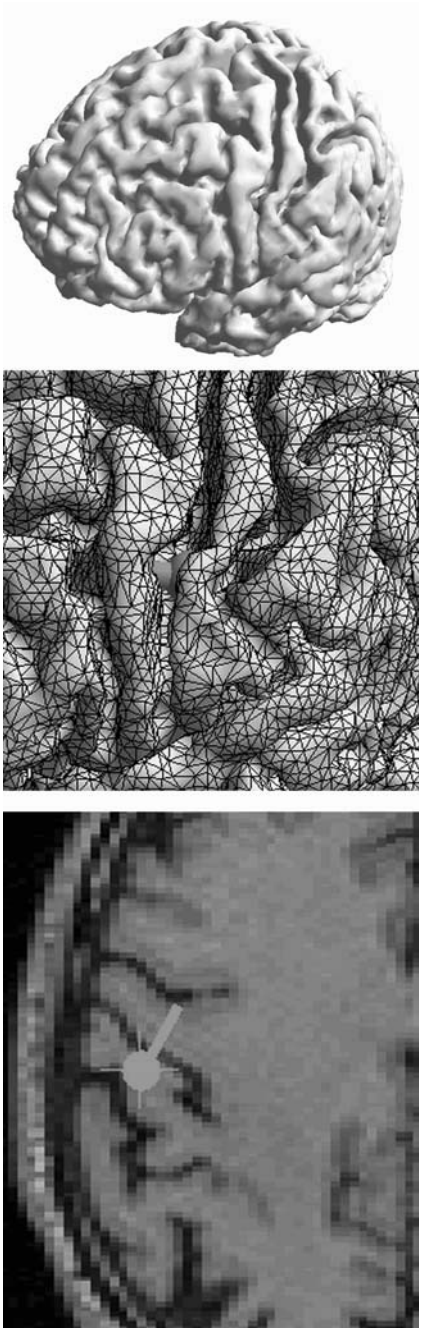



**Figure 12.8** Three-dimensional rendering of: (a) electrodes, potential map, color scale, time axis, BEM head model, and fitted dipole; (b) electrodes, potential map, MRI slice, and fitted dipole; and (c) electrodes, potential map, cortical sLORETA result with 15-mm source extension, and color scale.

#### 12.1.5.1 Comparison of Grid, Cortical, and Extended Cortical Sources

To demonstrate the effect of various source space constellations and of extended sources onto the results of CDR, a point source (dipole) was simulated and noise was added to achieve an SNR of 15. The simulated data was analyzed using sLORETA [34, 35] CDRs for the following source spaces: a 7-mm three-dimensional grid, a 3-mm cortex without source orientation constraint, a 3-mm cortex with cortical source orientation constraint, and extended sources of 20-mm patch size based on the same 3-mm cortex and its source orientations. Simulated dipole and reconstruction results are shown in Figure 12.9.

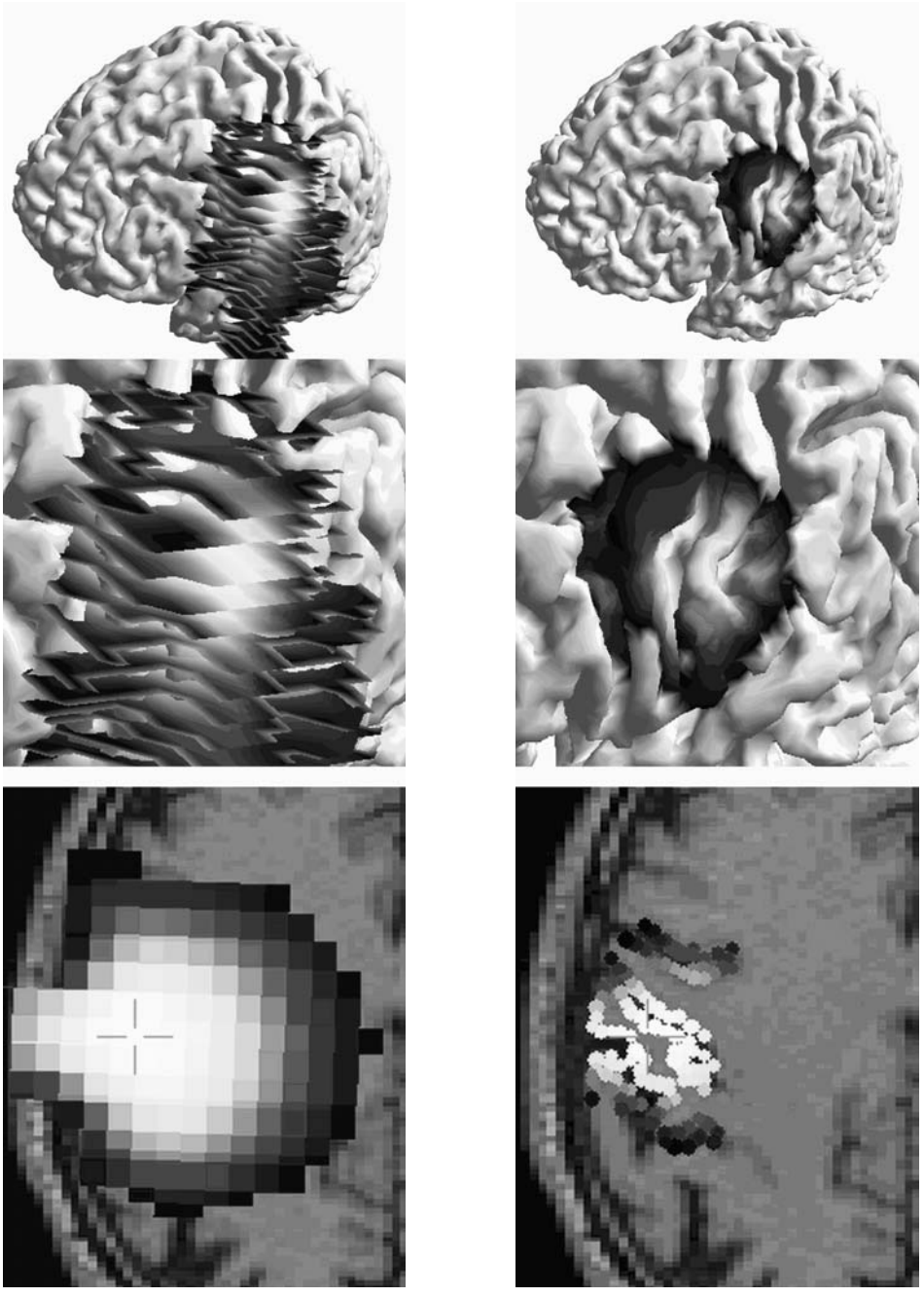
Moving from a grid to a cortical source space, using cortical orientations and extended sources, adds a priori information to the source reconstruction and delivers results that are even closer to reality. However, the interpretation of results



(a)

**Figure 12.9** Three-dimensional rendering (top and middle images) and orthogonal cuts (bottom image) of sLORETA analysis results for: (a) simulated dipole data using different source spaces: (b) three-dimensional grid, (c) cortex without orientation constraint, (d) cortex with orientation constraint, and (e) extended cortical sources with orientation constraint.

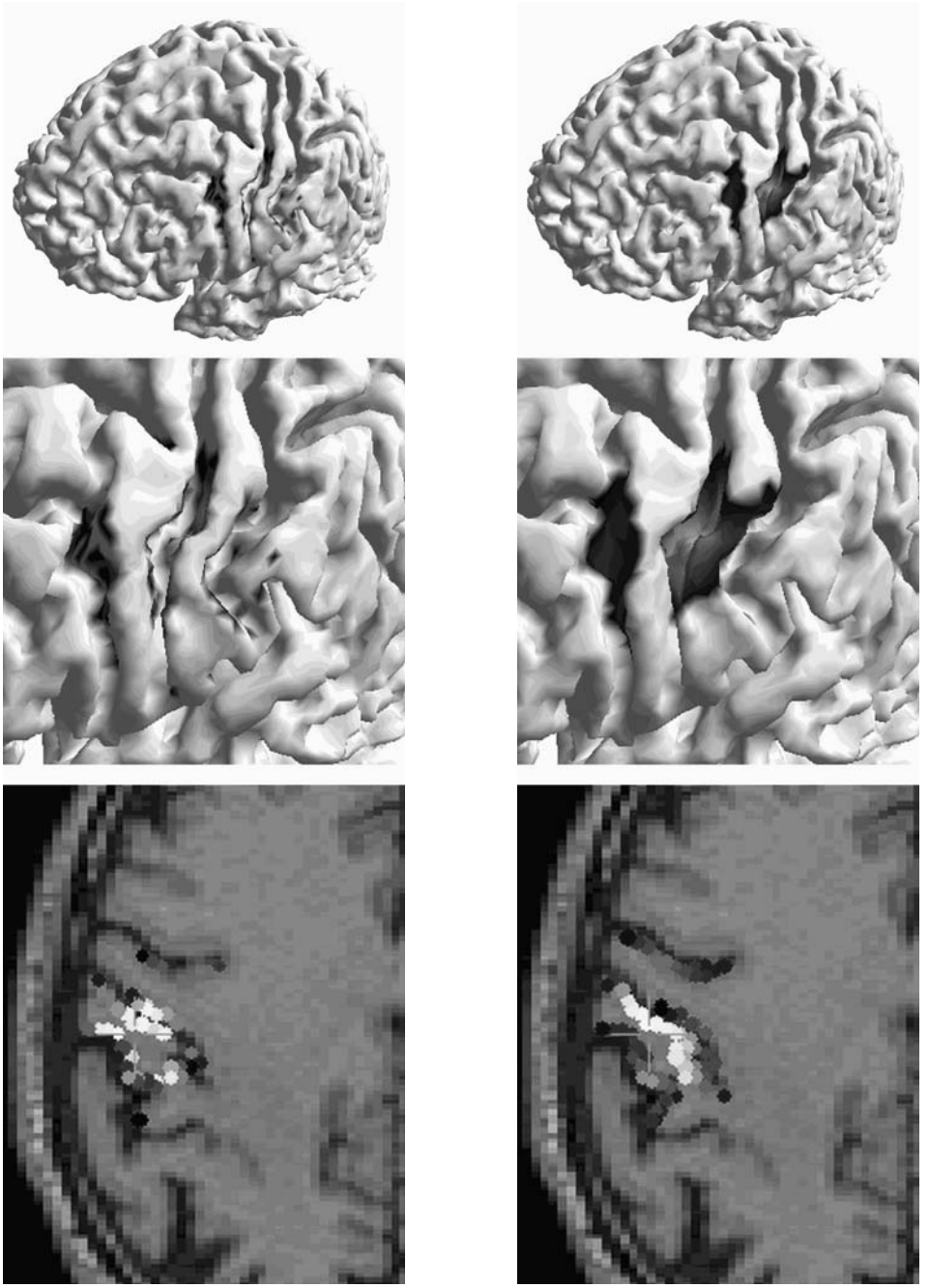
obtained using a cortical source space with cortical orientation constraint [Figure 12.9(d)] is especially difficult: It is hard to tell which of the several reconstructed



(b) (c)

**Figure 12.9** (continued)

patches on parallel walls of neighboring sulci are distinct sources or whether the fragmentation comes from the fact that the algorithm selects only sources with orientations that match the data. This interpretation problem vanishes when extended sources are used [Figure 12.9(e)].



(d) (e)

**Figure 12.9** (continued)

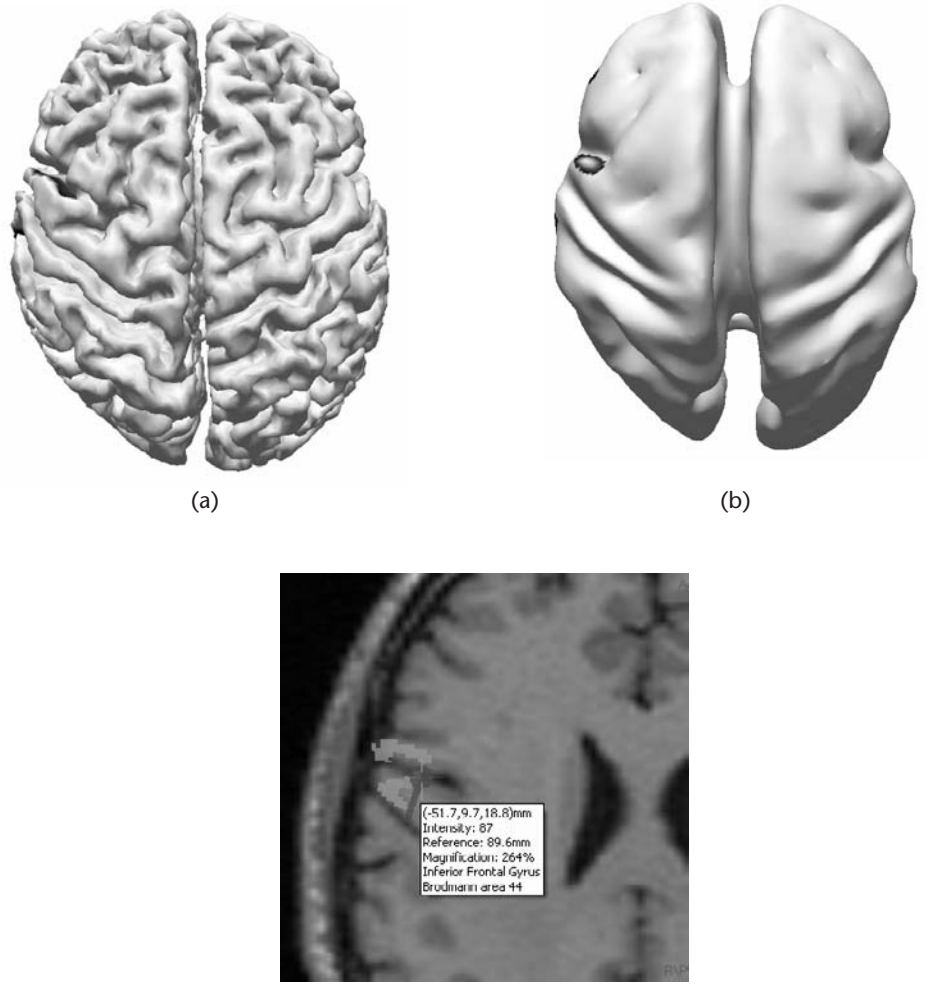
### 12.1.5.2 Flattening the Cortex

Flattening of the cortical sheet allows viewing activity that is hidden in a sulcus. Depending on how the cortical triangle mesh has been obtained, a restoration of the sphere-cortex homeomorphism may need to be performed first so that connections between opposing sulcal walls are removed [36]. Then, by administering forces that

aim to even out angles between neighboring triangles while, optionally, trying to preserve triangle angles or area and starting a relaxation process, the cortex can be inflated [37] (see Figure 12.10). If portions of the cortical mesh are cut before starting the process, flattening is possible as well.

#### 12.1.5.3 Talairach Coordinates and Brodmann Areas

By coregistering MRI data with a functional or anatomical atlas, source locations may be linked to Brodmann areas or anatomical features. The most common atlas-based coordinate system is the one introduced by [38]. Here, the brain is divided into 12 sections defined by the anterior commissure, the posterior commissure, the interhemispheric fissure, and the extent of the brain in each of the




(a) (b)

(c)

**Figure 12.10** (a, b) Three-dimensional rendering of cortex with CDR result on sulcal wall (a) before and (b) after inflation. (c) Fitted dipole, Brodmann area 44 according to [38], and information window (containing Talairach coordinates, image data intensity, distance from a predefined reference location, magnification factor, and anatomical and functional areas), overlaid onto MRI.

three coordinate axes. Either by computing the best-fit nonrigid transformation between the MRI and a template whose Talairach coordinates are known [39, 40], or by defining the same landmarks and extensions in the individual MRI [41], Talairach coordinates can be obtained for each brain voxel. Functional and anatomical atlas data available in Talairach coordinates [42] can then be overlaid onto the individual MRI and related to the sources of the EEG (see Figure 12.10). By means of this established subject-independent and brain size-independent coordinate system, source analysis results can easily be reported and pooled for meta-analysis.

## 12.2 Simultaneous EEG and fMRI

### 12.2.1 Introduction

The EEG is a reflection of the electrical activity of the brain, as recorded on the scalp. It offers a high temporal resolution, in the order of milliseconds. However, due to the volume conduction effects of the cortex and the cerebrospinal fluid, as well as the attenuating effects of the skull and scalp, the EEG has a poor spatial resolution [43]. Neural electrical dipoles that lie deeper within the brain or are oriented tangentially to the scalp surface have negligible contribution to the EEG [43]. Additionally, the electrical activity recorded by any given scalp electrode is not necessarily a reflection of the activity of the cortical regions directly underneath that electrode. Localization of electrical sources within the brain is therefore a highly ill-posed inverse problem. This is despite the fact that high-density EEGs can now be recorded by up to 256 scalp sensors.

Advances in fMRI in the past two decades have heralded an alternative method for studying the functional activity of the brain. Functional MRI is based on the hemodynamic and metabolic response of the brain, secondary to localized neuronal activation. More specifically, it images the contrast in the blood oxygenation level–dependent (BOLD) response of the brain during resting period and during a specific task under study. The advantage of fMRI is that it has a very high spatial resolution (<1 mm) as well as the ability to image deeper structures of the brain (unlike EEG, which is primarily a reflection of the activity of the neocortex). A complete functional image of the brain typically can be acquired every 2 to 3 seconds.

The fMRI image is based on a statistical comparison of the BOLD response between an idealized "resting" phase and a task execution phase. However, fMRI is not a direct measure of neural electrical activity but rather a reflection of secondary metabolic and hemodynamic changes arising as a consequence of such neural activity. Additionally, the BOLD response of the brain is much slower (typically 2.5 to 6 seconds) than the almost instantaneous electrical changes associated with neural activation. The simultaneous measurement of EEG and fMRI, therefore, offers the possibility of combining the best of both techniques as well as circumventing their respective disadvantages.

For applications requiring precise localization of the electrical activity within the brain, simultaneous fMRI evidence can better constrain the solution of the inverse problem of EEG source localization. Recently, fMRI has also been used to study various EEG phenomena such as visual alpha rhythms and evoked potentials.

Many of these EEG features have been empirically used for addressing clinical and research questions; however, often their genesis is poorly understood. Simultaneous EEG-fMRI offers the exciting possibility of imaging the brain regions associated with these EEG phenomena, as well as allowing us to make inferences about the large-scale neuronal pathways involved in generation of such EEG activity.

Additionally, EEG has the potential to alter the traditional methodology of generating fMRI images. As noted previously, the fMRI is based on a statistical comparison between multiple trials of an idealized "resting" phase and "task" phase. The statistical basis of this comparison relies on the behavioral responses of the individual; however, it is well known that behavioral changes do not necessarily generate exactly repeatable neuronal responses, and vice versa. Mental states such as attention, fatigue, and so forth play an important role in the underlying neuronal dynamics, a factor that current fMRI reconstruction methods largely ignore by relying solely on behavioral cues and outcomes. Because the EEG is a direct reflection of some of these mental states, its incorporation into the way fMRI images are generated could make them more reflective of the true neuronal dynamics.

The ultimate goal of combining EEG with fMRI is to exploit the complementary information in these two separate datasets to better understand the functional dynamics of the brain. This chapter aims to address the theoretical and practical considerations for recording and analyzing simultaneous EEG-fMRI, as well as some current and emerging applications.

### 12.2.2 Technical Challenges

The bore of the MRI scanner is a high-field-strength magnet, typically 1.0 Tesla to 4.0 Tesla. Additionally, as part of the fMRI imaging methodology, rapidly changing magnetic gradient fields and RF pulses are applied during image acquisition. As a result, recording EEGs inside the MRI scanner is especially challenging due to the strong electromotive forces induced during these rapidly changing applied fields, as well as moving conductor loops within the strong static magnetic field [44]. These induced currents are orders of magnitude larger than the EEG itself. Not only do they interfere with the EEG signal, they might cause severe injury to the subject due to localized heating and burns [45].

The presence of EEG electrodes on the scalp might cause distortions in the MRI images because of magnetic susceptibility resulting from the conductor elements of the electrodes and chemical shift artifacts resulting from the saline gel in the electrode–skin interface. However, sufficient technical advances have been made in the past decade to overcome most of these issues, to the extent that simultaneous EEG-fMRI can be safely recorded using specialized hardware, and the quality of both are comparable to independently acquired fMRI and EEG.

#### 12.2.2.1 Hardware Considerations

The quality of the acquired EEG and fMRI as well as patient safety can be substantially improved by observing some general principles aimed toward avoidance of current loops, minimization of movements (albeit very small ones) within the scan-

ner, and elimination of ferromagnetic materials in the EEG apparatus placed inside the scanner.

The first priority is patient safety. This can be ensured to a large extent by using specialized EEG recording caps, the electrodes of which are typically made of plastic rings with thin layers of silver/silver chloride or gold [46, 47]. Dual twisted connecting leads are used (to substantially reduce induced artifacts) that are made of materials such as carbon fiber [47]. All leads have resistors in series (5 to 10 kΩ) to reduce heating effects due to large induced currents [45]. Motion artifacts, which can lead to substantial distortions in the EEG, can be reduced by using methods to fixate the subject's head, such as vacuum cushions, inside the head cage of the MRI scanner. The wires running from the cap are typically taped down firmly. Some existing systems digitize the EEG signal inside the scanner room and carry the signal outside the room using a fiber optic cable. Hardware lowpass filters are added to each EEG channel to attenuate the imaging artifacts. The EEG amplifiers used in simultaneous EEG-fMRI must have a high bit resolution (24- to 32-bit digitization) to prevent saturation by the higher voltage artifacts and yet maintain sufficient dynamic resolution in the EEG. The sampling rates must be very high (2,000 to 10,000 Hz), to allow for software-based cancellation of imaging artifacts. The quality of the EEG recording can be further improved by turning off the helium pumps of the MR scanner, which helps in reducing high-frequency vibrations associated with the pump.

The purpose of the above measures is to ensure patient safety as well as to get the best possible EEG and fMRI recording. However, even with the best hardware settings, substantial artifacts are still induced in the recorded EEG, and specialized algorithms are required for their suppression.

#### 12.2.2.2 Imaging Artifacts

The RF magnetic fields as well as transient gradient magnetic pulses, applied during MR imaging, induce strong electromotive forces in the electrodes and wires of the EEG cap. These induced voltages are in the order of few hundred millivolts, whereas the underlying EEG is in the order of few microvolts. The earliest attempts at simultaneous EEG-fMRI tried to address this issue by employing "interleaved" acquisition, whereby a sparse MR pulse sequence would be used, and the EEG recorded in between the imaging intervals (and hence without the induced artifacts) would be used in the analysis [48]. Since the hemodynamic response to neuronal activation is typically delayed by a few seconds, an appropriate timing of EEG acquisition followed by fMRI acquisition can theoretically record the response to the same neural events. However, this method is not truly "simultaneous" EEG-fMRI and cannot be used in a variety of settings where simultaneous recordings are needed, such as in the study of epileptic spikes [49] or spontaneous fluctuations of various EEG rhythms [50].

The imaging artifacts have been shown to be linearly additive to the EEG [51] and therefore can be removed independent of other artifacts or phenomenon in the EEG. The most widely used imaging artifact removal method is the weighted average artifact subtraction [52]. In this method, an estimate of the imaging artifact is made, based on the previous couple of imaging pulses (typically 5 to 10), followed

by subtraction of the template from the current artifact period. Figure 12.11 depicts a few channels of EEG recorded inside a 3T scanner before and during an imaging sequence, as well as the results of artifact subtraction using this algorithm. A similar algorithm is based on subtraction of an adaptively modified template artifact based on the power spectra of the individual artifacts [53].

One important prerequisite for successful operation of most of these algorithms is precise synchronization of the scanner pulses with the EEG, which allows for phase locking between the individual artifacts and the template being subtracted. Recently, independent component analysis (ICA) has been proposed [54] as a method for removing imaging artifacts, obviating the need for scanner synchronization. However, irrespective of the algorithm used, residual baseline high-frequency noise (typically above 50 Hz) [55] remains in the EEG because of temporal jitter between the MR scanner and the EEG. For applications requiring analysis of averaged EEGs (evoked potentials or event-related potentials), this is usually not problematic (because the baseline noise cancels out on averaging). However, in applications where the ongoing EEG is the subject of analysis, these residual artifacts can be removed using a lowpass filter, but this comes at the expense of suppressing higher-frequency components of the EEG.

It should be noted, however, that good scanner synchronization itself can substantially reduce the high-frequency residual artifacts. Recently, a "stepping stone" sampling scheme has been proposed [56] whereby the EEG was sampled (during the image acquisition), when the artifacts were around the baseline levels. This methodology, when implemented with a high degree of scanner synchronization, has been

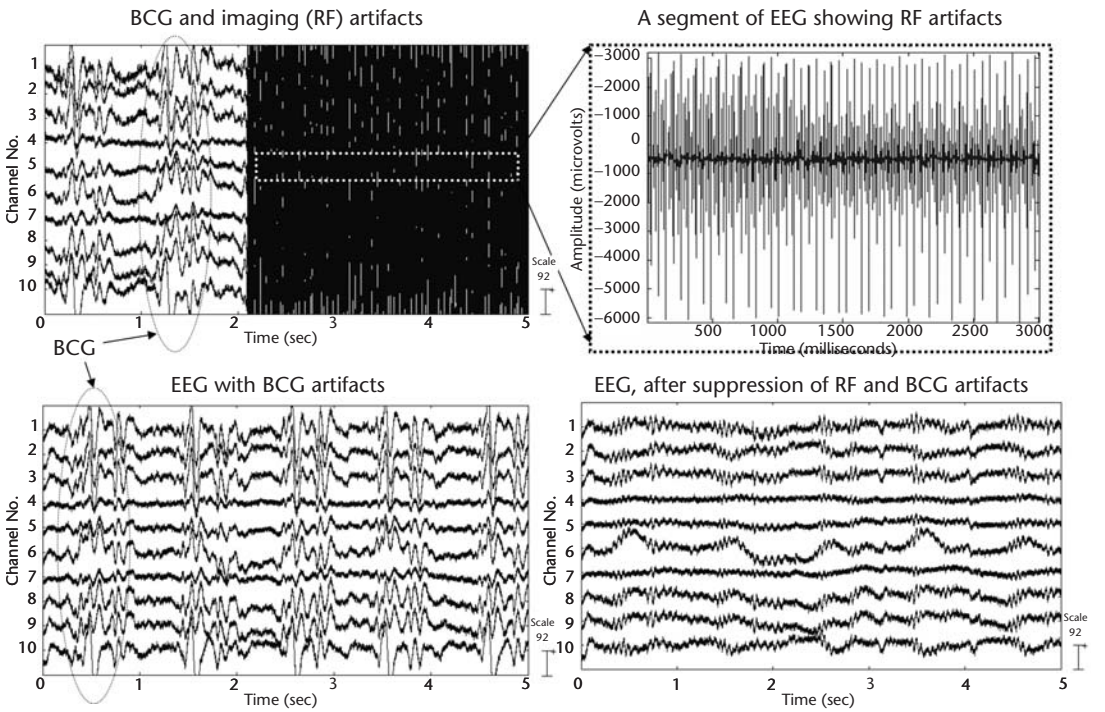



**Figure 12.11** Examples of EEGs recorded inside the MR scanner during and between imaging sequences.

demonstrated to result in substantially reduced higher-frequency imaging artifact residuals.

#### 12.2.2.3 Ballistocardiogram Artifacts

With each cardiac cycle, a characteristic artifact, known as the ballistocardiogram (BCG), shows up in the EEG recorded inside the static magnetic field of the scanner. These artifacts are believed to originate from the pulsatile movements of the scalp arteries [46, 57], resulting in movements of the scalp electrodes and to a lesser extent inducing a Hall effect voltage related to this flow [58]. Figure 12.11 shows a segment of the EEG recorded inside a 3T scanner, depicting BCG artifacts. The amplitude of the BCG artifact is much larger than the underlying EEG and increases with the field strength of the scanner magnet [59]. The frequency spectrum of this artifact (typically 0 to 12 Hz) [57, 60] overlaps strongly with the EEG, and it is somewhat nonstationary on a beat-by-beat basis, making it difficult to cancel out using a fixed template subtraction.

During recording, all methods aimed at minimizing electrode and wire movements help in substantially reducing the amplitude of the BCG, including the use of tight elastic bandages on the EEG cap, fixating the subject's head using vacuum cushions, firmly taping down the wires, and so forth. Adaptive filtering techniques [60] have been successfully implemented for artifact suppression, by using piezoelectric motion sensors placed on the scalp, which serve as reference signals for adaptive noise cancellation algorithms such as Wiener filters. However, this method is critically dependent on being able to record one or more reference signals, an option which might not be available in all hardware setups.

Other algorithms commonly used for suppression of this artifact are based on average artifact subtraction [57]. In this method, a template artifact is estimated by averaging the last few BCG artifacts (phase locked to a cardiac trigger signal such as a finger plethysmogram or electrocardiogram). This template is then subtracted from the current artifact period. This method needs a cardiac pulse signal, both for synchronization and subtraction. Separate artifact templates are adaptively estimated for each channel. Some recent modifications to this technique include subtraction of an amplitude-adapted dynamic template [61] and median filtering to eliminate outliers in estimation of the artifact template [62, 63].

ICA-based algorithms also have recently been demonstrated to be effective in suppression of BCG artifacts [64]. This method obviates the need for a reference template artifact as well as a cardiac pulse signal (such as EKG or finger pulse). However, as with all ICA-based methods, this assumes that the BCG artifact in each channel is a linear mixture of one or more independent "BCG sources."

Some recently developed techniques are based on the Teager energy operator [65], combined adaptive thresholding [66], and dilated discrete Hermite functions [67].

#### 12.2.2.4 Distortions in MR Images Because of EEG Hardware

The presence of the EEG cap and electrodes on the scalp could potentially cause magnetic susceptibility artifacts in the MRI images ("smearing" effects caused by

the presence of ferromagnetic materials in the area being imaged). However, with the use of specialized electrodes as mentioned in Section 12.2.2.1 (especially thin disk electrodes), these susceptibility artifacts in the MRI images are restricted to a few millimeters in depth, less than the thickness of the scalp and skull. Therefore, they do not interfere with the quality of an image of the cerebral cortex (and other deeper structures) [46, 47]. The use of carbon fiber leads further prevents distortions in the MR images [48]. The other source of artifacts in the MR images could potentially arise from "chemical shift" effects induced by the electrode gel. These artifacts are typically seen at the interface of fat and water in tissues and appear as dark or bright bands at the edges. The use of oil-based electrode gels should therefore be avoided. Figure 12.12 shows fMRI images of a patient (acquired with simultaneous EEG). Note that the electrode artifacts on the scalp do not affect the quality of the image of the cerebral cortex.

#### 12.2.2.5 Effect of MRI Environment on Neural Activity

It could be argued that the presence of a strong magnetic field could affect brain signals. Also, the environment of the MRI scanner, including the high-decibel noise during imaging, as well as the vibrations in the bore of the magnet, might have sufficient psychological effect to alter the EEG, as compared to similar recording sessions outside the scanner. The majority of scientific evidence suggests that there is no effect of the high-field-strength magnet on various EEG phenomena such as P300

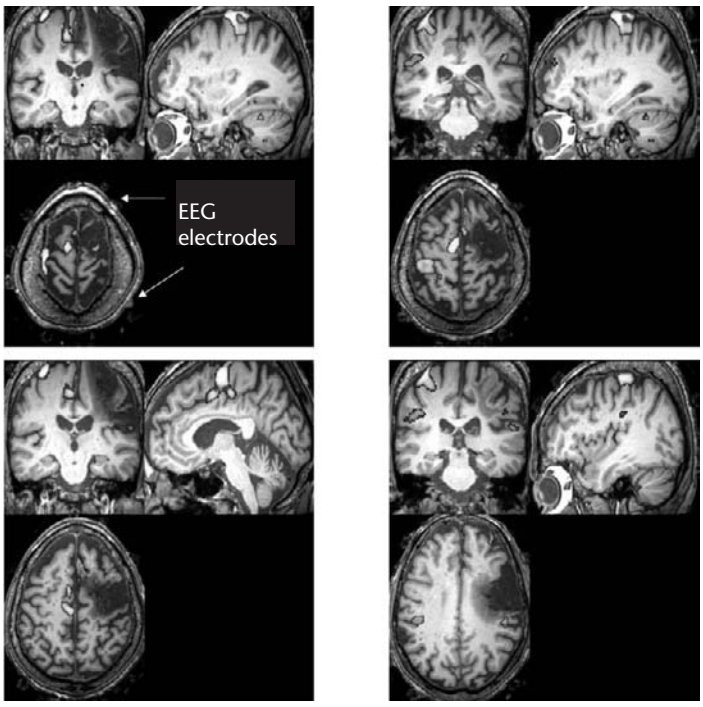



**Figure 12.12** Simultaneous EEG-fMRI data recorded from a subject performing left elbow flexions and extensions during poststroke rehabilitation (Johns Hopkins University). Note that the magnetic susceptibility artifacts underneath the EEG electrodes are restricted to the scalp and skull and do not distort the fMRI. (Courtesy of Johns Hopkins University.)

[68], VEPs [55], SSVEPs, or lateralized readiness potentials [69]. However, some studies in the past have suggested modified somatosensory evoked potentials and auditory evoked potentials [70, 71]. Further investigation is needed to analyze the effects of the scanner environment on spontaneous brain rhythms as well as evoked potentials and event-related potentials generated by different sensory modalities.

### 12.2.3 Using fMRI to Study EEG Phenomena

Since the first recording of the EEG by Hans Berger in the 1920s, efforts have been made to understand the significance and the genesis of various spontaneous rhythms as well as induced patterns in the EEG. These include the well-known alpha (8 to 12 Hz) and beta (16 to 25 Hz) rhythms, sleep spindles (12 to 16 Hz), interictal epileptic spikes, as well as evoked and event-related potentials. In most of these cases, inverse modeling approaches have been used to estimate the location of the generators of EEG current dipoles. However, due to volume conduction and the mixing effects of multiple dipoles, the localization of sources of EEG activity cannot be determined uniquely or with a high spatiotemporal resolution. Furthermore, these methods require strong a priori assumptions about the head model and impose restrictions on the location and number of possible dipoles, most of which are not easily verifiable [72]. fMRI, on the other hand, allows for imaging the functional activation of various anatomical regions of the brain with a high spatial resolution, including deeper structures such as the cerebellum and midbrain. Increasingly, fMRI is being used to investigate EEG phenomenon, such as brain rhythms, evoked potentials, or event-related potentials as well as conditions that can be monitored or classified using features of the EEG, such as epilepsy, sleep stages, or the like.

#### 12.2.3.1 fMRI Correlates of EEG Rhythms

In recent years various methods have been suggested for studying the relationship between fMRI and rhythms of the EEG. The most widely studied of these is the posterior alpha rhythm. The basis of these methods is to correlate the time course of the power of this frequency band (as observed in each EEG channel) with the BOLD response of each voxel. Power spectra are calculated using standard spectral estimation techniques such as the short time Fourier transform or wavelet analysis, followed by convolution with a universal hemodynamic response function (that characterizes the coupling of neuronal activation to the BOLD response). This time series is then correlated with the observed BOLD response of each voxel of the brain [50, 73–75]. Figure 12.13(a) depicts the results of such a study, showing correlation between spontaneous fluctuations of the posterior alpha rhythm and the BOLD signal between various regions of interest in the brain. Figure 12.13(b) shows the functional images obtained by this method.

Although this is simple and intuitive, each stage of this technique has been modified by other groups to overcome some of the drawbacks, not obvious at first glance. For instance, the EEG recorded by each channel is neither an exclusive result of the activity directly underneath the corresponding scalp electrode nor reflective of independent neuronal phenomenon. Each channel of the EEG is rather a

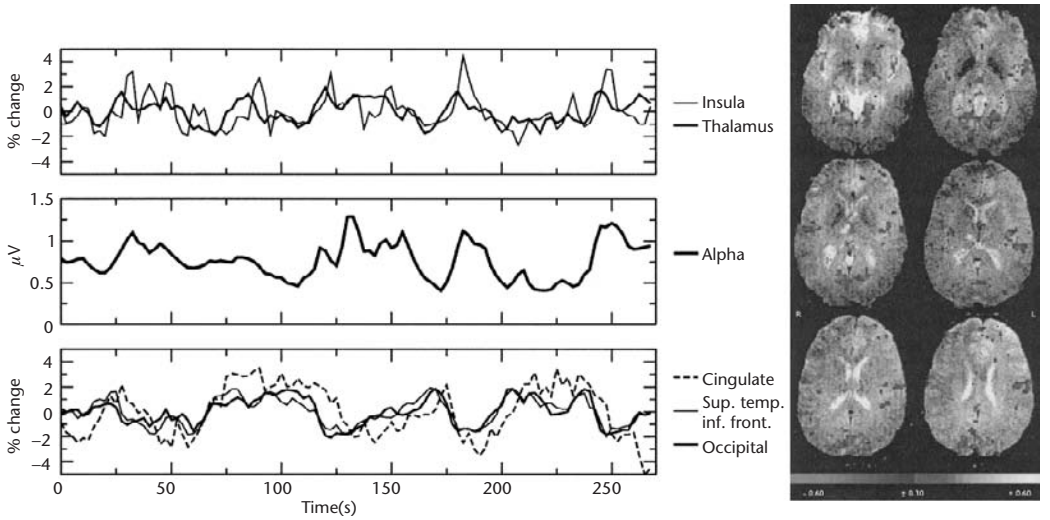



**Figure 12.13** (a) Time course of the average MRI percentage signal change for regions of interest in which the BOLD signal was positively (top) and negatively (bottom) correlated with alpha rhythm for a subject. At the center is the alpha power time course convolved with a hemodynamic response function, which was used as the independent response model to create the tomographic map of alpha activity. (b) Regions where the MR signal increased and decreased with elevations in alpha power. The bottom bar shows the Pearson correlation value between the signal intensity and alpha power modulation. (*From:* [50]. © 2002 Lippincott Williams & Wilkins. Reprinted with permission.)

weighted and superimposed activity of multiple neuronal current generators, which might be functionally and spatially separate. To study the fMRI correlates of the alpha rhythm arising from a particular region of interest (or as a result of a particular activity, such as attention modulation), it is necessary to account for, and reverse, this superposition and smearing effect as much as possible. Statistical methods such as ICA are proving to be promising in separating out these functionally independent activations, followed by extraction of the time course of their spectral power [76]. Alternatively, simple spatial filters such as the Laplacian or large Laplacian are able to somewhat localize the activity of the EEG to that originating from the cortical areas directly underneath the respective electrodes.

More recently, an alternative approach to this correlation analysis has been proposed, that of using the frequency band of interest as a regressor in the statistical model used for generating fMRI images.

Traditionally, fMRI images are based on statistical parametric mapping (SPM) and employ a combination of classical statistics and topological inference, describing voxel-specific responses to experimental conditions. A detailed overview of the physical and mathematical basis of fMRI is beyond the scope of this text and can be found in [77]. Briefly, fMRI data is first spatially processed and registered onto a common anatomical space. The responses in this space are characterized using the general linear model (GLM). The GLM serves to describe the responses using a convolution model of the standard hemodynamic response function (HRF). This tentatively explains the fact that BOLD signals are mathematically represented as the delayed vascular response to a neural excitation function.

To model the spatial nature of the imaging data, SPM techniques make use of random field theory. In simultaneously recorded EEG and fMRI, features of interest from the EEG can serve as regressors in the GLM. Traditionally, the regressors of the GLM model consist of behavioral and experimental conditions (appropriately convolved with a standardized HRF). When investigating the hemodynamic correlates of various spectral bands of the EEG, the GLM can be appropriately modified to incorporate the band power as a regressor in generating the statistical fMRI image. This approach has been used to study postmovement beta rebound [78], posterior alpha rhythm [73], and so forth. Figure 12.14 schematically depicts the various steps involved in generating the EEG regressors for the GLM model [73].

#### 12.2.3.2 Epilepsy: Spike-Correlated fMRI Analysis

Simultaneous EEG-fMRI is an emerging tool for studying the focus and spread of epileptic activity, as well as to gain a better understanding of its hemodynamic correlates. The methods used for epileptic spike–correlated fMRI analysis are somewhat similar to those described in the previous section, but with some marked differences. These arise from the fact that spike activity in the EEG is neither spatially nor temporally regular, and the "regressors" for fMRI therefore are not as simple as those for an EEG rhythm. Additionally, the HRF during epileptic activity

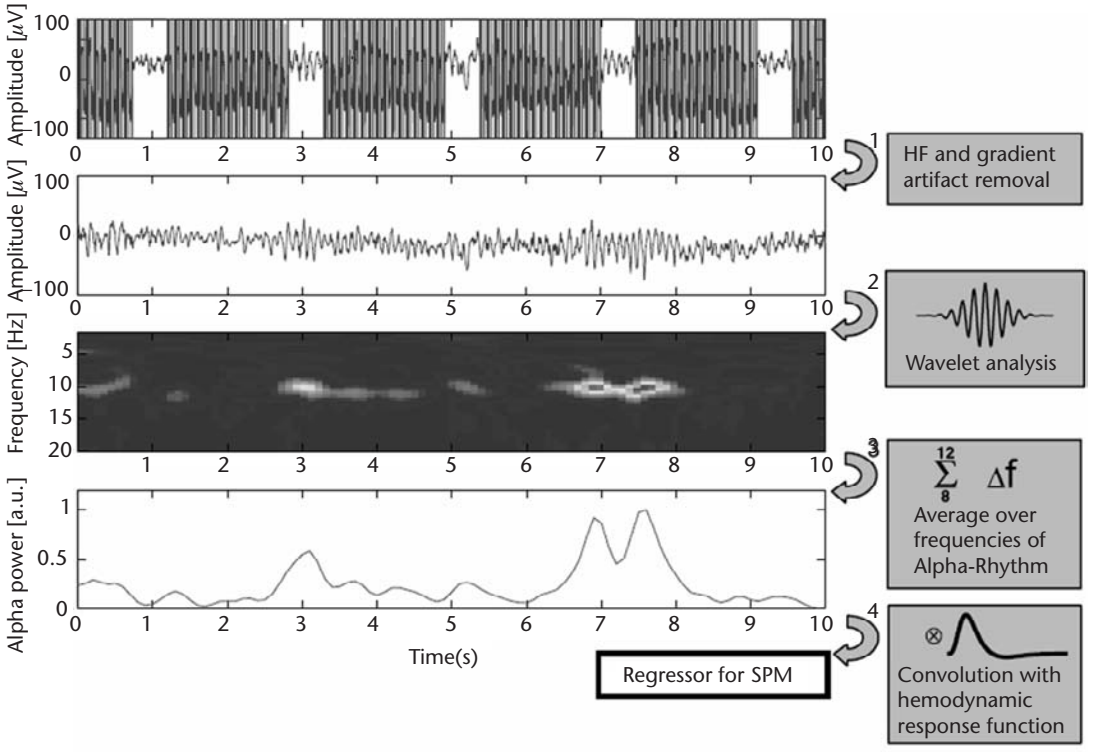



**Figure 12.14** Analysis of EEG acquired simultaneously with fMRI. This figure gives a schematic representation of the different steps (indicated by arrows) in the data analysis. Step 1, application of algorithms for MR artifact correction; step 2, time-frequency decomposition by wavelet analysis; step 3, estimation of the alpha power by averaging alpha-band frequencies; step 4, convolution with the hemodynamic response function to estimate a predictor for the BOLD signal. (*From:* [73]. © 2003 Elsevier. Reprinted with permission.)

is not similar to the standardized HRF, used for most fMRI analysis, and is the subject of numerous investigations [79–82].

One of the first objectives in using EEG-fMRI for the study of epileptic activity is to find which areas of the brain are activated by the physiological spikes in the EEG. It should be remembered that fMRI does not directly measure electrical activity of the neurons but the changes in blood oxygenation indirectly caused by this activity. The fMRI responses to EEG spikes are delayed and are dispersed by about 4 to 6 seconds. The responses can be modeled by convolution of the filtered EEG spike with an HRF. A simple model at a point $s$ in $D$-dimensional Euclidean space ($D = 3$ here) is a linear model

$$Y(s) = X\beta(s) + \sigma(s)\xi(s) \tag{12.10}$$

where $Y(s)$ is a column vector of $n$ observations at point $s$. $X$ is a design matrix incorporating the response to the neural excitations. In MATLAB, the matrix $X$ is obtained by convolving a column vector of 1s and 0s with the standard HRF. The 1s are placed at scan locations where the EEG data manifests a neural excitation (spike) and 0s elsewhere. $\beta(s)$ is an unknown coefficient, $\sigma(s)$ is a scalar standard deviation, and $\xi(s)$ is a column vector of temporally correlated Gaussian errors. The HRF can be modeled as a gamma function or as the difference between two gamma functions, whose parameters may be estimated as well, creating a nonlinear model [83–85].

The steps outlined above seem straightforward. However, the difficulty arises because: (1) each observation $Y(s)$ is an entire three-dimensional image, rather than a single value, and neighboring voxels tend to be correlated, and (2) all activations corresponding to EEG spikes may not have a direct correlation.

The analyses are generally carried out on a voxel-by-voxel basis. The parameter estimates are therefore suboptimal. The correlation from neighboring voxels is modeled by the variance in the term $\xi(s)$. The variance is reduced by spatial smoothing of the parametrized image data.

An advantage of using the GLM is that the design matrix can be extended to include columns that model effects of correlated noise in the fMRI data. The regressors added may be a vector that correlates the cardiac activity or other confounding effects such as low-frequency rhythms of the EEG. A primary confound used in most EEG-fMRI models incorporates the six rigid body transformation parameters (obtained while realigning the spatial data to standard stereotaxic coordinate space) [86].

The parameters of primary importance are the effects that are correlated to the physiological spikes in the EEG (i.e., corresponding to the first column of the design matrix) and their standard deviations. The key quantity for activation detection is their ratio, or $T$ statistic, $T(s)$. These parameters are not smoothed, because smoothing always increases bias as the cost for reducing noise. Smoothing is perhaps best reserved for parameters of secondary importance, such as temporal correlations or other ratios of variances and covariances.

The coefficient vector $\beta$ varies from voxel to voxel, and they are estimated separately for each voxel. The method of least-squares estimates $\beta$ by minimizing the sum of the squared residuals:

$$Q = (Y - X\beta)^T (Y - X\beta) \tag{12.11}$$

Differentiating with respect to $\beta$ and setting the derivative to zero yields

$$X^T (y - X\beta) = 0 \tag{12.12}$$

or

$$\hat{\beta} = (X^T X)^{-1} X^T Y \tag{12.13}$$

Using the formula for the variance of a linear transformation, the variance of the respective parameter estimates is obtained as

$$\begin{aligned} \text{var}(\hat{\beta}) &= \left[(X^T X)^{-1} X^T\right] \text{var}(Y) \left[(X^T X)^{-1} X^T\right]^T \\ &= \text{var}(Y)(X^T X)^{-1} \\ &= \sigma^2 (X^T X)^{-1} \end{aligned} \tag{12.14}$$

where $\sigma^2$ is the variance of the fMRI data at a particular voxel after removal of the temporal correlations.

The objective is to compare the spike-correlated fMRI to the rest state data (background EEG). Ignoring for simplicity any nuisance effects such as motion artifacts or cardiac effects, we have $X$ as an $n \times 2$ matrix ($n$ is the number of scans), with rows [1 0] for spike-correlated response and [0 1] for the background rest state. In this simple example, we write the $\beta$ vector as

$$\beta = [\beta_0 \beta_1]^T \tag{12.15}$$

Using a simple hypothesis testing, the activations are determined by rejecting the null hypothesis:

$$H_0 : \beta_0 = \beta_1 \tag{12.16}$$

These constraints on the parameters under the null hypothesis can be recast into the matrix form

$$[1 \ -1] \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \equiv C\beta = 0 \tag{12.17}$$

To check on this hypothesis, we obtain the unconstrained estimate $\hat{\beta}$ and the constrained estimate $\hat{\hat{\beta}}$ that satisfies the constraint $\mathbf{C}\hat{\hat{\beta}} = 0$. The matrix $\mathbf{C}$ is called the contrast matrix. The constrained estimate is obtained in terms of the unconstrained estimate using [87]

$$\hat{\hat{\beta}} = \hat{\beta} + \left(X^T X\right)^{-1} C^T \left[C\left(X^T X\right)^{-1} C^T\right]^{-1} C\hat{\beta} \tag{12.18}$$

In general, for number of parameters = $p$ and number of contrasts = $q$, the error variance $\sigma^2$ is estimated as

$$\hat{\sigma}^2 = \left(Y - X\hat{\beta}\right)\left(Y - X\hat{\beta}\right)\Big/(n - p) \tag{12.19}$$

which is $\chi^2$ distributed with $(n - p)$ degrees of freedom. It is always true that

$$Q_1 = \left(Y - X\hat{\beta}\right)^T \left(Y - X\hat{\beta}\right)$$

is less than

$$Q_0 = \left(Y - X\hat{\hat{\beta}}\right)^T \left(Y - X\hat{\hat{\beta}}\right) \tag{12.20}$$

because the $\hat{\beta}$'s are unconstrained.

When the null hypothesis is true (regions where there is no spike-correlated activation), an estimate of the error variance using $Q_1$ and $Q_0$ can be obtained as

$$\hat{\hat{\sigma}}^2 = \frac{(Q_1 - Q_0)}{q} \tag{12.21}$$

If $H_0$ is true, then the ratio

$$F \equiv \frac{\hat{\hat{\sigma}}^2}{\hat{\sigma}^2} = \frac{(Q_1 - Q_0)/q}{Q_0/(n - p)} \tag{12.22}$$

will be distributed according to an $F$-distribution with $q$ and $(n - p)$ degrees of freedom. For $(q = 1)$, as in the EEG-fMRI case illustrated, $F$ reduces to the square of the $t$-random variable with $n - p$ degrees of freedom.

If $H_0$ is true, then the numerator and denominator are both estimating $\sigma^2$, so the value of $F$ tends to be $\leq 1$. So a standard practice in EEG-fMRI analysis is to assume that $H_0$ is true, calculate the $F$-value, and compare the computed $F$-value against the critical value in an $F$-table with $q$ and $n - p$ degrees of freedom. If the computed value is larger than the critical threshold $F$-value, then one can reject the null hypothesis and accept the decision that the voxel shows a positive area of activation correlated to the EEG spikes. Figure 12.15 [49] shows an example of spike-correlated fMRI of an epileptic seizure.

#### 12.2.3.3 Evoked Potentials and Event-Related Potentials

Simultaneously recorded EEG and fMRI offers the possibility of mapping the neural substrates of evoked activity with a higher spatiotemporal resolution than is possible by either modality alone. Previously, various research groups have relied on sep-

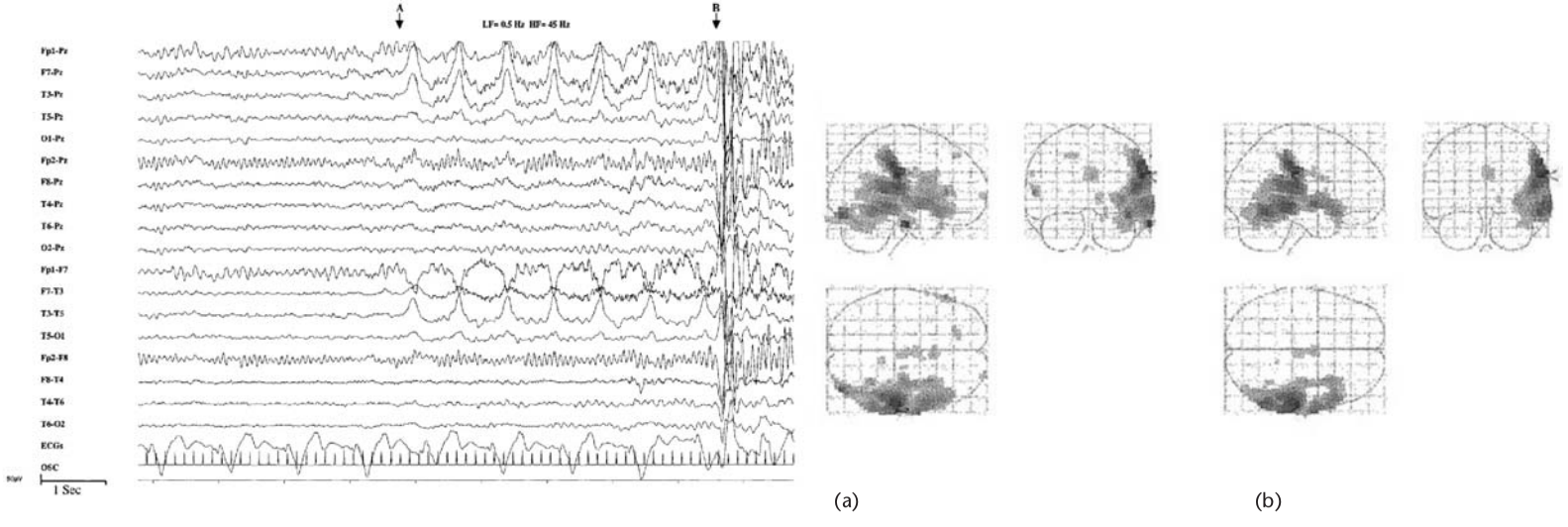



**Figure 12.15** (a, b) Spike-correlated fMRI of an epileptic episode. (*From:* [49]. © 2002 Elsevier. Reprinted with permission.)

arately acquired EEG and fMRI, recorded during similar experimental paradigms, to study a variety of evoked activity such as P300 potentials [68], N170 potentials [88], and visual spatial attention [89], to name a few. However, the evoked activity might not be similar across the two recording sessions, due to variations in mental conditions, the effect of the scanner environment, and the like. So simultaneous EEG-fMRI, either using interleaved acquisition or by employing artifact rejection algorithms, can obviate the major sources of variability. This approach has been used to map the temporal evolution of the source of VEPs by using fMRI data to constrain the EEG source localization problem [48]. Figure 12.16 depicts the millisecond-by-millisecond evolution of the neural sources of VEP activity when using simultaneous EEG-fMRI. This approach has been used to investigate auditory evoked potentials [90] and somatosensory evoked potentials as well [91]. Additionally, there is the possibility of using the evoked activity as a regressor to the GLM for fMRI analysis, similar to the methods described in Sections 12.2.3.1 and 12.2.3.2.

#### 12.2.3.4 Sleep Studies

Functional MRI studies of sleep (as well as mental vigilance) cannot rely solely on behavioral information but require some indicator of sleep (or vigilance) states. Sleep stages can be classified using ongoing EEG, and therefore fMRI studies of sleep have to rely on simultaneous EEG acquisition. This has been employed for investigating the neural correlates of REM sleep [92]. In such studies, "silent" imaging sequences are employed to prevent interruptions of the ongoing progression of the sleep cycle. This is an instance in which fMRI would not be possible without the aid of EEG.

### 12.2.4 EEG in Generation of Better Functional MR Images

Functional MRI relies on behavioral outcomes for statistical comparison of the BOLD signal between an idealized rest state and an experimental condition. However, there is a large variation in the neural activity during the "resting" phase, primarily due to attention, fatigue, motivation, and so forth. Markers in the EEG that are indicators of such mental states could potentially be used to build a better model to account for the spontaneous variability in the BOLD signal [74, 75]. The basic method would be similar to that outlined in Section 12.2.3.1. However, the sponta-

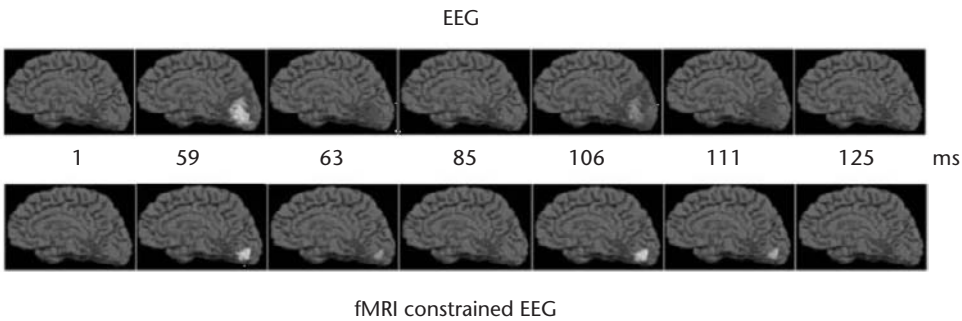



**Figure 12.16** Millisecond evolution of the sources of a VEP, as mapped using EEG alone and using fMRI-constrained EEG source localization. (*From:* [48]. © 2001 Elsevier. Reprinted with permission.)

neous fluctuations in EEG features that can be considered as vigilance/alertness monitors would be used to inform the fMRI model. In one such study [93] the posterior alpha rhythm (which is known to desynchronize with attention and increase in amplitude with mental fatigue and restfulness) has been used as a regressor in the GLM for fMRI image reconstruction. The main premise of these efforts is to deal with the fact that there is no idealized resting state, an assumption that is fundamental to most functional MR imaging techniques.

### 12.2.5 The Inverse EEG Problem: fMRI Constrained EEG Source Localization

The apparent appeal of combining EEG and fMRI is to get the best of both worlds, that is, higher temporal and spatial resolution attributed to each modality respectively. The inverse EEG source localization problem is highly ill posed, is heavily dependent on the forward model assumptions, and has infinite possible solutions to account for any given boundary condition. The use of BOLD response data to somehow constrain the solution of the inverse problem seems appealing and has generally yielded better results. But one needs to be mindful of the fact that EEG and BOLD changes are on different temporal scales, and more importantly, the presence of changes in one of them does not necessarily imply detectable changes in the other.

For instance, neural activities in deeper structures of the brain are easily detectable by fMRI but make minimal or no contribution to the EEG. Similarly, dipoles oriented tangentially to the scalp surfaces or opposing dipoles on either bank of a deep sulcus, for instance, result in negligible contributions to the EEG [94]. Conversely, one could record a large EEG contribution due to synchronous activity of only a few neurons but with minimal metabolic load and hence negligible contribution to the BOLD signal. Recent evidence from single-unit neuronal recordings using microelectrodes suggests that the substrates for neuronal activity and that of BOLD changes do not exactly match spatially. It is important to keep in mind the mismatches between the electrical and hemodynamic signals of the same neuronal event. However, incorporating fMRI data into the model used for solving the inverse EEG problem has the potential to improve spatial localization without compromising temporal resolution.

### 12.2.6 Ongoing and Future Directions

EEG and fMRI provide complementary information regarding neuronal dynamics, albeit at different spatial and temporal scales. Existing methodologies largely rely on using one of these two methodologies to study the other. However, recent methods that make joint inferences about neuronal activity from both electrical and hemodynamic data seem to offer additional benefits that traditional techniques cannot. These include using fMRI to inform the EEG forward model (for generating the inverse source localization problem), as well as using EEG to inform the fMRI forward model (by using EEG features as regressors in the GLM). A recent method that jointly analyzes both classes of data simultaneously is the N-PLS (multiway partial least squares) algorithm [95], which decomposes the multidimensional EEG and fMRI data into a sum of time-frequency "atoms." This is based on singular value decomposition of the covariance matrix between fMRI (spatial and temporal activ-

ity of each voxel) and EEG (spatial, temporal, and spectral activity of each channel). As compared to traditional fMRI, where voxels are analyzed independently of each other, this approach has the potential to take into account all of the fMRI voxels and EEG channels jointly.

The physiological basis of coupling between EEG (electrical activity) and BOLD (hemodynamic or metabolic activity) needs further investigation (and is indeed the subject of numerous studies). Most of these analysis methods rely on linear models, and nonlinearity in neuronal dynamics, especially as it relates to the interaction between EEG and BOLD, needs to be further investigated [96].

## Acknowledgments

M. Wagner thanks the editors, Shanbao Tong and Nitish V. Thakor, for the invitation to write this chapter and Manfred Fuchs and Jörn Kastner for helpful comments and ongoing discussions.

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CHAPTER 13

# Cortical Functional Mapping by High-Resolution EEG

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We present a methodology to assess cortical activity by estimating statistically significant sources using noninvasive high-resolution electroencephalography (HREEG). This implies the estimation of the cortical distributed sources of the HREEG data acquired during the execution of different tasks and the estimation of the cortical power spectra in the selected frequency bands relative to each task *A*nalyzed. The aim of the procedure developed is to assess straightforwardly significant differences between the cortical activities related to different experimental tasks. Such information is not appreciable by using conventional mapping procedures in the time domain. Furthermore, the same methodology allows us to separate from the cortical activity caused by the normal activity of the brain any statistically significant current density estimates related to the experimental task.

## 13.1 HREEG: An Overview

Information about brain activity can be obtained by measuring different physical variables arising from the brain processes, such as the increase in consumption of oxygen by the neural tissues or a variation of the electric potential over the scalp surface. All of these variables are connected in direct or indirect ways to the ongoing neural processes, and each variable has its own spatial and temporal resolution. The different neuroimaging techniques are thus confined to the spatiotemporal resolution offered by the variables being monitored. Human neocortical processes involve temporal and spatial scales spanning several orders of magnitude, from the rapidly shifting somatosensory processes characterized by a temporal scale of milliseconds and a spatial scale of few square millimeters to the memory processes, involving time periods of seconds and spatial scale of square centimeters. Today, no neuroimaging method allows a spatial resolution on a millimeter scale and a temporal resolution on a millisecond scale.

Electroencephalography is an interesting technique that presents a high temporal resolution, on the millisecond scale, adequate to follow the brain activity. However, this technique has a relatively modest spatial resolution, beyond the centimeter scale, because of the intersensor distances and the fundamental laws of electromagnetism [1].

The simultaneous activation of an entire population of neurons can generate an electric signal detectable on the head surface with electrodes placed on the scalp. To estimate the cortical activity, the EEG signal has to be measured at different scalp sites; the most common measurement system is the international montage 10-20.

Generally, the standard EEG analysis, using 20 to 30 electrodes, allows a spatial resolution of about 6 to 7 cm. HREEG is a technology used to increase the spatial resolution of the EEG potentials recorded on the scalp: In this case data is acquired using 64 to 128 electrodes and is then processed to remove the effects of attenuation caused by the low-conductivity structures of the head.

A key point of HREEG technologies is the availability of an accurate model of the head as a volume conductor by using anatomic MRI. These images are obtained by using the MRI facilities largely available in all research and clinical institutions worldwide. Reference landmarks such as nasion, inion, vertex, and preauricular points may be labeled using vitamin E pills as markers. T1-weighted MR images are typically used because they present maximal contrast between the structures of interest.

Contouring algorithms allow the segmentation of the principal tissues (scalp, skull, dura mater) from the MR images [2]. Separate surfaces of scalp, skull, dura mater, and cortical envelopes are extracted for each experimental subject, yielding a closed triangulated mesh. This procedure produces an initial description of the anatomic structure that uses several hundred thousand points—well too many for subsequent mathematical procedures. These structures are thus down-sampled and triangulated to produce scalp, skull, and dura mater geometrical models with about 1,000 to 1,300 triangles for each surface. These triangulations were found to be adequate to model the spatial structures of these head tissues.

A different number of triangles is used in the modeling of the cortical surface because its envelope is more convoluted than those of the scalp, skull, and dura mater structures. As many as 5,000 to 6,000 triangles may be used to model the cortical envelope for the purpose of following the spatial shape of the cerebral cortex. To allow coregistration with other geometrical information, the coordinates of the triangulated structures are referred to an orthogonal coordinate system ($x$, $y$, $z$) based on the positions of nasion and preauricular points extracted from the MR images. For instance, the midpoint of the line connecting the preauricular points can be set as the origin of the coordinate system, with the $y$ axis going through the right preauricular point, the $x$ axis lying on the plane determined by nasion and preauricular points (directed anteriorly), and the $z$ axis normal to this plane (directed upward). Once the model of the scalp surface has been generated, the integration of the electrode positions is accomplished by using the information about the sensor locations produced by the three-dimensional digitizer. The sensor positions on the scalp model are determined by using a nonlinear fitting technique. Figure 13.1 shows the results of the integration between the EEG scalp electrode positions and a realistic head model.

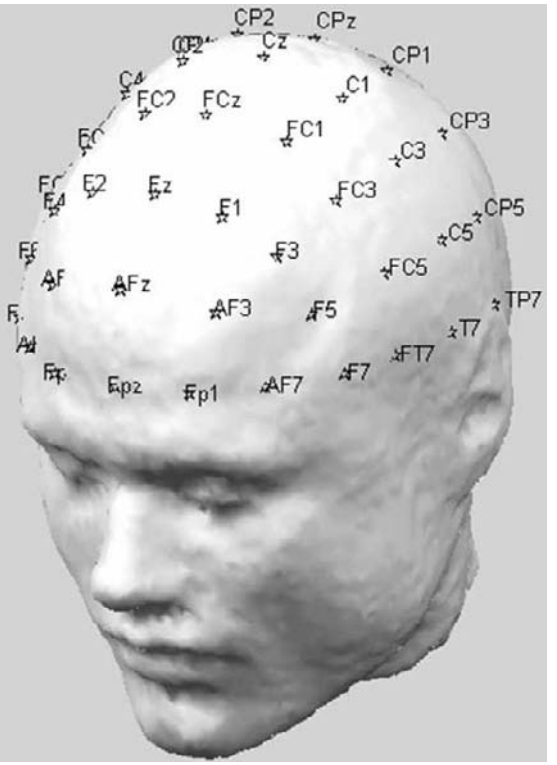



**Figure 13.1** Integration between EEG scalp electrode positions and a realistic head model generated using the T1-weighted MR images of the subject. When MRIs of a subject's head are not available, it is still possible to coregister the electrode positions employed for the EEG recording with an average head model, a standard head taken as an average of the MRIs of 150 subjects. Such an average head model is available from the McGill University Web site.

## 13.2 The Solution of the Linear Inverse Problem: The Head Models and the Cortical Source Estimation

The ultimate goal of any EEG recording is to produce information about the brain activity of a subject during a particular sensorimotor or cognitive task. When the EEG activity is mainly generated by circumscribed cortical sources (i.e., short-latency evoked potentials/magnetic fields), the locations and strengths of these sources can be reliably estimated by the dipole localization technique [3, 4]. In contrast, when the EEG activity is generated by extended cortical sources (i.e., event-related potentials/magnetic fields), the underlying cortical sources can be described using a distributed source model with spherical or realistic head models [5–7]. With this approach, typically thousands of equivalent current dipoles covering the cortical surface modeled and located at the triangle center are used, and their strengths are estimated using linear and nonlinear inverse procedures [7, 8]. Taking into account the measurement noise $n$, supposed to be normally distributed, an estimate of the dipole source configuration that generated a measured potential $\mathbf{b}$ can be obtained by solving the linear system:

$$\mathbf{Ax} + n = \mathbf{b} \tag{13.1}$$

where **A** is an $m \times n$ matrix with the number of rows equal to the number of sensors and the number of columns equal to the number of modeled sources. We denote with $A \cdot j$ the potential distribution over the $m$ sensors due to each unitary $j$th cortical dipole. The collection of all of the $m$-dimensional vectors $A \cdot j$, ($j = 1, \ldots, n$) describes how each dipole generates the potential distribution over the head model, and this collection is called the lead field matrix **A**. This is a strongly underdetermined linear system, in which the number of unknowns, the dimension of the vector **x**, is greater than the number of measurements **b** by about one order of magnitude. In this case, from the linear algebra we know that infinite solutions for the **x** dipole strength vector are available, explaining in the same way the data vector **b**. Furthermore, the linear system is ill-conditioned as a result of the substantial equivalence of several columns of the electromagnetic lead field matrix **A**. In fact, we know that each column of the lead field matrix arose from the potential distribution generated by the dipolar sources that are located in similar positions and have orientations along the cortical model used. Regularization of the inverse problem consists in attenuating the oscillatory modes generated by vectors that are associated with the smallest singular values of the lead field matrix **A**, introducing supplementary and a priori information on the sources to be estimated.

In the following, we use the term *source space* to characterize the vector space in which the "best" current strength solution **x** will be found. *Data space* is the vector space in which the vector **b** of the measured data is considered. The electrical lead field matrix **A** and the data vector **b** must be referenced consistently.

Before we proceed to the derivation of a possible solution for the problem, we recall a few definitions from algebra that will be useful. A more complete introduction to the theory of vector spaces is outside the scope of this chapter, and the interested readers could refer to related textbooks [9, 10]. In a vector space provided with a definition of an inner product $(\cdot, \cdot)$, it is possible to associate a value or modulus to a vector **b** by using the notation

$$(b, b) = \|b\| \tag{13.2}$$

Any symmetric positive definite matrix **M** is said to be a metric for the vector space furnished with the inner product $(\cdot, \cdot)$, and the squared modulus of a vector **b** in a space equipped with the norm $M$ is described by

$$\|b\|_M^2 = b^T M b \tag{13.3}$$

With this in mind, we now face the problem of deriving a general solution of the problem previously described under the assumption of the existence of two distinct metrics $N$ and $M$ for the source and the data space, respectively. Because the system is undetermined, infinite solutions exist. However, we are looking for a particular vector solution **x** that has the following properties: (1) It has the minimum residual in fitting the data vector **b** under the norm $M$ in the data space, and (2) it has the minimum strength in the source space under the norm $N$. To take into account these

properties, we have to solve the problem utilizing the Lagrange multiplier and minimizing the following functional that expresses the desired properties for the sources **x** [7, 11–14]:

$$\phi = \|Ax - b\|_M^2 + \lambda^2 \|x\|_N^2 \tag{13.4}$$

The solution of the problem depends on the adequacy of the data and source space metrics. Under the hypothesis of $M$ and $N$ positive definite, the solution is given by taking the derivative of the functional and setting it to zero. After a few straightforward computations the solution is

$$x = Gb \tag{13.5}$$

$$G = N^{-1}A'\left(AN^{-1}A' + \lambda M^{-1}\right)^{-1} \tag{13.6}$$

where **G** is called the *pseudoinverse matrix*, or the *inverse operator*, that maps the measured data **b** onto the source space. Note that the requirement of positive definite matrices for the metrics $N$ and $M$ allows us to consider their inverses. The last equation stated that the inverse operator **G** depends on the matrices $M$ and $N$ that describe the norm of the measurements and the source space, respectively. The metric $M$, characterizing the idea of closeness in the data space, can be particularized by taking into account the sensor noise levels by using the Mahalanobis distance [13].

If no a priori information is available for the solution of the linear inverse problem, the matrices $M$ and $N$ are set to the identity, and the minimum norm estimation is obtained [15]. However, it was recognized that in this particular application the solutions obtained with the minimum norm constraints are biased toward those sources that are located nearest to the sensors. In fact, there is a dependence of the distance on the law of potential (and magnetic field) generation, and this dependence tends to increase the activity of the more superficial sources while depressing the activity of the sources far from the sensors. The solution to this bias was obtained by taking into account a compensation factor for each dipole that equalizes the "visibility" of the dipole from the sensors. Such a technique, called *column norm normalization* was adopted largely by the scientists in this field. With the column norm normalization, the inverse of the resulting source metric is

$$\left(N^{-1}\right)_{ii} = \|A_{-1}\|^{-2} \tag{13.7}$$

in which $(N^{-1})_{ii}$ is the $i$th element of the inverse of the diagonal matrix **N** and $\|A_{-i}\|$, is the L2 norm of the $i$th column of the lead field matrix **A**. In this way, dipoles close to the sensors, and hence with a large $\|A_{-i}\|$, will be depressed in the solution of the inverse problem because their activations are not convenient from the point of view of the functional cost. The use of this definition of matrix **N** in the source estimation is known as the *weighted minimum norm* solution [5, 6].

## 13.3 Frequency-Domain Analysis: Cortical Power Spectra Computation

Here we propose an alternative procedure, which is based on the estimation of the spectral power of cortical signals based on the spectral power of scalp measurements. Let

$$b(t) = Ax(t) \tag{13.8}$$

be the vector of scalp measurements in a given time window and

$$b(f) = Ax(f) \tag{13.9}$$

be the corresponding vector of cortical estimates, if $\mathbf{b}(f)$ is the Fourier transform of $\mathbf{b}(t)$.

We also have

$$x(f) = Gb(f) \tag{13.10}$$

where **G** is the pseudo inverse of the lead field matrix **A**.

We define the matrix of cross-power spectral densities (CSDs) as the matrix whose element $(i, j)$ is the cross spectrum of the $i$th and $j$th channel of the signal. Thus,

$$CSD_b(f) = b(f)b^H(f) \tag{13.11}$$

where the $\mathbf{b}^H(f)$ is the conjugate-transposed (Hermitian) of $\mathbf{b}(f)$.

Analogously

$$CSD_x(f) = x(f)x^H(f) \tag{13.12}$$

Using the previous equation,

$$CSC_x(f) = x(f)x^H(f) = Gb(f)b^H(f)G^T = G\,CSD_b(f)G^T \tag{13.13}$$

If $\mathbf{b}(t)$ and $\mathbf{x}(t)$ are not deterministic signals, but rather we have several trials (realizations) of a stochastic process, the last equation holds if we substitute every CDS with its expected or estimated values.

In our case we need only the power spectral densities of estimated cortical sources; for this reason we need to compute only the diagonal of $CSD_x(f)$.

The estimation of the cortical activity returns a current density estimate for each of the about 3,000 to 5,000 dipoles constituting the modeled cortical source space. Each dipole presents a time-varying magnitude representing the spectral power variations generated during the course of the task. This rather large amount of data can be synthesized by computing the ensemble average of the magnitudes of all of the dipole belonging to the same cortical ROI. Each ROI was defined on the cortical model adopted in accordance with the BAs, which are regions of the cerebral cortex whose neurons share the same anatomic (and often also functional) properties. After a decade of neuroimaging studies, the different BAs have been assigned precise roles in the reception and analysis of sensory and motor commands, as well as in memory

processing. Actually, such areas are largely used in neuroscience as a reference system for sharing cortical activation patterns found with different neuroimaging techniques.

As a result of this anatomically guided data reduction, we pass from the analysis of about 3,000 time series to the evaluation of fewer than a hundred (the BAs located in both cerebral hemispheres). These BA waveforms, related to the increase or decrease of the spectral power of the cortical current density in the investigated frequency band, can be successively averaged across the subjects of the studied population. The grand-average waveforms describe the time behavior of the spectral power increase or decrease of the current density in the population during the task examined.

## 13.4 Statistical Analysis: A Method to Assess Differences Between Brain Activities During Different Experimental Tasks

When an experiment is being conducted for EEG measurements, typically the task is repeated by the subject a variable number of times (often called trials) in order to collect enough EEG data to allow a statistical validation of the results.

Let **S** be the matrix of the power spectra of the cortical sources computed, which has a dimension equal to the number of sources times the frequency bin used times the number of trials recorded. We compute the average of the power spectral values related to *i*th dipole within the *j*th frequency band of interest (theta, 3 to 7 Hz; alpha, 8 to 12 Hz; beta, 13 to 29 Hz; gamma, 30 to 40 Hz); this operation is repeated for each source and each frequency band. Thus, for each frequency band we have a matrix $\overline{\mathbf{S}}_j$ (with dimension sources **x** trials) which represent the distribution along the number of trials of the mean spectral power of each cortical sources.

The aim of the procedure is to find the differences between the cortical power *distributions* related to two different experimental tasks performed by the subject, say, task *A* and task *B*. For this reason, we compute the matrices $\overline{\mathbf{S}}_j$ related to the EEG data recorded during task *A* and task *B*, and we refer to them as $\overline{S_j^A}$ and $\overline{S_j^B}$. Then, we perform a statistical contrast between such spectral matrices $\overline{S_j^A}$ and $\overline{S_j^B}$ using appropriate univariate statistical tests (such as the Student's test with the correction for multiple comparisons).

For the *i*th source and the *j*th frequency band, we denote by $\mu_A^{i,j}$ and $\mu_B^{i,j}$ the mean values of the cortical power spectra distributions. In the following we want to verify the null hypothesis $H_0$ that such mean values are statistically similar:

$$H_0 : \mu_A^{i,j} = \mu_B^{i,j} \text{ and } H_A : \mu_A^{i,j} \neq \mu_B^{i,j},$$

where $H_A$ is the alternative hypothesis, that such differences are instead significantly different.

Under the assumptions that: (1) the two samples came from normal (Gaussian) populations, and (2) the populations have equal variances, Student's *t*-value for testing the previous hypotheses can be expressed as

$$t = \frac{\overline{X_A} - \overline{X_B}}{S_{\overline{X_A} - \overline{X_B}}} \tag{13.14}$$

The quantity $\overline{X_A} - \overline{X_B}$ is simply the difference between the two means, and $S_{\overline{X_A} - \overline{X_B}}$ is the standard error of the difference between the sample means. The last quantity is a statistic that can be calculated from the sample data, and it is an estimate of the standard error of the population, indicated by $\sigma_{\overline{X_A} - \overline{X_B}}$. It can be shown mathematically that the variance of the difference between two independent variables is equal to the sum of the variances of the two variables, so the standard error of the population could be computed as the sum of the standard error of the two groups as follows:

$$\sigma_{\overline{X_A} - \overline{X_B}} = \sigma_{\overline{X_A}} + \sigma_{\overline{X_B}}$$

Independence means that there is no correlation between the two variables A and B. Because $\sigma_{\overline{X}} = \sqrt{\frac{\sigma^2}{n}}$, where $n$ is the population dimension, we can write

$$\sigma_{\overline{X_A} - \overline{X_B}} = \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}} \tag{13.15}$$

Because the two-sample Student's $t$-test requires the homoscedasticity of the variances of the two samples (i.e., we assume that $\sigma_A^2 = \sigma_B^2 = \sigma^2$), we can write

$$\sigma_{\overline{X_A} - \overline{X_B}} = \sqrt{\frac{\sigma^2}{n_A} + \frac{\sigma^2}{n_B}} \tag{13.16}$$

Thus, to calculate the estimate of $\sigma_{\overline{X_A} - \overline{X_B}}$, an estimate of $\sigma^2$ is required. Denoting by $s_A^2$ and $s_B^2$ the statistically similar estimators of the variance $\sigma^2$, we compute the pooled variance $s_p^2$ to obtain the best estimate of $\sigma^2$

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2} \tag{13.17}$$

and

$$s_{\overline{X_A} - \overline{X_B}}^2 = \frac{s_p^2}{n_A} + \frac{s_p^2}{n_B} \tag{13.18}$$

Thus,

$$s_{\overline{X_A} - \overline{X_B}} = \sqrt{\frac{s_p^2}{n_A} + \frac{s_p^2}{n_B}} \tag{13.19}$$

Finally, we obtain

$$t = \frac{\overline{X_A} - \overline{X_B}}{\sqrt{\frac{s_p^2}{n_A} + \frac{s_p^2}{n_B}}} \tag{13.20}$$

The results of the previous statistical analysis can be represented on a realistic model of the cortex of the subjects for each of the frequency bands of interest. The model will show only the statistically significant cortical activations, pointing out the differences between the characteristic activations during the tasks being considered (see Figure 13.2).

The figure represents statistically significant spectral cortical activations represented on the model of the cortex related to the average head model provided by McGill University. The figure shows the areas of statistically significant spectral cortical activity occurring in the brain of a representative subject during the execution of task *A* as compared to the brain activity elicited by the execution of task *B*. Usually, task *B* is related to some "rest" state, whereas task *A* is related to the formal experiment being conducted. Generally, these statistical brain pictures could be generated for each frequency band investigated. In the different views of the cortical surface, grayscale is used to highlight the cortical zones in which the brain activity during task *A* is statistically significantly different from the cortical activity during task *B*. In contrast, the cortical areas that are activated in a similar way during tasks *A* and *B* for the particular subject are presented in gray. The dark gray indicates the maximum of the statistical differences between the cortical power spectra estimated during tasks *A* and *B* in the particular subject investigated, after the Bonferroni correction for multiple comparisons. The black is at the lowest level of statistical significance at 5% Bonferroni corrected.

Until now, the assumptions of the normality and homoscedasticity of the estimated spectra were assumed in order to perform the statistical analysis with the Student's test. It might be argued which test could be used in the case in which such

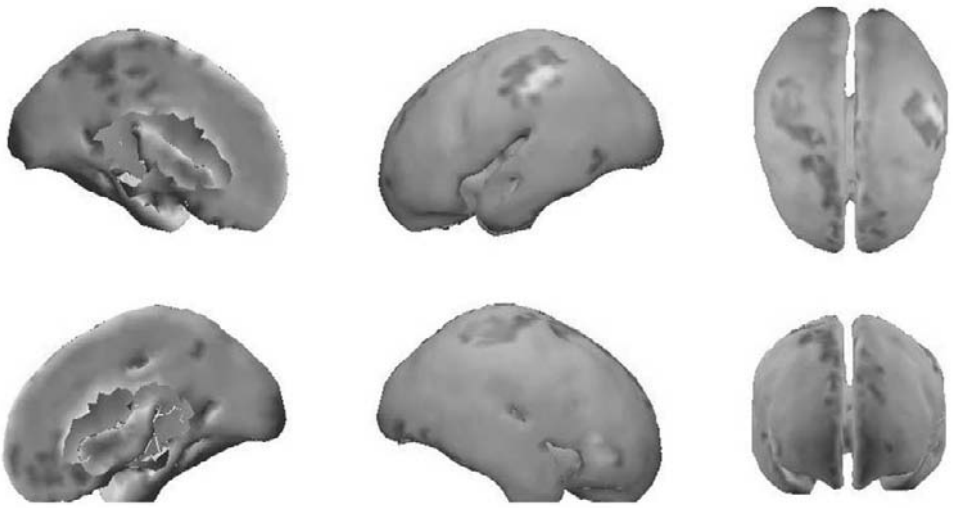


**Figure 13.2** Statistically significant spectral cortical activations represented on the model of the cortex related to the average head model provided by McGill University.

assumptions did not hold. In this respect the good news is that the robustness of the Student's test from the assumption of homoscedasticity and normality are generally very high, and hence we can use the test even though there is no precise information related to the Gaussian and the homoscedastic nature of the data [16]. However, the appropriate statistic to deal with the heteroscedastic case is the Sattertwaite or Cochran and Cox estimation of the standard error to insert in the formulation presented above. Such calculations can be found in a standard statistic textbook [16], but they lead to results very similar to those obtained with the standard Student's approach.

As a final statistical issue, it is well known that many univariate statistical tests (like those presented in this application, one for each cortical dipole modeled) can easily generate the appearance of false positive results (known as type I errors or *alpha inflation*). The large number of univariate tests performed results in statistically significant differences being found between two analyzed samples when no real differences exist. The usual conservative approach in this case is to use a Bonferroni correction, a procedure that simply defines as statistically significant at 5%, for instance, all of the statistical results that are still significant when their probability is divided by the number of univariate tests performed. Let $N$ be the number of univariate tests to be performed and $t_{0.05}$ be the statistical threshold for a single univariate $t$-test to perform for our contrast at a 5% level of statistical significance. We can state that the results will be statistically significant at the 5% level Bonferroni corrected ($t_{0.05\text{Bonf}}{}^*$) all of those cortical dipoles that present a $t$-value associated with a probability $p_{0.05}$ higher than

$$p_{0.05}{}^* = p_{0.05}/N \tag{13.21}$$

Usually, the Bonferroni correction is a very conservative measure of statistical differences between two groups during the execution of multiple univariate tests. We can verify this with a practical example: If the number of cortical dipoles to be tested on a simple realistic head model is about 3,000 ($N = 3{,}000$) and the number of EEG trials analyzed for the subject is 10 ($M = 10$), the value for a single univariate Student $t$-threshold will be $t_{0.05} = 2.26$. This means that during the statistical comparisons on each of the 3,000 dipoles, all of the $t$-values higher than 2.26 will be declared statistically significant at 5% (i.e., with a $p_{0.05}$). However, because we know that many false positives could occurs due to the execution of multiple $t$-tests, by adopting the Bonferroni procedure we will instead declare statistically significant at 5% Bonferroni corrected all of the statistical tests that returns a $p$ value higher than

$$P_{0.5\text{Bonf}} = 0.05/3000 = 0.000016$$

Here is a value of $t$ that generates such a probability for a population of 10 EEG trials:

$$t_{0.05}{}^* = 11$$

This means that by using the Bonferroni correction, we can declare as statistically significant at 5% Bonferroni corrected all of those cortical areas that generate a

$t$-value equal to or higher than the value needed to have a statistical significance in a single univariate test of $p = 0.00001$.

Due to the excessive severity of the Bonferroni correction, the scientific community has also shown interest in other methods that are less conservative in protecting against type I errors. Examples are the procedures for controlling the false discovery rate, described and discussed in several papers by Benjamini and coauthors [17, 18], or the Holm-Bonferroni procedure [19]. An example will shed light on the last procedure, usually employed in some source localization algorithms for EEG or MEG data.

Suppose that there are $k$ hypotheses to be tested and the overall type 1 error rate is $\alpha$. In our context, $k$ could be equal to 3,000 (one $t$-test for each cortical dipole), and the error rate is 5%. Execution of the multiple univariate tests results in a list of 3,000 $p$-values. The issue now is how to deal with such $p$-values by using the Holm-Bonferroni procedure. This procedure starts by ordering the $p$-values and comparing the smallest $p$-value to $\alpha/k$, the value of the Bonferroni correction to be adopted for only one $p$-value. If that $p$-value is less than $\alpha/k$, then that hypothesis can be rejected and the procedure started over again with the same $\alpha$. The procedure tests the remaining $k - 1$ hypotheses by ordering the $k - 1$ remaining $p$-values and comparing the smallest one to $\alpha/(k - 1)$. This procedure is iterated until the hypothesis with the smallest $p$-value cannot be rejected. At that point the procedure stops, and all hypotheses that were not rejected at previous steps are accepted. This procedure is obviously less severe than the simple application of the Bonferroni test on all of the $p$-values with the threshold level $\alpha/k$.

## 13.5 Group Analysis: The Extraction of Common Features Within the Population

In the previous paragraphs, we have considered the generation of a statistical representation of the cortical areas that differ in spectral power in a particular subject during the execution of a task $A$ as compared to a task $B$. We also discussed a way to validate the statistical results against the type I error due to the execution of multiple univariate tests. Here, we move to the problem of generating a power spectra cortical map representing the common statistical features for the analyzed population in a particular frequency band. In this respect it is mandatory that the statistical cortical activity estimated for each subject can be reported to a common cortical source space. This could be performed by using the Tailaraich transformation, as usually employed by scientists using fMRI.

After the statistically significant areas of cortical activation in all subjects have been reported on a common cortical source space, it is possible to generate a group representation of the results. We can again use color to indicate whether the activation of a single cortical voxel is significant in all of the population analyzed, or in all but one analyzed, and so forth.

Figure 13.3 contrasts the areas of common statistically significant activity during the execution of tasks $A$ and $B$ not by a single subject (as in Figure 13.2) but rather by the entire population analyzed. The brain again is shown from different perspectives, and the quantities mapped are the differences in spectral activation in

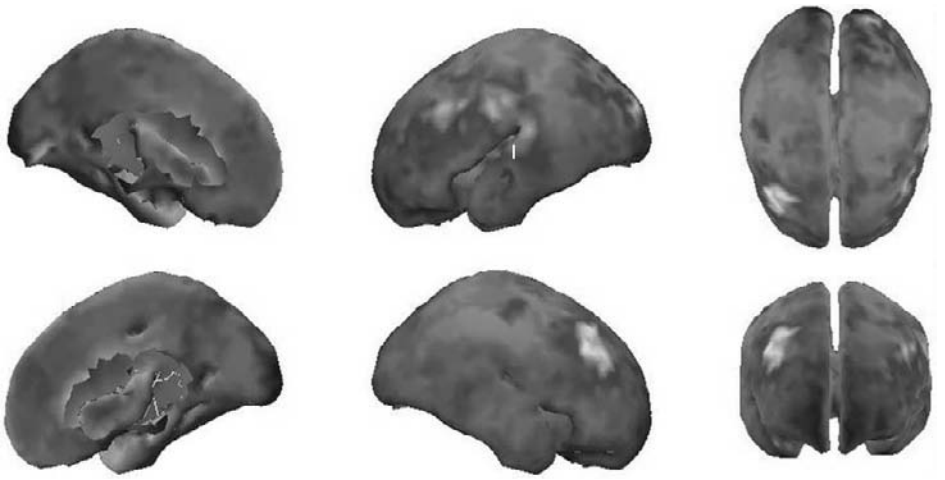


**Figure 13.3** Representation of the common spectral cortical activation within the analyzed experimental population using the average cortical model provided by McGill University as a common source space.

a particular frequency band during the execution of two tasks by a population. The yellow indicates the cortical areas where significant differences were found in the power spectra activity of all of the subjects, the red indicates the areas where differences were noted in all but one of the subjects, and so forth. The gray indicates areas in which the spectral activity is not common to all the subjects. All of the cortical spectral activities presented in color are statistically significant at 5% Bonferroni corrected for multiple comparisons.

## 13.6 Conclusions

The capabilities of the high-resolution EEG mapping techniques have reached levels where they are able to follow the dynamics of brain processes with a high temporal resolution and an appreciable spatial resolution, on the order of 1 or 2 square centimeters. Appropriate mathematical procedures are now able to return us information about the cortical sources active during the execution of a series of experimental tasks by a single subject or a group of subjects. Such procedures are quite similar to those employed for the past decade by scientists using the fMRI as a brain imaging device. In this chapter we briefly presented a body of techniques able to return useful information about the cortical areas where statistically significant brain activity occurs in the spectral domain during the execution of two tasks. Such techniques are relatively easy to implement and could constitute a valid support for the EEG data analysis of complex experiments involving several subjects and different experimental paradigms.

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CHAPTER 14

# Cortical Function Mapping with Intracranial EEG

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Although PET and fMRI images of human brain function have captured the imagination of both scientists and the lay public, they still offer only indirect and delayed measures of synaptic activity. EEG (and MEG) offer the only real-time noninvasive measures of human neuronal activity, and the multidimensional complexity of their signals potentially yields far more information about brain function than the scalar measure of neuronal metabolism. Advances in EEG/MEG signal analysis have allowed investigators to mine the riches of these signals to study not only which individual brain regions "light up" during cognitive operations, but also how complex cognitive operations can arise from the dynamic and cooperative interaction of distributed brain regions.

The strengths of EEG and MEG as tools for cortical function mapping, however, have often been offset by the inverse problem for identifying their signal sources. Under the unusual circumstances of patients undergoing surgery for epilepsy, this limitation can largely be circumvented by recording from electrodes that are surgically implanted in or on the surface of the brain. This chapter reviews how this and other advantages of intracranial EEG (iEEG) have been exploited in recent years to map human cortical function for both clinical and research purposes.

## 14.1 Strengths and Limitations of iEEG

The circumstances under which iEEG is clinically necessary are relatively few [1] and consist primarily of patients undergoing surgical treatment for intractable epilepsy, brain tumors, or vascular malformations of the brain. These circumstances account for the most important limitation of iEEG, which is that the recorded signals and their responses to functional brain activation may be affected by abnormalities of brain structure and neurophysiology that accompany brain disease. For example, iEEG recordings in patients with intractable epilepsy may be contaminated by epileptiform discharges or may be distorted by functional reorganization due to chronic seizures. Nevertheless, the improved spatial resolution of iEEG and its higher signal-to-noise ratio, particularly for high-frequency activity, provide an unprecedented opportunity for studying the spatial and temporal characteristics of

different electrophysiological indices of cortical activation and cortical network dynamics.

iEEG recordings not only have excellent temporal resolution, like all electrophysiological techniques, but they also can be considered to have a "mesoscopic" spatial resolution that is intermediate between the microscopic scale of single and multiunit recordings of neuronal activity, which are almost exclusively the purvey of animal researchers, and the macroscopic scale of EEG/MEG recordings. This unique ability of iEEG to see "both the forest and the trees" has made it particularly useful for studying how populations of cortical neurons organize their activity during perceptual and cognitive tasks.

## 14.2 Intracranial EEG Recording Methods

iEEG recordings can be made with a variety of electrode sizes and configurations. These include multicontact depth electrodes that may be implanted stereotactically into deep structures such as amygdala, hippocampus, orbital-frontal cortex, or cortex in the depths of the interhemispheric fissure. Information from CT, MRI, or cerebral angiograms may be used to avoid hemorrhage from blood vessels along the implant trajectory. An important advantage of this approach is that the holes that must be drilled in the skull for each electrode penetration are relatively small, reducing the likelihood of infection. Furthermore, a variety of contact sizes may be used, including microwires from which micro-EEG or local field potentials (LFPs) can be recorded. Many epilepsy surgery centers prefer this approach, even for studying cortical tissue on the lateral surface of the brain. However, comprehensive coverage of broad cortical regions requires an increasing number of penetrations, and when a large cortical region must be sampled, to localize a patient's seizure focus and map eloquent cortex in and around the potential zone of resection, for example, it may be more practical to use subdural ECoG.

Subdural ECoG (also called *epipial recording*) is performed with an array of electrodes that usually have a larger surface area (for example, 2.3-mm-diameter exposed surface) than depth electrodes have and are embedded in a soft silastic sheet in a variety of configurations. These may include strips that contain a single row of electrodes, usually up to 8 cm long with 1-cm center-to-center spacing (eight electrodes). Multiple electrode strips may be implanted through a single burr hole in order to sparsely sample a large cortical region. More comprehensive coverage of a particular cortical region usually requires two-dimensional arrays of electrodes, or grids. These grids vary in dimensions from $2 \times 4$ to $8 \times 8$ $cm^2$, but they can be customized to cover the cortical territory of interest. Because they cannot fit through burr holes, implantation of these grids requires a craniotomy (Figure 14.1), which is a more invasive procedure with a greater likelihood of complications.

Electrode placement is dictated solely by clinical concerns and is usually based on scalp EEG recordings of the patient's seizures and interictal epileptiform discharges, as well as the proximity of the estimated seizure focus to eloquent cortical areas, that is, motor, sensory, or language cortices. However, because the exact locations of the seizure focus and eloquent cortex are not known a priori, electrode coverage is not necessarily limited to cortical regions with abnormal structure or

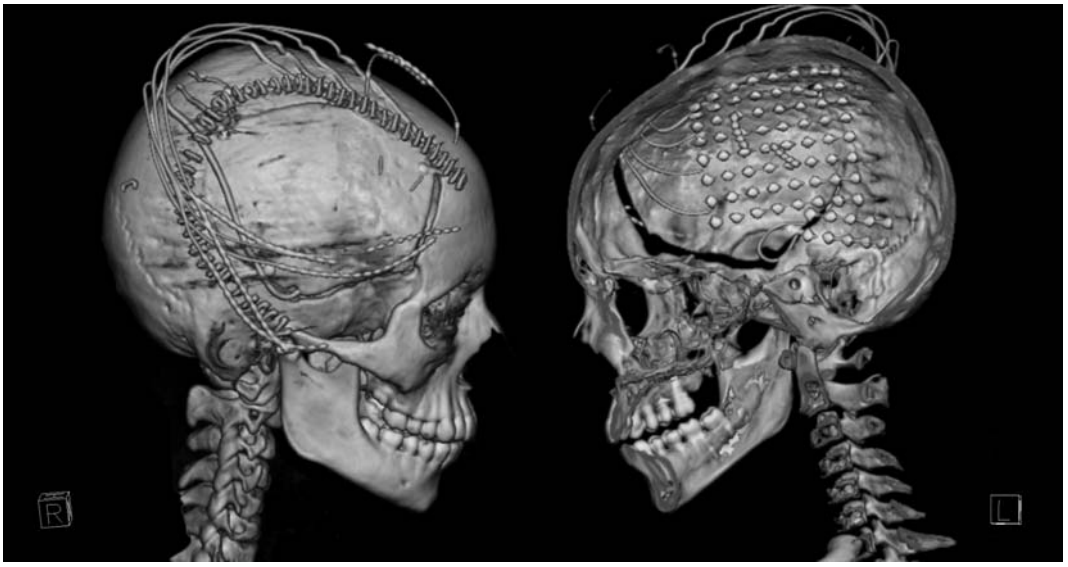



**Figure 14.1** Three-dimensional reconstruction of computed tomogram after implantation of subdural grids and depth electrodes. The left image (R = right projection) shows craniotomy with overlying staples and iEEG wires exiting the skull. The right image shows iEEG electrodes implanted over the right hemisphere and visible from a left projection (L) with the left half of the skull invisible.

neurophysiology. For example, when preoperative studies indicate a seizure focus in the frontal lobe but there is no lesion, electrode placement may include not only a large area of prefrontal cortex but also part of the parietal cortex so that sensorimotor cortex can be identified by electrocortical stimulation mapping (ESM) or somatosensory evoked potentials and thus spared in the event of a large prefrontal cortical excision.

Early studies of iEEG often required systems in which the head boxes, amplifiers, ADCs, and/or recording computers had to be built from scratch or extensively modified from existing scalp EEG systems. For more than a decade, commercially available systems for long-term video EEG monitoring have incorporated iEEG specifications that are adequate for clinical purposes, but until recently these specifications were not optimal for cortical function mapping with ERPs and/or other electrophysiological correlates of cortical activation. In the past few years, commercial systems have increased their capacity for the number of recorded channels, as well as the sampling rate per channel. For example, the Johns Hopkins Epilepsy Monitoring Unit uses a system (Stellate Systems, Montreal, Canada) capable of recording up to 128 channels at a sampling rate of 1,000 Hz. This is adequate for most clinical and research applications of iEEG, but it is not uncommon to implant more electrodes than can be recorded at once, requiring prioritization and cumbersome rotations of electrode montages. Furthermore, there is increasing interest in smaller, more closely spaced electrodes in order to obtain finer spatial maps of seizures and function and to achieve greater sensitivity to high-frequency activity (see Section 14.3.4.1).

Studies of ERPs and high-frequency (gamma) oscillations also require higher sampling rates and bit depths. The minimum specifications are 16-bit ADCs with a

per-channel sampling rate of 1,000 Hz. However, studies of low-amplitude, high-frequency components within both phase-locked and nonphase-locked responses would likely benefit from even greater specifications. In the near future, optimal system configurations for iEEG will likely be capable of 24-bit recordings of up to 256 or 512 channels at 3,000 Hz per channel. Until such systems are commercially available and considered necessary for clinical purposes, cortical function mapping with iEEG will likely require some degree of customization.

Another important consideration for cortical function mapping with iEEG is the need to record markers for events during tasks eliciting functional activation. Analysis of phase-locked, as well as nonphase-locked, iEEG responses requires a temporal reference point for averaging signal energy in the time and frequency domains. It is therefore necessary to record event markers with a high degree of temporal precision. Ideally, these markers should be recorded directly into the iEEG data acquisition stream, either in the iEEG channels themselves or in digital channels recorded on the same computer bus. Systems in which task events are recorded in parallel by a separate computer are fraught with synchronization problems and are inherently unreliable. Most commercially available video EEG systems do not have explicit capabilities for recording event markers other than those for seizures, but this capability can often be added to existing configurations with relatively simple modifications.

## 14.3 Localizing Cortical Function

Like noninvasive scalp EEG signals, iEEG signals can be analyzed from a variety of different perspectives in order to correlate signal changes with functional brain activation. These analyses can be divided into those that focus on signal changes in individual cortical regions and those that focus on interactions between cortical regions (see Section 14.4). In the former, signal analyses are motivated by a desire to localize functional activation to a particular cortical region and to measure the strength and timing of activation in this region with respect to controlled parameters of the sensory, motor, or cognitive task. In the latter, signal analyses are designed to identify and establish functional relationships between different cortical regions, often in hopes of inferring a network of cortical regions that are jointly responsible for carrying out functional tasks. In this section we discuss different methods for analyzing regional brain responses and present examples of the different kinds of responses obtained with these methods, as well as some considerations regarding their interpretation and application to cortical function mapping.

### 14.3.1 Analysis of Phase-Locked iEEG Responses

To localize functional activation at individual recording sites, iEEG signal analyses, like those of scalp EEG, have largely focused on the measurement of signal components that either are or are not phase-locked to a sensory task, motor output, or cognitive operation. This distinction between phase-locked and nonphase-locked components is based primarily on whether EEG or iEEG signals that are recorded

during multiple trials of a functional task are averaged across trials in the time domain or in the frequency domain.

Averaging EEG signals in the time domain extracts phase-locked signal components as ERPs and discards nonphase-locked components as noise. In contrast, averaging EEG signals in the frequency domain focuses on event-related changes in EEG spectral energy that may have both phase-locked and nonphase-locked components [2]. The latter approach is still temporally anchored to an event across multiple trials, but the result is not limited to phase-locked components, as with ERPs. In either approach, it is necessary to verify the significance of putative functional responses with respect to some reference, which is usually derived from a baseline period preceding the event under study.

The distinction between these types of responses, particularly the distinction between ERPs and nonphase-locked increases in signal energy (often termed *induced responses*), can easily be confused. Averaging in the time domain necessarily yields phase-locked responses. However, significant variability (jitter) in the latency (or phase) of ERPs can distort their appearance in time-averaged responses. High-frequency components are more susceptible to latency jitter, and there is usually more jitter of ERP components (and their corresponding cognitive processes) at longer latencies. ERPs with high-frequency (e.g., gamma) components are usually confined to early (<150 ms) latencies. On the other hand, ERPs at longer latencies (e.g., P300) usually consist of low-frequency components that are more resistant to jitter [3, 4]. Because averaging in the frequency domain does not require phase locking, it may be better suited to investigate cortical processes with longer or more variable latencies and to investigate high-frequency EEG responses at longer latencies. EEG responses may have different combinations of frequencies, latencies, and phase locking. Nevertheless, the distinction between phase-locked and nonphase-locked responses is often still a practical one. Furthermore, many studies suggest that these different classes of EEG responses may have distinct functional response properties [5–7].

### 14.3.2 Application of Phase-Locked iEEG Responses to Cortical Function Mapping

A substantial literature has already accumulated from iEEG investigations of phase-locked responses, that is, ERPs, under a variety of experimental cognitive paradigms. The most influential of these is the oddball paradigm in which infrequent stimuli of any sensory modality (auditory having been studied the most) are randomly presented in a stream of frequent stimuli. Detection of the infrequent stimuli may be performed either automatically or intentionally. Infrequent stimuli produce a variety of phase-locked responses with a positive polarity and a peak varying from 300 to 600 ms [8], and a vast number of scalp EEG studies have explored a variety of behavioral and clinical factors affecting the different components (e.g., P3a and P3b) of this ERP.

Because of the inherent uncertainty regarding the sources of scalp-recorded ERPs, a number of iEEG studies have investigated the brain structures responsible for different components of the P300. Insights from iEEG studies have also been supplemented by studies of the effects of focal lesions on the P3a and P3b [9, 10], as

well as by functional neuroimaging studies [11]. Converging data from these studies have demonstrated that the P3 and its component waveforms are generated by a widely distributed network of cortical regions that collectively take part in detection behavior. The P3a component, which is evoked by rare stimuli and likely represents an orienting response to unexpected but behaviorally salient stimuli (e.g., to screeching tires or a gunshot), is generated in dorsolateral prefrontal cortex, supramarginal gyrus, and cingulate cortex (Figure 14.2) [12]. In contrast, the P3b component, which is evoked by target stimuli during a voluntary detection task, is generated by ventrolateral prefrontal cortex, superior temporal sulcus, posterior superior parietal cortex, and medial temporal structures including hippocampus and perirhinal regions.

Another cognitive task that has been studied extensively with iEEG is discrimination and identification of complex visual stimuli such as real or pictured objects, faces, letters, or words. These tasks require processing within the functional-anatomic domain known as the "what" stream of visual information processing in temporal-occipital cortex. Although much has been learned about this system through basic investigations in nonhuman primates, and subsequently through functional neuroimaging studies in humans, human iEEG studies have provided vital information about the spatiotemporal dynamics of information processing in the ventral temporal-occipital stream [13–17].

iEEG studies of phase-locked responses have also yielded important insights about more basic sensory evoked potentials. For example, iEEG studies have been instrumental in determining the cortical generators of the somatosensory evoked potential [18–20], as well as the somatotopic organization of somatosensory cortex [19, 21–23]. Even the cortical networks responsible for pain perception have been studied with iEEG [24, 25]. A number of studies have also explored the neural substrates of a variety of ERPs elicited by functional activation of motor, premotor, and supplementary motor cortex [26].

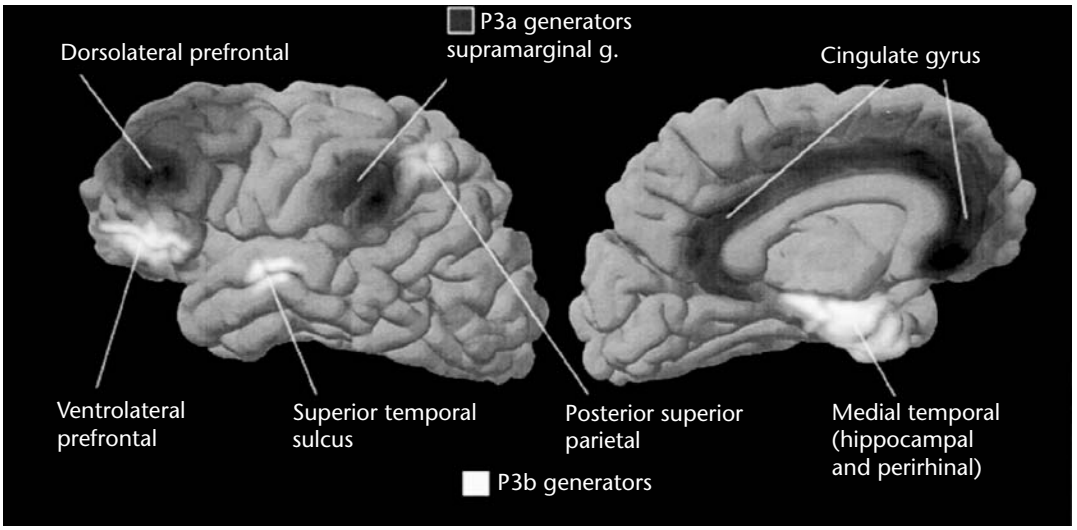



**Figure 14.2** Summary of areas where P3a and P3b were generated by simple rare stimuli, according to recordings at a cumulative ~4,000 iEEG sites. Lateral (left) and medial (right) views are shown. (*From:* [12]. © 1998 Elsevier Science Ireland Ltd. Reprinted with permission.)

### 14.3.3 Analysis of Nonphase-Locked Responses in iEEG

Compared to both scalp EEG and iEEG studies of phase-locked responses (ERPs), studies of nonphase-locked responses have been more recent. Early studies of nonphase-locked EEG activity focused on event-related power changes in relatively narrow frequency bands [27–29]. These studies primarily addressed well-defined narrowband oscillations such as the occipital alpha rhythm and the sensorimotor mu rhythm. Such prominent oscillations were assumed to require the synchronized activity of a large population of cortical neurons. Indeed, any potentials visible at the scalp likely require the summation of dendritic membrane potentials in a large population of neurons, and temporal synchronization at some spatial scale must be present for substantial summation to occur. Thus, event-related power changes in narrow frequency bands were thought to reflect the degree of synchronization among oscillating elements in cortical networks, and the terms *event-related desynchronization* (ERD) and *event-related synchronization* (ERS) were coined to refer to suppression or augmentation, respectively, of power in a particular frequency range [27]. More recent studies have sometimes avoided this terminology because it may not accurately reflect the neurophysiological mechanisms underlying all phenomena observed with this approach. Furthermore, significant event-related power changes are not limited to narrowband oscillations, as in alpha ERD. Nevertheless, the term *ERD/ERS* is still often used as a convenient, easily articulated, and widely recognized "nickname" for changes in signal energy that are time locked, but not necessarily phase locked, to an event and that are derived by the same general approach to signal analysis.

Most analyses of ERD/ERS in scalp EEG and iEEG no longer rely on quantification of signal energy in narrow frequency bands. Bandpass filtering requires a priori knowledge of the frequency bands where event-related signal changes will occur. There is not only considerable variability in these reactive frequency bands across subjects [30, 31], but there may also be significant variability across recording sites and functional tasks [32]. Although reactive frequency bands can be determined empirically by comparing power spectra in activated versus baseline EEG epochs, this requires a priori knowledge of the timing of functional brain activation. For these reasons, time-frequency analysis is the new standard for analyses of event-related EEG energy changes.

Time-frequency analysis consists of two basic steps: (1) time-frequency decomposition of the EEG signal, and (2) statistical analysis of event-related energy changes in time-frequency space. Both steps can be accomplished by a variety of methods.

#### 14.3.3.1 Time-Frequency Decomposition

The methods that have been used for time-frequency decomposition of EEG signals include Fourier transformation, wavelet transformation, complex demodulation, and matching pursuit decomposition, among others. For reasons discussed later, our laboratory has chosen to use matching pursuits with a dictionary of Gabor functions.

The time-frequency representation of the signal can be interpreted either as the integral transformation of the time series to time-frequency space or as a decompo-

sition of the time series into the sum of components localized in time and frequency. The short-time Fourier transform (STFT) uses specific window functions well localized in time to compute the time-frequency power distribution. The choice of the window function determines the time-frequency resolution of the result. In general, the time-frequency transforms are defined by window (i.e., kernel) functions that determine the properties of the transformation. Cohen [33] defined a general class of kernel functions useful for time-frequency analysis.

The wavelet analysis can also be represented in terms of a transformation with specific kernel functions, but instead of transforming the time series to time-frequency space, wavelets transform the time series to time-scale space, where the scale is a dilation parameter of the wavelet. The wavelet transform is thus a multiscale decomposition. Transformation of wavelets into frequency space provides the time-frequency representation of the signal. Both STFT and wavelet transformations can also be viewed as decompositions of signals into combinations of basis functions that are well localized in time and frequency. This can be conceptualized as dividing the time-frequency plane into time-frequency boxes with dimensions in time and frequency that reflect the limits of localization dictated by the uncertainty principle. This tailing of the time-frequency plane is predefined by the choice of window or mother wavelet.

The matching pursuit method also decomposes the signals into linear combinations of functions localized in time and frequency, but instead of using a limited number of orthogonal functions, it uses a large dictionary of nonorthogonal functions. In our laboratory we use a dictionary of translated, dilated, and modulated Gaussians (Gabor functions). These functions, often called *atoms*, are characterized by three independent parameters: time, scale, and frequency, thus providing overlapping tiling of the time-frequency plane and giving a much more robust representation of the signal:

$$g_\gamma(t) = \frac{1}{\sqrt{s}} g\left(\frac{t-u}{s}\right) e^{i\xi t} \text{ with } g(t) = 2^{1/4} e^{-\pi t^2} \text{ and } \gamma = (s,u,\xi) \tag{14.1}$$

where $\frac{1}{\sqrt{s}}$ normalizes the norm of $g_\gamma$ to 1, which allows the conservation of energy of the decomposition. The dictionary we use also includes Fourier and Dirac atoms.

The signal $f$ can be written as the sum of $m$ atoms $g_n$ and a residue:

$$f = \sum_{n=0}^{m-1} \left\langle R^n f, g_n \right\rangle g_n + R^m f \tag{14.2}$$

The matching pursuit method selects functions iteratively, in each step choosing $g_n$ toe maximize the inner product $\left\langle R^n f, g_n \right\rangle$, where $R^n f$ is the residual from the previous step. This results in an adaptive decomposition that approximates local patterns in the signal well, including short transients and longer rhythmic components. The original algorithm was described in 1993 by Mallat and Zhang [34] and was first applied to intracranial EEG five years later [35]. The time-frequency energy representation is obtained by summing the Wigner-Ville distributions of $g_n$. This ensures that the time-frequency representation does not include cross terms, which are the

main problem in transform-based time-frequency decompositions. In applications to event-related EEG responses, the time-frequency decomposition is usually performed for each trial separately and then averaged and subjected to statistical comparison with a "baseline" period.

Since the matching pursuits approach chooses functions from a nonorthogonal dictionary, it avoids biases resulting from identical tiling of the time-frequency plane, as in other methods. Furthermore, the choice of Gabor functions ensures the best possible time resolution allowed by the uncertainty principle. The original Mallat-Zhang algorithm uses a dyadic dictionary in which the scale variable $s$ [as defined in (14.1)] is restricted to be a power of two [i.e., $s = 2^j$ for $0 < j < \log_2(N)$]. This restriction results in a bias toward frequencies that are of the form $Fs/2^j$ and their multiples, where $Fs$ is the sampling frequency. To reduce the bias in determining the location of component functions in the decomposition, an additional step of searching on a finer grid is performed. Typically, this finer grid includes smaller intervals in time between functions with small scale and smaller intervals in frequency for functions with large scale. Durka et al. [36] introduced an alternative method using a stochastic dictionary where the dictionary functions are evenly distributed in time-frequency space. However, this method is computationally more expensive. In ERD/ERS applications where the results are computed by averaging the Wigner-Ville distributions from multiple trials, the results obtained using stochastic and dyadic dictionaries are not significantly different [37]. Typically, for ERD/ERS analyses we use the original Mallat-Zhang algorithm with a dyadic dictionary with subsampling in time and frequency for the two smallest and two largest scale octaves, respectively.

#### 14.3.3.2 Statistical Analysis of Nonphase-Locked Responses

To study event-related EEG responses in the context of functional brain activation, time-frequency decomposition of the EEG signal must be followed by statistical analysis of the time-frequency signal representations in order to determine whether there are energy changes during activation that are statistically different from changes that might otherwise occur without activation. For this purpose a reference or baseline interval may be chosen from a variety of possible sources. In reality there is no such time interval when one can be sure that the EEG signal contains no energy related to brain activation or cortical computation. This problem is analogous to the search for an "inactive" reference in EEG recordings, though the search here is in time instead of space. The best reference interval would be one under which all conditions are equal to the experiment except the functional task under study. The most frequently chosen reference interval is a time immediately preceding the stimulus or behavior that marks the onset of each trial or repetition of a functional task. The main potential pitfall of this approach is contamination of the baseline interval by brain activation or deactivation due to stimuli or responses from the previous trial, extraneous stimuli (e.g., people talking or moving nearby), or anticipation of regularly occurring stimuli (thus the need for jittered intertrial intervals).

An alternative approach is to take random samples of epochs throughout the recorded experiment to generate a surrogate distribution of energy estimates that is independent of the timing of the task. One can also record a long segment of EEG

while the subject is "at rest"; however, there may be significant changes in the subject's level of arousal between the time when the subject is at rest and when the experiment is performed. Long intervals between the experiment and the baseline condition may yield statistical results that are due to differences in conditions not related to the functional task; such differences may also occur immediately before or after an experiment.

Multiple trials of a functional task are recorded in order to achieve stable time-frequency estimates of EEG signal energy associated with the task and the reference interval, as mentioned above. The minimum number of trials that should be used is difficult to define, but 25 is a good place to start, and for many tasks a greater number of trials is desirable. The length of the reference interval is an important consideration and should be taken into account when designing the experimental task itself. Longer intervals will generally yield more stable estimates of "baseline conditions." It is important for the reference interval to capture not only the variability of energy across different frequencies but also its variability across time. However, if the reference interval is too long and the intertrial interval is too short, the reference interval may be contaminated by residual activation or deactivation from preceding trials.

Statistical analyses may be performed on time-frequency estimates of EEG signal energy using a variety of approaches. To use parametric methods such as *t*-tests, ANOVAs, or regression analyses, it is necessary for the energy estimates to approximate a normal distribution, and for this purpose the natural logarithm has been commonly used [38, 39]. The stability of power estimates, and thus the power of statistical tests, can be enhanced by decimating the time-frequency space, that is, reducing the time or frequency resolution of the estimates.

An important and frequently overlooked problem for time-frequency analyses of event-related EEG signal changes is that of accounting for multiple comparisons. In a typical event-related task, the poststimulus interval of interest may be 1 to 2 seconds, and the frequency spectrum of interest may extend from 1 to 200 Hz or beyond. The energy in each of the two-dimensional time-frequency "pixels" in this time-frequency plane is being tested for a significant difference from baseline, and if the time and frequency resolution are maximized, there will be an enormous number of statistical tests, or comparisons (e.g., 1,000 time divisions × 200 frequency divisions = 200,000 comparisons). However, if the threshold for statistical significance is the customary *p*-value of 0.05, the chance that a nonsignificant difference will be significant is 1 in 20, and for 200,000 comparisons we can expect that at least 10,000 time-frequency pixels will be deemed different from baseline when they are not.

The easiest but most conservative method to correct for multiple comparisons is the Bonferroni correction, which simply divides the *p*-value by the number of comparisons (e.g., corrected *p*-value = 0.05/200,000 = 0.00000025). It may be surprising that such an extreme threshold for statistical significance can be exceeded, but it is routinely surpassed in many pixels of a typical time-frequency analysis of iEEG ERD/ERS responses. Nevertheless, the Bonferroni correction is probably inappropriately conservative in the application at hand because it assumes that each statistical comparison is independent of all others, and event-related energy changes in adjacent pixels are not independent of each other. Instead, these energy changes are

usually observed in large clusters of time-frequency pixels, corresponding to different kinds of EEG responses already outlined above. Although we have recently begun to explore two-dimensional smoothing of energy estimates in the time-frequency plane, there are many ways in which dependencies between time-frequency pixels might be taken into account, including the false detection rate [40]. Another approach is to use nonparametric statistical tests [41].

### 14.3.4 Application of Nonphase-Locked Responses to Cortical Function Mapping

Functional brain activation is associated with a complex constellation of nonphase-locked responses (also known as ERD/ERS phenomena, as mentioned earlier) in a variety of frequency ranges (e.g., theta, alpha, beta, and gamma bands), with reactive frequencies varying across individual subjects, brain regions, and experimental conditions [30, 42]. These responses may be categorized according to the frequency ranges at which they are observed and their timing with respect to functional activation. For example, power suppression (ERD) is observed in alpha (8 to 12 Hz) and beta (13 to 30 Hz) frequency ranges before and during brief self-paced finger movements. Like suppression of the occipital alpha rhythm when the eyes are opened, this response suggests suppression of a resting rhythm [42]. Following a brief, self-paced movement, there is a prominent rebound beta ERS that is localized to somatotopically appropriate regions of sensorimotor cortex contralateral to the moving body part. This response has been interpreted to represent the reset of functional circuits responsible for the movement [42, 43]. In addition, during movement in one limb alpha or beta ERS may be observed over regions of sensorimotor cortex representing the other limbs that are not moved, suggesting a center-surround pattern of reciprocal activation and inhibition [42].

Most observations of nonphase-locked responses (ERD/ERS) in both scalp EEG and iEEG support the general view that functional activation of cortex is associated with power suppression (ERD) in low frequencies (i.e., alpha and beta) and power augmentation (ERS) in higher frequencies (i.e., gamma). Power augmentation in lower frequencies (i.e., usually beta, sometimes alpha) has been interpreted in some circumstances as a correlate of cortical inhibition or postactivation resetting of cortical networks. However, power augmentation may also be observed in theta and alpha frequencies at short latencies following cortical activation. This may reflect integrative mechanisms associated with memory [30, 44]. When it accompanies activation of sensory cortices, however, it often reflects phase-locked components of the response, that is, ERPs [45]. In both scalp EEG and iEEG studies of nonphase-locked responses, it may be useful to minimize these phase-locked components or to at least identify them as such. One method for doing this is to subtract the time-averaged ERP from the raw signal in each trial before averaging across trials in the frequency domain [32]. A common objection to this approach is that it may contaminate nonphase-locked responses with energy from phase-locked responses, especially if there is significant jitter in the phase-locked responses. In most cases the energy from phase-locked responses is so small and occurs at such low frequencies that contamination of nonphase-locked responses is likely negligible. However, to address this objection it may be useful to explicitly compare the

spatial-temporal distributions and other properties of responses obtained by the two approaches, that is, averaging across trials in time versus frequency domains [45].

#### 14.3.4.1 High-Frequency Nonphase-Locked Responses in iEEG

Interest in high-frequency EEG activity has been heightened in the past few decades by observations in animals of synchronized neuronal firing in gamma frequencies, usually around 40 Hz. Furthermore, synchronization of neuronal firing has been proposed as a fundamental mechanism of neural computation because it could theoretically serve as a temporal code that dynamically "binds" spatially segregated neurons (e.g., across cortical columns, areas, or even hemispheres) into assemblies representing higher-order stimulus properties [46–48]. The connection between these theories of synchronous neuronal firing and EEG activity in gamma frequencies has been supported by basic experiments demonstrating the synchronization of single units with LFP gamma oscillations [49, 50]. In addition, a large number of studies of gamma oscillations have been undertaken in humans using scalp EEG and, more recently, MEG [51]. Until recently, most of these studies focused on gamma frequencies in and around 40 Hz. The limited frequency range of these studies may have been due to a priori assumptions based on the band-limited gamma oscillations observed in animals, as well as a long-standing bias that no relevant EEG activity existed above 40 Hz. The latter bias may have arisen from the technical limitations of pen-based analog EEG recordings and from many years of clinical interpretation based on signals recorded with low-pass filters at or below 70 Hz.

The development of digital recording methods has made it possible to record EEG activity with a much broader spectrum than previous pen-based recordings, but this capability has not yet been widely exploited because the clinical relevance of high-frequency EEG activity is only now starting to be fully appreciated. A growing number of investigations using human intracranial EEG have indicated that there is clinically relevant EEG activity in frequencies far beyond the scope of traditional noninvasive EEG studies. These studies have demonstrated high-frequency EEG activity that is associated with both pathological and normal physiological neural activity. Pathological high-frequency iEEG activity has been extensively reported in association with epilepsy. For example, high-frequency oscillations, called "ripples," have been recorded with microwire depth electrodes from human hippocampus and entorhinal cortex in patients with intractable epilepsy [52]. These recordings have suggested that ripple oscillations at 80 to 200 Hz may be distinct from higher-frequency "fast ripples" (250 to 500 Hz), which are more prevalent in seizure foci [53]. However, iEEG recordings from larger subdural and depth electrodes that are routinely used in clinical practice have shown that high-frequency oscillations in the ripple frequency range are also associated with epileptogenic processes [54], and the spectral characteristics of high-frequency oscillations associated with epilepsy may depend on the electrode size, and thus the spatial scale, of iEEG recordings [55].

Advances in digital EEG recordings, combined with the clinical circumstances necessitating iEEG electrode implantation, have also allowed investigations of the role of high-frequency oscillations in association with physiological functional brain activation. These investigations have revealed event-related nonphase-locked

responses in gamma frequencies higher than the traditional 40 Hz gamma band previously studied in human scalp EEG and in early microelectrode recordings of animals [1]. These "high-gamma responses" (HGRs) have a characteristically broadband spectral profile that spans a large range of frequencies. The lower and upper frequency boundaries of these HGRs are quite variable, but they most commonly range from ~60 Hz to ~200 Hz, with the majority of event-related energy changes occurring between 80 and 150 Hz [37, 56].

HGRs were first described in detail in humans with subdural ECoG recordings of sensorimotor cortex [57], and in this and subsequent iEEG studies HGRs have exhibited functional response properties that distinguish them from both phase-locked responses (ERPs) and ERD/ERS phenomena previously observed with scalp EEG in other frequency bands [1]. For example, the temporal and spatial distributions of HGRs are often more discrete and/or functionally specific for task-related cortical activation than other electrophysiological responses. Furthermore, HGRs have been demonstrated in a great variety of functional-anatomic domains, including motor cortex [57–66], frontal eye fields [67], somatosensory cortex [68–71], auditory cortex [32, 45, 72, 73], visual cortex [74–78], olfactory cortex [79], and language cortex [78, 80–83]. This seemingly ubiquitous occurrence of HGRs during functional activation of neocortex has suggested the possibility that it is a general electrophysiological index of cortical processing.

The first published study of HGRs in human iEEG utilized a visually cued motor task to test whether event-related spectral changes could discriminate the well-known somatotopic organization of sensorimotor cortex [57, 84]. In addition to the predicted ERD/ERS in alpha and beta bands, a broadband energy increase was observed in frequencies ranging from 75 to 100 Hz. These HGRs were located within somatotopically defined regions of sensorimotor cortex [57] and corresponded well to ESM results for motor function. In addition, they were observed only during contralateral limb movements, whereas alpha and beta ERD were observed during both contralateral and ipsilateral limb movements. The temporal patterns of HGRs were also different from those of other ERD/ERS phenomena. HGRs occurred in briefer bursts limited to the onset and offset [85] of movement, and the latencies of these bursts covaried with movement onset/offset. These findings suggested that HGRs reflect movement initiation and execution. As in previous scalp EEG studies [42, 86], ERD/ERS in other frequencies appeared to reflect both movement planning and execution.

During self-paced finger and wrist movements, Ohara et al. [58] observed nonphase-locked gamma activity in S1 and M1 extending up to 90 Hz. High-gamma activity (60 to 90 Hz, in particular) was time locked to movement onset and was short-lived after movement onset. In addition, both low- and high-gamma ERS were observed only during contralateral movements. Pfurtscheller et al. [59] also observed broadband high-gamma activity (60 to 90 Hz) over sensorimotor cortex while subjects performed self-paced tongue and finger movements. The topographic pattern of high-gamma activity was more discrete and somatotopically specific than the more widespread mu (alpha) and beta ERD, and its temporal pattern was also briefer, corresponding to movement onset.

Miller et al. recently published the largest series to date of iEEG recordings in sensorimotor cortex [61]. In this study 22 subjects underwent iEEG recordings dur-

ing a variety of motor tasks, and the topographic patterns were compared for ERD in low frequencies (8 to 32 Hz) versus ERS in high-gamma frequencies (76 to 100 Hz). HGRs had a more focused spatial distribution than did ERD. In addition, iEEG recording sites with ERD/ERS had a somatotopic organization corresponding to the movement of different body parts and corresponded well to the results of electrocortical stimulation mapping. Although two other studies limited their iEEG signal analyses to frequencies below 60 Hz [62, 87], they too showed good correspondence between gamma responses and the results of ESM. Taken as a whole, these reports suggest that iEEG could be a useful tool for mapping motor function in patients undergoing surgical resections near or within motor cortex.

The first paper demonstrating the use of HGRs to map language cortex reported a patient with normal hearing and speech who was also fluent in sign language [80]. The spatiotemporal patterns of HGRs were compared during tasks with different modalities of input (visual versus auditory stimuli) and output (spoken versus signed responses). The location and timing of HGRs during these tasks were consistent with general principles of the functional neuroanatomy of human language and with the latencies of verbal and signed responses. In addition, HGRs frequently, but not always, corresponded to the results of electrocortical stimulation mapping. In contrast, alpha ERD was observed to occur in a broader spatial distribution and with temporal latencies and durations that less closely matched those of task performance [88]. HGRs thus appeared to be better suited for distinguishing the patterns of functional brain activation associated with different language tasks.

To further investigate the clinical utility of high-gamma ERS for mapping language cortex, the spatial patterns of HGR during picture naming were compared with ESM maps of naming and mouth movements responsible for verbal output in the same clinical subjects [81]. When sensitivity/specificity estimates were made for the 12 electrode sites with the greatest high-gamma ERS, the specificity of high-gamma ERS with respect to ESM (the "gold standard") was 84%, but its sensitivity was only 43%. Its relatively good specificity suggests that HGRs might be useful for constructing a preliminary functional map in order to identify cortical sites of lower priority for ESM mapping. However, its low sensitivity indicates that it cannot yet replace ESM even though ESM carries the added risk of stimulating seizures. This low sensitivity also raises the question as to whether current iEEG recording technology is sufficiently sensitive to the sources of HGRs: Is it possible that some HGRs are falling between the cracks of the standard 1-cm-spaced electrode arrays [89]?

iEEG recordings from auditory cortex have demonstrated HGRs during tone and speech discrimination [32], during an auditory oddball paradigm [72], and during an auditory sensory gating (P50) paradigm [45]. These studies have shown that nonphase-locked HGRs are distinct from phase-locked ERPs, which are usually composed of lower-frequency components and often have somewhat different spatial and temporal distributions. In general, HGRs during auditory stimulation have been observed in a relatively focused spatial distribution concentrated over posterior superior temporal gyrus in a spatial distribution similar to but not identical with the N100 of the auditory evoked response associated with onset of the stimuli. The onset of HGRs often coincide with the N100 but usually last longer. More importantly, however, the magnitudes of HGRs appear to be correlated with the degree of functional activation. For example, HGRs had a greater magnitude during discrimi-

nation of speech stimuli than during discrimination of tone stimuli, suggesting that HGR in auditory cortex depends on the complexity of acoustic stimuli and, by inference, on the amount of cortical processing necessary for the auditory discrimination [32]. In contrast, the amplitude of the N100 depends much less on the type of stimulus.

iEEG studies of auditory cortex have also highlighted the broadband spectral profile of HGRs. For example, power spectral analyses of HGRs during auditory discrimination indicated that the greatest energy increases occurred at 80 to 100 Hz but also extended up to 150 to 200 Hz [32, 37]. In a study by Edwards et al. [72], HGRs were reported between 60 and 250 Hz, centered at ~100 Hz, and in a study by Trautner et al. [45], HGRs extended to 200 Hz. Similar frequency responses have been observed in microelectrode recordings of auditory cortex in monkeys [90, 91]. In addition, studies in both humans and animals have shown that nonphase-locked responses in the traditional 40-Hz gamma band are more variable and less sensitive to functional activation of cortex [32, 91]. This may be due to variability in the upper boundary of frequencies at which event-related power suppression (ERD) occurs. If ERD extends into low-gamma frequencies, it may obscure any power augmentation in this frequency range. For this reason analysis of event-related 40-Hz responses that are based on narrowly bandpass-filtered signals may at times yield misleading results. Time-frequency analyses of iEEG recordings have shown that the lower boundary of HGR power augmentation may extend into 40-Hz frequencies, but the most consistent responses occur above 60 Hz. The greater reliability of ECoG power changes in higher-gamma frequencies has recently been reinforced by a study in which movement classification for different body parts was best for power at 70 to 150 Hz, which the authors called the "chi" band. The limited classification accuracy of lower-gamma frequencies (30 to 70 Hz) was interpreted to result from a superposition in the power spectrum of band-limited power suppression (ERD) at lower frequencies (ranging up to ~50 Hz) and an increase in power across all frequencies, obeying a power law.

The broadband spectral profiles of HGRs recorded with iEEG in humans are difficult to reconcile with earlier conceptualizations of gamma oscillations associated with functional activation. These ideas of gamma oscillations have been derived largely from observations of relatively band-limited responses (e.g., in and around 40 Hz) in microelectrode recordings of animals. Although accumulating observations in animals of higher-frequency, broader-band oscillations appear to be gradually extending the frequency range that is quoted for "gamma," gamma responses are still largely conceptualized as band-limited network oscillations. However, the broadband frequency response of HGRs observed with iEEG during functional activation would presumably require the summation of activity in multiple spatially overlapping neural populations or assemblies, each oscillating at different, perhaps overlapping or broadly tuned, frequencies [57, 92].

An alternative explanation for the broadband nature of HGRs is that they are the time-frequency representations of transient responses with a broad range of frequency components. Intertrial jitter in the latency of these transients presumably renders them invisible when averaged across trials in the time domain. Recent investigations of microelectrode recordings from macaque SII cortex during tactile stimulation have observed broadband HGRs with spectral profiles identical to those

recorded in human iEEG [93, 94]. Detailed time-frequency analyses of these HGRs using matching pursuits revealed that they are temporally tightly linked to neuronal spikes, although their precise generating mechanisms could not be determined [93]. In addition, LFP power in the high-gamma range was strongly correlated, both in its temporal profile and in its trial-by-trial variation, with the firing rate of the recorded neural population [94].

Whether the underlying signals giving rise to HGRs are oscillations or transients, the log power law of electrophysiological recordings would predict that activity in such a high-frequency range is much more likely to be recorded at the mesoscale of subdural ECoG if there is some degree of synchronization across a large population of neural generators. In a recent simulation of the generation of subdural ECoG HGRs by different firing patterns in the underlying cortical population, both an increase in firing rate and an increase in neuronal synchrony increased high-gamma power. However, ECoG high-gamma power was much more sensitive to increases in neuronal synchrony than in firing rate [94]. Thus, ECoG HGRs could index neuronal synchronization even if the underlying firing pattern is not a band-limited oscillation.

Synchronization across subpopulations of neurons has been hypothesized to constitute a temporal coding strategy for cortical computation that complements rate coding and plays a role in higher cortical functions such as attention [95, 96]. Several human iEEG studies have found an augmentation of high-gamma activity in association with attention [56, 66, 75, 77, 83, 97], as well as long-term memory [98], and working memory [99]. Regardless of whether HGRs reflect an increase in firing rate or an increase in neural synchronization, there is ample evidence from the aforementioned studies that HGRs likely reflect patterns of neural activity that are relevant to cortical computation, and that HGRs may serve as useful markers for cortical function mapping.

## 14.4 Cortical Network Dynamics

Although cortical function mapping has typically concentrated on localizing functional activation in discrete cortical regions, most of the higher cortical functions that are clinically relevant, for example, expressive and receptive language function, are understood to depend on the dynamic interplay between multiple cortical regions that are spatially distributed across different brain regions. For example, common language tasks may require the activation of Wernicke's area in posterior superior temporal and inferior parietal regions, Broca's area in inferior prefrontal regions, and a variety of other regions in frontal, parietal, temporal, and occipital lobes. To understand the functional role that each individual brain region plays, as well as the degree to which functions might be shared across networks of cortical regions, it would be useful to examine the interactions between cortical regions during their functional activation, preferably on a time scale that would allow inferences as to whether activation of the network evolves in a serial, parallel, or cascaded manner. To elucidate the dynamic structure of cortical networks supporting cognition, it would also be useful to test whether activity in one component of the network has a causal influence on activity in other components.

Many of the methods that have been developed to study the dynamics of cortical networks have been based on the idea that oscillatory activity plays a role in organizing the activity of neuronal assemblies in large-scale neural networks [100–103]. A functional task engaging such a network is expected to be accompanied by rapidly changing interrelationships and interactions between the oscillations generated in the various components of the network. To measure these interactions, investigators have studied temporal fluctuations in coherence [104, 105], brain electrical source analysis coherences [106], and oscillatory synchrony [107, 108]. Recently, attention has also focused on the directionality of interactions between brain regions. For this purpose various approaches have been adopted, such as calculations of evoked potential covariances [109], the imaginary part of coherency [110], adaptive phase estimation [111], and methods based on Granger causality [112–120].

### 14.4.1 Analysis of Causality in Cortical Networks

According to Granger causality, an observed time series $x_l(t)$ causes another series $x_k(t)$ if knowledge of $x_l(t)$'s past significantly improves prediction of $x_k(t)$. Methods based on this concept, extended to multichannel data, may be designed to determine the sources and targets for interactions among brain regions [121, 122], allowing one to study the dynamic architecture of brain networks participating in cognitive tasks. An important problem to be solved by multivariate causality analysis is whether causal interactions are direct or indirect (mediated by another site or by several sites). Granger causality itself does not answer the question. Thus, the directness of interactions may be inferred by related methods: combining directed transfer function (DTF) [119] with phase spectrum and cross-correlogram analyses [123], using partial direct coherence [112, 113, 120], or using a direct directed transfer function (dDTF) [124, 125].

dDTF makes use of partial coherence (which reveals the directness of interactions) [126, 127] and DTF (which reveals directionality of interactions), and has been widely used in investigations of activity flow in amnesic and Alzheimer's patients [128], in patients with spinal cord injuries [129], in investigations of the source of seizure onset in epileptic neural networks [130–132], in studies of wake-sleep transitions [133–135], in working memory [136], and during encoding and retrieval [137], as well as in animal behaviors [138]. DTF and related methods have also been employed to investigate causal influences in fMRI data [139–141] and have been used in a brain–computer interface [142]. However, DTF and dDTF are not designed to analyze very short data epochs, as needed to track the dynamics of cognitive processes. This particular limitation may be overcome when multiple trials of a particular cognitive task are available for analysis [143], in which case a modification of DTF, the short-time directed transfer function (SDTF), may be used [117, 144–147].

To combine the benefits of directionality, directness, and short-time windowing, Korzeniewska et al. [148] introduced a new estimator, the short-time direct directed transfer function (SdDTF). This function evaluates the directions, intensities, and spectral contents of direct causal interactions between signals and is also adapted for examining short-time epochs. These properties of SdDTF are expected

to make it an effective tool for analyzing nonstationary signals such as EEG activity accompanying cognitive processes.

The advantages of SdDTF are illustrated by a simple simulation of signals in three different channels changing over time. A schematic of the simulation is presented in Figure 14.3, and the results are shown in Figures 14.4 and 14.5. The signals were simulated in such a way that there was no flow (no causal relations) between channels during the first second. During the next second, flows from channel 1 to 2 ($1 \rightarrow 2$) and from 2 to 3 ($2 \rightarrow 3$) were simulated. There was again no flow during the third second, and during the last second, flows $3 \rightarrow 2$ and $2 \rightarrow 1$ were simulated. These signals were created as follows: During the first second, all signals contained the same spectral components between 86 and 96 Hz, but white noise was added separately to the signals in each channel. During the next second, when flows $1 \rightarrow 2$ and $2 \rightarrow 3$ were simulated, channel 1 contained the same signal used during the first second. Channel 2 contained the signal used in channel 1, albeit shifted later by 6 ms, and white noise of a different mean and variance was added to it. Channel 3 contained the signal used in channel 2, albeit shifted later by 12 ms, with additional white noise mean and variance different from what was added to all of the previously used signals. During the third second, all channels contained the same frequency components between 105 and 114 Hz and different components of white noise as in the first second of the simulation. During the last second, when flows of opposite directions were simulated ($3 \rightarrow 2$ and $2 \rightarrow 1$), channel 2 contained the signal used in channel 3, shifted by 6 ms, with additional white noise of different mean and variance than the previous ones. Likewise, channel 1 contained the signal used in channel 2, shifted by 12 ms, with additional white noise.

In the cross spectra shown in Figure 14.4(a), there is activity around 90 Hz during the first two seconds and activity around 110 Hz during the next two seconds for each pair of channels. Ordinary coherences, depicted in Figure 14.4(b), are also similar for each pair of channels. They show relationships for 86- to 96-Hz and 105- to 114-Hz components during the first and third epochs, as well as for noise compo-

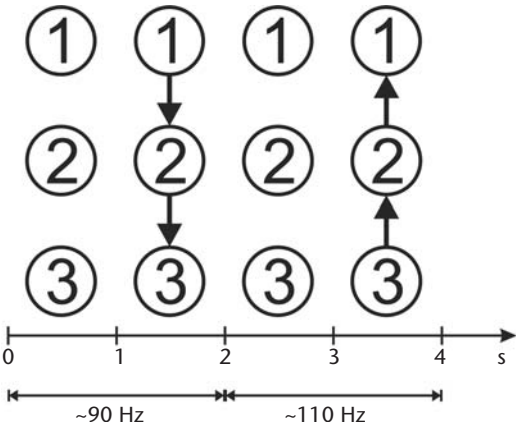



**Figure 14.3** Schematic of simulated model of activity flows. The horizontal axis is time in seconds. The frequencies of simulated activity flows are shown below for different time periods. Circled numbers represent different signals (recording sites), and arrows indicate direct flows of activity in the simulated signals (from one site to another). (*From:* [148]. © 2008 Wiley-Liss Inc. Reprinted with permission.)

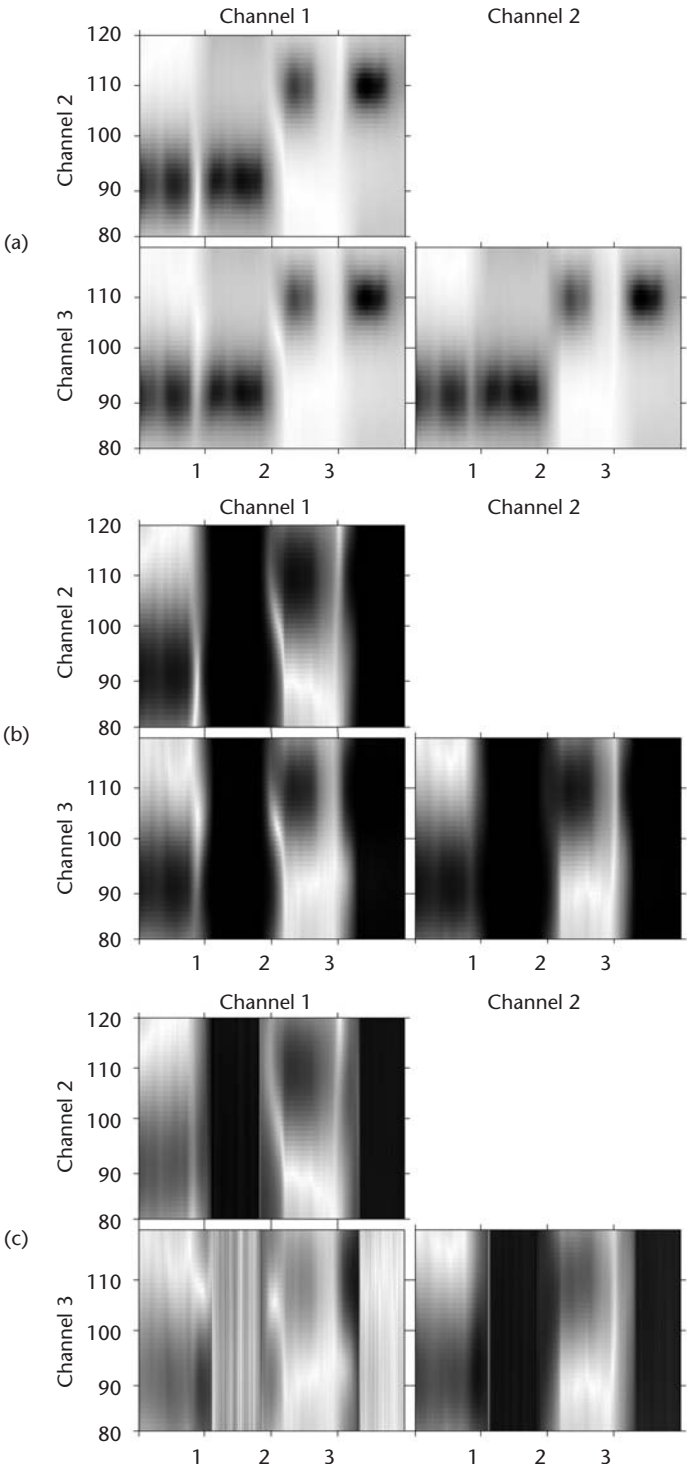



**Figure 14.4** Cross-spectra and coherences of three-channel MVAR model of simulated signals (as in Figure 14.3). In each plot the horizontal axis represents time in seconds (0 to 4 seconds), the vertical axis represents frequency (80 to 120 Hz), and the grayscale (white = minimum, black = maximum) represents the value of the calculated functions: (a) cross spectrum, (b) ordinary coherence, and (c) partial coherence. Each plot is for a pair of channels whose numbers are denoted at the top of the columns and the left sides of the rows. (*From:* [148]. © 2008 Wiley-Liss Inc. Reprinted with permission.)

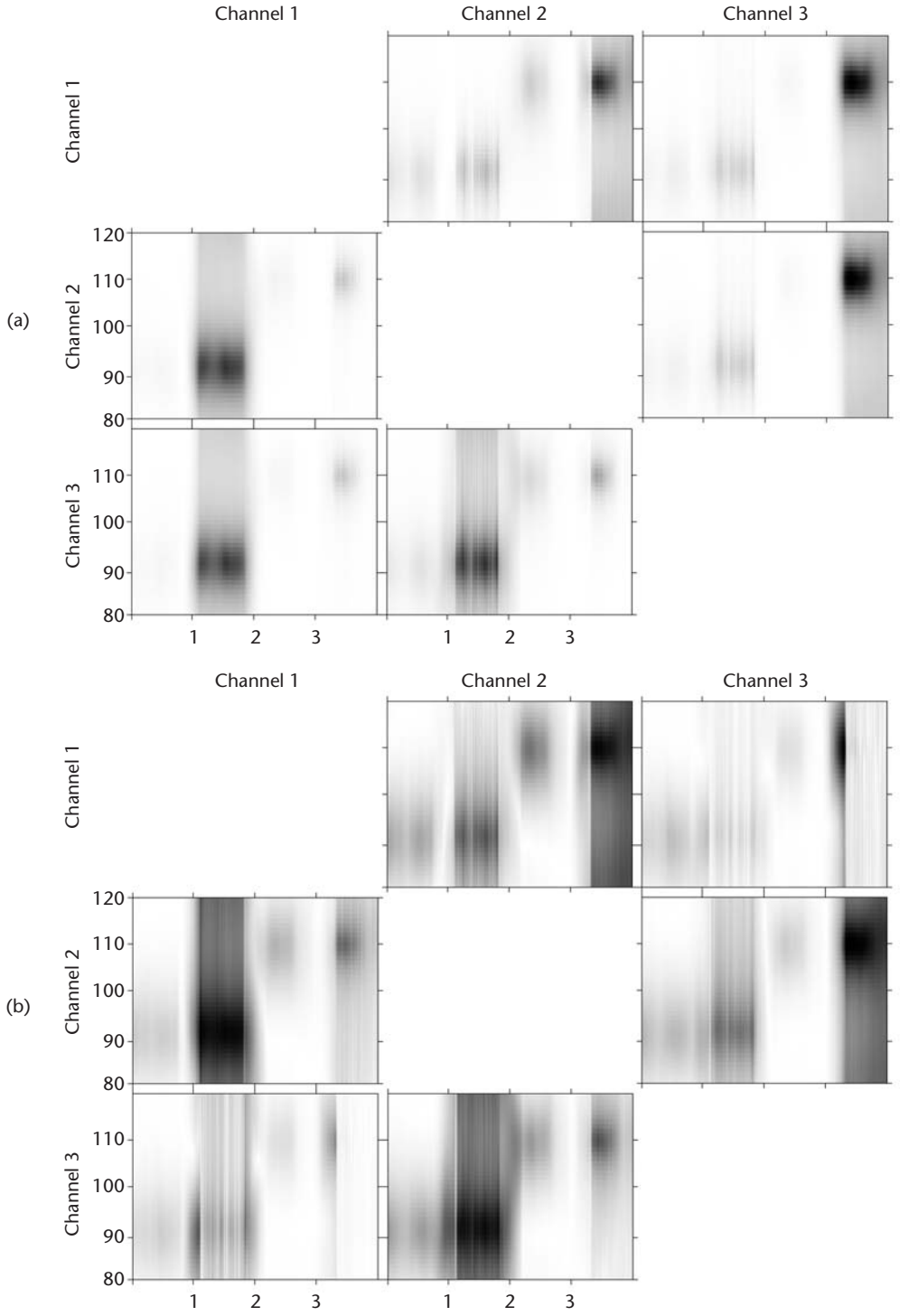



**Figure 14.5** SDTFs and SdDTFs of a three-channel MVAR model of simulated signals (as in Figure 14.3). Axes and scale as in Figure 14.4. (a) SDTFs for pairs of channels denoted as in Figure 14.4. The matrix is not symmetric; each plot shows SDTF for flows from the channel named at the top to the channel named to the left of the plot. (b) SdDTF plots for direct flows from the channel named above to the channel named to the left. (*From:* [148]. © 2008 Wiley-Liss Inc. Reprinted with permission.)

nents during the second and fourth epochs, when noise components were propagated across channels. However, partial coherences [Figure 14.4(c)] show relationships only between channels 1 and 2 and between channels 2 and 3 during the second and fourth epochs, when signals were directly related between these channels. Partial coherence is close to zero during the first and third epochs because the spectral components are common to all three channels and noise components are independent; thus, no direct relationships should be observed between channels. Partial coherence is close to zero for channels 1 and 3 during the second epoch because a noise component added to channel 2 was not present in channel 1, and it is also close to zero during the fourth epoch because a noise component added to channel 2 was not present in channel 3. Thus, there were no direct relationships between signals 1 and 3. The small patches in the plots of partial coherence for channels 1 and 3 are edge effects from analyzing windows with concatenated signals.

SDTF [Figure 14.5(a)] does not differentiate direct flows from indirect ones. There are visible flows $1 \rightarrow 2$ and $2 \rightarrow 3$ (direct flows), as well as $1 \rightarrow 3$ (indirect flow), during the second epoch, and flows $3 \rightarrow 2$ and $2 \rightarrow 1$ (direct flows), as well as $3 \rightarrow 1$ (indirect flow) during the fourth epoch. SdDTF plots [Figure 14.5(b)] illustrate the effect of multiplying SDTF by partial coherence. Only direct flows $1 \rightarrow 2$ and $2 \rightarrow 3$ are observed during the second epoch, and only direct flows $3 \rightarrow 2$ and $2 \rightarrow 1$ are observed during the fourth epoch. The indirect flows $1 \rightarrow 3$ and $3 \rightarrow 1$ seen in SDTF are eliminated in the SdDTF plots (the thin patch in $3 \rightarrow 1$ is an edge effect from concatenated signals), yielding a more precise estimate of the changing relationships between signals.

To evaluate the statistical significance of event-related changes in SdDTF, that is, event-related causality (ERC), new statistical methodology was developed for comparing prestimulus (baseline) with poststimulus SdDTF values. The main difference between this methodology and other statistical methods is that both the baseline and poststimulus epochs are treated as nonstationary (for more details, see [148]).

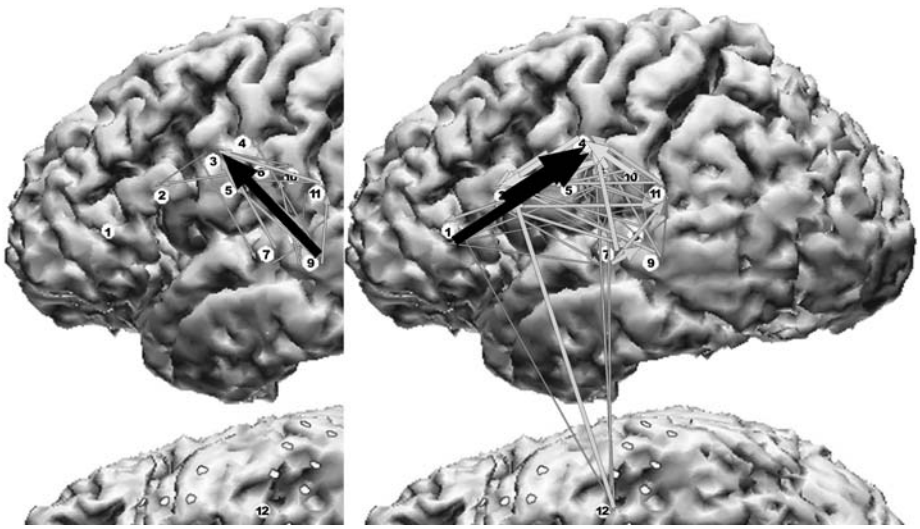
### 14.4.2 Application of ERC to Cortical Function Mapping

Application of ERC to human ECoG signals recorded during language tasks has yielded interesting observations that are generally consistent with the putative functional neuroanatomy and dynamics of human language. In particular, Korzeniewska et al. [148] applied ERC analyses to an auditory word repetition task in which the patient heard a series of spoken words and repeated each one aloud. Previous observations of event-related high-gamma activity during this and other language tasks led us to focus our ERC analyses on the causal interactions between signals in high-gamma frequencies [80, 81, 88].

Integrals of ERC calculated for high-gamma interactions (82 to 100 Hz) during auditory word repetition are illustrated in Figure 14.6. The magnitude of this integral is represented by the width of the arrow, and each arrow illustrates an increase of ERC in the frequency range 82 to 100 Hz. This frequency range was empirically derived based on the mean ERC over all time points and all pairs of analyzed channels. Its boundaries were defined by local minima of the averaged ERC. This was done in lieu of choosing an arbitrary frequency range in order to avoid artificial summation of flows related to different frequency bands.

During the word perception stage of word repetition [Figure 14.6(a)], the predominant increase in ERC for 82 to 100 Hz is observed from auditory association cortex to mouth/tongue motor cortex (E9 → E3). In addition, there are other, less prominent flows from auditory association cortex to mouth/tongue motor cortex (E9 → E5, E9 → E4, E7 → E3, and E8 → E3). Furthermore, there are also increases in flows from auditory association cortex to mouth/tongue motor cortex "via" supramarginal gyrus (BA40/Wernicke's area): E9 → E11 → E3. The second stage, the "spoken response" [Figure 14.6(b)], is characterized mainly by flow increases from Broca's area to tongue/mouth motor cortex (E2 → E4, E1 → E4), but also by smaller flow increases from Wernicke's area to mouth/tongue motor cortex (E7 → E4, E11 → E4) and from mouth/tongue motor cortex to Broca's area (E4 → E2, E3 → E2), perhaps reflecting the activation of feedback pathways while the patient speaks and hears her own spoken response.

Given the remaining uncertainties regarding the neural generators of high-gamma responses (see Section 14.3.4.1), the interpretation of ERC flows in high-gamma frequencies can only be provisional at this point. One can only speculate that if high-gamma activity at one recording site is generated by the synchronous neural firing (output) of a cortical population that projects to a separate, downstream population, it may have a time-dependent causal relationship with the high-gamma activity generated by the subsequent output of the downstream population. Although evidence for this and other potential interpretations is still lacking, the results that have been obtained with this and related methods to date illustrate the potential for iEEG and advanced signal analysis to study the dynamics of cortical networks at physiologically relevant temporal scales.




(a) (b)

**Figure 14.6** Integrals of ERC for the frequency range 82 to 100 Hz, calculated for two stages of an auditory word repetition task: (a) auditory perception (between stimulus onset and offset), and (b) verbal response (following the mean response onset). Arrows indicate the directionality of ERC, and the width and darkness of each arrow represent the magnitude of the ERC integral (only positive values are shown). (*From:* [148]. © 2008 Wiley-Liss Inc. Reprinted with permission.)

## 14.5 Future Applications of iEEG

Research utilizing quantitative analyses of iEEG recordings will continue to delineate the functional response properties of different electrophysiological correlates of cortical processing. These studies are expected to provide a basic foundation for future applications of iEEG and noninvasive electrophysiological recordings. High-gamma responses have now been observed in a variety of different functional neuroanatomic domains with relatively consistent functional response properties. This suggests that HGRs may serve as a general purpose index of cortical activation and information processing. However, much is still unknown about the neural substrates and functional response properties of this response, and the same can be said about the other electrophysiological indices that have been studied with iEEG and qEEG in general. Future applications of these responses may depend on a better understanding of their dependence on the cytoarchitectonics, functional connectivity, and types of processing in different cortical regions. Elucidation of these relationships will likely require further basic investigations in animals. Nevertheless, some applications of iEEG are moving forward without this information. For example, Miller et al. [60] recently showed that high-gamma ERS is sufficiently robust that it can be seen in single trials, for example, during a single handshake. This opens the door to real-time mapping of motor function. If the signal-to-noise ratio of single-trial HGRs is adequate, they could serve as a useful electrophysiological index for intraoperative brain mapping and for brain–computer interfaces. Indeed, a recent study has shown that high-gamma activity can be used to discriminate the direction of two-dimensional movements of a joystick [63].

The clinical applications of quantitative iEEG are currently limited to the relatively small number of patients undergoing epilepsy surgery. However, recent studies have demonstrated the feasibility of using MEG [71, 149] or even scalp EEG [150] to record high-gamma activity, suggesting that improved recording technology may soon greatly expand the clinical applications of high-gamma activity, as well as the utility of ERD/ERS, ERPs, and electrophysiological responses in general for exploring the neural mechanisms and functional dynamics of higher cognitive functions in humans.

Elucidation of functionally relevant patterns of organization in iEEG signals, like those of other complex electrophysiological signals, has required, and will continue to require, advanced signal processing methods implemented by a team of neuroscientists and biomedical engineers. Broader application of iEEG to the scientific study of human perception and cognition will require collaborations with cognitive scientists, experimental psychologists, and systems neuroscientists.

## Acknowledgments

The authors thank the editors, Nitish Thakor and Shanbao Tong, for their invitation to contribute this chapter, as well as the anonymous reviewer for valuable suggestions. Our research was supported by R01-NS40596.

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